2.3C: Organic Isomers

Isomers are molecules with the same chemical formula but have different structures, which creates different properties in the molecules.

Learning Objectives

- Give examples of isomers

Key Points

- Isomers are molecules with the same chemical formula but have different structures.
- Isomers differ in how their bonds are positioned to surrounding atoms.
- When the carbons are bound on the same side of the double bond, this is the cis configuration; if they are on opposite sides of the double bond, it is a trans configuration.
- Triglycerides, which show both cis and trans configurations, can occur as either saturated or unsaturated, depending upon how many hydrogen atoms they have attached to them.

Key Terms

- **fatty acid**: Any of a class of aliphatic carboxylic acids, of general formula CnH2n+1COOH, that occur combined with glycerol as animal or vegetable oils and fats.
- **isomer**: Any of two or more compounds with the same molecular formula but with different structure.

The three-dimensional placement of atoms and chemical bonds within organic molecules is central to understanding their chemistry. Molecules that share the same chemical formula but differ in the placement (structure) of their atoms and/or chemical bonds are known as isomers.

Structural Isomers

Structural isomers (such as butane and isobutane) differ in the placement of their covalent bonds. Both molecules have four carbons and ten hydrogens (C_4H_{10}), but the different arrangement of the atoms within the molecules leads to differences in their chemical properties. For example, due to their different chemical properties, butane is suited for use as a fuel for cigarette lighters and torches, whereas isobutane is suited for use as a refrigerant and a propellant in spray cans.

Geometric Isomers

Geometric isomers, on the other hand, have similar placements of their covalent bonds but differ in how these bonds are made to the surrounding atoms, especially in carbon-to-carbon double bonds. In the simple molecule butene (C_4H_8), the two methyl groups (CH_3) can be on either side of the double covalent bond central to the molecule. When the carbons are bound on the same side of the double bond, this is the cis configuration; if they are on opposite sides of the double bond, it is a trans configuration. In the trans configuration, the carbons form a more or less linear structure, whereas the carbons in the cis configuration make a bend (change in direction) of the carbon backbone.

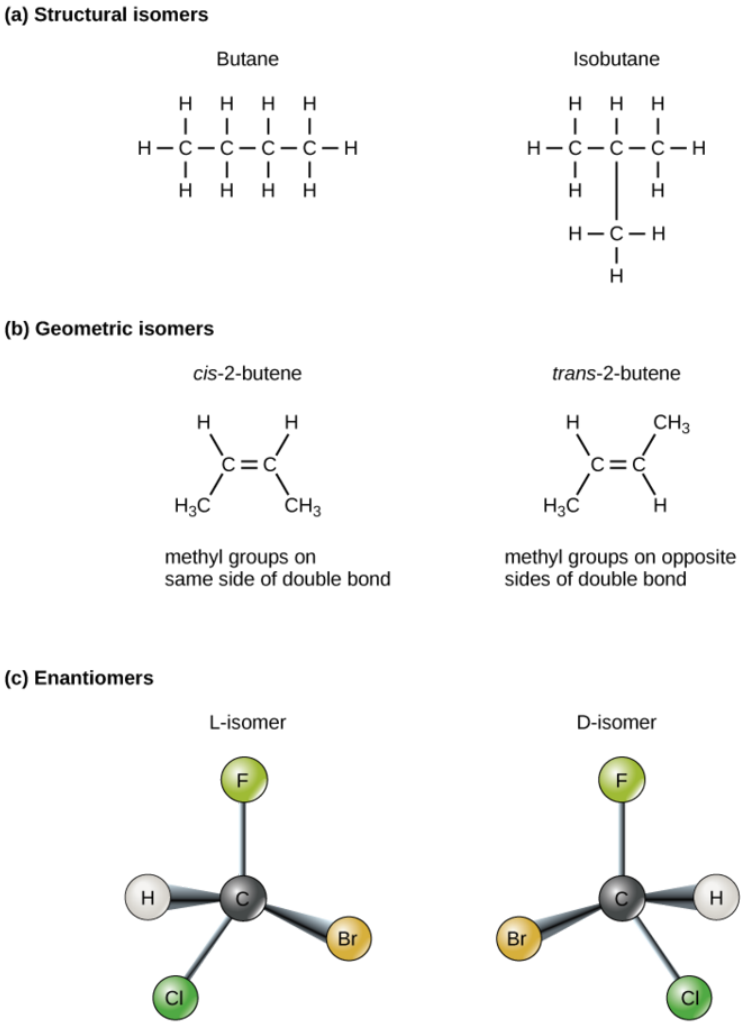


Figure 2.3C.1: **Isomers**: Molecules that have the same number and type of atoms arranged differently are called isomers. (a) Structural isomers have a different covalent arrangement of atoms. (b) Geometric isomers have a different arrangement of atoms around a double bond. (c) Enantiomers are mirror images of each other.

Cis or Trans Configurations

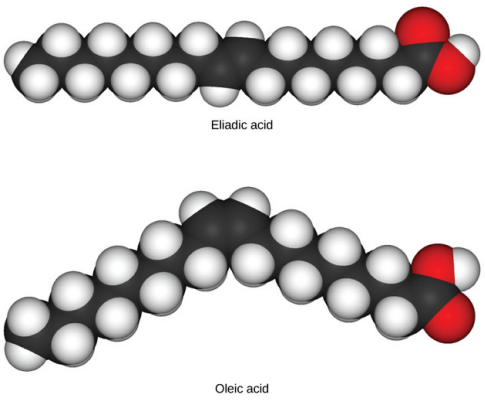


Figure 2.3C.1: **Cis and Trans Fatty Acids**: These space-filling models show a cis (oleic acid) and a trans (eliadic acid) fatty acid. Notice the bend in the molecule cause by the cis configuration.

In triglycerides (fats and oils), long carbon chains known as fatty acids may contain double bonds, which can be in either the cis or trans configuration. Fats with at least one double bond between carbon atoms are unsaturated fats. When some of these bonds are in the cis configuration, the resulting bend in the carbon backbone of the chain means that triglyceride

molecules cannot pack tightly, so they remain liquid (oil) at room temperature. On the other hand, triglycerides with trans double bonds (popularly called trans fats), have relatively linear fatty acids that are able to pack tightly together at room temperature and form solid fats.

In the human diet, trans fats are linked to an increased risk of cardiovascular disease, so many food manufacturers have reduced or eliminated their use in recent years. In contrast to unsaturated fats, triglycerides without double bonds between carbon atoms are called saturated fats, meaning that they contain all the hydrogen atoms available. Saturated fats are a solid at room temperature and usually of animal origin.

2.3D: Organic Enantiomers

Enantiomers share the same chemical structure and bonds but differ in the placement of atoms such that they are mirror images of each other.

Learning Objectives

- Give examples of enantiomers

Key Points

- Enantiomers are stereoisomers, a type of isomer where the order of the atoms in the two molecules is the same but their arrangement in space is different.
- Many molecules in the bodies of living beings are enantiomers; there is sometimes a large difference in the effects of two enantiomers on organisms.
- Enantiopure compounds refer to samples having, within the limits of detection, molecules of only one chirality.
- Compounds that are enantiomers of each other have the same physical properties except for the direction in which they rotate polarized light and how they interact with different optical isomers of other compounds.

Key Terms

- **enantiomer**: One of a pair of stereoisomers that is the mirror image of the other, but may not be superimposed on this other stereoisomer.
- **chirality**: The phenomenon in chemistry, physics, and mathematics in which objects are mirror images of each other, but are not identical.
- **stereoisomer**: one of a set of the isomers of a compound in which atoms are arranged differently about a chiral center; they exhibit optical activity

Enantiomers

Stereoisomers are a type of isomer where the order of the atoms in the two molecules is the same but their arrangement in space is different. The two main types of stereoisomerism are diastereomerism (including ‘cis-trans isomerism’) and optical isomerism (also known as ‘enantiomerism’ and ‘chirality’). Optical isomers are stereoisomers formed when asymmetric centers are present; for example, a carbon with four different groups bonded to it. Enantiomers are two optical isomers (i.e. isomers that are reflections of each other). Every stereocenter in one isomer has the opposite configuration in the other. They share the same chemical structure and chemical bonds, but differ in the three-dimensional placement of atoms so that they are mirror images, much as a person’s left and right hands are. Compounds that are enantiomers of each other have the same physical properties except for the direction in which they rotate polarized light and how they interact with different optical isomers of other compounds.

The amino acid alanine is example of an entantiomer. The two structures, D-alanine and L-alanine, are non-superimposable. In nature, only the L-forms of amino acids are used to make proteins. Some D forms of amino acids are seen in the cell walls of bacteria, but never in their proteins. Similarly, the D-form of glucose is the main product of photosynthesis and the L-form of the molecule is rarely seen in nature.

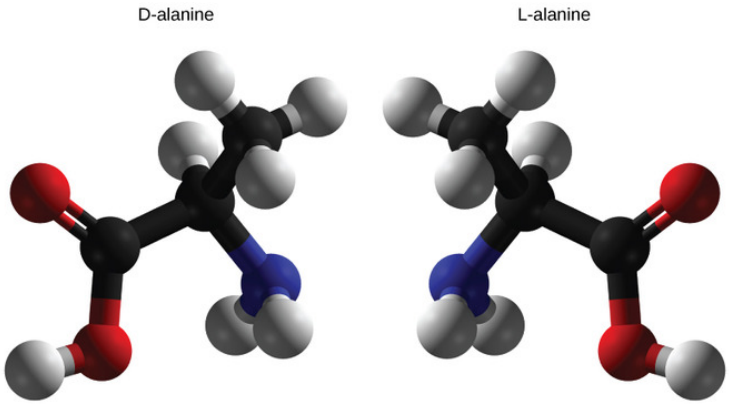


Figure 2.3D.1: **Enantiomers**: D-alanine and L-alanine are examples of enantiomers or mirror images. Only the L-forms of amino acids are used to make proteins.

Organic compounds that contain a chiral carbon usually have two non-superposable structures. These two structures are mirror images of each other and are, thus, commonly called enantiomorphs; hence, this structural property is now commonly referred to as enantiomerism. Enantiopure compounds refer to samples having, within the limits of detection, molecules of only one chirality.

Enantiomers of each other often show different chemical reactions with other substances that are also enantiomers. Since many molecules in the bodies of living beings are enantiomers themselves, there is sometimes a marked difference in the effects of two enantiomers on living beings. In drugs, for example, often only one of a drug's enantiomers is responsible for the desired physiologic effects, while the other enantiomer is less active, inactive, or sometimes even responsible for adverse effects. Owing to this discovery, drugs composed of only one enantiomer ("enantiopure") can be developed to enhance the pharmacological efficacy and sometimes do away with some side effects.

2.3E: Organic Molecules and Functional Groups

Functional groups are groups of molecules attached to organic molecules and give them specific identities or functions.

Learning Objectives

- Describe the importance of functional groups to organic molecules

Key Points

- Functional groups are collections of atoms that attach the carbon skeleton of an organic molecule and confer specific properties.
- Each type of organic molecule has its own specific type of functional group.
- Functional groups in biological molecules play an important role in the formation of molecules like DNA, proteins, carbohydrates, and lipids.
- Functional groups include: hydroxyl, methyl, carbonyl, carboxyl, amino, phosphate, and sulfhydryl.

Key Terms

- **hydrophobic**: lacking an affinity for water; unable to absorb, or be wetted by water
- **hydrophilic**: having an affinity for water; able to absorb, or be wetted by water

Location of Functional Groups

Functional groups are groups of atoms that occur within organic molecules and confer specific chemical properties to those molecules. When functional groups are shown, the organic molecule is sometimes denoted as “R.” Functional groups are found along the “carbon backbone” of macromolecules which is formed by chains and/or rings of carbon atoms with the occasional substitution of an element such as nitrogen or oxygen. Molecules with other elements in their carbon backbone are substituted hydrocarbons. Each of the four types of macromolecules—proteins, lipids, carbohydrates, and nucleic acids—has its own characteristic set of functional groups that contributes greatly to its differing chemical properties and its function in living organisms.

Properties of Functional Groups

A functional group can participate in specific chemical reactions. Some of the important functional groups in biological molecules include: hydroxyl, methyl, carbonyl, carboxyl, amino, phosphate, and sulfhydryl groups. These groups play an important role in the formation of molecules like DNA, proteins, carbohydrates, and lipids.

Classifying Functional Groups

Functional groups are usually classified as hydrophobic or hydrophilic depending on their charge or polarity. An example of a hydrophobic group is the non-polar methane molecule. Among the hydrophilic functional groups is the carboxyl group found in amino acids, some amino acid side chains, and the fatty acid heads that form triglycerides and phospholipids. This carboxyl group ionizes to release hydrogen ions (H^+) from the COOH group resulting in the negatively charged COO^- group; this contributes to the hydrophilic nature of whatever molecule it is found on. Other functional groups, such as the carbonyl group, have a partially negatively charged oxygen atom that may form hydrogen bonds with water molecules, again making the molecule more hydrophilic.

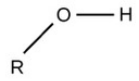
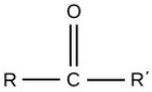
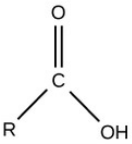
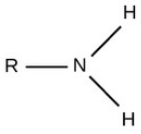
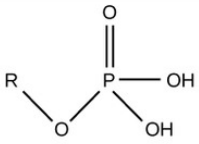
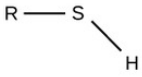
Functional Group	Structure	Properties
Hydroxyl		Polar
Methyl		Nonpolar
Carbonyl		Polar
Carboxyl		Charged, ionizes to release H^+. Since carboxyl groups can release H^+ ions into solution, they are considered acidic.
Amino		Charged, accepts H^+ to form NH_3^+. Since amino groups can remove H^+ from solution, they are considered basic.
Phosphate		Charged, ionizes to release H^+. Since phosphate groups can release H^+ ions into solution, they are considered acidic.
Sulfhydryl		Polar

Figure $2.3E.1$: **Examples of functional groups**: The functional groups shown here are found in many different biological molecules, where "R" is the organic molecule.

Hydrogen Bonds between Functional Groups

Hydrogen bonds between functional groups (within the same molecule or between different molecules) are important to the function of many macromolecules and help them to fold properly and maintain the appropriate shape needed to function correctly. Hydrogen bonds are also involved in various recognition processes, such as DNA complementary base pairing and the binding of an enzyme to its substrate.

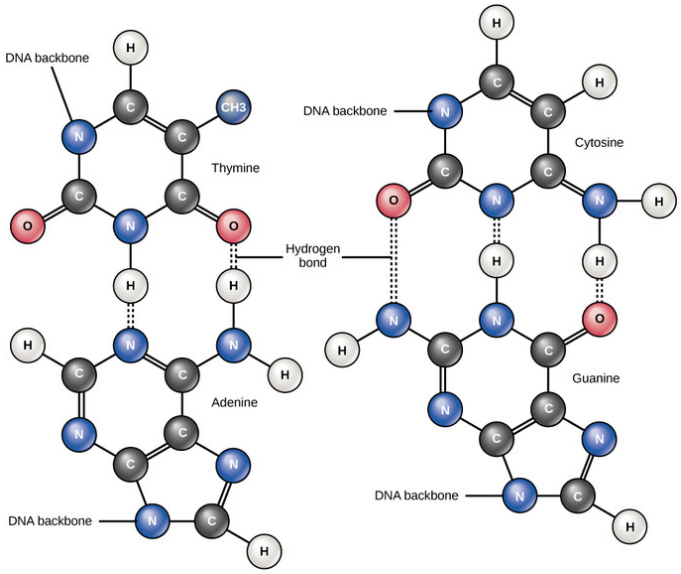


Figure 2.3E.1: **Hydrogen bonds in DNA**: Hydrogen bonds connect two strands of DNA together to create the double-helix structure.

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2.4: Synthesis of Biological Macromolecules

Topic hierarchy

2.4A: Types of Biological Macromolecules

Biological macromolecules, the large molecules necessary for life, include carbohydrates, lipids, nucleic acids, and proteins.

2.4B: Dehydration Synthesis

2.4C: Hydrolysis

2.4A: Types of Biological Macromolecules

Learning Objectives

- Identify the four major classes of biological macromolecules

Nutrients are the molecules that living organisms require for survival and growth but that animals and plants cannot synthesize themselves. Animals obtain nutrients by consuming food, while plants pull nutrients from soil.

Figure 2.4A.1: Sources of biological macromolecules: Foods such as bread, fruit, and cheese are rich sources of biological macromolecules.

Many critical nutrients are biological macromolecules. The term “macromolecule” was first coined in the 1920s by Nobel laureate Hermann Staudinger. Staudinger was the first to propose that many large biological molecules are built by covalently linking smaller biological molecules together.

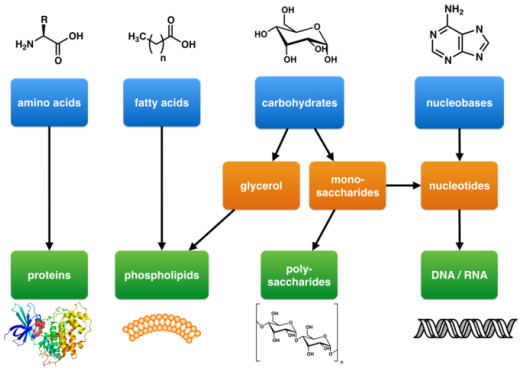
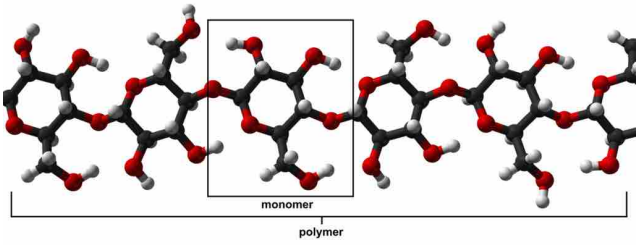


Figure 2.4A.2: Living organisms are made up of chemical building blocks: All organisms are composed of a variety of these biological macromolecules.

Monomers and Polymers

Biological macromolecules play a critical role in cell structure and function. Most (but not all) biological macromolecules are polymers, which are any molecules constructed by linking together many smaller molecules, called monomers. Typically all the monomers in a polymer tend to be the same, or at least very similar to each other, linked over and over again to build up the larger macromolecule. These simple monomers can be linked in many different combinations to produce complex biological polymers, just as a few types of Lego blocks can build anything from a house to a car.



Monomers and polymers: Many small monomer subunits combine to form this carbohydrate polymer.

Examples of these monomers and polymers can be found in the sugar you might put in your coffee or tea. Regular table sugar is the disaccharide sucrose (a polymer), which is composed of the monosaccharides fructose and glucose (which are monomers). If we were to string many carbohydrate monomers together we could make a polysaccharide like starch. The prefixes “mono-” (one), “di-” (two),and “poly-” (many) will tell you how many of the monomers have been joined together in a molecule.

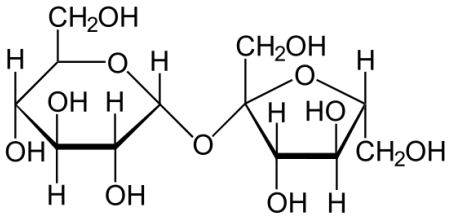


Figure 2.4A.3: The molecule sucrose (common table sugar): The carbohydrate monosaccharides (fructose and glucose) are joined to make the disaccharide sucrose.

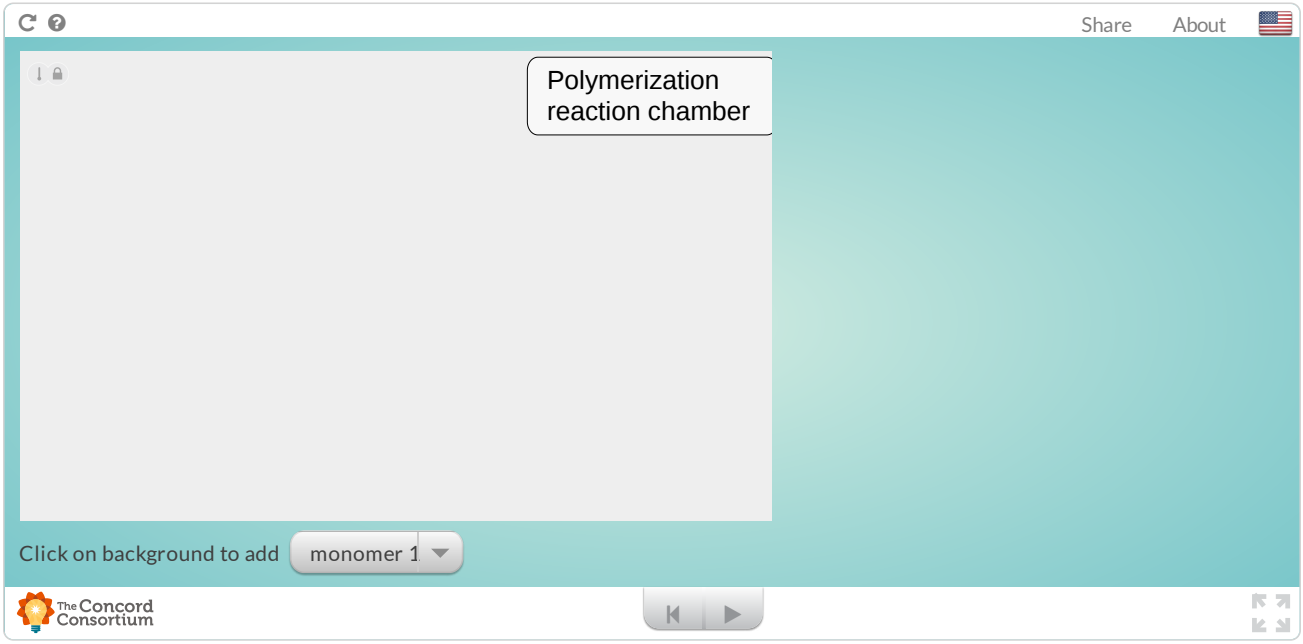
Biological macromolecules all contain carbon in ring or chain form, which means they are classified as organic molecules. They usually also contain hydrogen and oxygen, as well as nitrogen and additional minor elements.

Four Classes of Biological Macromolecules

There are four major classes of biological macromolecules:

1. carbohydrates
2. lipids
3. proteins
4. nucleic acids

Each of these types of macromolecules performs a wide array of important functions within the cell; a cell cannot perform its role within the body without many different types of these crucial molecules. In combination, these biological macromolecules make up the majority of a cell's dry mass. (Water molecules make up the majority of a cell's total mass.) All the molecules both inside and outside of cells are situated in a water-based (i.e., aqueous) environment, and all the reactions of biological systems are occurring in that same environment.



Carbohydrates, proteins, and nucleic acids are built from small molecular units that are connected to each other by strong covalent bonds. The small molecular units are called monomers (mono means one, or single), and they are linked together into long chains called polymers (poly means many, or multiple). Each different type of macromolecule, except lipids, is built from a different set of monomers that resemble each other in composition and size. Lipids are not polymers, because they are not built from monomers (units with similar composition).

Key Points

- Biological macromolecules are important cellular components and perform a wide array of functions necessary for the survival and growth of living organisms.
- The four major classes of biological macromolecules are carbohydrates, lipids, proteins, and nucleic acids.

Key Terms

- **polymer**: A relatively large molecule consisting of a chain or network of many identical or similar monomers chemically bonded to each other.
- **monomer**: A relatively small molecule that can form covalent bonds with other molecules of this type to form a polymer.

: Dehydration Synthesis

In dehydration synthesis, monomers combine with each other via covalent bonds to form polymers.

Learning Objectives

- Explain dehydration (or condensation) reactions

Key Points

- During dehydration synthesis, either the hydrogen of one monomer combines with the hydroxyl group of another monomer releasing a molecule of water, or two hydrogens from one monomer combine with one oxygen from the other monomer releasing a molecule of water.
- The monomers that are joined via dehydration synthesis reactions share electrons and form covalent bonds with each other.
- As additional monomers join via multiple dehydration synthesis reactions, this chain of repeating monomers begins to form a polymer.
- Complex carbohydrates, nucleic acids, and proteins are all examples of polymers that are formed by dehydration synthesis.
- Monomers like glucose can join together in different ways and produce a variety of polymers. Monomers like mononucleotides and amino acids join together in different sequences to produce a variety of polymers.

Key Terms

- **covalent bond**: A type of chemical bond where two atoms are connected to each other by the sharing of two or more electrons.
- **monomer**: A relatively small molecule which can be covalently bonded to other monomers to form a polymer.

Dehydration Synthesis

Most macromolecules are made from single subunits, or building blocks, called monomers. The monomers combine with each other via covalent bonds to form larger molecules known as polymers. In doing so, monomers release water molecules as byproducts. This type of reaction is known as dehydration synthesis, which means “to put together while losing water. ” It is also considered to be a condensation reaction since two molecules are condensed into one larger molecule with the loss of a smaller molecule (the water.)

In a dehydration synthesis reaction between two un-ionized monomers, such as monosaccharide sugars, the hydrogen of one monomer combines with the hydroxyl group of another monomer, releasing a molecule of water in the process. The removal of a hydrogen from one monomer and the removal of a hydroxyl group from the other monomer allows the monomers to share electrons and form a covalent bond. Thus, the monomers that are joined together are being *dehydrated* to allow for *synthesis* of a larger molecule.

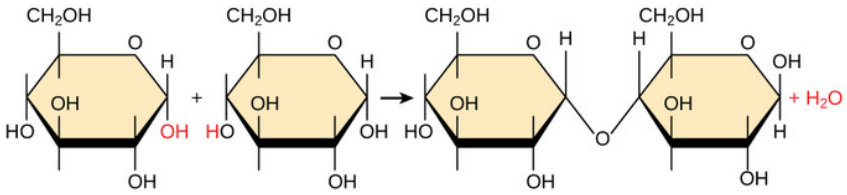


Figure 2.4B.1: **A dehydration synthesis reaction involving un-ionized moners..**: In the dehydration synthesis reaction between two molecules of glucose, a hydroxyl group from the first glucose is combined with a hydrogen from the second glucose, creating a covalent bond that links the two monomeric sugars (monosaccharides) together to form the dissacharide maltose. In the process, a water molecule is formed.

When the monomers are ionized, such as is the case with amino acids in an aqueous environment like cytoplasm, two hydrogens from the positively-charged end of one monomer are combined with an oxygen from the negatively-charged end of another monomer, again forming water, which is released as a side-product, and again joining the two monomers with a covalent bond.

Figure $2.4B.1$: **A dehydration synthesis reaction involving ionized monomers.**: In the dehydration synthesis reaction between two amino acids, with are ionized in aqueous environments like the cell, an oxygen from the first amino acid is combined with two hydrogens from the second amino acid, creating a covalent bond that links the two monomers together to form a dipeptide. In the process a water molecule is formed.

As additional monomers join via multiple dehydration synthesis reactions, the chain of repeating monomers begins to form a polymer. Different types of monomers can combine in many configurations, giving rise to a diverse group of macromolecules. Three of the four major classes of biological macromolecules (complex carbohydrates, nucleic acids, and proteins), are composed of monomers that join together via dehydration synthesis reactions. Complex carbohydrates are formed from monosaccharides, nucleic acids are formed from mononucleotides, and proteins are formed from amino acids.

There is great diversity in the manner by which monomers can combine to form polymers. For example, glucose monomers are the constituents of starch, glycogen, and cellulose. These three are polysaccharides, classified as carbohydrates, that have formed as a result of multiple dehydration synthesis reactions between glucose monomers. However, the manner by which glucose monomers join together, specifically locations of the covalent bonds between connected monomers and the orientation (stereochemistry) of the covalent bonds, results in these three different polysaccharides with varying properties and functions. In nucleic acids and proteins, the location and stereochemistry of the covalent linkages connecting the monomers do not vary from molecule to molecule, but instead the multiple kinds of monomers (five different monomers in nucleic acids, A, G, C, T, and U mononucleotides; 21 different amino acids monomers in proteins) are combined in a huge variety of sequences. Each protein or nucleic acid with a different sequence is a different molecule with different properties.

2.4C: Hydrolysis

Hydrolysis reactions result in the breakdown of polymers into monomers by using a water molecule and an enzymatic catalyst.

Learning Objectives

- Explain hydrolysis reactions

Key Points

- Hydrolysis reactions use water to breakdown polymers into monomers and is the opposite of dehydration synthesis, which forms water when synthesizing a polymer from monomers.
- Hydrolysis reactions break bonds and release energy.
- Biological macromolecules are ingested and hydrolyzed in the digestive tract to form smaller molecules that can be absorbed by cells and then further broken down to release energy.

Key Terms

- **enzyme**: a globular protein that catalyses a biological chemical reaction
- **hydrolysis**: A chemical process of decomposition involving the splitting of a bond by the addition of water.

Hydrolysis

Polymers are broken down into monomers in a process known as hydrolysis, which means “to split water,” a reaction in which a water molecule is used during the breakdown. During these reactions, the polymer is broken into two components. If the components are un-ionized, one part gains a hydrogen atom (H-) and the other gains a hydroxyl group (OH–) from a split water molecule. This is what happens when monosaccharides are released from complex carbohydrates via hydrolysis.

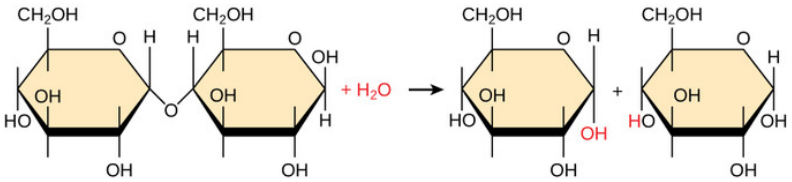


Figure 2.4C.1: **Hydrolysis reaction generating un-ionized products.**: In the hydrolysis reaction shown here, the disaccharide maltose is broken down to form two glucose monomers with the addition of a water molecule. One glucose gets a hydroxyl group at the site of the former covalent bond, the other glucose gets a hydrogen atom. This is the reverse of the dehydration synthesis reaction joining these two monomers.

If the components are ionized after the split, one part gains two hydrogen atoms and a positive charge, the other part gains an oxygen atom and a negative charge. This is what happens when amino acids are released from protein chains via hydrolysis.

image

Figure 2.4C.1: **Hydrolysis reaction generating ionized products.**: In the hydrolysis reaction shown here, the dipeptide is broken down to form two ionized amino acids with the addition of a water molecule. One amino acid gets an oxygen atom and a negative charge, the other amino acid gets two hydrogen atoms and a positive charge. This is the reverse of the dehydration synthesis reaction joining these two monomers.

These reactions are in contrast to dehydration synthesis (also known as condensation) reactions. In dehydration synthesis reactions, a water molecule is formed as a result of generating a covalent bond between two monomeric components in a larger polymer. In hydrolysis reactions, a water molecule is consumed as a result of breaking the covalent bond holding together two components of a polymer.

Dehydration and hydrolysis reactions are chemical reactions that are catalyzed, or "sped up," by specific enzymes; dehydration reactions involve the formation of new bonds, requiring energy, while hydrolysis reactions break bonds and release energy.

In our bodies, food is first hydrolyzed, or broken down, into smaller molecules by catalytic enzymes in the digestive tract. This allows for easy absorption of nutrients by cells in the intestine. Each macromolecule is broken down by a specific enzyme. For instance, carbohydrates are broken down by amylase, sucrase, lactase, or maltase. Proteins are broken down by the enzymes trypsin, pepsin, peptidase and others. Lipids are broken down by lipases. Once the smaller metabolites that result from these hydrolytic enzymezes are absorbed by cells in the body, they are further broken down by other enzymes. The breakdown of these macromolecules is an overall energy-releasing process and provides energy for cellular activities.

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CHAPTER OVERVIEW

20: PHYLOGENIES AND THE HISTORY OF LIFE

20.1: ORGANIZING LIFE ON EARTH

20.1A: PHYLOGENETIC TREES
Phylogenetic trees illustrate the hypothetical evolution of organisms and their relationship to other species.

20.1B: LIMITATIONS OF PHYLOGENETIC TREES
Limitations of phylogenetic trees include the inability to distinguish evolutionary time and relatedness between distinct species.

20.1C: THE LEVELS OF CLASSIFICATION
20.2: DETERMINING EVOLUTIONARY RELATIONSHIPS

20.2A: DISTINGUISHING BETWEEN SIMILAR TRAITS
20.2B: BUILDING PHYLOGENETIC TREES
20.3: PERSPECTIVES ON THE PHYLOGENETIC TREE

20.3A: LIMITATIONS TO THE CLASSIC MODEL OF PHYLOGENETIC TREES
20.3B: HORIZONTAL GENE TRANSFER
20.3C: ENDOSYMBIOTIC THEORY AND THE EVOLUTION OF EUKARYOTES
20.3D: WEB, NETWORK, AND RING OF LIFE MODELS

20.1: Organizing Life on Earth

20.1A: Phylogenetic Trees

Phylogenetic trees illustrate the hypothetical evolution of organisms and their relationship to other species.

20.1B: Limitations of Phylogenetic Trees

Limitations of phylogenetic trees include the inability to distinguish evolutionary time and relatedness between distinct species.

20.1C: The Levels of Classification

20.1A: Phylogenetic Trees

LEARNING OBJECTIVES

- Describe the various types of phylogenetic trees and how they organize life

Scientists use a tool called a phylogenetic tree, a type of diagram, to show the evolutionary pathways and connections among organisms. Scientists consider phylogenetic trees to be a hypothesis of the evolutionary past since one cannot go back to confirm the proposed relationships. In other words, a “tree of life”, as it is sometimes called, can be constructed to illustrate when different organisms evolved and to show the relationships among different organisms.

Unlike a taxonomic classification diagram, a phylogenetic tree can be read like a map of evolutionary history. Many phylogenetic trees have a single lineage at the base representing a common ancestor. Scientists call such trees ‘rooted,’ which means there is a single ancestral lineage (typically drawn from the bottom or left) to which all organisms represented in the diagram relate. Notice in the rooted phylogenetic tree that the three domains (Bacteria, Archaea, and Eukarya) diverge from a single point and branch off. The small branch that plants and animals (including humans) occupy in this diagram shows how recent and miniscule these groups are compared with other organisms. Unrooted trees don’t show a common ancestor but do show relationships among species.

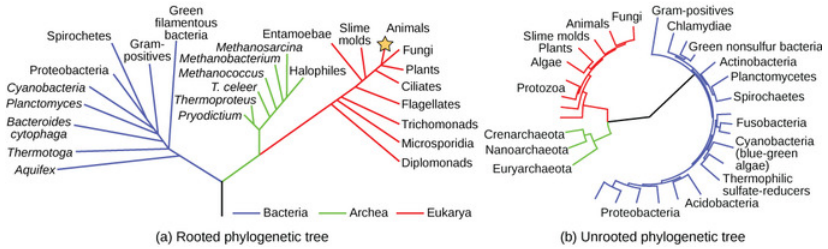

Figure 20.1A.1: **Phylogenetic trees**: Both of these phylogenetic trees shows the relationship of the three domains of life (Bacteria, Archaea, and Eukarya), but the (a) rooted tree attempts to identify when various species diverged from a common ancestor, while the (b) unrooted tree does not.

In a rooted tree, the branching indicates evolutionary relationships. The point where a split occurs, called a branch point, represents where a single lineage evolved into a distinct new one. A lineage that evolved early from the root and remains unbranched is called basal taxon. When two lineages stem from the same branch point, they are called sister taxa. A branch with more than two lineages is called a polytomy and serves to illustrate where scientists have not definitively determined all of the relationships. It is important to note that although sister taxa and polytomy do share an ancestor, it does not mean that the groups of organisms split or evolved from each other. Organisms in two taxa may have split apart at a specific branch point, but neither taxa gave rise to the other.

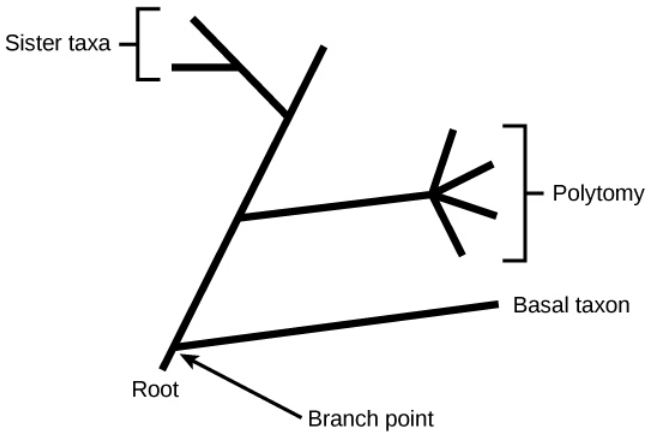

Figure 20.1A.1: **Rooted phylogenetic trees**: The root of a phylogenetic tree indicates that an ancestral lineage gave rise to all organisms on the tree. A branch point indicates where two lineages diverged. A lineage that evolved early and remains unbranched is a basal taxon. When two lineages stem from the same branch point, they are sister taxa. A branch with more than two lineages is a polytomy.

Rooted phylogenetic trees can serve as a pathway to understanding evolutionary history. The pathway can be traced from the origin of life to any individual species by navigating through the evolutionary branches between the two points. Also, by starting with a single species and tracing back towards the “trunk” of the tree, one can discover that species’ ancestors, as well as where lineages share a common ancestry. In addition, the tree can be used to study entire groups of organisms.

Another point to mention on phylogenetic tree structure is that rotation at branch points does not change the information. For example, if a branch point was rotated and the taxon order changed, this would not alter the information because the evolution of each taxon from the branch point was independent of the other.

Many disciplines within the study of biology contribute to understanding how past and present life evolved over time; together, these disciplines contribute to building, updating, and maintaining the “tree of life.” Information is used to organize and classify organisms based on evolutionary relationships in a scientific field called systematics. Data may be collected from fossils, from studying the structure of body parts or molecules used by an organism, and by DNA analysis. By combining data from many sources, scientists can put together the phylogeny of an organism. Since phylogenetic trees are hypotheses, they will continue to change as new types of life are discovered and new information is learned.

Key Points

- Rooted trees have a single lineage at the base representing a common ancestor that connects all organisms presented in a phylogenetic diagram.
- Branch points in a phylogenetic tree represent a split where a single lineage evolved into a distinct new one, while basal taxon depict unbranched lineages that evolved early from the root.
- Unrooted trees portray relationships among species, but do not depict their common ancestor.
- Phylogenetic trees are hypotheses and are, therefore, modified as data becomes available.
- Systematics uses data from fossils, the study of bodily structures, molecules used by a species, and DNA analysis to contribute to the building, updating, and maintaining of phylogenetic trees.

Key Terms

- **polytomy**: a section of a phylogeny in which the evolutionary relationships cannot be fully resolved to dichotomies
- **basal taxon**: a lineage, displayed using a phylogenetic tree, that evolved early from the root and from which no other branches have diverged
- **systematics**: research into the relationships of organisms; the science of systematic classification
- **phylogeny**: the visual representation of the evolutionary history of organisms; based on rigorous analyses

20.1B: Limitations of Phylogenetic Trees

Limitations of phylogenetic trees include the inability to distinguish evolutionary time and relatedness between distinct species.

Learning Objectives

- Identify the limitations of phylogenetic trees as representations of the organization of life

Key Points

- Closely-related species may not always look more alike, while groups that are not closely related yet evolved under similar conditions, may appear more similar to each other.
- In phylogenetic trees, branches do not usually account for length of time and only depict evolutionary order.
- Phylogenetic trees are like real trees in that they do not simply grow in only one direction after a new branch forms; the evolution of one organism does not necessarily signify the evolutionary end of another.

Key Terms

- **phenotypical**: of or pertaining to a phenotype: the appearance of an organism based on a multifactorial combination of genetic traits and environmental factors

Limitations of Phylogenetic Trees

It may be easy to assume that more closely-related organisms look more alike; while this is often the case, it is not always true. If two closely-related lineages evolved under significantly varied surroundings or after the evolution of a major new adaptation, it is possible for the two groups to appear more different than other groups that are not as closely related. For example, the phylogenetic tree shows that lizards and rabbits both have amniotic eggs, whereas frogs do not; yet lizards and frogs appear more similar than lizards and rabbits.

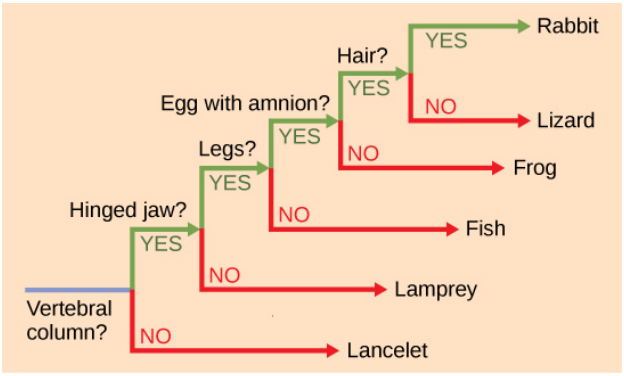

Figure 20.1*B*.1: **Limitations of phylogenetic trees**: This ladder-like phylogenetic tree of vertebrates is rooted by an organism that lacked a vertebral column. At each branch point, organisms with different characters are placed in different groups based on the characteristics they share.

Another aspect of phylogenetic trees is that, unless otherwise indicated, the branches do not account for length of time, only the evolutionary order. In other words, the length of a branch does not typically mean more time passed; nor does a short branch mean less time passed, unless specified on the diagram. A tree may not indicate how much time passed between the evolution of amniotic eggs and hair. What the tree does show is the order in which things took place. For example, the tree in the diagram shows that the oldest trait is the vertebral column, followed by hinged jaws, and so forth. Remember, any phylogenetic tree is a part of the greater whole and, as with a real tree, it does not grow in only one direction after a new branch develops. So, simply because a vertebral column evolved does not mean that invertebrate evolution ceased. It only means that a new branch formed. Also, groups that are not closely related, but evolve under similar conditions, may appear more phenotypically similar to each other than to a close relative.

20.1C: The Levels of Classification

Taxanomic classification divides species in a hierarchical system beginning with a domain and ending with a single species.

Learning Objectives

- Describe how taxonomic classification of organisms is accomplished and detail the levels of taxonomic classification from domain to species

Key Points

- Categories within taxonomic classification are arranged in increasing specificity.
- The most general category in taxonomic classification is domain, which is the point of origin for all species; all species belong to one of these domains: Bacteria, Archaea, and Eukarya.
- Within each of the three domains, we find kingdoms, the second category within taxonomic classification, followed by subsequent categories that include phylum, class, order, family, genus, and species.
- At each classification category, organisms become more similar because they are more closely related.
- As scientific technology advances, changes to the taxonomic classification of many species must be altered as inaccuracies in classifications are discovered and corrected.

Key Terms

- **binomial nomenclature**: the scientific system of naming each species of organism with a Latinized name in two parts
- **taxon**: any of the taxonomic categories such as phylum or subspecies
- **Linnaeus**: Swedish botanist, physician and zoologist who laid the foundations for the modern scheme of nomenclature; known as the “father of modern taxonomy”

The Levels of Classification

Taxonomy (which literally means “arrangement law”) is the science of classifying organisms to construct internationally-shared classification systems with each organism placed into more and more inclusive groupings. Think about how a grocery store is organized. One large space is divided into departments, such as produce, dairy, and meats. Then each department further divides into aisles, then each aisle into categories and brands, and then, finally, a single product. This organization from larger to smaller, more-specific categories is called a hierarchical system.

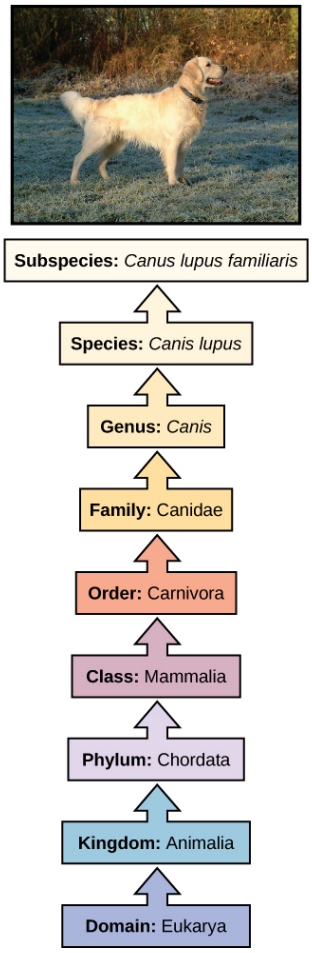

Figure 20.1C.1: **Hierarchical models**: The taxonomic classification system uses a hierarchical model to organize living organisms into increasingly specific categories. The common dog, Canis lupus familiaris, is a subspecies of Canis lupus, which also includes the wolf and dingo.

The taxonomic classification system (also called the Linnaean system after its inventor, Carl Linnaeus, a Swedish botanist, zoologist, and physician) uses a hierarchical model. Moving from the point of origin, the groups become more specific, until one branch ends as a single species. For example, after the common beginning of all life, scientists divide organisms into three large categories called domains: Bacteria, Archaea, and Eukarya. Within each domain is a second category called a kingdom. After kingdoms, the subsequent categories of increasing specificity are: phylum, class, order, family, genus, and species.

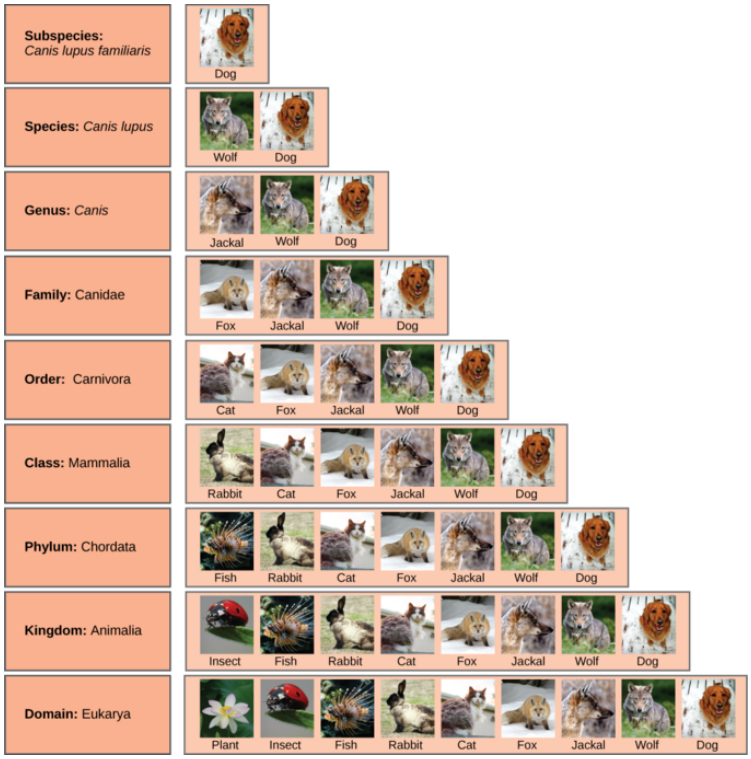

Figure 20.1C.1: **Levels in taxonomic classification**: At each sublevel in the taxonomic classification system, organisms become more similar. Dogs and wolves are the same species because they can breed and produce viable offspring, but they are different enough to be classified as different subspecies.

The kingdom Animalia stems from the Eukarya domain. The full name of an organism technically has eight terms. For dogs, it is: Eukarya, Animalia, Chordata, Mammalia, Carnivora, Canidae, *Canis*, and *lupus*. Notice that each name is capitalized except for species and that genus and species names are italicized. Scientists generally refer to an organism only by its genus and species, which is its two-word scientific name, in what is called binomial nomenclature. Therefore, the scientific name of the dog is *Canis lupus*. The name at each level is also called a taxon. In other words, dogs are in order Carnivora. Carnivora is the name of the taxon at the order level; Canidae is the taxon at the family level, and so forth. Organisms also have a common name that people typically use; in this case, dog. Note that the dog is additionally a subspecies: the “familiaris” in *Canis lupus familiaris*. Subspecies are members of the same species that are capable of mating and reproducing viable offspring, but they are considered separate subspecies due to geographic or behavioral isolation or other factors.

Dogs actually share a domain (Eukarya) with the widest diversity of organisms, including plants and butterflies. At each sublevel, the organisms become more similar because they are more closely related. Historically, scientists classified organisms using physical characteristics, but as DNA technology developed, more precise phylogenies have been determined.

Recent genetic analysis and other advancements have found that some earlier phylogenetic classifications do not align with the evolutionary past; therefore, changes and updates must be made as new discoveries occur. Recall that phylogenetic trees are hypotheses and are modified as data becomes available. In addition, classification historically has focused on grouping organisms mainly by shared characteristics and does not necessarily illustrate how the various groups relate to each other from an evolutionary perspective. For example, despite the fact that a hippopotamus resembles a pig more than a whale, the hippopotamus may be the closest living relative to the whale.

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20.2: Determining Evolutionary Relationships

Topic hierarchy

- 20.2A: Distinguishing between Similar Traits
- 20.2B: Building Phylogenetic Trees

20.2A: Distinguishing between Similar Traits

Similar traits can be either homologous structures that share an embryonic origin or analogous structures that share a function.

Learning Objectives

- Explain the difference between homologous and analogous structures

Key Points

- Organisms may be very closely related, even though they look quite different, due to a minor genetic change that caused a major morphological difference.
- Unrelated organisms may appear very similar because both organisms developed common adaptations that evolved within similar environmental conditions.
- To determine the phylogeny of an organism, scientists must determine whether a similarity is homologous or analogous.
- The advancement of DNA technology, the area of molecular systematics, describes the use of information on the molecular level, including DNA analysis.

Key Terms

- **analogous**: when similar similar physical features occur in organisms because of environmental constraints and not due to a close evolutionary relationship
- **homologous**: when similar physical features and genomes stem from developmental similarities that are based on evolution
- **phylogeny**: the evolutionary history of an organism
- **molecular systematics**: molecular phylogenetics is the analysis of hereditary molecular differences, mainly in DNA sequences, to gain information on an organism's evolutionary relationships

Two Options for Similarities

In general, organisms that share similar physical features and genomes tend to be more closely related than those that do not. Such features that overlap both morphologically (in form) and genetically are referred to as homologous structures; they stem from developmental similarities that are based on evolution. For example, the bones in the wings of bats and birds have homologous structures.

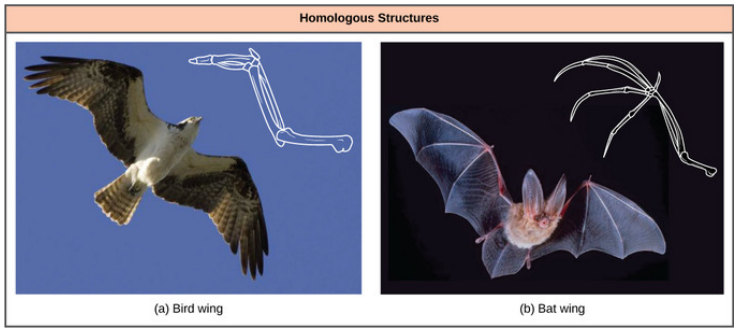

Figure 20.2A.1: **Homologous structures**: Bat and bird wings are homologous structures, indicating that bats and birds share a common evolutionary past.

Notice it is not simply a single bone, but rather a grouping of several bones arranged in a similar way. The more complex the feature, the more probable that any overlap is due to a common evolutionary past. Imagine two people from different countries both inventing a car with all the same parts and in exactly the same arrangement without any previous or shared knowledge. That outcome would be highly improbable. However, if two people both invented a hammer, it would be reasonable to conclude that both could have the original idea without the help of the other. The same relationship between complexity and shared evolutionary history is true for homologous structures in organisms.

Misleading Appearances

Some organisms may be very closely related, even though a minor genetic change caused a major morphological difference to make them look quite different. Similarly, unrelated organisms may be distantly related, but appear very similar. This usually happens because both organisms developed common adaptations that evolved within similar environmental conditions. When similar characteristics occur because of environmental constraints and not due to a close evolutionary relationship, it is called an analogy or homoplasy. For example, insects use wings to fly like bats and birds, but the wing structure and embryonic origin is completely different. These are called analogous structures.

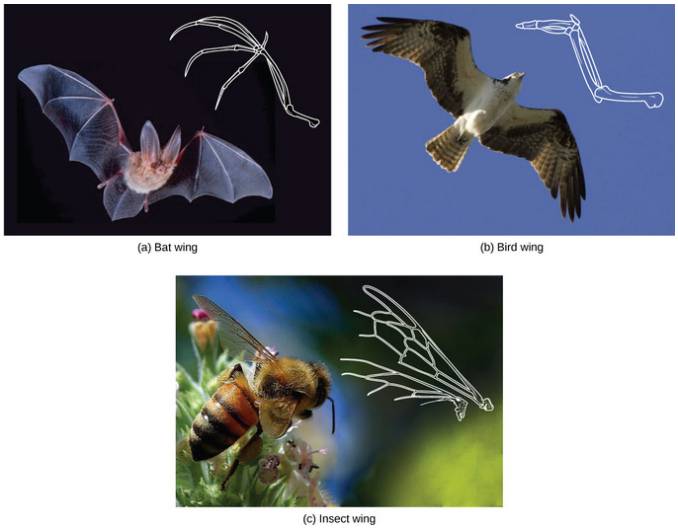

Figure 20.2*A*.1: **Analogous structures**: The (c) wing of a honeybee is similar in shape to a (b) bird wing and (a) bat wing, and it serves the same function. However, the honeybee wing is not composed of bones and has a distinctly-different structure and embryonic origin. These wing types (insect versus bat and bird) illustrate an analogy: similar structures that do not share an evolutionary history.

Similar traits can be either homologous or analogous. Homologous structures share a similar embryonic origin; analogous organs have a similar function. For example, the bones in the front flipper of a whale are homologous to the bones in the human arm. These structures are not analogous. The wings of a butterfly and the wings of a bird are analogous, but not homologous. Some structures are both analogous and homologous: the wings of a bird and the wings of a bat are both homologous and analogous. Scientists must determine which type of similarity a feature exhibits to decipher the phylogeny of the organisms being studied.

Molecular Comparisons

With the advancement of DNA technology, the area of molecular systematics, which describes the use of information on the molecular level including DNA analysis, has blossomed. New computer programs not only confirm many earlier classified organisms, but also uncover previously-made errors. As with physical characteristics, even the DNA sequence can be tricky to read in some cases. For some situations, two very closely-related organisms can appear unrelated if a mutation occurred that caused a shift in the genetic code. An insertion or deletion mutation would move each nucleotide base over one place, causing two similar codes to appear unrelated.

Sometimes two segments of DNA code in distantly-related organisms randomly share a high percentage of bases in the same locations, causing these organisms to appear closely related when they are not. For both of these situations, computer technologies have been developed to help identify the actual relationships. Ultimately, the coupled use of both morphologic and molecular information is more effective in determining phylogeny.

20.2B: Building Phylogenetic Trees

A phylogenetic tree sorts organisms into clades or groups of organisms that descended from a single ancestor using maximum parsimony.

Learning Objectives

- Describe the cladistics as a method used to create phylogenetic trees

Key Points

- Phylogenetic trees sort organisms into clades: groups of organisms that descended from a single ancestor.
- Organisms of a single clade are called a monophyletic group.
- Scientists use the phrase “descent with modification” because genetic changes occur even though related organisms have many of the same characteristics and genetic codes.
- A characteristic is considered a shared-ancestral character if it is found in the ancestor of a group and all of the organisms in the taxon or clade have that trait.
- If only some of the organisms have a certain trait, it is called a shared- derived character because this trait derived at some point, but does not include all of the ancestors in the clade.
- Scientists often use a concept called maximum parsimony, which means that events occurred in the simplest, most obvious way, to aid in the tremendous task of describing phylogenies accurately.

Key Terms

- **monophyletic**: of, pertaining to, or affecting a single phylum (or other taxon) of organisms
- **derived**: of, or pertaining to, conditions unique to the descendant species of a clade, and not found in earlier ancestral species
- **clades**: groups of organisms that descended from a single ancestor
- **ancestral**: of, pertaining to, derived from, or possessed by, an ancestor or ancestors; as, an ancestral estate
- **maximum parsimony**: the preferred phylogenetic tree is the tree that requires the least evolutionary change to explain some observed data

Building Phylogenetic Trees

After the homologous and analogous traits are sorted, scientists often organize the homologous traits using a system called cladistics. This system sorts organisms into clades: groups of organisms that descended from a single ancestor. For example, all of the organisms in the orange region evolved from a single ancestor that had amniotic eggs. Consequently, all of these organisms also have amniotic eggs and make a single clade, also called a monophyletic group. Clades must include all of the descendants from a branch point.

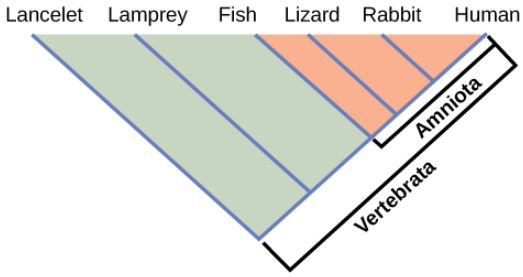

Figure 20.2B.1: **Common ancestors**: Lizards, rabbits, and humans all descend from a common ancestor that had an amniotic egg. Thus, lizards, rabbits, and humans all belong to the clade Amniota. Vertebrata is a larger clade that also includes fish and lamprey.

Clades can vary in size depending on which branch point is being referenced. The important factor is that all of the organisms in the clade or monophyletic group stem from a single point on the tree. This can be remembered because monophyletic breaks down into “mono,” meaning one, and “phyletic,” meaning evolutionary relationship. Notice in the various examples of clades how each clade comes from a single point, whereas the non-clade groups show branches that do not share a single point.

Shared Characteristics

Organisms evolve from common ancestors and then diversify. Scientists use the phrase “descent with modification” because even though related organisms have many of the same characteristics and genetic codes, changes occur. This pattern repeats as one goes through the phylogenetic tree of life:

1. A change in the genetic makeup of an organism leads to a new trait which becomes prevalent in the group.
2. Many organisms descend from this point and have this trait.
3. New variations continue to arise: some are adaptive and persist, leading to new traits.
4. With new traits, a new branch point is determined (go back to step 1 and repeat).

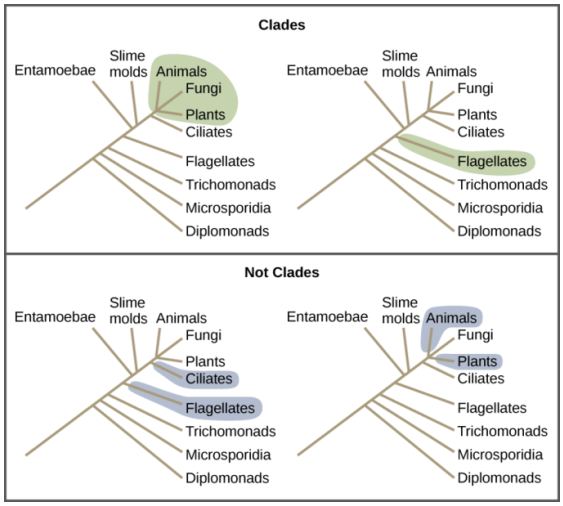

Figure 20.2B.1: **Examples of clades**: All the organisms within a clade stem from a single point on the tree. A clade may contain multiple groups, as in the case of animals, fungi, and plants, or a single group, as in the case of flagellates. Groups that diverge at a different branch point, or that do not include all groups in a single branch point, are not considered clades.

If a characteristic is found in the ancestor of a group, it is considered a shared-ancestral character because all of the organisms in the taxon or clade have that trait. Now, consider the amniotic egg characteristic in the same figure. Only some of the organisms have this trait; to those that do, it is called a shared-derived character because this trait derived at some point, but does not include all of the ancestors in the tree. The tricky aspect to shared-ancestral and shared-derived characters is the fact that these terms are relative. The same trait can be considered one or the other depending on the particular diagram being used. These terms help scientists distinguish between clades in the building of phylogenetic trees.

Choosing the Right Relationships

Imagine being the person responsible for organizing all of the items in a department store properly; an overwhelming task. Organizing the evolutionary relationships of all life on earth proves much more difficult: scientists must span enormous blocks of time and work with information from long-extinct organisms. Trying to decipher the proper connections, especially given the presence of homologies and analogies, makes the task of building an accurate tree of life extraordinarily difficult. Add to that the advancement of DNA technology, which now provides large quantities of genetic sequences to be used and analyzed. Taxonomy is a subjective discipline: many organisms have more than one connection to each other, so each taxonomist will decide the order of connections.

To aid in the tremendous task of describing phylogenies accurately, scientists often use a concept called maximum parsimony, which means that events occurred in the simplest, most obvious way. For example, if a group of people entered a forest preserve to go hiking, based on the principle of maximum parsimony, one could predict that most of the people would hike on established trails rather than forge new ones. For scientists deciphering evolutionary pathways, the same idea is used: the pathway of evolution probably includes the fewest major events that coincide with the evidence at hand. Starting with all of the homologous traits in a group of organisms, scientists look for the most obvious and simple order of evolutionary events that led to the occurrence of those traits.

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20.3: Perspectives on the Phylogenetic Tree

Topic hierarchy

- 20.3A: Limitations to the Classic Model of Phylogenetic Trees
- 20.3B: Horizontal Gene Transfer
- 20.3C: Endosymbiotic Theory and the Evolution of Eukaryotes
- 20.3D: Web, Network, and Ring of Life Models

20.3A: Limitations to the Classic Model of Phylogenetic Trees

The concepts of phylogenetic modeling are constantly changing causing limitations to the classic model to arise.

Learning Objectives

- Identify the limitations to the classic model of phylogenetic trees

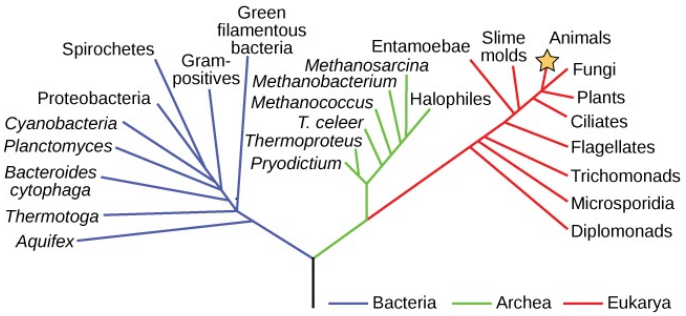
Key Points

- Charles Darwin sketched the first phylogenetic tree in 1837.
- A single trunk on a phylogenetic tree represents a common ancestor and the branches represent the divergence of species from this ancestor.
- Prokaryotes are assumed to evolve clonally in the classic tree model.
- Horizontal gene transfer is the transfer of genes between unrelated species and, as such, complicates the simple tree model.
- Ultimate gene transfer has provided theories of genome fusion between symbiotic or endosymbiotic organisms.

Key Terms

- **phylogenetic**: of, or relating to the evolutionary development of organisms
- **clonal**: pertaining to asexual reproduction
- **horizontal gene transfer**: the transfer of genetic material from one organism to another one that is not its offspring; especially common among bacteria

The concepts of phylogenetic modeling are constantly changing. It is one of the most dynamic fields of study in all of biology. Over the last several decades, new research has challenged scientists' ideas about how organisms are related. New models of these relationships have been proposed for consideration by the scientific community. Many phylogenetic trees have been shown as models of the evolutionary relationship among species. Phylogenetic trees originated with Charles Darwin, who sketched the first phylogenetic tree in 1837, which served as a pattern for subsequent studies for more than a century. The concept of a phylogenetic tree with a single trunk representing a common ancestor, with the branches representing the divergence of species from this ancestor, fits well with the structure of many common trees, such as the oak. However, evidence from modern DNA sequence analysis and newly-developed computer algorithms has caused skepticism about the validity of the standard tree model in the scientific community.



(a) Rooted phylogenetic tree

Figure 20.3A.1: **Tree of life**: The (a) concept of the "tree of life" goes back to an 1837 sketch by Charles Darwin. Like an (b) oak tree, the "tree of life" has a single trunk and many branches.

Classical thinking about prokaryotic evolution, included in the classic tree model, is that species evolve clonally. That is, they produce offspring themselves with only random mutations causing the descent into the variety of modern and extinct species known to science. This view is somewhat complicated in eukaryotes that reproduce sexually, but the laws of Mendelian genetics explain the variation in offspring, again, to be a result of a mutation within the species. The concept of genes being transferred between unrelated species was not considered as a possibility until relatively recently. Horizontal gene transfer (HGT), also known as lateral gene transfer, is the transfer of genes between unrelated species. HGT has been shown to be an ever-present phenomenon, with many evolutionists postulating a major role for this process in evolution, thus complicating the simple tree model. Genes have been shown to be passed between species which are only distantly

related using standard phylogeny, thus adding a layer of complexity to the understanding of phylogenetic relationships. Finally, as an example of the ultimate gene transfer, theories of genome fusion between symbiotic or endosymbiotic organisms have been proposed to explain an event of great importance: the evolution of the first eukaryotic cell, without which humans could not have come into existence.

20.3B: Horizontal Gene Transfer

Horizontal gene transfer (HGT) is the passing of genetic material between species by mechanisms other than from parent to offspring.

Learning Objectives

- Explain how horizontal gene transfer can make resolution of phylogenies difficult

Key Points

- It is thought that HGT is more prevalent in prokaryotes than eukaryotes, but that only about 2% of the prokaryotic genome may be transferred by this process.
- Many scientists believe that HGT and mutation appear to be (especially in prokaryotes) a significant source of genetic variation, which is the raw material for the process of natural selection.
- HGT in prokaryotes occurs by four different mechanisms: transformation, transduction, conjugation, and via gene transfer agents.
- HGT occurs in plants through transposons (jumping genes), which transfer between different species of plants.
- An example of HGT in animals is the transfer (through consumption) of fungal genes into insects called aphids, which allows the aphids the ability to make carotenoids on their own.

Key Terms

- **transformation**: the alteration of a bacterial cell caused by the transfer of DNA from another, especially if pathogenic
- **transduction**: horizontal gene transfer mechanism in prokaryotes where genes are transferred using a virus
- **conjugation**: the temporary fusion of organisms, especially as part of sexual reproduction

Horizontal Gene Transfer

Horizontal gene transfer (HGT) is the introduction of genetic material from one species to another species by mechanisms other than the vertical transmission from parent(s) to offspring. These transfers allow even distantly-related species (using standard phylogeny) to share genes, influencing their phenotypes. It is thought that HGT is more prevalent in prokaryotes, but that only about 2% of the prokaryotic genome may be transferred by this process. Some researchers believe these estimates are premature; the actual importance of HGT to evolutionary processes must be viewed as a work in progress. As the phenomenon is investigated more thoroughly, it may be revealed to be more common. Many evolutionists postulate a major role for this process in evolution, thus complicating the simple tree model. A number of scientists believe that HGT and mutation appear to be (especially in prokaryotes) a significant source of genetic variation, which is the raw material for the process of natural selection. These transfers may occur between any two species that share an intimate relationship, thus adding a layer of complexity to the understanding or resolution of phylogenetic relationships.

Summary of Mechanisms of Prokaryotic and Eukaryotic HGT

	Mechanism	Mode of Transmission	Example
Prokaryotes	transformation	DNA uptake	many prokaryotes
	transduction	bacteriophage (virus)	bacteria
	conjugation	pilus	many prokaryotes
	gene transfer agents	phage-like particles	purple non-sulfur bacteria
Eukaryotes	from food organisms	unknown	aphid
	jumping genes	transposons	rice and millet plants
	epiphytes/parasites	unknown	yew tree fungi
	from viral infections		

Figure $20.3B.1$: **Mechanisms of prokaryotic and eukaryotic horizontal gene transfer**: Horizontal gene transfer is the introduction of genetic material from one species to another species by mechanisms other than the vertical transmission from parent(s) to offspring. These transfers allow even distantly-related species (using standard phylogeny) to share genes, influencing their phenotypes. Examples of mechanisms of horizontal gene transfer are listed for both prokaryotic and eukaryotic organisms.

in Prokaryotes

echanism of HGT has been shown to be quite common in the prokaryotic domains of Bacteria and Archaea, significantly changing the way their evolution is viewed. These gene transfers between species are the major mechanism whereby bacteria acquire resistance to antibiotics. Classically, this type of transfer was thought to occur by three different mechanisms:

- Transformation: naked DNA is taken up by a bacteria.
- Transduction: genes are transferred using a virus.
- Conjugation: the use a hollow tube called a pilus to transfer genes between organisms.

More recently, a fourth mechanism of gene transfer between prokaryotes has been discovered. Small, virus-like particles called gene transfer agents (GTAs) transfer random genomic segments from one species of prokaryote to another. GTAs have been shown to be responsible for genetic changes, sometimes at a very high frequency compared to other evolutionary processes. The first GTA was characterized in 1974 using purple, non-sulfur bacteria. These GTAs, which are thought to be bacteriophages that lost the ability to reproduce on their own, carry random pieces of DNA from one organism to another. The ability of GTAs to act with high frequency has been demonstrated in controlled studies using marine bacteria. Gene transfer events in marine prokaryotes, either by GTAs or by viruses, have been estimated to be as high as 10^{13} per year in the Mediterranean Sea alone. GTAs and viruses are thought to be efficient HGT vehicles with a major impact on prokaryotic evolution.

HGT in Eukaryotes

Although it is easy to see how prokaryotes exchange genetic material by HGT, it was initially thought that this process was absent in eukaryotes. After all, prokaryotes are only single cells exposed directly to their environment, whereas the sex cells of multicellular organisms are usually sequestered in protected parts of the body. It follows from this idea that the gene transfers between multicellular eukaryotes should be more difficult. Indeed, it is thought that this process is rarer in eukaryotes and has a much smaller evolutionary impact than in prokaryotes. In spite of this fact, HGT between distantly-related organisms has been demonstrated in several eukaryotic species. It is possible that more examples will be discovered in the future.

In plants, gene transfer has been observed in species that cannot cross-pollinate by normal means. Transposons or “jumping genes” have been shown to transfer between rice and millet plant species. Furthermore, fungal species feeding on yew trees, from which the anti-cancer drug TAXOL® is derived from the bark, have acquired the ability to make taxol themselves; a clear example of gene transfer.

In animals, a particularly interesting example of HGT occurs within the aphid species. Aphids are insects that vary in color based on carotenoid content. Carotenoids are pigments made by a variety of plants, fungi, and microbes, which serve a variety of functions in animals who obtain these chemicals from their food. Humans require carotenoids to synthesize vitamin A and we obtain them by eating orange fruits and vegetables: carrots, apricots, mangoes, and sweet potatoes. On the other hand, aphids have acquired the ability to make the carotenoids on their own. According to DNA analysis, this ability is due to the transfer of fungal genes into the insect by HGT, presumably as the insect consumed fungi for food. A carotenoid enzyme called a desaturase is responsible for the red coloration seen in certain aphids. Furthermore, it has been shown that when this gene is inactivated by mutation, the aphids revert back to their more common green color.

(a) (b)

Figure 20.3B.1: **HGT within the aphid species**: (a) Red aphids get their color from red carotenoid pigment. Genes necessary to make this pigment are present in certain fungi. Scientists speculate that aphids acquired these genes through HGT after consuming fungi for food. If genes for making carotenoids are inactivated by mutation, the aphids revert back to (b) their green color. Red coloration makes the aphids much more conspicuous to predators, but evidence suggests that red aphids are more resistant to insecticides than green ones. Thus, red aphids may be more fit to survive in some environments than green ones.

20.3C: Endosymbiotic Theory and the Evolution of Eukaryotes

Genome fusion occurs during endosymbiosis, which is the mechanism proposed as responsible for the first eukaryotic cells.

Learning Objectives

- Describe the genome fusion hypothesis and its relationship to the evolution of eukaryotes

Key Points

- Two symbiotic organisms become endosymbiotic when one species is taken inside the cytoplasm of another species, resulting in genome fusion.
- Genome fusion, by endosymbiosis, between two species, one an Archaea and the other a Bacteria, has been proposed as responsible for the evolution of the first eukaryotic cells.
- Gram-negative bacteria are proposed to result from an endosymbiotic fusion of archaeal and bacterial species through a mechanism that has also been used to explain the double membranes found in mitochondria and chloroplasts.
- The nucleus-first hypothesis proposes the nucleus evolved in prokaryotes first, followed by a later fusion of the new eukaryote with bacteria that became mitochondria.
- The mitochondria-first hypothesis proposes mitochondria were first established in a prokaryotic host, which subsequently acquired a nucleus to become the first eukaryotic cell.
- The eukaryote-first hypothesis proposes prokaryotes actually evolved from eukaryotes by losing genes and complexity.

Key Terms

- **genome fusion**: a result of endosymbiosis when a genome consists of genes from both the endosymbiont and the host.
- **symbiotic**: of a relationship with mutual benefit between two individuals or organisms
- **endosymbiosis**: when one symbiotic species is taken inside the cytoplasm of another symbiotic species and both become endosymbiotic

Genome Fusion and the Evolution of Eukaryotes

Scientists believe the ultimate event in HGT (horizontal gene transfer) occurs through genome fusion between different species when two symbiotic organisms become endosymbiotic. This occurs when one species is taken inside the cytoplasm of another species, which ultimately results in a genome consisting of genes from both the endosymbiont and the host. This mechanism is an aspect of the Endosymbiont Theory, which is accepted by a majority of biologists as the mechanism whereby eukaryotic cells obtained their mitochondria and chloroplasts. However, the role of endosymbiosis in the development of the nucleus is more controversial. Nuclear and mitochondrial DNA are thought to be of different (separate) evolutionary origin, with the mitochondrial DNA being derived from the circular genomes of bacteria that were engulfed by ancient prokaryotic cells. Mitochondrial DNA can be regarded as the smallest chromosome. Interestingly enough, mitochondrial DNA is inherited only from the mother. The mitochondrial DNA degrades in sperm when the sperm degrades in the fertilized egg or, in other instances, when the mitochondria located in the flagellum of the sperm fails to enter the egg.

Within the past decade, the process of genome fusion by endosymbiosis has been proposed to be responsible for the evolution of the first eukaryotic cells. Using DNA analysis and a new mathematical algorithm called conditioned reconstruction (CR), it has been proposed that eukaryotic cells developed from an endosymbiotic gene fusion between two species: one an Archaea and the other a Bacteria. As mentioned, some eukaryotic genes resemble those of Archaea, whereas others resemble those from Bacteria. An endosymbiotic fusion event would clearly explain this observation. On the other hand, this work is new and the CR algorithm is relatively unsubstantiated, which causes many scientists to resist this hypothesis.

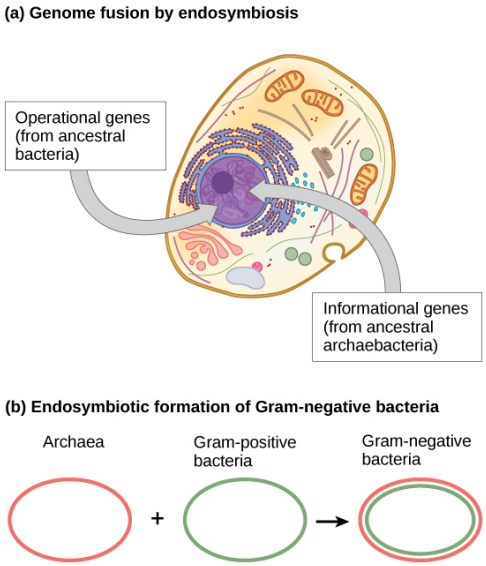


Figure $20.3C.1$: **Endosymbiosis in eukaryotes**: The theory that mitochondria and chloroplasts are endosymbiotic in origin is now widely accepted. More controversial is the proposal that (a) the eukaryotic nucleus resulted from the fusion of archaeal and bacterial genomes; and that (b) Gram-negative bacteria, which have two membranes, resulted from the fusion of Archaea and Gram-positive bacteria, each of which has a single membrane.

More recent work proposes that gram-negative bacteria, which are unique within their domain in that they contain two lipid bilayer membranes, did result from an endosymbiotic fusion of archaeal and bacterial species. The double membrane would be a direct result of endosymbiosis, with the endosymbiont picking up the second membrane from the host as it was internalized. This mechanism has also been used to explain the double membranes found in mitochondria and chloroplasts. A lot of skepticism still surrounds this hypothesis; the ideas are still debated within the biological science community.

There are several other competing hypotheses as to the origin of eukaryotes and the nucleus. One idea about how the eukaryotic nucleus evolved is that prokaryotic cells produced an additional membrane which surrounded the bacterial chromosome. Some bacteria have the DNA enclosed by two membranes; however, there is no evidence of a nucleolus or nuclear pores. Other proteobacteria also have membrane-bound chromosomes. If the eukaryotic nucleus evolved this way, we would expect one of the two types of prokaryotes to be more closely-related to eukaryotes. Another hypothesis, the nucleus-first hypothesis, proposes the nucleus evolved in prokaryotes first, followed by a later fusion of the new eukaryote with bacteria that became mitochondria. The mitochondria-first hypothesis, however, proposes mitochondria were first established in a prokaryotic host, which subsequently acquired a nucleus (by fusion or other mechanisms) to become the first eukaryotic cell. Most interestingly, the eukaryote-first hypothesis proposes prokaryotes actually evolved from eukaryotes by losing genes and complexity. All of these hypotheses are testable. Only time and more experimentation will determine which hypothesis is best supported by data.

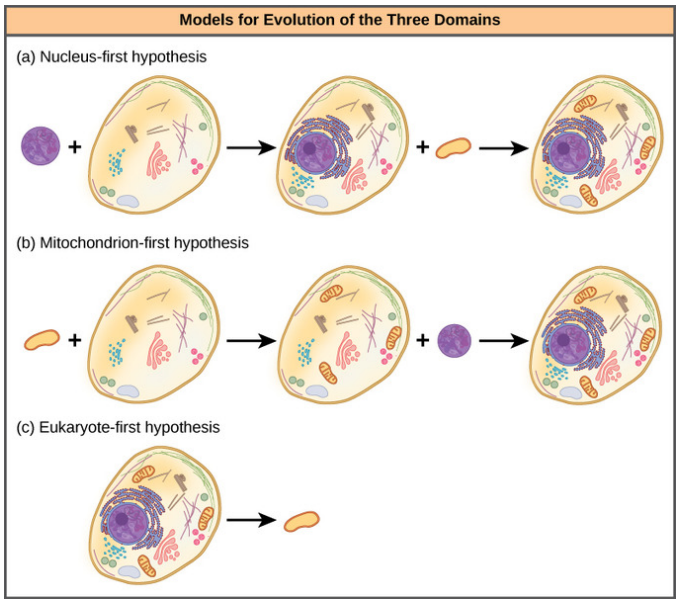


Figure $20.3C.1$: **Three hypotheses of eukaryotic and prokaryotic evolution**: Three alternate hypotheses of eukaryotic and prokaryotic evolution are (a) the nucleus-first hypothesis, (b) the mitochondrion-first hypothesis, and (c) the eukaryote-first hypothesis.

20.3D: Web, Network, and Ring of Life Models

To more accurately describe the phylogenetic relationships of life, web and ring models have been proposed as updates to tree models.

Learning Objectives

- Describe the web, network, and ring of life models of phylogenetic trees

Key Points

- A phylogenetic model that resembles a web or a network was proposed since eukaryotes evolved not from a single prokaryotic ancestor, but from a pool of many species that were sharing genes by HGT mechanisms.
- A phylogenetic model that resembles a ring was proposed in which species of all three domains, Archaea, Bacteria, and Eukarya, evolved from a single pool of gene-swapping prokaryotes.
- Phylogenetic models will continue to evolve as phylogeneticists remain highly skeptical of the current tree, web, and ring models.

Key Terms

- **web of life**: a phylogenetic model that resembles a web or a network more than a tree
- **ring of life**: a phylogenetic model where all three domains of life (Archaea, Bacteria, and Eukarya) evolved from a pool of primitive prokaryotes

The recognition of the importance of Horizontal gene transfer (HGT), especially in the evolution of prokaryotes, has caused some to propose abandoning the classic “tree of life” model. In 1999, a phylogenetic model that resembles a web or a network more than a tree was proposed. The hypothesis is that eukaryotes evolved not from a single prokaryotic ancestor, but from a pool of many species that were sharing genes by HGT mechanisms. Some individual prokaryotes were responsible for transferring the bacteria that caused mitochondrial development in the new eukaryotes, whereas other species transferred the bacteria that gave rise to chloroplasts. This model is often called the “web of life.” In an effort to save the tree analogy, some have proposed using the Ficus tree with its multiple trunks as a phylogenetic tree to represent the evolutionary role for HGT.

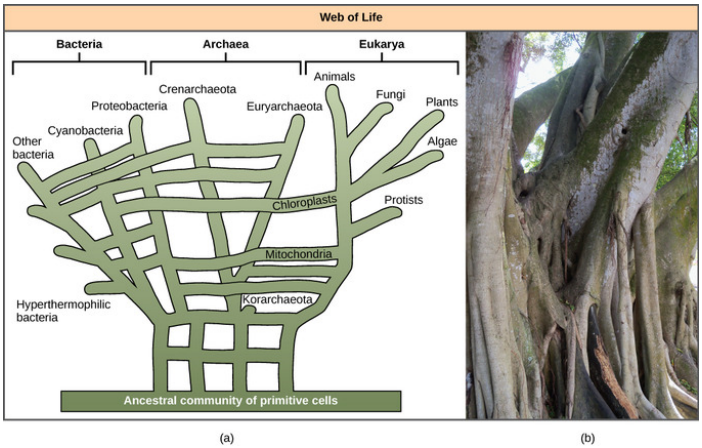


Figure 20.3D.1: **Phylogenetic web of life model**: In the (a) phylogenetic model proposed by W. Ford Doolittle, the “tree of life” arose from a community of ancestral cells, has multiple trunks, and has connections between branches where horizontal gene transfer has occurred. Visually, this concept is better represented by (b) the multi-trunked Ficus than by the single trunk of the oak, similar to the tree drawn by Darwin.

Others have proposed abandoning any tree-like model of phylogeny in favor of a ring structure. The ” ring of life ” is a phylogenetic model where all three domains of life evolved from a pool of primitive prokaryotes. Using the conditioned reconstruction algorithm, it proposes a ring-like model in which species of all three domains (Archaea, Bacteria, and Eukarya) evolved from a single pool of gene-swapping prokaryotes. This structure is proposed as the best fit for data from extensive DNA analyses; the ring model is the only one that adequately takes HGT and genomic fusion into account. However, phylogeneticists remain highly skeptical of this model.

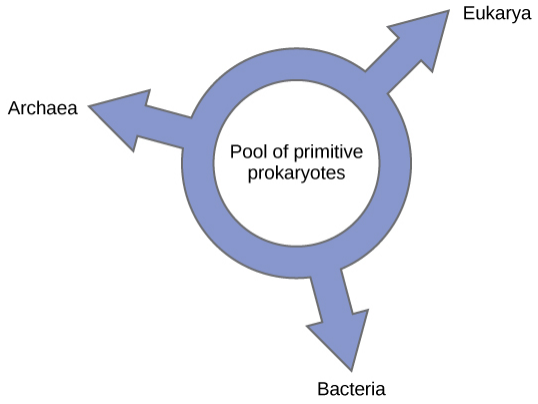


Figure 20.3D.1: **Phylogenetic ring of life model**: According to the "ring of life" phylogenetic model, the three domains of life evolved from a pool of primitive prokaryotes.

In summary, the "tree of life" model proposed by Darwin must be modified to include HGT. This does not mean a tree, web, or a ring will correlate completely to an accurate description of phylogenetic relationships of life. A consequence of the new thinking about phylogenetic models is the idea that Darwin's original conception of the phylogenetic tree is too simple, but made sense based on what was known at that time.

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CHAPTER OVERVIEW

21: VIRUSES

Thumbnail: Ebola virus. (Public Domain; CDC).

21.1: VIRAL EVOLUTION, MORPHOLOGY, AND CLASSIFICATION

21.1A: DISCOVERY AND DETECTION OF VIRUSES
21.1B: EVOLUTION OF VIRUSES
21.1C: VIRAL MORPHOLOGY
21.1D: VIRUS CLASSIFICATION
21.2: VIRUS INFECTIONS AND HOSTS

21.2A: STEPS OF VIRUS INFECTIONS
21.2B: THE LYTIC AND LYSOGENIC CYCLES OF BACTERIOPHAGES
21.2C: ANIMAL VIRUSES
21.2D: PLANT VIRUSES
21.3: PREVENTION AND TREATMENT OF VIRAL INFECTIONS

21.3A: VACCINES AND IMMUNITY
21.3B: VACCINES AND ANTI-VIRAL DRUGS FOR TREATMENT
21.4: PRIONS AND VIROIDS

21:4A: PRIONS AND VIROIDS

Topic hierarchy

21.1A: Discovery and Detection of Viruses

21.1B: Evolution of Viruses

21.1C: Viral Morphology

21.1D: Virus Classification

21.1A: Discovery and Detection of Viruses

Viruses are infectious particles about 100 times smaller than bacteria and can only be observed by electron microscopy.

Learning Objectives

- Describe how viruses were first discovered and how they are detected

Key Points

- Virions, single virus particles, are 20–250 nanometers in diameter.
- In the past, viruses were classified by the type of nucleic acid they contained, DNA or RNA, and whether they had single- or double-stranded nucleic acid.
- Molecular analysis of viral replicative cycles is now more routinely used to classify viruses.

Key Terms

- **virus**: a submicroscopic infectious organism, now understood to be a non-cellular structure consisting of a core of DNA or RNA surrounded by a protein coat
- **virion**: a single individual particle of a virus (the viral equivalent of a cell)

Discovery and Detection

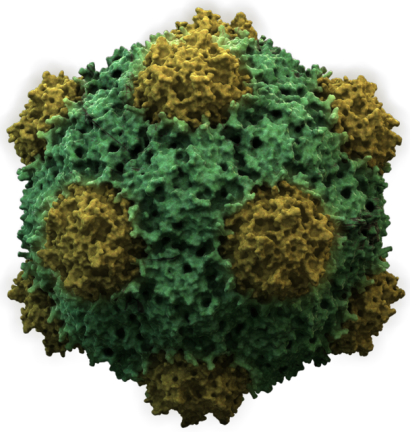


Figure 21.1*A*.1: **The structure of the icosahedral cowpea mosaic virus**: In the past, viruses were classified by the type of nucleic acid they contained, DNA or RNA, and whether they had single- or double-stranded nucleic acid.

Viruses were first discovered after the development of a porcelain filter, called the Chamberland-Pasteur filter, which could remove all bacteria visible in the microscope from any liquid sample. In 1886, Adolph Meyer demonstrated that a disease of tobacco plants, tobacco mosaic disease, could be transferred from a diseased plant to a healthy one via liquid plant extracts. In 1892, Dmitri Ivanowski showed that this disease could be transmitted in this way even after the Chamberland-Pasteur filter had removed all viable bacteria from the extract. Still, it was many years before it was proven that these "filterable" infectious agents were not simply very small bacteria, but were a new type of tiny, disease-causing particle.

Virions, single virus particles, are very small, about 20–250 nanometers in diameter. These individual virus particles are the infectious form of a virus outside the host cell. Unlike bacteria (which are about 100 times larger), we cannot see viruses with a light microscope, with the exception of some large virions of the poxvirus family. It was not until the development of the electron microscope in the late 1930s that scientists got their first good view of the structure of the tobacco mosaic virus (TMV) and other viruses. The surface structure of virions can be observed by both scanning and transmission electron microscopy, whereas the internal structures of the virus can only be observed in images from a transmission electron microscope. The use of these technologies has enabled the discovery of many viruses of all types of living organisms. They were initially grouped by shared morphology. Later, groups of viruses were classified by the type of nucleic acid they contained, DNA or RNA, and whether their nucleic acid was single- or double-stranded. More recently, molecular analysis of viral replicative cycles has further refined their classification.

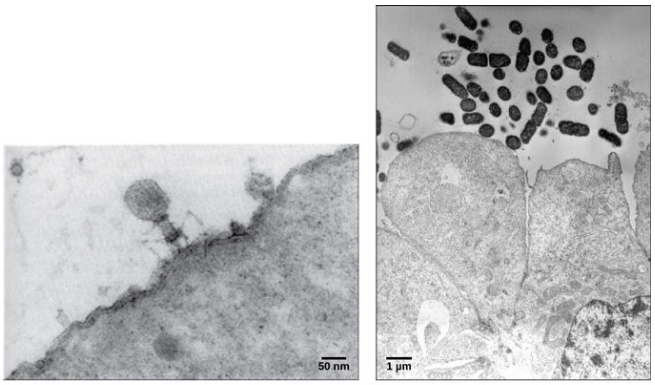


Figure 21.1A.1: **Examples of transmission electron micrographs of viruses**: In these transmission electron micrographs, (a) a virus is dwarfed by the bacterial cell it infects, while (b) these E. coli cells are dwarfed by cultured colon cells.

21.1B: Evolution of Viruses

The evolution of viruses is speculative as they do not fossilize; biochemical and genetic information is used to create virus histories.

Learning Objectives

- Describe the difficulties in determining the origin of viruses

Key Points

- Scientists agree that viruses don't have a single common ancestor, but have yet to agree on a single hypothesis about virus origins.
- The devolution or the regressive hypothesis suggests that viruses evolved from free-living cells.
- The escapist or the progressive hypothesis suggests that viruses originated from RNA and DNA molecules that escaped from a host cell.
- The self-replicating hypothesis posits a system of self-replication that most probably involves evolution alongside the host cells.

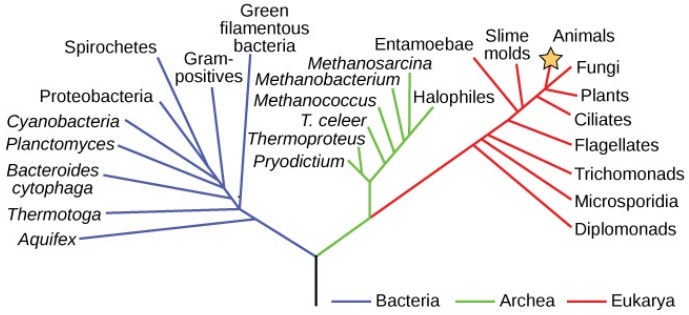
Key Terms

- **self-replicating**: able to generate a copy of itself
- **devolution**: degeneration (as opposed to evolution)

Evolution of Viruses

Although biologists have accumulated a significant amount of knowledge about how present-day viruses evolve, much less is known about how viruses originated in the first place. When exploring the evolutionary history of most organisms, scientists can look at fossil records and similar historic evidence. However, viruses do not fossilize, so researchers must conjecture by investigating how today's viruses evolve and by using biochemical and genetic information to create speculative virus histories.

While most findings agree that viruses don't have a single common ancestor, scholars have yet to find one hypothesis about virus origins that is fully accepted in the field. One possible hypothesis, called devolution or the regressive hypothesis, proposes to explain the origin of viruses by suggesting that viruses evolved from free-living cells. However, many components of how this process might have occurred are a mystery. A second hypothesis (called escapist or the progressive hypothesis) accounts for viruses having either an RNA or a DNA genome and suggests that viruses originated from RNA and DNA molecules that escaped from a host cell. A third hypothesis posits a system of self-replication similar to that of other self-replicating molecules, probably evolving alongside the cells they rely on as hosts; studies of some plant pathogens support this hypothesis.



(a) Rooted phylogenetic tree

Figure 21.1*B*.1: **Common ancestor tree of life**: This phylogenetic tree of the three domains of life (Bacteria, Archaea, and Eukarya) attempts to identify when various species diverged from a common ancestor. Finding a common ancestor for viruses has proven to be far more difficult, especially since they do not fossilize.

As technology advances, scientists may develop and refine further hypotheses to explain the origin of viruses. The emerging field called virus molecular systematics attempts to do just that through comparisons of sequenced genetic

material. These researchers hope to one day better understand the origin of viruses, a discovery that could lead to advances in the treatments for the ailments they produce.

21.1C: Viral Morphology

Viruses of all shapes and sizes consist of a nucleic acid core, an outer protein coating or capsid, and sometimes an outer envelope.

Learning Objectives

- Describe the relationship between the viral genome, capsid, and envelope

Key Points

- Viruses are classified into four groups based on shape: filamentous, isometric (or icosahedral), enveloped, and head and tail.
- Many viruses attach to their host cells to facilitate penetration of the cell membrane, allowing their replication inside the cell.
- Non-enveloped viruses can be more resistant to changes in temperature, pH, and some disinfectants than are enveloped viruses.
- The virus core contains the small single- or double-stranded genome that encodes the proteins that the virus cannot get from the host cell.

Key Terms

- **capsid**: the outer protein shell of a virus
- **envelope**: an enclosing structure or cover, such as a membrane
- **filamentous**: Having the form of threads or filaments
- **isometric**: of, or being a geometric system of three equal axes lying at right angles to each other (especially in crystallography)

Viral Morphology

Viruses are acellular, meaning they are biological entities that do not have a cellular structure. Therefore, they lack most of the components of cells, such as organelles, ribosomes, and the plasma membrane. A virion consists of a nucleic acid core, an outer protein coating or capsid, and sometimes an outer envelope made of protein and phospholipid membranes derived from the host cell. The capsid is made up of protein subunits called capsomeres. Viruses may also contain additional proteins, such as enzymes. The most obvious difference between members of viral families is their morphology, which is quite diverse. An interesting feature of viral complexity is that host and virion complexity are uncorrelated. Some of the most intricate virion structures are observed in bacteriophages, viruses that infect the simplest living organisms: bacteria.

Morphology

Viruses come in many shapes and sizes, but these are consistent and distinct for each viral family. In general, the shapes of viruses are classified into four groups: filamentous, isometric (or icosahedral), enveloped, and head and tail. Filamentous viruses are long and cylindrical. Many plant viruses are filamentous, including TMV (tobacco mosaic virus). Isometric viruses have shapes that are roughly spherical, such as poliovirus or herpesviruses. Enveloped viruses have membranes surrounding capsids. Animal viruses, such as HIV, are frequently enveloped. Head and tail viruses infect bacteria. They have a head that is similar to icosahedral viruses and a tail shape like filamentous viruses.

Many viruses use some sort of glycoprotein to attach to their host cells via molecules on the cell called viral receptors. For these viruses, attachment is a requirement for later penetration of the cell membrane, allowing them to complete their replication inside the cell. The receptors that viruses use are molecules that are normally found on cell surfaces and have their own physiological functions. Viruses have simply evolved to make use of these molecules for their own replication.

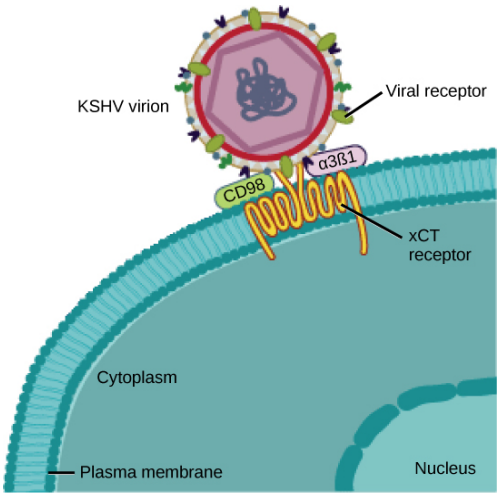


Figure 21.1C.1: **Example of a virus attaching to its host cell**: The KSHV virus binds the xCT receptor on the surface of human cells. This attachment allows for later penetration of the cell membrane and replication inside the cell.

Overall, the shape of the virion and the presence or absence of an envelope tell us little about what disease the virus may cause or what species it might infect, but they are still useful means to begin viral classification. Among the most complex virions known, the T4 bacteriophage, which infects the *Escherichia coli* bacterium, has a tail structure that the virus uses to attach to host cells and a head structure that houses its DNA. Adenovirus, a non-enveloped animal virus that causes respiratory illnesses in humans, uses glycoprotein spikes protruding from its capsomeres to attach to host cells. Non-enveloped viruses also include those that cause polio (poliovirus), plantar warts (papillomavirus), and hepatitis A (hepatitis A virus).

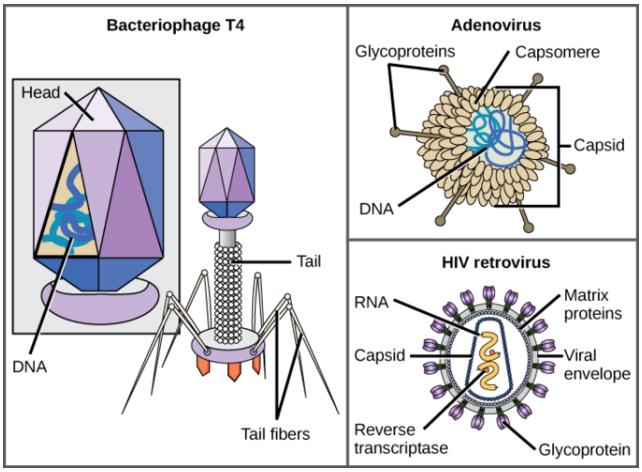


Figure 21.1C.1: **Examples of virus shapes**: Viruses can be either complex in shape or relatively simple. This figure shows three relatively-complex virions: the bacteriophage T4, with its DNA-containing head group and tail fibers that attach to host cells; adenovirus, which uses spikes from its capsid to bind to host cells; and HIV, which uses glycoproteins embedded in its envelope to bind to host cells.

Enveloped virions like HIV consist of nucleic acid and capsid proteins surrounded by a phospholipid bilayer envelope and its associated proteins. Glycoproteins embedded in the viral envelope are used to attach to host cells. Other envelope proteins include the matrix proteins that stabilize the envelope and often play a role in the assembly of progeny virions. Chicken pox, influenza, and mumps are examples of diseases caused by viruses with envelopes. Because of the fragility of the envelope, non-enveloped viruses are more resistant to changes in temperature, pH, and some disinfectants than are enveloped viruses.

Types of Nucleic Acid

Unlike nearly all living organisms that use DNA as their genetic material, viruses may use either DNA or RNA. The virus core contains the genome or total genetic content of the virus. Viral genomes tend to be small, containing only those genes that encode proteins that the virus cannot obtain from the host cell. This genetic material may be single- or double-

stranded. It may also be linear or circular. While most viruses contain a single nucleic acid, others have genomes that have several, called segments.

In DNA viruses, the viral DNA directs the host cell's replication proteins to synthesize new copies of the viral genome and to transcribe and translate that genome into viral proteins. DNA viruses cause human diseases, such as chickenpox, hepatitis B, and some venereal diseases, like herpes and genital warts.

RNA viruses contain only RNA as their genetic material. To replicate their genomes in the host cell, the RNA viruses encode enzymes that can replicate RNA into DNA, which cannot be done by the host cell. These RNA polymerase enzymes are more likely to make copying errors than DNA polymerases and, therefore, often make mistakes during transcription. For this reason, mutations in RNA viruses occur more frequently than in DNA viruses. This causes them to change and adapt more rapidly to their host. Human diseases caused by RNA viruses include hepatitis C, measles, and rabies.

21.1D: Virus Classification

Viruses are classified by factors such as their core content, capsid structure, presence of outer envelope, and how mRNA is produced.

Learning Objectives

- Describe how viruses are classified

Key Points

- The type of genetic material, either DNA or RNA, and whether its structure is single- or double-stranded, linear or circular, and segmented or non-segmented are factors for classification.
- Virus capsids can be classified as naked icosahedral, enveloped icosahedral, enveloped helical, naked helical, and complex.
- Virus can either have an envelope or not.
- A more recent system, the Baltimore classification scheme, groups viruses into seven classes according to how the mRNA is produced during the replicative cycle of the virus.

Key Terms

- **Baltimore classification**: a classification scheme that groups viruses into seven classes according to how the mRNA is produced during the replicative cycle of the virus
- **messenger RNA**: Messenger RNA (mRNA) is a molecule of RNA that encodes a chemical “blueprint” for a protein product.

Virus Classification

To understand the features shared among different groups of viruses, a classification scheme is necessary. However, most viruses are not thought to have evolved from a common ancestor, so the methods that scientists use to classify living things are not very useful. Biologists have used several classification systems in the past, based on the morphology and genetics of the different viruses. However, these earlier classification methods grouped viruses based on which features of the virus they were using to classify them. The most commonly-used classification method today is called the Baltimore classification scheme which is based on how messenger RNA (mRNA) is generated in each particular type of virus. The surface structure of virions can be observed by both scanning and transmission electron microscopy, whereas the internal structures of the virus can only be observed in images from a transmission electron microscope.

Past Systems of Classification

Viruses are classified in several ways: by factors such as their core content, the structure of their capsids, and whether they have an outer envelope. Viruses may use either DNA or RNA as their genetic material. The virus core contains the genome or total genetic content of the virus. Viral genomes tend to be small, containing only those genes that encode proteins that the virus cannot obtain from the host cell. This genetic material may be single- or double-stranded. It may also be linear or circular. While most viruses contain a single nucleic acid, others have genomes that have several, which are called segments. The type of genetic material (DNA or RNA) and its structure (single- or double-stranded, linear or circular, and segmented or non-segmented) are used to classify the virus core structures.

Virus Classification by Genome Structure and Core	
Core Classifications	**Examples**
RNA	Rabies virus, retroviruses
DNA	Herpesviruses, smallpox virus
Single-stranded	Rabies virus, retroviruses
Double-stranded	Herpesviruses, smallpox virus
Linear	Rabies virus, retroviruses, herpesviruses, smallpox virus
Circular	Papillomaviruses, many bacteriophages
Non-segmented: genome consists of a single segment of genetic material	Parainfluenza viruses
Segmented: genome is divided into multiple segments	Influenza viruses

Figure 21.1D.1: **Virus classification by genome structure and core**: The type of genetic material (DNA or RNA) and its structure (single- or double-stranded, linear or circular, and segmented or non-segmented) are used to classify the virus core structures.

Viruses can also be classified by the design of their capsids. Isometric viruses have shapes that are roughly spherical, such as poliovirus or herpesviruses. Enveloped viruses have membranes surrounding capsids. Animal viruses, such as HIV, are frequently enveloped. Head and tail viruses infect bacteria and have a head that is similar to icosahedral viruses and a tail shape like filamentous viruses. Capsids are classified as naked icosahedral, enveloped icosahedral, enveloped helical, naked helical, and complex. For example, the tobacco mosaic virus has a naked helical capsid. The adenovirus has an icosahedral capsid.

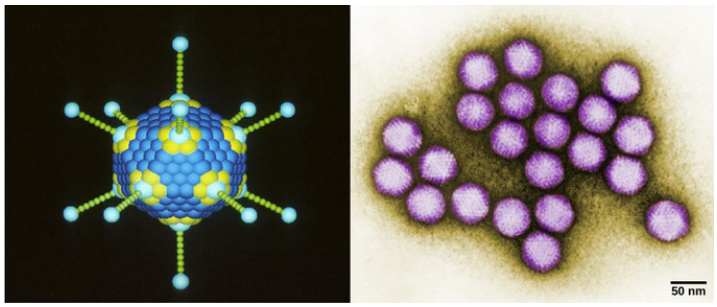


Figure 21.1D.1: **Adenovirus classification**: Adenovirus (left) is depicted with a double-stranded DNA genome enclosed in an icosahedral capsid that is 90–100 nm across. The virus, shown clustered in the micrograph (right), is transmitted orally and causes a variety of illnesses in vertebrates, including human eye and respiratory infections.

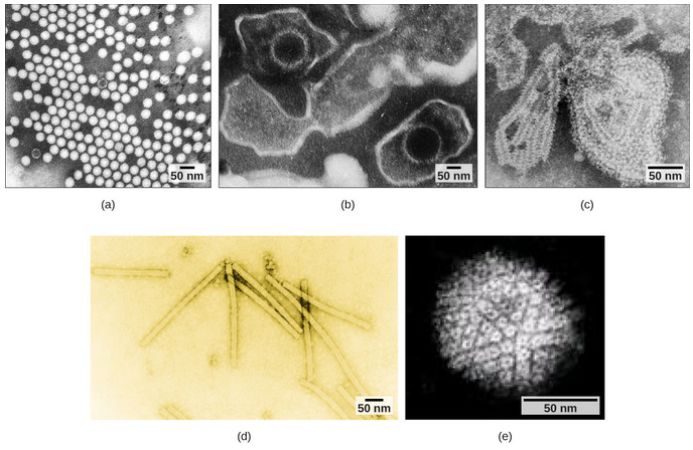
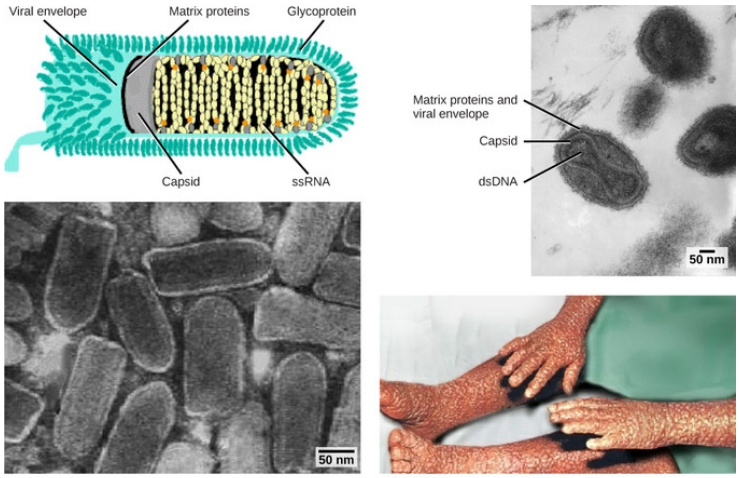


Figure 21.1D.1: **Transmission electron micrograph of viruses**: Transmission electron micrographs of various viruses show their structures. The capsid of the (a) polio virus is naked icosahedral; (b) the Epstein-Barr virus capsid is enveloped icosahedral; (c) the mumps virus capsid is an enveloped helix; (d) the tobacco mosaic virus capsid is naked helical; and (e) the herpesvirus capsid is complex.

Virus Classification by Capsid Structure	
Capsid Classification	**Examples**
Naked icosahedral	Hepatitis A virus, polioviruses
Enveloped icosahedral	Epstein-Barr virus, herpes simplex virus, rubella virus, yellow fever virus, HIV-1
Enveloped helical	Influenza viruses, mumps virus, measles virus, rabies virus
Naked helical	Tobacco mosaic virus
Complex with many proteins; some have combinations of icosahedral and helical capsid structures	Herpesviruses, smallpox virus, hepatitis B virus, T4 bacteriophage

Figure 21.1D.1: **Virus classification by capsid structure**: Viruses can also be classified by the design of their capsids which are classified as naked icosahedral, enveloped icosahedral, enveloped helical, naked helical, and complex.



(a) Rabies virus (b) Variola virus

Figure 21.1D.1: **Example of viruses classified by caspid design**: Viruses are classified based on their core genetic material and capsid design. (a) Rabies virus has a single-stranded RNA (ssRNA) core and an enveloped helical capsid, whereas (b) variola virus, the causative agent of smallpox, has a double-stranded DNA (dsDNA) core and a complex capsid.

Baltimore Classification

The most commonly-used system of virus classification was developed by Nobel Prize-winning biologist David Baltimore in the early 1970s. In addition to the differences in morphology and genetics mentioned above, the Baltimore classification scheme groups viruses according to how the mRNA is produced during the replicative cycle of the virus. Viruses can contain double-stranded DNA (dsDNA), single-stranded DNA (ssDNA), double-stranded RNA (dsRNA), single-stranded RNA with a positive polarity (ssRNA), ssRNA with a negative polarity, diploid (two copies) ssRNA, and partial dsDNA genomes. Positive polarity means that the genomic RNA can serve directly as mRNA and a negative polarity means that their sequence is complementary to the mRNA.

Baltimore Classification			
Group	**Characteristics**	**Mode of mRNA Production**	**Example**
I	Double-stranded DNA	mRNA is transcribed directly from the DNA template	Herpes simplex (herpesvirus)
II	Single-stranded DNA	DNA is converted to double-stranded form before RNA is transcribed	Canine parvovirus (parvovirus)
III	Double-stranded RNA	mRNA is transcribed from the RNA genome	Childhood gastroenteritis (rotavirus)
IV	Single stranded RNA (+)	Genome functions as mRNA	Common cold (picornavirus)
V	Single stranded RNA (-)	mRNA is transcribed from the RNA genome	Rabies (rhabdovirus)
VI	Single stranded RNA viruses with reverse transcriptase	Reverse transcriptase makes DNA from the RNA genome; DNA is then incorporated in the host genome; mRNA is transcribed from the incorporated DNA	Human immunodeficiency virus (HIV)
VII	Double stranded DNA viruses with reverse transcriptase	The viral genome is double-stranded DNA, but viral DNA is replicated through an RNA intermediate; the RNA may serve directly as mRNA or as a template to make mRNA	Hepatitis B virus (hepadnavirus)

Figure 21.1D.1: **Baltimore classification**: The Baltimore classification scheme, the most commonly used, was developed by Nobel Prize-winning biologist David Baltimore in the early 1970s. The scheme groups viruses according to how the mRNA is produced during the replicative cycle of the virus, in addition to the differences in morphology and genetics.

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21.2: Virus Infections and Hosts

Topic hierarchy

- 21.2A: Steps of Virus Infections
- 21.2B: The Lytic and Lysogenic Cycles of Bacteriophages
- 21.2C: Animal Viruses
- 21.2D: Plant Viruses

21.2A: Steps of Virus Infections

Viral infection involves the incorporation of viral DNA into a host cell, replication of that material, and the release of the new viruses.

Learning Objectives

- List the steps of viral replication and explain what occurs at each step

Key Points

- Viral replication involves six steps: attachment, penetration, uncoating, replication, assembly, and release.
- During attachment and penetration, the virus attaches itself to a host cell and injects its genetic material into it.
- During uncoating, replication, and assembly, the viral DNA or RNA incorporates itself into the host cell's genetic material and induces it to replicate the viral genome.
- During release, the newly-created viruses are released from the host cell, either by causing the cell to break apart, waiting for the cell to die, or by budding off through the cell membrane.

Key Terms

- **virion**: a single individual particle of a virus (the viral equivalent of a cell)
- **glycoprotein**: a protein with covalently-bonded carbohydrates
- **retrovirus**: a virus that has a genome consisting of RNA

Steps of Virus Infections

A virus must use cell processes to replicate. The viral replication cycle can produce dramatic biochemical and structural changes in the host cell, which may cause cell damage. These changes, called cytopathic (causing cell damage) effects, can change cell functions or even destroy the cell. Some infected cells, such as those infected by the common cold virus known as rhinovirus, die through lysis (bursting) or apoptosis (programmed cell death or "cell suicide"), releasing all progeny virions at once. The symptoms of viral diseases result from the immune response to the virus, which attempts to control and eliminate the virus from the body and from cell damage caused by the virus. Many animal viruses, such as HIV (Human Immunodeficiency Virus), leave the infected cells of the immune system by a process known as budding, where virions leave the cell individually. During the budding process, the cell does not undergo lysis and is not immediately killed. However, the damage to the cells that the virus infects may make it impossible for the cells to function normally, even though the cells remain alive for a period of time. Most productive viral infections follow similar steps in the virus replication cycle: attachment, penetration, uncoating, replication, assembly, and release.

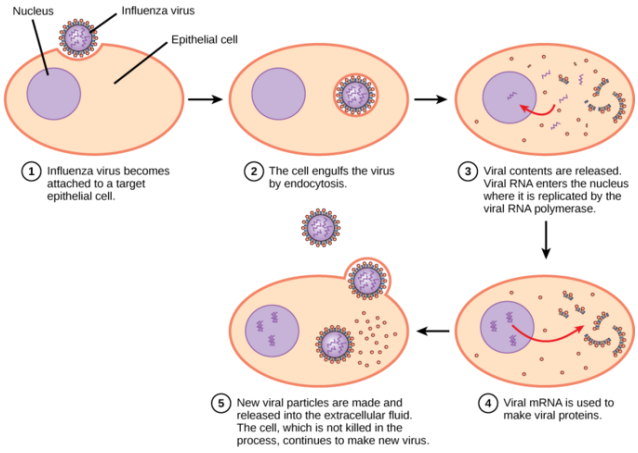

Figure 21.2A.1: **Pathway to viral infection**: In influenza virus infection, glycoproteins attach to a host epithelial cell. As a result, the virus is engulfed. RNA and proteins are made and assembled into new virions.

Attachment

A virus attaches to a specific receptor site on the host cell membrane through attachment proteins in the capsid or via glycoproteins embedded in the viral envelope. The specificity of this interaction determines the host (and the cells within the host) that can be infected by a particular virus. This can be illustrated by thinking of several keys and several locks where each key will fit only one specific lock.

Entry

The nucleic acid of bacteriophages enters the host cell naked, leaving the capsid outside the cell. Plant and animal viruses can enter through endocytosis, in which the cell membrane surrounds and engulfs the entire virus. Some enveloped viruses enter the cell when the viral envelope fuses directly with the cell membrane. Once inside the cell, the viral capsid is degraded and the viral nucleic acid is released, which then becomes available for replication and transcription.

Replication and Assembly

The replication mechanism depends on the viral genome. DNA viruses usually use host cell proteins and enzymes to make additional DNA that is transcribed to messenger RNA (mRNA), which is then used to direct protein synthesis. RNA viruses usually use the RNA core as a template for synthesis of viral genomic RNA and mRNA. The viral mRNA directs the host cell to synthesize viral enzymes and capsid proteins, and to assemble new virions. Of course, there are exceptions to this pattern. If a host cell does not provide the enzymes necessary for viral replication, viral genes supply the information to direct synthesis of the missing proteins. Retroviruses, such as HIV, have an RNA genome that must be reverse transcribed into DNA, which then is incorporated into the host cell genome.

To convert RNA into DNA, retroviruses must contain genes that encode the virus-specific enzyme reverse transcriptase, which transcribes an RNA template to DNA. Reverse transcription never occurs in uninfected host cells; the needed enzyme, reverse transcriptase, is only derived from the expression of viral genes within the infected host cells. The fact that HIV produces some of its own enzymes not found in the host has allowed researchers to develop drugs that inhibit these enzymes. These drugs, including the reverse transcriptase inhibitor AZT, inhibit HIV replication by reducing the activity of the enzyme without affecting the host's metabolism. This approach has led to the development of a variety of drugs used to treat HIV and has been effective at reducing the number of infectious virions (copies of viral RNA) in the blood to non-detectable levels in many HIV-infected individuals.

Egress

The last stage of viral replication is the release of the new virions produced in the host organism. They are then able to infect adjacent cells and repeat the replication cycle. As you have learned, some viruses are released when the host cell dies, while other viruses can leave infected cells by budding through the membrane without directly killing the cell.

21.2B: The Lytic and Lysogenic Cycles of Bacteriophages

Bacteriophages, viruses that infect bacteria, may undergo a lytic or lysogenic cycle.

Learning Objectives

- Describe the lytic and lysogenic cycles of bacteriophages

Key Points

- Viruses are species specific, but almost every species on Earth can be affected by some form of virus.
- The lytic cycle involves the reproduction of viruses using a host cell to manufacture more viruses; the viruses then burst out of the cell.
- The lysogenic cycle involves the incorporation of the viral genome into the host cell genome, infecting it from within.

Key Terms

- **latency**: The ability of a pathogenic virus to lie dormant within a cell.
- **bacteriophage**: A virus that specifically infects bacteria.
- **lytic cycle**: The normal process of viral reproduction involving penetration of the cell membrane, nucleic acid synthesis, and lysis of the host cell.
- **lysogenic cycle**: A form of viral reproduction involving the fusion of the nucleic acid of a bacteriophage with that of a host, followed by proliferation of the resulting prophage.

Different Hosts and Their Viruses

Viruses are often very specific as to which hosts and which cells within the host they will infect. This feature of a virus makes it specific to one or a few species of life on earth. So many different types of viruses exist that nearly every living organism has its own set of viruses that try to infect its cells. Even the smallest and simplest of cells, prokaryotic bacteria, may be attacked by specific types of viruses.

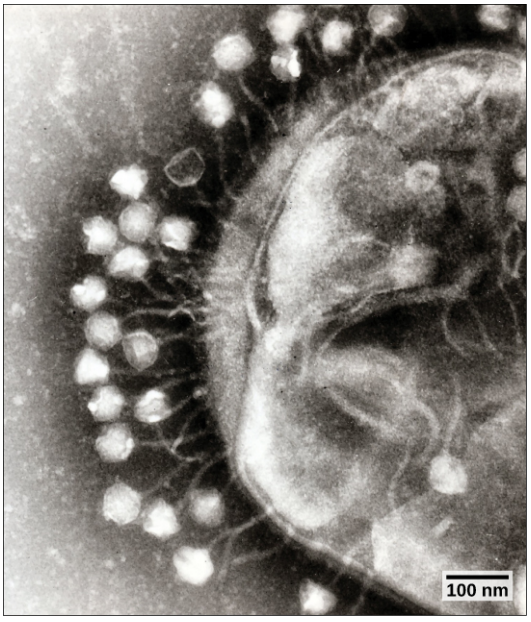

Figure $21.2B.1$: **Bacteriophage**: This transmission electron micrograph shows bacteriophages attached to a bacterial cell.

Bacteriophages

Bacteriophages are viruses that infect bacteria. Bacteriophages may have a lytic cycle or a lysogenic cycle, and a few viruses are capable of carrying out both. When infection of a cell by a bacteriophage results in the production of new virions, the infection is said to be productive.

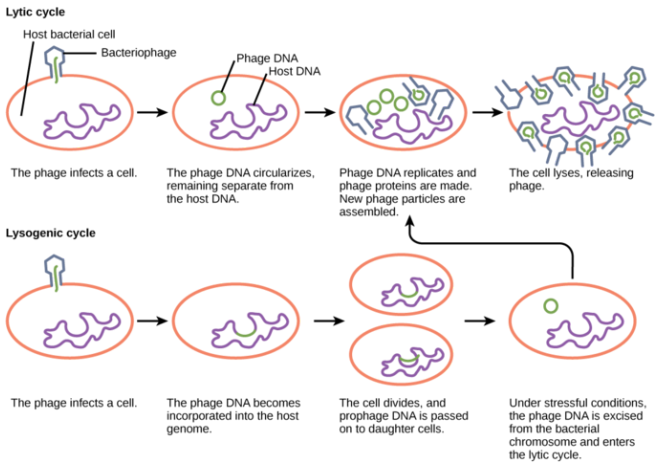

Figure 21.2B.1: **Lytic versus lysogenic cycle**: A temperate bacteriophage has both lytic and lysogenic cycles. In the lytic cycle, the phage replicates and lyses the host cell. In the lysogenic cycle, phage DNA is incorporated into the host genome, where it is passed on to subsequent generations. Environmental stressors such as starvation or exposure to toxic chemicals may cause the prophage to excise and enter the lytic cycle.

Lytic Cycle

With lytic phages, bacterial cells are broken open (lysed) and destroyed after immediate replication of the virion. As soon as the cell is destroyed, the phage progeny can find new hosts to infect. An example of a lytic bacteriophage is T4, which infects *E. coli* found in the human intestinal tract. Lytic phages are more suitable for phage therapy.

Some lytic phages undergo a phenomenon known as lysis inhibition, where completed phage progeny will not immediately lyse out of the cell if extracellular phage concentrations are high.

Lysogenic Cycle

In contrast, the lysogenic cycle does not result in immediate lysing of the host cell. Those phages able to undergo lysogeny are known as temperate phages. Their viral genome will integrate with host DNA and replicate along with it fairly harmlessly, or may even become established as a plasmid. The virus remains dormant until host conditions deteriorate, perhaps due to depletion of nutrients; then, the endogenous phages (known as prophages) become active. At this point they initiate the reproductive cycle, resulting in lysis of the host cell. As the lysogenic cycle allows the host cell to continue to survive and reproduce, the virus is reproduced in all of the cell's offspring. An example of a bacteriophage known to follow the lysogenic cycle and the lytic cycle is the phage lambda of E. coli.

Latency Period

Viruses that infect plant or animal cells may also undergo infections where they are not producing virions for long periods. An example is the animal herpes viruses, including herpes simplex viruses, which cause oral and genital herpes in humans. In a process called latency, these viruses can exist in nervous tissue for long periods of time without producing new virions, only to leave latency periodically and cause lesions in the skin where the virus replicates. Even though there are similarities between lysogeny and latency, the term lysogenic cycle is usually reserved to describe bacteriophages.

21.2C: Animal Viruses

Animal viruses have their genetic material copied by a host cell after which they are released into the environment to cause disease.

Learning Objectives

- Describe various animal viruses and the diseases they cause

Key Points

- Animal viruses may enter a host cell by either receptor -mediated endocytosis or by changing shape and entering the cell through the cell membrane.
- Viruses cause diseases in humans and other animals; they often have to run their course before symptoms disappear.
- Examples of viral animal diseases include hepatitis C, chicken pox, and shingles.

Key Terms

- **receptor-mediated endocytosis**: a process by which cells internalize molecules (endocytosis) by the inward budding of plasma membrane vesicles containing proteins with receptor sites specific to the molecules being internalized

Animal Viruses

Animal viruses, unlike the viruses of plants and bacteria, do not have to penetrate a cell wall to gain access to the host cell. Non-enveloped or “naked” animal viruses may enter cells in two different ways. When a protein in the viral capsid binds to its receptor on the host cell, the virus may be taken inside the cell via a vesicle during the normal cell process of receptor-mediated endocytosis. An alternative method of cell penetration used by non-enveloped viruses is for capsid proteins to undergo shape changes after binding to the receptor, creating channels in the host cell membrane. The viral genome is then “injected” into the host cell through these channels in a manner analogous to that used by many bacteriophages. Enveloped viruses also have two ways of entering cells after binding to their receptors: receptor-mediated endocytosis and fusion. Many enveloped viruses enter the cell by receptor-mediated endocytosis in a fashion similar to some non-enveloped viruses. On the other hand, fusion only occurs with enveloped virions. These viruses, which include HIV among others, use special fusion proteins in their envelopes to cause the envelope to fuse with the plasma membrane of the cell, thus releasing the genome and capsid of the virus into the cell cytoplasm.

After making their proteins and copying their genomes, animal viruses complete the assembly of new virions and exit the cell. Using the example of HIV, enveloped animal viruses may bud from the cell membrane as they assemble themselves, taking a piece of the cell’s plasma membrane in the process. On the other hand, non-enveloped viral progeny, such as rhinoviruses, accumulate in infected cells until there is a signal for lysis or apoptosis, and all virions are released together.

Animal viruses are associated with a variety of human diseases. Some of them follow the classic pattern of acute disease, where symptoms worsen for a short period followed by the elimination of the virus from the body by the immune system with eventual recovery from the infection. Examples of acute viral diseases are the common cold and influenza. Other viruses cause long-term chronic infections, such as the virus causing hepatitis C, whereas others, like herpes simplex virus, cause only intermittent symptoms. Still other viruses, such as human herpes viruses 6 and 7, which in some cases can cause the minor childhood disease roseola, often successfully cause productive infections without causing any symptoms at all in the host; these patients have an asymptomatic infection.

In hepatitis C infections, the virus grows and reproduces in liver cells, causing low levels of liver damage. The damage is so low that infected individuals are often unaware that they are infected, with many infections only detected by routine blood work on patients with risk factors such as intravenous drug use. Since many of the symptoms of viral diseases are caused by immune responses, a lack of symptoms is an indication of a weak immune response to the virus. This allows the virus to escape elimination by the immune system and persist in individuals for years, while continuing to produce low levels of progeny virions in what is known as a chronic viral disease. Chronic infection of the liver by this virus leads to a much greater chance of developing liver cancer, sometimes as much as 30 years after the initial infection.

As mentioned, herpes simplex virus can remain in a state of latency in nervous tissue for months, even years. As the virus "hides" in the tissue and makes few if any viral proteins, there is nothing for the immune response to act against; immunity to the virus slowly declines. Under certain conditions, including various types of physical and psychological stress, the latent herpes simplex virus may be reactivated and undergo a lytic replication cycle in the skin, causing the lesions associated with the disease. Once virions are produced in the skin and viral proteins are synthesized, the immune response is again stimulated and resolves the skin lesions in a few days by destroying viruses in the skin. As a result of this type of replicative cycle, appearances of cold sores and genital herpes outbreaks only occur intermittently, even though the viruses remain in the nervous tissue for life. Latent infections are common with other herpes viruses as well, including the varicella-zoster virus that causes chickenpox. After having a chickenpox infection in childhood, the varicella-zoster virus can remain latent for many years and reactivate in adults to cause the painful condition known as "shingles".

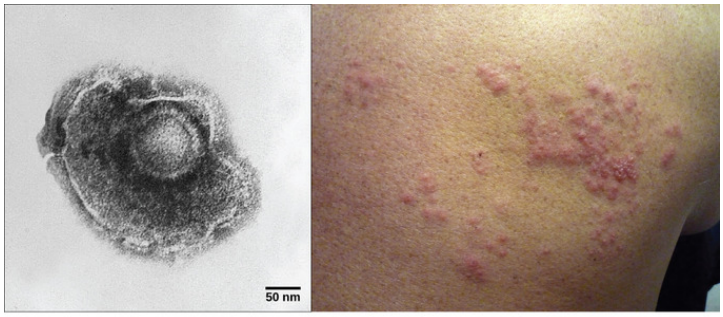

Figure 21.2C.1: **Chicken pox virus**: (a) Varicella-zoster, the virus that causes chickenpox, has an enveloped icosahedral capsid visible in this transmission electron micrograph. Its double-stranded DNA genome incorporates into the host DNA and reactivates after latency in the form of (b) shingles, often exhibiting a rash.

21.2D: Plant Viruses

Plant viruses can cause damage to stems, leaves, and fruits and can have a major impact on the economy because of food supply disruptions.

Learning Objectives

- Give examples of plant viruses and explain why they can be costly to the economy

Key Points

- Plants have cell walls which protect them from viruses entering their cells, so some type of damage must occur in order for them to become infected.
- When viruses are passed between plants, it is called horizontal transmission; when they are passed from the parent plant to the offspring, it is called vertical transmission.
- Symptoms of plant virus infection include malformed leaves, black streaks on the stems, discoloration of the leaves and fruits, and ring spots.
- Plant viruses can cause major disruptions to crop growth, which in turn can have a major impact on the economy.

Key Terms

- **horizontal transmission**: the transmission of an infectious agent, such as bacterial, fungal, or viral infection, between members of the same species that are not in a parent-child relationship
- **vertical transmission**: the transmission of an infection or other disease from the female of the species to the offspring

Plant Viruses

Plant viruses, like other viruses, contain a core of either DNA or RNA. As plant viruses have a cell wall to protect their cells, their viruses do not use receptor-mediated endocytosis to enter host cells as is seen with animal viruses. For many plant viruses to be transferred from plant to plant, damage to some of the plants' cells must occur to allow the virus to enter a new host. This damage is often caused by weather, insects, animals, fire, or human activities such as farming or landscaping. Additionally, plant offspring may inherit viral diseases from parent plants.

Plant viruses can be transmitted by a variety of vectors: through contact with an infected plant's sap, by living organisms such as insects and nematodes, and through pollen. When plant viruses are transferred between different plants, this is known as horizontal transmission; when they are inherited from a parent, this is called vertical transmission.

Symptoms of viral diseases vary according to the virus and its host. One common symptom is hyperplasia: the abnormal proliferation of cells that causes the appearance of plant tumors known as galls. Other viruses induce hypoplasia, or decreased cell growth, in the leaves of plants, causing thin, yellow areas to appear. Still other viruses affect the plant by directly killing plant cells; a process known as cell necrosis. Other symptoms of plant viruses include malformed leaves, black streaks on the stems of the plants, altered growth of stems, leaves, or fruits, and ring spots, which are circular or linear areas of discoloration found in a leaf.

Figure 21.2D.1: **Oak tree galls**: Galls are abnormal plant growth or swellings comprised of plant tissue. Galls are usually found on foliage or twigs. These unusual deformities are caused by plant growth-regulating chemicals or stimuli produced by an insect or other arthropod pest species. The chemicals produced by these causal organisms interfere with normal plant cell growth.

Plant viruses can seriously disrupt crop growth and development, significantly affecting our food supply. They are responsible for poor crop quality and quantity globally, and can bring about huge economic losses annually. Other viruses may damage plants used in landscaping. Some viruses that infect agricultural food plants include the name of the plant they infect, such as tomato spotted wilt virus, bean common mosaic virus, and cucumber mosaic virus. In plants used for landscaping, two of the most common viruses are peony ring spot and rose mosaic virus. There are far too many plant viruses to discuss each in detail, but symptoms of bean common mosaic virus result in lowered bean production and stunted, unproductive plants. In the ornamental rose, the rose mosaic disease causes wavy yellow lines and colored splotches on the leaves of the plant.

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21.3: Prevention and Treatment of Viral Infections

Topic hierarchy

- 21.3A: Vaccines and Immunity
- 21.3B: Vaccines and Anti-Viral Drugs for Treatment

21.3A: Vaccines and Immunity

Vaccinations prevent viruses from spreading by building immunity to the virus.

Learning Objectives

- Explain how vaccination protects vaccinated individuals and the community

Key Points

- Vaccinations are prepared with live viruses, killed viruses, or molecular subunits of the virus.
- A live vaccine consists of a small dose of the active virus.
- A killed vaccine contains the inactivated virus.
- It is possible, though rare, for live vaccines to cause the disease they're used to prevent.
- Live vaccines are made by growing the virus in a lab, which causes mutations that allow them to grow better in the lab than in the host, thereby inhibiting their ability to cause disease.
- Even though live vaccines are designed to cause few symptoms, back mutations can occur and cause the virus to readapt to the host and the disease to spread.

Key Terms

- **vaccination**: inoculation in order to protect against a particular disease or strain of disease; causes a primary immune response without illness, allowing the secondary response to destroy subsequent infection
- **live vaccine**: consists of an active microbe (virus or bacteria)
- **killed vaccine**: (inactivated vaccine) consists of virus particles which are grown in culture and then killed using a method such as with heat or formaldehyde

Vaccines for Prevention

While we do have limited numbers of effective antiviral drugs, such as those used to treat HIV and influenza, the primary method of controlling viral disease is by vaccination, which is intended to prevent outbreaks by building immunity to a virus or virus family. Vaccines may be prepared using live viruses, killed viruses, or molecular subunits of the virus. The killed viral vaccines and subunit viruses are both incapable of causing disease.

Figure 21.3A.1: **Vaccinations**: Vaccinations are designed to boost immunity to a virus to prevent infection.

Live viral vaccines are designed in the laboratory to cause few symptoms in recipients while giving them protective immunity against future infections. Polio was one disease that represented a milestone in the use of vaccines. Mass immunization campaigns in the 1950s and 1960s significantly reduced the incidence of the disease, which caused muscle paralysis in children and generated great fear in the general population when regional epidemics occurred. The success of the polio vaccine paved the way for the routine dispensation of childhood vaccines against measles, mumps, rubella, chickenpox, and other diseases.

The danger of using live vaccines, which are usually more effective than killed vaccines, is low, but significant since the possibility that these viruses will revert to their disease-causing form by back mutations is still present. Live vaccines are usually made by attenuating (weakening) the "wild-type" (disease-causing) virus by growing it in the laboratory in tissues or at temperatures different from what the virus is accustomed to in the host. Adaptations to these new cells or

temperatures induce mutations in the genomes of the virus, allowing it to grow better in the laboratory while inhibiting its ability to cause disease when reintroduced into conditions found in the host. These attenuated viruses still cause infection, but since they do not grow very well, they allow the immune response to develop in time to prevent major disease. Back mutations occur when the vaccine undergoes mutations in the host such that it readapts to the host and can again cause disease, which can then be spread to other humans in an epidemic. This type of scenario happened as recently as 2007 in Nigeria where mutations in a polio vaccine led to an epidemic of polio in that country.

Some vaccines are in continuous development because certain viruses, such as influenza and HIV, have a high mutation rate compared to other viruses and normal host cells. With influenza, mutations in the surface molecules of the virus help the organism evade the protective immunity that may have been obtained in a previous influenza season, making it necessary for individuals to get vaccinated every year. Other viruses, such as those that cause the childhood diseases measles, mumps, and rubella, mutate so infrequently that the same vaccine is used year after year.

21.3B: Vaccines and Anti-Viral Drugs for Treatment

Vaccines and anti-viral drugs can be used to inhibit the virus and reduce symptoms in individuals suffering from viral infections.

Learning Objectives

- Give examples of treatments with anti-viral drugs

Key Points

- Vaccines can boost an individual's immune response and control viruses, such as Ebola and rabies, before they become deadly.
- Anti-viral drugs inhibit the virus by blocking the actions of its proteins; they are used to control and reduce symptoms for viral diseases.
- Tamiflu can reduce flu symptoms by inhibiting the enzyme neuraminidase, which blocks the virus from spreading to uninfected cells.
- Anti-HIV drugs inhibit and control viral replication at many different phases of the HIV replication cycle, so patients taking these drugs have a higher survival rate.
- Viruses can develop resistance to individual anti-viral drugs.
- The treatment of HIV involves a mixture of different drugs (fusion inhibitors, reverse transcriptase inhibitors, integrase inhibitors, and protease inhibitors) in a cocktail; viruses have greater difficulty gaining resistance to multiple drugs.

Key Terms

- **virion**: a single individual particle of a virus (the viral equivalent of a cell)
- **anti-viral drug**: a class of medication, such as antibiotics, that inhibits the virus by blocking the actions of one or more of its proteins
- **Ebola virus**: an extremely contagious virus of African origin that causes Ebola fever, spread through contact with bodily fluids or secretions of infected persons and by airborne particles

Vaccines and Anti-viral Drugs for Treatment

In some cases, vaccines can be used to treat an active viral infection. The concept behind this is that by giving the vaccine, immunity is boosted without adding more disease-causing virus. In the case of rabies, a fatal neurological disease transmitted via the saliva of rabies virus-infected animals, the progression of the disease from the time of the animal bite to the time it enters the central nervous system may be two weeks or longer. This is enough time to vaccinate an individual who suspects that they have been bitten by a rabid animaL; their boosted immune response is sufficient to prevent the virus from entering nervous tissue. Thus, the potentially-fatal neurological consequences of the disease are averted; the individual only has to recover from the infected bite. This approach is also being used for the treatment of Ebola virus, one of the fastest and most deadly viruses on earth. Transmitted by bats and great apes, this disease can cause death in 70–90 percent of infected humans within two weeks. Using newly-developed vaccines that boost the immune response in this way, there is hope that affected individuals will be better able to control the virus, potentially saving a greater percentage of infected persons from a rapid and very painful death.

Another way of treating viral infections is the use of antiviral drugs. These drugs often have limited success in curing viral disease, but in many cases, they have been used to control and reduce symptoms for a wide variety of viral diseases. For most viruses, these drugs can inhibit the virus by blocking the actions of one or more of its proteins. It is important that the targeted proteins be encoded by viral genes and that these molecules are not present in a healthy host cell. In this way, viral growth is inhibited without damaging the host. There are large numbers of antiviral drugs available to treat infections, some specific for a particular virus and others that can affect multiple viruses.

Antivirals have been developed to treat genital herpes (herpes simplex II) and influenza. For genital herpes, drugs such as acyclovir can reduce the number and duration of episodes of active viral disease during which patients develop viral lesions in their skin cells. As the virus remains latent in nervous tissue of the body for life, this drug is not curative, but can make the symptoms of the disease more manageable. For influenza, drugs like Tamiflu (oseltamivir) can reduce the

duration of “flu” symptoms by one or two days, but the drug does not prevent symptoms entirely. Tamiflu works by inhibiting an enzyme (viral neuraminidase) that allows new virions to leave their infected cells. Thus, Tamiflu inhibits the spread of virus from infected to uninfected cells. Other antiviral drugs, such as Ribavirin, have been used to treat a variety of viral infections, although its mechanism of action against certain viruses remains unclear.

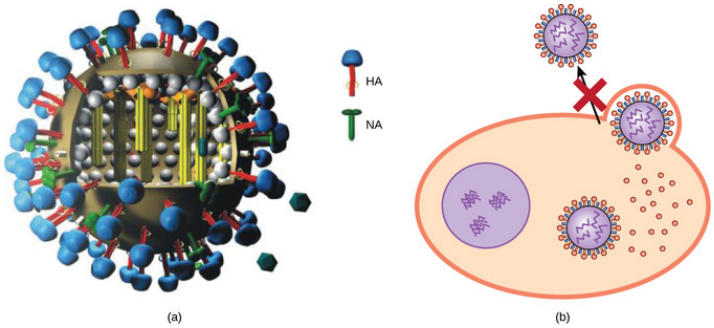

Figure $21.3B.1$: **Tamiflu**: (a) Tamiflu inhibits a viral enzyme called neuraminidase (NA) found in the influenza viral envelope. (b) Neuraminidase cleaves the connection between viral hemagglutinin (HA), also found in the viral envelope, and glycoproteins on the host cell surface. Inhibition of neuraminidase prevents the virus from detaching from the host cell, thereby blocking further infection.

Anti-HIV Drugs

By far, the most successful use of antivirals has been in the treatment of the retrovirus HIV, which causes a disease that, if untreated, is usually fatal within 10–12 years after infection. Anti-HIV drugs have been able to control viral replication to the point that individuals receiving these drugs survive for a significantly longer time than the untreated.

Anti-HIV drugs inhibit viral replication at many different phases of the HIV replicative cycle. Drugs have been developed that inhibit the fusion of the HIV viral envelope with the plasma membrane of the host cell (fusion inhibitors), the conversion of its RNA genome into double-stranded DNA (reverse transcriptase inhibitors), the integration of the viral DNA into the host genome (integrase inhibitors), and the processing of viral proteins (protease inhibitors).

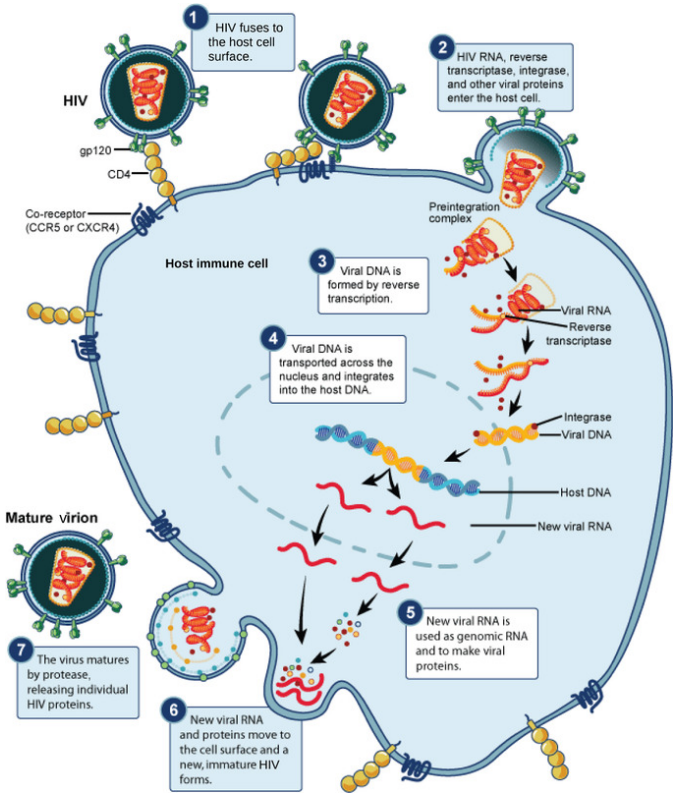

Figure 21.3B.1: **HIV**: HIV, an enveloped, icosahedral virus, attaches to the CD4 receptor of an immune cell and fuses with the cell membrane. Viral contents are released into the cell where viral enzymes convert the single-stranded RNA genome into DNA and incorporate it into the host genome.

When any of these drugs are used individually, the high mutation rate of the virus allows it to easily and rapidly develop resistance to the drug, limiting the drug's effectiveness. The breakthrough in the treatment of HIV was the development of HAART, highly-active anti-retroviral therapy, which involves a mixture of different drugs, sometimes called a drug "cocktail." By attacking the virus at different stages of its replicative cycle, it is much more difficult for the virus to develop resistance to multiple drugs at the same time. Still, even with the use of combination HAART therapy, there is concern that, over time, the virus will develop resistance to this therapy. Thus, new anti-HIV drugs are constantly being developed with the hope of continuing the battle against this highly fatal virus.

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21.4: Prions and Viroids

Topic hierarchy

- 21:4A: Prions and Viroids

21:4A: Prions and Viroids

Prions are infectious particles that contain no nucleic acids, and viroids are small plant pathogens that do not encode proteins.

Learning Objectives

- Describe prions and viroids and their basic properties

Key Points

- The prion appears to be the first infectious agent found whose transmission is not reliant upon genes made of DNA or RNA.
- An infectious structural variant of a normal cellular protein called PrP (prion protein) is known to cause spongiform encephalopathies.
- Prions have been implicated in fatal neurodegenerative diseases, such as kuru in humans and bovine spongiform encephalopathy (BSE) in cattle.
- Loss of motor control and unusual behaviors are common symptoms of individuals with kuru and BSE; symptoms are usually followed by death.
- Viroids do not have a capsid or outer envelope and can reproduce only within a host cell.
- Viroids are not known to cause any human diseases, but they are responsible for crop failures and the loss of millions of dollars in agricultural revenue each year.

Key Terms

- **prion**: a self-propagating misfolded conformer of a protein that is responsible for a number of diseases that affect the brain and other neural tissue
- **proteinaceous**: of, pertaining to, or consisting of protein
- **viroid**: plant pathogens that consist of just a short section of RNA, but without the protein coat typical of viruses

Prions

Prions, so-called because they are proteinaceous, are infectious particles, smaller than viruses, that contain no nucleic acids (neither DNA nor RNA). Historically, the idea of an infectious agent that did not use nucleic acids was considered impossible, but pioneering work by Nobel Prize-winning biologist Stanley Prusiner has convinced the majority of biologists that such agents do indeed exist.

Fatal neurodegenerative diseases, such as kuru in humans and bovine spongiform encephalopathy (BSE) in cattle (commonly known as “mad cow disease”), were shown to be transmitted by prions. The disease was spread by the consumption of meat, nervous tissue, or internal organs between members of the same species. Kuru, native to humans in Papua New Guinea, was spread from human to human via ritualistic cannibalism. BSE, originally detected in the United Kingdom, spread between cattle by the practice of including cattle nervous tissue in feed for other cattle. Individuals with kuru and BSE show symptoms of loss of motor control and unusual behaviors, such as uncontrolled bursts of laughter with kuru, followed by death. Kuru was controlled by inducing the population to abandon its ritualistic cannibalism.

On the other hand, BSE was initially thought to affect only cattle. Cattle that died of BSE had developed lesions or “holes” in the brain, causing the brain tissue to resemble a sponge. Later on in the outbreak, however, it was shown that a similar encephalopathy in humans known as variant Creutzfeldt-Jakob disease (CJD) could be acquired from eating beef from animals with BSE, sparking bans by various countries on the importation of British beef and causing considerable economic damage to the British beef industry. BSE still exists in various areas. Although a rare disease, individuals that acquire CJD are difficult to treat. The disease spreads from human to human by blood, so many countries have banned blood donation from regions associated with BSE.

The cause of spongiform encephalopathies, such as kuru and BSE, is an infectious structural variant of a normal cellular protein called PrP (prion protein). It is this variant that constitutes the prion particle. PrP exists in two forms: PrP^{c}, the normal form of the protein, and PrP^{sc}, the infectious form. Once introduced into the body, the PrP^{sc} contained within the prion binds to PrP^{c} and converts it to PrP^{sc}. This leads to an exponential increase of the PrP^{sc} protein, which aggregates.

PrP^{sc} is folded abnormally; the resulting conformation (shape) is directly responsible for the lesions seen in the brains of infected cattle. Thus, although not without some detractors among scientists, the prion appears to be an entirely new form of infectious agent; the first one found whose transmission is not reliant upon genes made of DNA or RNA.

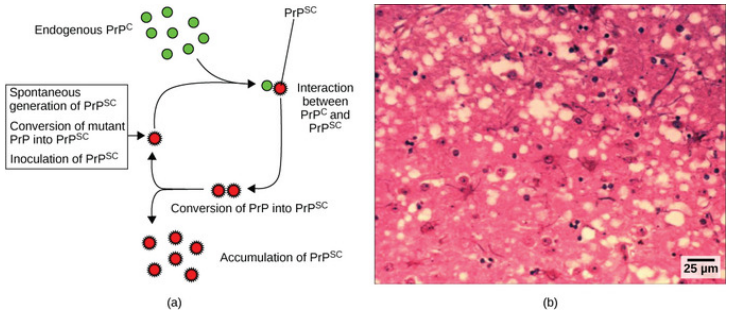

Figure 21.1: **Example of the formation of a prion**: (a) Endogenous normal prion protein (PrPc) is converted into the disease-causing form (PrPsc) when it encounters this variant form of the protein. PrPsc may arise spontaneously in brain tissue, especially if a mutant form of the protein is present, or it may occur via the spread of misfolded prions consumed in food into brain tissue. (b) This prion-infected brain tissue, visualized using light microscopy, shows the vacuoles that give it a spongy texture, typical of transmissible spongiform encephalopathies.

Viroids

Viroids are plant pathogens: small, single-stranded, circular RNA particles that are much simpler than a virus. They do not have a capsid or outer envelope, but, as with viruses, can reproduce only within a host cell. Viroids do not, however, manufacture any proteins. They produce only a single, specific RNA molecule. Human diseases caused by viroids have yet to be identified.

Viroid-infected plants are responsible for crop failures and the loss of millions of dollars in agricultural revenue each year. Some of the plants they infect include potatoes, cucumbers, tomatoes, chrysanthemums, avocados, and coconut palms.

Figure 21.1: **Potatoes infected by a viroid**: These potatoes have been infected by the potato spindle tuber viroid (PSTV). It is typically spread when infected knives are used to cut healthy potatoes, which are then planted.

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CHAPTER OVERVIEW

22: PROKARYOTES: BACTERIA AND ARCHAEA

Thumbnail: Scanning electron micrograph of neutrophil ingesting methicillin-resistant Staphylococcus aureus bacteria. (Public Domain; NIAID/NIH).

22.1: PROKARYOTIC DIVERSITY

22.1A: CLASSIFICATION OF PROKARYOTES
22.1B: THE ORIGINS OF ARCHAEA AND BACTERIA
22.1C: EXTREMOPHILES AND BIOFILMS
22.2: STRUCTURE OF PROKARYOTES

22.2A: BASIC STRUCTURES OF PROKARYOTIC CELLS
22.2B: PROKARYOTIC REPRODUCTION
22.3: PROKARYOTIC METABOLISM

22.3A: ENERGY AND NUTRIENT REQUIREMENTS FOR PROKARYOTES
22.3B: THE ROLE OF PROKARYOTES IN ECOSYSTEMS
22.4: BACTERIAL DISEASES IN HUMANS

22.4A: HISTORY OF BACTERIAL DISEASES
Infectious diseases, spread from person to person by bacteria, are among the leading causes of death despite advances in medical research.

22.4B: BIOFILMS AND DISEASE
22.4C: ANTIBIOTICS: ARE WE FACING A CRISIS?
Excessive use of antibiotics in animals or as imprudent medical treatments has resulted in the propagation of antibiotic-resistant bacteria.

22.4D: BACTERIAL FOODBORNE DISEASES
Foodborne diseases can be associated with bacteria-caused illnesses in both animal and plant-based food sources.

22.5: BENEFICIAL PROKARYOTES

22.5A: SYMBIOSIS BETWEEN BACTERIA AND EUKARYOTES
22.5B: EARLY BIOTECHNOLOGY: CHEESE, BREAD, WINE, BEER, AND YOGURT
22.5C: PROKARYOTES AND ENVIRONMENTAL BIOREMEDIATION

: Prokaryotic Diversity

Topic hierarchy

- 22.1A: Classification of Prokaryotes
- 22.1B: The Origins of Archaea and Bacteria
- 22.1C: Extremophiles and Biofilms

22.1A: Classification of Prokaryotes

Prokaryotic organisms were the first living things on earth and still inhabit every environment, no matter how extreme.

Learning Objectives

- Discuss the origins of prokaryotic organisms in terms of the geologic timeline

Key Points

- All living things can be classified into three main groups called domains; these include the Archaea, the Bacteria, and the Eukarya.
- Prokaryotes arose during the Precambrian Period 3.5 to 3.8 billion years ago.
- Prokaryotic organisms can live in every type of environment on Earth, from very hot, to very cold, to super haline, to very acidic.
- The domains Bacteria and Archaea are the ones containing prokaryotic organisms.
- The Archaea are prokaryotes that inhabit extreme environments, such as inside of volcanoes, while Bacteria are more common organisms, such as *E. coli*.

Key Terms

- **prokaryote**: an organism whose cell (or cells) are characterized by the absence of a nucleus or any other membrane-bound organelles
- **domain**: in the three-domain system, the highest rank in the classification of organisms, above kingdom: Bacteria, Archaea, and Eukarya
- **archaea**: a taxonomic domain of single-celled organisms lacking nuclei, formerly called archaebacteria, but now known to differ fundamentally from bacteria

Evolution of Prokaryotes

In the recent past, scientists grouped living things into five kingdoms (animals, plants, fungi, protists, and prokaryotes) based on several criteria such as: the absence or presence of a nucleus and other membrane-bound organelles, the absence or presence of cell walls, multicellularity, etc. In the late 20^{th} century, the pioneering work of Carl Woese and others compared sequences of small-subunit ribosomal RNA (SSU rRNA) which resulted in a more fundamental way to group organisms on earth. Based on differences in the structure of cell membranes and in rRNA, Woese and his colleagues proposed that all life on earth evolved along three lineages, called domains. The domain Bacteria comprises all organisms in the kingdom Bacteria, the domain Archaea comprises the rest of the prokaryotes, and the domain Eukarya comprises all eukaryotes, including organisms in the kingdoms Animalia, Plantae, Fungi, and Protista.

The current model of the evolution of the first, living organisms is that these were some form of prokaryotes, which may have evolved out of protobionts. In general, the eukaryotes are thought to have evolved later in the history of life. However, some authors have questioned this conclusion, arguing that the current set of prokaryotic species may have evolved from more complex eukaryotic ancestors through a process of simplification. Others have argued that the three domains of life arose simultaneously, from a set of varied cells that formed a single gene pool.

Two of the three domains, Bacteria and Archaea, are prokaryotic. Based on fossil evidence, prokaryotes were the first inhabitants on Earth, appearing 3.5 to 3.8 billion years ago during the Precambrian Period. These organisms are abundant and ubiquitous; that is, they are present everywhere. In addition to inhabiting moderate environments, they are found in extreme conditions: from boiling springs to permanently frozen environments in Antarctica; from salty environments like the Dead Sea to environments under tremendous pressure, such as the depths of the ocean; and from areas without oxygen, such as a waste management plant, to radioactively-contaminated regions, such as Chernobyl. Prokaryotes reside in the human digestive system and on the skin, are responsible for certain illnesses, and serve an important role in the preparation of many foods.

Figure 22.1*A*.1: **Prokaryotes in extreme environments**: Certain prokaryotes can live in extreme environments such as the Morning Glory pool, a hot spring in Yellowstone National Park. The spring's vivid blue color is from the prokaryotes that thrive in its very hot waters.

22.1B: The Origins of Archaea and Bacteria

Archaea are believed to have evolved from gram-positive bacteria and can occupy more extreme environments.

Learning Objectives

- Distinguish bacteria from archaea in terms of their origins

Key Points

- The first prokaryotes were adapted to the extreme conditions of early earth.
- It has been proposed that archaea evolved from gram-positive bacteria as a response to antibiotic selection pressures.
- Microbial mats and stromatolites represent some of the earliest prokaryotic formations that have been found.

Key Terms

- **stromatolite**: a laminated, columnar, rock-like structure built over geologic time by microorganisms such as cyanobacteria
- **gram-positive**: that is stained violet by Gram's method due to the presence of a peptidoglycan cell wall
- **sacculus**: a small sac
- **indel**: either an insertion or deletion mutation in the genetic code

Prokaryotes, the First Inhabitants of Earth

When and where did life begin? What were the conditions on earth when life began? Prokaryotes were the first forms of life on earth, existing for billions of years before plants and animals appeared. The earth and its moon are thought to be about 4.54 billion years old. This estimate is based on evidence from radiometric dating of meteorite material together with other substrate material from earth and the moon. Early earth had a very different atmosphere (contained less molecular oxygen) than it does today and was subjected to strong radiation; thus, the first organisms would have flourished where they were more protected, such as in ocean depths or beneath the surface of the earth. Also at this time, strong volcanic activity was common on Earth. It is probable that these first organisms, the first prokaryotes, were adapted to very high temperatures. Early earth was prone to geological upheaval and volcanic eruption, and was subject to bombardment by mutagenic radiation from the sun. The first organisms were prokaryotes that could withstand these harsh conditions.

Although probable prokaryotic cell fossils date to almost 3.5 billion years ago, most prokaryotes do not have distinctive morphologies; fossil shapes cannot be used to identify them as Archaea. Instead, chemical fossils of unique lipids are more informative because such compounds do not occur in other organisms. Some publications suggest that archaean or eukaryotic lipid remains are present in shales dating from 2.7 billion years ago. Such lipids have also been detected in Precambrian formations. The oldest such traces come from the Isua district of west Greenland, which include earth's oldest sediments, formed 3.8 billion years ago. The archaeal lineage may be the most ancient that exists on earth.

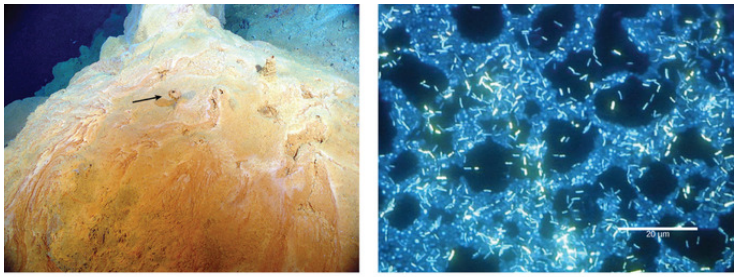
Within prokaryotes, archaeal cell structure is most similar to that of gram-positive bacteria, largely because both have a single lipid bilayer and usually contain a thick sacculus of varying chemical composition. In phylogenetic trees based upon different gene / protein sequences of prokaryotic homologs, the archaeal homologs are more closely related to those of Gram-positive bacteria. Archaea and gram-positive bacteria also share conserved indels in a number of important proteins, such as Hsp70 and glutamine synthetase.

It has been proposed that the archaea evolved from gram-positive bacteria in response to antibiotic selection pressure. This is suggested by the observation that archaea are resistant to a wide variety of antibiotics that are primarily produced by gram-positive bacteria and that these antibiotics primarily act on the genes that distinguish archaea from bacteria. The evolution of Archaea in response to antibiotic selection, or any other competitive selective pressure, could also explain their adaptation to extreme environments (such as high temperature or acidity) as the result of a search for unoccupied niches to escape from antibiotic-producing organisms.

Microbial Mats

Microbial mats or large biofilms may represent the earliest forms of life on earth; there is fossil evidence of their presence starting about 3.5 billion years ago. A microbial mat is a multi-layered sheet of prokaryotes that includes mostly bacteria,

but also archaea. Microbial mats are a few centimeters thick, typically growing where different types of materials interface, mostly on moist surfaces. The various types of prokaryotes that comprise the mats use different metabolic pathways, which is the reason for their various colors. Prokaryotes in a microbial mat are held together by a glue-like sticky substance that they secrete called extracellular matrix.



(a) (b)

Figure 22.1B.1: **Microbial mat**: This (a) microbial mat, about one meter in diameter, grows over a hydrothermal vent in the Pacific Ocean in a region known as the "Pacific Ring of Fire." The mat helps retain microbial nutrients. Chimneys, such as the one indicated by the arrow, allow gases to escape. (b) In this micrograph, bacteria are visualized using fluorescence microscopy.

The first microbial mats likely obtained their energy from chemicals found near hydrothermal vents. A hydrothermal vent is a breakage or fissure in the earth's surface that releases geothermally-heated water. With the evolution of photosynthesis about 3 billion years ago, some prokaryotes in microbial mats came to use a more widely-available energy source, sunlight, whereas others were still dependent on chemicals from hydrothermal vents for energy and food.

Stromatolites

Fossilized microbial mats represent the earliest record of life on earth. A stromatolite is a sedimentary structure formed when minerals are precipitated out of water by prokaryotes in a microbial mat. Stromatolites form layered rocks made of carbonate or silicate. Although most stromatolites are artifacts from the past, there are places on earth where stromatolites are still forming. For example, growing stromatolites have been found in the Anza-Borrego Desert State Park in San Diego County, California.

(a) (b)

Figure 22.1B.1: **Stromatolites**: (a) These living stromatolites are located in Shark Bay, Australia. (b) These fossilized stromatolites, found in Glacier National Park, Montana, are nearly 1.5 billion years old.

22.1C: Extremophiles and Biofilms

Prokaryotes are well adapted to living in all types of conditions, including extreme ones, and prefer to live in colonies called biofilms.

Learning Objectives

- Discuss the distinguishing features of extremophiles and the environments that produce biofilms

Key Points

- Prokaryotes live in all environments, no matter how extreme they may be.
- Bacteria that prefer very salty environments are called halophiles, while those that live in very acidic environments are called acidophiles.
- An example of a habitat that halophiles can colonize is the Dead Sea, a body of water that is 10 times saltier than regular ocean water.
- A biofilm is a microbial community held together in a gummy-textured matrix that consists primarily of polysaccharides secreted by the organisms.
- Biofilms can be found clogging pipes, on kitchen counters, or even on the surface of one's teeth.

Key Terms

- **extremophile**: an organism that lives under extreme conditions of temperature, salinity, etc; commercially important as a source of enzymes that operate under similar conditions
- **halophile**: an organism that lives and thrives in an environment of high salinity, often requiring such an environment; a form of extremophile
- **alkaliphile**: any organism that lives and thrives in an alkaline environment, such as a soda lake; a form of extremophile

Microbes Are Adaptable: Life in Moderate and Extreme Environments

Some organisms have developed strategies that allow them to survive harsh conditions. Prokaryotes thrive in a vast array of environments; some grow in conditions that would seem very normal to us, whereas others are able to thrive and grow under conditions that would kill a plant or animal. Almost all prokaryotes have a cell wall: a protective structure that allows them to survive in both hyper- and hypo-osmotic conditions. Some soil bacteria are able to form endospores that resist heat and drought, thereby allowing the organism to survive until favorable conditions recur. These adaptations, along with others, allow bacteria to be the most abundant life form in all terrestrial and aquatic ecosystems.

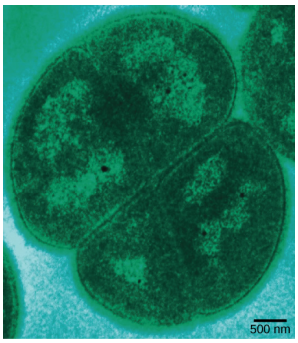


Figure 22.1C.1: **Bacteria and radiation tolerance**: Deinococcus radiodurans, visualized in this false color transmission electron micrograph, is a prokaryote that can tolerate very high doses of ionizing radiation. It has developed DNA repair mechanisms that allow it to reconstruct its chromosome even if it has been broken into hundreds of pieces by radiation or heat.

Other bacteria and archaea are adapted to grow under extreme conditions and are called extremophiles, meaning "lovers of extremes." Extremophiles have been found in all kinds of environments: the depth of the oceans, hot springs, the Arctic and the Antarctic, in very dry places, deep inside earth, in harsh chemical environments, and in high radiation environments, just to mention a few. These organisms give us a better understanding of prokaryotic diversity and raise the possibility of finding new prokaryotic species that may lead to the discovery of new therapeutic drugs or have industrial

applications. Because they have specialized adaptations that allow them to live in extreme conditions, many extremophiles cannot survive in moderate environments. There are many different groups of extremophiles. They are identified based on the conditions in which they grow best. Several habitats are extreme in multiple ways. For example, a soda lake is both salty and alkaline, so organisms that live in a soda lake must be both alkaliphiles and halophiles. Other extremophiles, like radioresistant organisms, do not prefer an extreme environment (in this case, one with high levels of radiation), but have adapted to survive in it.

Prokaryotes in the Dead Sea

One example of a very harsh environment is the Dead Sea, a hypersaline basin that is located between Jordan and Israel. Hypersaline environments are essentially concentrated seawater. In the Dead Sea, the sodium concentration is 10 times higher than that of seawater. The water also contains high levels of magnesium (about 40 times higher than in seawater) that would be toxic to most living things. Iron, calcium, and magnesium, elements that form divalent ions (Fe^{2+}, Ca^{2+}, and Mg^{2+}), produce what is commonly referred to as "hard" water. Taken together, the high concentration of divalent cations, the acidic pH (6.0), and the intense solar radiation flux make the Dead Sea a unique, and uniquely hostile, ecosystem.

Figure 22.1C.1: **Halophile habitats**: (a) The Dead Sea is hypersaline. Nevertheless, salt-tolerant bacteria thrive in this sea. (b) These halobacteria cells can form salt-tolerant bacterial mats.

The Ecology of Biofilms

Until a couple of decades ago, microbiologists used to think of prokaryotes as isolated entities living apart. This model, however, does not reflect the true ecology of prokaryotes, most of which prefer to live in communities where they can interact. A biofilm is a microbial community held together in a gummy-textured matrix that consists primarily of polysaccharides secreted by the organisms, together with some proteins and nucleic acids. Biofilms grow attached to surfaces. Some of the best-studied biofilms are composed of prokaryotes, although fungal biofilms have also been described, as well as some composed of a mixture of fungi and bacteria.

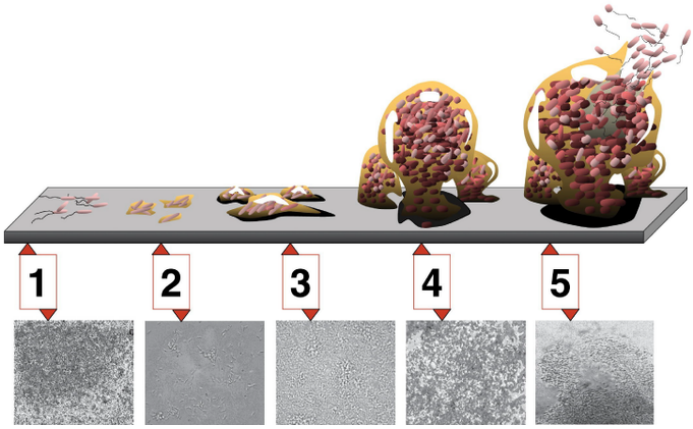


Figure 22.1C.1: **Biofilm Development**: Five stages of biofilm development are shown. During stage 1, initial attachment, bacteria adhere to a solid surface via weak van der Waals interactions. During stage 2, irreversible attachment, hairlike appendages called pili permanently anchor the bacteria to the surface. During stage 3, maturation I, the biofilm grows through cell division and recruitment of other bacteria. An extracellular matrix composed primarily of polysaccharides holds the biofilm together. During stage 4, maturation II, the biofilm continues to grow and takes on a more complex shape. During stage 5, dispersal, the biofilm matrix is partly broken down, allowing some bacteria to escape and colonize another surface. Micrographs of a Pseudomonas aeruginosa biofilm in each of the stages of development are shown.

Biofilms are present almost everywhere: they can cause the clogging of pipes and readily colonize surfaces in industrial settings. In recent, large-scale outbreaks of bacterial contamination of food, biofilms have played a major role. They also colonize household surfaces, such as kitchen counters, cutting boards, sinks, and toilets, as well as places on the human body, such as the surfaces of our teeth.

Interactions among the organisms that populate a biofilm, together with their protective exopolysaccharidic (EPS) environment, make these communities more robust than free-living, or planktonic, prokaryotes. The sticky substance that holds bacteria together also excludes most antibiotics and disinfectants, making biofilm bacteria hardier than their planktonic counterparts. Overall, biofilms are very difficult to destroy because they are resistant to many common forms of sterilization.

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: Structure of Prokaryotes

Topic hierarchy

- 22.2A: Basic Structures of Prokaryotic Cells
- 22.2B: Prokaryotic Reproduction

22.2A: Basic Structures of Prokaryotic Cells

Prokaryotes, found in both Domain Archaea and Bacteria, are unicellular organisms that lack membrane-bound organelles and a defined nucleus.

Learning Objectives

- Describe the basic structure of a typical prokaryote

Key Points

- Prokaryotic cells lack a defined nucleus, but have a region in the cell, termed the nucleoid, in which a single chromosomal, circular, double-stranded DNA molecule is located.
- Archaeal membranes have replaced the fatty acids of bacterial membranes with isoprene; some archaeal membranes are monolayer rather than bilayer.
- Prokaryotes can be further classified based on the composition of the cell wall in terms of the amount of peptidoglycan present.
- Gram-positive organisms typically lack the outer membrane found in gram-negative organisms and contain a large amount of peptidoglycan in the cell wall, roughly 90%.
- Gram-negative bacteria have a relatively thin cell wall composed of a few layers of peptidoglycan.
- Gram-negative bacteria have a relatively thin cell wall composed of a few layers of peptidoglycan.

Key Terms

- **nucleoid**: the irregularly-shaped region within a prokaryote cell where the genetic material is localized
- **plasmid**: a circle of double-stranded DNA that is separate from the chromosomes, which is found in bacteria and protozoa
- **osmotic pressure**: the hydrostatic pressure exerted by a solution across a semipermeable membrane from a pure solvent

The Prokaryotic Cell

Prokaryotes are unicellular organisms that lack organelles or other internal membrane-bound structures. Therefore, they do not have a nucleus, but, instead, generally have a single chromosome: a piece of circular, double-stranded DNA located in an area of the cell called the nucleoid. Most prokaryotes have a cell wall outside the plasma membrane.

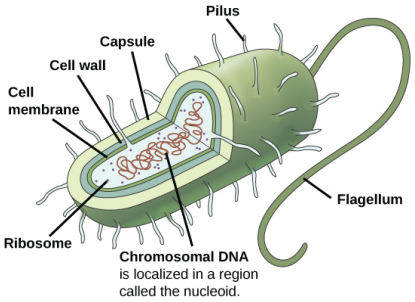

Figure $22.2A.1$: **Prokaryotic cell structure**: The features of a typical prokaryotic cell are shown.

The composition of the cell wall differs significantly between the domains Bacteria and Archaea, the two domains of life into which prokaryotes are divided. The composition of their cell walls also differs from the eukaryotic cell walls found in plants (cellulose) or fungi and insects (chitin). The cell wall functions as a protective layer and is responsible for the organism's shape. Some bacteria have a capsule outside the cell wall. Other structures are present in some prokaryotic species, but not in others. For example, the capsule found in some species enables the organism to attach to surfaces, protects it from dehydration and attack by phagocytic cells, and increases its resistance to our immune responses. Some species also have flagella used for locomotion and pili used for attachment to surfaces. Plasmids, which consist of extra-chromosomal DNA, are also present in many species of bacteria and archaea.

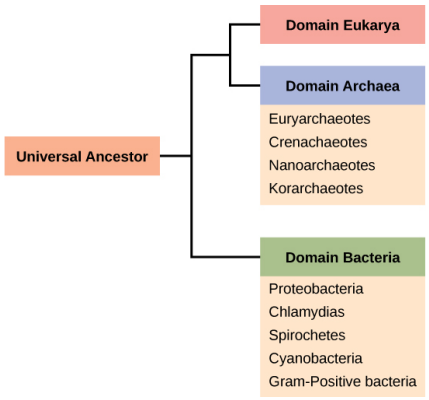

Figure 22.2A.1: **Domains of life**: Bacteria and Archaea are both prokaryotes, but differ enough to be placed in separate domains. An ancestor of modern Archaea is believed to have given rise to Eukarya, the third domain of life. Archaeal and bacterial phyla are shown; the evolutionary relationship between these phyla is still open to debate.

The Plasma Membrane

The plasma membrane is a thin lipid bilayer (6 to 8 nanometers) that completely surrounds the cell and separates the inside from the outside. Its selectively-permeable nature keeps ions, proteins, and other molecules within the cell, preventing them from diffusing into the extracellular environment, while other molecules may move through the membrane. The general structure of a cell membrane is a phospholipid bilayer composed of two layers of lipid molecules. In archaeal cell membranes, isoprene (phytanyl) chains linked to glycerol replace the fatty acids linked to glycerol in bacterial membranes. Some archaeal membranes are lipid monolayers instead of bilayers.

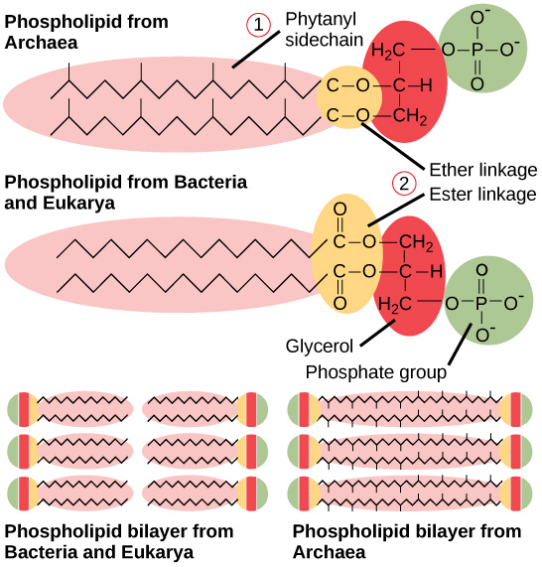

Figure 22.2A.1: **Plasma membrane structure**: Archaeal phospholipids differ from those found in Bacteria and Eukarya in two ways. First, they have branched phytanyl sidechains instead of linear ones. Second, an ether bond instead of an ester bond connects the lipid to the glycerol.

The Cell Wall

The cytoplasm of prokaryotic cells has a high concentration of dissolved solutes. Therefore, the osmotic pressure within the cell is relatively high. The cell wall is a protective layer that surrounds some cells and gives them shape and rigidity. It is located outside the cell membrane and prevents osmotic lysis (bursting due to increasing volume). The chemical composition of the cell walls varies between archaea and bacteria. It also varies between bacterial species.

Bacterial cell walls contain peptidoglycan composed of polysaccharide chains that are cross-linked by unusual peptides containing both L- and D-amino acids, including D-glutamic acid and D-alanine. Proteins normally have only L-amino acids; as a consequence, many of our antibiotics work by mimicking D-amino acids and, therefore, have specific effects on

bacterial cell wall development. There are more than 100 different forms of peptidoglycan. S-layer (surface layer) proteins are also present on the outside of cell walls of both archaea and bacteria.

Bacteria are divided into two major groups: gram-positive and gram-negative, based on their reaction to gram staining. Note that all gram-positive bacteria belong to one phylum; bacteria in the other phyla (Proteobacteria, Chlamydias, Spirochetes, Cyanobacteria, and others) are gram-negative. The gram-staining method is named after its inventor, Danish scientist Hans Christian Gram (1853–1938). The different bacterial responses to the staining procedure are ultimately due to cell wall structure. Gram-positive organisms typically lack the outer membrane found in gram-negative organisms. Up to 90 percent of the cell wall in gram-positive bacteria is composed of peptidoglycan, with most of the rest composed of acidic substances called teichoic acids. Teichoic acids may be covalently linked to lipids in the plasma membrane to form lipoteichoic acids. Lipoteichoic acids anchor the cell wall to the cell membrane. Gram-negative bacteria have a relatively thin cell wall composed of a few layers of peptidoglycan (only 10 percent of the total cell wall), surrounded by an outer envelope containing lipopolysaccharides (LPS) and lipoproteins. This outer envelope is sometimes referred to as a second lipid bilayer. The chemistry of this outer envelope is very different, however, from that of the typical lipid bilayer that forms plasma membranes.

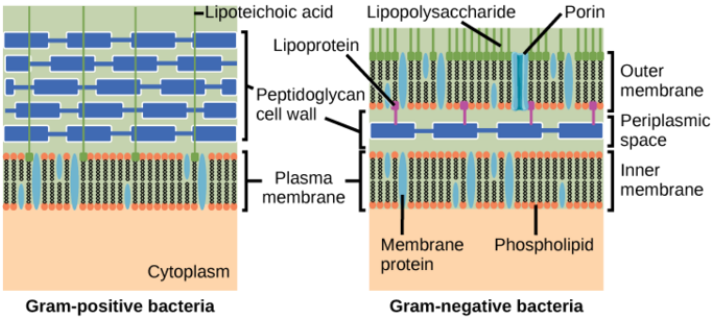

Figure 22.2A.1: **Gram-positive and gram-negative bacteria**: Bacteria are divided into two major groups: gram-positive and gram-negative. Both groups have a cell wall composed of peptidoglycan: in gram-positive bacteria, the wall is thick, whereas in gram-negative bacteria, the wall is thin. In gram-negative bacteria, the cell wall is surrounded by an outer membrane that contains lipopolysaccharides and lipoproteins. Porins, proteins in this cell membrane, allow substances to pass through the outer membrane of gram-negative bacteria. In gram-positive bacteria, lipoteichoic acid anchors the cell wall to the cell membrane.

22.2B: Prokaryotic Reproduction

Prokaryotes reproduce asexually by binary fission; they can also exchange genetic material by transformation, transduction, and conjugation.

Learning Objectives

- Distinguish among the types of reproduction in prokaryotes

Key Points

- Binary fission is a type of reproduction in which the chromosome is replicated and the resultant prokaryote is an exact copy of the parental prokaryate, thus leaving no opportunity for genetic diversity.
- Transformation is a type of prokaryotic reproduction in which a prokaryote can take up DNA found within the environment that has originated from other prokaryotes.
- Transduction is a type of prokaryotic reproduction in which a prokaryote is infected by a virus which injects short pieces of chromosomal DNA from one bacterium to another.
- Conjugation is a type of prokaryotic reproduction in which DNA is transferred between prokaryotes by means of a pilus.

Key Terms

- **transformation**: the alteration of a bacterial cell caused by the transfer of DNA from another, especially if pathogenic
- **transduction**: horizontal gene transfer mechanism in prokaryotes where genes are transferred using a virus
- **binary fission**: the process whereby a cell divides asexually to produce two daughter cells
- **conjugation**: the temporary fusion of organisms, especially as part of sexual reproduction
- **pilus**: a hairlike appendage found on the cell surface of many bacteria

Reproduction

Reproduction in prokaryotes is asexual and usually takes place by binary fission. The DNA of a prokaryote exists as as a single, circular chromosome. Prokaryotes do not undergo mitosis; rather the chromosome is replicated and the two resulting copies separate from one another, due to the growth of the cell. The prokaryote, now enlarged, is pinched inward at its equator and the two resulting cells, which are clones, separate. Binary fission does not provide an opportunity for genetic recombination or genetic diversity, but prokaryotes can share genes by three other mechanisms.

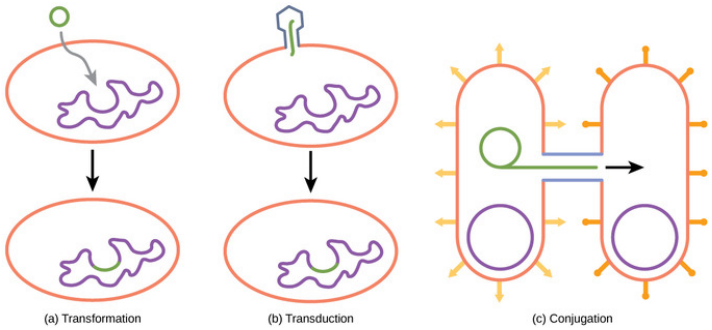

Figure 22.2B.1: **Modes of prokaryote reproduction**: Besides binary fission, there are three other mechanisms by which prokaryotes can exchange DNA. In (a) transformation, the cell takes up prokaryotic DNA directly from the environment. The DNA may remain separate as plasmid DNA or be incorporated into the host genome. In (b) transduction, a bacteriophage injects DNA into the cell that contains a small fragment of DNA from a different prokaryote. In (c) conjugation, DNA is transferred from one cell to another via a mating bridge that connects the two cells after the pilus draws the two bacteria close enough to form the bridge.

In transformation, the prokaryote takes in DNA found in its environment that is shed by other prokaryotes. If a nonpathogenic bacterium takes up DNA for a toxin gene from a pathogen and incorporates the new DNA into its own chromosome, it, too, may become pathogenic. In transduction, bacteriophages, the viruses that infect bacteria, sometimes also move short pieces of chromosomal DNA from one bacterium to another. Transduction results in a recombinant organism. Archaea are not affected by bacteriophages, but instead have their own viruses that translocate genetic material from one individual to another. In conjugation, DNA is transferred from one prokaryote to another by means of a pilus, which brings the organisms into contact

with one another. The DNA transferred can be in the form of a plasmid or as a hybrid, containing both plasmid and chromosomal DNA.

Reproduction can be very rapid: a few minutes for some species. This short generation time, coupled with mechanisms of genetic recombination and high rates of mutation, result in the rapid evolution of prokaryotes, allowing them to respond to environmental changes (such as the introduction of an antibiotic) very rapidly.

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Topic hierarchy

- 22.3A: Energy and Nutrient Requirements for Prokaryotes
- 22.3B: The Role of Prokaryotes in Ecosystems

22.3A: Energy and Nutrient Requirements for Prokaryotes

Prokaryotes need a source of energy, a source of carbon, macronutrients, and micronutrients to survive.

Learning Objectives

- Summarize what prokaryotes need to remain alive and functioning

Key Points

- The main components of the organic compounds in a prokaryotic cell are macronutrients (such as carbon, hydrogen, oxygen, nitrogen, phosphorus, and sulfur) that make up important biomolecules, including proteins and nucleic acids.
- Nutrients that are needed by prokaryotes in small quantities to perform cellular functions, including certain electron transport chain reactions, are known as micronutrients (e.g., iron, boron, chromium, and manganese).
- Autotrophic prokaryotes are able to fix inorganic compounds, such as carbon dioxide, to obtain carbon, while heterotrophic prokaryotes use organic compounds as their source of carbon.

Key Terms

- **macronutrient**: any of the elements required in large amounts by all living things
- **chemotroph**: an organism that obtains energy by the oxidation of electron-donating molecules in the environment
- **micronutrient**: a mineral, vitamin, or other substance that is essential, even in very small quantities, for growth or metabolism

Needs of Prokaryotes

The diverse environments and ecosystems on Earth have a wide range of conditions in terms of temperature, available nutrients, acidity, salinity, and energy sources. Prokaryotes are very well equipped to make their living out of a vast array of nutrients and conditions. To live, prokaryotes need a source of energy, a source of carbon, and some additional nutrients.

Macronutrients

Cells are essentially a well-organized assemblage of macromolecules and water. Recall that macromolecules are produced by the polymerization of smaller units called monomers. For cells to build all of the molecules required to sustain life, they need certain substances, collectively called nutrients. When prokaryotes grow in nature, they obtain their nutrients from the environment. Nutrients that are required in large amounts are called macronutrients, whereas those required in smaller or trace amounts are called micronutrients. Just a handful of elements are considered macronutrients: carbon, hydrogen, oxygen, nitrogen, phosphorus, and sulfur. (A mnemonic for remembering these elements is the acronym CHONPS.)

Why are these macronutrients needed in large amounts? They are the components of organic compounds in cells. Carbon is the major element in all macromolecules: carbohydrates, proteins, nucleic acids, lipids, and many other compounds. Carbon accounts for about 50 percent of the composition of the cell. Nitrogen represents 12 percent of the total dry weight of a typical cell and is a component of proteins, nucleic acids, and other cell constituents. Most of the nitrogen available in nature is either atmospheric nitrogen (N_2) or another inorganic form. Diatomic (N_2) nitrogen, however, can be converted into an organic form only by certain organisms, called nitrogen-fixing organisms. Both hydrogen and oxygen are part of many organic compounds and of water. Phosphorus is required by all organisms for the synthesis of nucleotides and phospholipids. Sulfur is part of the structure of some amino acids such as cysteine and methionine. It is also present in several vitamins and coenzymes. Other important macronutrients are potassium (K), magnesium (Mg), calcium (Ca), and sodium (Na). Although these elements are required in smaller amounts, they are very important for the structure and function of the prokaryotic cell.

Micronutrients

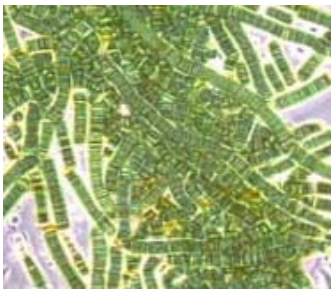

Figure 22.3A.1: **Filaments of photosynthetic cyanobacteria**: Cyanobacteria are an example of phototrophic prokaryotes.

In addition to these macronutrients, prokaryotes require various metallic elements in small amounts. These are referred to as micronutrients or trace elements. For example, iron is necessary for the function of the cytochromes involved in electron-transport reactions. Some prokaryotes require other elements (such as boron (B), chromium (Cr), and manganese (Mn)) primarily as enzyme cofactors.

The Ways in Which Prokaryotes Obtain Energy

Prokaryotes can use different sources of energy to assemble macromolecules from smaller molecules. Phototrophs (or phototrophic organisms) obtain their energy from sunlight. Chemotrophs (or chemosynthetic organisms) obtain their energy from chemical compounds. Chemotrophs that can use organic compounds as energy sources are called chemoorganotrophs. Those that can also use inorganic compounds as energy sources are called chemolithotrophs.

The Ways in Which Prokaryotes Obtain Carbon

Just as prokaryotes can use different sources of energy, they can also utilize different sources of carbon compounds. Recall that organisms that are able to fix inorganic carbon are called autotrophs. Autotrophic prokaryotes synthesize organic molecules from carbon dioxide. In contrast, heterotrophic prokaryotes obtain carbon from organic compounds. To make the picture more complex, the terms that describe how prokaryotes obtain energy and carbon can be combined. Thus, photoautotrophs use energy from sunlight and carbon from carbon dioxide and water, whereas chemoheterotrophs obtain energy and carbon from an organic chemical source. Chemolithoautotrophs obtain their energy from inorganic compounds, while building their complex molecules from carbon dioxide. Table 1 summarizes carbon and energy sources in prokaryotes.

Carbon and Energy Sources in Prokaryotes				
Energy Sources			**Carbon Sources**	
Light	Chemicals		Carbon dioxide	Organic compounds
Phototrophs	Chemotrophs		Autotrophs	Heterotrophs
	Organic chemicals	Inorganic chemicals		
	Chemo-organotrophs	Chemolithotrophs		

Figure 22.3A.1: **Table 1. Carbon and energy sources in prokaryotes**: This table summarizes the types of energy and carbon sources for different types of prokaryotes.

22.3B: The Role of Prokaryotes in Ecosystems

Prokaryotes play vital roles in the movement of carbon dioxide and nitrogen in the carbon and nitrogen cycles.

Learning Objectives

- Give examples of the beneficial roles played by prokaryotes in different ecosystems

Key Points

- Carbon and nitrogen are both macronutrients that are necessary for life on earth; prokaryotes play vital roles in their cycles.
- The carbon cycle is maintained by prokaryotes that remove carbon dioxide and return it to the atmosphere.
- Prokaryotes play a major role in the nitrogen cycle by fixing atomspheric nitrogen into ammonia that plants can use and by converting ammonia into other forms of nitrogen sources.

Key Terms

- **carbon cycle**: the physical cycle of carbon through the earth's biosphere, geosphere, hydrosphere, and atmosphere that includes such processes as photosynthesis, decomposition, respiration and carbonification
- **nitrogen cycle**: the natural circulation of nitrogen, in which atmospheric nitrogen is converted to nitrogen oxides and deposited in the soil, where it is used by organisms or decomposed back to elemental nitrogen
- **nitrogen fixation**: the conversion of atmospheric nitrogen into ammonia and organic derivatives, by natural means, especially by microorganisms in the soil, into a form that can be assimilated by plants

Role of Prokaryotes in Ecosystems

Prokaryotes are ubiquitous: There is no niche or ecosystem in which they are not present. Prokaryotes play many roles in the environments they occupy, but the roles they play in the carbon and nitrogen cycles are vital to life on earth.

Prokaryotes and the Carbon Cycle

Carbon is one of the most important macronutrients. Prokaryotes play an important role in the carbon cycle. Carbon is cycled through earth's major reservoirs: land, the atmosphere, aquatic environments, sediments and rocks, and biomass. The movement of carbon is via carbon dioxide, which is removed from the atmosphere by land plants and marine prokaryotes and is returned to the atmosphere via the respiration of chemoorganotrophic organisms, including prokaryotes, fungi, and animals. Although the largest carbon reservoir in terrestrial ecosystems is in rocks and sediments, that carbon is not readily available.

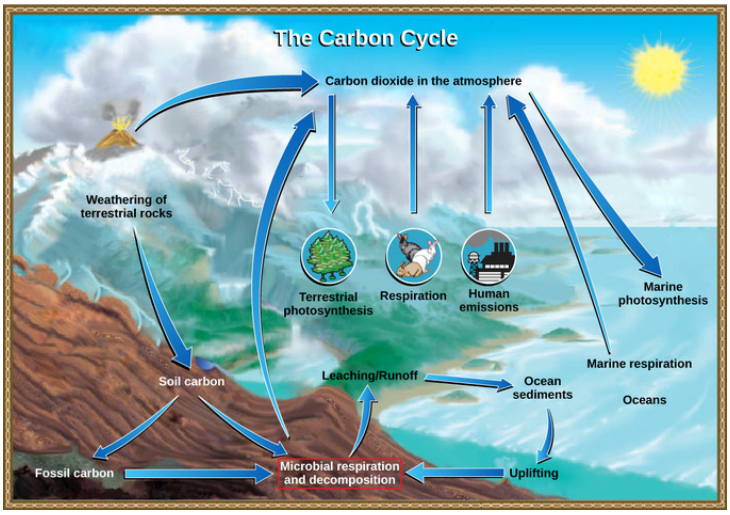

Figure 22.3B.1: **Carbon cycle**: Prokaryotes play a significant role in continuously moving carbon through the biosphere.

A large amount of available carbon is found in land plants, which are producers that use carbon dioxide from the air to synthesize carbon compounds. Related to this, one very significant source of carbon compounds is humus, which is a mixture of organic materials from dead plants and prokaryotes that have resisted decomposition. Consumers such as animals use organic compounds generated by producers, releasing carbon dioxide to the atmosphere. Then, bacteria and fungi, collectively called decomposers, carry out the breakdown (decomposition) of plants and animals and their organic compounds. The most important contributor of carbon dioxide to the atmosphere is microbial decomposition of dead material (dead animals, plants, and humus).

In aqueous environments and their anoxic sediments, there is another carbon cycle taking place. In this case, the cycle is based on one-carbon compounds. In anoxic sediments, prokaryotes, mostly archaea, produce methane (CH_4). This methane moves into the zone above the sediment, which is richer in oxygen and supports bacteria called methane oxidizers that oxidize methane to carbon dioxide, which then returns to the atmosphere.

Prokaryotes and the Nitrogen Cycle

Nitrogen is a very important element for life because it is part of proteins and nucleic acids. As a macronutrient in nature, it is recycled from organic compounds to ammonia, ammonium ions, nitrate, nitrite, and nitrogen gas by myriad processes, many of which are carried out solely by prokaryotes; they are key to the nitrogen cycle. The largest pool of nitrogen available in the terrestrial ecosystem is gaseous nitrogen from the air, but this nitrogen is not usable by plants, which are primary producers. Gaseous nitrogen is transformed, or "fixed," into more-readily available forms such as ammonia through the process of nitrogen fixation by natural means, especially by microorganisms (prokayotes) in the soil. Ammonia can then be used by plants or converted to other forms.

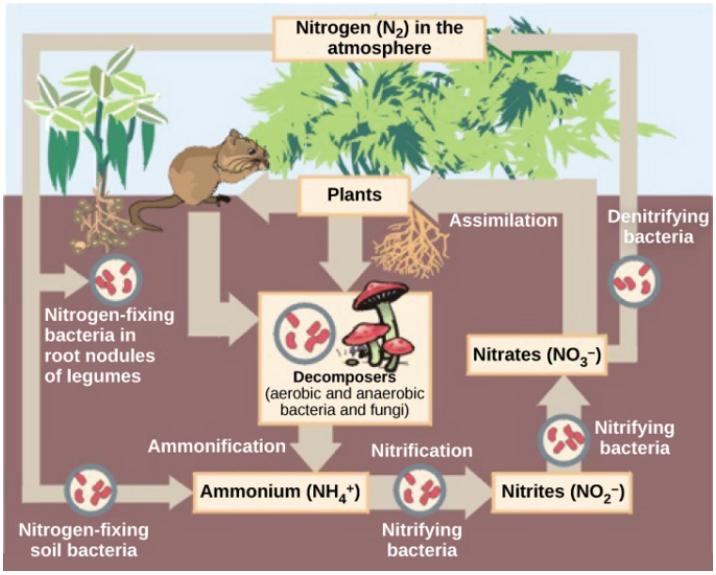

Figure $22.3B.1$: **Nitrogen cycle**: Prokaryotes play a key role in the nitrogen cycle.

Another source of ammonia is ammonification, the process by which ammonia is released during the decomposition of nitrogen-containing organic compounds. Ammonia released to the atmosphere, however, represents only 15 percent of the total nitrogen released; the rest is as N_2 and N_2O. Ammonia is catabolized anaerobically by some prokaryotes, yielding N_2 as the final product. Nitrification is the conversion of ammonium to nitrite and nitrate. Nitrification in soils is carried out by bacteria belonging to the genera *Nitrosomas*, *Nitrobacter*, and *Nitrospira*. The bacteria perform the reverse process, the reduction of nitrate from the soils to gaseous compounds such as N_2O, NO, and N_2, a process called denitrification.

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22.4: Bacterial Diseases in Humans

22.4A: History of Bacterial Diseases

Infectious diseases, spread from person to person by bacteria, are among the leading causes of death despite advances in medical research.

22.4B: Biofilms and Disease

22.4C: Antibiotics: Are We Facing a Crisis?

Excessive use of antibiotics in animals or as imprudent medical treatments has resulted in the propagation of antibiotic-resistant bacteria.

22.4D: Bacterial Foodborne Diseases

Foodborne diseases can be associated with bacteria-caused illnesses in both animal and plant-based food sources.

22.4A: History of Bacterial Diseases

LEARNING OBJECTIVES

- Give examples of historical, new, and re-emerging bacterial diseases in humans

There are records about infectious diseases as far back as 3000 B.C.E. A number of significant pandemics caused by bacteria have been documented over several hundred years. Some of the most memorable pandemics led to the decline of cities and nations. In the 21st century, infectious diseases remain among the leading causes of death worldwide, despite advances made in medical research and treatments in recent decades. A disease spreads when the pathogen that causes it is passed from one person to another. For a pathogen to cause disease, it must be able to reproduce in the host's body and damage the host in some way.

The Plague of Athens

In 430 B.C.E., the Plague of Athens killed one-quarter of the Athenian troops that were fighting in the great Peloponnesian War and weakened Athens' dominance and power. The plague impacted people living in overcrowded Athens as well as troops aboard ships that had to return to Athens. The source of the plague may have been identified recently when researchers from the University of Athens were able to use DNA from teeth recovered from a mass grave. The scientists identified nucleotide sequences from a pathogenic bacterium, *Salmonella enterica* serovar typhi, which causes typhoid fever. This disease is commonly seen in overcrowded areas and has caused epidemics throughout recorded history.

Figure 22.4*A*.1: **Salmonella enterica serovar typhi**: Salmonella enterica serovar typhi, the causative agent of typhoid fever, is a gram-negative, rod-shaped gamma protobacterium. Typhoid fever, which is spread through feces, causes intestinal hemorrhage, high fever, delirium and dehydration. Today, between 16 and 33 million cases of this re-emerging disease occur annually, resulting in over 200,000 deaths. Carriers of the disease can be asymptomatic. In a famous case in the early 1900s, a cook named Mary Mallon unknowingly spread the disease to over fifty people, three of whom died. Other Salmonella serotypes cause food poisoning.

Bubonic Plagues

From 541 to 750 C.E.., an outbreak of what was likely a bubonic plague (the Plague of Justinian), eliminated one-quarter to one-half of the human population in the eastern Mediterranean region. The population in Europe dropped by 50 percent during this outbreak. The bubonic plague would strike Europe more than once.

One of the most devastating pandemics was the Black Death (1346 to 1361) that is believed to have been another outbreak of bubonic plague caused by the bacterium *Yersinia pestis*. It is thought to have originated initially in China and spread along the Silk Road, a network of land and sea trade routes, to the Mediterranean region and Europe, carried by rat fleas living on black rats that were always present on ships. The Black Death reduced the world's population from an estimated 450 million to about 350 to 375 million. Bubonic plague struck London hard again in the mid-1600s. In modern times, approximately 1,000 to 3,000 cases of plague arise globally each year. Although contracting bubonic plague before antibiotics meant almost certain death, the bacterium responds to several types of modern antibiotics; mortality rates from plague are now very low.

Figure 22.4A.1: **Bubonic plague**: The (a) Great Plague of London killed an estimated 200,000 people, or about twenty percent of the city's population. The causative agent, the (b) bacterium Yersinia pestis, is a gram-negative, rod-shaped bacterium from the class Gamma Proteobacteria. The disease is transmitted through the bite of an infected flea, which is infected by a rodent. Symptoms include swollen lymph nodes, fever, seizure, vomiting of blood, and (c) gangrene.

Migration of Diseases to New Populations

Over the centuries, Europeans tended to develop genetic immunity to endemic infectious diseases, but when European conquerors reached the western hemisphere, they brought with them disease-causing bacteria and viruses, which triggered epidemics that completely devastated populations of Native Americans who had no natural resistance to many European diseases. It has been estimated that up to 90 percent of Native Americans died from infectious diseases after the arrival of Europeans, making conquest of the New World a foregone conclusion.

Emerging and Re-emerging Diseases

The distribution of a particular disease is dynamic. Therefore, changes in the environment, the pathogen, or the host population can dramatically impact the spread of a disease. According to the World Health Organization (WHO), an emerging disease is one that has appeared in a population for the first time, or that may have existed previously, but is rapidly increasing in incidence or geographic range. This definition also includes re-emerging diseases that were previously under control. Approximately 75 percent of recently-emerging infectious diseases affecting humans are zoonotic diseases. Zoonoses, diseases that primarily infect animals and are transmitted to humans, are of both viral and bacterial origins. Brucellosis is an example of a prokaryotic zoonosis that is re-emerging in some regions. Necrotizing fasciitis (commonly known as flesh-eating bacteria) has been increasing in virulence for the last 80 years, for unknown reasons.

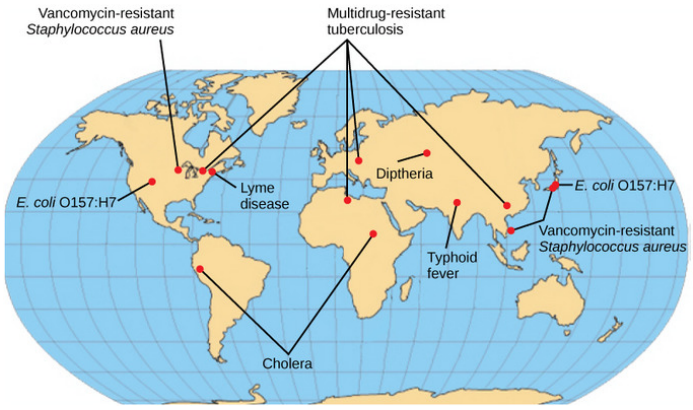

Figure 22.4A.1: **Regions of bacterial disease emergence**: The map shows regions where bacterial diseases are emerging or reemerging.

Some of the currently-emerging diseases are not actually new, but are diseases that were catastrophic in the past. They devastated populations, became dormant for a while, but have re-emerged, sometimes more virulent than before. Such was the case with bubonic plague. Other diseases, like tuberculosis, were never eradicated, but were under control in some regions of the world until re-emerging, mostly in urban centers with high concentrations of immunocompromised people. The WHO has identified certain diseases whose worldwide re-emergence should be monitored. Among these are two viral diseases (dengue fever and yellow fever) and three bacterial diseases (diphtheria, cholera, and bubonic plague). The war against infectious diseases has no foreseeable end.

Key Points

- A pathogen must be able to reproduce in the host's body and damage the host in some way to cause disease.
- Before antibiotics, contracting plagues usually meant death; however, most bacterium associated with these plagues respond to modern antibiotics; mortality rates from these diseases are now very low.
- Emerging diseases include those that have appeared in a population for the first time or that may have existed previously, but are rapidly spreading; this also includes re-emerging diseases that were previously under control.
- The spread of disease can be impacted dramatically by changes in the environment, the pathogen, or the host population.

Key Terms

- **zoonosis**: an animal disease that can be transmitted to humans
- **plague**: an epidemic or pandemic caused by any pestilence
- **pathogen**: any organism or substance, especially a microorganism, capable of causing disease, such as bacteria, viruses, protozoa, or fungi

22.4B: Biofilms and Disease

Biofilms, complex colonies of bacteria acting as a unit in their release of toxins, are highly resistant to antibiotics and host defense.

Learning Objectives

- Give examples of the roles played by biofilms in human diseases

Key Points

- Once a biofilm infection is established, it is very difficult to eradicate because biofilms exhibit great resistance to most methods used to control microbial growth, including antibiotics.
- Biofilms are able to grow anywhere there is an optimal combination of moisture, nutrients, and a surface.
- Biofilms are responsible for diseases such as infections in patients and readily settle within wounds and burns; they can also easily colonize medical devices and other surfaces where sterility is vital for health.

Key Terms

- **biofilm**: a thin film of mucus created by and containing a colony of bacteria and other microorganisms
- **nosocomial**: contracted in a hospital, or arising from hospital treatment

Biofilms and Disease

Biofilms are complex colonies of bacteria (often containing several species) that exchange chemical signals to coordinate the release of toxins that will attack the host. Once established, they are very difficult to destroy as they are highly resistant to antimicrobial treatments and host defense. Biofilms form when microorganisms adhere to the surface of some object in a moist environment and begin to reproduce. They grow virtually everywhere in almost any environment where there is a combination of moisture, nutrients, and a surface. Biofilms are responsible for diseases such as infections in patients with cystic fibrosis, Legionnaires' disease, and otitis media. They produce dental plaque and colonize catheters, prostheses, transcutaneous and orthopedic devices, contact lenses, and internal devices such as pacemakers. They also form in open wounds and burned tissue. In healthcare environments, biofilms grow on hemodialysis machines, mechanical ventilators, shunts, and other medical equipment. In fact, 65 percent of all infections acquired in the hospital (nosocomial infections) are attributed to biofilms. Biofilms are also related to diseases contracted from food because they colonize the surfaces of vegetable leaves and meat, as well as food-processing equipment that is not adequately cleaned.

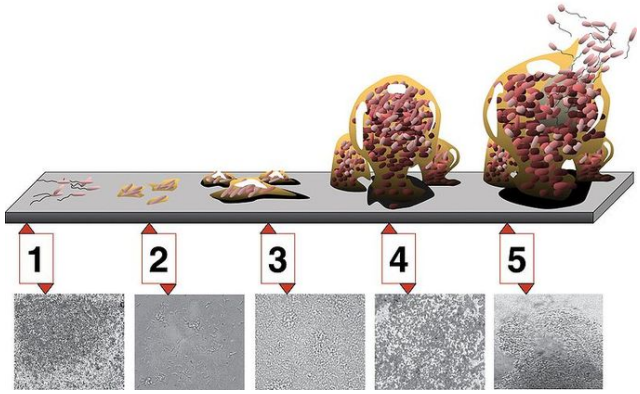

Figure 22.4B.1: **The Five Stages of Biofilm Development**: Stage 1: initial attachment; stage 2: irreversible attachment; stage 3: maturation I; stage 4: maturation II; stage 5: dispersion. Each stage of development in the diagram is paired with a photomicrograph of a developing Pseudomonas aeruginosa biofilm. All photomicrographs are shown at the same scale.

Biofilm infections develop gradually and often do not cause immediate symptoms. They are rarely resolved by host defense mechanisms. Once an infection by a biofilm is established, it is very difficult to eradicate because biofilms tend to be resistant to most of the methods used to control microbial growth, including antibiotics. Biofilms respond poorly or only temporarily to antibiotics. It has been said that they can resist up to 1,000 times the antibiotic concentrations used to

e same bacteria when they are free-living or planktonic. An antibiotic dose that large would harm the patient; re, scientists are working on new ways to eradicate biofilms.

22.4C: Antibiotics: Are We Facing a Crisis?

LEARNING OBJECTIVES

- Discuss antibiotic resistance.

The word antibiotic comes from the Greek word “anti” meaning “against” and “bios” meaning “life.” An antibiotic is a chemical, produced either by microbes or synthetically, that is hostile to the growth of other organisms. Today’s news and other media often address concerns about an antibiotic crisis. Are the antibiotics that easily treated bacterial infections in the past becoming obsolete? Are there new “superbugs”: bacteria that have evolved to become more resistant to our arsenal of antibiotics? Is this the beginning of the end of antibiotics? All these questions challenge the healthcare community.

One of the main causes of resistant bacteria is the abuse of antibiotics. The imprudent and excessive use of antibiotics has resulted in the natural selection of resistant forms of bacteria. The antibiotic kills most of the infecting bacteria; therefore, only the resistant forms remain. These resistant forms reproduce, resulting in an increase in the proportion of resistant forms over non-resistant ones. Another major misuse of antibiotics is in patients with colds or the flu, for which antibiotics are useless. There is also the excessive use of antibiotics in livestock along with the routine use of antibiotics in animal feed, both of which promote bacterial resistance. In the United States, 70 percent of the antibiotics produced are fed to animals. Because they are given to livestock in low doses, the probability of resistance developing is maximized. These resistant bacteria are readily transferred to humans.

One of the Superbugs: MRSA

The imprudent use of antibiotics has paved the way for bacteria to expand populations of resistant forms. For example, *Staphylococcus aureus*, often called “staph,” is a common bacterium that can live in the human body and is usually easily treated with antibiotics. A very dangerous strain, however, methicillin-resistant *Staphylococcus aureus* (MRSA) has made the news over the past few years. This strain is resistant to many commonly-used antibiotics, including methicillin, amoxicillin, penicillin, and oxacillin. MRSA can cause infections of the skin, but it can also infect the bloodstream, lungs, urinary tract, or sites of injury. While MRSA infections are common among people in healthcare facilities, they have also appeared in healthy people who have not been hospitalized, but who live or work in tight populations (like military personnel and prisoners). Researchers have expressed concern about the way this latter source of MRSA targets a much younger population than those residing in care facilities. The Journal of the American Medical Association (JAMA) reported that, among MRSA-afflicted persons in healthcare facilities, the average age is 68, whereas people with “community-associated MRSA” (CA-MRSA) have an average age of 23.

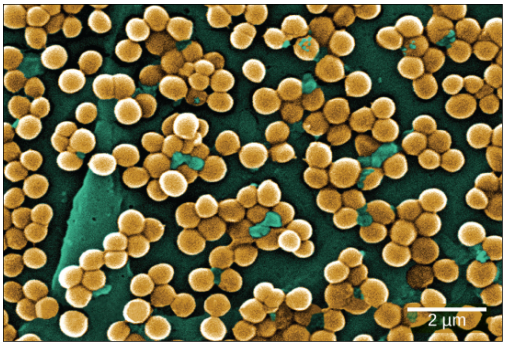

Figure 22.4C.1: **MRSA, a superbug**: This scanning electron micrograph shows methicillin-resistant Staphylococcus aureus bacteria, commonly known as MRSA. S. aureus is not always pathogenic, but can cause diseases such as food poisoning and skin and respiratory infections.

In summary, the medical community is facing an antibiotic crisis. Some scientists believe that after years of being protected from bacterial infections by antibiotics, we may be returning to a time in which a simple bacterial infection could again devastate the human population. Researchers are developing new antibiotics, but it takes many years of research and clinical trials, plus financial investments in the millions of dollars, to generate an effective and approved drug.

Key Points

- In antibiotic resistance, antibiotics will kill most of the infecting bacteria leaving behind only the resistant forms, which reproduce, resulting in an increase in the proportion of resistant forms over non-resistant ones.
- Cold and flu treatments and the medication of livestock are examples of antibiotic misuse responsible for bacterial resistance.
- Methicillin-resistant *Staphylococcus aureus* (MRSA) is an example of a dangerous antibiotic-resistant strain of bacteria that can infect sick, as well as healthy people.
- Due to the growing resistance to antibiotics, scientists believe that we may be returning to a time in which a simple bacterial infection could again detrimentally impact human populations.

Key Terms

- **antibiotic**: any substance that can destroy or inhibit the growth of bacteria and similar microorganisms

Reference

Naimi, TS, LeDell, KH, Como-Sabetti, K, et al. Comparison of community- and health care-associated methicillin-resistant *Staphylococcus aureus* infection. JAMA 290 (2003): 2976–84, doi: 10.1001/jama.290.22.2976.

22.4D: Bacterial Foodborne Diseases

LEARNING OBJECTIVES

- Give examples of bacterial foodborne diseases in humans

Prokaryotes are everywhere. They readily colonize the surface of any type of material. Food is not an exception. Most of the time, prokaryotes colonize food and food-processing equipment in the form of a biofilm. Outbreaks of bacterial infection related to food consumption are common. A foodborne disease (colloquially called "food poisoning") is an illness resulting from the consumption of pathogenic bacteria, viruses, or other parasites that contaminate food. Although the United States has one of the safest food supplies in the world, the U.S. Centers for Disease Control and Prevention (CDC) has reported that "76 million people get sick, more than 300,000 are hospitalized, and 5,000 Americans die each year from foodborne illness."

The characteristics of foodborne illnesses have changed over time. In the past, sporadic cases of botulism, the potentially fatal disease produced by a toxin from the anaerobic bacterium *Clostridium botulinum*, were relatively common. Some of the sources for this bacterium were non-acidic canned foods, homemade pickles, and processed meat and sausages. The can, jar, or package created a suitable anaerobic environment where *Clostridium* could grow. However, proper sterilization and canning procedures have reduced the incidence of this disease.

While people may tend to think of foodborne illnesses as associated with animal-based foods, most cases are now linked to produce. There have been serious, produce-related outbreaks associated with raw spinach in the United States and with vegetable sprouts in Germany. These types of outbreaks have become more common. The raw spinach outbreak in 2006 was produced by the bacterium *E. coli* serotype O157:H7. A serotype is a strain of bacteria that carries a set of similar antigens on its cell surface. There are often many different serotypes of a bacterial species. Most *E. coli* are not particularly dangerous to humans, but serotype O157:H7 can cause bloody diarrhea and is potentially fatal.

All types of food can potentially be contaminated with bacteria. Recent outbreaks of *Salmonella* reported by the CDC occurred in foods as diverse as peanut butter, alfalfa sprouts, and eggs. A deadly outbreak in Germany in 2010 was caused by *E. coli* contamination of vegetable sprouts. The strain that caused the outbreak was found to be a new serotype not previously involved in other outbreaks, which indicates that *E. coli* is continuously evolving.

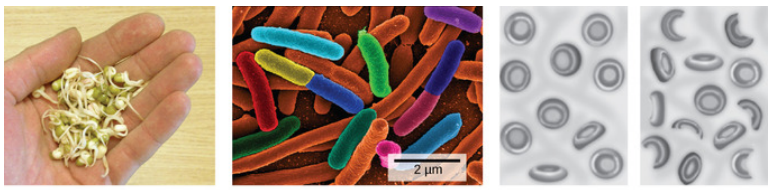

Figure 22.4*D*.1: **Bacterial illnesses from food**: (a) Vegetable sprouts grown at an organic farm were the cause of an (b) E. coli outbreak that killed 32 people and sickened 3,800 in Germany in 2011. The strain responsible, E. coli O104:H4, produces Shiga toxin, a substance that inhibits protein synthesis in the host cell. The toxin (c) destroys red blood cells, resulting in bloody diarrhea. Deformed red blood cells clog the capillaries of the kidney, which can lead to kidney failure, as happened to 845 patients in the 2011 outbreak. Kidney failure is usually reversible, but some patients experience kidney problems years later.

Key Points

- Food and food-processing equipment are usually colonized by biofilms.
- A foodborne disease is an illness resulting from the consumption of pathogenic bacteria, viruses, or other parasites that contaminate animal or plant-based food.
- Proper sterilization techniques and canning procedures have reduced the incidence of botulism.
- *E. coli* outbreaks have become more common as new strains continue to evolve.

Key Terms

- **serotype**: a group of microorganisms characterized by a specific set of antigens

tulism: poisoning caused by the toxin from Clostridium botulinum, a type of anaerobic bacteria that grows in roperly-prepared food

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22.5: Beneficial Prokaryotes

Topic hierarchy

- 22.5A: Symbiosis between Bacteria and Eukaryotes
- 22.5B: Early Biotechnology: Cheese, Bread, Wine, Beer, and Yogurt
- 22.5C: Prokaryotes and Environmental Bioremediation

22.5A: Symbiosis between Bacteria and Eukaryotes

Prokaryotes fix nitrogen into a form that can be used by eukaryotes.

Learning Objectives

- Explain the need for nitrogen fixation and how it is accomplished

Key Points

- Prokaryotes perform biological nitrogen fixation (BNF) to convert nitrogen gas from the atmosphere into ammonia, which can be used by eukaryotes to form important biomolecules such as amino acids and nucleic acids.
- Although most nitrogen fixation is performed by prokaryotes, abiotic processes, such as industrial processes and lightning, can also fix nitrogen.
- Some bacteria form symbiotic relationships with legumes, which provide oxygen-free nodules on their roots for nitrogen fixation to occur; this process allows ammonia to form naturally from atomspheric nitrogen to act as fertilizer for soils.

Key Terms

- **abiotic**: nonliving, inanimate, characterized by the absence of life; of inorganic matter
- **nitrogen fixation**: the conversion of atmospheric nitrogen into ammonia and organic derivatives, by natural means, especially by microorganisms in the soil, into a form that can be assimilated by plants
- **legume**: a large family of herbs, shrubs, and trees that bear nodules on the roots that contain nitrogen-fixing bacteria

Cooperation between Bacteria and Eukaryotes: Nitrogen Fixation

Nitrogen is a very important element for living things because it is part of the nucleotides and amino acids that are the building blocks of nucleic acids and proteins, respectively. Nitrogen is usually the most limiting element in terrestrial ecosystems. Atmospheric nitrogen, N_2, provides the largest pool of available nitrogen. However, eukaryotes cannot use atmospheric, gaseous nitrogen to synthesize macromolecules. Fortunately, nitrogen can be "fixed," meaning it is converted into ammonia (NH_3) either biologically or abiotically. Abiotic nitrogen fixation occurs as a result of lightning or by industrial processes.

Bacterial Nitrogen Fixation

Biological nitrogen fixation (BNF) is exclusively carried out by prokaryotes: soil bacteria, cyanobacteria, and *Frankia spp.* (filamentous bacteria interacting with actinorhizal plants such as alder, bayberry, and sweet fern). After photosynthesis and cellular respiration, BNF is the second most important biological process on Earth. The equation representing the process is as follows, where P_i stands for inorganic phosphate: $N_2 + 16ATP + 8^- + 8H^+ \; 2NH_3 + 16ADP + 16P_i + H_2$

The total fixed nitrogen through BNF is about 100 to 180 million metric tons per year. Biological processes contribute 65 percent of the nitrogen used in agriculture.

Types of Bacteria

Cyanobacteria are the most important nitrogen fixers in aquatic environments. In soil, members of the genus Clostridium are examples of free-living, nitrogen-fixing bacteria. Other bacteria live symbiotically with legume plants, providing the most important source of BNF. Symbionts may fix more nitrogen in soils than free-living organisms by a factor of 10. Soil bacteria, collectively called rhizobia, are able to symbiotically interact with legumes to form nodules: specialized structures where nitrogen fixation occurs. Nitrogenase, the enzyme that fixes nitrogen, is inactivated by oxygen, so the nodule provides an oxygen-free area for nitrogen fixation to take place. This process provides a natural and inexpensive plant fertilizer as it converts (reduces) atmospheric nitrogen to ammonia, which is easily usable by plants. The use of legumes is an excellent alternative to chemical fertilization and is of special interest to sustainable agriculture, which seeks to minimize the use of chemicals and conserve natural resources. Through symbiotic nitrogen fixation, the plant benefits from using an endless source of nitrogen: the atmosphere. Bacteria benefit from using photosynthates (carbohydrates

ed during photosynthesis) from the plant and having a protected niche. Additionally, the soil benefits from being ly fertilized. Therefore, the use of rhizobia as biofertilizers is a sustainable practice.

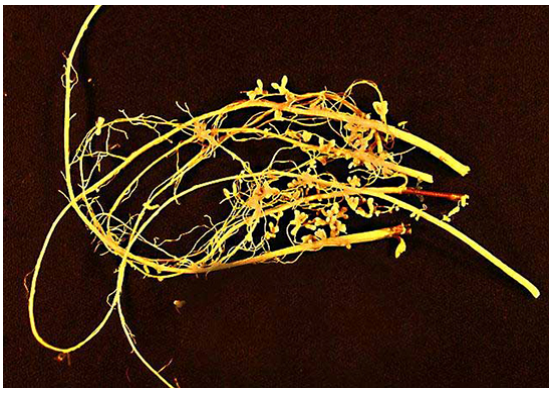

Figure 22.5A.1: **Location of Nitrogen Fixation**: Soybean (Glycine max) is a legume that interacts symbiotically with the soil bacterium Bradyrhizobium japonicum to form specialized structures on the roots called nodules where nitrogen fixation occurs.

Some legumes, like soybeans, are also key sources of agricultural protein. Some of the most important legumes are soybean, peanuts, peas, chickpeas, and beans. Other legumes, such as alfalfa, are used to feed cattle.

22.5B: Early Biotechnology: Cheese, Bread, Wine, Beer, and Yogurt

Some of the earliest biotechnology used prokaryotes for the production of food products such as cheese, bread, wine, beer, and yogurt.

Learning Objectives

- Discuss the origins of food biotechnology as indicated by the production of cheese, bread, wine, beer, and yogurt

Key Points

- Prokaryotes and other microbes are beneficial to some food production by transforming textures, providing flavors, producing ethanol, and providing protection from unwanted microbes.
- Bacteria breakdown proteins and fats into a complex mix of amino acids, amines, and fatty acids; this processing alters the food product.
- Many food production processes rely on the fermentation of prokaryotes and other microbes to produce the desired flavors; in the case of beer and wine, they also affect the desired amount of ethanol.

Key Terms

- **fermentation**: an anaerobic biochemical reaction, in yeast, for example, in which enzymes catalyze the conversion of sugars to alcohol or acetic acid with the evolution of carbon dioxide
- **biotechnology**: the use of living organisms (especially microorganisms) in industrial, agricultural, medical, and other technological applications

Early Biotechnology: Cheese, Bread, Wine, Beer, and Yogurt

According to the United Nations Convention on Biological Diversity, biotechnology is “any technological application that uses biological systems, living organisms, or derivatives thereof, to make or modify products or processes for specific use. ” The concept of “specific use” involves some sort of commercial application. Genetic engineering, artificial selection, antibiotic production, and cell culture are current topics of study in biotechnology. However, humans have used prokaryotes before the term biotechnology was even coined. Some of the products are as simple as cheese, bread, wine, beer, and yogurt,which employ both bacteria and other microbes, such as yeast.

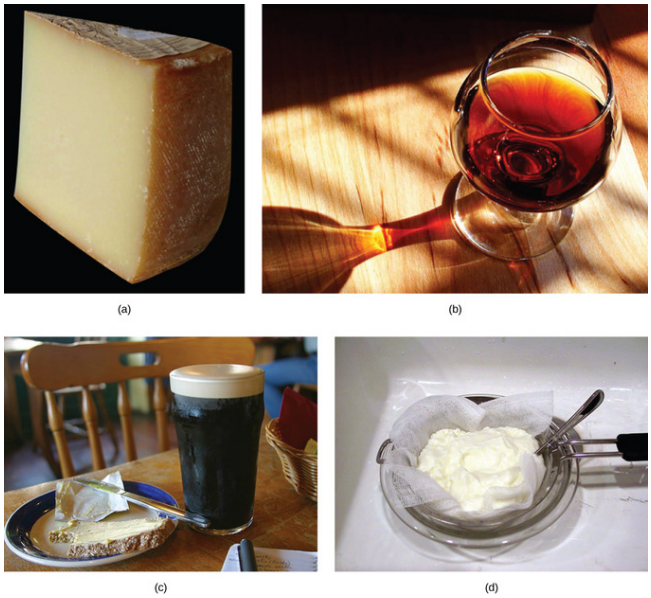

Figure $22.5B.1$: **Products made using prokaryotes**: Some of the products derived from the use of prokaryotes in early biotechnology include (a) cheese, (b) wine, (c) beer and bread, and (d) yogurt.

Cheese production began around 4,000–7,000 years ago when humans began to breed animals and process their milk. Fermentation, in this case, preserves nutrients because milk will spoil relatively quickly, but when processed as cheese, it

is more stable. A required step in cheese-making is separating the milk into solid curds and liquid whey. This usually is done by acidifying the milk and adding rennet. The acidification can be accomplished directly by the addition of an acid like vinegar, but usually starter bacteria are employed instead. These starter bacteria convert milk sugars into lactic acid. The same bacteria (and the enzymes they produce) also play a large role in the eventual flavor of aged cheeses. Most cheeses are made with starter bacteria from the *Lactococci*, *Lactobacilli*, or *Streptococci* families. As a cheese ages, microbes and enzymes transform texture and intensify flavor. This transformation is largely a result of the breakdown of casein proteins and milkfat into a complex mix of amino acids, amines, and fatty acids. Some cheeses have additional bacteria or molds intentionally introduced before or during aging. In traditional cheesemaking, these microbes might already be present in the aging room; they are simply allowed to settle and grow on the stored cheeses. More often today, prepared cultures are used, giving more consistent results and putting fewer constraints on the environment where the cheese ages.

Records of brewing beer date back about 6,000 years to the Sumerians. Evidence indicates that the Sumerians discovered fermentation by chance. Wine has been produced for about 4,500 years. The production of beer and wine use microbes, including both yeast and bacteria, to produce ethanol during fermentation as well as provide flavor to the beverage. Similarly, bread is one of the oldest prepared foods. Bread-making also uses the fermentation of yeast and some bacteria for leavening and flavor. Additionally, evidence suggests that cultured milk products, such as yogurt, have existed for at least 4,000 years. These products use prokaryotes (as with cheese) to provide flavor and to protect the food product from other unwanted microbes.

22.5C: Prokaryotes and Environmental Bioremediation

Bioremediation occurs when prokaryotes clean up a polluted environment through the natural breakdown of pollutants.

Learning Objectives

- Give examples of the use of prokaryotes in enviromental bioremediation

Key Points

- To clean up oil spills, bacteria are introduced to the area of the spill where they break down the hydrocarbons of the oil into carbon dioxide; this is an example of bioremediation.
- Toxic metals, such as mercury (II), can be converted into nontoxic forms, such as mercury (0), by bacteria.
- Using natural organisms as examples, scientists can engineer bacteria for improved bioremediation of desired pollutants.
- Bioremediation can remove oil, some pesticides, fertilizers, and toxic chemicals, such as arsenic, from the environment.

Key Terms

- **bioremediation**: the use of biological organisms, usually microorganisms, to remove contaminants, especially from soil or polluted water
- **biotransformation**: the changes (both chemical and physical) that occur to a substance (especially a drug) by the actions of enzymes within an organism

Using Prokaryotes to Clean up Our Planet: Bioremediation

Microbial bioremediation is the use of prokaryotes (or microbial metabolism) to remove pollutants. Bioremediation has been used to remove agricultural chemicals (pesticides, fertilizers) that leach from soil into groundwater and the subsurface. Certain toxic metals and oxides, such as selenium and arsenic compounds, can also be removed from water by bioremediation. The reduction of SeO_4^{-2} to SeO_3^{-2} and to Se^0 (metallic selenium) is a method used to remove selenium ions from water. Mercury is an example of a toxic metal that can be removed from an environment by bioremediation. As an active ingredient of some pesticides, mercury is used in industry and is also a by-product of certain processes, such as battery production. Methyl mercury is usually present in very low concentrations in natural environments, but it is highly toxic because it accumulates in living tissues. Several species of bacteria can carry out the biotransformation of toxic mercury into nontoxic forms. These bacteria, such as *Pseudomonas aeruginosa*, can convert Hg^{+2} into elemental Hg^0, which is nontoxic to humans.

One of the most useful and interesting examples of the use of prokaryotes for bioremediation purposes is the cleanup of oil spills. The importance of prokaryotes to petroleum bioremediation has been demonstrated in several oil spills in recent years, such as the Exxon Valdez spill in Alaska (1989), the Prestige oil spill in Spain (2002), the spill into the Mediterranean from a Lebanon power plant (2006), and, more recently, the BP oil spill in the Gulf of Mexico (2010). To clean up these spills, bioremediation is promoted by the addition of inorganic nutrients that help bacteria to grow. Hydrocarbon-degrading bacteria feed on hydrocarbons in the oil droplet, breaking down the hydrocarbons. Some species, such as *Alcanivorax borkumensis*, produce surfactants that solubilize the oil, whereas other bacteria degrade the oil into carbon dioxide. In the case of oil spills in the ocean, ongoing, natural bioremediation tends to occur if there are oil-consuming bacteria in the ocean prior to the spill. In addition to naturally occurring oil-degrading bacteria, humans select and engineer bacteria that possess the same capability with increased efficacy and the spectrum of hydrocarbon compounds that can be processed. Under ideal conditions, it has been reported that up to 80 percent of the non-volatile components in oil can be degraded within one year of the spill. Other oil fractions containing aromatic and highly-branched hydrocarbon chains are more difficult to remove and remain in the environment for longer periods of time.

(a) (b)

Figure 22.5C.1: **Bioremediation in the Exxon Valdez oil spill**: (a) Cleaning up oil after the Valdez spill in Alaska, workers hosed oil from beaches and then used a floating boom to corral the oil, which was finally skimmed from the water surface. Some species of bacteria are able to solubilize and degrade the oil. (b) One of the most catastrophic consequences of oil spills is the damage to fauna.

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CHAPTER OVERVIEW

23: PROTISTS

23.1: EUKARYOTIC ORIGINS

23.1A: EARLY EUKARYOTES
23.1B: CHARACTERISTICS OF EUKARYOTIC DNA
23.1C: ENDOSYMBIOSIS AND THE EVOLUTION OF EUKARYOTES
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23.3: GROUPS OF PROTISTS

23.3A: EXCAVATA
23.3B: CHROMALVEOLATA: ALVEOLATES
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23.3D: RHIZARIA
Rhizaria are a supergroup of protists, typically amoebas, that are characterized by the presence of needle-like pseudopodia.

23.3E: ARCHAEPLASTIDA
Archaeplastida are a supergroup of protists that comprise red and green algae, which include unicellular, multicellular, and colonial forms.

23.3F: AMOEBOZOA AND OPISTHOKONTA
23.4: ECOLOGY OF PROTISTS

23.4A: PROTISTS AS PRIMARY PRODUCERS, FOOD SOURCES, AND SYMBIONTS
23.4B: PROTISTS AS HUMAN PATHOGENS
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23.1: Eukaryotic Origins

Topic hierarchy

- 23.1A: Early Eukaryotes
- 23.1B: Characteristics of Eukaryotic DNA
- 23.1C: Endosymbiosis and the Evolution of Eukaryotes
- 23.1D: The Evolution of Mitochondria
- 23.1E: The Evolution of Plastids

23.1A: Early Eukaryotes

Protists are eukaryotes that first appeared approximately 2 billion years ago with the rise of atmospheric oxygen levels.

Learning Objectives

- Discuss the origins of eukaryotes in terms of the geologic time line

Key Points

- On a geological time line, protists are among the first organisms that evolved after prokaryotes.
- Today's eukaryotes evolved from a common ancestor with the following features: a nucleus that divided via mitosis, DNA associated with histones, a cytoskeleton and endomembrane system, the ability to make cilia/flagella.
- Protists vary widely in size, from single cells approximately 10 µm in size to multicellular seaweeds that are visible with the naked eye.

Key Terms

- **cyanobacteria**: photosynthetic prokaryotic microorganisms, of phylum Cyanobacteria, once known as blue-green algae
- **aerobic**: living or occurring only in the presence of oxygen
- **endomembrane**: all the membraneous components inside a eukaryotic cell, including the nuclear envelope, endoplastic reticulum, and Golgi apparatus

Origins of Eukaryotes

Humans have been familiar with macroscopic organisms (organisms big enough to see with the unaided eye) since before there was a written history. It is likely that most cultures distinguished between animals and land plants, but most probably included the macroscopic fungi as plants. Therefore, it became an interesting challenge to deal with the world of microorganisms once microscopes were developed a few centuries ago. Many different naming schemes were used over the last couple of centuries, but it has become the most common practice to refer to eukaryotes that are not land plants, animals, or fungi as protists.

Most protists are microscopic, unicellular organisms that are abundant in soil, freshwater, brackish, and marine environments. They are also common in the digestive tracts of animals and in the vascular tissues of plants. Others invade the cells of other protists, animals, and plants. Not all protists are microscopic. Some have huge, macroscopic cells, such as the plasmodia (giant amoebae) of myxomycete slime molds or the marine green alga *Caulerpa*, which can have single cells that can be several meters in size. Some protists are multicellular, such as the red, green, and brown seaweeds. It is among the protists that one finds the wealth of ways that organisms can grow. They are among the first organisms to evolve with the rise of eukaryotes.

Eukaryotes in a Geological Time Frame

The oldest fossil evidence of eukaryotes, cells measuring 10 µm or greater, is about 2 billion years old. All fossils older than this appear to be prokaryotes. It is probable that today's eukaryotes are descended from an ancestor that had a prokaryotic cellular organization. The last common ancestor (LCA) of today's Eukarya had several characteristics that included: cells with nuclei that divided mitotically and contained linear chromosomes where the DNA was associated with histones; a cytoskeleton and endomembrane system; and the ability to make cilia/flagella during at least part of its life cycle. The LCA was aerobic because it had mitochondria that were the result of an aerobic alpha-proteobacterium that lived inside a host cell. Whether this host had a nucleus at the time of the initial symbiosis remains unknown. The LCA may have had a cell wall for at least part of its life cycle, but more data are needed to confirm this hypothesis. Today's eukaryotes are very diverse in their shapes, organization, life cycles, and number of cells per individual.

While today's atmosphere is about one-fifth molecular oxygen (O_2), geological evidence shows that it originally lacked O_2. Without oxygen, aerobic respiration would not be expected; living things would have relied on fermentation instead. At some point before about 3.5 billion years ago, some prokaryotes evolved the ability to photosynthesize. Cyanobacteria used water as a hydrogen source and released O_2 as a waste product. Originally, oxygen-rich environments were probably localized around places where cyanobacteria were active, but by about 2 billion years ago, geological evidence shows that

oxygen was building up to higher concentrations in the atmosphere. Oxygen levels similar to today's levels only arose within the last 700 million years. Recall that the first fossils that we believe to be eukaryotes date to about 2 billion years old, so they appeared as oxygen levels were increasing.

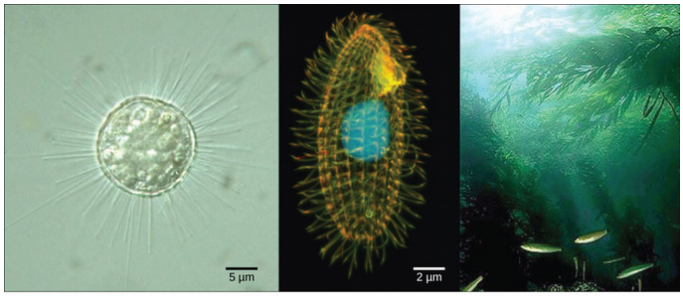

Figure 23.1*A*.1: **Protist varieties**: Protists range from the microscopic, single-celled (a) Acanthocystis turfacea and the (b) ciliate Tetrahymena thermophila, both visualized here using light microscopy, to the enormous, multicellular (c) kelps (Chromalveolata) that extend for hundreds of feet in underwater "forests. "

23.1B: Characteristics of Eukaryotic DNA

Eukaryotes, having probably evolved from prokaryotes, have more complex traits in both cell and DNA organization.

Learning Objectives

- Compare and contrast prokaryotic DNA to eukaryotic DNA

Key Points

- Prokaryotic genomic DNA is attached to the plasma membrane in the form of a nucleoid, in contrast to eukaryotic DNA, which is located in a nucleus.
- Eukaryotic DNA is linear, compacted into chromosomes by histones, and has telomeres at each end to protect from deterioration.
- Prokaryotes contain circular DNA in addition to smaller, transferable DNA plasmids.
- Eukaryotic cells contain mitochondrial DNA in addition to nuclear DNA.
- Eukaryotes separate replicated chromosomes by mitosis, using cytoskeletal proteins, whereas prokaryotes divide more simply via binary fission.

Key Terms

- **telomere**: either of the repetitive nucleotide sequences at each end of a eukaryotic chromosome, which protect the chromosome from degradation
- **plasmid**: a circle of double-stranded DNA that is separate from the chromosomes, which is found in bacteria and protozoa

Characteristics of Eukaryotic DNA compared to Prokaryotic DNA

Prokaryotic cells are known to be much less complex than eukaryotic cells since eukaryotic cells are considered to be present at a later point of evolution. It is probable that eukaryotic cells evolved from prokaryotic cells. Differences in complexity can be seen at the cellular level.

The single characteristic that is both necessary and sufficient to define an organism as a eukaryote is a nucleus surrounded by a nuclear envelope with nuclear pores. All extant eukaryotes have cells with nuclei; most of a eukaryotic cell's genetic material is contained within the nucleus. In contrast, prokaryotic DNA is not contained within a nucleus, but rather is attached to the plasma membrane and contained in the form of a nucleoid, an irregularly-shaped region that is not surrounded by a nuclear membrane.

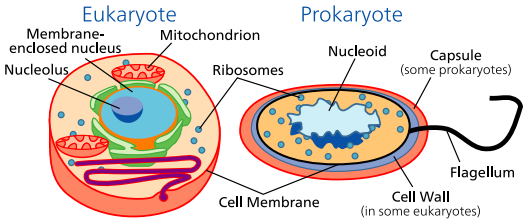

Figure 23.1B.1: **Cellular location of eukaryotic and prokaryotic DNA**: Eukaryotic DNA is stored in a nucleus, whereas prokaryotic DNA is in the cytoplasm in the form of a nucleoid.

Eukaryotic DNA is packed into bundles of chromosomes, each consisting of a linear DNA molecule coiled around basic (alkaline) proteins called histones, which wind the DNA into a more compact form. Prokaryotic DNA is found in circular, non-chromosomal form. In addition, prokaryotes have plasmids, which are smaller pieces of circular DNA that can replicate separately from prokaryotic genomic DNA. Because of the linear nature of eukaryotic DNA, repeating non-coding DNA sequences called telomeres are present on either end of the chromosomes as protection from deterioration.

Mitosis, a process of nuclear division wherein replicated chromosomes are divided and separated using elements of the cytoskeleton, is universally present in eukaryotes. The cytoskeleton contains structural and motility components called actin microfilaments and microtubules. All extant eukaryotes have these cytoskeletal elements. Prokaryotes on the other

hand undergo binary fission in a process where the DNA is replicated, then separates to two poles of the cell, and, finally, the cell fully divides.

A major DNA difference between eukaryotes and prokaryotes is the presence of mitochondrial DNA (mtDNA) in eukaryotes. Because eukaryotes have mitochondria and prokaryotes do not, eukaryotic cells contain mitochondrial DNA in addition to DNA contained in the nucleus and ribosomes. The mtDNA is composed of significantly fewer base pairs than nuclear DNA and encodes only a few dozen genes, depending on the organism.

23.1C: Endosymbiosis and the Evolution of Eukaryotes

Eukaryotes may have been a product of one cell engulfing another and evolving over time until the separate cells became a single organism.

Learning Objectives

- Describe the general concept of endosymbiosis and the evolution of eukaryotes

Key Points

- Endosymbiosis is the concept of one cell engulfing another and both cells benefiting from the relationship.
- Endosymbiosis was originally considered after the observation of the similarity between plant chloroplasts and free-living cyanobacteria.
- Peroxisomes may have been the first endosymbionts, caused by the increasing amount of atmospheric oxygen at that point in geological time.
- Over time, endosymbionts may have transferred some of their DNA to the host nucleus, thus becoming dependent on the host for survival and completing full integration into a single organism.

Key Terms

- **cyanobacteria**: photosynthetic prokaryotic microorganisms, of phylum Cyanobacteria, once known as blue-green algae
- **peroxisome**: a eukaryotic organelle that is the source of the enzymes that catalyze the production and breakdown of hydrogen peroxide and are responsible for the oxidation of long-chain fatty acids
- **endosymbiont**: an organism that lives within the body or cells of another organism

Endosymbiosis and the Evolution of Eukaryotes

To fully understand eukaryotic organisms, it is necessary to understand that all extant eukaryotes are descendants of a chimeric organism that was a composite of a host cell and the cell(s) of an alpha-proteobacterium that “took up residence” inside the host. This major theme in the origin of eukaryotes is known as endosymbiosis, where one cell engulfs another such that the engulfed cell survives and both cells benefit. Over many generations, a symbiotic relationship can result in two organisms that depend on each other so completely that neither could survive on its own. Endosymbiotic events probably contributed to the origin of the last common ancestor (LCA) of today’s eukaryotes.

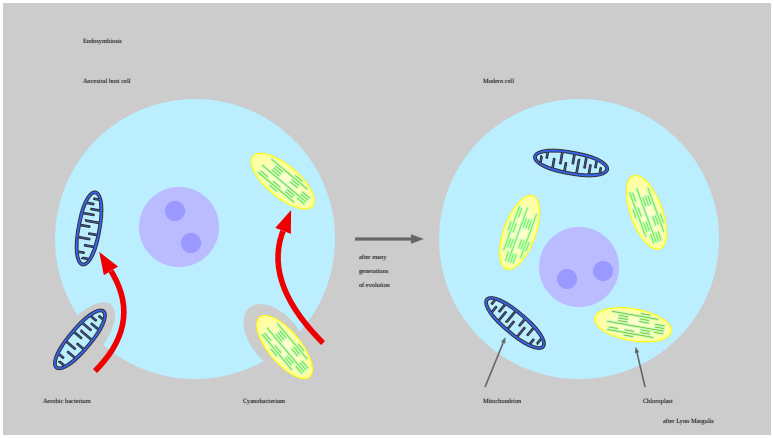

Figure 23.1C.1: **Endosymbiosis**: Modern eukaryotic cells evolved from more primitive cells that engulfed bacteria with useful properties, such as energy production. Combined, the once-independent organisms flourished and evolved into a single organism.

Endosymbiotic Theory

The endosymbiotic theory was first articulated by the Russian botanist Konstantin Mereschkowski in 1905. Mereschkowski was familiar with work by botanist Andreas Schimper, who had observed in 1883 that the division of chloroplasts in green plants closely resembled that of free-living cyanobacteria. Schimper had tentatively proposed that green plants arose from a

symbiotic union of two organisms. Ivan Wallin extended the idea of an endosymbiotic origin to mitochondria in the 1920s. These theories were initially dismissed or ignored. More detailed electron microscopic comparisons between cyanobacteria and chloroplasts combined with the discovery that plastids (organelles associated with photosynthesis) and mitochondria contain their own DNA led to a resurrection of the idea in the 1960s. The endosymbiotic theory was advanced and substantiated with microbiological evidence by Lynn Margulis in 1967.

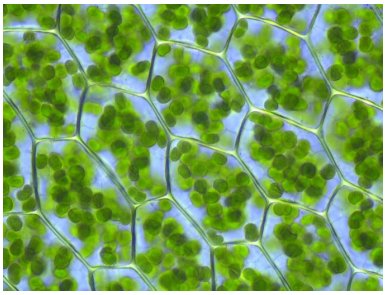

Figure $23.1C.1$: **Chloroplasts in plants**: A eukaryote with mitochondria engulfed a cyanobacterium in an event of serial primary endosymbiosis, creating a lineage of cells with both organelles. These cyanobacteria have become chloroplasts in modern plant cells. The cyanobacterial endosymbiont already had a double membrane.

In 1981 she argued that eukaryotic cells originated as communities of interacting entities, including endosymbiotic spirochetes that developed into eukaryotic flagella and cilia. This last idea has not received much acceptance because flagella lack DNA and do not show ultrastructural similarities to bacteria or archaea. According to Margulis and Dorion Sagan, “Life did not take over the globe by combat, but by networking” (*i.e.*, by cooperation). The possibility that the peroxisome organelles may have an endosymbiotic origin has also been considered, although they lack DNA. Christian de Duve proposed that they may have been the first endosymbionts, allowing cells to withstand growing amounts of free molecular oxygen in the earth’s atmosphere. However, it now appears that they may be formed *de novo*, contradicting the idea that they have a symbiotic origin.

It is believed that over millennia these endosymbionts transferred some of their own DNA to the host cell’s nucleus during the evolutionary transition from a symbiotic community to an instituted eukaryotic cell (called “serial endosymbiosis”). This hypothesis is thought to be possible because it is known today from scientific observation that transfer of DNA occurs between bacteria species, even if they are not closely related. Bacteria can take up DNA from their surroundings and have a limited ability to incorporate it into their own genome.

23.1D: The Evolution of Mitochondria

Mitochondria are energy-producing organelles that are thought to have once been a type of free-living alpha-proteobacterium.

Learning Objectives

- Explain the relationship between endosymbiosis and mitochondria to the evolution of eukaryotes

Key Points

- Eukaryotic cells contain varying amounts of mitochondria, depending on the cells' energy needs.
- Mitochondria have many features that suggest they were formerly independent organisms, including their own DNA, cell-independent division, and physical characteristics similar to alpha-proteobacteria.
- Some mitochondrial genes transferred to the nuclear genome over time, yet mitochondria retained some genetic material for reasons not completely understood.
- The hypothesized transfer of genes from mitochondria to the host cell's nucleus likely explains why mitochondria are not able to survive outside the host cell.

Key Terms

- **crista**: cristae (singular crista) are the internal compartments formed by the inner membrane of a mitochondrion
- **vacuole**: a large, membrane-bound, fluid-filled compartment in a cell's cytoplasm
- **endosymbiosis**: when one symbiotic species is taken inside the cytoplasm of another symbiotic species and both become endosymbiotic

Relationship between Endosymbiosis and Mitochondria

One of the major features distinguishing prokaryotes from eukaryotes is the presence of mitochondria. Eukaryotic cells contain anywhere from one to several thousand mitochondria, depending on the cell's level of energy consumption. Each mitochondrion measures between 1 to 10 µm in length and exists in the cell as an organelle that can be ovoid to worm-shaped to intricately branched. Mitochondria arise from the division of existing mitochondria. They may fuse together. They move around inside the cell by interactions with the cytoskeleton. However, mitochondria cannot survive outside the cell. As the amount of oxygen increased in the atmosphere billions of years ago and as successful aerobic prokaryotes evolved, evidence suggests that an ancestral cell with some membrane compartmentalization engulfed a free-living aerobic prokaryote, specifically an alpha-proteobacterium, thereby giving the host cell the ability to use oxygen to release energy stored in nutrients. Alpha-proteobacteria are a large group of bacteria that includes species symbiotic with plants, disease organisms that can infect humans via ticks, and many free-living species that use light for energy. Several lines of evidence support the derivation of mitochondria from this endosymbiotic event. Most mitochondria are shaped like alpha-proteobacteria and are surrounded by two membranes, which would result when one membrane-bound organism engulfs another into a vacuole. The mitochondrial inner membrane involves substantial infoldings called cristae that resemble the textured, outer surface of alpha-proteobacteria. The matrix and inner membrane are rich with enzymes necessary for aerobic respiration.

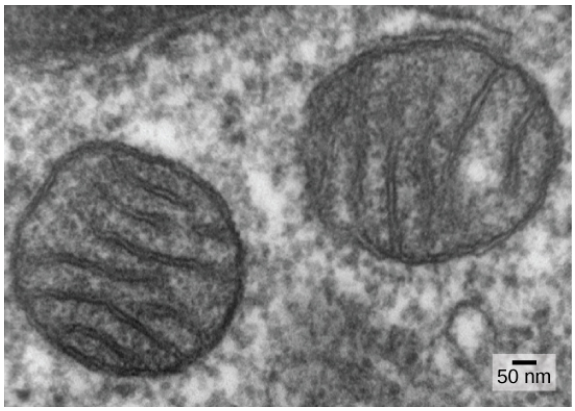

Figure 23.1D.1: **Micrograph of mammaliam mitochondria**: In this transmission electron micrograph of mitochondria in a mammalian lung cell, the cristae, infoldings of the mitochondrial inner membrane, can be seen in cross-section.

Mitochondria divide independently by a process that resembles binary fission in prokaryotes. Specifically, mitochondria are not formed *de novo* by the eukaryotic cell; they reproduce within the cell and are distributed between two cells when cells divide. Therefore, although these organelles are highly integrated into the eukaryotic cell, they still reproduce as if they are independent organisms within the cell. However, their reproduction is synchronized with the activity and division of the cell. Mitochondria have their own circular DNA chromosome that is stabilized by attachments to the inner membrane and carries genes similar to genes expressed by alpha-proteobacteria. Mitochondria also have special ribosomes and transfer RNAs that resemble these components in prokaryotes. These features all support that mitochondria were once free-living prokaryotes.

Mitochondrial Genes

Mitochondria that carry out aerobic respiration have their own genomes, with genes similar to those in alpha-proteobacteria. However, many of the genes for respiratory proteins are located in the nucleus. When these genes are compared to those of other organisms, they appear to be of alpha-proteobacterial origin. Additionally, in some eukaryotic groups, such genes are found in the mitochondria, whereas in other groups, they are found in the nucleus. This has been interpreted as evidence that genes have been transferred from the endosymbiont chromosome to the host genome. This loss of genes by the endosymbiont is probably one explanation why mitochondria cannot live without a host.

Despite the transfer of genes between mitochondria and the nucleus, mitochondria retain much of their own independent genetic material. One possible explanation for mitochondria retaining control over some genes is that it may be difficult to transport hydrophobic proteins across the mitochondrial membrane as well as ensure that they are shipped to the correct location, which suggests that these proteins must be produced within the mitochondria. Another possible explanation is that there are differences in codon usage between the nucleus and mitochondria, making it difficult to be able to fully transfer the genes. A third possible explanation is that mitochondria need to produce their own genetic material so as to ensure metabolic control in eukaryotic cells, which indicates that mtDNA directly influences the respiratory chain and the reduction/oxidation processes of the mitochondria.

23.1E: The Evolution of Plastids

Plastids may derive from cyanobacteria engulfed via endosymbiosis by early eukaryotes, giving cells the ability to conduct photosynthesis.

Learning Objectives

- Explain the relationship between endosymbiosis and plastids to the evolution of eukaryotes

Key Points

- Chloroplasts, chromoplasts, and leucoplasts are each a type of plastid.
- Plastids in eukaryotes derive from primary endosymbiosis with ancient cyanobacteria.
- Chlorarachniophytes are a type of algae that resulted from secondary endosymbiosis, when a eukaryote engulfed a green alga (which itself was a product of primary endosymbiosis with a cyanobacterium).
- Plastids share several features with mitochondria, including having their own DNA and the ability to replicate by binary fission.

Key Terms

- **chloroplast**: an organelle found in the cells of green plants and photosynthetic algae where photosynthesis takes place
- **thylakoid**: a folded membrane within plant chloroplasts from which grana are made, used in photosynthesis
- **plastid**: any of various organelles found in the cells of plants and algae, often concerned with photosynthesis

Plastids

Some groups of eukaryotes are photosynthetic: their cells contain, in addition to the standard eukaryotic organelles, another kind of organelle called a plastid. There are three type of plastids: chloroplasts, chromoplasts, and leucoplasts. Chloroplasts are plastids that conduct photosynthesis. Chromoplasts are plastids that synthesize and store pigments. Leucoplasts are plastids located in the non-synthetic tissues of a plant (*e.g.*, roots) and generally store non-pigment molecules.

Like mitochondria, plastids appear to have a primary endosymbiotic origin, but differ in that they derive from cyanobacteria rather than alpha-proteobacteria. Cyanobacteria are a group of photosynthetic bacteria with all the conventional structures of prokaryotes. Unlike most prokaryotes, however, they have extensive, internal membrane-bound compartments called thylakoids, which contain chlorophyll and are the site of the light-dependent reactions of photosynthesis. In addition to thylakoids, chloroplasts found in eukaryotes have a circular DNA chromosome and ribosomes similar to those of cyanobacteria. Each chloroplast is surrounded by two membranes, suggestive of primary endosymbiosis. The outer membrane surrounding the plastid is thought have derived from the vacuole in the host, while the inner membrane is thought to have derived from the plasma membrane of the endosymbiont.

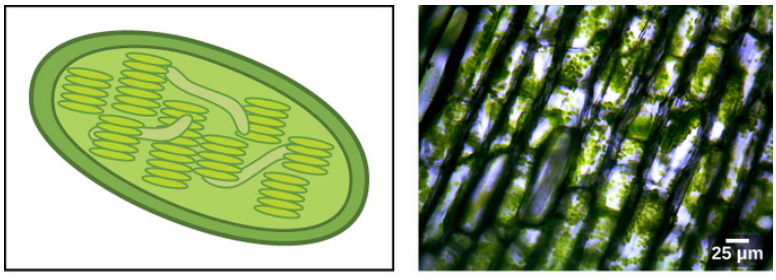

Figure 23.1E.1: **Chloroplast**: (a) This chloroplast cross-section illustrates its elaborate inner membrane organization. Stacks of thylakoid membranes compartmentalize photosynthetic enzymes and provide scaffolding for chloroplast DNA. (b) The chloroplasts can be seen as small green spheres.

There is also, as with the case of mitochondria, strong evidence that many of the genes of the endosymbiont transferred to the nucleus. Plastids, like mitochondria, cannot live independently outside the host. In addition, like mitochondria, plastids derive from the binary fission of other plastids. Researchers have suggested that the endosymbiotic event that led to

Archaeplastida (land plants, red and green algae) occurred 1 to 1.5 billion years ago, at least 500 million years after the fossil record suggests the presence of eukaryotes.

Secondary Endosymbiosis in Chlorarachniophytes

Endosymbiosis involves one cell engulfing another to produce, over time, a co-evolved relationship in which neither cell could survive alone. The chloroplasts of red and green algae, for instance, are derived from the engulfment of a photosynthetic cyanobacterium by an early prokaryote. This leads to the question of the possibility of a cell containing an endosymbiont to become engulfed itself, resulting in a secondary endosymbiosis. Not all plastids in eukaryotes derive directly from primary endosymbiosis. Some of the major groups of algae became photosynthetic by secondary endosymbiosis; that is, by taking in either green algae or red algae as endosymbionts. Numerous microscopic and genetic studies support this conclusion; secondary plastids are surrounded by three or more membranes; some secondary plastids even have clear remnants of the nucleus of endosymbiotic algae.

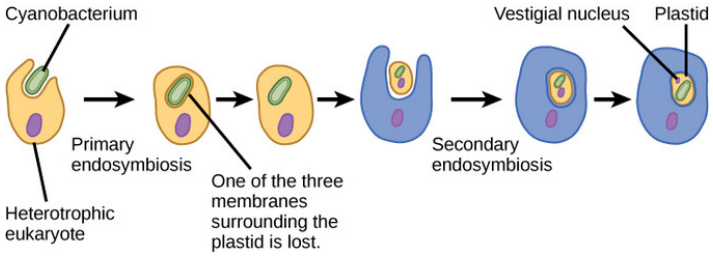

Figure 23.1E.1: **Primary and secondary endosymbiosis**: The hypothesized process of endosymbiotic events leading to the evolution of chlorarachniophytes is shown. In a primary endosymbiotic event, a heterotrophic eukaryote consumed a cyanobacterium. In a secondary endosymbiotic event, the cell resulting from primary endosymbiosis was consumed by a second cell. The resulting organelle became a plastid in modern chlorarachniophytes.

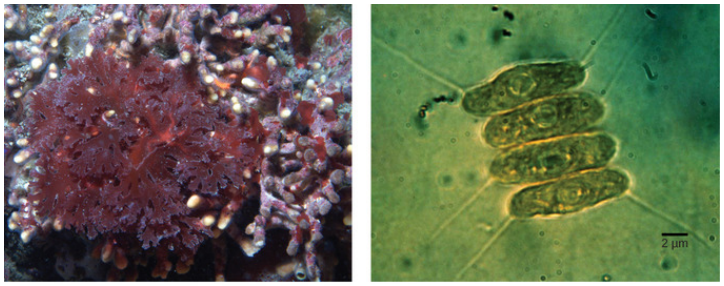

Figure 23.1E.1: **Red and green algae**: (a) Red algae and (b) green algae (visualized by light microscopy) share similar DNA sequences with photosynthetic cyanobacteria. Scientists speculate that, in a process called endosymbiosis, an ancestral prokaryote engulfed a photosynthetic cyanobacterium that evolved into modern-day chloroplasts.

Molecular and morphological evidence suggest that the chlorarachniophyte protists are derived from a secondary endosymbiotic event. Chlorarachniophytes are rare algae indigenous to tropical seas and sand. These protists are thought to have originated when a eukaryote engulfed a green alga, the latter of which had already established an endosymbiotic relationship with a photosynthetic cyanobacterium. Several lines of evidence support that chlorarachniophytes evolved from secondary endosymbiosis. The chloroplasts contained within the green algal endosymbionts are capable of photosynthesis, making chlorarachniophytes photosynthetic. The green algal endosymbiont also exhibits a stunted vestigial nucleus. In fact, it appears that chlorarachniophytes are the products of a recent (on the scale of evolution) secondary endosymbiotic event. The plastids of chlorarachniophytes are surrounded by four membranes: the first two correspond to the inner and outer membranes of the photosynthetic cyanobacterium, the third corresponds to the green alga, and the fourth corresponds to the vacuole that surrounded the green alga when it was engulfed by the chlorarachniophyte ancestor.

The process of secondary endosymbiosis is not unique to chlorarachniophytes. In fact, secondary endosymbiosis of green algae also led to euglenid protists, whereas secondary endosymbiosis of red algae led to the evolution of dinoflagellates, apicomplexans, and stramenopiles.

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: Characteristics of Protists

Topic hierarchy

- 23.2A: Cell Structure, Metabolism, and Motility
- 23.2B: Protist Life Cycles and Habitats

23.2A: Cell Structure, Metabolism, and Motility

Protists are an incredibly diverse set of eukaryotes of various sizes, cell structures, metabolisms, and methods of motility.

Learning Objectives

- Describe the metabolism and structure of protists, explaining the structures that provide their motility

Key Points

- Protist cells may contain a single nucleus or many nuclei; they range in size from microscopic to thousands of meters in area.
- Protists may have animal-like cell membranes, plant-like cell walls, or may be covered by a pellicle.
- Some protists are heterotrophs and ingest food by phagocytosis, while other types of protists are photoautotrophs and store energy via photosynthesis.
- Most protists are motile and generate movement with cilia, flagella, or pseudopodia.

Key Terms

- **amorphous**: lacking a definite form or clear shape
- **multinucleate**: having more than one nucleus
- **pellicle**: cuticle, the hard protective outer layer of certain life forms
- **taxis**: the movement of an organism in response to a stimulus; similar to kinesis, but more direct
- **phagocytosis**: the process where a cell incorporates a particle by extending pseudopodia and drawing the particle into a vacuole of its cytoplasm
- **phagosome**: a membrane-bound vacuole within a cell containing foreign material captured by phagocytosis

Cell Structure

The cells of protists are among the most elaborate and diverse of all cells. Most protists are microscopic and unicellular, but some true multicellular forms exist. A few protists live as colonies that behave in some ways as a group of free-living cells and in other ways as a multicellular organism. Still other protists are composed of enormous, multinucleate, single cells that look like amorphous blobs of slime, or in other cases, similar to ferns. Many protist cells are multinucleated; in some species, the nuclei are different sizes and have distinct roles in protist cell function.

Single protist cells range in size from less than a micrometer to thousands of square meters (giant kelp). Animal-like cell membranes or plant-like cell walls envelope protist cells. In other protists, glassy silica-based shells or pellicles of interlocking protein strips encase the cells. The pellicle functions like a flexible coat of armor, preventing the protist from external damage without compromising its range of motion.

Metabolism

Protists exhibit many forms of nutrition and may be aerobic or anaerobic. Protists that store energy by photosynthesis belong to a group of photoautotrophs and are characterized by the presence of chloroplasts. Other protists are heterotrophic and consume organic materials (such as other organisms) to obtain nutrition. Amoebas and some other heterotrophic protist species ingest particles by a process called phagocytosis in which the cell membrane engulfs a food particle and brings it inward, pinching off an intracellular membranous sac, or vesicle, called a food vacuole. The vesicle containing the ingested particle, the phagosome, then fuses with a lysosome containing hydrolytic enzymes to produce a phagolysosome, which breaks down the food particle into small molecules that diffuse into the cytoplasm for use in cellular metabolism. Undigested remains ultimately exit the cell via exocytosis.

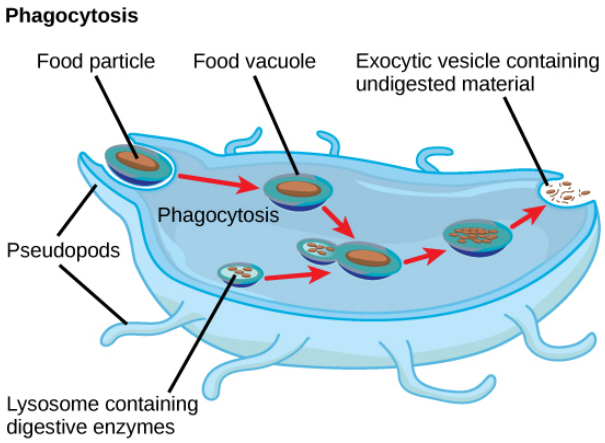

Figure 23.2A.1: **Protist metabolism**: The stages of phagocytosis include the engulfment of a food particle, the digestion of the particle using enzymes contained within a lysosome, and the expulsion of undigested materials from the cell.

Subtypes of heterotrophs, called saprobes, absorb nutrients from dead organisms or their organic wastes. Some protists function as mixotrophs, obtaining nutrition by photoautotrophic or heterotrophic routes, depending on whether sunlight or organic nutrients are available.

Motility

The majority of protists are motile, but different types of protists have evolved varied modes of movement. Protists such as euglena have one or more flagella, which they rotate or whip to generate movement. Paramecia are covered in rows of tiny cilia that they beat to swim through liquids. Other protists, such at amoebae, form cytoplasmic extensions called pseudopodia anywhere on the cell, anchor the pseudopodia to a surface, and pull themselves forward. Some protists can move toward or away from a stimulus; a movement referred to as taxis. Protists accomplish phototaxis, movement toward light, by coupling their locomotion strategy with a light-sensing organ.

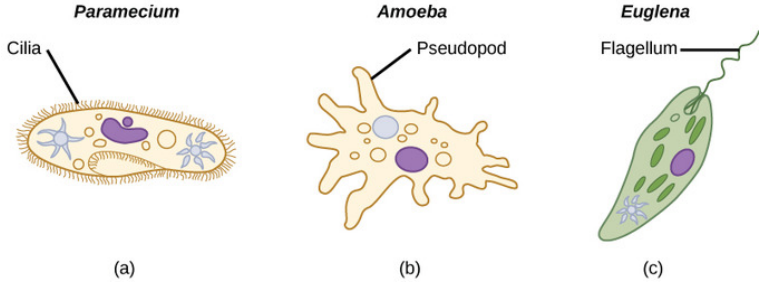

Figure 23.2A.1: **Different types of motility in protists**: Protists use various methods for transportation. (a) A paramecium waves hair-like appendages called cilia. (b) An amoeba uses lobe-like pseudopodia to anchor itself to a solid surface and pull itself forward. (c) Euglena uses a whip-like tail called a flagellum.

23.2B: Protist Life Cycles and Habitats

Protists live in a wide variety of habitats, including most bodies of water, as parasites in both plants and animals, and on dead organisms.

Learning Objectives

- Describe the habitats and life cycles of various protists

Key Points

- Slime molds are categorized on the basis of their life cycles into plasmodial or cellular types, both of which end their life cycle in the form of dispersed spores.
- Plasmodial slime molds form a single-celled, multinucleate mass, whereas cellular slime molds form an aggregated mass of separate amoebas that are able to migrate as a unified whole.
- Slimes molds feed primarily on bacteria and fungi and contribute to the decomposition of dead plants.

Key Terms

- **haploid**: of a cell having a single set of unpaired chromosomes
- **sporangia**: an enclosure in which spores are formed (also called a fruiting body)
- **plasmodium**: a mass of cytoplasm, containing many nuclei, created by the aggregation of amoeboid cells of slime molds during their vegetative phase
- **diploid**: of a cell, having a pair of each type of chromosome, one of the pair being derived from the ovum and the other from the spermatozoon

Life Cycle of Slime Molds

Protist life cycles range from simple to extremely elaborate. Certain parasitic protists have complicated life cycles and must infect different host species at different developmental stages to complete their life cycle. Some protists are unicellular in the haploid form and multicellular in the diploid form, which is a strategy also employed by animals. Other protists have multicellular stages in both haploid and diploid forms, a strategy called alternation of generations that is also used by plants.

Plasmodial slime molds

The slime molds are categorized on the basis of their life cycles into plasmodial or cellular types. Plasmodial slime molds are composed of large, multinucleate cells and move along surfaces like an amorphous blob of slime during their feeding stage. The slime mold glides along, lifting and engulfing food particles, especially bacteria. Upon maturation, the plasmodium takes on a net-like appearance with the ability to form fruiting bodies, or sporangia, during times of stress. Meiosis produces haploid spores within the sporangia. Spores disseminate through the air or water to potentially land in more favorable environments. If this occurs, the spores germinate to form amoeboid or flagellate haploid cells that can combine with each other and produce a diploid zygotic slime mold to complete the life cycle.

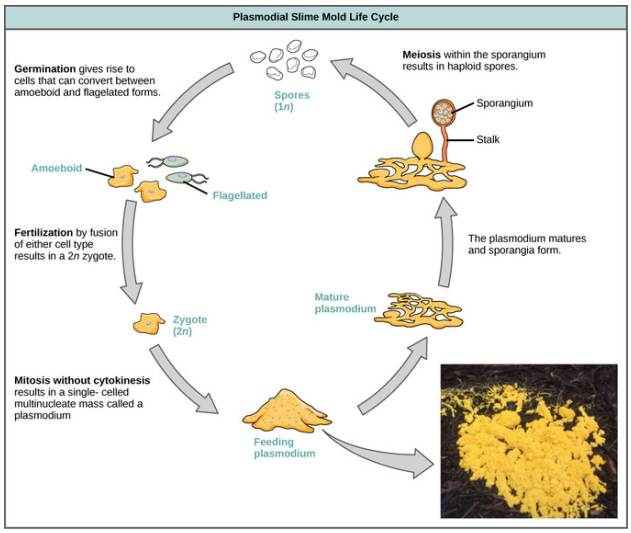

Figure 23.2B.1: **Plasmodial slime mold life cycle**: Haploid spores develop into amoeboid or flagellated forms, which are then fertilized to form a diploid, multinucleate mass called a plasmodium. This plasmodium is net-like and, upon maturation, forms a sporangium on top of a stalk. The sporangium forms haploid spores through meiosis, after which the spores disseminate, germinate, and begin the life cycle anew. The brightly-colored plasmodium in the inset photo is a single-celled, multinucleate mass.

Cellular slime molds

The cellular slime molds function as independent amoeboid cells when nutrients are abundant. When food is depleted, cellular slime molds aggregate into a mass of cells that behaves as a single unit called a slug. Some cells in the slug contribute to a 2–3-millimeter stalk, which dries up and dies in the process. Cells atop the stalk form an asexual fruiting body that contains haploid spores. As with plasmodial slime molds, the spores are disseminated and can germinate if they land in a moist environment. One representative genus of the cellular slime molds is *Dictyostelium*, which commonly exists in the damp soil of forests.

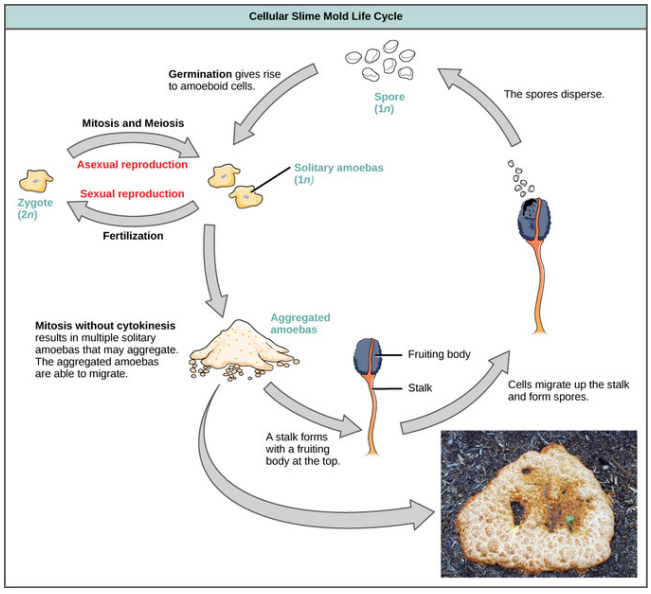

Figure 23.2B.1: **Cellular slime mold life cycle**: Cellular slime molds may engage in two forms of life cycles: as solitary amoebas when nutrients are abundant or as aggregated amoebas (inset photo) when nutrients are scarce. In aggregate form, some individuals contribute to the formation of a stalk, on top of which sits a fruiting body full of spores that disseminate and germinate in the proper moist environment.

Habitats of Various Protists

There are over 100,000 described living species of protists. Nearly all protists exist in some type of aquatic environment, including freshwater and marine environments, damp soil, and even snow. Paramecia are a common example of aquatic protists. Due to their abundance and ease of use as research organisms, they are often subjects of study in classrooms and laboratories. In addition to aquatic protists, several protist species are parasites that infect animals or plants and, therefore, live in their hosts. Amoebas can be human parasites and can cause dysentery while inhabiting the small intestine. Other protist species live on dead organisms or their wastes and contribute to their decay. Approximately 1000 species of slime mold thrive on bacteria and fungi within rotting trees and other plants in forests around the world, contributing to the life cycle of these ecosystems.

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23.3: Groups of Protists

Topic hierarchy

- 23.3A: Excavata
- 23.3B: Chromalveolata: Alveolates
- 23.3C: Chromalveolata: Stramenopiles
- 23.3D: Rhizaria

Rhizaria are a supergroup of protists, typically amoebas, that are characterized by the presence of needle-like pseudopodia.

- 23.3E: Archaeplastida

Archaeplastida are a supergroup of protists that comprise red and green algae, which include unicellular, multicellular, and colonial forms.

- 23.3F: Amoebozoa and Opisthokonta

23.3A: Excavata

Excavata, defined by a feeding groove that is “excavated” from one side, includes Diplomonads, Parabasalids and Euglenozoans.

Learning Objectives

- Describe characteristics of Excavates, including Diplomonads, Parabasalids and Euglenozoans

Key Points

- Excavata are a supergroup of protists that are defined by an asymmetrical appearance with a feeding groove that is “excavated” from one side; it includes various types of organisms which are parasitic, photosynthetic and heterotrophic predators.
- Excavata includes the protists: Diplomonads, Parabasalids and Euglenozoans.
- Diplomonads are defined by the presence of a nonfunctional, mitochrondrial-remnant organelle called a mitosome.
- Parabasalids are characterized by a semi-functional mitochondria referred to as a hydrogenosome; they are comprised of parasitic protists, such as *Trichomonas vaginalis*.
- Euglenozoans can be classified as mixotrophs, heterotrophs, autotrophs, and parasites; they are defined by their use of flagella for movement.

Key Terms

- **mitosome**: an organelle found within certain unicellular eukaryotes which lack mitochondria
- **hydrogenosome**: a membrane-bound organelle found in ciliates, trichomonads, and fungi which produces molecular hydrogen and ATP
- **kinetoplast**: a disk-shaped mass of circular DNA inside a large mitochondrion, found specifically in protozoa of the class Kinetoplastea

Excavata

Many of the protist species classified into the supergroup Excavata are asymmetrical, single-celled organisms with a feeding groove “excavated” from one side. This supergroup includes heterotrophic predators, photosynthetic species, and parasites. Its subgroups are the diplomonads, parabasalids, and euglenozoans.

Diplomonads

Among the Excavata are the diplomonads, which include the intestinal parasite, *Giardia lamblia*. Until recently, these protists were believed to lack mitochondria. Mitochondrial remnant organelles, called mitosomes, have since been identified in diplomonads, but these mitosomes are essentially nonfunctional. Diplomonads exist in anaerobic environments and use alternative pathways, such as glycolysis, to generate energy. Each diplomonad cell has two identical nuclei and uses several flagella for locomotion.

Giardia lamblia

The mammalian intestinal parasite *Giardia lamblia*,visualized here using scanning electron microscopy, is a waterborne protist that causes severe diarrhea when ingested.

Parabasalids

A second Excavata subgroup, the parabasalids, also exhibits semi-functional mitochondria. In parabasalids, these structures function anaerobically and are called hydrogenosomes because they produce hydrogen gas as a byproduct. Parabasalids move with flagella and membrane rippling. *Trichomonas vaginalis*, a parabasalid that causes a sexually-transmitted disease in humans, employs these mechanisms to transit through the male and female urogenital tracts. *T. vaginalis* causes trichomoniasis, which appears in an estimated 180 million cases worldwide each year. Whereas men rarely exhibit symptoms during an infection with this protist, infected women may become more susceptible to secondary infection with human immunodeficiency virus (HIV) or genital wart virus infection, which causes over 90% of cervical cancer. Pregnant women infected with *T. vaginalis* are at an increased risk of serious complications, such as pre-term delivery.

Euglenozoans

Euglenozoans includes parasites, heterotrophs, autotrophs, and mixotrophs, ranging in size from 10 to 500 µm. Euglenoids move through their aquatic habitats using two long flagella that guide them toward light sources sensed by a primitive ocular organ called an eyespot. The familiar genus, *Euglena*, encompasses some mixotrophic species that display a photosynthetic capability only when light is present. In the dark, the chloroplasts of *Euglena* shrink up and temporarily cease functioning; the cells, instead, take up organic nutrients from their environment.

The human parasite, *Trypanosoma brucei*, belongs to a different subgroup of Euglenozoa, the kinetoplastids. The kinetoplastid subgroup is named after the kinetoplast, a DNA mass carried within the single, oversized mitochondrion possessed by each of these cells. This subgroup includes several parasites, collectively called trypanosomes, which cause devastating human diseases by infecting an insect species during a portion of their life cycle. *T. brucei* develops in the gut of the tsetse fly after the fly bites an infected human or other mammalian host. The parasite then travels to the insect salivary glands to be transmitted to another human or other mammal when the infected tsetse fly consumes another blood meal. *T. brucei* is common in central Africa and is the causative agent of African sleeping sickness, a disease associated with severe chronic fatigue and coma; it can be fatal if left untreated.

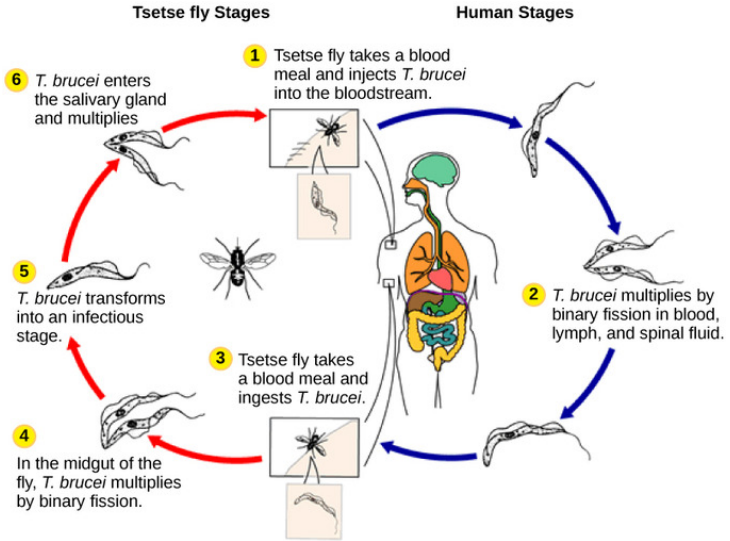


Figure 23.3A.1: **Life cycle of *Trypanosoma brucei***: *Trypanosoma brucei*, the causative agent of sleeping sickness, spends part of its life cycle in the tsetse fly and part in humans.

23.3B: Chromalveolata: Alveolates

Alveolates are defined by the presence of an alveolus beneath the cell membrane and include dinoflagellates, apicomplexans and ciliates.

Learning Objectives

- Evaluate traits associated with protists classified as alveolates which include dinoflagellates, apicomplexans, and ciliates

Key Points

- Alveolates are classified under the group Chromalveolata which developed as a result of a secondary endosymbiotic event.
- Dinoflagellates are defined by their flagella structure which lays perpendicular and fits into the cellulose plates of the dinoflagellate, promoting a spinning motion.
- Apicomplexans are defined by the asymmetrical distribution of their microtubules, fibrin, and vacuoles; they include the parasitic protist *Plasmodium* which causes malaria.
- Ciliates are defined by the presence of cilia (such as the oral groove in the *Paramecium)*, which beat synchronously to aid the organism in locomotion and obtaining nutrients.
- Ciliates are defined by the presence of cilia, which beat synchronously, to aid the organism in locomotion and obtaining nutrients, such as the oral groove in the *Paramecium*.

Key Terms

- **osmoregulation**: the homeostatic regulation of osmotic pressure in the body in order to maintain a constant water content
- **plastid**: any of various organelles found in the cells of plants and algae, often concerned with photosynthesis
- **conjugation**: the temporary fusion of organisms, especially as part of sexual reproduction

Chromalveolata

Current evidence suggests that species classified as chromalveolates are derived from a common ancestor that engulfed a photosynthetic red algal cell, which itself had already evolved chloroplasts from an endosymbiotic relationship with a photosynthetic prokaryote. Therefore, the ancestor of chromalveolates is believed to have resulted from a secondary endosymbiotic event. However, some chromalveolates appear to have lost red alga-derived plastid organelles or lack plastid genes altogether. Therefore, this supergroup should be considered a hypothesis-based working group that is subject to change and can be subdivided into alveolates and stramenopiles.

Alveolates

A large body of data supports that the alveolates are derived from a shared common ancestor. The alveolates are named for the presence of an alveolus, or membrane-enclosed sac, beneath the cell membrane. The exact function of the alveolus is unknown, but it may be involved in osmoregulation. The alveolates are further categorized into the dinoflagellates, the apicomplexans, and the ciliates.

Dinoflagellates

Dinoflagellates exhibit extensive morphological diversity and can be photosynthetic, heterotrophic, or mixotrophic. Many dinoflagellates are encased in interlocking plates of cellulose with two perpendicular flagella that fit into the grooves between the cellulose plates. One flagellum extends longitudinally and a second encircles the dinoflagellate. Together, the flagella contribute to the characteristic spinning motion of dinoflagellates. These protists exist in freshwater and marine habitats; they are a component of plankton.

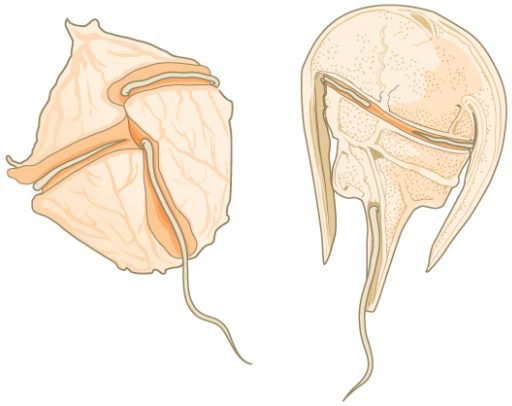


Figure 23.3B.1: **Dinoflagellates**: The dinoflagellates exhibit great diversity in shape. Many are encased in cellulose armor and have two flagella that fit in grooves between the plates. Movement of these two perpendicular flagella causes a spinning motion.

Some dinoflagellates generate light, called bioluminescence, when they are jarred or stressed. Large numbers of marine dinoflagellates (billions or trillions of cells per wave) can emit light and cause an entire breaking wave to twinkle or take on a brilliant blue color. For approximately 20 species of marine dinoflagellates, population explosions (called blooms) during the summer months can tint the ocean with a muddy red color. This phenomenon is called a red tide and results from the abundant red pigments present in dinoflagellate plastids. In large quantities, these dinoflagellate species secrete an asphyxiating toxin that can kill fish, birds, and marine mammals. Red tides can be massively detrimental to commercial fisheries; humans who consume these protists may become poisoned.

Figure 23.3B.1: **Bioluminescence**: Bioluminescence is emitted from dinoflagellates in a breaking wave, as seen from the New Jersey coast.

Apicomplexans

The apicomplexan protists are so named because their microtubules, fibrin, and vacuoles are asymmetrically distributed at one end of the cell in a structure called an apical complex. The apical complex is specialized for entry and infection of host cells. Indeed, all apicomplexans are parasitic. This group includes the genus *Plasmodium*, which causes malaria in humans. Apicomplexan life cycles are complex, involving multiple hosts and stages of sexual and asexual reproduction.

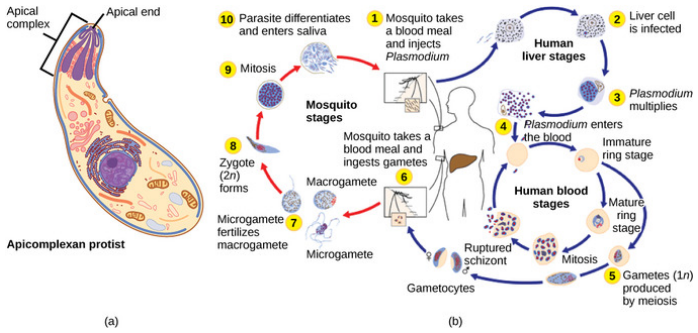


Figure 23.3B.1: **Parasitic apicomplexans**: (a) Apicomplexans are parasitic protists. They have a characteristic apical complex that enables them to infect host cells. (b) Plasmodium, the causative agent of malaria, has a complex life cycle typical of apicomplexans.

Ciliates

The ciliates, which include *Paramecium* and *Tetrahymena*, are a group of protists 10 to 3,000 micrometers in length that are covered in rows, tufts, or spirals of tiny cilia. By beating their cilia synchronously or in waves, ciliates can coordinate directed movements and ingest food particles. Certain ciliates have fused cilia-based structures that function like paddles, funnels, or fins. Ciliates also are surrounded by a pellicle, providing protection without compromising agility. The genus *Paramecium* includes protists that have organized their cilia into a plate-like primitive mouth called an oral groove, which is used to capture and digest bacteria. Food captured in the oral groove enters a food vacuole where it combines with digestive enzymes. Waste particles are expelled by an exocytic vesicle that fuses at a specific region on the cell membrane: the anal pore. In addition to a vacuole-based digestive system, *Paramecium* also uses contractile vacuoles: osmoregulatory vesicles that fill with water as it enters the cell by osmosis and then contract to squeeze water from the cell.

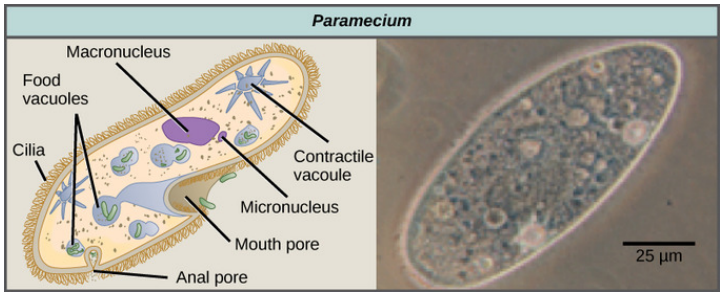


Figure 23.3B.1: **Paramecium**: Paramecium has a primitive mouth (called an oral groove) to ingest food and an anal pore to excrete it. Contractile vacuoles allow the organism to excrete excess water. Cilia enable the organism to move.

Paramecium has two nuclei, a macronucleus and a micronucleus, in each cell. The micronucleus is essential for sexual reproduction, whereas the macronucleus directs asexual binary fission and all other biological functions. The process of sexual reproduction in *Paramecium* underscores the importance of the micronucleus to these protists. *Paramecium* and most other ciliates reproduce sexually by conjugation. This process begins when two different mating types of *Paramecium* make physical contact and join with a cytoplasmic bridge. The diploid micronucleus in each cell then undergoes meiosis to produce four haploid micronuclei. Three of these degenerate in each cell, leaving one micronucleus that then undergoes mitosis, generating two haploid micronuclei. The cells each exchange one of these haploid nuclei and move away from each other. A similar process occurs in bacteria that have plasmids. Fusion of the haploid micronuclei generates a completely novel diploid pre-micronucleus in each conjugative cell. This pre-micronucleus undergoes three rounds of mitosis to produce eight copies, while the original macronucleus disintegrates. Four of the eight pre-micronuclei become full-fledged micronuclei, whereas the other four perform multiple rounds of DNA replication and then become new macronuclei. Two cell divisions then yield four new paramecia from each original conjugative cell.

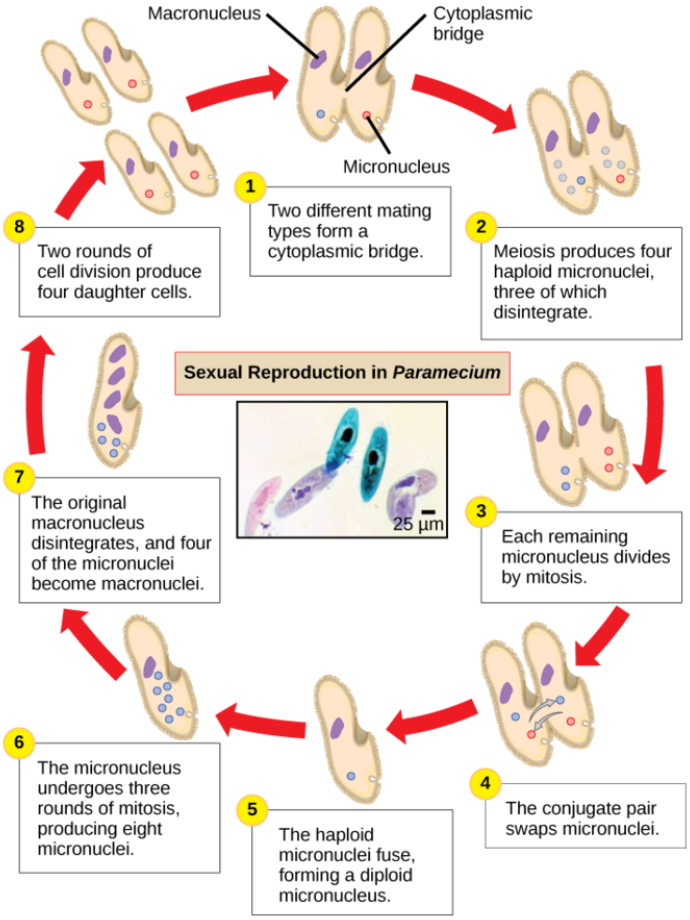


Figure 23.3$B.1$: ***Paramecium*: sexual reproduction**: The complex process of sexual reproduction in *Parameciumcreates* eight daughter cells from two original cells. Each cell has a macronucleus and a micronucleus. During sexual reproduction, the macronucleus dissolves and is replaced by a micronucleus.

23.3C: Chromalveolata: Stramenopiles

Stramenophiles include photosynthetic marine algae and heterotrophic protists such as diatoms, brown and golden algae, and oomycetes.

Learning Objectives

- Describe characteristics of the following Stramenophiles: diatoms, brown algae, golden algae, and oomycetes

Key Points

- Stramenophiles, also referred to as heterokonts, are a subclass of chromalveolata, and are identified by the presence of a “hairy” flagellum.
- Diatoms, present in both freshwater and marine plankton, are unicellular photosynthetic protists that are characterized by the presence of a cell wall composed of silicon dioxide that displays intricate patterns.
- Golden algae, present in both freshwater and marine plankton communities, are unicellular photosynthetic protists characterized by the presence of carotenoids (yellow-orange photosynthetic pigments).
- Oomycetes, commonly referred to as water molds, are characterized by their fungus-like morphology, a cellulose-based cell wall, and a filamentous network used for nutrient uptake.
- Oomycetes, commonly referred to as water molds, are characterized by their fungus-like morphology, a cellulose-based cell wall and a filamentous network used for nutrient uptake.

Key Terms

- **stipe**: the stem of a kelp
- **raphe**: a ridge or seam on an organ, bodily tissue, or other structure, especially at the join between two halves or sections
- **saprobe**: an organism that lives off of dead or decaying organic material

Chromalveolates

Current evidence suggests that chromalveolates have an ancestor which resulted from a secondary endosymbiotic event. The species which fall under the classification of chromalveolates have evolved from a common ancestor that engulfed a photosynthetic red algal cell. This red algal cell had previously evolved chloroplasts from an endosymbiotic relationship with a photosynthetic prokaryote. Chromalveolates include very important photosynthetic organisms, such as diatoms, brown algae, and significant disease agents in animals and plants. The chromalveolates can be subdivided into alveolates and stramenopiles.

Stramenopiles

A subgroup of chromalveolates, the stramenopiles, also referred to as heterokonts, includes photosynthetic marine algae and heterotrophic protists. The unifying feature of this group is the presence of a textured, or “hairy,” flagellum. Many stramenopiles also have an additional flagellum that lacks hair-like projections. Members of this subgroup range in size from single-celled diatoms to the massive and multicellular kelp.

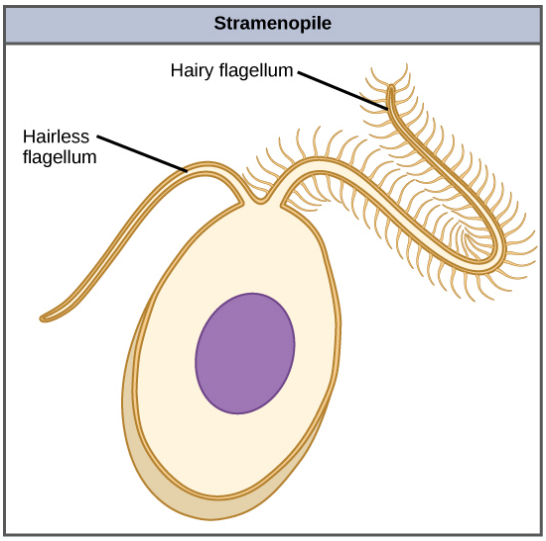


Figure $23.3C.1$: **Stramenophile structure**: This stramenopile cell has a single hairy flagellum and a secondary smooth flagellum.

Diatoms

The diatoms are unicellular photosynthetic protists that encase themselves in intricately patterned, glassy cell walls composed of silicon dioxide in a matrix of organic particles. These protists are a component of freshwater and marine plankton. Most species of diatoms reproduce asexually, although some instances of sexual reproduction and sporulation also exist. Some diatoms exhibit a slit in their silica shell called a raphe. By expelling a stream of mucopolysaccharides from the raphe, the diatom can attach to surfaces or propel itself in one direction.

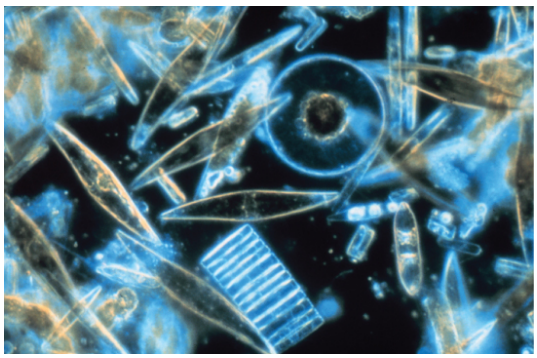


Figure $23.3C.1$: **Diatoms**: Assorted diatoms, visualized here using light microscopy, live among annual sea ice in McMurdo Sound, Antarctica. Diatoms range in size from 2 to 200 µm.

During periods of nutrient availability, diatom populations bloom to numbers greater than can be consumed by aquatic organisms. The excess diatoms die and sink to the sea floor where they are not easily reached by saprobes that feed on dead organisms. As a result, the carbon dioxide that the diatoms had consumed and incorporated into their cells during photosynthesis is not returned to the atmosphere. In general, this process by which carbon is transported deep into the ocean is described as the biological carbon pump because carbon is “pumped” to the ocean depths where it is inaccessible to the atmosphere as carbon dioxide. The biological carbon pump is a crucial component of the carbon cycle that helps to maintain lower atmospheric carbon dioxide levels.

Golden Algae

Like diatoms, golden algae are largely unicellular, although some species can form large colonies. Their characteristic gold color results from their extensive use of carotenoids, a group of photosynthetic pigments that are generally yellow or orange in color. Golden algae are found in both freshwater and marine environments, where they form a major part of the plankton community.

Brown Algae

The brown algae are primarily marine, multicellular organisms that are known colloquially as seaweeds. Giant kelps are a type of brown algae. Some brown algae have evolved specialized tissues that resemble terrestrial plants, with root-like holdfasts, stem-like stipes, and leaf-like blades that are capable of photosynthesis. The stipes of giant kelps are enormous, extending in some cases for 60 meters. A variety of algal life cycles exists, but the most complex is alternation of generations in which both haploid and diploid stages involve multicellularity. For instance, compare this life cycle to that of humans. In humans, haploid gametes produced by meiosis (sperm and egg) combine in fertilization to generate a diploid zygote that undergoes many rounds of mitosis to produce a multicellular embryo and then a fetus. However, the individual sperm and egg themselves never become multicellular beings. In the brown algae genus *Laminaria*, haploid spores develop into multicellular gametophytes, which produce haploid gametes that combine to produce diploid organisms that then become multicellular organisms with a different structure from the haploid form. Terrestrial plants also have evolved alternation of generations.

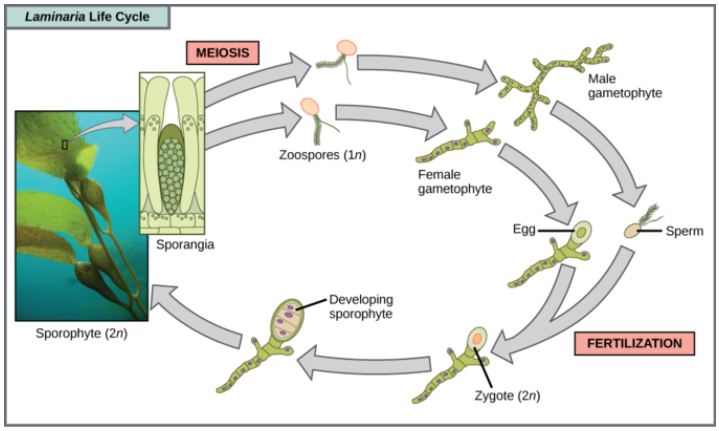


Figure 23.3C.1: **Brown algae life cycle**: Several species of brown algae, such as the *Laminaria* shown here, have evolved life cycles in which both the haploid (gametophyte) and diploid (sporophyte) forms are multicellular. The gametophyte is different in structure from the sporophyte.

Oomycetes

The water molds, oomycetes ("egg fungus"), were so-named based on their fungus-like morphology, but molecular data have shown that the water molds are not closely related to fungi. The oomycetes are characterized by a cellulose-based cell wall and an extensive network of filaments that allow for nutrient uptake. As diploid spores, many oomycetes have two oppositely-directed flagella (one hairy and one smooth) for locomotion. The oomycetes are non-photosynthetic and include many saprobes and parasites. The saprobes appear as white fluffy growths on dead organisms. Most oomycetes are aquatic, but some parasitize terrestrial plants. One plant pathogen is *Phytophthora infestans*, the causative agent of late blight of potatoes, such as occurred in the nineteenth century Irish potato famine.

Figure 23.3C.1: **Oomycete**: A saprobic oomycete engulfs a dead insect.

23.3D: Rhizaria

Learning Objectives

- Describe characteristics associated with Rhizaria

The Rhizaria supergroup includes many of the amoebas, most of which have threadlike or needle-like pseudopodia. Pseudopodia function to trap and engulf food particles and to direct movement in rhizarian protists. These pseudopods project outward from anywhere on the cell surface and can anchor to a substrate. The protist then transports its cytoplasm into the pseudopod, thereby moving the entire cell. This type of motion, called cytoplasmic streaming, is used by several diverse groups of protists as a means of locomotion or as a method to distribute nutrients and oxygen.

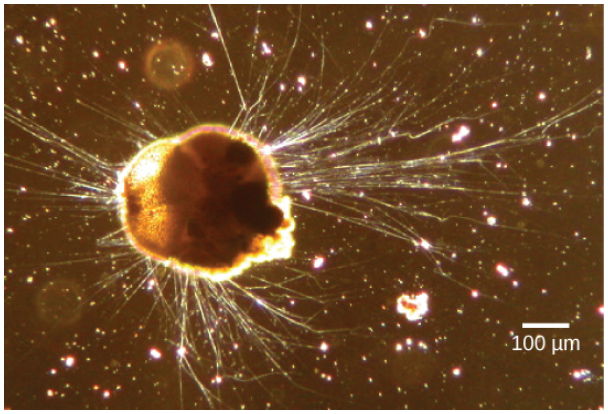


Figure 23.3D.1: **Ammonia tepida**: Ammonia tepida, a Rhizaria species viewed here using phase contrast light microscopy, exhibits many threadlike pseudopodia.

Forams

Foraminiferans, or forams, are unicellular heterotrophic protists, ranging from approximately 20 micrometers to several centimeters in length; they occasionally resemble tiny snails. As a group, the forams exhibit porous shells, called tests, that are built from various organic materials and typically hardened with calcium carbonate. The tests may house photosynthetic algae, which the forams can harvest for nutrition. Foram pseudopodia extend through the pores and allow the forams to move, feed, and gather additional building materials. Foraminiferans are also useful as indicators of pollution and changes in global weather patterns.

The life-cycle involves an alternation between haploid and diploid phases. The haploid phase initially has a single nucleus, and divides to produce gametes with two flagella. The diploid phase is multinucleate, and after meiosis fragments to produce new organisms. The benthic forms has multiple rounds of asexual reproduction between sexual generations.

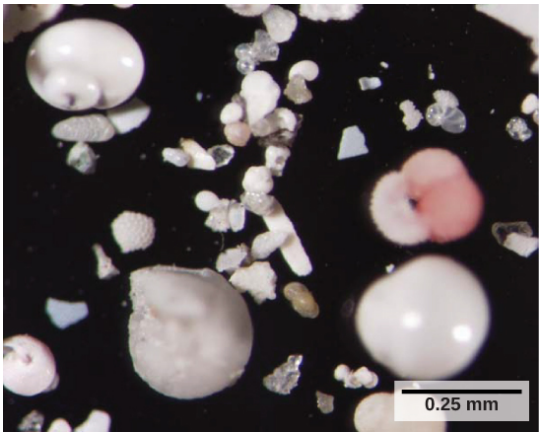


Figure 23.3D.1: **Forams**: These shells from foraminifera sank to the sea floor.

Radiolarians

A second subtype of Rhizaria, the radiolarians, exhibit intricate exteriors of glassy silica with radial or bilateral symmetry. Radiolarians display needle-like pseudopods that are supported by microtubules which radiate outward from the cell bodies of these protists and function to catch food particles. The shells of dead radiolarians sink to the ocean floor, where they may accumulate in 100 meter-thick depths. Preserved, sedimented radiolarians are very common in the fossil record.

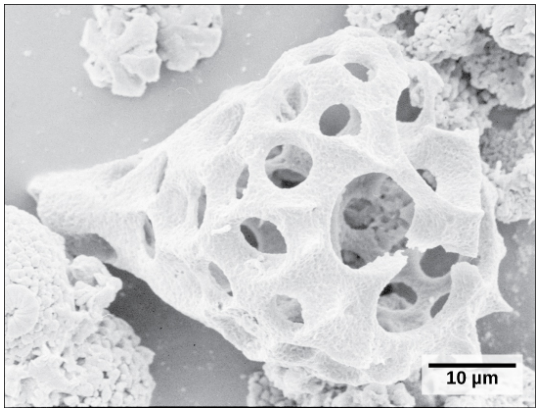


Figure $23.3D.1$: **Radiolarian shell**: This fossilized radiolarian shell was imaged using a scanning electron microscope.

Key Points

- The needle-like pseudopodia are used to carry out a process called cytoplasmic streaming which is a means of locomotion or distributing nutrients and oxygen.
- Two major subclassifications of Rhizaria include Forams and Radiolarians.
- Forams are characterized as unicellular heterotrophic protists that have porous shells, referred to as tests, which can contain photosynthetic algae that the foram can use as a nutrient source.
- Radiolarians are characterized by a glassy silica exterior that displays either bilateral or radial symmetry.

Key Terms

- **pseudopodia**: temporary projections of eukaryotic cells
- **test**: the external calciferous shell of a foram

23.3E: Archaeplastida

Learning Objectives

- Describe the relationship between red algae, green algae, and land plants

Red algae and green algae are included in the supergroup Archaeplastida. It is well documented that land plants evolved from a common ancestor of these protists; their closest relatives are found within this group. Molecular evidence supports that all Archaeplastida are descendants of an endosymbiotic relationship between a heterotrophic protist and a cyanobacterium. The red and green algae include unicellular, multicellular, and colonial forms

Red Algae

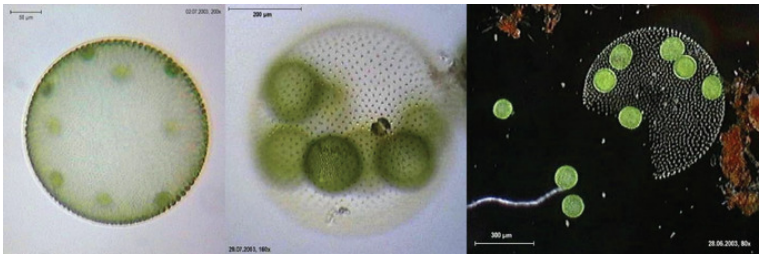
Red algae, or rhodophytes, are primarily multicellular, lack flagella, and range in size from microscopic, unicellular protists to large, multicellular forms grouped into the informal seaweed category. The red algae life cycle is an alternation of generations. Some species of red algae contain phycoerythrins, photosynthetic accessory pigments that are red in color and outcompete the green tint of chlorophyll, making these species appear as varying shades of red. Other protists classified as red algae lack phycoerythrins and are parasites. Red algae are common in tropical waters where they have been detected at depths of 260 meters. Other red algae exist in terrestrial or freshwater environments.

Green Algae: Chlorophytes and Charophytes

The most abundant group of algae is the green algae. The green algae exhibit similar features to the land plants, particularly in terms of chloroplast structure. It is well supported that this group of protists share a relatively-recent common ancestors with land plants. The green algae are subdivided into the chlorophytes and the charophytes. The charophytes are the closest-living relatives of land plants, resembling them in morphology and reproductive strategies. Charophytes are common in wet habitats where their presence often signals a healthy ecosystem.

The chlorophytes exhibit great diversity of form and function. Chlorophytes primarily inhabit freshwater and damp soil; they are a common component of plankton. *Chlamydomonas* is a simple, unicellular chlorophyte with a pear-shaped morphology and two opposing, anterior flagella that guide this protist toward light sensed by its eyespot. More complex chlorophyte species exhibit haploid gametes and spores that resemble *Chlamydomonas*.

The chlorophyte *Volvox* is one of only a few examples of a colonial organism, which behaves in some ways like a collection of individual cells, but in other ways like the specialized cells of a multicellular organism. Volvox colonies contain 500 to 60,000 cells, each with two flagella, contained within a hollow, spherical matrix composed of a gelatinous glycoprotein secretion. Individual *Volvox* cells move in a coordinated fashion and are interconnected by cytoplasmic bridges. Only a few of the cells reproduce to create daughter colonies, an example of basic cell specialization in this organism.



Volvox aureus

Volvox aureus is a green alga in the supergroup Archaeplastida. This species exists as a colony, consisting of cells immersed in a gel-like matrix and intertwined with each other via hair-like cytoplasmic extensions.

True multicellular organisms, such as the sea lettuce, *Ulva*, are represented among the chlorophytes. In addition, some chlorophytes exist as large, multinucleate, single cells. Species in the genus *Caulerpa* exhibit flattened, fern-like foliage and can reach lengths of 3 meters. *Caulerpa* species undergo nuclear division, but their cells do not complete cytokinesis, remaining instead as massive and elaborate single cells.

Caulerpa taxifolia

Caulerpa taxifolia is a chlorophyte consisting of a single cell containing potentially thousands of nuclei.

Key Points

- Archaeplastida are typically associated with their relationship to land plants; in addition, molecular evidence shows that Archaeplastida evolved from an endosymbiotic relationship between a heterotrophic protist and a cyanobacterium.
- Red algae (rhodophytes), are classified as Archaeplastida and are most often characterized by the presence of the red pigment phycoerythrin; however, there are red algae that lack phycoerythrins and can be classified as parasites.
- Red algae typically exist as multicellular protists that lack flagella; however, they can also exist as unicellular organisms.
- Green algae are the most abundant group of algae and can be further classified as chlorophytes and charophytes.
- Charophytes are the green algae which resemble land plants and are their closest living relative.
- Chlorophytes are the green algae which exhibit a wide range of forms; they can be unicellular, multicellular, or colonial.

Key Terms

- **endosymbiotic**: that lives within a body or cells of another organism
- **plankton**: a generic term for all the organisms that float in the sea

23.3F: Amoebozoa and Opisthokonta

Amoebozoa are a type of protist that is characterized by the presence of pseudopodia which they use for locomotion and feeding.

Learning Objectives

- Describe characteristics of Amoebozoa

Key Points

- Amoebozoa (amoebas) can live in either marine and fresh water or in soil.
- Amoebozoa are characterized by the presence of pseudopodia, which are extensions that can be either tube-like or flat lobes and are used for locomotion and feeding.
- Amooebozoa can be further divided into subclassifications that include slime molds; these can be found as both plasmodial and cellular types.
- Plasmodial slime molds are characterized by the presence of large, multinucleate cells that have the ability to glide along the surface and engulf food particles as they move.
- Cellular molds are characterized by the presence of independent amoeboid cells during times of nutrient abundancy and the development of a cellular mass, called a slug, during times of nutrient depletion.
- Archamoebae, Flabellinea, and Tubulinea are also groups of Amoebozoa; their defining characteristics include: Archamoebae lack mitochondria; Flabellinea flatten during locomotion and lack a shell and flagella; Tubulinea have a rough cylindrical form during locomotion with cylindrical pseudopodia.

Key Terms

- **rhizaria**: a species-rich supergroup of mostly unicellular eukaryotes that for the most part are amoeboids with filose, reticulose, or microtubule-supported pseudopods
- **plasmodium**: a mass of cytoplasm, containing many nuclei, created by the aggregation of amoeboid cells of slime molds during their vegetative phase
- **sporangia**: an enclosure in which spores are formed (also called a fruiting body)

Amoebozoa

Protists are eukaryotic organisms that are classified as unicellular, colonial, or multicellular organisms that do not have specialized tissues. This identifying property sets protists apart from other organisms within the Eukarya domain. The amoebozoans are classified as protists with pseudopodia which are used in locomotion and feeding. Amoebozoans live in marine environments, fresh water, or in soil. In addition to the defining pseudopodia, they also lack a shell and do not have a fixed body. The pseudopodia which are characteristically exhibited include extensions which can be tube-like or flat lobes, rather than the hair-like pseudopodia of rhizarian amoeba. Rhizarian amoeba are amoeboids with filose, reticulose, or microtubule-supported pseudopods and include the groups: Cercozoa, Foraminifera, and Radiolaria and are classified as bikonts. The Amoebozoa include several groups of unicellular amoeba-like organisms that are free-living or parasites that are classified as unikonts. The best known and most well-studied member of this group is the slime mold. Additional members include the Archamoebae, Tubulinea, and Flabellinea.

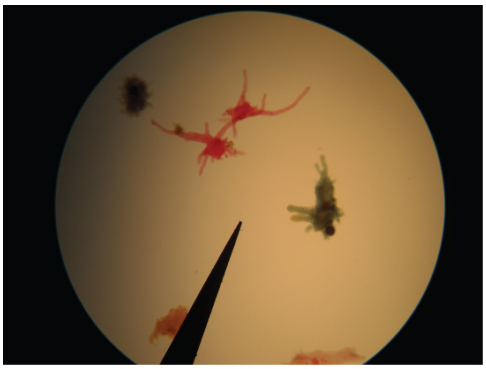


Figure 23.3F.1: Pseudopodia structures: Amoebae with tubular and lobe-shaped pseudopodia, such as the ones seen under this microscope, would be morphologically classified as amoebozoans.

Slime Molds

A subset of the amoebozoans, the slime molds, has several morphological similarities to fungi that are thought to be the result of convergent evolution. For instance, during times of stress, some slime molds develop into spore -generating fruiting bodies, similar to fungi.

The slime molds are categorized on the basis of their life cycles into plasmodial or cellular types. Plasmodial slime molds are composed of large, multinucleate cells that move along surfaces like an amorphous blob of slime during their feeding stage. Food particles are lifted and engulfed into the slime mold as it glides along. Upon maturation, the plasmodium takes on a net-like appearance with the ability to form fruiting bodies, or sporangia, during times of stress. Haploid spores are produced by meiosis within the sporangia. These spores can be disseminated through the air or water to potentially land in more favorable environments. If this occurs, the spores germinate to form ameboid or flagellate haploid cells that can combine with each other and produce a diploid zygotic slime mold to complete the life cycle.

Figure 23.3F.1: **Badhamia utricularis**: Badhamia utricularis: an example of a plasmodial slime mold with the ability to form a fruiting body.

The cellular slime molds function as independent amoeboid cells when nutrients are abundant. When food is depleted, cellular slime molds pile onto each other into a mass of cells that behaves as a single unit called a slug. Some cells in the slug contribute to a 2–3-millimeter stalk, drying up and dying in the process. Cells atop the stalk form an asexual fruiting

body that contains haploid spores. As with plasmodial slime molds, the spores are disseminated and can germinate if they land in a moist environment. One representative genus of the cellular slime molds is Dictyostelium, which commonly exists in the damp soil of forests.

Figure 23.3F.1: **Plasmodial slime mold: Physarum polycephalum**: Physarum polycephalum is an example of a cellular slime mold.

Archamoebae, Flabellinea, and Tubulinea

The Archamoebae are a group of Amoebozoa distinguished by the absence of mitochondria. They include genera that are internal parasites or commensals of animals (*Entamoeba* and *Endolimax*). A few species are human pathogens, causing diseases such as amoebic dysentery. The other genera of archamoebae live in freshwater habitats and are unusual among amoebae in possessing flagella. Most have a single nucleus and flagellum, but the giant amoeba, *Pelomyxa*, has many of each.

The Tubulinea are a major grouping of Amoebozoa, including most of the larger and more familiar amoebae like *Amoeba*, *Arcella*, and *Difflugia*. During locomotion, most Tubulinea have a roughly cylindrical form or produce numerous cylindrical pseudopods. Each cylinder advances by a single central stream of cytoplasm, granular in appearance, and has no subpseudopodia. This distinguishes them from other amoeboid groups, although in some members this is not the normal type of locomotion.

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Topic hierarchy

23.4A: Protists as Primary Producers, Food Sources, and Symbionts

23.4B: Protists as Human Pathogens

23.4C: Protists as Plant Pathogens

23.4A: Protists as Primary Producers, Food Sources, and Symbionts

Protists function as sources of food for organisms on land and sea.

Learning Objectives

- Give examples of how protists act as primary producers

Key Points

- Photosynthetic protists serve as producers of nutrition for other organisms.
- Protists like zooxanthellae have a symbiotic relationship with coral reefs; the protists act as a food source for coral and the coral provides shelter and compounds for photosynthesis for the protists.
- Protists feed a large portion of the world's aquatic species and conduct a quarter of the world's photosynthesis.
- Protists help land-dwelling animals such as cockroaches and termites digest cellulose.

Key Terms

- **zooxanthella**: an animal of the genus Symbiodinium, a yellow dinoflagellate, notably found in coral reefs
- **primary producer**: an autotroph organism that produces complex organic matter using photosynthesis or chemosynthesis

Primary Producers/Food Sources

Protists function in various ecological niches. Some protist species are essential components of the food chain and are generators of biomass.

Protists are essential sources of nutrition for many other organisms. In some cases, as in plankton, protists are consumed directly. Alternatively, photosynthetic protists serve as producers of nutrition for other organisms. For instance, photosynthetic dinoflagellates called zooxanthellae use sunlight to fix inorganic carbon. In this symbiotic relationship, these protists provide nutrients for the coral polyps that house them, giving corals a boost of energy to secrete a calcium carbonate skeleton. In turn, the corals provide the protists with a protected environment and the compounds needed for photosynthesis. This type of symbiotic relationship is important in nutrient-poor environments. Without dinoflagellate symbionts, corals lose algal pigments in a process called coral bleaching and they eventually die. This explains why reef-building corals do not reside in waters deeper than 20 meters: insufficient light reaches those depths for dinoflagellates to photosynthesize.

Figure 23.4A.1: **Corals and dinoflagellates**: Coral polyps obtain nutrition through a symbiotic relationship with dinoflagellates.

The protists themselves and their products of photosynthesis are essential, directly or indirectly, to the survival of organisms ranging from bacteria to mammals. As primary producers, protists feed a large proportion of the world's aquatic species. (On land, terrestrial plants serve as primary producers.) In fact, approximately one-quarter of the world's photosynthesis is conducted by protists, particularly dinoflagellates, diatoms, and multicellular algae.

Figure 23.4*A*. 1: **Protists and aquatic organisms**: Virtually all aquatic organisms depend directly or indirectly on protists for food.

Protists do not only create food sources for sea-dwelling organisms. Certain anaerobic parabasalid species exist in the digestive tracts of termites and wood-eating cockroaches where they contribute an essential step in the digestion of cellulose ingested by these insects as they bore through wood.

23.4B: Protists as Human Pathogens

Many protists exist as parasites that infect and cause diseases in their hosts.

Learning Objectives

- Identify the effects on humans of protist pathogens

Key Points

- The protist parasite *Plasmodium* must colonize both a mosquito and a vertebrate; *P. falciparum*, which is responsible for 50 percent of malaria cases, is transmitted to humans by the African malaria mosquito, *Anopheles gambiae*.
- When *P. falciparum* infects and destroys red blood cells, they burst, and parasitic waste leaks into the blood stream, causing deliruim, fever, and anemia in patients.
- *Trypanosoma brucei* is responsible for African sleeping sickness which the human immune system is unable to guard against since it has thousands of possible antigens and changes surface glycoproteins with each infectious cycle.
- Another *Trypanosoma* species, *T. cruzi*, inhabits the heart and digestive system tissues, causing malnutrition and heart failure.

Key Terms

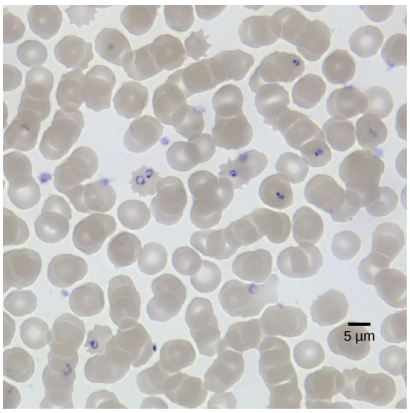
- **Trypanosoma**: infects a variety of hosts and cause various diseases, including the fatal African sleeping sickness in humans
- **plasmodium**: parasitic protozoa that must colonize a mosquito and a vertebrate to complete its life cycle
- **pathogen**: any organism or substance, especially a microorganism, capable of causing disease, such as bacteria, viruses, protozoa, or fungi

Human Pathogens

A pathogen is anything that causes disease. Parasites live in or on an organism and harm that organism. A significant number of protists are pathogenic parasites that must infect other organisms to survive and propagate. Protist parasites include the causative agents of malaria, African sleeping sickness, and waterborne gastroenteritis in humans.

Plasmodium Species

Members of the genus *Plasmodium* must colonize both a mosquito and a vertebrate to complete their life cycle. In vertebrates, the parasite develops in liver cells and goes on to infect red blood cells, bursting from and destroying the blood cells with each asexual replication cycle. Of the four *Plasmodium* species known to infect humans, *P. falciparum* accounts for 50 percent of all malaria cases and is the primary cause of disease-related fatalities in tropical regions of the world. In 2010, it was estimated that malaria caused between one and one-half million deaths, mostly in African children. During the course of malaria, *P. falciparum* can infect and destroy more than one-half of a human's circulating blood cells, leading to severe anemia. In response to waste products released as the parasites burst from infected blood cells, the host immune system mounts a massive inflammatory response with episodes of delirium-inducing fever as parasites lyse red blood cells, spilling parasitic waste into the bloodstream. *P. falciparum* is transmitted to humans by the African malaria mosquito, *Anopheles gambiae*. Techniques to kill, sterilize, or avoid exposure to this highly-aggressive mosquito species are crucial to malaria control.



Plasmodium

Red blood cells are shown to be infected with *P. falciparum*, the causative agent of malaria. In this light microscopic image taken using a 100× oil immersion lens, the ring-shaped *P. falciparum*stains purple.

Trypanosomes

Trypanosoma brucei, the parasite that is responsible for African sleeping sickness, confounds the human immune system by changing its thick layer of surface glycoproteins with each infectious cycle. The glycoproteins are identified by the immune system as foreign antigens and a specific antibody defense is mounted against the parasite. However, *T. brucei* has thousands of possible antigens; with each subsequent generation, the protist switches to a glycoprotein coating of a different molecular structure. In this way, *T. brucei* is capable of replicating continuously without the immune system ever succeeding in clearing the parasite. Without treatment, *T. brucei* attacks red blood cells, causing the patient to lapse into a coma and eventually die. During epidemic periods, mortality from the disease can be high. Greater surveillance and control measures lead to a reduction in reported cases; some of the lowest numbers reported in 50 years (fewer than 10,000 cases in all of sub-Saharan Africa) have happened since 2009.

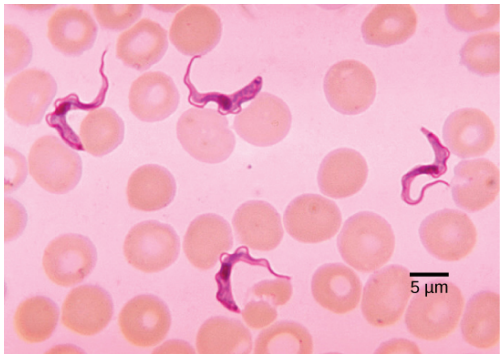


Figure $23.4B.1$: **Trypanosomes**: Trypanosomes are shown among red blood cells.

In Latin America, another species, *T. cruzi*, is responsible for Chagas disease. *T. cruzi* infections are mainly caused by a blood-sucking bug. The parasite inhabits heart and digestive system tissues in the chronic phase of infection, leading to malnutrition and heart failure due to abnormal heart rhythms. An estimated 10 million people are infected with Chagas disease; it caused 10,000 deaths in 2008.

23.4C: Protists as Plant Pathogens

Many protists act as parasites that prey on plants or as decomposers that feed on dead organisms.

Learning Objectives

- Describe the ways in which protists act as decomposers and the actions of parasitic protists on plants

Key Points

- *Plasmopara viticola* causes downy mildew in grape plants, resulting in stunted growth and withered, discolored leaves.
- Since downy mildew has a higher incidence in the late summer, planting early in the season can reduce the threat of downy mildew; fungicides are also somewhat effective at preventing downy mildew.
- *Phytophthora infestans* causes potato late blight (potato stalks and stems decay into black slime) and was responsible for the Irish potato famine in the nineteenth century.
- Protist saprobes feed on dead organisms, which returns inorganic nutrients to soil and water.

Key Terms

- **saprobe**: an organism that lives off of dead or decaying organic material
- **oomycete**: fungus-like filamentous unicellular protists; the water molds
- **downy mildew**: plant disease caused by oomycetes; causes stunted growth in plants as well as discolored, withered leaves

Plant Parasites

Protist parasites prey on terrestrial plants and include agents that cause massive destruction to food crops. The oomycete *Plasmopara viticola* parasitizes grape plants, which causes a disease called downy mildew. Grape plants infected with *P. viticola* appear stunted and have discolored, withered leaves. The spread of downy mildew nearly collapsed the French wine industry in the nineteenth century. They are easily controlled once discovered, so careful monitoring of susceptible hosts is key because if left unaddressed, the organism can quickly spread and completely overwhelm the host species

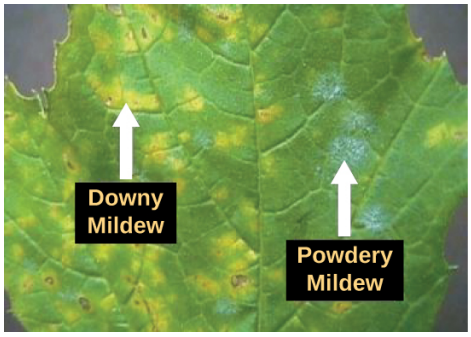


Figure 23.4C.1: **Downy mildew**: Both downy and powdery mildews on this grape leaf are caused by an infection of P. viticola.

Because the downy mildew pathogen does not overwinter in midwestern fields, crop rotations and tillage practices do not affect disease development. The pathogen tends to become established in late summer. Therefore, planting early season varieties may further reduce the threat posed by downy mildew. Fungicides can also be applied to control downy mildew. Broad spectrum protectant fungicides such as chlorothalonil, mancozeb, and fixed copper are somewhat effective in protecting against downy mildew infection.

Phytophthora infestans is an oomycete responsible for potato late blight. This disease causes potato stalks and stems to decay into black slime. Widespread potato blight caused by *P. infestans* led to the well-known Irish potato famine in the nineteenth century that claimed the lives of approximately one million people and resulted in the emigration of at least one million more from Ireland. Late blight continues to plague potato crops in certain parts of the United States and Russia, wiping out as much as 70 percent of crops when no pesticides are applied.

Figure 23.4C.1: **Potato Late Blight**: These unappetizing remnants result from an infection with P. infestans, the causative agent of potato late blight.

Agents of Decomposition

The fungus-like protist saprobes are specialized to absorb nutrients from non-living organic matter, such as dead organisms or their wastes. For instance, many types of oomycetes grow on dead animals or algae. Saprobic protists have the essential function of returning inorganic nutrients to the soil and water. This process allows for new plant growth, which in turn generates sustenance for other organisms along the food chain. Indeed, without saprobe species, such as protists, fungi, and bacteria, life would cease to exist as all organic carbon became "tied up" in dead organisms.

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CHAPTER OVERVIEW

24: FUNGI

24.1: CHARACTERISTICS OF FUNGI

24.1A: CHARACTERISTICS OF FUNGI
24.1B: FUNGI CELL STRUCTURE AND FUNCTION
24.1C: FUNGI REPRODUCTION
24.2: ECOLOGY OF FUNGI

24.2A: FUNGI HABITAT, DECOMPOSITION, AND RECYCLING
24.2B: MUTUALISTIC RELATIONSHIPS WITH FUNGI AND FUNGIVORES
24.3: CLASSIFICATIONS OF FUNGI

24.3A: CHYTRIDIOMYCOTA: THE CHYTRIDS
24.3B: ZYGOMYCOTA - THE CONJUGATED FUNGI
Zygomycota, a small group in the fungi kingdom, can reproduce asexually or sexually, in a process called conjugation.

24.3C: ASCOMYCOTA - THE SAC FUNGI
Most fungi belong to the Phylum Ascomycota, which uniquely forms of an ascus, a sac-like structure that contains haploid ascospores.

24.3D: BASIDIOMYCOTA: THE CLUB FUNGI
24.3E: DEUTEROMYCOTA - THE IMPERFECT FUNGI
Phylum Deuteromycota is a polyphyletic group of asexually-reproducing fungi that do not display a sexual phase; they are known as imperfect.

24.3F: GLOMEROMYCOTA
Glomeromycetes are an important group of fungi that live in close symbiotic association with the roots of trees and plants.

24.4: FUNGAL PARASITES AND PATHOGENS

24.4A: FUNGI AS PLANT, ANIMAL, AND HUMAN PATHOGENS
24.5: IMPORTANCE OF FUNGI IN HUMAN LIFE

24.5A: IMPORTANCE OF FUNGI IN HUMAN LIFE

: Characteristics of Fungi

Topic hierarchy

- 24.1A: Characteristics of Fungi
- 24.1B: Fungi Cell Structure and Function
- 24.1C: Fungi Reproduction

24.1A: Characteristics of Fungi

Fungi, latin for mushroom, are eukaryotes which are responsible for decomposition and nutrient cycling through the environment.

Learning Objectives

- Describe the role of fungi in the ecosystem

Key Points

- Fungi are more closely related to animals than plants.
- Fungi are heterotrophic: they use complex organic compounds as sources of energy and carbon, not photosynthesis.
- Fungi multiply either asexually, sexually, or both.
- The majority of fungi produce spores, which are defined as haploid cells that can undergo mitosis to form multicellular, haploid individuals.
- Fungi interact with other organisms by either forming beneficial or mutualistic associations (mycorrhizae and lichens) or by causing serious infections.

Key Terms

- **mycorrhiza**: a symbiotic association between a fungus and the roots of a vascular plant
- **spore**: a reproductive particle, usually a single cell, released by a fungus, alga, or plant that may germinate into another
- **lichen**: any of many symbiotic organisms, being associations of fungi and algae; often found as white or yellow patches on old walls, etc.
- **Ascomycota**: a taxonomic division within the kingdom Fungi; those fungi that produce spores in a microscopic sporangium called an ascus
- **heterotrophic**: organisms that use complex organic compounds as sources of energy and carbon

Introduction to Fungi

The word fungus comes from the Latin word for mushrooms. Indeed, the familiar mushroom is a reproductive structure used by many types of fungi. However, there are also many fungi species that don't produce mushrooms at all. Being eukaryotes, a typical fungal cell contains a true nucleus and many membrane-bound organelles. The kingdom Fungi includes an enormous variety of living organisms collectively referred to as Ascomycota, or true Fungi. While scientists have identified about 100,000 species of fungi, this is only a fraction of the 1.5 million species of fungus probably present on earth. Edible mushrooms, yeasts, black mold, and the producer of the antibiotic penicillin, *Penicillium notatum*, are all members of the kingdom Fungi, which belongs to the domain Eukarya.

Figure 24.1*A*. 1: **Examples of fungi**: Many species of fungus produce the familiar mushroom (a) which is a reproductive structure. This (b) coral fungus displays brightly-colored fruiting bodies. This electron micrograph shows (c) the spore-bearing structures of Aspergillus, a type of toxic fungi found mostly in soil and plants.

Fungi, once considered plant-like organisms, are more closely related to animals than plants. Fungi are not capable of photosynthesis: they are heterotrophic because they use complex organic compounds as sources of energy and carbon. Some fungal organisms multiply only asexually, whereas others undergo both asexual reproduction and sexual reproduction with alternation of generations. Most fungi produce a large number of spores, which are haploid cells that can undergo mitosis to form multicellular, haploid individuals. Like bacteria, fungi play an essential role in ecosystems

because they are decomposers and participate in the cycling of nutrients by breaking down organic and inorganic materials to simple molecules.

Fungi often interact with other organisms, forming beneficial or mutualistic associations. For example most terrestrial plants form symbiotic relationships with fungi. The roots of the plant connect with the underground parts of the fungus forming mycorrhizae. Through mycorrhizae, the fungus and plant exchange nutrients and water, greatly aiding the survival of both species Alternatively, lichens are an association between a fungus and its photosynthetic partner (usually an alga). Fungi also cause serious infections in plants and animals. For example, Dutch elm disease, which is caused by the fungus *Ophiostoma ulmi*, is a particularly devastating type of fungal infestation that destroys many native species of elm (*Ulmus* sp.) by infecting the tree's vascular system. The elm bark beetle acts as a vector, transmitting the disease from tree to tree. Accidentally introduced in the 1900s, the fungus decimated elm trees across the continent. Many European and Asiatic elms are less susceptible to Dutch elm disease than American elms.

In humans, fungal infections are generally considered challenging to treat. Unlike bacteria, fungi do not respond to traditional antibiotic therapy because they are eukaryotes. Fungal infections may prove deadly for individuals with compromised immune systems.

Fungi have many commercial applications. The food industry uses yeasts in baking, brewing, and cheese and wine making. Many industrial compounds are byproducts of fungal fermentation. Fungi are the source of many commercial enzymes and antibiotics.

24.1B: Fungi Cell Structure and Function

Fungi are unicellular or multicellular thick-cell-walled heterotroph decomposers that eat decaying matter and make tangles of filaments.

Learning Objectives

- Describe the physical structures associated with fungi

Key Points

- Fungal cell walls are rigid and contain complex polysaccharides called chitin (adds structural strength) and glucans.
- Ergosterol is the steroid molecule in the cell membranes that replaces the cholesterol found in animal cell membranes.
- Fungi can be unicellular, multicellular, or dimorphic, which is when the fungi is unicellular or multicellular depending on environmental conditions.
- Fungi in the morphological vegetative stage consist of a tangle of slender, thread-like hyphae, whereas the reproductive stage is usually more obvious.
- Fungi like to be in a moist and slightly acidic environment; they can grow with or without light or oxygen.
- Fungi are saprophyte heterotrophs in that they use dead or decomposing organic matter as a source of carbon.

Key Terms

- **glucan**: any polysaccharide that is a polymer of glucose
- **ergosterol**: the functional equivalent of cholesterol found in cell membranes of fungi and some protists, as well as, the steroid precursor of vitamin D2
- **mycelium**: the vegetative part of any fungus, consisting of a mass of branching, threadlike hyphae, often underground
- **hypha**: a long, branching, filamentous structure of a fungus that is the main mode of vegetative growth
- **septum**: cell wall division between hyphae of a fungus
- **thallus**: vegetative body of a fungus
- **saprophyte**: any organism that lives on dead organic matter, as certain fungi and bacteria
- **chitin**: a complex polysaccharide, a polymer of N-acetylglucosamine, found in the exoskeletons of arthropods and in the cell walls of fungi; thought to be responsible for some forms of asthma in humans

Cell Structure and Function

Fungi are eukaryotes and have a complex cellular organization. As eukaryotes, fungal cells contain a membrane-bound nucleus where the DNA is wrapped around histone proteins. A few types of fungi have structures comparable to bacterial plasmids (loops of DNA). Fungal cells also contain mitochondria and a complex system of internal membranes, including the endoplasmic reticulum and Golgi apparatus.

Unlike plant cells, fungal cells do not have chloroplasts or chlorophyll. Many fungi display bright colors arising from other cellular pigments, ranging from red to green to black. The poisonous *Amanita muscaria* (fly agaric) is recognizable by its bright red cap with white patches. Pigments in fungi are associated with the cell wall. They play a protective role against ultraviolet radiation and can be toxic.

Figure 24.1B.1: **The poisonous *Amanita muscaria***: The poisonous *Amanita muscaria* is native to temperate and boreal regions of North America.

The rigid layers of fungal cell walls contain complex polysaccharides called chitin and glucans. Chitin, also found in the exoskeleton of insects, gives structural strength to the cell walls of fungi. The wall protects the cell from desiccation and predators. Fungi have plasma membranes similar to other eukaryotes, except that the structure is stabilized by ergosterol: a steroid molecule that replaces the cholesterol found in animal cell membranes. Most members of the kingdom Fungi are nonmotile.

Growth

The vegetative body of a fungus is a unicellular or multicellular thallus. Dimorphic fungi can change from the unicellular to multicellular state depending on environmental conditions. Unicellular fungi are generally referred to as yeasts. *Saccharomyces cerevisiae* (baker's yeast) and *Candida* species (the agents of thrush, a common fungal infection) are examples of unicellular fungi.

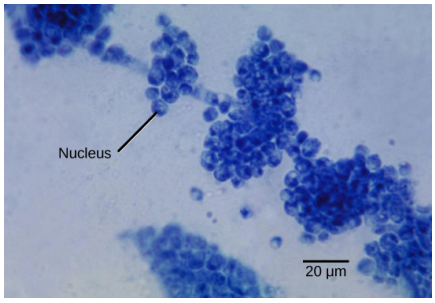

Figure 24.1B.1: **Example of a unicellular fungus**: *Candida albicans* is a yeast cell and the agent of candidiasis and thrush. This organism has a similar morphology to coccus bacteria; however, yeast is a eukaryotic organism (note the nucleus).

Most fungi are multicellular organisms. They display two distinct morphological stages: the vegetative and reproductive. The vegetative stage consists of a tangle of slender thread-like structures called hyphae (singular, hypha), whereas the reproductive stage can be more conspicuous. The mass of hyphae is a mycelium. It can grow on a surface, in soil or decaying material, in a liquid, or even on living tissue. Although individual hyphae must be observed under a microscope, the mycelium of a fungus can be very large, with some species truly being "the fungus humongous." The giant *Armillaria solidipes* (honey mushroom) is considered the largest organism on Earth, spreading across more than 2,000 acres of underground soil in eastern Oregon; it is estimated to be at least 2,400 years old.

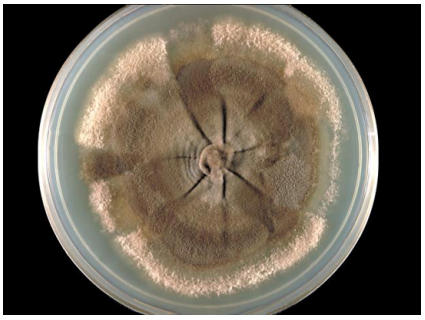

Figure 24.1B.1: **Example of a mycelium of a fungus**: The mycelium of the fungus *Neotestudina rosati* can be pathogenic to humans. The fungus enters through a cut or scrape and develops a mycetoma, a chronic subcutaneous infection.

Most fungal hyphae are divided into separate cells by endwalls called septa (singular, septum) (a, c). In most phyla of fungi, tiny holes in the septa allow for the rapid flow of nutrients and small molecules from cell to cell along the hypha. They are described as perforated septa. The hyphae in bread molds (which belong to the Phylum Zygomycota) are not separated by septa. Instead, they are formed by large cells containing many nuclei, an arrangement described as coenocytic hyphae (b). Fungi thrive in environments that are moist and slightly acidic; they can grow with or without light.

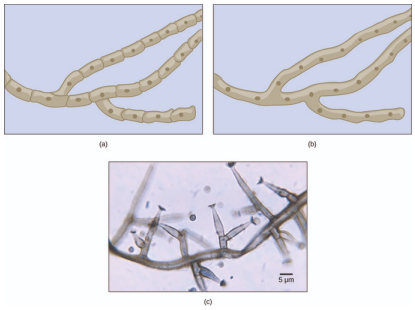

Figure 24.1B.1: **Division of hyphae into separate cells**: Fungal hyphae may be (a) septated or (b) coenocytic (coeno- = "common"; -cytic = "cell") with many nuclei present in a single hypha. A bright field light micrograph of (c) *Phialophora richardsiae* shows septa that divide the hyphae.

Nutrition

Like animals, fungi are heterotrophs: they use complex organic compounds as a source of carbon, rather than fix carbon dioxide from the atmosphere as do some bacteria and most plants. In addition, fungi do not fix nitrogen from the atmosphere. Like animals, they must obtain it from their diet. However, unlike most animals, which ingest food and then digest it internally in specialized organs, fungi perform these steps in the reverse order: digestion precedes ingestion. First, exoenzymes are transported out of the hyphae, where they process nutrients in the environment. Then, the smaller molecules produced by this external digestion are absorbed through the large surface area of the mycelium. As with animal cells, the polysaccharide of storage is glycogen rather than the starch found in plants.

Fungi are mostly saprobes (saprophyte is an equivalent term): organisms that derive nutrients from decaying organic matter. They obtain their nutrients from dead or decomposing organic matter, mainly plant material. Fungal exoenzymes are able to break down insoluble polysaccharides, such as the cellulose and lignin of dead wood, into readily-absorbable glucose molecules. The carbon, nitrogen, and other elements are thus released into the environment. Because of their varied metabolic pathways, fungi fulfill an important ecological role and are being investigated as potential tools in bioremediation.

Some fungi are parasitic, infecting either plants or animals. Smut and Dutch elm disease affect plants, whereas athlete's foot and candidiasis (thrush) are medically important fungal infections in humans.

24.1C: Fungi Reproduction

Fungi can reproduce asexually by fragmentation, budding, or producing spores, or sexually with homothallic or heterothallic mycelia.

Learning Objectives

- Describe the mechanisms of sexual and asexual reproduction in fungi

Key Points

- New colonies of fungi can grow from the fragmentation of hyphae.
- During budding, a bulge forms on the side of the cell; the bud ultimately detaches after the nucleus divides mitotically.
- Asexual spores are genetically identical to the parent and may be released either outside or within a special reproductive sac called a sporangium.
- Adverse environmental conditions often cause sexual reproduction in fungi.
- Mycelium can either be homothallic or heterothallic when reproducing sexually.
- Fungal sexual reproduction includes the following three stages: plasmogamy, karyogamy, and gametangia.

Key Terms

- **homothallic**: male and female reproductive structures are present in the same plant or fungal mycelium
- **gametangium**: an organ or cell in which gametes are produced that is found in many multicellular protists, algae, fungi, and the gametophytes of plants
- **spore**: a reproductive particle, usually a single cell, released by a fungus, alga, or plant that may germinate into another
- **sporangium**: a case, capsule, or container in which spores are produced by an organism
- **karyogamy**: the fusion of two nuclei within a cell
- **plasmogamy**: stage of sexual reproduction joining the cytoplasm of two parent mycelia without the fusion of nuclei

Reproduction

Fungi reproduce sexually and/or asexually. Perfect fungi reproduce both sexually and asexually, while imperfect fungi reproduce only asexually (by mitosis).

In both sexual and asexual reproduction, fungi produce spores that disperse from the parent organism by either floating on the wind or hitching a ride on an animal. Fungal spores are smaller and lighter than plant seeds. The giant puffball mushroom bursts open and releases trillions of spores. The huge number of spores released increases the likelihood of landing in an environment that will support growth.

Figure 24.1C.1: **The release of fungal spores**: The (a) giant puff ball mushroom releases (b) a cloud of spores when it reaches maturity.

Asexual Reproduction

Fungi reproduce asexually by fragmentation, budding, or producing spores. Fragments of hyphae can grow new colonies. Mycelial fragmentation occurs when a fungal mycelium separates into pieces with each component growing into a separate

mycelium. Somatic cells in yeast form buds. During budding (a type of cytokinesis), a bulge forms on the side of the cell, the nucleus divides mitotically, and the bud ultimately detaches itself from the mother cell.

The most common mode of asexual reproduction is through the formation of asexual spores, which are produced by one parent only (through mitosis) and are genetically identical to that parent. Spores allow fungi to expand their distribution and colonize new environments. They may be released from the parent thallus, either outside or within a special reproductive sac called a sporangium.

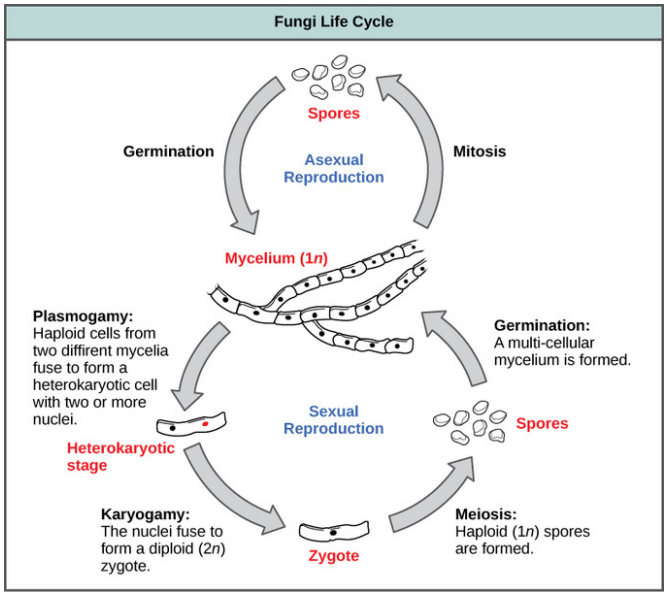

Figure 24.1C.1: **Types of fungal reproduction**: Fungi may utilize both asexual and sexual stages of reproduction; sexual reproduction often occurs in response to adverse environmental conditions.

There are many types of asexual spores. Conidiospores are unicellular or multicellular spores that are released directly from the tip or side of the hypha. Other asexual spores originate in the fragmentation of a hypha to form single cells that are released as spores; some of these have a thick wall surrounding the fragment. Yet others bud off the vegetative parent cell. Sporangiospores are produced in a sporangium.

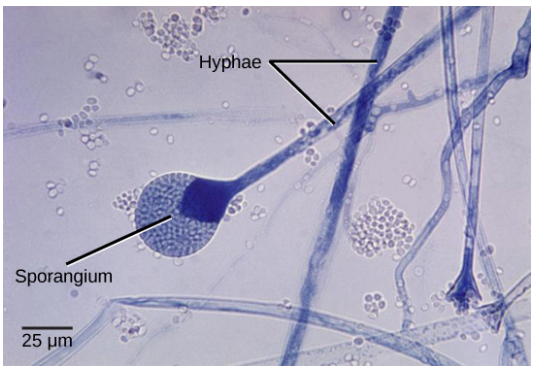

Figure 24.1C.1: **Release of spores from a sporangium**: This bright field light micrograph shows the release of spores from a sporangium at the end of a hypha called a sporangiophore. The organism depicted is a Mucor sp. fungus: a mold often found indoors.

Sexual Reproduction

Sexual reproduction introduces genetic variation into a population of fungi. In fungi, sexual reproduction often occurs in response to adverse environmental conditions. Two mating types are produced. When both mating types are present in the same mycelium, it is called homothallic, or self-fertile. Heterothallic mycelia require two different, but compatible, mycelia to reproduce sexually.

Although there are many variations in fungal sexual reproduction, all include the following three stages. First, during plasmogamy (literally, “marriage or union of cytoplasm”), two haploid cells fuse, leading to a dikaryotic stage where two

haploid nuclei coexist in a single cell. During karyogamy (“nuclear marriage”), the haploid nuclei fuse to form a diploid zygote nucleus. Finally, meiosis takes place in the gametangia (singular, gametangium) organs, in which gametes of different mating types are generated. At this stage, spores are disseminated into the environment.

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24.2: Ecology of Fungi

Topic hierarchy

- 24.2A: Fungi Habitat, Decomposition, and Recycling
- 24.2B: Mutualistic Relationships with Fungi and Fungivores

24.2A: Fungi Habitat, Decomposition, and Recycling

Fungi are the major decomposers of nature; they break down organic matter which would otherwise not be recycled.

Learning Objectives

- Explain the roles played by fungi in decomposition and recycling

Key Points

- Aiding the survival of species from other kingdoms through the supply of nutrients, fungi play a major role as decomposers and recyclers in the wide variety of habitats in which they exist.
- Fungi provide a vital role in releasing scarce, yet biologically-essential elements, such as nitrogen and phosphorus, from decaying matter.
- Their mode of nutrition, which involves digestion before ingestion, allows fungi to degrade many large and insoluble molecules that would otherwise remain trapped in a habitat.

Key Terms

- **decomposer**: any organism that feeds off decomposing organic material, especially bacterium or fungi
- **exoenzyme**: any enzyme, generated by a cell, that functions outside of that cell
- **saprobe**: an organism that lives off of dead or decaying organic material

Fungi & Their Roles as Decomposers and Recyclers

Fungi play a crucial role in the balance of ecosystems. They colonize most habitats on earth, preferring dark, moist conditions. They can thrive in seemingly-hostile environments, such as the tundra. However, most members of the Kingdom Fungi grow on the forest floor where the dark and damp environment is rich in decaying debris from plants and animals. In these environments, fungi play a major role as decomposers and recyclers, making it possible for members of the other kingdoms to be supplied with nutrients and to live.

The food web would be incomplete without organisms that decompose organic matter. Some elements, such as nitrogen and phosphorus, are required in large quantities by biological systems; yet, they are not abundant in the environment. The action of fungi releases these elements from decaying matter, making them available to other living organisms. Trace elements present in low amounts in many habitats are essential for growth, but would remain tied up in rotting organic matter if fungi and bacteria did not return them to the environment via their metabolic activity.

Figure 24.2A.1: **Fungi as decomposers**: Fungi are an important part of ecosystem nutrient cycles. These bracket fungi growing on the side of a tree are the fruiting structures of a basidiomycete. They receive their nutrients through their hyphae, which invade and decay the tree trunk.

The ability of fungi to degrade many large and insoluble molecules is due to their mode of nutrition. As seen earlier, digestion precedes ingestion. Fungi produce a variety of exoenzymes to digest nutrients. These enzymes are either released into the substrate or remain bound to the outside of the fungal cell wall. Large molecules are broken down into small molecules, which are transported into the cell by a system of protein carriers embedded in the cell membrane. Because the

movement of small molecules and enzymes is dependent on the presence of water, active growth depends on a relatively-high percentage of moisture in the environment.

As saprobes, fungi help maintain a sustainable ecosystem for the animals and plants that share the same habitat. In addition to replenishing the environment with nutrients, fungi interact directly with other organisms in beneficial, but sometimes damaging, ways.

Figure 24.2*A*.1: **Fungi: beneficial & pathogenic**: Shelf fungi, so called because they grow on trees in a stack, attack and digest the trunk or branches of a tree. While some shelf fungi are found only on dead trees, others can parasitize living trees, causing eventual death. They are considered serious tree pathogens.

24.2B: Mutualistic Relationships with Fungi and Fungivores

Members of Kingdom Fungi form ecologically beneficial mutualistic relationships with cyanobateria, plants, and animals.

Learning Objectives

- Describe mutualistic relationships with fungi

Key Points

- Mutualistic relationships are those where both members of an association benefit; Fungi form these types of relationships with various other Kingdoms of life.
- Mycorrhiza, formed from an association between plant roots and primitive fungi, help increase a plant's nutrient uptake; in return, the plant supplies the fungi with photosynthesis products for their metabolic use.
- In lichen, fungi live in close proximity with photosynthetic cyanobateria; the algae provide fungi with carbon and energy while the fungi supplies minerals and protection to the algae.
- Mutualistic relationships between fungi and animals involves numerous insects; Arthropods depend on fungi for protection, while fungi receive nutrients in return and ensure a way to disseminate the spores into new environments.

Key Terms

- **mycorrhiza**: a symbiotic association between a fungus and the roots of a vascular plant
- **lichen**: any of many symbiotic organisms, being associations of fungi and algae; often found as white or yellow patches on old walls, etc.
- **thallus**: vegetative body of a fungus

Mutualistic Relationships

Symbiosis is the ecological interaction between two organisms that live together. However, the definition does not describe the quality of the interaction. When both members of the association benefit, the symbiotic relationship is called mutualistic. Fungi form mutualistic associations with many types of organisms, including cyanobacteria, plants, and animals.

Fungi & Plant Mutualism

Mycorrhiza, which comes from the Greek words "myco" meaning fungus and "rhizo" meaning root, refers to the association between vascular plant roots and their symbiotic fungi. About 90 percent of all plant species have mycorrhizal partners. In a mycorrhizal association, the fungal mycelia use their extensive network of hyphae and large surface area in contact with the soil to channel water and minerals from the soil into the plant, thereby increasing a plant's nutrient uptake. In exchange, the plant supplies the products of photosynthesis to fuel the metabolism of the fungus.

Mycorrhizae display many characteristics of primitive fungi: they produce simple spores, show little diversification, do not have a sexual reproductive cycle, and cannot live outside of a mycorrhizal association. There are a number of types of mycorrhizae. Ectomycorrhizae ("outside" mycorrhiza) depend on fungi enveloping the roots in a sheath (called a mantle) and a Hartig net of hyphae that extends into the roots between cells. The fungal partner can belong to the Ascomycota, Basidiomycota, or Zygomycota. In a second type, the Glomeromycete fungi form vesicular–arbuscular interactions with arbuscular mycorrhiza (sometimes called endomycorrhizae). In these mycorrhiza, the fungi form arbuscules that penetrate root cells and are the site of the metabolic exchanges between the fungus and the host plant. The arbuscules (from the Latin for "little trees") have a shrub-like appearance. Orchids rely on a third type of mycorrhiza. Orchids are epiphytes that form small seeds without much storage to sustain germination and growth. Their seeds will not germinate without a mycorrhizal partner (usually a Basidiomycete). After nutrients in the seed are depleted, fungal symbionts support the growth of the orchid by providing necessary carbohydrates and minerals. Some orchids continue to be mycorrhizal throughout their lifecycle.

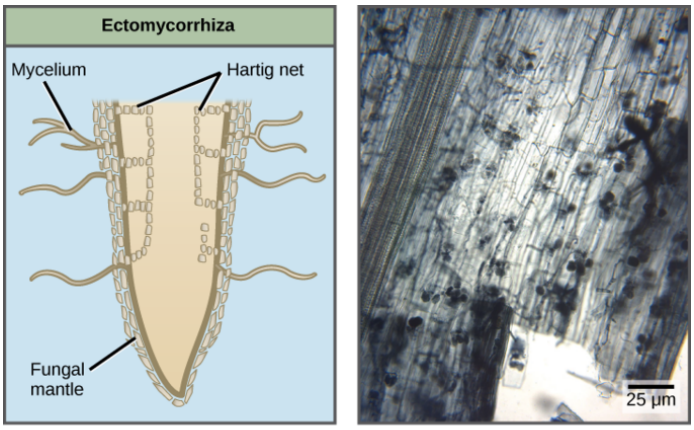

Figure 24.2B.1: **Mycorrhizal fungi**: (a) Ectomycorrhiza and (b) arbuscular mycorrhiza have different mechanisms for interacting with the roots of plants.

Lichens

Lichens display a range of colors and textures. They can survive in the most unusual and hostile habitats. They cover rocks, gravestones, tree bark, and the ground in the tundra where plant roots cannot penetrate. Lichens can survive extended periods of drought: they become completely desiccated and then rapidly become active once water is available again. Lichens fulfill many ecological roles, including acting as indicator species, which allow scientists to track the health of a habitat because of their sensitivity to air pollution.

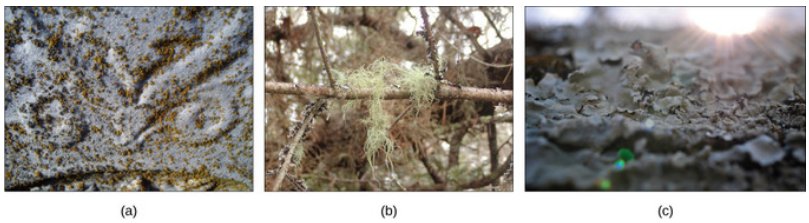

Figure 24.2B.1: **Lichen: fungi and cyanobateria**: Lichens have many forms. They may be (a) crust-like, (b) hair-like, or (c) leaf-like.

Lichens are not a single organism, but, rather, an example of a mutualism in which a fungus (usually a member of the Ascomycota or Basidiomycota phyla) lives in close contact with a photosynthetic organism (a eukaryotic alga or a prokaryotic cyanobacterium). Generally, neither the fungus nor the photosynthetic organism can survive alone outside of the symbiotic relationship. The body of a lichen, referred to as a thallus, is formed of hyphae wrapped around the photosynthetic partner. The photosynthetic organism provides carbon and energy in the form of carbohydrates. Some cyanobacteria fix nitrogen from the atmosphere, contributing nitrogenous compounds to the association. In return, the fungus supplies minerals and protection from dryness and excessive light by encasing the algae in its mycelium. The fungus also attaches the symbiotic organism to the substrate.

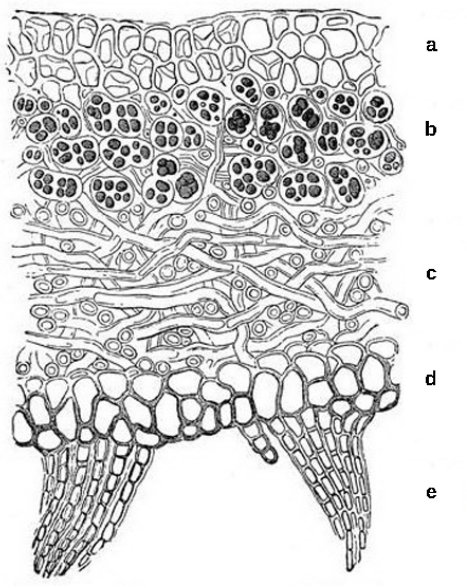

Figure 24.2B.1: **Thallus of lichen**: This cross-section of a lichen thallus shows the (a) upper cortex of fungal hyphae, which provides protection; the (b) algal zone where photosynthesis occurs, the (c) medulla of fungal hyphae, and the (d) lower cortex, which also provides protection and may have (e) rhizines to anchor the thallus to the substrate.

The thallus of lichens grows very slowly, expanding its diameter a few millimeters per year. Both the fungus and the alga participate in the formation of dispersal units for reproduction. Lichens produce soredia, clusters of algal cells surrounded by mycelia. Soredia are dispersed by wind and water and form new lichens.

Fungi & Animal Mutualism

Fungi have evolved mutualisms with numerous insects. Arthropods (jointed, legged invertebrates, such as insects) depend on the fungus for protection from predators and pathogens, while the fungus obtains nutrients and a way to disseminate spores into new environments. The association between species of Basidiomycota and scale insects is one example. The fungal mycelium covers and protects the insect colonies. The scale insects foster a flow of nutrients from the parasitized plant to the fungus. In a second example, leaf-cutting ants of Central and South America literally farm fungi. They cut disks of leaves from plants and pile them up in gardens. Fungi are cultivated in these disk gardens, digesting the cellulose in the leaves that the ants cannot break down. Once smaller sugar molecules are produced and consumed by the fungi, the fungi in turn become a meal for the ants. The insects also patrol their garden, preying on competing fungi. Both ants and fungi benefit from the association. The fungus receives a steady supply of leaves and freedom from competition, while the ants feed on the fungi they cultivate.

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24.3: Classifications of Fungi

Topic hierarchy

24.3A: Chytridiomycota: The Chytrids

24.3B: Zygomycota - The Conjugated Fungi

Zygomycota, a small group in the fungi kingdom, can reproduce asexually or sexually, in a process called conjugation.

24.3C: Ascomycota - The Sac Fungi

Most fungi belong to the Phylum Ascomycota, which uniquely forms of an ascus, a sac-like structure that contains haploid ascospores.

24.3D: Basidiomycota: The Club Fungi

24.3E: Deuteromycota - The Imperfect Fungi

Phylum Deuteromycota is a polyphyletic group of asexually-reproducing fungi that do not display a sexual phase; they are known as imperfect.

24.3F: Glomeromycota

Glomeromycetes are an important group of fungi that live in close symbiotic association with the roots of trees and plants.

24.3A: Chytridiomycota: The Chytrids

Chytrids are the most primitive group of fungi and the only group that possess gametes with flagella.

Learning Objectives

- Describe the ecology and reproduction of chytrids

Key Points

- The first recognizable chytrids appeared more than 500 million years ago during the late pre-Cambrian period.
- Like protists, chytrids usually live in aquatic environments, but some species live on land.
- Some chytrids are saprobes while others are parasites that may be harmful to amphibians and other animals.
- Chytrids reproduce both sexually and asexually, which leads to the production of zoospores.
- Chytrids have chitin in their cell walls; one unique group also has cellulose along with chitin.
- Chytrids are mostly unicellular, but multicellular organisms do exist.

Key Terms

- **chytridiomycete**: an organism of the phylum Chytridiomycota
- **zoospore**: an asexual spore of some algae and fungi
- **flagellum**: a flagellum is a lash-like appendage that protrudes from the cell body of certain prokaryotic and eukaryotic cells
- **coenocytic**: a multinucleate cell that can result from multiple nuclear divisions without their accompanying cytokinesis

Chytridiomycota: The Chytrids

The kingdom Fungi contains five major phyla, which were established according to their mode of sexual reproduction or use of molecular data. The Phylum Chytridiomycota (chytrids) is one of the five true phyla of fungi. There is only one class in the Phylum Chytridiomycota, the Chytridiomycetes. The chytrids are the simplest and most primitive Eumycota, or true fungi. The evolutionary record shows that the first, recognizable chytrids appeared during the late pre-Cambrian period, more than 500 million years ago. Like all fungi, chytrids have chitin in their cell walls, but one group of chytrids has both cellulose and chitin in the cell wall. Most chytrids are unicellular; a few form multicellular organisms and hyphae, which have no septa between cells (coenocytic). They reproduce both sexually and asexually; the asexual spores are called diploid zoospores. Their gametes are the only fungal cells known to have a flagellum.

The ecological habitat and cell structure of chytrids have much in common with protists. Chytrids usually live in aquatic environments, although some species live on land. Some species thrive as parasites on plants, insects, or amphibians, while others are saprobes. Some chytrids cause diseases in many species of amphibians, resulting in species decline and extinction. An example of a harmful parasitic chytrid is *Batrachochytrium dendrobatidis*, which is known to cause skin disease. Another chytrid species, *Allomyces*, is well characterized as an experimental organism. Its reproductive cycle includes both asexual and sexual phases. *Allomyces* produces diploid or haploid flagellated zoospores in a sporangium.

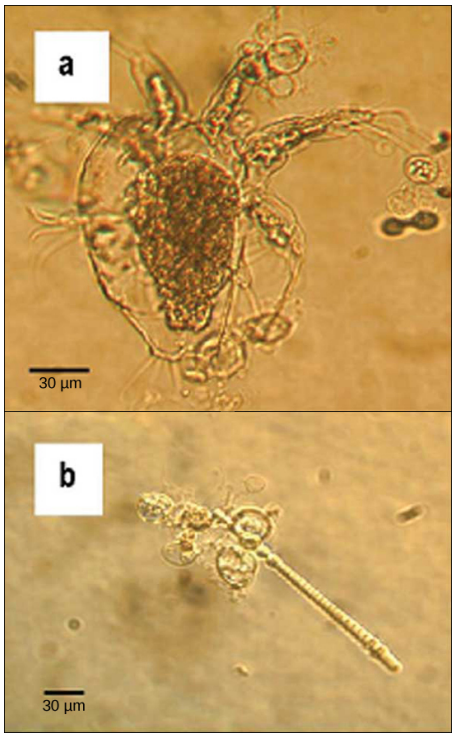

Figure 24.3A.1: **Parasitic chytrids**: The chytrid *Batrachochytrium dendrobatidis*is seen in these light micrographs as transparent spheres growing on (a) a freshwater arthropod and (b) algae. This chytrid causes skin diseases in many species of amphibians, resulting in species decline and extinction.

24.3B: Zygomycota - The Conjugated Fungi

LEARNING OBJECTIVES

- Describe the ecology and reproduction of Zygomycetes

The zygomycetes are a relatively small group in the fungi kingdom and belong to the Phylum Zygomycota. They include the familiar bread mold, *Rhizopus stolonifer*, which rapidly propagates on the surfaces of breads, fruits, and vegetables. They are mostly terrestrial in habitat, living in soil or on plants and animals. Most species are saprobes meaning they live off decaying organic material. Some are parasites of plants, insects, and small animals, while others form symbiotic relationships with plants. Zygomycetes play a considerable commercial role. The metabolic products of other species of *Rhizopus* are intermediates in the synthesis of semi-synthetic steroid hormones.

Zygomycetes have a thallus of coenocytic hyphae in which the nuclei are haploid when the organism is in the vegetative stage. The fungi usually reproduce asexually by producing sporangiospores. The black tips of bread mold, *Rhizopus stolonifer*, are the swollen sporangia packed with black spores. When spores land on a suitable substrate, they germinate and produce a new mycelium.

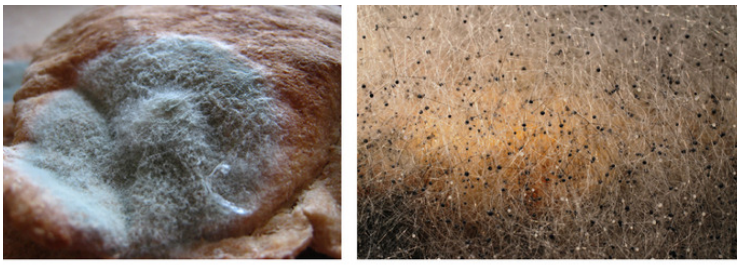

Figure $24.3B.1$: **Sporangia of bread mold**: Sporangia grow at the end of stalks, which appear as (a) white fuzz seen on this bread mold, *Rhizopus stolonifer*. The (b) tips of bread mold are the spore-containing sporangia.

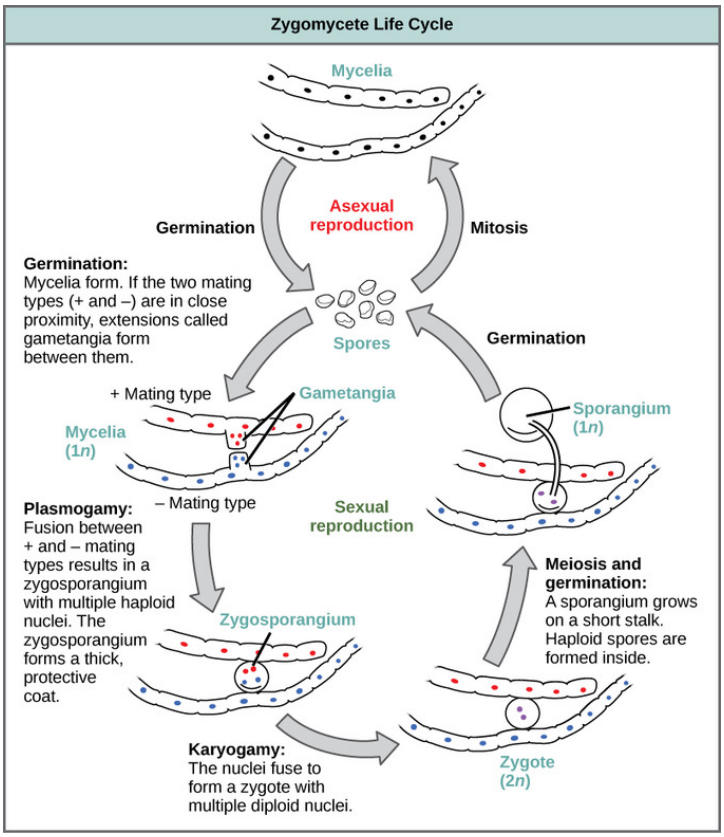

Figure 24.3B.1: **Zygomycete life cycle**: Zygomycetes have asexual and sexual life cycles. In the sexual life cycle, plus and minus mating types conjugate to form a zygosporangium.

Sexual reproduction starts when conditions become unfavorable. Two opposing mating strains (type + and type –) must be in close proximity for gametangia (singular: gametangium) from the hyphae to be produced and fuse, leading to karyogamy. The developing diploid zygospores have thick coats that protect them from desiccation and other hazards. They may remain dormant until environmental conditions become favorable. When the zygospore germinates, it undergoes meiosis and produces haploid spores, which will, in turn, grow into a new organism. This form of sexual reproduction in fungi is called conjugation (although it differs markedly from conjugation in bacteria and protists), giving rise to the name “conjugated fungi”.

Key Points

- Most zygomycota are saprobes, while a few species are parasites.
- Zygomycota usually reproduce asexually by producing sporangiospores.
- Zygomycota reproduce sexually when environmental conditions become unfavorable.
- To reproduce sexually, two opposing mating strains must fuse or conjugate, thereby, sharing genetic content and creating zygospores.
- The resulting diploid zygospores remain dormant and protected by thick coats until environmental conditions have improved.
- When conditions become favorable, zygospores undergo meiosis to produce haploid spores, which will eventually grow into a new organism.

Key Terms

- **zygomycete**: an organism of the phylum Zygomycota
- **karyogamy**: the fusion of two nuclei within a cell
- **zygospore**: a spore formed by the union of several zoospores
- **conjugation**: the temporary fusion of organisms, especially as part of sexual reproduction

24.3C: Ascomycota - The Sac Fungi

LEARNING OBJECTIVES

- Describe the ecology and the reproduction of Ascomycetes

The majority of known fungi belong to the Phylum Ascomycota, which is characterized by the formation of an ascus (plural, asci), a sac-like structure that contains haploid ascospores. Many ascomycetes are of commercial importance. Some play a beneficial role, such as the yeasts used in baking, brewing, and wine fermentation, plus truffles and morels, which are held as gourmet delicacies. *Aspergillus oryzae* is used in the fermentation of rice to produce sake. Other ascomycetes parasitize plants and animals, including humans. For example, fungal pneumonia poses a significant threat to AIDS patients who have a compromised immune system. Ascomycetes not only infest and destroy crops directly, they also produce poisonous secondary metabolites that make crops unfit for consumption. Filamentous ascomycetes produce hyphae divided by perforated septa, allowing streaming of cytoplasm from one cell to the other. Conidia and asci, which are used respectively for asexual and sexual reproductions, are usually separated from the vegetative hyphae by blocked (non-perforated) septa.

Asexual reproduction is frequent and involves the production of conidiophores that release haploid conidiospores. Sexual reproduction starts with the development of special hyphae from either one of two types of mating strains. The “male” strain produces an antheridium (plural: antheridia) and the “female” strain develops an ascogonium (plural: ascogonia). At fertilization, the antheridium and the ascogonium combine in plasmogamy without nuclear fusion. Special ascogenous hyphae arise, in which pairs of nuclei migrate: one from the “male” strain and one from the “female” strain. In each ascus, two or more haploid ascospores fuse their nuclei in karyogamy. During sexual reproduction, thousands of asci fill a fruiting body called the ascocarp. The diploid nucleus gives rise to haploid nuclei by meiosis. The ascospores are then released, germinate, and form hyphae that are disseminated in the environment and start new mycelia.

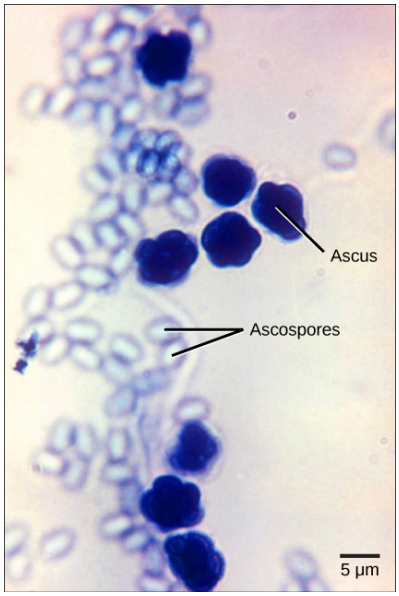

Figure 24.3C.1: **Release of ascospores**: The bright field light micrograph shows ascospores being released from asci in the fungus Talaromyces flavus var. flavus.

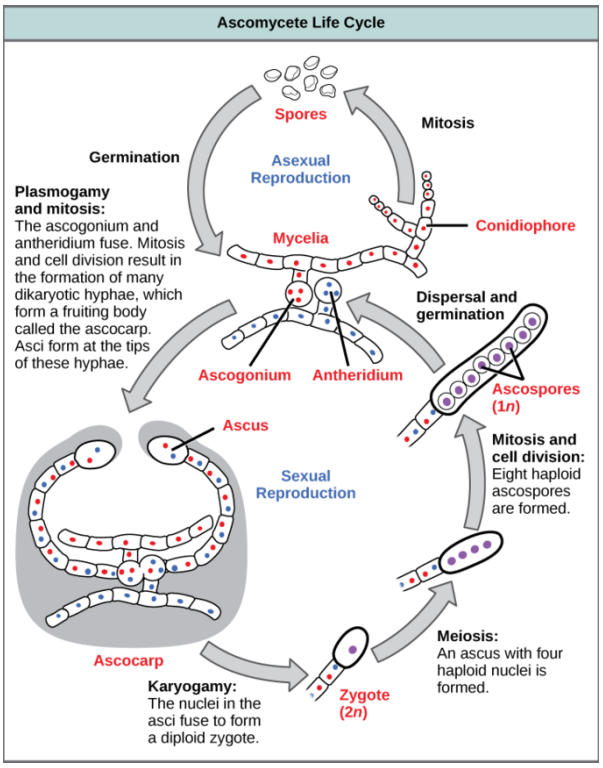

Figure 24.3C.1: **Lifecycle of an ascomycete**: The lifecycle of an ascomycete is characterized by the production of asci during the sexual phase. The haploid phase is the predominant phase of the life cycle.

Key Points

- Ascomycota fungi are the yeasts used in baking, brewing, and wine fermentation, plus delicacies such as truffles and morels.
- Ascomycetes are filamentous and produce hyphae divided by perforated septa.
- Ascomycetes frequently reproduce asexually which leads to the production of conidiophores that release haploid conidiospores.
- Two types of mating strains, a “male” strain which produces an antheridium and a “female” strain which develops an ascogonium, are required for sexual reproduction.
- The antheridium and the ascogonium combine in plasmogamy at the time of fertilization, followed by nuclei fusion in the asci.
- In the ascocarp, a fruiting body, thousands of asci undergo meiosis to generate haploid ascospores ready to be released to the world.

Key Terms

- **plasmogamy**: stage of sexual reproduction joining the cytoplasm of two parent mycelia without the fusion of nuclei
- **Ascomycota**: a taxonomic division within the kingdom Fungi; those fungi that produce spores in a microscopic sporangium called an ascus
- **ascus**: a sac-shaped cell present in ascomycete fungi; it is a reproductive cell in which meiosis and an additional cell division produce eight spores
- **ascospore**: a sexually-produced spore from the ascus of an Ascomycetes fungus
- **conidia**: asexual, non-motile spores of a fungus, named after the Greek word for dust, conia; also known as conidiospores and mitospores
- **antheridia**: a haploid structure or organ producing and containing male gametes (called antherozoids or sperm) present in lower plants like mosses and ferns, primitive vascular psilotophytes, and fungi
- **ascogonium**: a haploid structure or organ producing and containing female gametes in certain Ascomycota fungi
- **ascocarp**: the sporocarp of an ascomycete, typically bowl-shaped

- **ascomycete**: any fungus of the phylum Ascomycota, characterized by the production of a sac, or ascus, which contains non-motile spores

24.3D: Basidiomycota: The Club Fungi

The basidiomycota are mushroom-producing fungi with developing, club-shaped fruiting bodies called basidia on the gills under its cap.

Learning Objectives

- Describe the ecology and reproduction of the Basidiomycota

Key Points

- The majority of edible fungi belong to the Phylum Basidiomycota.
- The basidiomycota includes shelf fungus, toadstools, and smuts and rusts.
- Unlike most fungi, basidiomycota reproduce sexually as opposed to asexually.
- Two different mating strains are required for the fusion of genetic material in the basidium which is followed by meiosis producing haploid basidiospores.
- Mycelia of different mating strains combine to produce a secondary mycelium that contains haploid basidiospores in what is called the dikaryotic stage, where the fungi remains until a basidiocarp (mushroom) is generated with the developing basidia on the gills under its cap.

Key Terms

- **basidiocarp**: a fruiting body that protrudes from the ground, known as a mushroom, which has a developing basidia on the gills under its cap
- **basidiomycete**: a fungus of the phylum Basidiomycota, which produces sexual spores on a basidium
- **Basidiomycota**: a taxonomic division within the kingdom Fungi: 30,000 species of fungi that produce spores from a basidium
- **basidium**: a small structure, shaped like a club, found in the Basidiomycota phylum of fungi, that bears four spores at the tips of small projections
- **basidiospore**: a sexually-reproductive spore produced by fungi of the phylum Basidiomycota

Basidiomycota: The Club Fungi

The fungi in the Phylum Basidiomycota are easily recognizable under a light microscope by their club-shaped fruiting bodies called basidia (singular, basidium), which are the swollen terminal cell of a hypha. The basidia, which are the reproductive organs of these fungi, are often contained within the familiar mushroom, commonly seen in fields after rain, on the supermarket shelves, and growing on your lawn. These mushroom-producing basidiomyces are sometimes referred to as “gill fungi” because of the presence of gill-like structures on the underside of the cap. The “gills” are actually compacted hyphae on which the basidia are borne. This group also includes shelf fungus, which cling to the bark of trees like small shelves. In addition, the basidiomycota includes smuts and rusts, which are important plant pathogens, and toadstools. Most edible fungi belong to the Phylum Basidiomycota; however, some basidiomycetes produce deadly toxins. For example, *Cryptococcus neoformans* causes severe respiratory illness.

Figure 24.3D.1: **Fruiting bodies of a basidiomycete**: The fruiting bodies of a basidiomycete form a ring in a meadow, commonly called "fairy ring." The best-known fairy ring fungus has the scientific name *Marasmius oreades*. The body of this fungus, its mycelium, is underground and grows outward in a circle. As it grows, the mycelium depletes the soil of nitrogen, causing the mycelia to grow away from the center, leading to the "fairy ring" of fruiting bodies where there is adequate soil nitrogen.

The lifecycle of basidiomycetes includes alternation of generations. Spores are generally produced through sexual reproduction, rather than asexual reproduction. The club-shaped basidium carries spores called basidiospores. In the basidium, nuclei of two different mating strains fuse (karyogamy), giving rise to a diploid zygote that then undergoes meiosis. The haploid nuclei migrate into basidiospores, which germinate and generate monokaryotic hyphae. The mycelium that results is called a primary mycelium. Mycelia of different mating strains can combine and produce a secondary mycelium that contains haploid nuclei of two different mating strains. This is the dikaryotic stage of the basidiomyces lifecyle and it is the dominant stage. Eventually, the secondary mycelium generates a basidiocarp, which is a fruiting body that protrudes from the ground; this is what we think of as a mushroom. The basidiocarp bears the developing basidia on the gills under its cap.

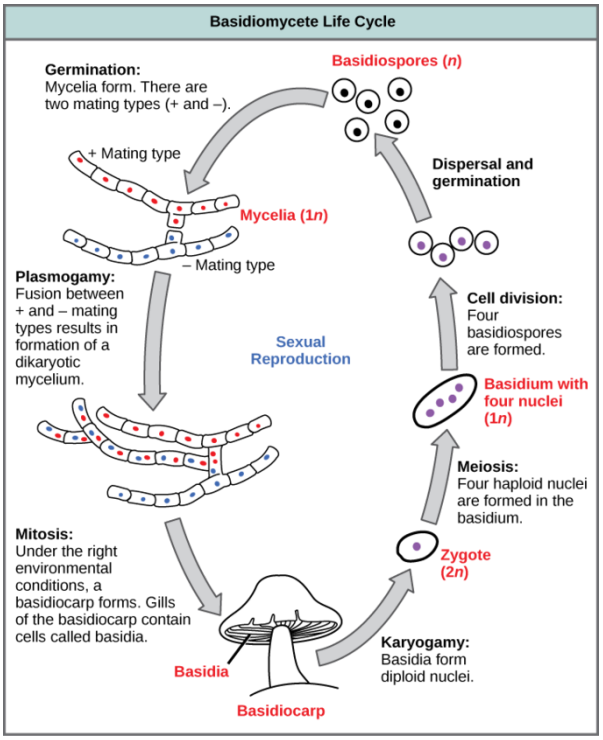

Figure 24.3D.1: Lifecycle of a basidiomycete: The lifecycle of a basidiomycete alternates generation with a prolonged stage in which two nuclei (dikaryon) are present in the hyphae.

24.3E: Deuteromycota - The Imperfect Fungi

LEARNING OBJECTIVES

- Describe the ecology and reproduction of the Deuteromycota

Imperfect fungi are those that do not display a sexual phase. They are classified as belonging to the form Phylum Deuteromycota. Deuteromycota is a polyphyletic group where many species are more closely related to organisms in other phyla than to each other; hence it cannot be called a true phylum and must, instead, be given the name form phylum. Since they do not possess the sexual structures that are used to classify other fungi, they are less well described in comparison to other divisions. Most members live on land, with a few aquatic exceptions. They form visible mycelia with a fuzzy appearance and are commonly known as mold. Molecular analysis shows that the closest group to the deuteromycetes is the ascomycetes. In fact, some species, such as *Aspergillus*, which were once classified as imperfect fungi, are now classified as ascomycetes.

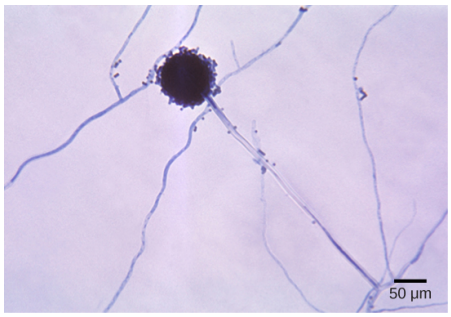

Figure 24.3E.1: **Example of an imperfect fungus**: Aspergillus niger is an imperfect fungus commonly found as a food contaminant. The spherical structure in this light micrograph is a conidiophore.

Reproduction of Deuteromycota is strictly asexual, occuring mainly by production of asexual conidiospores. Some hyphae may recombine and form heterokaryotic hyphae. Genetic recombination is known to take place between the different nuclei.

Imperfect fungi have a large impact on everyday human life. The food industry relies on them for ripening some cheeses. The blue veins in Roquefort cheese and the white crust on Camembert are the result of fungal growth. The antibiotic penicillin was originally discovered on an overgrown Petri plate on which a colony of *Penicillium* fungi killed the bacterial growth surrounding it. Many imperfect fungi cause serious diseases, either directly as parasites (which infect both plants and humans), or as producers of potent toxic compounds, as seen in the aflatoxins released by fungi of the genus *Aspergillus*.

Key Points

- Deuteromycota do not possess the sexual structures that are used to classify other fungi.
- Most deuteromycota live on land; they form visible mycelia with a fuzzy appearance called mold.
- Recombination of genetic material is known to take place between the different nuclei after some hyphae recombine.

Key Terms

- **deuteromycete**: an organism of the phylum Deuteromycota
- **Deuteromycota**: a taxonomic morphological group within the kingdom Fungi; the fungi have no sexual reproduction
- **polyphyletic**: having multiple ancestral sources; referring to a taxon that does not contain the most recent common ancestor of its members
- **conidiospore**: a unicellular spore produced asexually by a fungus

24.3F: Glomeromycota

LEARNING OBJECTIVES

- Describe the ecology and reproduction of Glomeromycetes

In the kingdom Fungi, the Glomeromycota is a newly-established phylum comprised of about 230 species that live in close association with the roots of trees and plants. Fossil records indicate that trees and their root symbionts share a long evolutionary history. It appears that most members of this family form arbuscular mycorrhizae: the hyphae interact with the root cells forming a mutually-beneficial association where the plants supply the carbon source and energy in the form of carbohydrates to the fungus while the fungus supplies essential minerals from the soil to the plant. This association is termed biotrophic. The Glomeromycota species that have arbuscular mycorrhizal are terrestrial and widely distributed in soils worldwide where they form symbioses with the roots of the majority of plant species. They can also be found in wetlands, including salt-marshes, and are associated with epiphytic plants.

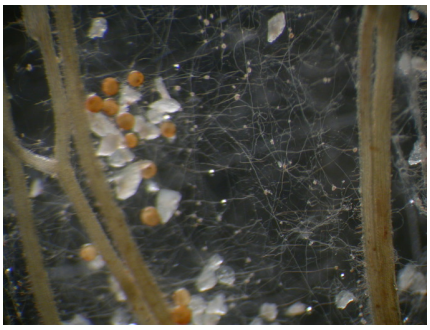

Figure $24.3F.1$: **Glyomeromycetes and tree roots**: This image illustrates the bitrophic relationship between a glomeromycota (Gigaspora margarita) and the roots of a plant (Lotus corniculatus).

The glomeromycetes do not reproduce sexually and cannot survive without the presence of plant roots. They have coenocytic hyphae and reproduce asexually, producing glomerospores. The biochemical and genetic characterization of the Glomeromycota has been hindered by their biotrophic nature, which impedes laboratory culturing. This obstacle was eventually surpassed with the use of root cultures. With the advent of molecular techniques, such as gene sequencing, the phylogenetic classification of Glomeromycota has become clearer. The first mycorrhizal gene to be sequenced was the small-subunit ribosomal RNA (SSU rRNA). This gene is highly conserved and commonly used in phylogenetic studies so it was isolated from spores of each taxonomic group. Using a molecular clock approach based on the substitution rates of SSU sequences, scientists were able to estimate the time of divergence of the fungi. This analysis shows that all glomeromycetes probably descended from a common ancestor 462 and 353 million years ago, making them a monophyletic lineage. A long-held theory is that Glomeromycota were instrumental in the colonization of land by plants.

Key Points

- Most glomeromycetes form arbuscular mycorrhizae, a type of symbiotic relationship between a fungus and plant roots; the plants supply a source of energy to the fungus while the fungus supplies essential minerals to the plant.
- Glomeromycota that have arbuscular mycorrhizal are mostly terrestrial, but can also be found in wetlands.
- The glomeromycetes reproduce asexually by producing glomerospores and cannot survive without the presence of plant roots.
- DNA analysis shows that all glomeromycetes probably descended from a common ancestor 462 and 353 million years ago.
- The classification of fungi as Glomeromycota has been redefined with adoption of molecular techniques.

Key Terms

- **biotrophic**: describing a parasite that needs its host to stay alive
- **arbuscular mycorrhizae**: a type of symbiotic relationship between a fungus and the roots of a plant where the plants supply a source of energy to the fungus while the fungus supplies essential minerals to the plant
- **glomeromycete**: an organism of the phylum Glomeromycota

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24.4: Fungal Parasites and Pathogens

Topic hierarchy

- 24.4A: Fungi as Plant, Animal, and Human Pathogens

24.4A: Fungi as Plant, Animal, and Human Pathogens

From crop and food spoilage to severe infections in animal species, fungal parasites and pathogens are wide spread and difficult to treat.

Learning Objectives

- Give examples of fungi that are plant and animal parasites and pathogens

Key Points

- In plants, fungi can destroy plant tissue directly or through the production of potent toxins, which usually ends in host death and can even lead to ergotism in animals like humans.
- During mycosis, fungi, like dermatophytes, successfully attack hosts directly by colonizing and destroying their tissues.
- Examples of fungal parasites and pathogens in animals that cause mycoses include *Batrachochytrium dendrobatidis*, *Geomyces destructans*, and *Histoplasma capsulatum*.
- Systemic mycoses, such as valley fever, Histoplasmosis, or pulmonary disease, are fungal diseases that spread to internal organs and commonly enter the body through the respiratory system.
- Opportunistic mycoses, fungal infections that are common in all environments, mainly take advantage of individuals who have a compromised immune system, such as AIDS patients.
- Fungi can also cause mycetismus, a disease caused by the ingestion of toxic mushrooms that leads to poisoning.

Key Terms

- **mycosis**: a fungal disease caused by infection and direct damage
- **dermatophyte**: a parasitic fungus that secretes extracellular enzymes that break down keratin, causing infections the skin, such as jock itch and athlete's foot
- **aflatoxin**: toxic, carcinogenic compounds released by fungi of the genus Aspergillus; contaminate nut and grain harvests
- **ergot**: any fungus in the genus Claviceps which are parasitic on grasses

Fungal Parasites and Pathogens

The production of sufficient good-quality crops is essential to human existence. Plant diseases have ruined crops, bringing widespread famine. Many plant pathogens are fungi that cause tissue decay and eventual death of the host. In addition to destroying plant tissue directly, some plant pathogens spoil crops by producing potent toxins. Fungi are also responsible for food spoilage and the rotting of stored crops. For example, the fungus *Claviceps purpurea* causes ergot, a disease of cereal crops (especially of rye). Although the fungus reduces the yield of cereals, the effects of the ergot's alkaloid toxins on humans and animals are of much greater significance. In animals, the disease is referred to as ergotism. The most common signs and symptoms are convulsions, hallucinations, gangrene, and loss of milk in cattle. The active ingredient of ergot is lysergic acid, which is a precursor of the drug LSD. Smuts, rusts, and powdery or downy mildew are other examples of common fungal pathogens that affect crops.

Figure 24.4A.1: **Fungal pathogens**: Some fungal pathogens include (a) green mold on grapefruit, (b) powdery mildew on a zinnia, (c) stem rust on a sheaf of barley, and (d) grey rot on grapes.

Aflatoxins are toxic, carcinogenic compounds released by fungi of the genus *Aspergillus*. Periodically, harvests of nuts and grains are tainted by aflatoxins, leading to massive recall of produce. This sometimes ruins producers and causes food shortages in developing countries.

Animal and Human Parasites and Pathogens

Fungi can affect animals, including humans, in several ways. A mycosis is a fungal disease that results from infection and direct damage. Fungi attack animals directly by colonizing and destroying tissues. Mycotoxicosis is the poisoning of humans (and other animals) by foods contaminated by fungal toxins (mycotoxins). Mycetismus describes the ingestion of preformed toxins in poisonous mushrooms. In addition, individuals who display hypersensitivity to molds and spores develop strong and dangerous allergic reactions. Fungal infections are generally very difficult to treat because, unlike bacteria, fungi are eukaryotes. Antibiotics only target prokaryotic cells, whereas compounds that kill fungi also harm the eukaryotic animal host.

Many fungal infections are superficial; that is, they occur on the animal's skin. Termed cutaneous ("skin") mycoses, they can have devastating effects. For example, the decline of the world's frog population in recent years may be caused by the fungus *Batrachochytrium dendrobatidis*, which infects the skin of frogs and presumably interferes with gaseous exchange. Similarly, more than a million bats in the United States have been killed by white-nose syndrome, which appears as a white ring around the mouth of the bat. It is caused by the cold-loving fungus *Geomyces destructans*, which disseminates its deadly spores in caves where bats hibernate. Mycologists are researching the transmission, mechanism, and control of *G. destructans* to stop its spread.

Fungi that cause the superficial mycoses of the epidermis, hair, and nails rarely spread to the underlying tissue. These fungi are often misnamed "dermatophytes", from the Greek words dermis meaning skin and phyte meaning plant, although they are not plants. Dermatophytes are also called "ringworms" because of the red ring they cause on skin. They secrete extracellular enzymes that break down keratin (a protein found in hair, skin, and nails), causing conditions such as athlete's foot and jock itch. These conditions are usually treated with over-the-counter topical creams and powders; they are easily cleared. More persistent superficial mycoses may require prescription oral medications.

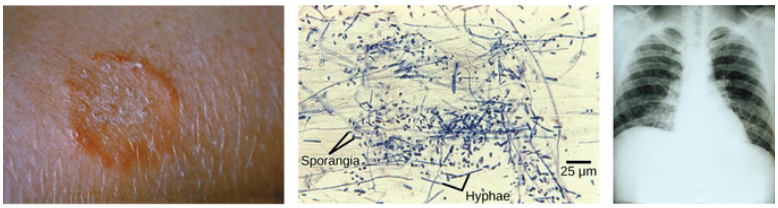

Figure 24.4A.1: **Mycosis infection**: (a) Ringworm presents as a red ring on skin; (b) *Trichophyton violaceum*, shown in this bright field light micrograph, causes superficial mycoses on the scalp; (c) *Histoplasma capsulatum* is an ascomycete that infects airways and causes symptoms similar to influenza.

Systemic mycoses spread to internal organs, most commonly entering the body through the respiratory system. For example, coccidioidomycosis (valley fever) is commonly found in the southwestern United States where the fungus resides in the dust. Once inhaled, the spores develop in the lungs and cause symptoms similar to those of tuberculosis. Histoplasmosis is caused by the dimorphic fungus *Histoplasma capsulatum*. It also causes pulmonary infections. In rarer cases, it causes swelling of the membranes of the brain and spinal cord. Treatment of these and many other fungal diseases requires the use of antifungal medications that have serious side effects.

Opportunistic mycoses are fungal infections that are either common in all environments or are part of the normal biota. They mainly affect individuals who have a compromised immune system. Patients in the late stages of AIDS suffer from opportunistic mycoses that can be life threatening. The yeast *Candida sp.*, a common member of the natural biota, can grow unchecked and infect the vagina or mouth (oral thrush) if the pH of the surrounding environment, the person's immune defenses, or the normal population of bacteria are altered.

Mycetismus can occur when poisonous mushrooms are eaten. It causes a number of human fatalities during mushroom-picking season. Many edible fruiting bodies of fungi resemble highly-poisonous relatives. Amateur mushroom hunters are cautioned to carefully inspect their harvest and avoid eating mushrooms of doubtful origin.

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: Importance of Fungi in Human Life

Topic hierarchy

- 24.5A: Importance of Fungi in Human Life

24.5A: Importance of Fungi in Human Life

Fungi play important roles in many aspects of human life, including medicine, food, and farming.

Learning Objectives

- Explain the important roles fungi play in human life

Key Points

- The majority of grasses and trees require a mycorrhizal relationship with fungi to survive.
- Yeasts have been used for thousands of years in the production of beer, wine, and bread.
- Fungi not only directly produce substances that humans use as medicine, but they are also versatile tools in the vast field of medical research.
- Some fungi attack insects and, therefore, can be used as natural pesticides.

Key Terms

- **inoculant**: the active material used in an inoculation
- **ergot**: any fungus in the genus Claviceps which are parasitic on grasses
- **immunosuppressant**: capable of immunosuppression, or the reduction of immune system efficacy

Importance of Fungi in Human Life

Although we often think of fungi as organisms that cause disease and rot food, fungi are important to human life on many levels. They influence the well-being of human populations on a large scale because they are part of the nutrient cycle in ecosystems. They also have other ecosystem uses, such as pesticides.

Biological Insecticides

As animal pathogens, fungi help to control the population of damaging pests. These fungi are very specific to the insects they attack; they do not infect animals or plants. Fungi are currently under investigation as potential microbial insecticides, with several already on the market. For example, the fungus *Beauveria bassiana* is a pesticide being tested as a possible biological control agent for the recent spread of emerald ash borer.

Figure 24.5*A*. 1: **Fungal Insecticides**: The emerald ash borer is an insect that attacks ash trees. It is in turn parasitized by a pathogenic fungus that holds promise as a biological insecticide. The parasitic fungus appears as white fuzz on the body of the insect.

Farming

The mycorrhizal relationship between fungi and plant roots is essential for the productivity of farm land. Without the fungal partner in root systems, 80–90 percent of trees and grasses would not survive. Mycorrhizal fungal inoculants are available as soil additives from gardening supply stores and are promoted by supporters of organic agriculture.

Food

Fungi figure prominently in the human diet. Morels, shiitake mushrooms, chanterelles, and truffles are considered delicacies. The meadow mushroom, *Agaricus campestris*, appears in many dishes. Molds of the genus *Penicillium*ripen many cheeses. They originate in the natural environment such as the caves of Roquefort, France, where wheels of sheep milk cheese are stacked to capture the molds responsible for the blue veins and pungent taste of the cheese.

Figure 24.5A.1: **Morel mushroom**: The morel mushroom is an ascomycete much appreciated for its delicate taste.

Fermentation of grains to produce beer and of fruits to produce wine is an ancient art that humans in most cultures have practiced for millennia. Ancient humans acquired wild yeasts from the environment and used them to ferment sugars into CO_2 and ethanol under anaerobic conditions. It is now possible to purchase isolated strains of wild yeasts from different wine-making regions. Louis Pasteur was instrumental in developing a reliable strain of brewer's yeast, *Saccharomyces cerevisiae*, for the French brewing industry in the late 1850s.

Saccharomyces cerevisiae, also know as baker's yeast, is an important ingredient in bread, a food that has been considered a staple of human life for thousands of years. Before isolated yeast became available in modern times, humans simply let the dough collect yeast from the air and rise over a period of hours or days. A small piece of this leavened dough was saved and used as a starter (source of the same yeast) for the next batch, much in the same way sourdough bread is made today.

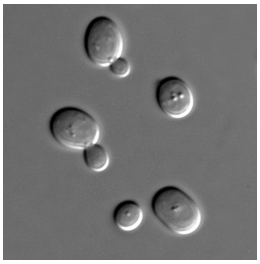

Figure 24.5A.1: **Saccharomyces cerevisiae**: The yeast Saccharomyces cerevisiae is approximately 5 µm in diameter and is important for the production of wine, beer, and bread. The yeast also has many applications in medical research.

Medicine

Many secondary metabolites of fungi are of great commercial importance. Fungi naturally produce antibiotics to kill or inhibit the growth of bacteria, limiting their competition in the natural environment. Important antibiotics, such as penicillin and the cephalosporins, can be isolated from fungi. Valuable drugs isolated from fungi include the immunosuppressant drug cyclosporine (which reduces the risk of rejection after organ transplant), the precursors of steroid hormones, and ergot alkaloids used to stop bleeding. Psilocybin is a compound found in fungi such as *Psilocybe*

semilanceata and *Gymnopilus junonius*, which have been used for their hallucinogenic properties by various cultures for thousands of years.

As simple eukaryotic organisms, fungi are important model research organisms. Many advances in modern genetics were achieved by the use of the red bread mold *Neurospora crassa*. Additionally, many important genes originally discovered in *S. cerevisiae* served as a starting point in discovering analogous human genes. As a eukaryotic organism, the yeast cell produces and modifies proteins in a manner similar to human cells, as opposed to the bacterium *Escherichia coli*, which lacks the internal membrane structures and enzymes to tag proteins for export. This makes yeast a much better organism for use in recombinant DNA technology experiments. Like bacteria, yeasts grow easily in culture, have a short generation time, and are amenable to genetic modification.

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CHAPTER OVERVIEW

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25.1: Early Plant Life

Topic hierarchy

- 25.1A: Early Plant Life
- 25.1B: Evolution of Land Plants
- 25.1C: Plant Adaptations to Life on Land
- 25.1D: Sporophytes and Gametophytes in Seedless Plants
- 25.1E: Structural Adaptations for Land in Seedless Plants
- 25.1F: The Major Divisions of Land Plants

25.1A: Early Plant Life

A diverse array of seedless plants still populate and thrive in the world today, particularly in moist environments.

Learning Objectives

- Describe the pervasiveness of seedless plants during the history of the kingdom Plantae

Key Points

- Non-vascular seedless plants, or bryophytes, are the group of plants that are the closest extant relative of early terrestrial plants.
- The vast majority of terrestrial plants today are seed plants, which tend to be better adapted to the arid land environment.
- Seedless plants are classified into three main catagories: green algae, seedless non- vascular plants, and seedless vascular plants.

Key Terms

- **vascular plant**: any plant possessing vascular tissue (xylem and phloem), including ferns, conifers, and flowering plants
- **bryophyte**: seedless, nonvascular plants that are the closest extant relative of early terrestrial plants

Introduction to Early Plant Life

An incredible variety of seedless plants populates the terrestrial landscape. Mosses may grow on a tree trunk and horsetails may display their jointed stems and spindly leaves across the forest floor. Today, however, seedless plants represent only a small fraction of the plants in our environment. The kingdom Plantae constitutes a large and varied group of organisms with more than 300,000 species of cataloged plants. Of these, more than 260,000 are seed plants. However, three hundred million years ago, seedless plants dominated the landscape and grew in the enormous swampy forests of the Carboniferous period. Their decomposition created large deposits of coal that we mine today.

Figure 25.1*A*. 1: **Horsetails are seedless plants**: Seedless plants, like these horsetails (*Equisetum*sp.), thrive in damp, shaded environments under a tree canopy where dryness is rare.

Current evolutionary thought holds that all plants, green algae as well as land dwellers, are monophyletic; that is, they are descendants of a single common ancestor. The evolutionary transition from water to land imposed severe constraints on plants. They had to develop strategies: to avoid drying out, to disperse reproductive cells in air, for structural support, and for capturing and filtering sunlight. While seed plants developed adaptations that allowed them to populate even the most arid habitats on Earth, full independence from water did not happen in all plants. Most seedless plants still require a moist environment.

Seedless plants are classified into three main categories: green algae, seedless non-vascular plants, and seedless vascular plants. Seedless non-vascular plants (bryophytes), such as mosses, are the group of plants that are the closest extant relative of early terrestrial plants. Seedless vascular plants include horsetails and ferns.

25.1B: Evolution of Land Plants

The geologic periods of the Paleozoic are marked by changes in the plant life that inhabited the earth.

Learning Objectives

- Summarize the development of adaptations in land plants

Key Points

- Land plants first appeared during the Ordovician period, more than 500 million years ago.
- The evolution of plants occurred by a stepwise development of physical structures and reproductive mechanisms such as vascular tissue, seed production, and flowering.
- Paleobotonists trace the evolution of plant morphology through a study of the fossil record in the context of the surrounding geological sediments.

Key Terms

- **Paleobotany**: the branch of paleontology or paleobiology dealing with the recovery and identification of plant remains from geological contexts
- **mycorrhiza**: a symbiotic association between a fungus and the roots of a vascular plant

Evolution of Land Plants

No discussion of the evolution of plants on land can be undertaken without a brief review of the timeline of the geological eras. The early era, known as the Paleozoic, is divided into six periods. It starts with the Cambrian period, followed by the Ordovician, Silurian, Devonian, Carboniferous, and Permian. The major event to mark the Ordovician, more than 500 million years ago, was the colonization of land by the ancestors of modern land plants. Fossilized cells, cuticles, and spores of early land plants have been dated as far back as the Ordovician period in the early Paleozoic era. The evolution of plants occurred by a gradual development of novel structures and reproduction mechanisms. Embryo protection developed prior to the development of vascular plants which, in turn, evolved before seed plants and flowering plants. The oldest-known vascular plants have been identified in deposits from the Devonian. One of the richest sources of information is the Rhynie chert, a sedimentary rock deposit found in Rhynie, Scotland, where embedded fossils of some of the earliest vascular plants have been identified.

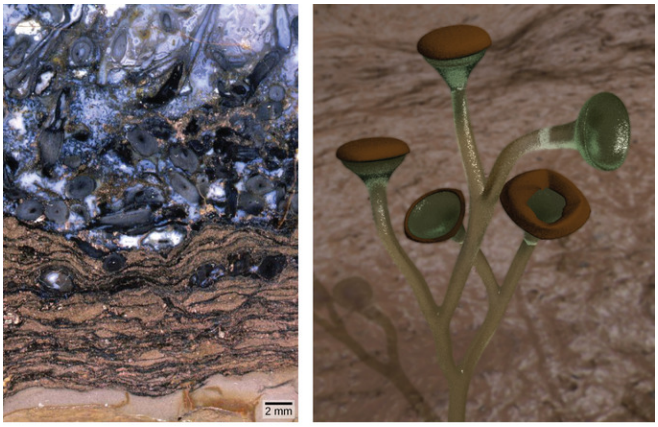

Figure $25.1B.1$: **The Rhynie chert sedimentary rock deposit**: This Rhynie chert contains fossilized material from vascular plants. The area inside the circle contains bulbous underground stems called corms and root-like structures called rhizoids.

image

Gradual evolution of land plants: The adaptation of plants to life on land occurred gradually through the stepwise development of physical structures and reproduction mechanisms

How organisms acquired traits that allow them to colonize new environments, and how the contemporary ecosystem is shaped, are fundamental questions of evolution. Paleobotany (the study of extinct plants) addresses these questions through the analysis of fossilized specimens retrieved from field studies, reconstituting the morphology of organisms that disappeared long ago. Paleobotanists trace the evolution of plants by following the modifications in plant morphology, which sheds light on the connection between existing plants by identifying common ancestors that display the same traits. This field seeks to find transitional species that bridge gaps in the path to the development of modern organisms. Paleobotanists collect fossil specimens in the field and place them in the context of the geological sediments and other fossilized organisms surrounding them.

Paleobotanists distinguish between extinct species, as fossils, and extant species, which are still living. The extinct vascular plants, classified as zosterophylls and trimerophytes, most probably lacked true leaves and roots, forming low vegetation mats similar in size to modern-day mosses, although some trimetophytes could reach one meter in height. The later genus *Cooksonia*, which flourished during the Silurian, has been extensively studied from well-preserved examples. Imprints of *Cooksonia* show slender, branching stems ending in what appear to be sporangia. From the recovered specimens, it is not possible to establish for certain whether *Cooksonia*possessed vascular tissues. Fossils indicate that by the end of the Devonian period, ferns, horsetails, and seed plants populated the landscape, giving rising to trees and forests. This luxuriant vegetation helped enrich the atmosphere in oxygen, making it easier for air-breathing animals to colonize dry land. Plants also established early symbiotic relationships with fungi, creating mycorrhizae: a relationship in which the fungal network of filaments increases the efficiency of the plant root system. The plants provide the fungi with byproducts of photosynthesis.

25.1C: Plant Adaptations to Life on Land

Plants adapted to the dehydrating land environment through the development of new physical structures and reproductive mechanisms.

Learning Objectives

- Discuss how lack of water in the terrestrial environment led to significant adaptations in plants

Key Points

- While some plants remain dependent on a moist and humid environment, many have adapted to a more arid climate by developing tolerance or resistance to drought conditions.
- Alternation of generations describes a life cycle in which an organism has both haploid (1n) and diploid (2n) multicellular stages, although in different species the haploid or diploid stage can be dominant.
- The life on land presents significant challenges for plants, including the potential for desiccation, mutagenic radiation from the sun, and a lack of buoyancy from the water.

Key Terms

- **desiccation tolerance**: the ability of an organism to withstand or endure extreme dryness, or drought-like condition
- **alternation of generation**: the life cycle of plants with a multicellular sporophyte, which is diploid, that alternates with a multicellular gametophyte, which is haploid

Plant Adaptations to Life on Land

As organisms adapted to life on land, they had to contend with several challenges in the terrestrial environment. The cell ‘s interior is mostly water: in this medium, small molecules dissolve and diffuse and the majority of the chemical reactions of metabolism take place. Desiccation, or drying out, is a constant danger for organisms exposed to air. Even when parts of a plant are close to a source of water, the aerial structures are prone to desiccation. Water also provides buoyancy to organisms. On land, plants need to develop structural support in a medium that does not give the same lift as water. The organism is also subject to bombardment by mutagenic radiation because air does not filter out the ultraviolet rays of sunlight. Additionally, the male gametes must reach the female gametes using new strategies because swimming is no longer possible. As such, both gametes and zygotes must be protected from desiccation. Successful land plants have developed strategies to face all of these challenges. Not all adaptations appeared at once; some species never moved very far from the aquatic environment, although others went on to conquer the driest environments on Earth.

Despite these survival challenges, life on land does offer several advantages. First, sunlight is abundant. Water acts as a filter, altering the spectral quality of light absorbed by the photosynthetic pigment chlorophyll. Second, carbon dioxide is more readily available in air than water since it diffuses faster in air. Third, land plants evolved before land animals; therefore, until dry land was also colonized by animals, no predators threatened plant life. This situation changed as animals emerged from the water and fed on the abundant sources of nutrients in the established flora. In turn, plants developed strategies to deter predation: from spines and thorns to toxic chemicals.

Early land plants, like the early land animals, did not live far from an abundant source of water and developed survival strategies to combat dryness. One of these strategies is called desiccation tolerance. Many mosses can dry out to a brown and brittle mat, but as soon as rain or a flood makes water available, mosses will absorb it and are restored to their healthy green appearance. Another strategy is to colonize environments where droughts are uncommon. Ferns, which are considered an early lineage of plants, thrive in damp and cool places such as the understory of temperate forests. Later, plants moved away from moist or aquatic environments and developed resistance to desiccation, rather than tolerance. These plants, like cacti, minimize the loss of water to such an extent they can survive in extremely dry environments.

The most successful adaptation solution was the development of new structures that gave plants the advantage when colonizing new and dry environments. Four major adaptations are found in all terrestrial plants: the alternation of generations, a sporangium in which the spores are formed, a gametangium that produces haploid cells, and apical meristem tissue in roots and shoots. The evolution of a waxy cuticle and a cell wall with lignin also contributed to the success of

land plants. These adaptations are noticeably lacking in the closely-related green algae, which gives reason for the debate over their placement in the plant kingdom.

Alternation of Generations

Alternation of generations describes a life cycle in which an organism has both haploid and diploid multicellular stages (n represents the number of copies of chromosomes). Haplontic refers to a lifecycle in which there is a dominant haploid stage (1n), while diplontic refers to a lifecycle in which the diploid (2n) is the dominant life stage. Humans are diplontic. Most plants exhibit alternation of generations, which is described as haplodiplodontic. The haploid multicellular form, known as a gametophyte, is followed in the development sequence by a multicellular diploid organism: the sporophyte. The gametophyte gives rise to the gametes (reproductive cells) by mitosis. This can be the most obvious phase of the life cycle of the plant, as in the mosses. In fact, the sporophyte stage is barely noticeable in lower plants (the collective term for the plant groups of mosses, liverworts, and lichens). Alternatively, the gametophyte stage can occur in a microscopic structure, such as a pollen grain, in the higher plants (a common collective term for the vascular plants). Towering trees are the diplontic phase in the life cycles of plants such as sequoias and pines.

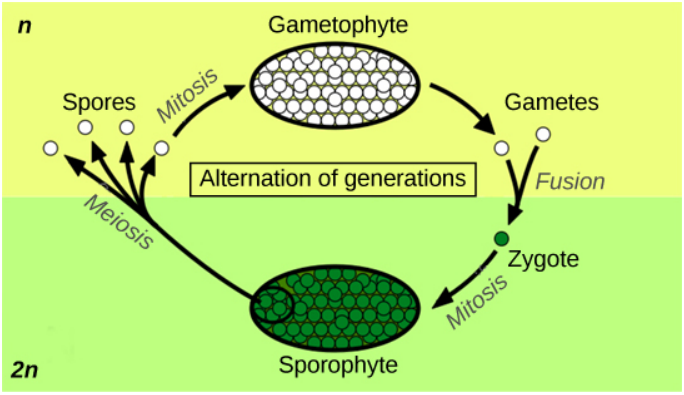

Figure $25.1C.1$: **Alternation of generations of plants**: Plants exhibit an alternation of generations between a 1n gametophyte and 2n sporophyte.

Protection of the embryo is a major requirement for land plants. The vulnerable embryo must be sheltered from desiccation and other environmental hazards. In both seedless and seed plants, the female gametophyte provides protection and nutrients to the embryo as it develops into the new generation of sporophyte. This distinguishing feature of land plants gave the group its alternate name of embryophytes.

25.1D: Sporophytes and Gametophytes in Seedless Plants

Sporophytes (2n) undergo meiosis to produce spores that develop into gametophytes (1n) which undergo mitosis.

Learning Objectives

- Describe the role of the sporophyte and gametophyte in plant reproduction

Key Points

- The diploid stage of a plant (2n), the sporophyte, bears a sporangium, an organ that produces spores during meiosis.
- Homosporous plants produce one type of spore which develops into a gametophyte (1n) with both male and female organs.
- Heterosporous plants produce separate male and female gametophytes, which produce sperm and eggs, respectively.
- In seedless plants, male gametangia (antheridium) release sperm, which can then swim to and fertilize an egg at the female gametangia (archegonia); this mode of reproduction is replaced by pollen production in seed plants.

Key Terms

- **gametophyte**: a plant (or the haploid phase in its life cycle) that produces gametes by mitosis in order to produce a zygote
- **gametangium**: an organ or cell in which gametes are produced that is found in many multicellular protists, algae, fungi, and the gametophytes of plants
- **sporopollenin**: a combination of biopolymers observed in the tough outer layer of the spore and pollen wall
- **syngamy**: the fusion of two gametes to form a zygote
- **sporophyte**: a plant (or the diploid phase in its life cycle) that produces spores by meiosis in order to produce gametophytes

Sporangia in Seedless Plants

The sporophyte of seedless plants is diploid and results from syngamy (fusion) of two gametes. The sporophyte bears the sporangia (singular, sporangium): organs that first appeared in the land plants. The term “sporangia” literally means “spore in a vessel”: it is a reproductive sac that contains spores. Inside the multicellular sporangia, the diploid sporocytes, or mother cells, produce haploid spores by meiosis, where the 2n chromosome number is reduced to 1n (note that many plant sporophytes are polyploid: for example, durum wheat is tetraploid, bread wheat is hexaploid, and some ferns are 1000-ploid). The spores are later released by the sporangia and disperse in the environment.

Figure 25.1*D*.1: **Sporangia**: Spore-producing sacs called sporangia grow at the ends of long, thin stalks in this photo of the moss Esporangios bryum.

Two different spore-forming methods are used in land plants, resulting in the separation of sexes at different points in the lifecycle. Seedless, non- vascular plants produce only one kind of spore and are called homosporous. The gametophyte phase (1n) is dominant in these plants. After germinating from a spore, the resulting gametophyte produces both male and female gametangia, usually on the same individual. In contrast, heterosporous plants produce two morphologically different types of spores. The male spores are called microspores, because of their smaller size, and develop into the male gametophyte; the comparatively larger megaspores develop into the female gametophyte. Heterospory is observed in a few seedless vascular plants and in all seed plants.

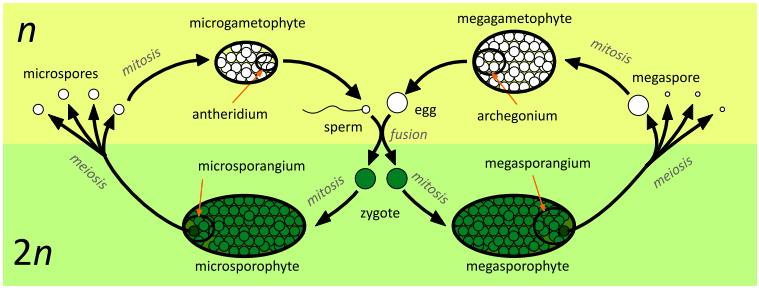

Figure 25.1D.1: **Lifecycle of heterosporous plants**: Heterosporous plants produce two morphologically different types of spores: microspores, which develop into the male gametophyte, and megaspores, which develop into the female gametophyte.

When the haploid spore germinates in a hospitable environment, it generates a multicellular gametophyte by mitosis. The gametophyte supports the zygote formed from the fusion of gametes and the resulting young sporophyte (vegetative form). The cycle then begins anew.

The spores of seedless plants are surrounded by thick cell walls containing a tough polymer known as sporopollenin. This complex substance is characterized by long chains of organic molecules related to fatty acids and carotenoids: hence the yellow color of most pollen. Sporopollenin is unusually resistant to chemical and biological degradation. In seed plants, which use pollen to transfer the male sperm to the female egg, the toughness of sporopollenin explains the existence of well-preserved pollen fossils. Sporopollenin was once thought to be an innovation of land plants; however, the green algae, Coleochaetes, also forms spores that contain sporopollenin.

Gametangia in Seedless Plants

Gametangia (singular, gametangium) are organs observed on multicellular haploid gametophytes. In the gametangia, precursor cells give rise to gametes by mitosis. The male gametangium (antheridium) releases sperm. Many seedless plants produce sperm equipped with flagella that enable them to swim in a moist environment to the archegonia: the female gametangium. The embryo develops inside the archegonium as the sporophyte. Gametangia are prominent in seedless plants, but are replaced by pollen grains in seed-producing plants.

25.1E: Structural Adaptations for Land in Seedless Plants

Plants developed a series of organs and structures to facilitate life on dry land independent from a constant source of water.

Learning Objectives

- Discuss the primary structural adaptations made by plants to living on land

Key Points

- Many plants developed a vascular system: to distribute water from the roots (via the xylem) and sugars from the shoots (via the phloem) throughout the entire plant.
- An apical meristem enables elongation of the shoots and roots, allowing a plant to access additional space and resources.
- Because of the waxy cuticle covering leaves to prevent water loss, plants evolved stomata, or pores on the leaves, which open and close to regulate traffic of gases and water vapor.
- Plants evolved pathways for the synthesis of complex organic molecules, called secondary metabolites, for protection from both UV lights and predators.

Key Terms

- **phloem**: a vascular tissue in land plants primarily responsible for the distribution of sugars and nutrients manufactured in the shoot
- **stoma**: a pore found in the leaf and stem epidermis used for gaseous exchange
- **xylem**: a vascular tissue in land plants primarily responsible for the distribution of water and minerals taken up by the roots; also the primary component of wood
- **meristem**: the plant tissue composed of totipotent cells that allows plant growth

Land Plant Adaptations

As plants adapted to dry land and became independent from the constant presence of water in damp habitats, new organs and structures made their appearance. Early land plants did not grow more than a few inches off the ground, competing for light on these low mats. By developing a shoot and growing taller, individual plants captured more light. Because air offers substantially less support than water, land plants incorporated more rigid molecules in their stems (and later, tree trunks).

Apical Meristems

Shoots and roots of plants increase in length through rapid cell division in a tissue called the apical meristem, which is a small zone of cells found at the shoot tip or root tip. The apical meristem is made of undifferentiated cells that continue to proliferate throughout the life of the plant. Meristematic cells give rise to all the specialized tissues of the organism. Elongation of the shoots and roots allows a plant to access additional space and resources: light, in the case of the shoot, and water and minerals, in the case of roots. A separate meristem, called the lateral meristem, produces cells that increase the diameter of tree trunks.

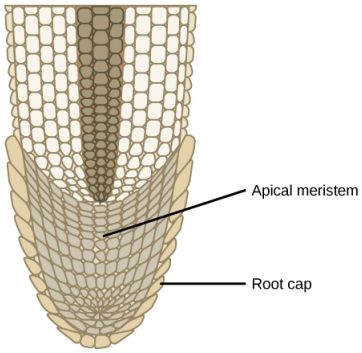

Figure 25.1E.1: **Apical meristem**: Addition of new cells in a root occurs at the apical meristem. Subsequent enlargement of these cells causes the organ to grow and elongate. The root cap protects the fragile apical meristem as the root tip is pushed through the soil by cell elongation.

Vascular structures

In small plants such as single-celled algae, simple diffusion suffices to distribute water and nutrients throughout the organism. However, for plants to develop larger forms, the evolution of vascular tissue for the distribution of water and solutes was a prerequisite. The vascular system contains xylem and phloem tissues. Xylem conducts water and minerals absorbed from the soil up to the shoot, while phloem transports food derived from photosynthesis throughout the entire plant. A root system evolved to take up water and minerals from the soil, while anchoring the increasingly taller shoot in the soil.

Additional land plant adaptations

In land plants, a waxy, waterproof cover called a cuticle protects the leaves and stems from desiccation. However, the cuticle also prevents intake of carbon dioxide needed for the synthesis of carbohydrates through photosynthesis. To overcome this, stomata, or pores, that open and close to regulate traffic of gases and water vapor, appeared in plants as they moved away from moist environments into drier habitats.

Water filters ultraviolet-B (UVB) light, which is harmful to all organisms, especially those that must absorb light to survive. This filtering does not occur for land plants. This presented an additional challenge to land colonization, which was met by the evolution of biosynthetic pathways for the synthesis of protective flavonoids and other compounds: pigments that absorb UV wavelengths of light and protect the aerial parts of plants from photodynamic damage.

Plants cannot avoid being eaten by animals. Instead, they synthesize a large range of poisonous secondary metabolites: complex organic molecules such as alkaloids, whose noxious smells and unpleasant taste deter animals. These toxic compounds can also cause severe diseases and even death, thus discouraging predation. Humans have used many of these compounds for centuries as drugs, medications, or spices. In contrast, as plants co-evolved with animals, the development of sweet and nutritious metabolites lured animals into providing valuable assistance in dispersing pollen grains, fruit, or seeds. Plants have been enlisting animals to be their helpers in this way for hundreds of millions of years.

25.1F: The Major Divisions of Land Plants

Land plants, or embryophytes, are classified by the presence or absence of vascular tissue and how they reproduce (with or without seeds).

Learning Objectives

- Identify the major divisions of land plants

Key Points

- Non- vascular plants, or bryophytes, appeared early in plant evolution and reproduce without seeds; they include mosses, liverworts, and hornworts.
- Vascular plants are subdivided into two classes: seedless plants, which probably evolved first (including lycophytes and pterophytes), and seed plants.
- Seed-producing plants include gymnosperms, which produce “naked” seeds, and angiosperms, which reproduce by flowering.

Key Terms

- **spermatophyte**: any plant that bears seeds rather than spores
- **embryophyte**: any member of the subkingdom Embryophyta; most land plants
- **bryophyte**: seedless, nonvascular plants that are the closest extant relative of early terrestrial plants

The Major Divisions of Land Plants

The green algae, known as the charophytes, and land plants are grouped together into a subphylum called the Streptophytina and are, therefore, called Streptophytes. Land plants, which are called embryophytes, are classified into two major groups according to the absence or presence of vascular tissue. Plants that lack vascular tissue, which is formed of specialized cells for the transport of water and nutrients, are referred to as non-vascular plants or bryophytes. Non-vascular embryophytes probably appeared early in land plant evolution and are all seedless. These plants include liverworts, mosses, and hornworts.

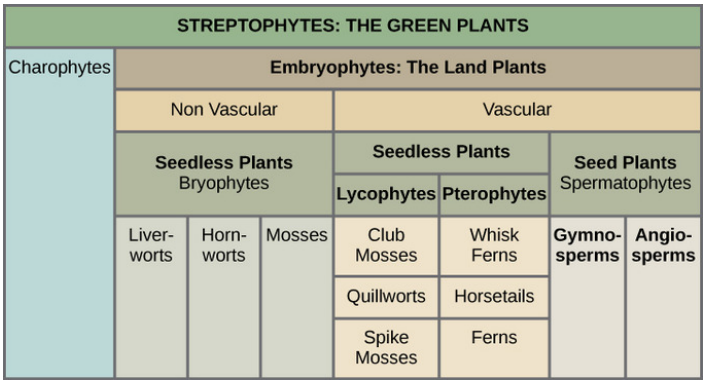

Figure 25.1F.1: **Major divisions of land plants**: Land plants are categorized by presence or absence of vascular tissue and their reproduction with or without the use of seeds.

In contrast, vascular plants developed a network of cells, called xylem and phloem, that conduct water and solutes throughout the plant. The first vascular plants appeared in the late Ordovician period of the Paleozoic Era (approximately 440-485 million years ago). These early plants were probably most similar to modern day lycophytes, which include club mosses (not to be confused with the mosses), and pterophytes, which include ferns, horsetails, and whisk ferns. Lycophytes and pterophytes are both referred to as seedless vascular plants because they do not produce any seeds.

The seed producing plants, or spermatophytes, form the largest group of all existing plants, dominating the landscape. Seed-producing plants include gymnosperms, most notably conifers, which produce “naked seeds,” and the most successful of all modern-day plants, angiosperms, which are the flowering plants. Angiosperms protect their seeds inside chambers at the center of a flower; the walls of the chamber later develop into a fruit.

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25.2: Green Algae: Precursors of Land Plants

Topic hierarchy

- 25.2A: Streptophytes and Reproduction of Green Algae
- 25.2B: Charales

25.2A: Streptophytes and Reproduction of Green Algae

Land plants and closely-related green algae (charophytes) are classified as Streptophytes; the remaining green algae are chlorophytes.

Learning Objectives

- Discuss the general similarities of green algae and land plants

Key Points

- There is a diverse array of green algae including single-celled or multicellular species, which can reproduce both sexually or asexually.
- The classification of green algae is challenging because they bear many of the structural and biochemical traits of plants.
- Species of green algae that are closely related to embryophytes are classified as charophytes while the remaining green algae are classified as chlorophytes.
- Like plants, charophytes have chlorophyll a and b, store carbohydrates as starch, have cell walls consisting of cellulose, and undergo similar cell-division processes.
- Charophytes have unique reproductive organs that differ considerably from that of other algae.

Key Terms

- **streptophytes**: a subphylum consisting of several orders of green algae and embryophytes
- **Charophyta**: a division of green algae that includes the closest relatives of the embryophyte plants
- **Chlorophyta**: a division of green algae that are considered more distantly related to plants

Streptophytes

Until recently, all photosynthetic eukaryotes were considered members of the kingdom Plantae. The brown, red, and gold algae, however, have been reassigned to the Protista kingdom. This is because, apart from their ability to capture light energy and fix CO_2, they lack many structural and biochemical traits that distinguish plants from protists. The position of green algae is more ambiguous. Green algae include unicellular and colonial flagellates, most with two flagella per cell, as well as various colonial, coccoid, and filamentous forms, along with macroscopic seaweeds, all of which add to the ambiguity of green algae classification since plants are multicellular.

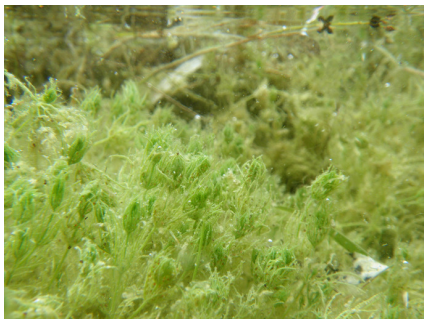

Figure 25.2A.1: **Chara vulgaris**: A representative charophyte alga is Chara vulgris, or common stonewort, which is a multicellular branching species that can grow up to 120m long.

Green algae contain the same carotenoids and chlorophyll a and b as land plants, whereas other algae have different accessory pigments and types of chlorophyll molecules in addition to chlorophyll a. Both green algae and land plants also store carbohydrates as starch. Cells in green algae divide along cell plates called phragmoplasts and their cell walls are layered with cellulose in the same manner as the cell walls of embryophytes. Consequently, land plants (embryophytes) and closely-related green algae (Charophyta) are now part of a new monophyletic group called Streptophyta. The remaining green algae, which are more distantly related to plants, belong to a group called Chlorophyta that includes more than 7000 different species that live in fresh or brackish water, in seawater, or in snow patches.

The Charophyta are a division of green algae that includes the closest relatives of the embryophyte plants. Charophyta are a small but important group of plants which show marked differences from both the Thallophyta and the Bryophyta. They are all specialized water plants. The reproductive organs consist of antheridia and oogonia, although the structure of these organs differs considerably from the corresponding organs in the Algae.

25.2B: Charales

Algae in the order Charales live in fresh water and are often considered the closest-living relatives of embryophytes.

Learning Objectives

- Identify the principle features of charophyte algae

Key Points

- The structure of charophyte algae consists of a thallus, which is the main stem, and branches that arise from nodes which bear both male and female reproductive structures.
- Although charophyte algae do not exhibit alteration of generations, they share a number of adaptations to life on land with embryophytes, including the encasement of eggs in protective enclosures.
- As new DNA sequence analysis techniques develop, revisions may need to be made in our understanding of plant evolution, such as indications that green algae in the order of Zygnematales may be more-closely related to embryophytes than is Charales.

Key Terms

- **Charales**: green algae in the division Charophyta which are green plants believed to be the closest relatives of the green land plants
- **sporopollenin**: a combination of biopolymers observed in the tough outer layer of the spore and pollen wall

Charales

Green algae in the order Charales, and the coleochaetes, microscopic green algae that enclose their spores in sporopollenin, are considered the closest-living relatives of embryophytes. The Charales can be traced as far back as 420 million years. They live in a range of fresh water habitats and vary in size from as small as a few millimeters to as large as a meter in length. A representative species of Charales is Chara, which is often called muskgrass or skunkweed because of its unpleasant smell.

Figure $25.2B.1$: **Charophyte algae**: A representative charophyte alga, Chara, is a noxious weed in Florida, where it clogs waterways.

In Charales, large cells form the thallus: the main stem of the alga. Branches arising from the nodes are made of smaller cells. Male and female reproductive structures are found on the nodes; the sperm have flagella. Unlike land plants, Charales do not undergo alternation of generations in their lifecycle. Like embryophytes, Charales exhibit a number of traits that are significant in their adaptation to land life. They produce the compounds lignin and sporopollenin. They form plasmodesmata, which are microscopic channels that connect the cytoplasm of adjacent cells. The egg and, later, the zygote, form in a protected chamber on the parent plant.

New information from recent, extensive DNA sequence analysis of green algae indicates that the Zygnematales are more closely-related to the embryophytes than the Charales. The Zygnematales include the familiar genus Spirogyra. As techniques in DNA analysis improve and new information on comparative genomics arises, the phylogenetic connections

between species will probably continue to change. Clearly, plant biologists have yet to solve the mystery of the origin of land plants.

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25.3: Bryophytes

Topic hierarchy

- 25.3A: Bryophytes
- 25.3B: Liverworts and Hornworts
- 25.3C: Mosses

25.3A: Bryophytes

Bryophytes (liverworts, mosses, and hornworts) are non-vascular plants that appeared on earth over 450 million years ago.

Learning Objectives

- Describe the characteristics of bryophytes

Key Points

- Bryophytes are the closest-living relative of early terrestrial plants; liverworts were the first Bryophytes, probably appearing during the Ordovician period.
- Bryophytes fossil formation is improbable since they do not possess lignin.
- Bryophytes thrive in mostly-damp habitats; however, some species can live in deserts while others can inhabit hostile environments such as the tundra.
- Bryophytes are nonvascular because they do not have tracheids; instead, water and nutrients circulate inside specialized conducting cells.
- In a bryophyte, all the vegetative organs belong to the gametophyte, which is the dominant and most familiar form; the sporophyte appears for only a short period.
- The sporophyte is dependent on the gametophyte and remains permanently attached to it in order to gain nutrition and protection.

Key Terms

- **bryophyte**: seedless, nonvascular plants that are the closest extant relative of early terrestrial plants
- **tracheid**: elongated cells in the xylem of vascular plants that serve in the transport of water and mineral salts
- **sporangium**: a case, capsule, or container in which spores are produced by an organism

Bryophytes

Bryophytes are the group of seedles plants that are the closest-extant relative of early terrestrial plants. The first bryophytes (liverworts) probably appeared in the Ordovician period, about 450 million years ago. However, because they lack of lignin and other resistant structures, bryophyte fossil formation is improbable and the fossil record is poor. Some spores protected by sporopollenin have survived and are attributed to early bryophytes. By the Silurian period, however, vascular plants had spread through the continents. This compelling fact is used as evidence that non-vascular plants must have preceded the Silurian period.

More than 25,000 species of bryophytes thrive in mostly-damp habitats, although some live in deserts. They constitute the major flora of inhospitable environments like the tundra where their small size and tolerance to desiccation offer distinct advantages. They generally lack lignin and do not have actual tracheids (xylem cells specialized for water conduction). Rather, water and nutrients circulate inside specialized conducting cells. Although the term non-tracheophyte is more accurate, bryophytes are commonly called non-vascular plants.

In a bryophyte, all the conspicuous vegetative organs, including the photosynthetic leaf-like structures, the thallus, stem, and the rhizoid that anchors the plant to its substrate, belong to the haploid organism, or gametophyte. The sporophyte is barely noticeable. Thus, the gametophyte is the dominant and most familiar form; the sporophyte appears for only a short period. The gametes formed by bryophytes swim with a flagellum. The sporangium, the multicellular sexual reproductive structure, is present in bryophytes and absent in the majority of algae. The sporophyte embryo also remains attached to the parent plant, which protects and nourishes it. This is a characteristic of land plants. The bryophytes are divided into three phyla: the liverworts (Hepaticophyta), the hornworts (Anthocerotophyta), and the mosses (true Bryophyta).

Figure 25.3A.1: **Moss**: Mosses (true bryophyta) are one of the three kinds of bryophytes (along with liverworts and hornworts). This image shows a moss growing on a dry stone wall.

25.3B: Liverworts and Hornworts

Liverworts and hornworts are both bryophytes, but aspects of their structures and development are different.

Learning Objectives

- Describe the distinguishing traits of hornworts and liverworts

Key Points

- The leaves of liverworts are lobate green structures similar to the lobes of the liver, while hornworts have narrow, pipe-like structures.
- The gametophyte stage is the dominant stage in both liverworts and hornworts; however, liverwort sporophytes do not contain stomata, while hornwort sporophytes do.
- The life cycle of liverworts and hornworts follows alternation of generations: spores germinate into gametophytes, the zygote develops into a sporophyte that releases spores, and then spores produce new gametophytes.
- Liverworts develop short, small sporophytes, whereas hornworts develop long, slender sporophytes.
- To aid in spore dispersal, liverworts utilize elaters, whereas hornworts utilize pseudoelaters.
- Liverworts and hornworts can reproduce asexually through the fragmentation of leaves into gemmae that disperse and develop into gametophytes.

Key Terms

- **alternation of generation**: the life cycle of plants with a multicellular sporophyte, which is diploid, that alternates with a multicellular gametophyte, which is haploid
- **pseudoelater**: single-celled structure that aids in spore dispersal
- **gemmae**: small, intact, complete pieces of plant that are produced in a cup on the surface of the thallus and develop into gametophytes through asexual reproduction

Liverworts and Hornworts

Liverworts

Liverworts (Hepaticophyta) are viewed as the plants most closely related to the ancestor that moved to land. Liverworts have colonized every terrestrial habitat on earth and diversified to more than 7000 existing species. Liverwort gametophytes (the dominant stage of the life cycle) form lobate green structures. The shape of these leaves are similar to the lobes of the liver; hence, providing the origin of the name given to the phylum. Openings that allow the movement of gases may be observed in liverworts. However, these are not stomata because they do not actively open and close. The plant takes up water over its entire surface and has no cuticle to prevent desiccation.

Figure $25.3B.1$: **Liverworts**: A liverwort, *Lunularia cruciata*, displays its lobate, flat thallus. The organism in the photograph is in the dominant gametophyte stage.

The liverwort's life cycle begins with the release of haploid spores from the sporangium that developed on the sporophyte. Spores disseminated by wind or water germinate into flattened thalli gametophytes attached to the substrate by thin, single-celled filaments. Male and female gametangia develop on separate, individual plants. Once released, male gametes swim with the aid of their flagella to the female gametangium (the archegonium), and fertilization ensues. The zygote grows into

a small sporophyte still attached to the parent gametophyte and develops spore-producing cells and elaters. The spore-producing cells undergo meiosis to form spores, which disperse (with the help of elaters), giving rise to new gametophytes. Thus, the life cycle of liverworts follows the pattern of alternation of generations.

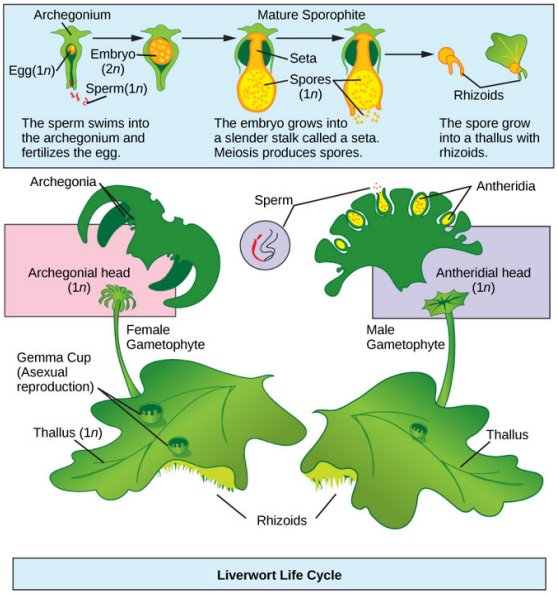

Figure $25.3B.1$: **Liverwort Life Cycle**: The life cycle of a typical liverwort follows the pattern of alternation of generations. Spores are released from sporophytes and form the gametophyte. Male gametes fertilize female gametes to form a zygote, which grows into a sporophyte. This sporophyte disperses spores with the help of elaters; the process begins again.

Liverwort plants can also reproduce asexually by the breaking of branches or the spreading of leaf fragments called gemmae. In this latter type of reproduction, the gemmae (small, intact, complete pieces of plant that are produced in a cup on the surface of the thallus) are splashed out of the cup by raindrops. The gemmae then land nearby and develop into gametophytes.

Hornworts

The hornworts (Anthocerotophyta) belong to the broad bryophyte group that have colonized a variety of habitats on land, although they are never far from a source of moisture. The short, blue-green gametophyte is the dominant phase of the lifecycle of a hornwort. The narrow, pipe-like sporophyte is the defining characteristic of the group. The sporophytes emerge from the parent gametophyte and continue to grow throughout the life of the plant. Stomata appear in the hornworts and are abundant on the sporophyte. Photosynthetic cells in the thallus contain a single chloroplast. Meristem cells at the base of the plant keep dividing and adding to its height. Many hornworts establish symbiotic relationships with cyanobacteria that fix nitrogen from the environment.

Figure $25.3B.1$: **Hornworts**: Unlike liverworts, hornworts grow a tall and slender sporophyte.

The life cycle of hornworts also follows the general pattern of alternation of generations and has a similar life cycle to liverworts. The gametophytes grow as flat thalli on the soil with embedded gametangia. Flagellated sperm swim to the

archegonia and fertilize eggs. However, unlike liverworts, the zygote develops into a long and slender sporophyte that eventually splits open, releasing spores. Additionally, thin cells called pseudoelaters surround the spores and help propel them further in the environment. Unlike the elaters observed in liverworts, the hornwort pseudoelaters are single-celled structures. The haploid spores germinate and produce the next generation of gametophytes. Like liverworts, some hornworts may also produce asexually through fragmentation.

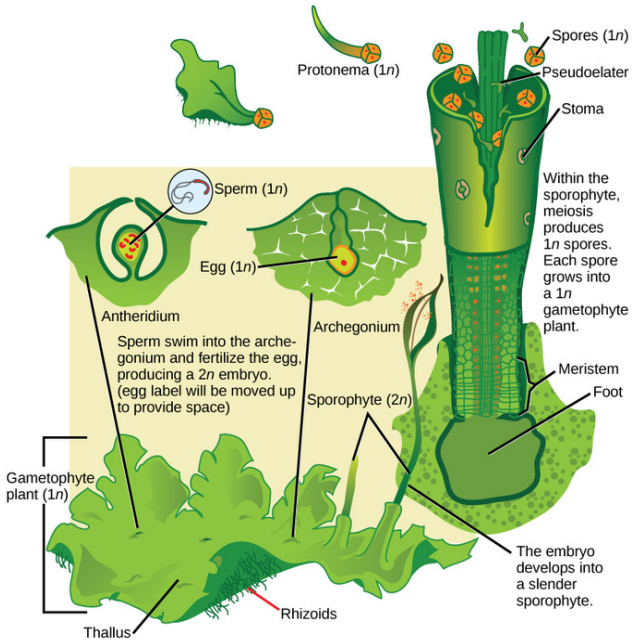

Figure $25.3B.1$: **Life Cycle of Hornworts**: The life cycle of hornworts is similar to that of liverworts. Both follow the pattern of alternation of generations. However, liverworts develop a small sporophyte, whereas hornworts develop a long, slender sporophyte. Liverworts also disperse their spores with the help of elaters, while hornworts utilize pseudoelaters to aid in spore dispersal.

25.3C: Mosses

Mosses are bryophytes that live in many environments and are characterized by their short flat leaves, root-like rhizoids, and peristomes.

Learning Objectives

- Describe the distinguishing traits of mosses

Key Points

- Mosses slow down erosion, store moisture and soil nutrients, and provide shelter for small animals and food for larger herbivores.
- Mosses have green, flat structures that resemble true leaves, which absorb water and nutrients; some mosses have small branches.
- Mosses have traits that are adaptations to dry land, such as stomata present on the stems of the sporophyte.
- Mosses are anchored to the substrate by rhizoids, which originate from the base of the gametophyte.
- The moss life cycle follows the pattern of alternation of generations where gametophytes form male and female gametophores, which fertilize to form the sporophyte; spores are released from the sporophyte to produce new gametophytes.
- The concentric tissue around the mouth of the capsule is made of triangular, close-fitting units that open and close to release spores, and the peristome increases the spread of spores after the tip of the capsule falls off at dispersal.

Key Terms

- **peristome**: one or two rings of tooth-like appendages surrounding the opening of the capsule of many mosses that aid in spreading spores
- **rhizoid**: a rootlike structure that acts as support and anchors the plant to its substrate
- **seta**: the stalk of a moss sporangium, or occasionally in a liverwort

Mosses

More than 10,000 species of mosses have been cataloged. Their habitats vary from the tundra, where they are the main vegetation, to the understory of tropical forests. In the tundra, the mosses' shallow rhizoids allow them to fasten to a substrate without penetrating the frozen soil. Mosses slow down erosion, store moisture and soil nutrients, and provide shelter for small animals as well as food for larger herbivores, such as the musk ox. Mosses are very sensitive to air pollution and are used to monitor air quality. They are also sensitive to copper salts. Such salts are a common ingredient of compounds marketed to eliminate mosses from lawns.

Mosses form diminutive gametophytes, which are the dominant phase of the life cycle. Green, flat structures resembling true leaves, but lacking vascular tissue are attached in a spiral to a central stalk or seta. The plants absorb water and nutrients directly through these leaf-like structures. The seta (plural, setae) contains tubular cells that transfer nutrients from the base of the sporophyte (the foot) to the sporangium. Some mosses have small branches. Some primitive traits of green algae, such as flagellated sperm, are still present in mosses that are dependent on water for reproduction. Other features of mosses are adaptations to dry land. For example, stomata are present on the stems of the sporophyte and a primitive vascular system runs up the sporophyte's stalk. Additionally, mosses are anchored to the substrate, whether it is soil, rock, or roof tiles, by multicellular rhizoids. These structures are precursors of roots. They originate from the base of the gametophyte, but are not the major route for the absorption of water and minerals. The lack of a true root system explains why it is so easy to rip moss mats from a tree trunk.

Figure 25.3C.1: **Setae**: This photograph shows the long slender stems, called setae, connected to capsules of the moss Thamnobryum alopecurum.

The moss life cycle follows the pattern of alternation of generations. The most familiar structure is the haploid gametophyte, which germinates from a haploid spore and forms first a protonema: usually, a tangle of single-celled filaments that hug the ground. Cells akin to an apical meristem actively divide and give rise to a gametophore, consisting of a photosynthetic stem and foliage-like structures. Rhizoids form at the base of the gametophore. Gametangia of both sexes develop on separate gametophores. The male organ (the antheridium) produces many sperm, whereas the archegonium (the female organ) forms a single egg. At fertilization, the sperm swims down the neck to the venter and unites with the egg inside the archegonium. The zygote, protected by the archegonium, divides and grows into a sporophyte, still attached by its foot to the gametophyte.

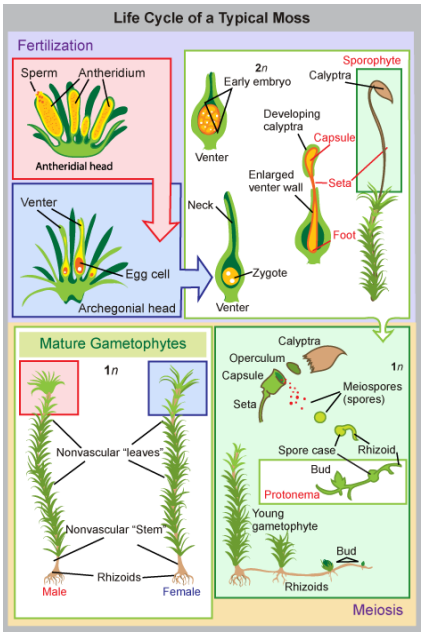

Figure 25.3C.1: **Life cycle of mosses**: The alternation of generations cycle begins when the gametophyte germinates from a haploid spore and forms a protonema. Apical meristem-like cells divide and give rise to the gametophores. The archegonium (female organ) and antheridium (male organ) develop on separate gametophores. After fertilization, the zygote divides and grows into a sporophyte, which stays attached to the gametophyte. Spores released from the sporophyte germinate and produce gametophytes; the process begins again.

A structure called a peristome increases the spread of spores after the tip of the capsule falls off at dispersal. The concentric tissue around the mouth of the capsule is made of triangular, close-fitting units, a little like “teeth”; these open and close depending on moisture levels, periodically releasing spores.

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25.4: Seedless Vascular Plants

Topic hierarchy

- 25.4A: Seedless Vascular Plants
- 25.4B: Vascular Tissue: Xylem and Phloem
- 25.4C: The Evolution of Roots in Seedless Plants
- 25.4D: Ferns and Other Seedless Vascular Plants
- 25.4E: The Importance of Seedless Vascular Plants

25.4A: Seedless Vascular Plants

Seedless vascular plants, which reproduce and spread through spores, are plants that contain vascular tissue, but do not flower or seed.

Learning Objectives

- Evaluate the evolution of seedless vascular plants

Key Points

- The life cycle of seedless vascular plants alternates between a diploid sporophyte and a haploid gametophyte phase.
- Seedless vascular plants reproduce through unicellular, haploid spores instead of seeds; the lightweight spores allow for easy dispersion in the wind.
- Seedless vascular plants require water for sperm motility during reproduction and, thus, are often found in moist environments.

Key Terms

- **gametophyte**: a plant (or the haploid phase in its life cycle) that produces gametes by mitosis in order to produce a zygote
- **sporophyte**: a plant (or the diploid phase in its life cycle) that produces spores by meiosis in order to produce gametophytes
- **tracheophyte**: any plant possessing vascular tissue (xylem and phloem), including ferns, conifers, and flowering plants

Seedless Vascular Plants

The vascular plants, or tracheophytes, are the dominant and most conspicuous group of land plants. They contain tissue that transports water and other substances throughout the plant. More than 260,000 species of tracheophytes represent more than 90 percent of the earth's vegetation. By the late Devonian period, plants had evolved vascular tissue, well-defined leaves, and root systems. With these advantages, plants increased in height and size and were able to spread to all habitats.

Seedless vascular plants are plants that contain vascular tissue, but do not produce flowers or seeds. In seedless vascular plants, such as ferns and horsetails, the plants reproduce using haploid, unicellular spores instead of seeds. The spores are very lightweight (unlike many seeds), which allows for their easy dispersion in the wind and for the plants to spread to new habitats. Although seedless vascular plants have evolved to spread to all types of habitats, they still depend on water during fertilization, as the sperm must swim on a layer of moisture to reach the egg. This step in reproduction explains why ferns and their relatives are more abundant in damp environments, including marshes and rainforests. The life cycle of seedless vascular plants is an alternation of generations, where the diploid sporophyte alternates with the haploid gametophyte phase. The diploid sporophyte is the dominant phase of the life cycle, while the gametophyte is an inconspicuous, but still-independent, organism. Throughout plant evolution, there is a clear reversal of roles in the dominant phase of the life cycle.

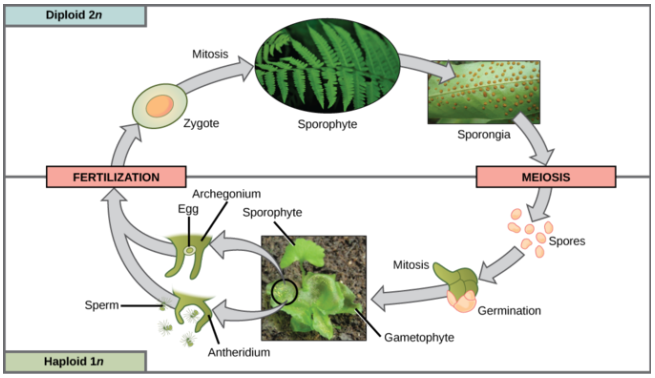

Figure 25.4A.1: **Life cycle of a fern**: This life cycle of a fern shows alternation of generations with a dominant sporophyte stage.

B: Vascular Tissue: Xylem and Phloem

Xylem and phloem form the vascular system of plants to transport water and other substances throughout the plant.

Learning Objectives

- Describe the functions of plant vascular tissue

Key Points

- Xylem transports and stores water and water-soluble nutrients in vascular plants.
- Phloem is responsible for transporting sugars, proteins, and other organic molecules in plants.
- Vascular plants are able to grow higher than other plants due to the rigidity of xylem cells, which support the plant.

Key Terms

- **xylem**: a vascular tissue in land plants primarily responsible for the distribution of water and minerals taken up by the roots; also the primary component of wood
- **phloem**: a vascular tissue in land plants primarily responsible for the distribution of sugars and nutrients manufactured in the shoot
- **tracheid**: elongated cells in the xylem of vascular plants that serve in the transport of water and mineral salts

Vascular Tissue: Xylem and Phloem

The first fossils that show the presence of vascular tissue date to the Silurian period, about 430 million years ago. The simplest arrangement of conductive cells shows a pattern of xylem at the center surrounded by phloem. Together, xylem and phloem tissues form the vascular system of plants.

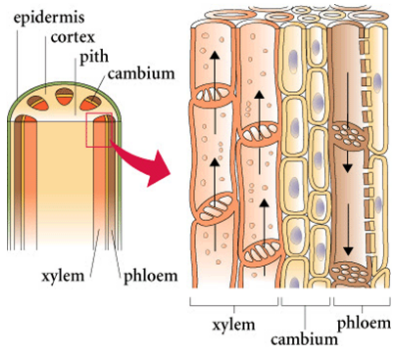

Figure $25.4B.1$: **Xylem and phloem**: Xylem and phloem tissue make up the transport cells of stems. The direction of water and sugar transportation through each tissue is shown by the arrows.

Xylem is the tissue responsible for supporting the plant as well as for the storage and long-distance transport of water and nutrients, including the transfer of water-soluble growth factors from the organs of synthesis to the target organs. The tissue consists of vessel elements, conducting cells, known as tracheids, and supportive filler tissue, called parenchyma. These cells are joined end-to-end to form long tubes. Vessels and tracheids are dead at maturity. Tracheids have thick secondary cell walls and are tapered at the ends. It is the thick walls of the tracheids that provide support for the plant and allow it to achieve impressive heights. Tall plants have a selective advantage by being able to reach unfiltered sunlight and disperse their spores or seeds further away, thus expanding their range. By growing higher than other plants, tall trees cast their shadow on shorter plants and limit competition for water and precious nutrients in the soil. The tracheids do not have end openings like the vessels do, but their ends overlap with each other, with pairs of pits present. The pit pairs allow water to pass horizontally from cell to cell.

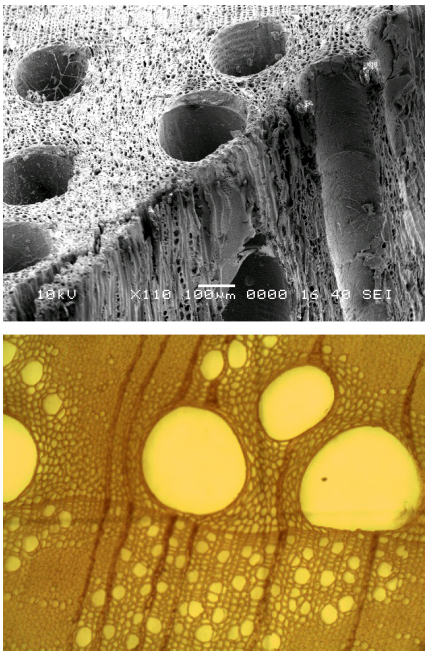

Figure $25.4B.1$: **Tracheids and vessel elements**: Tracheids (top) and vessel elements (bottom) are the water conducting cells of xylem tissue.

Phloem tissue is responsible for translocation, which is the transport of soluble organic substances, for example, sugar. The substances travel along sieve elements, but other types of cells are also present: the companion cells, parenchyma cells, and fibers. The end walls, unlike vessel members in xylem, do not have large openings. The end walls, however, are full of small pores where cytoplasm extends from cell to cell. These porous connections are called sieve plates. Despite the fact that their cytoplasm is actively involved in the conduction of food materials, sieve-tube members do not have nuclei at maturity. The activity of the sieve tubes is controlled by companion cells through plasmadesmata.

25.4C: The Evolution of Roots in Seedless Plants

Roots support plants by anchoring them to soil, absorbing water and minerals, and storing products of photosynthesis.

Learning Objectives

- Explain how roots provide support for plants

Key Points

- There are two main types of root systems: tap root systems consist of one main root that grows down vertically with smaller lateral roots growing off of the main root, while fibrous root systems form a dense network of roots near the soil surface.
- Roots can be modified to store food or starches and to provide additional support for plants; many vegetables, such as carrots, are modified roots.
- A zone of cell division, a zone of elongation, and a zone of maturation and differentiation make up a root tip, where the root cells divide, grow, and differentiate into specialized cells.
- The vascular system of roots is surrounded by an epidermis, which regulates materials that enter the root's vascular system.

Key Terms

- **endodermis**: in a plant stem or root, a cylinder of cells that separates the outer cortex from the central core and controls the flow of water and minerals within the plant
- **suberin**: a waxy material found in bark that can repel water
- **pericycle**: in a plant root, the cylinder of plant tissue between the endodermis and phloem

Roots: Support for the Plant

Roots are not well preserved in the fossil record. Nevertheless, it seems that roots appeared later in evolution than vascular tissue. The development of an extensive network of roots represented a significant new feature of vascular plants. Roots provided seed plants with three major functions: anchoring the plant to the soil, absorbing water and minerals and transporting them upwards, and storing the products of photosynthesis. Importantly, roots are modified to absorb moisture and exchange gases. In addition, while most roots are underground, some plants have adventitious roots, which emerge above the ground from the shoot.

Types of Root Systems

There are mainly two types of root systems. Dicots (flowering plants with two embryonic seed leaves) have a tap root system while monocots (flowering plants with one embryonic seed leaf) have a fibrous root system. A tap root system has a main root that grows down vertically from which many smaller lateral roots arise. Dandelions are a good example; their tap roots usually break off when trying to pull these weeds; they can regrow another shoot from the remaining root.

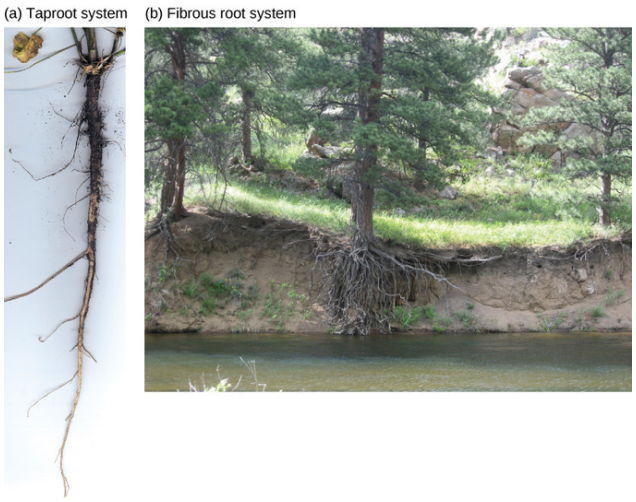

Figure 25.4C.1: **Root types**: (a) Tap root systems have a main root that grows down, while (b) fibrous root systems consist of many small roots.

A tap root system penetrates deep into the soil. In contrast, a fibrous root system is located closer to the soil surface, forming a dense network of roots that also helps prevent soil erosion (lawn grasses are a good example, as are wheat, rice, and corn). In addition, some plants actually have a combination of tap root and fibrous roots. Plants that grow in dry areas often have deep root systems, whereas plants growing in areas with abundant water tend to have shallower root systems.

Root Growth and Anatomy

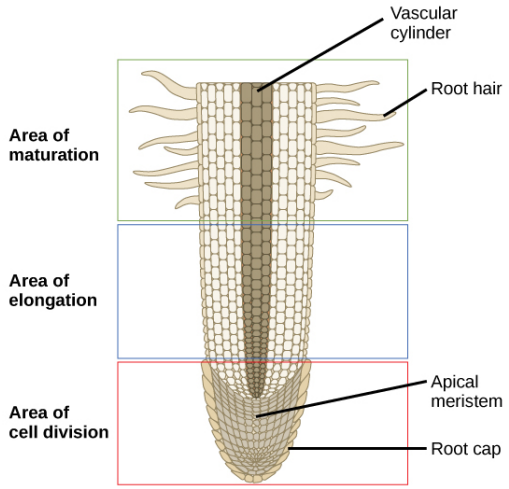

Figure 25.4C.1: **Zones on a root tip**: A longitudinal view of the root reveals the zones of cell division, elongation, and maturation. Cell division occurs in the apical meristem.

Root growth begins with seed germination. When the plant embryo emerges from the seed, the radicle of the embryo forms the root system. The tip of the root is protected by the root cap, a structure exclusive to roots and unlike any other plant structure. The root cap is continuously replaced because it gets damaged easily as the root pushes through soil. The root tip can be divided into three zones: a zone of cell division, a zone of elongation, and a zone of maturation and differentiation. The zone of cell division is closest to the root tip; it is made up of the actively-dividing cells of the root meristem. The zone of elongation is where the newly-formed cells increase in length, thereby lengthening the root. Beginning at the first root hair is the zone of cell maturation where the root cells begin to differentiate into special cell types. All three zones are in the first centimeter or so of the root tip.

Figure 25.4C.1: **Modified roots**: Many vegetables are modified roots, such as radishes and carrots, which store energy in the form of starches and sugars.

The vascular tissue in the root is arranged in the inner portion of the root, which is called the vascular cylinder. A layer of cells, known as the endodermis, separates the vascular tissue from the ground tissue in the outer portion of the root. The endodermis is exclusive to roots, serving as a checkpoint for materials entering the root's vascular system. A waxy substance called suberin is present on the walls of the endodermal cells. This waxy region, known as the Casparian strip, forces water and solutes to cross the plasma membranes of endodermal cells instead of slipping between the cells. This ensures that only materials required by the root pass through the endodermis, while toxic substances and pathogens are generally excluded. The outermost cell layer of the root's vascular tissue is the pericycle, an area that can give rise to lateral roots. In dicot roots, the xylem and phloem of the stele are arranged alternately in an X shape, whereas in monocot roots, the vascular tissue is arranged in a ring around the pith.

Root Modifications

Root structures may be modified for specific purposes. For example, some roots are bulbous and store starch. Aerial roots and prop roots are two forms of aboveground roots that provide additional support to anchor the plant. Tap roots, such as carrots, turnips, and beets, are examples of roots that are modified for food storage.

25.4D: Ferns and Other Seedless Vascular Plants

Ferns, club mosses, horsetails, and whisk ferns are seedless vascular plants that reproduce with spores and are found in moist environments.

Learning Objectives

- Identify types of seedless vascular plants

Key Points

- Club mosses, which are the earliest form of seedless vascular plants, are lycophytes that contain a stem and microphylls.
- Horsetails are often found in marshes and are characterized by jointed hollow stems with whorled leaves.
- Photosynthesis occurs in the stems of whisk ferns, which lack roots and leaves.
- Most ferns have branching roots and form large compound leaves, or fronds, that perform photosynthesis and carry the reproductive organs of the plant.

Key Terms

- **sorus**: a cluster of sporangia associated with a fern leaf
- **lycophyte**: a tracheophyte subdivision of the Kingdom Plantae; the oldest extant (living) vascular plant division at around 410 million years old
- **sporangia**: enclosures in which spores are formed

Ferns and Other Seedless Vascular Plants

Water is required for fertilization of seedless vascular plants; most favor a moist environment. Modern-day seedless tracheophytes include lycophytes and monilophytes.

Phylum Lycopodiophyta: Club Mosses

The club mosses, or phylum Lycopodiophyta, are the earliest group of seedless vascular plants. They dominated the landscape of the Carboniferous, growing into tall trees and forming large swamp forests. Today's club mosses are diminutive, evergreen plants consisting of a stem (which may be branched) and microphylls (leaves with a single unbranched vein). The phylum Lycopodiophyta consists of close to 1,200 species, including the quillworts (Isoetales), the club mosses (Lycopodiales), and spike mosses (Selaginellales), none of which are true mosses or bryophytes.

Lycophytes follow the pattern of alternation of generations seen in the bryophytes, except that the sporophyte is the major stage of the life cycle. The gametophytes do not depend on the sporophyte for nutrients. Some gametophytes develop underground and form mycorrhizal associations with fungi. In club mosses, the sporophyte gives rise to sporophylls arranged in strobili, cone-like structures that give the class its name. Lycophytes can be homosporous or heterosporous.

Figure $25.4D.1$: **Strobili of club mosses**: In some club mosses such as *Lycopodium clavatum*, sporangia are arranged in clusters called strobili.

Phylum Monilophyta: Class Equisetopsida (Horsetails)

Horsetails, whisk ferns, and ferns belong to the phylum Monilophyta, with horsetails placed in the Class Equisetopsida. The single extant genus *Equisetum* is the survivor of a large group of plants, which produced large trees, shrubs, and vines in the swamp forests in the Carboniferous. The plants are usually found in damp environments and marshes.

The stem of a horsetail is characterized by the presence of joints or nodes, hence the old name Arthrophyta (arthro- = "joint"; -phyta = "plant"). Leaves and branches come out as whorls from the evenly-spaced joints. The needle-shaped leaves do not contribute greatly to photosynthesis, the majority of which takes place in the green stem.

Figure $25.4D.1$: **Leaves of a horsetail**: The whorls of green structures at the joints are actually stems. The leaves are barely noticeable as brown rings just above each joint. Horsetails were once used as scrubbing brushes and so were called scouring rushes.

Silica collects in the epidermal cells, contributing to the stiffness of horsetail plants. Underground stems known as rhizomes anchor the plants to the ground. Modern-day horsetails are homosporous and produce bisexual gametophytes.

Phylum Monilophyta: Class Psilotopsida (Whisk Ferns)

While most ferns form large leaves and branching roots, the whisk ferns, Class Psilotopsida, lack both roots and leaves, which were probably lost by reduction. Photosynthesis takes place in their green stems; small yellow knobs form at the tip of the branch stem and contain the sporangia. Whisk ferns were considered an early pterophytes. However, recent comparative DNA analysis suggests that this group may have lost both leaves and roots through evolution and is more closely related to ferns.

Phylum Monilophyta: Class Polypodiopsida (Ferns)

With their large fronds, ferns are the most-readily recognizable seedless vascular plants. More than 20,000 species of ferns live in environments ranging from tropics to temperate forests. Although some species survive in dry environments, most ferns are restricted to moist, shaded places. Ferns made their appearance in the fossil record during the Devonian period and expanded during the Carboniferous.

The dominant stage of the life cycle of a fern is the sporophyte, which typically consists of large compound leaves called fronds. Fronds fulfill a double role; they are photosynthetic organs that also carry reproductive structure. The stem may be buried underground as a rhizome from which adventitious roots grow to absorb water and nutrients from the soil, or they may grow above ground as a trunk in tree ferns. Adventitious organs are those that grow in unusual places, such as roots growing from the side of a stem. Most ferns produce the same type of spores and are, therefore, homosporous. The diploid

sporophyte is the most conspicuous stage of the life cycle. On the underside of its mature fronds, sori (singular, sorus) form as small clusters where sporangia develop. Sporangia in a sorus produce spores by meiosis and release them into the air. Those that land on a suitable substrate germinate and form a heart-shaped gametophyte, which is attached to the ground by thin filamentous rhizoids. The inconspicuous gametophyte harbors both sex gametangia. Flagellated sperm are released and swim on a wet surface to where the egg is fertilized. The newly-formed zygote grows into a sporophyte that emerges from the gametophyte, growing by mitosis into the next generation sporophyte.

Figure 25.4*D*.1: **Sori on a fern frond**: Sori appear as small bumps on the underside of a fern frond.

25.4E: The Importance of Seedless Vascular Plants

Seedless vascular plants provide many benefits to life in ecosystems, including food and shelter and, to humans, fuel and medicine.

Learning Objectives

- Explain the beneficial roles of seedless vascular plants

Key Points

- Mosses and liverworts provide food and shelter for other organisms in otherwise barren or hostile environments.
- The level of pollution in an environment can be determined by the disappearance of mosses, which absorb the pollutants with moisture through their entire surfaces.
- Dried peat moss is used as a renewable resource for fuel.
- Ferns prevent soil erosion, promote topsoil formation, restore nitrogen to aquatic habitats by harboring cyanobacteria, make good house plants, and have been used as food and for medicinal remedies.
- Coal, a major fuel source and contributor to global warming, was deposited by the seedless vascular plants of the Carboniferous period.

Key Terms

- **bioindicator**: any species that acts as a biological indicator of the health of an environment
- **pharmacopoeia**: an official book describing medicines or other pharmacological substances, especially their use, preparation, and regulation
- **sphagnum**: any of various widely-distributed mosses, of the genus Sphagnum, which slowly decompose to form peat; often used for fuel

The Importance of Seedless Vascular Plants

Mosses and liverworts are often the first macroscopic organisms to colonize an area, both in a primary succession (where bare land is settled for the first time by living organisms) or in a secondary succession (where soil remains intact after a catastrophic event wipes out many existing species). Their spores are carried by the wind, birds, or insects. Once mosses and liverworts are established, they provide food and shelter for other species. In a hostile environment, such as the tundra where the soil is frozen, bryophytes grow well because they do not have roots and can dry and rehydrate rapidly once water is again available. Mosses are at the base of the food chain in the tundra biome. Many species, from small insects to musk oxen and reindeer, depend on mosses for food. In turn, predators feed on the herbivores, which are the primary consumers. Some reports indicate that bryophytes make the soil more amenable to colonization by other plants. Because they establish symbiotic relationships with nitrogen-fixing cyanobacteria, mosses replenish the soil with nitrogen.

At the end of the nineteenth century, scientists observed that lichens and mosses were becoming increasingly rare in urban and suburban areas. Since bryophytes have neither a root system for absorption of water and nutrients, nor a cuticle layer that protects them from desiccation, pollutants in rainwater readily penetrate their tissues; they absorb moisture and nutrients through their entire exposed surfaces. Therefore, pollutants dissolved in rainwater penetrate plant tissues readily and have a larger impact on mosses than on other plants. The disappearance of mosses can be considered a bioindicator for the level of pollution in the environment.

Ferns contribute to the environment by promoting the weathering of rock, accelerating the formation of topsoil, and slowing down erosion by spreading rhizomes in the soil. The water ferns of the genus *Azolla* harbor nitrogen-fixing cyanobacteria and restore this important nutrient to aquatic habitats.

Seedless plants have historically played a role in human life through uses as tools, fuel, and medicine. Dried peat moss, *Sphagnum*, is commonly used as fuel in some parts of Europe and is considered a renewable resource. *Sphagnum* bogs are cultivated with cranberry and blueberry bushes. The ability of Sphagnum to hold moisture makes the moss a common soil conditioner. Florists use blocks of Sphagnum to maintain moisture for floral arrangements.

Figure 25.4E.1: **Plants as a renewable resource for fuel**: Sphagnum acutifolium is dried peat moss and can be used as fuel.

The attractive fronds of ferns make them a favorite ornamental plant. Because they thrive in low light, they are well suited as house plants. More importantly, fiddleheads are a traditional spring food of Native Americans in the Pacific Northwest and are popular as a side dish in French cuisine. The licorice fern, *Polypodium glycyrrhiza*, is part of the diet of the Pacific Northwest coastal tribes, owing in part to the sweetness of its rhizomes. It has a faint licorice taste and serves as a sweetener. The rhizome also figures in the pharmacopoeia of Native Americans for its medicinal properties and is used as a remedy for sore throat.

Figure 25.4E.1: **Fiddlehead ferns as food**: A chicken dish with fiddlehead ferns as a side is shown. Native Americans traditionally cook fiddleheads with meals during the spring.

By far the greatest impact of seedless vascular plants on human life, however, comes from their extinct progenitors. The tall club mosses, horsetails, and tree-like ferns that flourished in the swampy forests of the Carboniferous period gave rise to large deposits of coal throughout the world. Coal provided an abundant source of energy during the Industrial Revolution, which had tremendous consequences on human societies, including rapid technological progress and growth of large cities, as well as the degradation of the environment. Coal is still a prime source of energy and also a major contributor to global warming.

Figure 25.4E.1: **Carboniferous period plants**: This drawing depicts the tall mosses and tree-like ferns of the Carboniferous period that deposited the large amounts of coal throughout the world.

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CHAPTER OVERVIEW

26: SEED PLANTS

26.1: EVOLUTION OF SEED PLANTS

26.1A: THE EVOLUTION OF SEED PLANTS AND ADAPTATIONS FOR LAND
26.1B: EVOLUTION OF GYMNOSPERMS
26.1C: EVOLUTION OF ANGIOSPERMS
26.2: GYMNOSPERMS

26.2A: CHARACTERISTICS OF GYMNOSPERMS
26.2B: LIFE CYCLE OF A CONIFER
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26.4: THE ROLE OF SEED PLANTS

26.4A: HERBIVORY AND POLLINATION
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26.1: Evolution of Seed Plants

Topic hierarchy

- 26.1A: The Evolution of Seed Plants and Adaptations for Land
- 26.1B: Evolution of Gymnosperms
- 26.1C: Evolution of Angiosperms

26.1A: The Evolution of Seed Plants and Adaptations for Land

The evolution of seeds allowed plants to reproduce independently of water; pollen allows them to disperse their gametes great distances.

Learning Objectives

- Recognize the significance of seed plant evolution

Key Points

- Plants are used for food, textiles, medicines, building materials, and many other products that are important to humans.
- The evolution of seeds allowed plants to decrease their dependency upon water for reproduction.
- Seeds contain an embryo that can remain dormant until conditions are favorable when it grows into a diploid sporophyte.
- Seeds are transported by the wind, water, or by animals to encourage reproduction and reduce competition with the parent plant.

Key Terms

- **seed**: a fertilized ovule, containing an embryonic plant
- **sporophyte**: a plant (or the diploid phase in its life cycle) that produces spores by meiosis in order to produce gametophytes
- **pollen**: microspores produced in the anthers of flowering plants

Evolution of Seed Plants

The lush palms on tropical shorelines do not depend upon water for the dispersal of their pollen, fertilization, or the survival of the zygote, unlike mosses, liverworts, and ferns of the terrain. Seed plants, such as palms, have broken free from the need to rely on water for their reproductive needs. They play an integral role in all aspects of life on the planet, shaping the physical terrain, influencing the climate, and maintaining life as we know it. For millennia, human societies have depended upon seed plants for nutrition and medicinal compounds; and more recently, for industrial by-products, such as timber and paper, dyes, and textiles. Palms provide materials including rattans, oils, and dates. Wheat is grown to feed both human and animal populations. The fruit of the cotton boll flower is harvested as a boll, with its fibers transformed into clothing or pulp for paper. The showy opium poppy is valued both as an ornamental flower and as a source of potent opiate compounds.

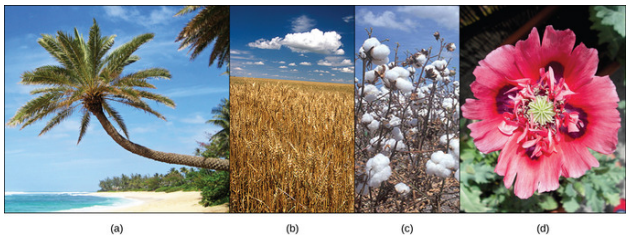

Figure 26.1*A*.1: **Seed plants dominate the landscape**: Seed plants dominate the landscape and play an integral role in human societies. (a) Palm trees grow along the shoreline; (b) wheat is a crop grown in most of the world; (c) the flower of the cotton plant produces fibers that are woven into fabric; (d) the potent alkaloids of the beautiful opium poppy have influenced human life both as a medicinal remedy and as a dangerously-addictive drug.

Seeds and Pollen as an Evolutionary Adaptation to Dry Land

Unlike bryophyte and fern spores (which are haploid cells dependent on moisture for rapid development of gametophytes), seeds contain a diploid embryo that will germinate into a sporophyte. Storage tissue to sustain growth and a protective coat give seeds their superior evolutionary advantage. Several layers of hardened tissue prevent desiccation, freeing reproduction from the need for a constant supply of water. Furthermore, seeds remain in a state of dormancy induced by desiccation and the hormone abscisic acid until conditions for growth become favorable. Whether blown by the wind,

floating on water, or carried away by animals, seeds are scattered in an expanding geographic range, thus avoiding competition with the parent plant.

Pollen grains are male gametophytes carried by wind, water, or a pollinator. The whole structure is protected from desiccation and can reach the female organs without dependence on water. Male gametes reach female gametophyte and the egg cell gamete though a pollen tube: an extension of a cell within the pollen grain. The sperm of modern gymnosperms lack flagella, but in cycads and the Gingko, the sperm still possess flagella that allow them to swim down the pollen tube to the female gamete; however, they are enclosed in a pollen grain.

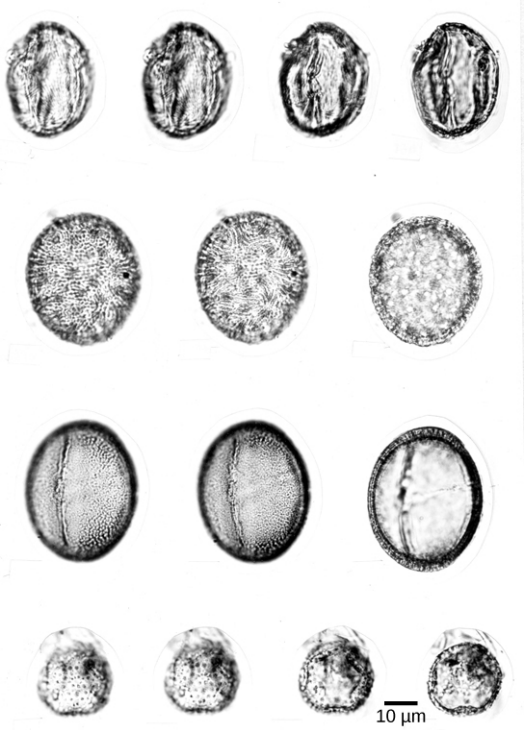

Figure 26.1A.1: **Fossilized pollen grains**: This fossilized pollen is from a Buckbean fen core found in Yellowstone National Park, Wyoming. The pollen is magnified 1,054 times.

26.1B: Evolution of Gymnosperms

Seed ferns gave rise to the gymnosperms during the Devonian Period, allowing them to adapt to dry conditions.

Learning Objectives

- Explain how and why gymnosperms became the dominant plant group during the Permian period

Key Points

- Seed ferns were the first seed plants, protecting their reproductive parts in structures called cupules.
- Seed ferns gave rise to the gymnosperms during the Paleozoic Era, about 390 million years ago.
- Gymnosperms include the gingkoes and conifers and inhabit many ecosystems, such as the taiga and the alpine forests, because they are well adapted for cold weather.
- True seed plants became more numerous and diverse during the Carboniferous period around 319 million years ago; an explosion that appears to be due to a whole genome duplication event.

Key Terms

- **cupule**: any small structure shaped like a cup
- **gymnosperm**: any plant, such as a conifer, whose seeds are not enclosed in an ovary
- **mutualism**: any interaction between two species that benefits both

Evolution of Gymnosperms

Figure 26.1B.1: **Seed ferns**: This fossilized leaf is from *Glossopteris*, a seed fern that thrived during the Permian age (290–240 million years ago).

The fossil plant *Elkinsia polymorpha*, a "seed fern" from the Devonian period (about 400 million years ago) is considered the earliest seed plant known to date. Seed ferns produced their seeds along their branches without specialized structures. What makes them the first true seed plants is that they developed structures called cupules to enclose and protect the ovule (the female gametophyte and associated tissues) which develops into a seed upon fertilization. Seed plants resembling modern tree ferns became more numerous and diverse in the coal swamps of the Carboniferous period. This appears to have been the result of a whole genome duplication event around 319 million years ago.

Figure 26.1B.1: **Gymnosperms of the taiga**: This boreal forest (taiga) has low-lying plants and conifer trees, as these plants are better suited to the colder, dryer conditions.

Fossil records indicate the first gymnosperms (progymnosperms) most likely originated in the Paleozoic era, during the middle Devonian period about 390 million years ago. Following the wet Mississippian and Pennsylvanian periods, which were dominated by giant fern trees, the Permian period was dry. This gave a reproductive edge to seed plants, which are better

adapted to survive dry spells. The Ginkgoales, a group of gymnosperms with only one surviving species, the Gingko biloba, were the first gymnosperms to appear during the lower Jurassic. Gymnosperms expanded in the Mesozoic era (about 240 million years ago), supplanting ferns in the landscape, and reaching their greatest diversity during this time. It has been suggested that during the mid-Mesozoic era, pollination of some extinct groups of gymnosperms was performed by extinct species of scorpionflies that had a specialized proboscis for feeding on pollination drops. The scorpionflies probably engaged in pollination mutualisms with gymnosperms, long before the similar and independent coevolution of nectar-feeding insects on angiosperms.

The Jurassic period was as much the age of the cycads (palm-tree-like gymnosperms) as the age of the dinosaurs. Gingkoales and the more familiar conifers also dotted the landscape. Although angiosperms (flowering plants) are the major form of plant life in most biomes, gymnosperms still dominate some ecosystems, such as the taiga (boreal forests) and the alpine forests at higher mountain elevations because of their adaptation to cold and dry growth conditions.

C: Evolution of Angiosperms

Angiosperms, which evolved in the Cretaceous period, are a diverse group of plants which protect their seeds within an ovary called a fruit.

Learning Objectives

- Discuss the evolution and adaptations of angiosperms

Key Points

- Angiosperms evolved during the late Cretaceous Period, about 125-100 million years ago.
- Angiosperms have developed flowers and fruit as ways to attract pollinators and protect their seeds, respectively.
- Flowers have a wide array of colors, shapes, and smells, all of which are for the purpose of attracting pollinators.
- Once the egg is fertilized, it grows into a seed that is protected by a fleshy fruit.
- As angiosperms evolved in the Cretaceous period, many modern groups of insects also appeared, including pollinating insects that drove the evolution of angiosperms; in many instances, flowers and their pollinators have coevolved.
- Angiosperms did not evolve from gymnosperms, but instead evolved in parallel with the gymnosperms; however, it is unclear as to what type of plant actually gave rise to angiosperms.

Key Terms

- **clade**: a group of animals or other organisms derived from a common ancestor species
- **angiosperm**: a plant whose ovules are enclosed in an ovary
- **basal angiosperm**: the first flowering plants to diverge from the ancestral angiosperm, including a single species of shrub from New Caledonia, water lilies and some other aquatic plants, and woody aromatic plants

Evolution of Angiosperms

Undisputed fossil records place the massive appearance and diversification of angiosperms in the middle to late Mesozoic era. Angiosperms (“seed in a vessel”) produce a flower containing male and/or female reproductive structures. Fossil evidence indicates that flowering plants first appeared in the Lower Cretaceous, about 125 million years ago, and were rapidly diversifying by the Middle Cretaceous, about 100 million years ago. Earlier traces of angiosperms are scarce. Fossilized pollen recovered from Jurassic geological material has been attributed to angiosperms. A few early Cretaceous rocks show clear imprints of leaves resembling angiosperm leaves. By the mid-Cretaceous, a staggering number of diverse, flowering plants crowd the fossil record. The same geological period is also marked by the appearance of many modern groups of insects, including pollinating insects that played a key role in ecology and the evolution of flowering plants.

Figure 26.1C.1: **Fossil evidence of angiosperms**: This leaf imprint shows a Ficus speciosissima, an angiosperm that flourished during the Cretaceous period. A large number of pollinating insects also appeared during this same time.

Although several hypotheses have been offered to explain this sudden profusion and variety of flowering plants, none have garnered the consensus of paleobotanists (scientists who study ancient plants). New data in comparative genomics and paleobotany have, however, shed some light on the evolution of angiosperms. Rather than being derived from gymnosperms, angiosperms form a sister clade (a species and its descendents) that developed in parallel with the gymnosperms. The two innovative structures of flowers and fruit represent an improved reproductive strategy that served to protect the embryo, while increasing genetic variability and range. Paleobotanists debate whether angiosperms evolved from small woody bushes, or were basal angiosperms related to tropical grasses. Both views draw support from cladistic studies. The so-called woody magnoliid hypothesis (which proposes that the early ancestors of angiosperms were shrubs) also offers molecular biological evidence.

The most primitive living angiosperm is considered to be *Amborellatrichopoda*, a small plant native to the rainforest of New Caledonia, an island in the South Pacific. Analysis of the genome of *A. trichopoda* has shown that it is related to all existing flowering plants and belongs to the oldest confirmed branch of the angiosperm family tree. A few other angiosperm groups, known as basal angiosperms, are viewed as primitive because they branched off early from the phylogenetic tree. Most modern angiosperms are classified as either monocots or eudicots based on the structure of their leaves and embryos. Basal angiosperms, such as water lilies, are considered more primitive because they share morphological traits with both monocots and eudicots.

Flowers and Fruits as an Evolutionary Adaptation

Angiosperms produce their gametes in separate organs, which are usually housed in a flower. Both fertilization and embryo development take place inside an anatomical structure that provides a stable system of sexual reproduction largely sheltered from environmental fluctuations. Flowering plants are the most diverse phylum on Earth after insects; flowers come in a bewildering array of sizes, shapes, colors, smells, and arrangements. Most flowers have a mutualistic pollinator, with the distinctive features of flowers reflecting the nature of the pollination agent. The relationship between pollinator and flower characteristics is one of the great examples of coevolution.

Figure 26.1C.1: **Coevolution of flowers and pollinators**: Many flowers have coevolved with particular pollinators, such that the flower is uniquely structured for the mouthparts of the pollinator. It often has features considered attractive to its particular pollinator.

Following fertilization of the egg, the ovule grows into a seed. The surrounding tissues of the ovary thicken, developing into a fruit that will protect the seed and often ensure its dispersal over a wide geographic range. Not all fruits develop from an ovary; such structures are “false fruits.” Like flowers, fruit can vary tremendously in appearance, size, smell, and taste. Tomatoes, walnut shells and avocados are all examples of fruit. As with pollen and seeds, fruits also act as agents of dispersal. Some may be carried away by the wind. Many attract animals that will eat the fruit and pass the seeds through their digestive systems, then deposit the seeds in another location. Cockleburs are covered with stiff, hooked spines that can hook into fur (or clothing) and hitch a ride on an animal for long distances. The cockleburs that clung to the velvet trousers of an enterprising Swiss hiker, George de Mestral, inspired his invention of the loop and hook fastener he named Velcro.

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26.2: Gymnosperms

Topic hierarchy

- 26.2A: Characteristics of Gymnosperms
- 26.2B: Life Cycle of a Conifer
- 26.2C: Diversity of Gymnosperms

26.2A: Characteristics of Gymnosperms

Gymnosperms are seed plants that have evolved cones to carry their reproductive structures.

Learning Objectives

- Discuss the type of seeds produced by gymnosperms

Key Points

- Gymnosperms produce both male and female cones, each making the gametes needed for fertilization; this makes them heterosporous.
- Megaspores made in cones develop into the female gametophytes inside the ovules of gymnosperms, while pollen grains develop from cones that produce microspores.
- Conifer sperm do not have flagella but rather move by way of a pollen tube once in contact with the ovule.

Key Terms

- **ovule**: the structure in a plant that develops into a seed after fertilization; the megasporangium of a seed plant with its enclosing integuments
- **sporophyll**: the equivalent to a leaf in ferns and mosses that bears the sporangia
- **heterosporous**: producing both male and female gametophytes

Characteristics of Gymnosperms

Gymnosperms are seed plants adapted to life on land; thus, they are autotrophic, photosynthetic organisms that tend to conserve water. They have a vascular system (used for the transportation of water and nutrients) that includes roots, xylem, and phloem. The name gymnosperm means “naked seed,” which is the major distinguishing factor between gymnosperms and angiosperms, the two distinct subgroups of seed plants. This term comes from the fact that the ovules and seeds of gymnosperms develop on the scales of cones rather than in enclosed chambers called ovaries.

Gymnosperms are older than angiosperms on the evolutionary scale. They are found far earlier in the fossil record than angiosperms. As will be discussed in subsequent sections, the various environmental adaptations gymnosperms have represent a step on the path to the most successful (diversity-wise) clade (monophyletic branch).

Gymnosperm Reproduction and Seeds

Gymnosperms are sporophytes (a plant with two copies of its genetic material, capable of producing spores). Their sporangia (receptacle in which sexual spores are formed) are found on sporophylls, plated scale-like structures that together make up cones. The female gametophyte develops from the haploid (meaning one set of genetic material) spores that are contained within the sporangia. Like all seed plants, gymnosperms are heterosporous: both sexes of gametophytes develop from different types of spores produced by separate cones. One type of cone is the small pollen cone, which produces microspores that subsequently develop into pollen grains. The other type of cones, the larger “ovulate” cones, make megaspores that develop into female gametophytes called ovules. Incredibly, this whole sexual process can take three years: from the production of the two sexes of gametophytes, to bringing the gametophytes together in the process of pollination, and finally to forming mature seeds from fertilized ovules. After this process is completed, the individual sporophylls separate (the cone breaks apart) and float in the wind to a habitable place. This is concluded with germination and the formation of a seedling. Conifers have sperm that do not have flagella, but instead are conveyed to the egg via a pollen tube. It is important to note that the seeds of gymnosperms are not enclosed in their final state upon the cone.

Figure 26.2A.1: **Female cone of Tamarack pine**: The female cone of *Pinus tontorta*, the Tamarack Pine, showing the rough scales. This is the cone that produces ovules.

Figure 26.2A.1: **Male cone of Tamarack pine**: The male cone of *Pinus tontorta*, the Tamarack pine, showing the close proximity of the scales. This is the cone that produces pollen.

26.2B: Life Cycle of a Conifer

Conifers are monoecious plants that produce both male and female cones, each making the necessary gametes used for fertilization.

Learning Objectives

- Describe the life cycle of a gymnosperm

Key Points

- Male cones give rise to microspores, which produce pollen grains, while female cones give rise to megaspores, which produce ovules.
- The pollen tube develops from the pollen grain to initiate fertilization; the pollen grain divides into two sperm cells by mitosis; one of the sperm cells unites with the egg cell during fertilization.
- Once the ovule is fertilized, a diploid sporophyte is produced, which gives rise to the embryo enclosed in a seed coat of tissue from the parent plant.
- Fetilization and seed development can take years; the seed that is formed is made up of three tissues: the seed coat, the gametophyte, and the embryo.

Key Terms

- **megaspore**: the larger spore of a heterosporous plant, typically producing a female gametophyte
- **microspore**: a small spore, as contrasted to the larger megaspore, which develops into male gametophytes
- **monoecious**: having the male (stamen) and female (carpel) reproductive organs on the same plant rather than on separate plants

Life Cycle of a Conifer

Pine trees are conifers (cone bearing) and carry both male and female sporophylls on the same mature sporophyte. Therefore, they are monoecious plants. Like all gymnosperms, pines are heterosporous, generating two different types of spores: male microspores and female megaspores. In the male cones (staminate cones), the microsporocytes give rise to pollen grains by meiosis. In the spring, large amounts of yellow pollen are released and carried by the wind. Some gametophytes will land on a female cone. Pollination is defined as the initiation of pollen tube growth. The pollen tube develops slowly as the generative cell in the pollen grain divides into two haploid sperm cells by mitosis. At fertilization, one of the sperm cells will finally unite its haploid nucleus with the haploid nucleus of an egg cell.

Female cones (ovulate cones) contain two ovules per scale. One megaspore mother cell (megasporocyte) undergoes meiosis in each ovule. Three of the four cells break down leaving only a single surviving cell which will develop into a female multicellular gametophyte. It encloses archegonia (an archegonium is a reproductive organ that contains a single large egg). Upon fertilization, the diploid egg will give rise to the embryo, which is enclosed in a seed coat of tissue from the parent plant. Fertilization and seed development is a long process in pine trees: it may take up to two years after pollination. The seed that is formed contains three generations of tissues: the seed coat that originates from the sporophyte tissue, the gametophyte that will provide nutrients, and the embryo itself.

In the life cycle of a conifer, the sporophyte (2n) phase is the longest phase. The gametophytes (1n), microspores and megaspores, are reduced in size. This phase may take more than one year between pollination and fertilization while the pollen tube grows towards the megasporocyte (2n), which undergoes meiosis into megaspores. The megaspores will mature into eggs (1n).

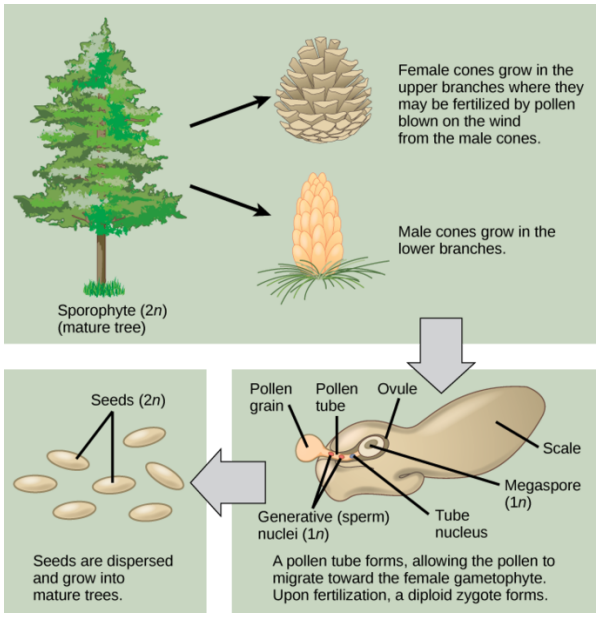

Figure $26.2B.1$: **Life cycle of a conifer**: This image shows the life cycle of a conifer. Pollen from male cones moves up into upper branches where it fertilizes female cones.

26.2C: Diversity of Gymnosperms

Gymnosperms are a diverse group of plants the protect their seeds with cones and do not produce flowers or fruits.

Learning Objectives

- Give examples showing the diversity of gymnosperms

Key Points

- Gymnosperms consist of four main phyla: the Coniferophyta, Cycadophyta, Gingkophyta and Gnetophyta.
- Conifers are the dominant plant of the gymnosperms, having needle-like leaves and living in areas where the weather is cold and dry.
- Cycads live in warm climates, have large, compound leaves, and are unusual in that they are pollinated by beetles rather than wind.
- *Gingko biloba* is the only remaining species of the Gingkophyta and is usually resistant to pollution.
- Gnetophytes are the gymnosperms believed to be most closely related to the angiosperms because of the presence of vessel elements within their stems.

Key Terms

- **tracheid**: elongated cells in the xylem of vascular plants that serve in the transport of water and mineral salts
- **angiosperm**: a plant whose ovules are enclosed in an ovary
- **conifer**: a plant belonging to the conifers; a cone-bearing seed plant with vascular tissue, usually a tree

Diversity of Gymnosperms

Modern gymnosperms are classified into four phyla. The first three (the Coniferophyta, Cycadophyta, and Gingkophyta) are similar in their production of secondary cambium (cells that generate the vascular system of the trunk or stem and are partially specialized for water transportation) and their pattern of seed development. However, these three phyla are not closely related phylogenetically to each other. The fourth phylum (the Gnetophyta) are considered the closest group to angiosperms because they produce true xylem tissue.

Coniferophytes

Conifers are the dominant phylum of gymnosperms, with the most variety of species. They are typically tall trees that usually bear scale-like or needle-like leaves. Water evaporation from leaves is reduced by their thin shape and the thick cuticle. Snow slides easily off needle-shaped leaves, keeping the load light and decreasing breaking of branches. Adaptations to cold and dry weather explain the predominance of conifers at high altitudes and in cold climates. Conifers include familiar evergreen trees such as pines, spruces, firs, cedars, sequoias, and yews. A few species are deciduous, losing their leaves in fall. The European larch and the tamarack are examples of deciduous conifers. Many coniferous trees are harvested for paper pulp and timber. The wood of conifers is more primitive than the wood of angiosperms; it contains tracheids, but no vessel elements, and is, therefore, referred to as “soft wood.”

Figure 26.2C.1: **Diversity of conifers**: Conifers are the dominant form of vegetation in cold or arid environments and at high altitudes. Shown here are the (a) evergreen spruce Picea sp., (b) juniper Juniperus sp., (c) sequoia Sequoia Semervirens, which is a deciduous gymnosperm, and (d) the tamarack Larix larcinia. Notice the yellow leaves of the tamarack.

Cycads

Cycads thrive in mild climates. They are often mistaken for palms because of the shape of their large, compound leaves. Cycads bear large cones and may be pollinated by beetles rather than wind, which is unusual for a gymnosperm (). They dominated the landscape during the age of dinosaurs in the Mesozoic, but only a hundred or so species persisted to modern times. Cycads face possible extinction; several species are protected through international conventions. Because of their attractive shape, they are often used as ornamental plants in gardens in the tropics and subtropics.

Figure 26.2C.1: **Cycad leaves**: This *Encephalartos ferox* cycad has large cones and broad, fern-like leaves.

Gingkophytes

The single surviving species of the gingkophytes group is the *Gingko biloba*. Its fan-shaped leaves, unique among seed plants because they feature a dichotomous venation pattern, turn yellow in autumn and fall from the tree. For centuries, *G. biloba* was cultivated by Chinese Buddhist monks in monasteries, which ensured its preservation. It is planted in public spaces because it is unusually resistant to pollution. Male and female organs are produced on separate plants. Typically, gardeners plant only male trees because the seeds produced by the female plant have an off-putting smell of rancid butter.

Gingko biloba

Gingko biloba is the only surviving species of the phylum Gingkophyta. This plate from the 1870 book Flora Japonica, Sectio Prima (Tafelband) depicts the leaves and fruit of Gingko biloba, as drawn by Philipp Franz von Siebold and Joseph Gerhard Zuccarini.

Gnetophytes

Gnetophytes are the closest relative to modern angiosperms and include three dissimilar genera of plants: *Ephedra*, *Gnetum*, and *Welwitschia*. Like angiosperms, they have broad leaves. In tropical and subtropical zones, gnetophytes are vines or small shrubs. *Ephedra* occurs in dry areas of the West Coast of the United States and Mexico. *Ephedra's* small, scale-like leaves are the source of the compound ephedrine, which is used in medicine as a potent decongestant. Because ephedrine is similar to amphetamines, both in chemical structure and neurological effects, its use is restricted to prescription drugs. Like angiosperms, but unlike other gymnosperms, all gnetophytes possess vessel elements in their xylem.

Figure 26.2C.1: **Gnetophytes**: (a) *Ephedra viridis*, known by the common name Mormon tea, grows on the West Coast of the United States and Mexico. (b) *Gnetum gnemon* grows in Malaysia. (c) The large *Welwitschia mirabilis* can be found in the Namibian desert.

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26.3: Angiosperms

Topic hierarchy

- 26.3A: Angiosperm Flowers
- 26.3B: Angsiosperm Fruit
- 26.3C: The Life Cycle of an Angiosperm
- 26.3D: Diversity of Angiosperms

26.3A: Angiosperm Flowers

Flowers are modified leaves containing the reproductive organs of angiospems; their pollination is usually accomplished by animals or wind.

Learning Objectives

- Describe the main parts of a flower and their purposes

Key Points

- Sepals, petals, carpels, and stamens are structures found in all flowers.
- To attract pollinators, petals usually exhibit vibrant colors; however, plants that depend on wind pollination contain flowers that are small and light.
- Carpels protect the female gametophytes and megaspores.
- The stigma is the structure where pollen is deposited and is connected to the ovary through the style.
- The anther, which comprises the stamen, is the site of microspore production and their development into pollen.

Key Terms

- **sepal**: a part of an angiosperm, and one of the component parts of the calyx; collectively the sepals are called the calyx (plural calyces), the outermost whorl of parts that form a flower
- **corolla**: an outermost-but-one whorl of a flower, composed of petals, when it is not the same in appearance as the outermost whorl (the calyx); it usually comprises the petal, which may be fused
- **stamen**: in flowering plants, the structure in a flower that produces pollen, typically consisting of an anther and a filament
- **carpel**: one of the individual female reproductive organs in a flower composed of an ovary, a style, and a stigma; also known as the gynoecium

Flowers

Flowers are modified leaves, or sporophylls, organized around a central stalk. Although they vary greatly in appearance, all flowers contain the same structures: sepals, petals, carpels, and stamens. The peduncle attaches the flower to the plant. A whorl of sepals (collectively called the calyx) is located at the base of the peduncle and encloses the unopened floral bud. Sepals are usually photosynthetic organs, although there are some exceptions. For example, the corolla in lilies and tulips consists of three sepals and three petals that look virtually identical. Petals, collectively the corolla, are located inside the whorl of sepals and often display vivid colors to attract pollinators. Flowers pollinated by wind are usually small, feathery, and visually inconspicuous. Sepals and petals together form the perianth. The sexual organs (carpels and stamens) are located at the center of the flower.

Styles, stigmas, and ovules constitute the female organ: the gynoecium or carpel. Flower structure is very diverse. Carpels may be singular, multiple, or fused. Multiple fused carpels comprise a pistil. The megaspores and the female gametophytes are produced and protected by the thick tissues of the carpel. A long, thin structure called a style leads from the sticky stigma, where pollen is deposited, to the ovary, enclosed in the carpel. The ovary houses one or more ovules, each of which will develop into a seed upon fertilization. The male reproductive organs, the stamens (collectively called the androecium), surround the central carpel. Stamens are composed of a thin stalk called a filament and a sac-like structure called the anther. The filament supports the anther, where the microspores are produced by meiosis and develop into pollen grains.

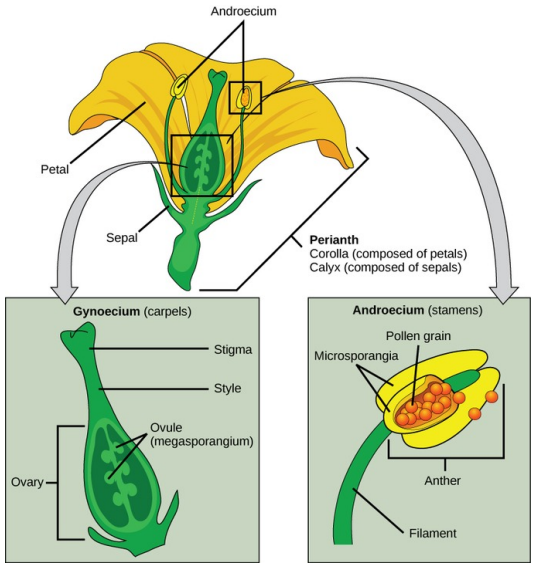

Figure 26.3A.1: **Structure of flowers**: This image depicts the structure of a flower. Perfect flowers produce both male and female floral organs. The flower shown has only one carpel, but some flowers have a cluster of carpels. Together, all the carpels make up the gynoecium.

26.3B: Angsiosperm Fruit

A fertilized, fully grown, and ripened ovary containing a seed forms what we know as fruit, important seed dispersal agents for plants.

Learning Objectives

- Recall the evolutionary advantage of fruits

Key Points

- Scientists classify fruit in many different categories that include descriptions, such as mature, fleshy, and dry; only a few are actually classified as being fleshy and sweet.
- Some fruit are developed from ovaries, while others develop from the pericarp, from clusters of flowers, or from separate ovaries in a single flower.
- Fruit are vital dispersal agents for plants; their unique shapes and features evolved to take advantage of specific dispersal modes.
- Dispersal methods of seeds within fruit include wind, water, herbivores, and animal fur.

Key Terms

- **fruit**: the seed-bearing part of a plant, often edible, colorful, and fragrant, produced from a floral ovary after fertilization
- **pericarp**: the outermost layer, or skin, of a ripe fruit or ovary
- **hypanthium**: the bowl-shaped part of a flower on which the sepals, petals, and stamens are borne

Fruit

In botany, a fertilized, fully-grown, and ripened ovary is a fruit. As the seed develops, the walls of the ovary in which it forms thicken and form the fruit, enlarging as the seeds grow. Many foods commonly-called vegetables are actually fruit. Eggplants, zucchini, string beans, and bell peppers are all technically fruit because they contain seeds and are derived from the thick ovary tissue. Acorns are nuts and winged, maple whirligigs (whose botanical name is samara) are also fruit. Botanists classify fruit into more than two dozen different categories, only a few of which are actually fleshy and sweet.

Mature fruit can be fleshy or dry. Fleshy fruit include the familiar berries, peaches, apples, grapes, and tomatoes. Rice, wheat, and nuts are examples of dry fruit. Another distinction is that not all fruits are derived from the ovary. For instance, strawberries are derived from the receptacle, while apples are derived from the pericarp, or hypanthium. Some fruits are derived from separate ovaries in a single flower, such as the raspberry. Other fruits, such as the pineapple, form from clusters of flowers. Additionally, some fruits, like watermelon and oranges, have rinds.

Regardless of how they are formed, fruits are an agent of seed dispersal. The variety of shapes and characteristics reflect the mode of dispersal, whether it be wind, water, or animals. Wind carries the light dry fruit of trees and dandelions. Water transports floating coconuts. Some fruits attract herbivores with color or perfume, or as food. Once eaten, tough, undigested seeds are dispersed through the herbivore's feces. Other fruits have burs and hooks to cling to fur and hitch rides on animals.

Figure 26.3B.1: **Wind dispersal**: The winged shape of *Alsomitra macrocarpa's* seeds allow them to use wind for dispersal. They can, therefore, glide for great distances.

Figure 26.3B.1: **Fruit dispersal**: A fruit's distinctive shape and specialized characteristics will determine its dispersal mechanism.

26.3C: The Life Cycle of an Angiosperm

Angiosperms are seed-producing plants that generate male and female gametophytes, which allow them to carry out double fertilization.

Learning Objectives

- Explain the life cycle of an angiosperm, including cross-pollination and the ways in which it takes place

Key Points

- Microspores develop into pollen grains, which are the male gametophytes, while megaspores form an ovule that contains the female gametophytes.
- In the ovule, the megasporocyte undergoes meiosis, generating four megaspores; three small and one large; only the large megaspore survives and produces the female gametophyte (embryo sac).
- When the pollen grain reaches the stigma, it extends its pollen tube to enter the ovule and deposits two sperm cells in the embryo sac.
- The two available sperm cells allow for double fertilization to occur, which results in a diploid zygote (the future embryo) and a triploid cell (the future endosperm), which acts as a food store.
- Some species are hermaphroditic (stamens and pistils are contained on a single flower), some species are monoecious (stamens and pistils occur on separate flowers, but the same plant), and some are dioecious (staminate and pistillate flowers occur on separate plants).

Key Terms

- **cotyledon**: the leaf of the embryo of a seed-bearing plant; after germination it becomes the first leaves of the seedling
- **heterosporous**: producing both male and female gametophytes
- **synergid**: either of two nucleated cells at the top of the embryo sac that aid in the production of the embryo; helper cells

The Life Cycle of an Angiosperm

The adult, or sporophyte, phase is the main phase of an angiosperm's life cycle. As with gymnosperms, angiosperms are heterosporous. Therefore, they generate microspores, which will produce pollen grains as the male gametophytes, and megaspores, which will form an ovule that contains female gametophytes. Inside the anthers' microsporangia, male gametophytes divide by meiosis to generate haploid microspores, which, in turn, undergo mitosis and give rise to pollen grains. Each pollen grain contains two cells: one generative cell that will divide into two sperm and a second cell that will become the pollen tube cell.

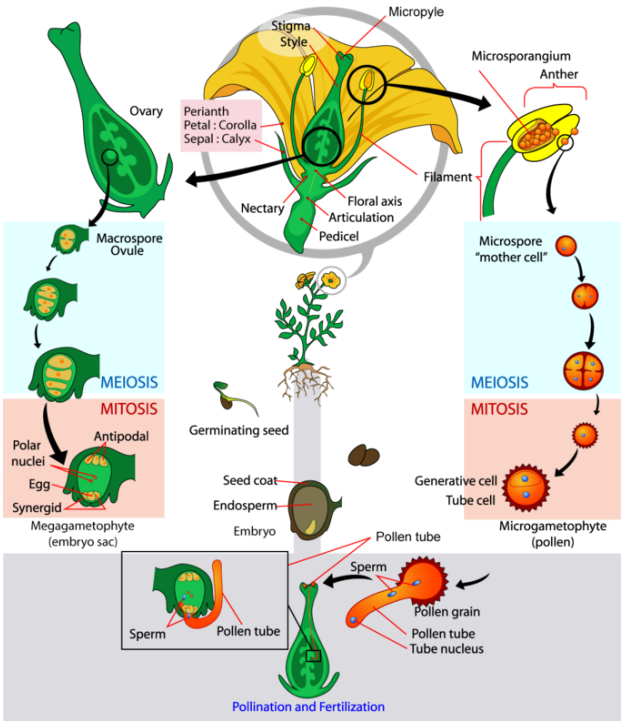

Figure 26.3C.1: **Life cycle of angiosperms**: The life cycle of an angiosperm is shown. Anthers and carpels are structures that shelter the actual gametophytes: the pollen grain and embryo sac. Double fertilization is a process unique to angiosperms.

The ovule, sheltered within the ovary of the carpel, contains the megasporangium protected by two layers of integuments and the ovary wall. Within each megasporangium, a megasporocyte undergoes meiosis, generating four megaspores: three small and one large. Only the large megaspore survives; it produces the female gametophyte referred to as the embryo sac. The megaspore divides three times to form an eight-cell stage. Four of these cells migrate to each pole of the embryo sac; two come to the equator and will eventually fuse to form a 2n polar nucleus. The three cells away from the egg form antipodals while the two cells closest to the egg become the synergids.

The mature embryo sac contains one egg cell, two synergids (“helper” cells), three antipodal cells, and two polar nuclei in a central cell. When a pollen grain reaches the stigma, a pollen tube extends from the grain, grows down the style, and enters through the micropyle, an opening in the integuments of the ovule. The two sperm cells are deposited in the embryo sac.

A double fertilization event then occurs. One sperm and the egg combine, forming a diploid zygote, the future embryo. The other sperm fuses with the 2n polar nuclei, forming a triploid cell that will develop into the endosperm, which is tissue that serves as a food reserve. The zygote develops into an embryo with a radicle, or small root, and one (monocot) or two (dicot) leaf-like organs called cotyledons. This difference in the number of embryonic leaves is the basis for the two major groups of angiosperms: the monocots and the eudicots. Seed food reserves are stored outside the embryo in the form of complex carbohydrates, lipids, or proteins. The cotyledons serve as conduits to transmit the broken-down food reserves from their storage site inside the seed to the developing embryo. The seed consists of a toughened layer of integuments forming the coat, the endosperm with food reserves, and the well-protected embryo at the center.

Figure $26.3C.1$: **The fruit of the Aesculus or Horse Chestnut tree**: These seeds are enclosed a protective outer covering called the seed coat, usually with some stored food. After fertilization and some growth in the angiosperm, the ripened ovule is produced. The formation of the seed completes the process of reproduction in seed plants (started with the development of flowers and pollination), with the embryo developed from the zygote and the seed coat from the integuments of the ovule.

Some species of angiosperms are hermaphroditic (stamens and pistils are contained on a single flower), some species are monoecious (stamens and pistils occur on separate flowers, but the same plant), and some are dioecious (staminate and pistillate flowers occur on separate plants). Both anatomical and environmental barriers promote cross-pollination mediated by a physical agent (wind or water) or an animal, such as an insect or bird. Cross-pollination increases genetic diversity in a species.

26.3D: Diversity of Angiosperms

Angiosperm diversity is divided into two main groups, monocot and dicots, based primarily on the number of cotyledons they possess.

Learning Objectives

- Explain how angiosperm diversity is classified

Key Points

- Angiosperm are flowering plants that are classified based on characteristics that include (but are not limited to) cotyledon structure, pollen grains, as well as flower and vascular tissue arrangement.
- Basal angiosperms, classified separately, contain features found in both monocots and dicots, as they are believed to have originated before the separation of these two main groups.
- Monocots contain a single cotyledon and have veins that run parallel to the length of their leaves; their flowers are arranged in three to six-fold symmetry.
- Dicots have flowers arranged in whorls, two cotyledons, and a vein arrangement that forms networks within their leaves.
- Monocots do not contain any true woody tissue while dicots can be herbacious or woody and have vascular tissue that forms a ring in the stem.

Key Terms

- **dicot**: a plant whose seedlings have two cotyledons; a dicotyledon
- **angiosperm**: a plant whose ovules are enclosed in an ovary
- **monocot**: one of two major groups of flowering plants (or angiosperms) that are traditionally recognized; seedlings typically have one cotyledon (seed-leaf)
- **cotyledon**: the leaf of the embryo of a seed-bearing plant; after germination it becomes the first leaves of the seedling
- **basal angiosperm**: the first flowering plants to diverge from the ancestral angiosperm, including a single species of shrub from New Caledonia, water lilies and some other aquatic plants, and woody aromatic plants

Diversity of Angiosperms

Angiosperms are classified in a single phylum: the Anthophyta. Modern angiosperms appear to be a monophyletic group, which means that they originated from a single ancestor. Flowering plants are divided into two major groups according to the structure of the cotyledons and pollen grains, among others. Monocots include grasses and lilies while eudicots or dicots form a polyphyletic group. However, many species exhibit characteristics that belong to either group; as such, the classification of a plant as a monocot or a eudicot is not always clearly evident. Basal angiosperms are a group of plants that are believed to have branched off before the separation into monocots and eudicots because they exhibit traits from both groups. They are categorized separately in many classification schemes. The Magnoliidae (magnolia trees, laurels, and water lilies) and the Piperaceae (peppers) belong to the basal angiosperm group.

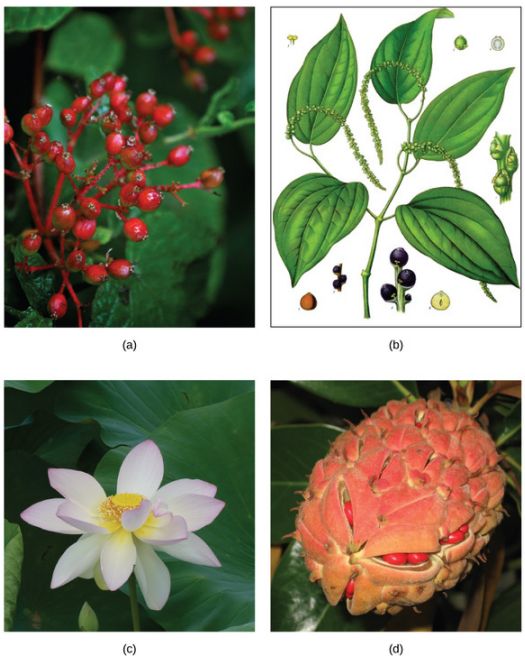

Figure 26.3D.1: **Examples of basal angiosperms**: The (a) common spicebush belongs to the Laurales, the same family as cinnamon and bay laurel. The fruit of (b) the *Piper nigrum* plant is black pepper, the main product that was traded along spice routes. Notice the small, unobtrusive, clustered flowers. (c) Lotus flowers, *Nelumbo nucifera*, have been cultivated since ancient times for their ornamental value; the root of the lotus flower is eaten as a vegetable. The red seeds of (d) a magnolia tree, characteristic of the final stage, are just starting to appear.

Basal Angiosperms

Examples of basal angiosperms include the Magnoliidae, Laurales, Nymphaeales, and the Piperales. Members in these groups all share traits from both monocot and dicot groups. The Magnoliidae are represented by the magnolias: tall trees bearing large, fragrant flowers that have many parts and are considered archaic. Laurel trees produce fragrant leaves and small, inconspicuous flowers. The Laurales grow mostly in warmer climates and are small trees and shrubs. Familiar plants in this group include the bay laurel, cinnamon, spice bush, and avocado tree. The Nymphaeales are comprised of the water lilies, lotus, and similar plants; all species thrive in freshwater biomes and have leaves that float on the water surface or grow underwater. Water lilies are particularly prized by gardeners and have graced ponds and pools for thousands of years. The Piperales are a group of herbs, shrubs, and small trees that grow in the tropical climates. They have small flowers without petals that are tightly arranged in long spikes. Many species are the source of prized fragrance or spices; for example, the berries of *Piper nigrum* are the familiar black peppercorns that are used to flavor many dishes.

Monocots

Plants in the monocot group are primarily identified as such by the presence of a single cotyledon in the seedling. Other anatomical features shared by monocots include veins that run parallel to the length of the leaves and flower parts that are arranged in a three- or six-fold symmetry. True woody tissue is rarely found in monocots. In palm trees, vascular and parenchyma tissues produced by the primary and secondary thickening of meristems form the trunk. The pollen from the first angiosperms was monosulcate, containing a single furrow or pore through the outer layer. This feature is still seen in the modern monocots. Vascular tissue of the stem is not arranged in any particular pattern. The root system is mostly adventitious and unusually positioned, with no major tap root. The monocots include familiar plants such as the true lilies (which are the origin of their alternate name: Liliopsida), orchids, grasses, and palms. Many important crops are monocots, such as rice and other cereals, corn, sugar cane, and tropical fruits like bananas and pineapples.

Figure 26.3D.1: **Monocots and Dicots: major crops of the world**: The world's major crops are flowering plants. (a) Rice, (b) wheat, and (c) bananas are monocots, while (d) cabbage, (e) beans, and (f) peaches are dicots.

Eudicots

Eudicots, or true dicots, are characterized by the presence of two cotyledons in the developing shoot. Veins form a network in leaves, while flower parts come in four, five, or many whorls. Vascular tissue forms a ring in the stem whereas in monocots, vascular tissue is scattered in the stem. Eudicots can be herbaceous (like grasses), or produce woody tissues. Most eudicots produce pollen that is trisulcate or triporate, with three furrows or pores. The root system is usually anchored by one main root developed from the embryonic radicle. Eudicots comprise two-thirds of all flowering plants.

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26.4: The Role of Seed Plants

Topic hierarchy

- 26.4A: Herbivory and Pollination
- 26.4B: The Importance of Seed Plants in Human Life
- 26.4C: Biodiversity of Plants

26.4A: Herbivory and Pollination

The diversity of plants can be attributed to pollination and herbivory, both examples of coevolution between animals and plants.

Learning Objectives

- Describe the interaction of plants and animals in achieving pollination

Key Points

- Herbivory is believed to have been as much a driving force in the evolution of plant diversity as pollination.
- Coevolution between herbivores and plants is commonly seen in nature; for example, plants have developed unique ways to fight off herbivores while, in turn, herbivores have developed specialized features to get around these defenses.
- Plants have developed unique pollination adaptations, such as the ability to capture the wind or attract specific classes of animals.
- Birds, insects, bats, lemurs, and lizards can act as pollinators; each is attracted to a specific plant adaptation, which has been developed to attract a suitable pollinator.
- Any disruption between pollinator and plant interactions, such as the extinction of a species, can lead to the collapse of an ecosystem and/or the demise of an agricultural industry.

Key Terms

- **coevolution**: the evolution of organisms of two or more species in which each adapts to changes in the other
- **pollination**: the transfer of pollen from an anther to a stigma that is carried out by insects, birds, bats, and the wind
- **herbivory**: the consumption of living plant tissue by animals

Animal & Plant Interactions

Angiosperm diversity is due in part to multiple interactions with animals. Herbivory has favored the development of defense mechanisms in plants and avoidance of those defense mechanisms in animals. Pollination (the transfer of pollen to a carpel) is mainly carried out by wind and animals; therefore, angiosperms have evolved numerous adaptations to capture the wind or attract specific classes of animals.

Coevolution of flowering plants and insects is a hypothesis that has received much attention and support, especially because both angiosperms and insects diversified at about the same time in the middle Mesozoic. Many authors have attributed the diversity of plants and insects to pollination and herbivory, which is the consumption of plants by insects and other animals. This is believed to have been as much a driving force as pollination.

Herbivory

Coevolution of herbivores and plant defenses is observed in nature. Unlike animals, most plants cannot outrun predators or use mimicry to hide from hungry animals. A sort of arms race exists between plants and herbivores. To “combat” herbivores, some plant seeds (such as acorn and unripened persimmon) are high in alkaloids and, therefore, unsavory to some animals. Other plants are protected by bark, although some animals have developed specialized mouth pieces to tear and chew vegetal material. Spines and thorns deter most animals, except for mammals with thick fur; some birds have specialized beaks to get past such defenses.

(a) (b)

Figure 26.4A.1: **Plant defenses from herbivory**: (a) Spines and (b) thorns are examples of plant defenses.

Herbivory has been used by seed plants for their own benefit in a display of mutualistic relationships. The dispersal of fruit by animals is the most striking example. The plant offers to the herbivore a nutritious source of food in return for spreading the plant's genetic material to a wider area.

An extreme example of collaboration between an animal and a plant is the case of acacia trees and ants. The trees support the insects with shelter and food. In return, ants discourage herbivores, both invertebrates and vertebrates, by stinging and attacking leaf-eating organisms.

Pollination

Grasses are a successful group of flowering plants that are wind pollinated. They produce large amounts of powdery pollen carried over large distances by the wind. The flowers are small and wisp-like. Large trees such as oaks, maples, and birches are also wind pollinated.

More than 80 percent of angiosperms depend on animals for pollination: the transfer of pollen from the anther to the stigma. Consequently, plants have developed many adaptations to attract pollinators. The specificity of specialized plant structures that target animals can be very surprising. It is possible, for example, to determine the type of pollinator favored by a plant just from the flower's characteristics. Many bird or insect-pollinated flowers secrete nectar, a sugary liquid. They also produce both fertile pollen for reproduction and sterile pollen rich in nutrients for birds and insects. Butterflies and bees can detect ultraviolet light. Flowers that attract these pollinators usually display a pattern of low ultraviolet reflectance that helps them quickly locate the flower's center to collect nectar while being dusted with pollen. Large, red flowers with little smell and a long funnel shape are preferred by hummingbirds who have good color perception, a poor sense of smell, and need a strong perch. White flowers that open at night attract moths. Other animals (such as bats, lemurs, and lizards) can also act as pollinating agents. Any disruption to these interactions, such as the disappearance of bees as a consequence of colony collapse disorders, can lead to disaster for agricultural industries that depend heavily on pollinated crops.

Figure 26.4A.1: **Animal-aided pollination**: As a bee collects nectar from a flower, it is dusted by pollen, which it then disperses to other flowers.

26.4B: The Importance of Seed Plants in Human Life

Human life has become dependent on plants for the qualities and developments that they provide, which include medicine and food production.

Learning Objectives

- Explain the importance of seed plants to humans

Key Points

- Providing much of the nutritional values that humans need, seed plants are the foundation of human diets across the world.
- Wood, paper, textiles, and dyes are just a few examples of plant uses in everyday human life.
- Traditionally, humans have also used plants as ornamental species through their use as decorations and as inspiration in the arts.
- As medicinal sources, plants are vital to humans, as many modern drugs have been derived from secondary plant metabolites; ancient societies also depended on them for their curative properties.
- The important relationship that human cultures have developed with plants can be studied through the field of ethnobotany.

Key Terms

- **ethnobotany**: the scientific study of the relationships between people and plants
- **pharmacognosy**: the branch of pharmacology that studies medical substances that are extracted from natural sources, including drugs derived from plants and herbs used for medicinal purposes
- **husbandry**: the raising of livestock and the cultivation of crops; agriculture

The Importance of Seed Plants in Human Life

Seed plants are cultivated for their beauty and smells, as well as their importance in the development of medicines. Plants are also the foundation of human diets across the world. Many societies eat, almost exclusively, vegetarian fare and depend solely on seed plants for their nutritional needs. A few crops (rice, wheat, and potatoes) dominate the agricultural landscape. Many crops were developed during the agricultural revolution when human societies made the transition from nomadic hunter–gatherers to horticulture and agriculture. Cereals, rich in carbohydrates, provide the staple of many human diets. In addition, beans and nuts supply proteins. Fats are derived from crushed seeds, as is the case for peanut and rapeseed (canola) oils, or fruits such as olives. Animal husbandry also requires large amounts of crops.

Figure 26.4B.1: **Importance of plants to humans**: Humans rely on plants for a variety of reasons. (a) Cacao beans were introduced in Europe from the New World, where they were used by Mesoamerican civilizations. Combined with sugar, another plant product, chocolate is a popular food. (b) Flowers like the tulip are cultivated for their beauty. (c) Quinine, extracted from cinchona trees, is used to treat malaria, to reduce fever, and to alleviate pain. (d) This violin is made of wood.

Staple crops are not the only food derived from seed plants. Fruits and vegetables provide nutrients, vitamins, and fiber. Sugar, to sweeten dishes, is produced from the monocot sugarcane and the eudicot sugar beet. Drinks are made from infusions of tea leaves, chamomile flowers, crushed coffee beans, or powdered cocoa beans. Spices come from many different plant parts: saffron and cloves are stamens and buds, black pepper and vanilla are seeds, the bark of a bush in the Laurales family supplies cinnamon, and the herbs that flavor many dishes come from dried leaves and fruit, such as the pungent red chili pepper. The volatile oils of flowers and bark provide the scent of perfumes. Additionally, no discussion of seed plant contribution to human diet would be complete without the mention of alcohol. Fermentation of plant-derived sugars and starches is used to produce alcoholic beverages in all societies. In some cases, the beverages are derived from the fermentation of sugars from fruit, as with wines, and, in other cases, from the fermentation of carbohydrates derived from seeds, as with beers.

Seed plants have many other uses, including providing wood as a source of timber for construction, fuel, and material to build furniture. Most paper is derived from the pulp of coniferous trees. Fibers of seed plants, such as cotton, flax, and hemp, are woven into cloth. Textile dyes, such as indigo, were mostly of plant origin until the advent of synthetic chemical dyes. Lastly, it is more difficult to quantify the benefits of ornamental seed plants. These grace private and public spaces, adding beauty and serenity to human lives and inspiring painters and poets alike.

The medicinal properties of plants have been known to human societies since ancient times. There are references to the use of plants' curative properties in Egyptian, Babylonian, and Chinese writings from 5,000 years ago. Many modern synthetic therapeutic drugs are derived or synthesized de novo from plant secondary metabolites. It is important to note that the same plant extract can be a therapeutic remedy at low concentrations, become an addictive drug at higher doses, and can potentially kill at high concentrations.

Ethnobotany

The relatively new field of ethnobotany studies the interaction between a particular culture and the plants native to the region. Seed plants have a large influence on day-to-day human life. Not only are plants the major source of food and medicine, they also influence many other aspects of society, from clothing to industry. The medicinal properties of plants were recognized early on in human cultures. From the mid-1900s, synthetic chemicals began to supplant plant-based remedies.

Pharmacognosy is the branch of pharmacology that focuses on medicines derived from natural sources. With massive globalization and industrialization, there is a concern that much human knowledge of plants and their medicinal purposes will disappear with the cultures that fostered them. This is where ethnobotanists come in. To learn about and understand the use of plants in a particular culture, an ethnobotanist must bring in knowledge of plant life and an understanding and appreciation of diverse cultures and traditions. The Amazon forest is home to an incredible diversity of vegetation and is considered an untapped resource of medicinal plants; yet, both the ecosystem and its indigenous cultures are threatened with extinction.

To become an ethnobotanist, a person must acquire a broad knowledge of plant biology, ecology, and sociology. Not only are the plant specimens studied and collected, but also the stories, recipes, and traditions that are linked to them. For ethnobotanists, plants are not viewed solely as biological organisms to be studied in a laboratory; they are seen as an integral part of human culture. The convergence of molecular biology, anthropology, and ecology make the field of ethnobotany a truly multidisciplinary science.

26.4C: Biodiversity of Plants

Plant biodiversity, vital to ecosystems, food crops, and medicine production, is threatened by habitat destruction and species extinction.

Learning Objectives

- Discuss the threats to plant biodiversity

Key Points

- Plant biodiversity is invaluable because it balances ecosystems, protects watersheds, mitigates erosion, moderates climate, and provides shelter for animals.
- Threats to plant biodiversity include the increasing human population, pollution, deforestation, and species extinction.
- Plant extinction is progressing at an alarming rate; this, in turn, affects other species, which also become extinct because they depend on the delicate ecological balance.Efforts to preserve plant biodiversity currently include heirloom seed collections and barcoding DNA analysis.

Key Terms

- **biodiversity**: the diversity (number and variety of species) of plant and animal life within a region
- **barcoding**: a taxonomic method that uses a short genetic marker in an organism's DNA to identify it as belonging to a particular species
- **heirloom seed**: seeds which are not of agricultural importance yet hold traditional importance; these seeds are kept in seed banks and are still maintained by some gardeners and farmers

Threats to Plant Biodiversity

Plants play a key role in ecosystems. They are a source of food and medicinal compounds while also providing raw materials for many industries. Rapid deforestation and industrialization, however, threaten plant biodiversity. In turn, this threatens the ecosystem.

Biodiversity of plants ensures a resource for new food crops and medicines. Plant life balances ecosystems, protects watersheds, mitigates erosion, moderates climate, and provides shelter for many animal species. Threats to plant diversity, however, come from many angles. The explosion of the human population, especially in tropical countries where birth rates are highest and economic development is in full swing, is leading to human encroachment into forested areas. To feed the larger population, humans need to obtain arable land which leads to massive clearing of trees. The need for more energy to power larger cities and economic growth results in the construction of dams, the consequent flooding of ecosystems, and increased emissions of pollutants. Other threats to tropical forests come from poachers who log trees for their precious wood. Ebony and Brazilian rosewood, both on the endangered list, are examples of tree species driven almost to extinction by indiscriminate logging.

Figure 26.4C.1: **Indiscriminate logging**: Indiscriminate logging, which leads to the clearing of whole habitats, has become a severe threat to plant biodiversity and has led to species extinction.

The number of plant species becoming extinct is increasing at an alarming rate. Because ecosystems are in a delicate balance and because seed plants maintain close symbiotic relationships with animals, whether predators or pollinators, the disappearance of a single plant can lead to the extinction of connected animal species. A real and pressing issue is that many plant species have not yet been cataloged; their place in the ecosystem is unknown. These unknown species are threatened by logging, habitat destruction, and loss of pollinators. They may become extinct before we have the chance to begin to understand the possible impacts resulting from their disappearance. Efforts to preserve biodiversity take several lines of action, from preserving heirloom seeds to barcoding species. Heirloom seeds come from plants that were traditionally grown in human populations, as opposed to the seeds used for large-scale agricultural production. Barcoding is a technique in which one or more short gene sequences, taken from a well-characterized portion of the genome, are used to identify a species through DNA analysis.

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CHAPTER OVERVIEW

27: INTRODUCTION TO ANIMAL DIVERSITY

27.1: FEATURES OF THE ANIMAL KINGDOM

27.1A: CHARACTERISTICS OF THE ANIMAL KINGDOM
27.1B: COMPLEX TISSUE STRUCTURE
27.1C: ANIMAL REPRODUCTION AND DEVELOPMENT
27.2: FEATURES USED TO CLASSIFY ANIMALS

27.2A: ANIMAL CHARACTERIZATION BASED ON BODY SYMMETRY
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27.1: Features of the Animal Kingdom

Topic hierarchy

- 27.1A: Characteristics of the Animal Kingdom
- 27.1B: Complex Tissue Structure
- 27.1C: Animal Reproduction and Development

27.1A: Characteristics of the Animal Kingdom

The animal kingdom is very diverse, but animals share many common characteristics, such as methods of development and reproduction.

Learning Objectives

- Describe the methods used to classify animals

Key Points

- Animals vary in complexity and are classified based on anatomy, morphology, genetic makeup, and evolutionary history.
- All animals are eukaryotic, multicellular organisms, and most animals have complex tissue structure with differentiated and specialized tissue.
- Animals are heterotrophs; they must consume living or dead organisms since they cannot synthesize their own food and can be carnivores, herbivores, omnivores, or parasites.
- Most animals are motile for at least some stages of their lives, and most animals reproduce sexually.

Key Terms

- **body plan**: an assemblage of morphological features shared among many members of a phylum-level group
- **heterotroph**: an organism that requires an external supply of energy in the form of food, as it cannot synthesize its own
- **extant**: still in existence; not extinct

Introduction: Features of the Animal Kingdom

Animal evolution began in the ocean over 600 million years ago with tiny creatures that probably do not resemble any living organism today. Since then, animals have evolved into a highly-diverse kingdom. Although over one million extant (currently living) species of animals have been identified, scientists are continually discovering more species as they explore ecosystems around the world. The number of extant species is estimated to be between 3 and 30 million.

But what is an animal? While we can easily identify dogs, birds, fish, spiders, and worms as animals, other organisms, such as corals and sponges, are not as easy to classify. Animals vary in complexity, from sea sponges to crickets to chimpanzees, and scientists are faced with the difficult task of classifying them within a unified system. They must identify traits that are common to all animals as well as traits that can be used to distinguish among related groups of animals. The animal classification system characterizes animals based on their anatomy, morphology, evolutionary history, features of embryological development, and genetic makeup. This classification scheme is constantly developing as new information about species arises. Understanding and classifying the great variety of living species help us better understand how to conserve the diversity of life on earth.

Even though members of the animal kingdom are incredibly diverse, most animals share certain features that distinguish them from organisms in other kingdoms. All animals are eukaryotic, multicellular organisms, and almost all animals have a complex tissue structure with differentiated and specialized tissues. Most animals are motile, at least during certain life stages. All animals require a source of food and are, therefore, heterotrophic: ingesting other living or dead organisms. This feature distinguishes them from autotrophic organisms, such as most plants, which synthesize their own nutrients through photosynthesis. As heterotrophs, animals may be carnivores, herbivores, omnivores, or parasites. Most animals reproduce sexually with the offspring passing through a series of developmental stages that establish a fixed body plan. The body plan refers to the morphology of an animal, determined by developmental cues.

(a) (b)

Figure 27.1A.1: **Heterotrophs**: All animals are heterotrophs that derive energy from food. The (a) black bear is an omnivore, eating both plants and animals. The (b) heartworm Dirofilaria immitis is a parasite that derives energy from its hosts. It spends its larval stage in mosquitoes and its adult stage infesting the heart of dogs and other mammals.

27.1B: Complex Tissue Structure

Animals, besides Parazoa (sponges), are characterized by specialized tissues such as muscle, nerve, connective, and epithelial tissues.

Learning Objectives

- List the various specialized tissue types found in animals and describe their functions

Key Points

- Animal cells don't have cell walls; their cells may be embedded in an extracellular matrix and have unique structures for intercellular communication.
- Animals have nerve and muscle tissues, which provide coordination and movement; these are not present in plants and fungi.
- Complex animal bodies demand connective tissues made up of organic and inorganic materials that provide support and structure.
- Animals are also characterized by epithelial tissues, like the epidermis, which function in secretion and protection.
- The animal kingdom is divided into Parazoa (sponges), which do not contain true specialized tissues, and Eumetazoa (all other animals), which do contain true specialized tissues.

Key Terms

- **Parazoa**: a taxonomic subkingdom within the kingdom Animalia; the sponges
- **Eumetazoa**: a taxonomic subkingdom, within kingdom Animalia; all animals except the sponges
- **epithelial tissue**: one of the four basic types of animal tissue, which line the cavities and surfaces of structures throughout the body, and also form many glands

Complex Tissue Structure

As multicellular organisms, animals differ from plants and fungi because their cells don't have cell walls; their cells may be embedded in an extracellular matrix (such as bone, skin, or connective tissue); and their cells have unique structures for intercellular communication (such as gap junctions). In addition, animals possess unique tissues, absent in fungi and plants, which allow coordination (nerve tissue) and motility (muscle tissue). Animals are also characterized by specialized connective tissues that provide structural support for cells and organs. This connective tissue constitutes the extracellular surroundings of cells and is made up of organic and inorganic materials. In vertebrates, bone tissue is a type of connective tissue that supports the entire body structure. The complex bodies and activities of vertebrates demand such supportive tissues. Epithelial tissues cover, line, protect, and secrete; these tissues include the epidermis of the integument: the lining of the digestive tract and trachea. They also make up the ducts of the liver and glands of advanced animals.

The animal kingdom is divided into Parazoa (sponges) and Eumetazoa (all other animals). As very simple animals, the organisms in group Parazoa ("beside animal") do not contain true specialized tissues. Although they do possess specialized cells that perform different functions, those cells are not organized into tissues. These organisms are considered animals since they lack the ability to make their own food. Animals with true tissues are in the group Eumetazoa ("true animals"). When we think of animals, we usually think of Eumetazoans, since most animals fall into this category.

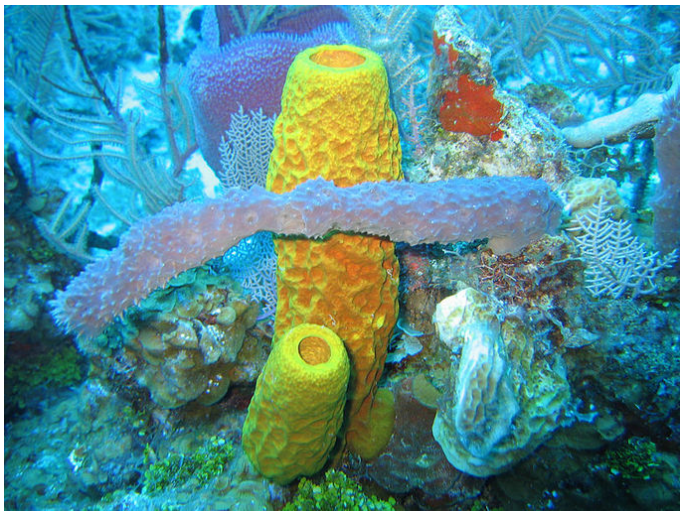

Figure 27.1B.1: **Sponges**: Sponges, such as those in the Caribbean Sea, are classified as Parazoans because they are very simple animals that do not contain true specialized tissues.

The different types of tissues in true animals are responsible for carrying out specific functions for the organism. This differentiation and specialization of tissues is part of what allows for such incredible animal diversity. For example, the evolution of nerve tissues and muscle tissues has resulted in animals' unique ability to rapidly sense and respond to changes in their environment. This allows animals to survive in environments where they must compete with other species to meet their nutritional demands.

27.1C: Animal Reproduction and Development

Most animals undergo sexual reproduction and have similar forms of development dictated by Hox genes.

Learning Objectives

- Explain the processes of animal reproduction and embryonic development

Key Points

- Most animals reproduce through sexual reproduction, but some animals are capable of asexual reproduction through parthenogenesis, budding, or fragmentation.
- Following fertilization, an embryo is formed, and animal tissues organize into organ systems; some animals may also undergo incomplete or complete metamorphosis.
- Cleavage of the zygote leads to the formation of a blastula, which undergoes further cell division and cellular rearrangement during a process called gastrulation, which leads to the formation of the gastrula.
- During gastrulation, the digestive cavity and germ layers are formed; these will later develop into certain tissue types, organs, and organ systems during a process called organogenesis.
- Hox genes are responsible for determining the general body plan, such as the number of body segments of an animal, the number and placement of appendages, and animal head-tail directionality.
- Hox genes, similar across most animals, can turn on or off other genes by coding transcription factors that control the expression of numerous other genes.

Key Terms

- **metamorphosis**: a change in the form and often habits of an animal after the embryonic stage during normal development
- **Hox gene**: genes responsible for determining the general body plan, such as the number of body segments of an animal, the number and placement of appendages, and animal head-tail directionality
- **blastula**: a 6-32-celled hollow structure that is formed after a zygote undergoes cell division

Animal Reproduction and Development

Most animals are diploid organisms (their body, or somatic, cells are diploid) with haploid reproductive (gamete) cells produced through meiosis. The majority of animals undergo sexual reproduction. This fact distinguishes animals from fungi, protists, and bacteria where asexual reproduction is common or exclusive. However, a few groups, such as cnidarians, flatworms, and roundworms, undergo asexual reproduction, although nearly all of those animals also have a sexual phase to their life cycle.

Processes of Animal Reproduction and Embryonic Development

During sexual reproduction, the haploid gametes of the male and female individuals of a species combine in a process called fertilization. Typically, the small, motile male sperm fertilizes the much larger, sessile female egg. This process produces a diploid fertilized egg called a zygote.

Some animal species (including sea stars and sea anemones, as well as some insects, reptiles, and fish) are capable of asexual reproduction. The most common forms of asexual reproduction for stationary aquatic animals include budding and fragmentation where part of a parent individual can separate and grow into a new individual. In contrast, a form of asexual reproduction found in certain insects and vertebrates is called parthenogenesis where unfertilized eggs can develop into new offspring. This type of parthenogenesis in insects is called haplodiploidy and results in male offspring. These types of asexual reproduction produce genetically identical offspring, which is disadvantageous from the perspective of evolutionary adaptability because of the potential buildup of deleterious mutations. However, for animals that are limited in their capacity to attract mates, asexual reproduction can ensure genetic propagation.

After fertilization, a series of developmental stages occur during which primary germ layers are established and reorganize to form an embryo. During this process, animal tissues begin to specialize and organize into organs and organ systems, determining their future morphology and physiology. Some animals, such as grasshoppers, undergo incomplete

metamorphosis, in which the young resemble the adult. Other animals, such as some insects, undergo complete metamorphosis where individuals enter one or more larval stages that may differ in structure and function from the adult. In complete metamorphosis, the young and the adult may have different diets, limiting competition for food between them. Regardless of whether a species undergoes complete or incomplete metamorphosis, the series of developmental stages of the embryo remains largely the same for most members of the animal kingdom.

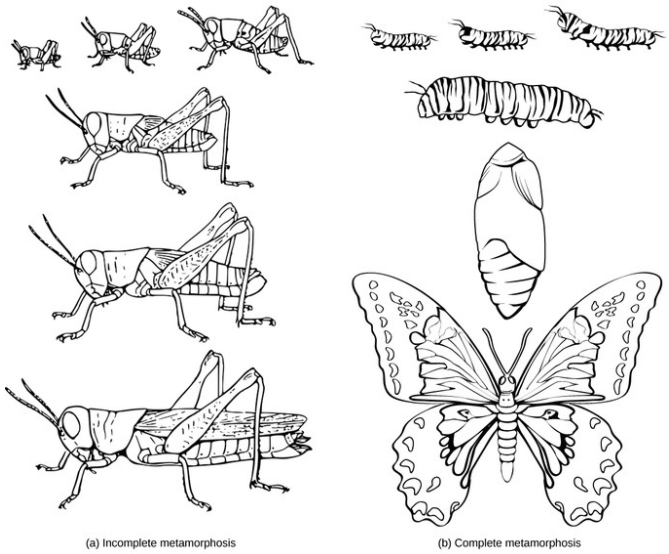

Figure 27.1C.1: **Incomplete and complete metamorphosis**: (a) The grasshopper undergoes incomplete metamorphosis. (b) The butterfly undergoes complete metamorphosis.

The process of animal development begins with the cleavage, or series of mitotic cell divisions, of the zygote. Three cell divisions transform the single-celled zygote into an eight-celled structure. After further cell division and rearrangement of existing cells, a 6–32-celled hollow structure called a blastula is formed. Next, the blastula undergoes further cell division and cellular rearrangement during a process called gastrulation. This leads to the formation of the next developmental stage, the gastrula, in which the future digestive cavity is formed. Different cell layers (called germ layers) are formed during gastrulation. These germ layers are programed to develop into certain tissue types, organs, and organ systems during a process called organogenesis.

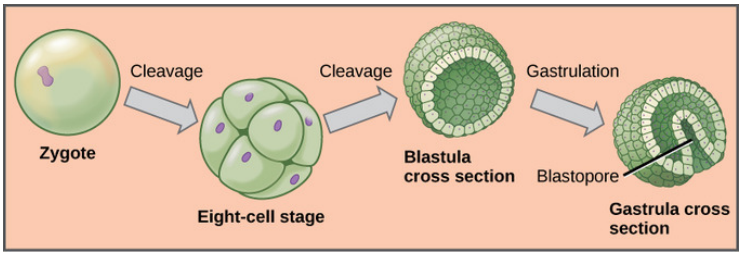

Figure 27.1C.1: **Embryonic development**: During embryonic development, the zygote undergoes a series of mitotic cell divisions, or cleavages, to form an eight-cell stage, then a hollow blastula. During a process called gastrulation, the blastula folds inward to form a cavity in the gastrula.

The Role of Homeobox (Hox) Genes in Animal Development

Since the early 19th century, scientists have observed that many animals, from the very simple to the complex, shared similar embryonic morphology and development. Surprisingly, a human embryo and a frog embryo, at a certain stage of embryonic development, appear remarkably similar. For a long time, scientists did not understand why so many animal species looked similar during embryonic development, but were very different as adults. Near the end of the 20th century, a particular class of genes that dictate developmental direction was discovered. These genes that determine animal structure are called “homeotic genes.” They contain DNA sequences called homeoboxes, with specific sequences referred to as Hox genes. This family of genes is responsible for determining the general body plan: the number of body segments of an animal, the number and placement of appendages, and animal head-tail directionality. The first Hox genes to be sequenced

were those from the fruit fly (*Drosophila melanogaster*). A single Hox mutation in the fruit fly can result in an extra pair of wings or even appendages growing from the "wrong" body part.

There are many genes that play roles in the morphological development of an animal, but Hox genes are so powerful because they can turn on or off large numbers of other genes. Hox genes do this by coding transcription factors that control the expression of numerous other genes. Hox genes are homologous in the animal kingdom: the genetic sequences and their positions on chromosomes are remarkably similar across most animals (e.g., worms, flies, mice, humans) because of their presence in a common ancestor. Hox genes have undergone at least two duplication events during animal evolution: the additional genes allowed more complex body types to evolve.

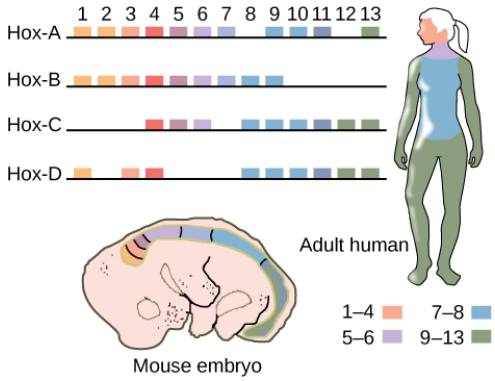

Figure 27.1C.1: **Hox genes**: Hox genes are highly-conserved genes encoding transcription factors that determine the course of embryonic development in animals. In vertebrates, the genes have been duplicated into four clusters: Hox-A, Hox-B, Hox-C, and Hox-D. Genes within these clusters are expressed in certain body segments at certain stages of development. Shown here is the homology between Hox genes in mice and humans. Note how Hox gene expression, as indicated with orange, pink, blue, and green shading, occurs in the same body segments in both the mouse and the human.

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: Features Used to Classify Animals

Topic hierarchy

- 27.2A: Animal Characterization Based on Body Symmetry
- 27.2B: Animal Characterization Based on Features of Embryological Development

27.2A: Animal Characterization Based on Body Symmetry

Animals can be classified by three types of body plan symmetry: radial symmetry, bilateral symmetry, and asymmetry.

Learning Objectives

- Differentiate among the ways in which animals can be characterized by body symmetry

Key Points

- Animals with radial symmetry have no right or left sides, only a top or bottom; these species are usually marine organisms like jellyfish and corals.
- Most animals are bilaterally symmetrical with a line of symmetry dividing their body into left and right sides along with a "head" and "tail" in addition to a top and bottom.
- Only sponges (phylum Porifera) have asymmetrical body plans.
- Some animals start life with one type of body symmetry, but develop a different type as adults; for example, sea stars are classified as bilaterally symmetrical even though their adult forms are radially symmetrical.

Key Terms

- **sagittal plane**: divides the body into right and left halves
- **radial symmetry**: a form of symmetry wherein identical parts are arranged in a circular fashion around a central axis
- **bilateral symmetry**: having equal arrangement of parts (symmetry) about a vertical plane running from head to tail

Animal Characterization Based on Body Symmetry

At a very basic level of classification, true animals can be largely divided into three groups based on the type of symmetry of their body plan: radially symmetrical, bilaterally symmetrical, and asymmetrical. Only a few animal groups display radial symmetry, while asymmetry is a unique feature of phyla Porifera (sponges). All types of symmetry are well suited to meet the unique demands of a particular animal's lifestyle.

Radial Symmetry

Radial symmetry is the arrangement of body parts around a central axis, like rays on a sun or pieces in a pie. Radially symmetrical animals have top and bottom surfaces, but no left and right sides, or front and back. The two halves of a radially symmetrical animal may be described as the side with a mouth ("oral side") and the side without a mouth ("aboral side"). This form of symmetry marks the body plans of animals in the phyla Ctenophora (comb jellies) and Cnidaria (corals, sea anemones, and other jellies). Radial symmetry enables these sea creatures, which may be sedentary or only capable of slow movement or floating, to experience the environment equally from all directions.

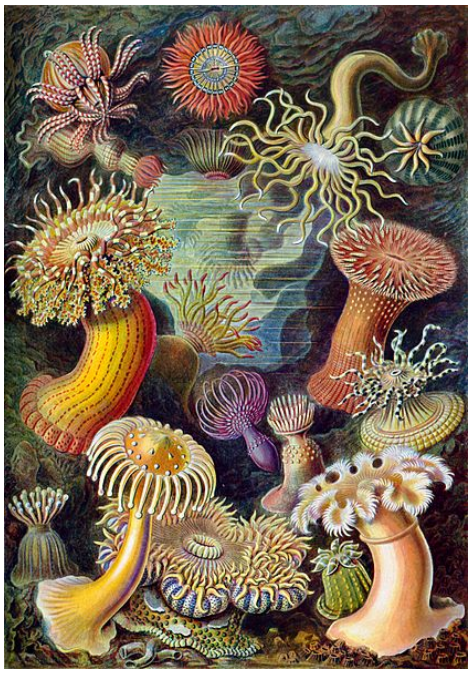

Figure 27.2A.1: **Radial symmetry**: Some organisms, like sea anemones (phylum Cnidaria), have radial symmetry.

Bilateral Symmetry

Bilateral symmetry involves the division of the animal through a sagittal plane, resulting in two mirror-image, right and left halves, such as those of a butterfly, crab, or human body. Animals with bilateral symmetry have a "head" and "tail" (anterior vs. posterior), front and back (dorsal vs. ventral), and right and left sides. All true animals, except those with radial symmetry, are bilaterally symmetrical. The evolution of bilateral symmetry and, therefore, the formation of anterior and posterior (head and tail) ends promoted a phenomenon called cephalization, which refers to the collection of an organized nervous system at the animal's anterior end. In contrast to radial symmetry, which is best suited for stationary or limited-motion lifestyles, bilateral symmetry allows for streamlined and directional motion. In evolutionary terms, this simple form of symmetry promoted active mobility and increased sophistication of resource-seeking and predator-prey relationships.

Figure 27.2A.1: **Bilateral symmetry**: This monarch butterfly demonstrates bilateral symmetry down the sagittal plane, with the line of symmetry running from ventral to dorsal and dividing the body into two left and right halves.

Animals in the phylum Echinodermata (such as sea stars, sand dollars, and sea urchins) display radial symmetry as adults, but their larval stages exhibit bilateral symmetry. This is termed secondary radial symmetry. They are believed to have evolved from bilaterally symmetrical animals; thus, they are classified as bilaterally symmetrical.

Figure 27.2A.1: **Secondary radial symmetry in echinoderms**: The larvae of echinoderms (sea stars, sand dollars, and sea urchins) have bilateral symmetry as larvae, but develop radial symmetry as full adults.

Asymmetry

Only members of the phylum Porifera (sponges) have no body plan symmetry. There are some fish species, such as flounder, that lack symmetry as adults. However, the larval fish are bilaterally symmetrical.

27.2B: Animal Characterization Based on Features of Embryological Development

Animals may be characterized by the presence of a coelom, formation of the mouth, and type of cell cleavage during embryonic development.

Learning Objectives

- Explain the ways in which animals can be characterized by features of embryological development

Key Points

- Diploblasts contain two germ layers (inner endoderm and outer ectoderm), while triploblasts contain three germ layers (endoderm, mesoderm, and ectoderm).
- The endoderm becomes the digestive and respiratory tracts; the ectoderm becomes the outer epithelial covering of the body surface and the central nervous system; and the mesoderm becomes all muscle tissues, connective tissues, and most other organs.
- Triploblasts can be further categorized into those without a coelom (acoelomates), those with a true coelom (eucoelomates), and those with “false” coeloms (pseudocoelomates).
- Bilaterally symmetrical, tribloblastic eucoelomates can be divided into protostomes, those animals that develop a mouth first, and deuterstomes, those animals that develop an anus first and a mouth second.
- In protostomes, the coelom forms when the mesoderm splits through the process of schizocoely, while in deuterostomes, the coelom forms when the mesoderm pinches off through the process of enterocoely.
- Protostomes undergo spiral cleavage, while deuterostomes undergo radial cleavage.

Key Terms

- **protostome**: any animal in which the mouth is derived first from the embryonic blastopore (“mouth first”)
- **deuterostome**: Any animal in which the initial pore formed during gastrulation becomes the anus, and the second pore becomes the mouth
- **diploblast**: a blastula in which there are two primary germ layers: the ectoderm and endoderm
- **triploblast**: a blastula in which there are three primary germ layers: the ectoderm, mesoderm, and endoderm; formed during gastrulation of the blastula
- **acoelomate**: any animal without a coelom, or body cavity
- **coelomate**: any animal possessing a fluid-filled cavity within which the digestive system is suspended.
- **schizocoely**: the process by which protostome animal embryos develop; it occurs when a coelom (body cavity) is formed by splitting the mesodermal embryonic tissue
- **enterocoely**: the process by which deuterostome animal embryos develop; the coelom forms from pouches “pinched” off of the digestive tract

Animal Characterization Based on Features of Embryological Development

Most animal species undergo a separation of tissues into germ layers during embryonic development. These germ layers are formed during gastrulation, developing into the animal’s specialized tissues and organs. Animals develop either two or three embryonic germs layers. Radially-symmetrical animals are diploblasts, developing two germ layers: an inner layer (endoderm) and an outer layer (ectoderm). Diploblasts have a non-living layer between the endoderm and ectoderm. Bilaterally-symmetrical animals are called triploblasts, developing three tissue layers: an inner layer (endoderm), an outer layer (ectoderm), and a middle layer (mesoderm).

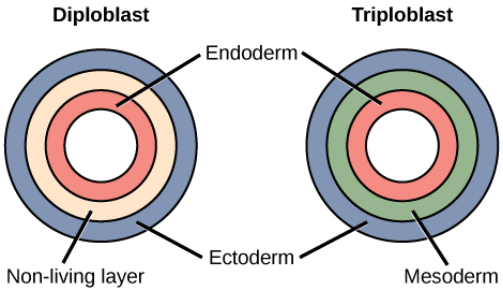

Figure 27.2B.1: **Germ development in embryogenesis**: During embryogenesis, diploblasts develop two embryonic germ layers: an ectoderm and an endoderm. Triploblasts develop a third layer, the mesoderm, between the endoderm and ectoderm

Germ Layers

Each of the three germ layers in a blastula, or developing ball of cells, becomes particular body tissues and organs. The endoderm gives rise to the stomach, intestines, liver, pancreas, and the lining of the digestive tract, as well as to the lining of the trachea, bronchi, and lungs of the respiratory tract. The ectoderm develops into the outer epithelial covering of the body surface and the central nervous system. The mesoderm, the third germ layer forming between the endoderm and ectoderm in triploblasts, gives rise to all muscle tissues (including the cardiac tissues and muscles of the intestines), connective tissues such as the skeleton and blood cells, and most other visceral organs such as the kidneys and the spleen.

Presence or Absence of a Coelom

Triploblasts can be differentiated into three categories: those that do not develop an internal body cavity called a coelom (acoelomates), those with a true coelom (eucoelomates), and those with "false" coeloms (pseudocoelomates).

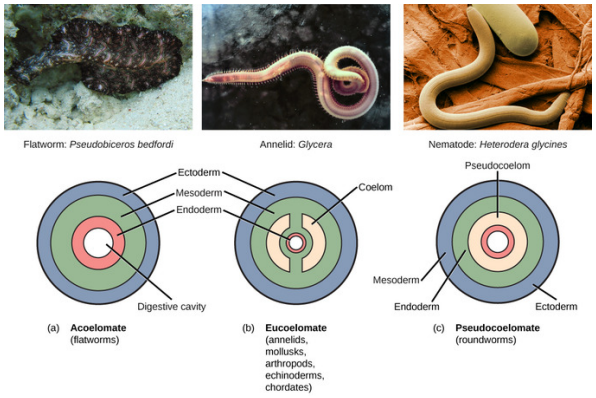

Figure 27.2B.1: **Differentiation in triploblasts**: Triploblasts may be (a) acoelomates, (b) eucoelomates, or (c) pseudocoelomates. Acoelomates have no body cavity. Eucoelomates have a body cavity within the mesoderm, called a coelom, which is lined with mesoderm. Pseudocoelomates also have a body cavity, but it is sandwiched between the endoderm and mesoderm.

Acoelomates

Triploblasts that do not develop a coelom are called acoelomates: their mesoderm region is completely filled with tissue. Flatworms in the phylum Platyhelminthes are acoelomates.

Eucoelomates

Eucoelomates (or coelomates) have a true coelom that arises entirely within the mesoderm germ layer and is lined by an epithelial membrane. This coelomic cavity represents a fluid-filled space that lies between the visceral organs and the body wall. It houses the digestive system, kidneys, reproductive organs, and heart, and it contains the circulatory system. The epithelial membrane also lines the organs within the coelom, connecting and holding them in position while allowing them some free motion. Annelids, mollusks, arthropods, echinoderms, and chordates are all eucoelomates. The coelom also provides space for the diffusion of gases and nutrients, as well as body flexibility and improved animal motility. The

coelom also provides cushioning and shock absorption for the major organ systems, while allowing organs to move freely for optimal development and placement.

Pseudocoelomates

The pseudocoelomates have a coelom derived partly from mesoderm and partly from endoderm. Although still functional, these are considered false coeloms. The phylum Nematoda (roundworms) is an example of a pseudocoelomate.

Embryonic Development of the Mouth

Bilaterally symmetrical, tribloblastic eucoelomates can be further divided into two groups based on differences in their early embryonic development. These two groups are separated based on which opening of the digestive cavity develops first: mouth (protostomes) or anus (deuterostomes). The word protostome comes from the Greek word meaning “mouth first. ” The protostomes include arthropods, mollusks, and annelids. Deuterostome originates from the word meaning “mouth second. ” Deuterostomes include more complex animals such as chordates, but also some simple animals such as echinoderms.

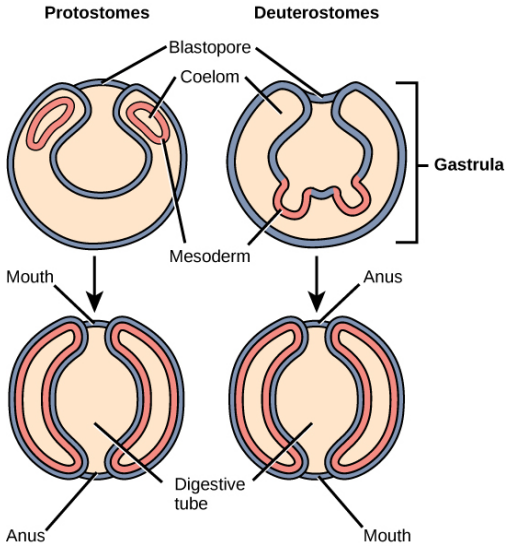

Figure 27.2B.1: **Early embryonic development in eucoelomates**: Eucoelomates can be divided into two groups based on their early embryonic development. In protostomes, part of the mesoderm separates to form the coelom in a process called schizocoely. In deuterostomes, the mesoderm pinches off to form the coelom in a process called enterocoely.

Development of the Coelom

The coelom of most protostomes is formed through a process called schizocoely, when a solid mass of the mesoderm splits apart and forms the hollow opening of the coelom. Deuterostomes differ in that their coelom forms through a process called enterocoely, when the mesoderm develops as pouches that are pinched off from the endoderm tissue. These pouches eventually fuse to form the mesoderm, which then gives rise to the coelom.

Embryonic Cleavage

Protostomes undergo spiral cleavage: the cells of one pole of the embryo are rotated and, thus, misaligned with respect to the cells of the opposite pole. This spiral cleavage is due to the oblique angle of the cleavage. Protostomes also undergo determinate cleavage: the developmental fate of each embryonic cell is pre-determined. Deuterostomes undergo radial cleavage where the cleavage axes are either parallel or perpendicular to the polar axis, resulting in the alignment of the cells between the two poles. Unlike protostomes, deuterostomes undergo indeterminate cleavage: cells remain undifferentiated until a later developmental stage. This characteristic of deuterostomes is reflected in the existence of familiar embryonic stem cells, which have the ability to develop into any cell type.

Topic hierarchy

27.3A: Constructing an Animal Phylogenetic Tree

27.3B: Molecular Analyses and Modern Phylogenetic Trees

27.3A: Constructing an Animal Phylogenetic Tree

Phylogenetic trees are constructed according to the evolutionary relationships that exist between organisms based on homologous traits.

Learning Objectives

- Describe the information needed to construct a phylogenetic tree of animals

Key Points

- Phylogenetic trees are constructed using various data derived from studies on homologous traits, analagous traits, and molecular evidence that can be used to establish relationships using polymeric molecules (DNA, RNA, and proteins).
- Evolutionary relationships between animal phyla, or Metazoa, are based on the the presence or absence of differentiated tissues, referred to as Eumetazoa or Parazoa, respectively.
- Eumetazoa can be further classified into categories that are based on whether they have radial or bilateral symmetry, referred to as Radiata or Bilateria, respectively.

Key Terms

- **orthologous**: having been separated by a speciation event
- **homoplasy**: a correspondence between the parts or organs of different species acquired as the result of parallel evolution or convergence

Constructing an Animal Phylogenetic Tree

Evolutionary trees, or phylogeny, is the formal study of organisms and their evolutionary history with respect to each other. Phylogenetic trees are most-commonly used to depict the relationships that exist between species. In particular, they clarify whether certain traits are homologous (found in the common ancestor as a result of divergent evolution) or homoplasy (sometimes referred to as analogous: a character that is not found in a common ancestor, but whose function developed independently in two or more organisms through convergent evolution). Evolutionary trees are diagrams that show various biological species and their evolutionary relationships. They consist of branches that flow from lower forms of life to the higher forms of life.

Evolutionary trees differ from taxonomy which is an ordered division of organisms into categories based on a set of characteristics used to assess similarities and differences. Evolutionary trees involve biological classification and use morphology to show relationships. Phylogeny is evolutionary history shown by the relationships found when comparing polymeric molecules such as RNA, DNA, or proteins of various organisms. The evolutionary pathway is analyzed by the sequence similarity of these polymeric molecules. This is based on the assumption that the similarities of sequence result from having fewer evolutionary divergences than others. The evolutionary tree is constructed by aligning the sequences; the length of the branch is proportional to the amount of amino acid differences between the sequences.

Phylogenetic systematics informs the construction of phylogenetic trees based on shared characters. Comparing nucleic acids or other molecules to infer relationships is a valuable tool for tracing an organism’s evolutionary history. The ability of molecular trees to encompass both short and long periods of time is hinged on the ability of genes to evolve at different rates, even in the same evolutionary lineage. For example, the DNA that codes for rRNA changes relatively slowly, so comparisons of DNA sequences in these genes are useful for investigating relationships between taxa that diverged a long time ago. Interestingly, 99% of the genes in humans and mice are detectably orthologous, and 50% of our genes are orthologous with those of yeast. The hemoglobin B genes in humans and in mice are orthologous. These genes serve similar functions, but their sequences have diverged since the time that humans and mice had a common ancestor.

Evolutionary pathways relating the members of a family of proteins may be deduced by examination of sequence similarity. This approach is based on the notion that sequences that are more similar to one another have had less evolutionary time to diverge than have sequences that are less similar. Evolutionary trees are used today for DNA hybridization, which determines the percentage difference of genetic material between two similar species. If there is a high resemblance of DNA between the two species, then the species are closely related. If only a small percentage is identical, then they are distantly related.

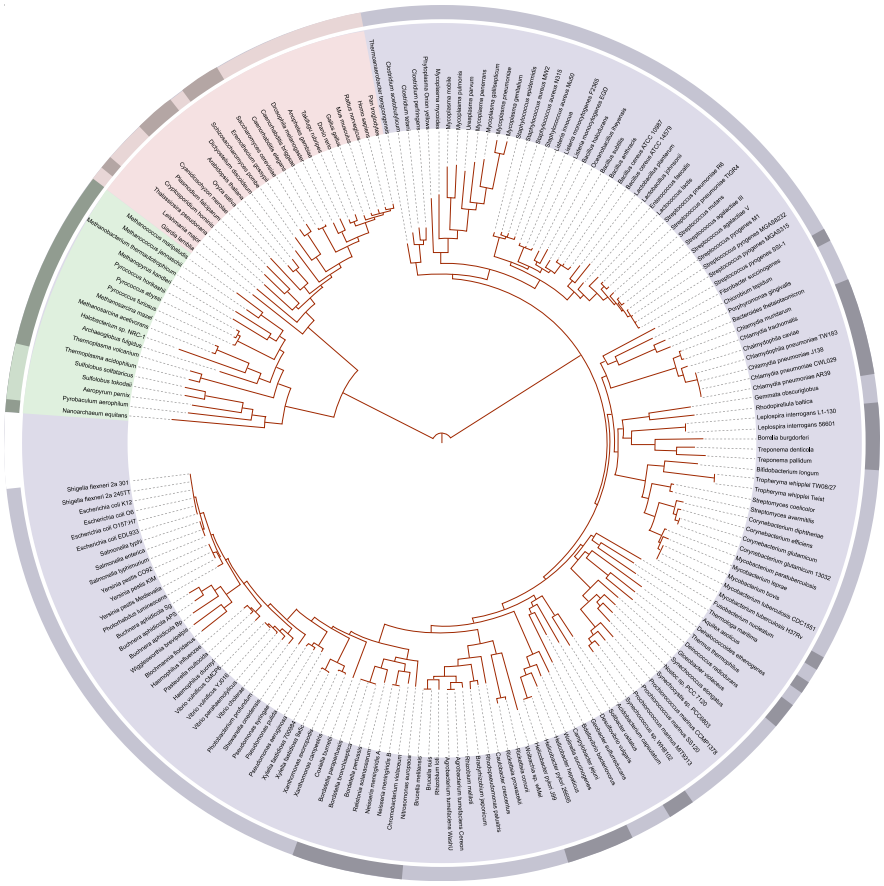

Figure 27.3*A*.1: **Phylogenetic tree of life**: A phylogenetic tree of life, showing the relationship between species whose genomes had been sequenced as of 2006. The very center represents the last universal ancestor of all life on earth. The different colors represent the three domains of life: pink represents eukaryota (animals, plants, and fungi); blue represents bacteria; and green represents archaea.

Animal Phyla

The current understanding of evolutionary relationships between animal, or Metazoa, phyla begins with the distinction between "true" animals with true differentiated tissues, called Eumetazoa, and animal phyla that do not have true differentiated tissues (such as the sponges), called Parazoa. Both Parazoa and Eumetazoa evolved from a common ancestral organism that resembles the modern-day protists called choanoflagellates. These protist cells strongly resemble sponge choanocyte cells.

Eumetazoa are subdivided into radially-symmetrical animals and bilaterally-symmetrical animals and are classified into clade Radiata or Bilateria, respectively. The cnidarians and ctenophores are animal phyla with true radial symmetry. All other Eumetazoa are members of the Bilateria clade. The bilaterally-symmetrical animals are further divided into deuterostomes (including chordates and echinoderms) and two distinct clades of protostomes (including ecdysozoans and lophotrochozoans). Ecdysozoa includes nematodes and arthropods; named for a commonly-found characteristic among the group: exoskeletal molting (termed ecdysis). Lophotrochozoa is named for two structural features, each common to certain phyla within the clade. Some lophotrochozoan phyla are characterized by a larval stage called trochophore larvae, and other phyla are characterized by the presence of a feeding structure called a lophophore.

27.3B: Molecular Analyses and Modern Phylogenetic Trees

The process of establishing relationships between organisms is increasingly becoming more accurate due to advances in molecular analysis.

Learning Objectives

- Distinguish between morphological and molecular data in creating phylogenetic trees of animals

Key Points

- The construction of phylogenetic trees is now based on similarities and differences within the molecular sources used for analysis which include DNA, RNA, and proteins.
- The ability to use molecular sources as a basis of phylogenetic tree construction has allowed for determination of previously-unknown evolutionary relationships between organisms.
- In addition to the establishment of new relationships within phylogenetic trees, the ability to use molecular sources for analysis has also created an emergence of new phlyums that were previously classified in larger groups.
- Besides identifying molecular similarities and differences between organisms, by assigning a constant mutation rate to a sequence and performing a sequence alignment, it is possible to determine when two organisms diverged from one another.

Key Terms

- **monophyletic**: of, pertaining to, or affecting a single phylum (or other taxon) of organisms

Modern Advances in Phylogenetic Understanding Come from Molecular Analyses

The phylogenetic groupings are continually being debated and refined by evolutionary biologists. Each year, new evidence emerges that further alters the relationships described by a phylogenetic tree diagram. Previously, phylogenetic trees were constructed based on homologous and analogous morphology; however, with the advances in molecular biology, construction of phylogenetic trees is increasingly performed using data derived from molecular analyses.

Many evolutionary relationships in the modern tree have only recently been determined due to molecular evidence. Nucleic acid and protein analyses have informed the construction of the modern phylogenetic animal tree. These data come from a variety of molecular sources, such as mitochondrial DNA, nuclear DNA, ribosomal RNA (rRNA), and certain cellular proteins. Evolutionary trees can be made by the determination of sequence information of similar genes in different organisms. Sequences that are similar to each other frequently are considered to have less time to diverge, while less similar sequences have more evolutionary time to diverge. The evolutionary tree is created by aligning sequences and having each branch length proportional to the amino acid differences of the sequences. Furthermore, by assigning a constant mutation rate to a sequence and performing a sequence alignment, it is possible to calculate the approximate time when the sequence of interest diverged into monophyletic groups.

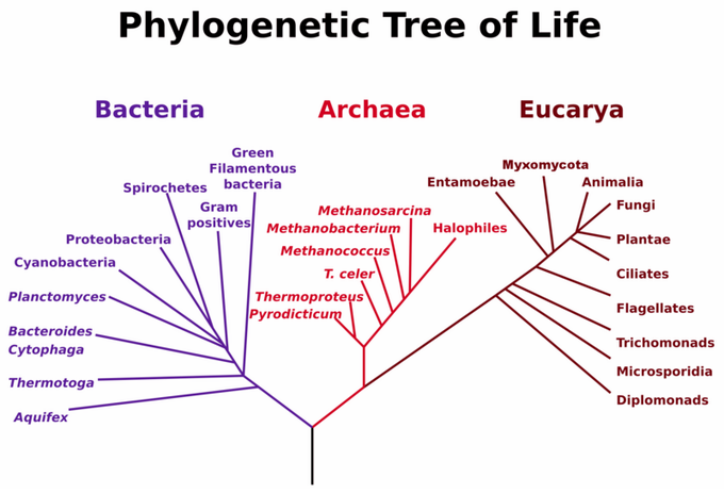

Figure 27.3B.1: **Phlyogenetic tree of life**: Advances in molecular biology and analysis of polymeric molecules such as DNA, RNA, and proteins have contributed to the development of phylogenetic trees.

Sequence alignments can be performed on a variety of sequences. For constructing an evolutionary tree from proteins, for example, the sequences are aligned and then compared. rRNA (ribosomal RNA) is typically used to compare organisms since rRNA has a slower mutation rate and is a better source for evolutionary tree construction. This is best supported by research of Dr. Carl Woese that was conducted in the late 1970s. Since the ribosomes are critical to the function of living organisms, they are not easily changed through the process of evolution. Taking advantage of this fact, Dr. Woese compared the minuscule differences in the sequences of ribosomes among a great array of bacteria and showed that they were not all related.

For example, a previously-classified group of animals called lophophorates, which included brachiopods and bryozoans, were long-thought to be primitive deuterostomes. Extensive molecular analysis using rRNA data found these animals to be protostomes, more closely related to annelids and mollusks. This discovery allowed for the distinction of the protostome clade: the lophotrochozoans. Molecular data have also shed light on some differences within the lophotrochozoan group. Some scientists believe that the phyla Platyhelminthes and Rotifera within this group should actually belong to their own group of protostomes termed Platyzoa.

Molecular research similar to the discoveries that brought about the distinction of the lophotrochozoan clade has also revealed a dramatic rearrangement of the relationships between mollusks, annelids, arthropods, and nematodes; a new ecdysozoan clade was formed. Due to morphological similarities in their segmented body types, annelids and arthropods were once thought to be closely related. However, molecular evidence has revealed that arthropods are actually more closely related to nematodes, now comprising the ecdysozoan clade, and annelids are more closely related to mollusks, brachiopods, and other phyla in the lophotrochozoan clade. These two clades now make up the protostomes.

Another change to former phylogenetic groupings because of molecular analyses includes the emergence of an entirely new phylum of worm called Acoelomorpha. These acoel flatworms were long thought to belong to the phylum Platyhelminthes because of their similar “flatworm” morphology. However, molecular analyses revealed this to be a false relationship and originally suggested that acoels represented living species of some of the earliest divergent bilaterians. More recent research into the acoelomorphs has called this hypothesis into question and suggested a closer relationship with deuterostomes. The placement of this new phylum remains disputed, but scientists agree that with sufficient molecular data, their true phylogeny will be determined.

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27.4: The Evolutionary History of the Animal Kingdom

Topic hierarchy

- 27.4A: Pre-Cambrian Animal Life
- 27.4B: The Cambrian Explosion of Animal Life
- 27.4C: Post-Cambrian Evolution and Mass Extinctions

27.4A: Pre-Cambrian Animal Life

Early animal life (Ediacaran biota) evolved from protists during the pre-Cambrian period, which is also known as the Ediacaran period.

Learning Objectives

- Describe the types of animals found in the Ediacaran period

Key Points

- The pre-Cambrian period (Ediacaran period) took place between 635-543 million years ago.
- Early animal life, called Ediacaran biota, evolved from protists; it was previously believed early animal life included only tiny, sessile, soft-bodied sea creatures, but scientific evidence suggests more complex animals lived during this time.
- Sponge-like fossils believed to represent the oldest animals with hard body parts, named *Coronacollina acula*, date back as far as 560 million years.
- The fossils of the earliest animal species ever found were small, one-centimeter long, sponge-like creatures, dating before 650 million years, which predates the Ediacaran period.
- The discovery of the fossils of the earliest animal species provided evidence that animals may have evolved before the Ediacaran period during the Cryogenian period.

Key Terms

- **Ediacaran period**: period from about 635-543 million years ago; the final period of the late Proterozoic Neoproterozoic Era
- **choanoflagellate**: any of a group of flagellate protozoa thought to be the closest unicellular ancestors of animals
- **Coronacollina acula**: sponge-like fossils believed to represent the oldest animals with hard body parts that date back as far as 560 million years

Pre-Cambrian Animal Life

The time before the Cambrian period is known as the Ediacaran period (between 635-543 million years ago), the final period of the late Proterozoic Neoproterozoic Era. It is believed that early animal life, termed Ediacaran biota, evolved from protists at this time. Some protist species called choanoflagellates closely resemble the choanocyte cells in the simplest animals, sponges. In addition to their morphological similarity, molecular analyses have revealed similar sequence homologies in their DNA.

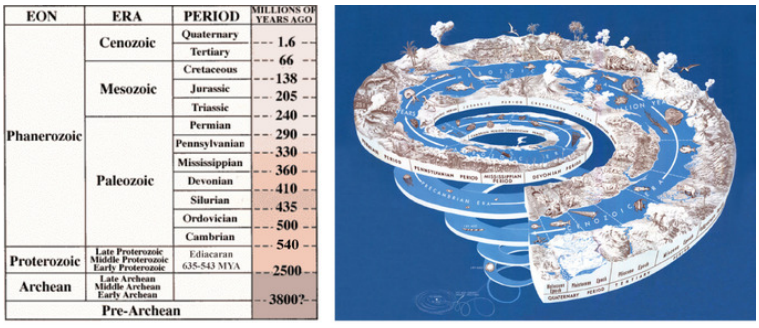

Figure 27.4A.1: **Earth's history**: (a) Earth's history is divided into eons, eras, and periods. The Ediacaran period was the final period of the Proterozoic Era which ended in the Cambrian period of the Phanerozoic Era. (b) Stages on the geological time scale are represented as a spiral.

The earliest life comprising Ediacaran biota was long believed to include only tiny, sessile, soft-bodied sea creatures. However, recently there has been increasing scientific evidence suggesting that more varied and complex animal species lived during this time, and possibly even before the Ediacaran period.

Fossils believed to represent the oldest animals with hard body parts were recently discovered in South Australia. These sponge-like fossils, named *Coronacollina acula*, date back as far as 560 million years. They are believed to show the existence

of hard body parts and spicules that extended 20–40 cm from the main body (estimated about 5 cm long). Other organisms, such as *Cyclomedusa* and *Dickinsonia*, also evolved during the Ediacaran period.

(a) (b)

Figure 27.4A.1: **Fossils from Ediacaran period**: Fossils of (a) *Cyclomedusa* and (b) *Dickinsonia* that evolved during the Ediacaran period.

Another recent fossil discovery may represent the earliest animal species ever found. While the validity of this claim is still under investigation, these primitive fossils appear to be small, one-centimeter long, sponge-like creatures. These fossils from South Australia date back 650 million years, actually placing the putative animal before the great ice age extinction event that marked the transition between the Cryogenian period and the Ediacaran period. Until this discovery, most scientists believed that there was no animal life prior to the Ediacaran period. Many scientists now believe that animals may, in fact, have evolved during the Cryogenian period.

27.4B: The Cambrian Explosion of Animal Life

During the Cambrian period, the most rapid evolution of new animal species occurred, but the cause of this explosion is still unknown.

Learning Objectives

- Compare the theories that attempt to explain the Cambrian Explosion

Key Points

- Echinoderms, mollusks, worms, chordates, and arthropods (including arthropods called trilobites which were the one of the first species to exhibit a sense of vision) developed during the Cambrian period.
- Environmental changes such as rising levels of atmospheric oxygen and an increase in oceanic calcium concentrations may have caused The Cambrian Explosion.
- A continental shelf with numerous shallow pools that provided the necessary living space for larger numbers of different types of animals to co-exist may have caused the Cambrian Explosion.
- The Cambrian Explosion may have been a result of ecological relationships between species, such as changes in the food web, competition for food and space, and predator-prey relationships.
- The evolution of Hox control genes resulting in animal complexity and flexibility may have provided the necessary opportunities for increases in possible animal morphologies.

Key Terms

- **Ordovician period**: covers the time between 485-443 million years ago; followed the Cambrian period
- **Cambrian explosion**: the relatively rapid appearance (over a period of many millions of years), around 530 million years ago, of most major animal phyla as demonstrated in the fossil record

The Cambrian Explosion of Animal Life

The Cambrian period, occurring between approximately 542–488 million years ago, marks the most rapid evolution of new animal phyla and animal diversity in earth's history. It is believed that most of the animal phyla in existence today had their origins during this time, often referred to as the Cambrian explosion. Echinoderms, mollusks, worms, arthropods, and chordates arose during this period. One of the most dominant species during the Cambrian period was the trilobite, an arthropod that was among the first animals to exhibit a sense of vision.

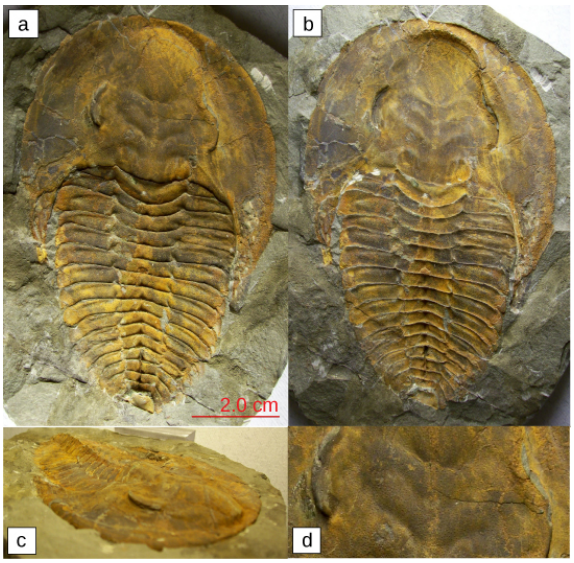

Figure $27.4B.1$: **Trilobites**: These fossils (a–d) belong to trilobites, extinct arthropods that appeared in the early Cambrian period 525 million years ago and disappeared from the fossil record during a mass extinction at the end of the Permian period about 250 million years ago.

Figure 27.4*B*.1: **Cambrian period**: An artist's rendition depicts some organisms from the Cambrian period.

The causes of the Cambrian explosion are still debated. There are many theories that attempt to answer this question. Environmental changes may have created a more suitable environment for animal life. Examples of these changes include rising atmospheric oxygen levels and large increases in oceanic calcium concentrations that preceded the Cambrian period. Some scientists believe that an expansive, continental shelf with numerous shallow lagoons or pools provided the necessary living space for larger numbers of different types of animals to co-exist. There is also support for theories that argue that ecological relationships between species, such as changes in the food web, competition for food and space, and predator-prey relationships, were primed to promote a sudden, massive coevolution of species. Yet other theories claim genetic and developmental reasons for the Cambrian explosion. The morphological flexibility and complexity of animal development afforded by the evolution of Hox control genes may have provided the necessary opportunities for increases in possible animal morphologies at the time of the Cambrian period. Theories that attempt to explain why the Cambrian explosion happened must be able to provide valid reasons for the massive animal diversification, as well as explain why it happened when it did. There is evidence that both supports and refutes each of the theories described above. The answer may very well be a combination of these and other theories.

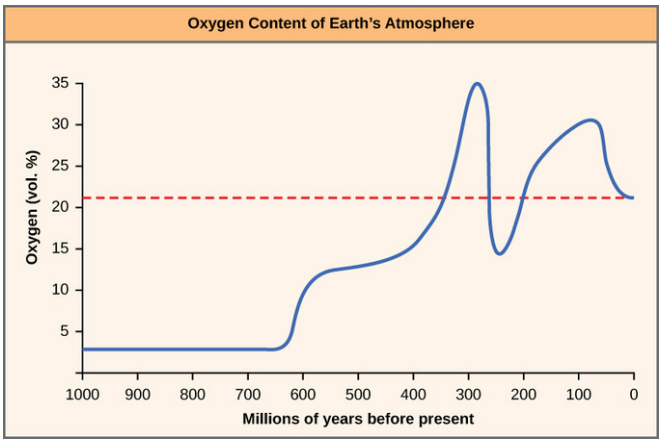

Figure 27.4*B*.1: **Earth's oxygen concentration**: The oxygen concentration in earth's atmosphere rose sharply around 300 million years ago.

Unresolved questions about the animal diversification that took place during the Cambrian period remain. For example, we do not understand how the evolution of so many species occurred in such a short period of time. Was there really an "explosion" of life at this particular time? Some scientists question the validity of this idea because there is increasing evidence to suggest that more animal life existed prior to the Cambrian period and that other similar species' so-called explosions (or radiations) occurred later in history as well. Furthermore, the vast diversification of animal species that appears to have begun during the Cambrian period continued well into the following Ordovician period. Despite some of

these arguments, most scientists agree that the Cambrian period marked a time of impressively-rapid animal evolution and diversification that is unmatched elsewhere during history.

27.4C: Post-Cambrian Evolution and Mass Extinctions

The post-Cambrian era was characterized by animal evolution and diversity where mass extinctions were followed by adaptive radiations.

Learning Objectives

- Differentiate among the causes of mass extinctions and their effects on animal life

Key Points

- During the Ordovician period, plant life first appeared on land, which allowed aquatic animals to move on to land.
- Periods of mass extinction caused by cataclysmic events like volcanic eruptions and meteor strikes have erased many genetic lines and created room for new species.
- The largest mass extinction event in earth's history, which occurred at the end of the Permian period, resulted in a loss of roughly 95 percent of the existing species at that time.
- The disappearance of some dominant species of Permian reptiles and the warm and stable climate that followed made it possible for the dinosaurs to emerge and diversify.
- Another mass extinction event caused by a meteor strike and volcanic ash eruption occurred at the end of the Cretaceous period, bringing the Mesozoic Era to an end and pushing dinosaurs into extinction.
- The disappearance of dinosaurs led to the dominance of plants, which created new niches for birds, insects, and mammals; animal diversity was also brought on by the creation of continents, islands, and mountains.

Key Terms

- **Cenozoic**: a geologic era about between 65 million years ago to the present when the continents moved to their current position and modern plants and animals evolved
- **mass extinction**: a sharp decrease in the total number of species in a relatively short period of time
- **Cretaceous**: the last geologic period within the Mesozoic era from about 146 to 65 million years ago; ended with a large mass extinction

Post-Cambrian Evolution and Mass Extinctions

The periods that followed the Cambrian during the Paleozoic Era were marked by further animal evolution and the emergence of many new orders, families, and species. As animal phyla continued to diversify, new species adapted to new ecological niches. During the Ordovician period, which followed the Cambrian period, plant life first appeared on land. This change allowed formerly-aquatic animal species to invade land, feeding directly on plants or decaying vegetation. Continual changes in temperature and moisture throughout the remainder of the Paleozoic Era due to continental plate movements encouraged the development of new adaptations to terrestrial existence in animals, such as limbs in amphibians and epidermal scales in reptiles.

Changes in the environment often create new niches (living spaces) that contribute to rapid speciation and increased diversity. On the other hand, cataclysmic events, such as volcanic eruptions and meteor strikes that obliterate life, can result in devastating losses of diversity. Such periods of mass extinction have occurred repeatedly in the evolutionary record of life, erasing some genetic lines while creating room for others to evolve into the empty niches left behind. The end of the Permian period (and the Paleozoic Era) was marked by the largest mass extinction event in Earth's history, a loss of roughly 95 percent of the extant species at that time. Some of the dominant phyla in the world's oceans, such as the trilobites, disappeared completely. On land, the disappearance of some dominant species of Permian reptiles made it possible for a new line of reptiles to emerge: the dinosaurs. The warm and stable climatic conditions of the ensuing Mesozoic Era promoted an explosive diversification of dinosaurs into every conceivable niche in land, air, and water. Plants, too, radiated into new landscapes and empty niches, creating complex communities of producers and consumers, some of which became extremely large on the abundant food available.

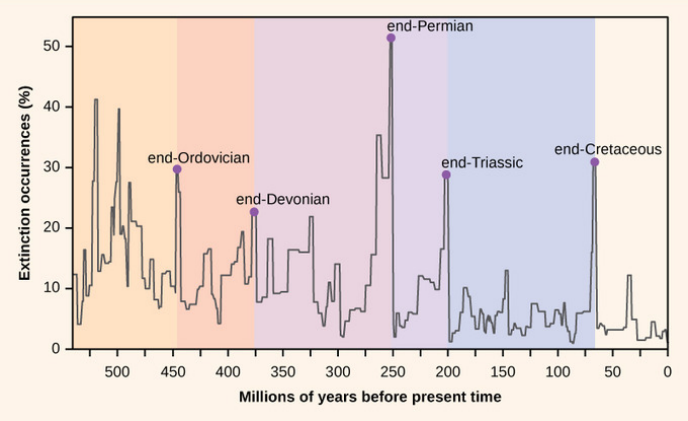

Figure 27.4C.1: **Mass extinctions**: Mass extinctions have occurred repeatedly over geological time.

Another mass extinction event occurred at the end of the Cretaceous period, bringing the Mesozoic Era to an end. Skies darkened and temperatures fell as a large meteor impact expelled tons of volcanic ash, blocking incoming sunlight. Plants died, herbivores and carnivores starved, and the mostly cold-blooded dinosaurs ceded their dominance of the landscape to more warm-blooded mammals. In the following Cenozoic Era, mammals radiated into terrestrial and aquatic niches once occupied by dinosaurs. Birds, the warm-blooded offshoots of one line of the ruling reptiles, became aerial specialists. The appearance and dominance of flowering plants in the Cenozoic Era created new niches for insects, as well as for birds and mammals. Changes in animal species diversity during the late Cretaceous and early Cenozoic were also promoted by a dramatic shift in earth's geography, as continental plates slid over the crust into their current positions, leaving some animal groups isolated on islands and continents or separated by mountain ranges or inland seas from other competitors. Early in the Cenozoic, new ecosystems appeared, with the evolution of grasses and coral reefs. Late in the Cenozoic, further extinctions followed by speciation occurred during ice ages that covered high latitudes with ice and then retreated, leaving new open spaces for colonization.

CHAPTER OVERVIEW

28: INVERTEBRATES

28.1: PHYLUM PORIFERA

28.1A: PHYLUM PORIFERA
28.1B: MORPHOLOGY OF SPONGES
28.1C: PHYSIOLOGICAL PROCESSES IN SPONGES
28.2: PHYLUM CNIDARIA

28.2A: PHYLUM CNIDARIA
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28.3: SUPERPHYLUM LOPHOTROCHOZOA

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28.5A: PHYLUM ECHINODERMATA
28.5B: CLASSES OF ECHINODERMS
Echinoderms consist of five distinct classes: sea stars, sea cucumbers, sea urchins and sand dollars, brittle stars, and sea lillies.

28.5C: PHYLUM CHORDATA

28.1: Phylum Porifera

Topic hierarchy

- 28.1A: Phylum Porifera
- 28.1B: Morphology of Sponges
- 28.1C: Physiological Processes in Sponges

28.1A: Phylum Porifera

Sponges lack true tissues, have no body symmetry, and are sessile; types are classified based on presence and composition of spicules.

Learning Objectives

- Explain the position of the phylum Porifera in the phylogenetic tree of invertebrates

Key Points

- As larvae, sponges are able to swim, but as adults, they are sessile, spending their life attached to a substrate.
- Although the majority of sponges live in marine habitats, one family, the Spongillidae, is found in fresh water.
- Calcarea, Hexactinellida, Demospongiae, and Homoscleromorpha make up the four classes of sponges; each type is classified based on the presence or composition of its spicules or spongin.
- Most sponges reproduce sexually; however, some can reproduce through budding and the regeneration of fragments.
- The majority of sponges are filter-feeders, but a few species are carnivorous due to the nutrient -poor environment in which they are found.

Key Terms

- **parazoan**: include only one phylum known as the sponges
- **endosymbiont**: an organism that lives within the body or cells of another organism
- **spongin**: a horny, sulfur-containing protein related to keratin that forms the skeletal structure of certain classes of sponges
- **spicule**: a sharp, needle-like piece
- **holdfast**: a root-like structure that anchors aquatic sessile organisms, such as seaweed, other sessile algae, stalked crinoids, benthic cnidarians, and sponges, to the substrate

Introduction

The invertebrates, or Invertebrata, are animals that do not contain bony structures such as the cranium and vertebrae. The simplest of all the invertebrates are the Parazoans, which include only the phylum Porifera. Phylum Porifera (“pori” = pores, “fera” = bearers) are popularly known as sponges. Sponge larvae are able to swim; however, adults are non-motile and spend their life attached to a substratum through a holdfast. The majority of sponges are marine, living in seas and oceans. There is, however, one family of fresh water sponges (Family Spongillidae). The great majority of the marine species can be found in ocean habitats ranging from tidal zones to depths exceeding 8,800 m (5.5 mi).

Sponges are classified within four classes: calcareous sponges (Calcarea), glass sponges (Hexactinellida), demosponges (Demospongiae), and the recently-recognized, encrusting sponges (Homoscleromorpha). The presence and composition of spicules and spongin are the differentiating characteristics between the classes of sponges. Demosponges, which contain spongin and may or may not have spicules, constitute about 90% of all known sponge species, including all freshwater ones, and have the widest range of habitats. Calcareous sponges, which have calcium carbonate spicules and, in some species, calcium carbonate exoskeletons, are restricted to relatively shallow marine waters where production of calcium carbonate is easiest. They contain no spongin. Hemoscleromorpha sponges tend to be massive or encrusting in form and have a very simple structure with very little variation in spicule form (all spicules tend to be very small). Hexactinellid sponges have sturdy lattice-like internal skeletons made up of fused spicules of silica; they tend to be more-or-less cup-shaped.

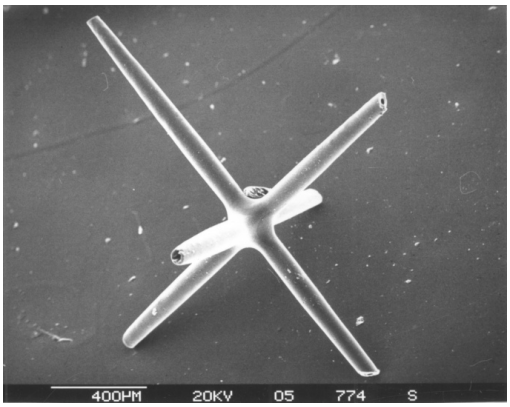

Figure 28.1*A*.1: **Sponge Spicule**: Sponges are classified based on the presence and types of spicules they contain.

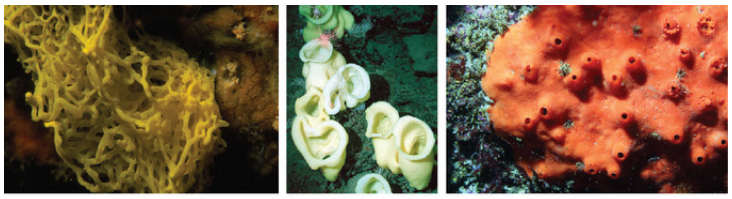

Figure 28.1*A*.1: **Types of sponges**: (a) *Clathrina clathrus* belongs to class Calcarea, (b) *Staurocalyptus* spp. (common name: yellow Picasso sponge) belongs to class *Hexactinellida*, and (c) *Acarnus erithacus* belongs to class Demospongia.

Unlike Protozoans, the Poriferans are multicellular. However, unlike higher metazoans, the cells that make up a sponge are not organized into tissues. Therefore, sponges lack true tissues and organs; in addition, they have no body symmetry. Sponges do, however, have specialized cells that perform specific functions. The shapes of their bodies are adapted for maximal efficiency of water flow through the central cavity, where nutrients are deposited, and leaves through a hole called the osculum. Many sponges have internal skeletons of spongin and/or spicules of calcium carbonate or silica. Primarily, their body consists of a thin sheet of cells over a frame (skeleton). As their name suggests, Poriferans are characterized by the presence of minute pores called ostia on their body.

Since water is vital to sponges for excretion, feeding, and gas exchange, their body structure facilitates the movement of water through the sponge. Structures such as canals, chambers, and cavities enable water to move through the sponge to nearly all body cells.

Most species use sexual reproduction, releasing sperm cells into the water to fertilize ova that in some species are released and in others are retained by the “mother. ” The fertilized eggs form larvae which swim off in search of places to settle. Sponges are also known for regenerating from fragments that are broken off, although this only works if the fragments include the right types of cells. A few species reproduce by budding. When conditions deteriorate, such as when temperatures drop, many freshwater species and a few marine ones produce gemmules: “survival pods” of unspecialized cells that remain dormant until conditions improve. They then either form completely new sponges or recolonize the skeletons of their parents.

Most of the approximately 5,000–10,000 known species of sponges are filter-feeders, feeding on bacteria and other food particles in the water. However, a few species of sponge that live in food-poor environments have become carnivores that prey mainly on small crustaceans. Other species host photosynthesizing micro-organisms as endosymbionts; these alliances often produce more food and oxygen than they consume.

28.1B: Morphology of Sponges

Instead of true tissues or organs, sponges have specialized cells that are in charge of important bodily functions and processes.

Learning Objectives

- Explain the various cell forms and bodily functions of sponges

Key Points

- Although sponges do not have organized tissue, they depend on specialized cells, such as choanocytes, porocytes, amoebocytes, and pinacocytes, for specialized functions within their bodies.
- The mesohyl acts as a type of endoskeleton, helping to maintain the tubular shape of sponges.
- Porocytes control the amount of water that enters pores into the spongocoel, while choanocytes, which are flagellated cells, aid the movement of water through the sponge, thereby helping the sponge to trap and ingest food particles.
- Amoebocytes carry out several special functions: they deliver nutrients from choanocytes to other cells, give rise to eggs for sexual reproduction, deliver phagocytized sperm from choanocytes to eggs, and can transform into other cell types.
- Collencytes, lophocytes, sclerocytes, and spongocytes are examples of cells that are derived from amoebocytes; these cells manage other vital functions in the body of sponges.

Key Terms

- **choanocyte**: any of the cells in sponges that contain a flagellum and are used to control the movement of water
- **spongocoel**: the large, central cavity of sponges
- **osculum**: an opening in a sponge from which water is expelled
- **mesohyl**: the gelatinous matrix within a sponge

Morphology of Sponges

The morphology of the simplest sponges takes the shape of a cylinder with a large central cavity, the spongocoel, occupying the inside of the cylinder. Water can enter into the spongocoel from numerous pores in the body wall. Water entering the spongocoel is extruded via a large, common opening called the osculum. However, sponges exhibit a range of diversity in body forms, including variations in the size of the spongocoel, the number of osculi, and where the cells that filter food from the water are located.

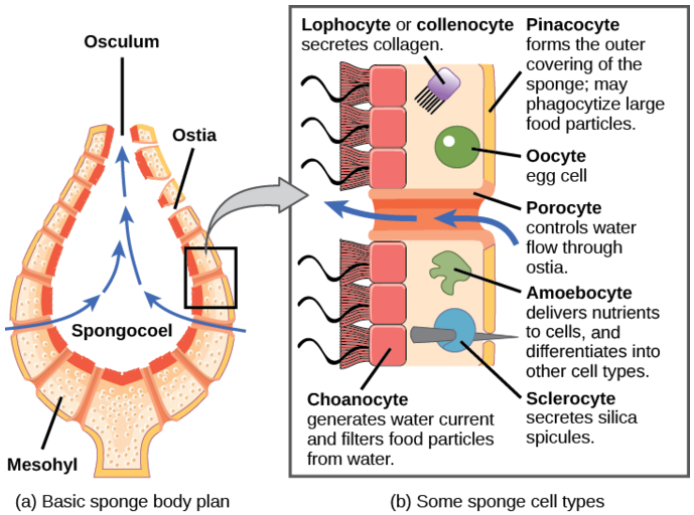

Figure 28.1B.1: **Sponge morphology**: The sponge's (a) basic body plan is a cylinder shape with a large central cavity. The specialized cell types in sponges (b) each perform a distinct function.

While sponges (excluding the Hexactinellids) do not exhibit tissue-layer organization, they do have different cell types that perform distinct functions. Pinacocytes, which are epithelial-like cells, form the outermost layer of sponges, enclosing a

jelly-like substance called mesohyl. Mesohyl is an extracellular matrix consisting of a collagen -like gel with suspended cells that perform various functions. The gel-like consistency of mesohyl acts as an endoskeleton, maintaining the tubular morphology of sponges. In addition to the osculum, sponges have multiple pores called ostia on their bodies that allow water to enter the sponge. In some sponges, ostia are formed by porocytes: single, tube-shaped cells that act as valves to regulate the flow of water into the spongocoel. In other sponges, ostia are formed by folds in the body wall of the sponge.

Choanocytes (“collar cells”) are present at various locations, depending on the type of sponge; however, they always line the inner portions of some space through which water flows: the spongocoel in simple sponges; canals within the body wall in more complex sponges; and chambers scattered throughout the body in the most complex sponges. Whereas pinacocytes line the outside of the sponge, choanocytes tend to line certain inner portions of the sponge body that surround the mesohyl. The structure of a choanocyte is critical to its function, which is to generate a water current through the sponge and to trap and ingest food particles by phagocytosis. Note that there is a similarity in appearance between the sponge choanocyte and choanoflagellates (Protista). This similarity suggests that sponges and choanoflagellates are closely related and probably share a recent, common ancestry.

The cell body is embedded in mesohyl. It contains all organelles required for normal cell function, but protruding into the “open space” inside of the sponge is a mesh-like collar composed of microvilli with a single flagellum in the center of the column. The cumulative effect of the flagella from all choanocytes aids the movement of water through the sponge: drawing water into the sponge through the numerous ostia, into the spaces lined by choanocytes, and eventually out through the osculum (or osculi). Meanwhile, food particles, including waterborne bacteria and algae, are trapped by the sieve-like collar of the choanocytes, slide down into the body of the cell, are ingested by phagocytosis, and become encased in a food vacuole. Finally, choanocytes will differentiate into sperm for sexual reproduction; they will become dislodged from the mesohyl, leaving the sponge with expelled water through the osculum.

The second crucial cells in sponges are called amoebocytes (or archaeocytes), named for the fact that they move throughout the mesohyl in an amoeba-like fashion. Amoebocytes have a variety of functions: delivering nutrients from choanocytes to other cells within the sponge; giving rise to eggs for sexual reproduction (which remain in the mesohyl); delivering phagocytized sperm from choanocytes to eggs; and differentiating into more-specific cell types. Some of these more-specific cell types include collencytes and lophocytes, which produce the collagen-like protein to maintain the mesohyl; sclerocytes, which produce spicules in some sponges; and spongocytes, which produce the protein spongin in the majority of sponges. These cells produce collagen to maintain the consistency of the mesohyl.

28.1C: Physiological Processes in Sponges

Sponges are sessile, feed by phagocytosis, and reproduce sexually and asexually; all major functions are regulated by water flow diffusion.

Learning Objectives

- Summarize the physiological processes of sponges

Key Points

- Choanocytes trap bacteria and other food particles from water flowing within the sponge: in through the ostia and out through the osculum; particles are ingested by phagocytosis.
- Sponges reproduce by sexual and asexual methods, which include fragmentation or budding; the production of gemmules is another asexual reproduction method, but is found only in freshwater sponges.
- Sponges are monoecious; depending on the species, production of gametes may be continuous through the year or dependent on water temperature.
- In nature, sponges are sessile as adults; however, under laboratory conditions, sponge cells are capable of localized creeping movements through organizational plasticity.
- Gas exchange, circulation, and excretion are other major body functions in the sponge; these are achieved through the diffusion of water through the sponge body.

Key Terms

- **oocyte**: a cell that develops into an egg or ovum; a female gametocyte
- **gemmule**: a small gemma or bud of dormant embryonic cells produced by some freshwater sponges
- **phagocytosis**: the process where a cell incorporates a particle by extending pseudopodia and drawing the particle into a vacuole of its cytoplasm

Physiological Processes in Sponges

Sponges, despite being simple organisms, regulate their different physiological processes through a variety of mechanisms. These mechanisms regulate metabolism, reproduction, and locomotion.

Metabolism

Sponges lack complex digestive, respiratory, circulatory, reproductive, and nervous systems. Their food is trapped when water passes through the ostia and out through the osculum. Bacteria smaller than 0.5 microns in size are trapped by choanocytes, which are the principal cells engaged in nutrition, and are ingested by phagocytosis. Particles that are larger than the ostia may be phagocytized by pinacocytes. In some sponges, amoebocytes transport food from cells that have ingested food particles to those that do not. For this type of digestion, in which food particles are digested within individual cells, the sponge draws water through diffusion. The limit of this type of digestion is that food particles must be smaller than individual cells.

All other major body functions in the sponge (gas exchange, circulation, excretion) are performed by diffusion between the cells that line the openings within the sponge and the water that is passing through those openings. All cell types within the sponge obtain oxygen from water through diffusion. Likewise, carbon dioxide is released into seawater by diffusion. In addition, nitrogenous waste produced as a by-product of protein metabolism is excreted via diffusion by individual cells into the water as it passes through the sponge.

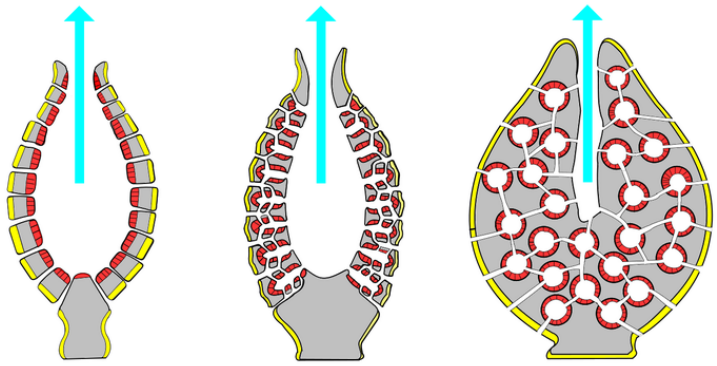

Figure 28.1C.1: **Water flow in a sponge**: In a sponge, water enters through the body pores and exits in the direction of the osculum (direction of blue arrow). This diffusion of water through the body supports major functions in the sponge.

Reproduction

Sponges reproduce by sexual, as well as, asexual methods. The typical means of asexual reproduction is either fragmentation (where a piece of the sponge breaks off, settles on a new substrate, and develops into a new individual) or budding (a genetically-identical outgrowth from the parent eventually detaches or remains attached to form a colony). An atypical type of asexual reproduction is found only in freshwater sponges, occurring through the formation of gemmules. Gemmules are environmentally-resistant structures produced by adult sponges wherein the typical sponge morphology is inverted. In gemmules, an inner layer of amoebocytes is surrounded by a layer of collagen (spongin) that may be reinforced by spicules. The collagen that is normally found in the mesohyl becomes the outer protective layer. In freshwater sponges, gemmules may survive hostile environmental conditions such as changes in temperature. They serve to recolonize the habitat once environmental conditions stabilize. Gemmules are capable of attaching to a substratum and generating a new sponge. Since gemmules can withstand harsh environments, are resistant to desiccation, and remain dormant for long periods, they are an excellent means of colonization for a sessile organism.

Sexual reproduction in sponges occurs when gametes are generated. Sponges are monoecious (hermaphroditic), which means that one individual can produce both gametes (eggs and sperm) simultaneously. In some sponges, production of gametes may occur throughout the year, whereas other sponges may show sexual cycles depending upon water temperature. Sponges may also become sequentially hermaphroditic, producing oocytes first and spermatozoa later. Oocytes arise by the differentiation of amoebocytes and are retained within the spongocoel, whereas spermatozoa result from the differentiation of choanocytes and are ejected via the osculum. Ejection of spermatozoa may be a timed and coordinated event, as seen in certain species. Spermatozoa carried along by water currents can fertilize the oocytes borne in the mesohyl of other sponges. Early larval development occurs within the sponge; free-swimming larvae are then released via the osculum.

Locomotion

Sponges are generally sessile as adults and spend their lives attached to a fixed substratum. They do not show movement over large distances as do free-swimming marine invertebrates. However, sponge cells are capable of creeping along substrata via organizational plasticity. Under experimental conditions, researchers have shown that sponge cells spread on a physical support demonstrate a leading edge for directed movement. It has been speculated that this localized creeping movement may help sponges adjust to microenvironments near the point of attachment. It must be noted, however, that this pattern of movement has been documented in laboratories, but it remains to be observed in natural sponge habitats.

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28.2: Phylum Cnidaria

Topic hierarchy

- 28.2A: Phylum Cnidaria
- 28.2B: Class Anthozoa
- 28.2C: Class Scyphozoa
- 28.2D: Class Cubozoa and Class Hydrozoa

28.2A: Phylum Cnidaria

Cnidarians are diploblastic, have organized tissue, undergo extracellular digestion, and use cnidocytes for protection and to capture prey.

Learning Objectives

- Describe the fundamental anatomy of a Cnidarian

Key Points

- Cnidarians have two distinct morphological body plans known as polyp, which are sessile as adults, and medusa, which are mobile; some species exhibit both body plans in their lifecycle.
- All cnidarians have two membrane layers in the body: the epidermis and the gastrodermis; between both layers they have the mesoglea, which is a connective layer.
- Cnidarians carry out extracellular digestion, where enzymes break down the food particles and cells lining the gastrovascular cavity absorb the nutrients.
- Cnidarians have an incomplete digestive system with only one opening; the gastrovascular cavity serves as both a mouth and an anus.
- The nervous system of cnidarians, responsible for tentacle movement, drawing of captured prey to the mouth, digestion of food, and expulsion of waste, is composed of nerve cells scattered across the body.
- Anthozoa, Scyphozoa, Cubozoa, and Hydrozoa make up the four different classes of Cnidarians.

Key Terms

- **diploblastic**: having two embryonic germ layers (the ectoderm and the endoderm)
- **cnidocyte**: a capsule, in certain cnidarians, containing a barbed, threadlike tube that delivers a paralyzing sting

Introduction to Phylum Cnidaria

Phylum Cnidaria includes animals that show radial or biradial symmetry and are diploblastic: they develop from two embryonic layers. Nearly all (about 99 percent) cnidarians are marine species.

Cnidarians contain specialized cells known as cnidocytes ("stinging cells"), which contain organelles called nematocysts (stingers). These cells are present around the mouth and tentacles, serving to immobilize prey with toxins contained within the cells. Nematocysts contain coiled threads that may bear barbs. The outer wall of the cell has hairlike projections called cnidocils, which are sensitive to touch. When touched, the cells are known to fire coiled threads that can either penetrate the flesh of the prey or predators of cnidarians, or ensnare it. These coiled threads release toxins into the target that can often immobilize prey or scare away predators ().

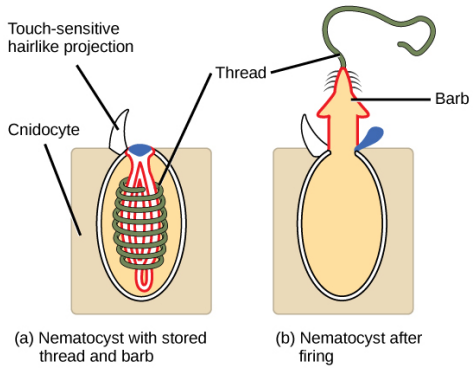

Figure 28.2A.1: **Cnidocytes**: Animals from the phylum Cnidaria have stinging cells called cnidocytes. Cnidocytes contain large organelles called (a) nematocysts that store a coiled thread and barb. When hairlike projections on the cell surface are touched, (b) the thread, barb, and a toxin are fired from the organelle.

Animals in this phylum display two distinct morphological body plans: polyp or "stalk" and medusa or "bell". An example of the polyp form is *Hydra* spp.; perhaps the most well-known medusoid animals are the jellies (jellyfish). Polyp forms are

as adults, with a single opening to the digestive system (the mouth) facing up with tentacles surrounding it. Medusa are motile, with the mouth and tentacles hanging down from an umbrella-shaped bell.

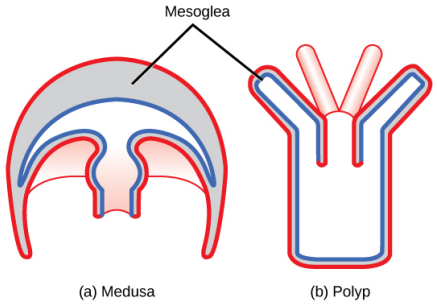

Figure 28.2A.1: **Cnidarian morphology**: Cnidarians have two distinct body plans, the medusa (a) and the polyp (b). All cnidarians have two membrane layers, with a jelly-like mesoglea between them.

Some cnidarians are polymorphic, having two body plans during their life cycle. An example is the colonial hydroid called an *Obelia*. The sessile polyp form has, in fact, two types of polyps. The first is the gastrozooid, which is adapted for capturing prey and feeding; the other type of polyp is the gonozooid, adapted for the asexual budding of medusa. When the reproductive buds mature, they break off and become free-swimming medusa, which are either male or female (dioecious). The male medusa makes sperm, whereas the female medusa makes eggs. After fertilization, the zygote develops into a blastula and then into a planula larva. The larva is free swimming for a while, but eventually attaches and a new colonial reproductive polyp is formed.

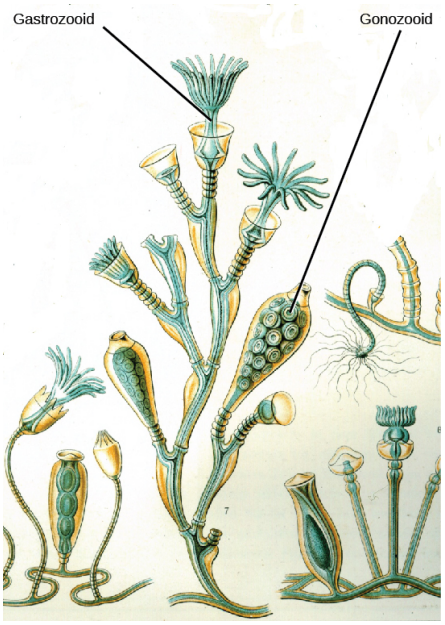

Figure 28.2A.1: **Types of polyps in *Obelia***: The sessile form of *Obelia geniculate* has two types of polyps: gastrozooids, which are adapted for capturing prey, and gonozooids, which bud to produce medusae asexually.

All cnidarians show the presence of two membrane layers in the body that are derived from the endoderm and ectoderm of the embryo. The outer layer (from ectoderm) is called the epidermis and lines the outside of the animal, whereas the inner layer (from endoderm) is called the gastrodermis and lines the digestive cavity. Between these two membrane layers is a non-living, jelly-like mesoglea connective layer. In terms of cellular complexity, cnidarians show the presence of differentiated cell types in each tissue layer: nerve cells, contractile epithelial cells, enzyme-secreting cells, and nutrient-absorbing cells, as well as the presence of intercellular connections. However, the development of organs or organ systems is not advanced in this phylum.

The nervous system is primitive, with nerve cells scattered across the body. This nerve net may show the presence of groups of cells in the form of nerve plexi (singular: plexus) or nerve cords. The nerve cells show mixed characteristics of motor as well as sensory neurons. The predominant signaling molecules in these primitive nervous systems are chemical

es, which perform both excitatory and inhibitory functions. Despite the simplicity of the nervous system, it nates the movement of tentacles, the drawing of captured prey to the mouth, the digestion of food, and the expulsion of waste.

The cnidarians perform extracellular digestion in which the food is taken into the gastrovascular cavity, enzymes are secreted into the cavity, and the cells lining the cavity absorb nutrients. The gastrovascular cavity has only one opening that serves as both a mouth and an anus; this is termed an incomplete digestive system. Cnidarian cells exchange oxygen and carbon dioxide by diffusion between cells in the epidermis with water in the environment, and between cells in the gastrodermis with water in the gastrovascular cavity. The lack of a circulatory system to move dissolved gases limits the thickness of the body wall, necessitating a non-living mesoglea between the layers. There is no excretory system or organs; nitrogenous wastes simply diffuse from the cells into the water outside the animal or in the gastrovascular cavity. There is also no circulatory system, so nutrients must move from the cells that absorb them in the lining of the gastrovascular cavity through the mesoglea to other cells.

The phylum Cnidaria contains about 10,000 described species divided into four classes: Anthozoa, Scyphozoa, Cubozoa, and Hydrozoa. The anthozoans, the sea anemones and corals, are all sessile species, whereas the scyphozoans (jellyfish) and cubozoans (box jellies) are swimming forms. The hydrozoans contain sessile forms and swimming colonial forms like the Portuguese Man O' War.

28.2B: Class Anthozoa

Members of the class Anthozoa display only polyp morphology and have cnidocyte-covered tentacles around their mouth opening.

Learning Objectives

- Identify the adaptive features of anthozoa

Key Points

- Anthozoans include sea anemones, sea pens, and corals.
- The pharynx of anthozoans (ingesting as well as egesting food) leads to the gastrovascular cavity, which is divided by mesenteries.
- In Anthozoans, gametes are produced by the polyp; if they fuse, they will give rise to a free-swimming planula larva, which will become sessile once it finds an optimal substrate.
- Sea anemonies and coral are examples of anthozoans that form unique mutualistic relationships with other animal species; both sea anemonies and coral benefit from food availability provided by their partners.

Key Terms

- **mesentery**: in invertebrates, it describes any tissue that divides the body cavity into partitions
- **cnidocyte**: a capsule, in certain cnidarians, containing a barbed, threadlike tube that delivers a paralyzing sting
- **hermatypic**: of a coral that is a species that builds coral reefs

Class Anthozoa

The class Anthozoa includes all cnidarians that exhibit a polyp body plan only; in other words, there is no medusa stage within their life cycle. Examples include sea anemones, sea pens, and corals, with an estimated number of 6,100 described species. Sea anemones are usually brightly colored and can attain a size of 1.8 to 10 cm in diameter. These animals are usually cylindrical in shape and are attached to a substrate.

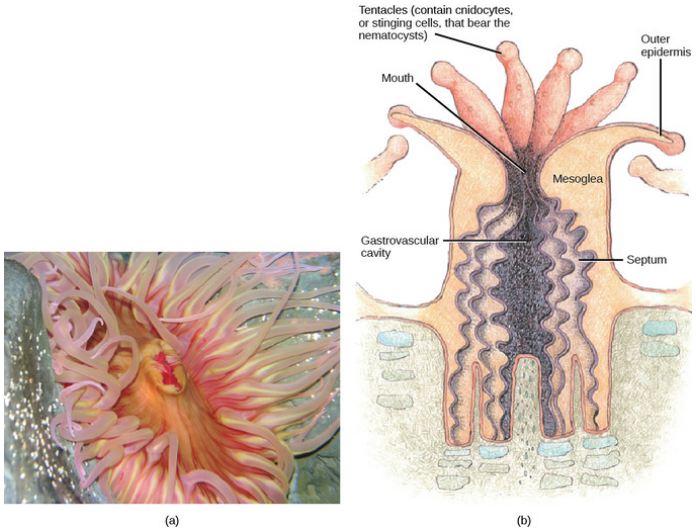

Figure 28.2B.1: **Anthozoans**: The sea anemone (a), like all anthozoans, has only a polyp body plan (b).

The mouth of a sea anemone is surrounded by tentacles that bear cnidocytes. They have slit-like mouth openings and a pharynx, which is the muscular part of the digestive system that serves to ingest as well as egest food. It may extend for up to two-thirds the length of the body before opening into the gastrovascular cavity. This cavity is divided into several chambers by longitudinal septa called mesenteries. Each mesentery consists of one ectodermal and one endodermal cell layer with the mesoglea sandwiched in between. Mesenteries do not divide the gastrovascular cavity completely; the smaller cavities coalesce at the pharyngeal opening. The adaptive benefit of the mesenteries appears to be an increase in surface area for absorption of nutrients and gas exchange.

Sea anemones feed on small fish and shrimp, usually by immobilizing their prey using the cnidocytes. Some sea anemones establish a mutualistic relationship with hermit crabs by attaching to the crab's shell. In this relationship, the anemone gets food particles from prey caught by the crab, while the crab is protected from the predators by the stinging cells of the anemone. Anemone fish, or clownfish, are able to live in the anemone since they are immune to the toxins contained within the nematocysts. Another type of anthozoan that forms an important mutualistic relationship is reef building coral. These hermatypic corals rely on a symbiotic relationship with zooxanthellae. The coral gains photosynthetic capability, while the zooxanthellae benefit by using nitrogenous waste and carbon dioxide produced by the cnidarian host.

Anthozoans remain polypoid throughout their lives. They can reproduce asexually by budding or fragmentation, or sexually by producing gametes. Both gametes are produced by the polyp, which can fuse to give rise to a free-swimming planula larva. The larva settles on a suitable substratum and develops into a sessile polyp.

28.2C: Class Scyphozoa

Scyphozoans are free-swimming, polymorphic, dioecious, and carnivorous cnidarians with a prominent medusa morphology.

Learning Objectives

- Explain the key features of scyphozoa

Key Points

- Scyphozoans have a ring of muscles that lines the dome of their bodies; these structures provide them with the contractile force they need to swim through water.
- Scyphozoans have separate sexes and form planula larvae through external fertilization.
- Jellies exhibit the polyp form, known as a scyphistoma, after their larvae settle on a substrate; these forms will later bud-off and transform into their more prominenent medusa forms.

Key Terms

- **dioecious**: having the male and female reproductive organs on separate parts (of the same species)
- **rhopalia**: small sensory structures found within Scyphozoa that are characterized by clusters of neurons that can be used to sense light
- **scyphistoma**: the polypoid form of scyphozoans
- **nematocyst**: a capsule, in certain cnidarians, containing a barbed, threadlike tube that delivers a paralyzing sting

Class Scyphozoa

Class Scyphozoa, an exclusively marine class of animals with about 200 known species, includes all the jellies. The defining characteristic of this class is that the medusa is the prominent stage in the life cycle, although there is a polyp stage present. Members of this species range from 2 to 40 cm in length, but the largest scyphozoan species, *Cyanea capillata*, can reach a size of 2 m across. Scyphozoans display a characteristic bell-like morphology.

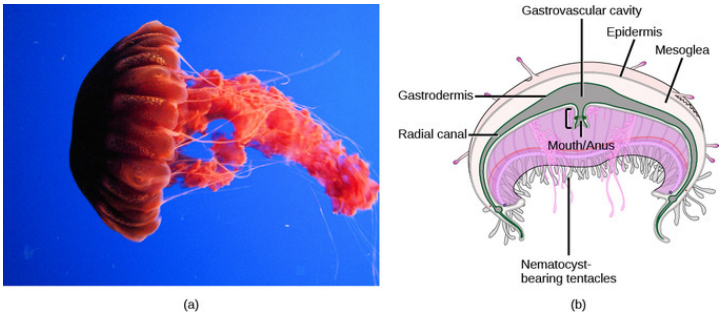

Figure 28.2C.1: **Scyphozoans**: For jellyfish (a), and all other scyphozoans, the medusa (b) is the most prominent of the two life stages.

In the jellyfish, a mouth opening, surrounded by tentacles bearing nematocysts, is present on the underside of the animal. Scyphozoans live most of their life cycle as free-swimming, solitary carnivores. The mouth leads to the gastrovascular cavity, which may be sectioned into four interconnected sacs, called diverticuli. In some species, the digestive system may be further branched into radial canals. Like the septa in anthozoans, the branched gastrovascular cells serves to increase the surface area for nutrient absorption and diffusion; thus, more cells are in direct contact with the nutrients in the gastrovascular cavity.

In scyphozoans, nerve cells are scattered over the entire body. Neurons may even be present in clusters called rhopalia. These animals possess a ring of muscles lining the dome of the body, which provides the contractile force required to swim through water. Scyphozoans are dioecious animals, having separate sexes. The gonads are formed from the gastrodermis with gametes expelled through the mouth. Planula larvae are formed by external fertilization; they settle on a substratum in a polypoid form known as scyphistoma. These forms may produce additional polyps by budding or may transform into the medusoid form. The life cycle of these animals can be described as polymorphic because they exhibit both a medusal and polypoid body plan at some point.

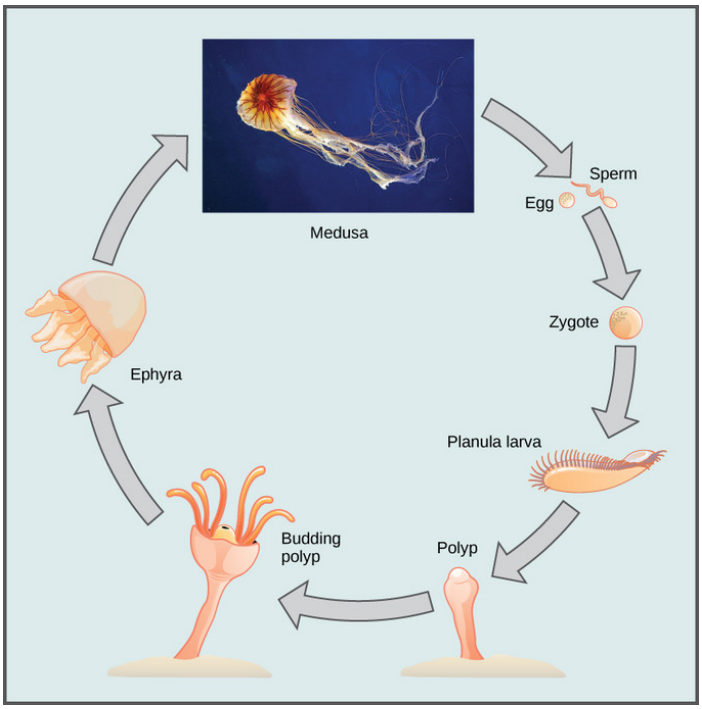

Figure 28.2C.1: **Lifecycle of a jellyfish**: The lifecycle of a jellyfish includes two stages: the medusa stage and the polyp stage. The polyp reproduces asexually by budding,while the medusa reproduces sexually.

28.2D: Class Cubozoa and Class Hydrozoa

Cubozoans live as box-shaped medusae while Hydrozoans are true polymorphs and can be found as colonial or solitary organisms.

Learning Objectives

- Distinguish between cubozoa and hydrozoa cnidarians

Key Points

- Cubozoans differ from Scyphozoans in their arrangement of tentacles; they are also known for their box-shaped medusa.
- Out of all cnidarians, cubozoans are the most venomous.
- Hydrozoans are polymorphs, existing as solitary polyps, solitary medusae, or as colonies.
- Hydrozoans are unique from all other cnidarians in that their gonads are derived from epidermal tissue.

Key Terms

- **hydroid**: any of many colonial coelenterates that exist mainly as a polyp; a hydrozoan

Class Cubozoa

Class Cubozoa includes jellies that have a box-shaped medusa: a bell that is square in cross-section; hence, they are colloquially known as “box jellyfish.” These species may achieve sizes of 15–25 cm. Cubozoans display overall morphological and anatomical characteristics that are similar to those of the scyphozoans. A prominent difference between the two classes is the arrangement of tentacles. This is the most venomous group of all the cnidarians.

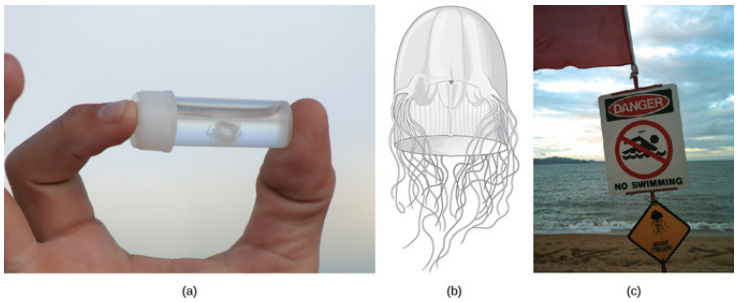

Figure 28.2D.1: **Cubozoans**: The (a) tiny cubazoan jelly *Malo kingi* is thimble shaped and, like all cubozoan jellies, (b) has four muscular pedalia to which the tentacles attach. *M. kingi* is one of two species of jellies known to cause Irukandji syndrome, a condition characterized by excruciating muscle pain, vomiting, increased heart rate, and psychological symptoms. Two people in Australia, where Irukandji jellies are most-commonly found, are believed to have died from Irukandji stings. (c) A sign on a beach in northern Australia warns swimmers of the danger.

The cubozoans contain muscular pads called pedalia at the corners of the square bell canopy, with one or more tentacles attached to each pedalium. These animals are further classified into orders based on the presence of single or multiple tentacles per pedalium. In some cases, the digestive system may extend into the pedalia. Nematocysts may be arranged in a spiral configuration along the tentacles; this arrangement helps to effectively subdue and capture prey. Cubozoans exist in a polypoid form that develops from a planula larva. These polyps show limited mobility along the substratum. As with scyphozoans, they may bud to form more polyps to colonize a habitat. Polyp forms then transform into the medusoid forms.

Class Hydrozoa

Hydrozoa includes nearly 3,200 species; most are marine, although some freshwater species are known. Animals in this class are polymorphs: most exhibit both polypoid and medusoid forms in their lifecycle, although this is variable.

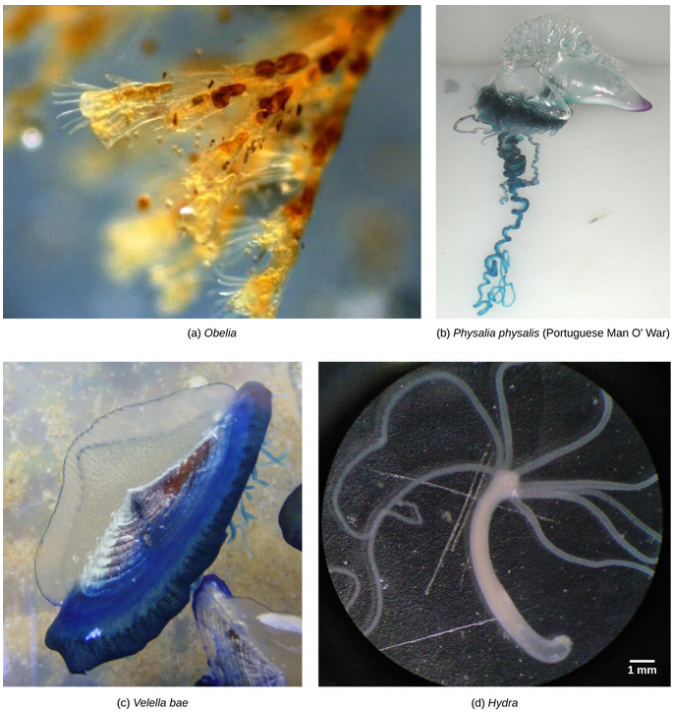

Figure 28.2D.1: **Hydrozoans**: (a) *Obelia*, (b) *Physalia physalis*, known as the Portuguese Man O‘ War, (c) *Velella bae*, and (d) *Hydra* have different body shapes, but all belong to the family Hydrozoa.

The polyp form in these animals often shows a cylindrical morphology with a central gastrovascular cavity lined by the gastrodermis. The gastrodermis and epidermis have a simple layer of mesoglea sandwiched between them. A mouth opening, surrounded by tentacles, is present at the oral end of the animal. Many hydrozoans form colonies that are composed of a branched colony of specialized polyps that share a gastrovascular cavity, such as in the colonial hydroid *Obelia*. Colonies may also be free-floating and contain medusoid and polypoid individuals in the colony as in *Physalia* (the Portuguese Man O’ War) or *Velella* (By-the-wind sailor). Other species are solitary polyps (*Hydra*) or solitary medusae (*Gonionemus*). The true characteristic shared by all these diverse species is that their gonads for sexual reproduction are derived from epidermal tissue, whereas in all other cnidarians they are derived from gastrodermal tissue.

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28.3: Superphylum Lophotrochozoa

Topic hierarchy

- 28.3A: Superphylum Lophotrochozoa
- 28.3B: Phylum Platyhelminthes
- 28.3C: Phylum Rotifera
- 28.3D: Phylum Nemertea
- 28.3E: Phylum Mollusca
- 28.3F: Classification of Phylum Mollusca
- 28.3G: Phylum Annelida

28.3A: Superphylum Lophotrochozoa

The Lophotrochozoa are protostomes possessing a blastopore, an early form of a mouth; they include the trochozoans and the lophophorata.

Learning Objectives

- Describe the phylogenetic position and basic features of lophotrochozoa

Key Points

- Lophotrochozoa have a blastopore, which is an involution of the ectoderm that forms a rudimentary mouth opening to the alimentary canal, a condition called protostomy or “first mouth”.
- The Lophotrochozoa are comprised of the trochozoans and the lophophorata, although the exact relationships between the different phyla are not clearly determined.
- Lophophores are characterized by the presence of the lophophore, a set of ciliated tentacles surrounding the mouth; they include the flatworms and several other phyla whose relationships are upheld by genetic evidence.
- Trochophore larvae are distinguished from the lophophores by two bands of cilia around the body; they include the Nemertea, Mollusca, Sipuncula, and Annelida.
- The lophotrochozoans have a mesoderm layer positioned between the ectoderm and endoderm and are bilaterally symmetrical, which signals the beginning of cephalization, the concentration of nervous tissues and sensory organs in the head of the organism.

Key Terms

- **blastopore**: the opening into the archenteron
- **lophophore**: a feeding organ of brachiopods, bryozoans, and phoronids
- **cephalization**: an evolutionary trend in which the neural and sense organs become centralized at one end (the head) of an animal

Lophotrochozoans

Animals belonging to superphylum Lophotrochozoa are protostomes: the blastopore (or the point of involution of the ectoderm or outer germ layer) becomes the mouth opening to the alimentary canal. This is called protostomy or “first mouth.” In protostomy, solid groups of cells split from the endoderm or inner germ layer to form a central mesodermal layer of cells. This layer multiplies into a band which then splits internally to form the coelom; this protostomic coelom is termed schizocoelom.

As lophotrochozoans, the organisms in this superphylum possess either lophophore or trochophore larvae. The exact relationships between the different phyla are not entirely certain. The lophophores include groups that are united by the presence of the lophophore, a set of ciliated tentacles surrounding the mouth. Lophophorata include the flatworms and several other phyla, including the Bryozoa, Entoprocta, Phoronida, and Brachiopoda. These clades are upheld when RNA sequences are compared. Trochophore larvae are characterized by two bands of cilia around the body. Previously, these were treated together as the Trochozoa, together with the arthropods, which do not produce trochophore larvae, but were considered close relatives of the annelids because they are both segmented. However, they show a number of important differences. Arthropods are now placed separately among the Ecdysozoa. The Trochozoa include the Nemertea, Mollusca, Sipuncula, and Annelida.

The lophotrochozoans are triploblastic, possessing an embryonic mesoderm sandwiched between the ectoderm and endoderm found in the diploblastic cnidarians. These phyla are also bilaterally symmetrical: a longitudinal section will divide them into right and left sides that are symmetrical. They also show the beginning of cephalization: the evolution of a concentration of nervous tissues and sensory organs in the head of the organism, which is where it first encounters its environment.

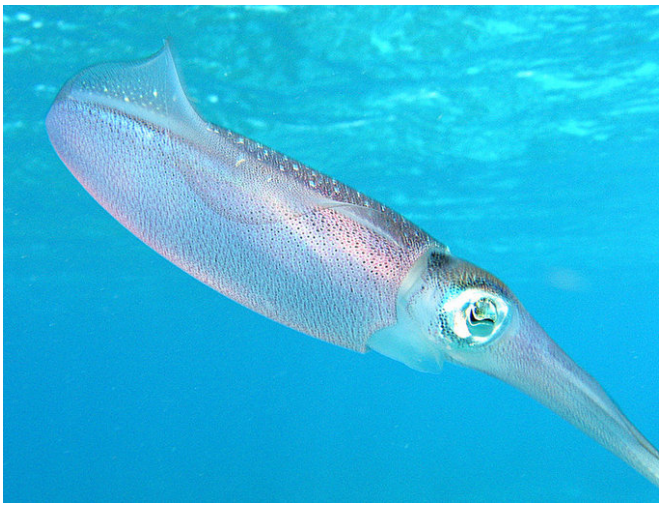

Figure 28.3A.1: **Lophotrochozoans**: The Caribbean Reef Squid or Sepioteuthis sepioidea is a complex lophotrochozoan. Species in this group have bilateral symmetry.

28.3B: Phylum Platyhelminthes

The Platyhelminthes are flatworms that lack a coelom; many are parasitic; all lack either a circulatory or respiratory system.

Learning Objectives

- Differentiate among the classes of platyhelminthes

Key Points

- The Platyhelminthes are acoelomate flatworms: their bodies are solid between the outer surface and the cavity of the digestive system.
- Most flatworms have a gastrovascular cavity rather than a complete digestive system; the same cavity used to bring in food is used to expel waste materials.
- Platyhelminthes are either predators or scavengers; many are parasites that feed on the tissues of their hosts.
- Flatworms posses a simple nervous system, no circulatory or respiratory system, and most produce both eggs and sperm, with internal fertilization.
- Platyhelminthes are divided into four classes: Turbellaria, free-living marine species; Monogenea, ectoparasites of fish; Trematoda, internal parasites of humans and other species; and Cestoda (tapeworms), which are internal parasites of many vertebrates.
- In flatworms, digested materials are taken into the cells of the gut lining by phagocytosis, rather than being processed internally.

Key Terms

- **acoelomate**: any animal without a coelom, or body cavity
- **ectoparasite**: a parasite that lives on the surface of a host organism
- **scolex**: the structure at the rear end of a tapeworm which, in the adult, has suckers and hooks by which it attaches itself to a host
- **proglottid**: any of the segments of a tapeworm; they contain both male and female reproductive organs

Phylum Platyhelminthes

Phylum Platyhelminthes is composed of the flatworms: acoelomate organisms that include many free-living and parasitic forms. Most of the flatworms are classified in the superphylum Lophotrochozoa, which also includes the mollusks and annelids. The Platyhelminthes consist of two lineages: the Catenulida and the Rhabditophora. The Catenulida, or “chain worms” is a small clade of just over 100 species. These worms typically reproduce asexually by budding. However, the offspring do not fully detach from the parents; therefore, they resemble a chain. The remaining flatworms discussed here are part of the Rhabditophora.

Many flatworms are parasitic, including important parasites of humans. Flatworms have three embryonic tissue layers that give rise to surfaces that cover tissues (from ectoderm), internal tissues (from mesoderm), and line the digestive system (from endoderm). The epidermal tissue is a single layer cells or a layer of fused cells (syncytium) that covers a layer of circular muscle above a layer of longitudinal muscle. The mesodermal tissues include mesenchymal cells that contain collagen and support secretory cells that secrete mucus and other materials at the surface. The flatworms are acoelomates: their bodies are solid between the outer surface and the cavity of the digestive system.

Physiological Processes of Flatworms

The free-living species of flatworms are predators or scavengers. Parasitic forms feed on the tissues of their hosts. Most flatworms have a gastrovascular cavity rather than a complete digestive system; in such animals, the “mouth” is also used to expel waste materials from the digestive system. Some species also have an anal opening. The gut may be a simple sac or highly branched. Digestion is extracellular, with digested materials taken in to the cells of the gut lining by phagocytosis. One group, the cestodes, lacks a digestive system. Flatworms have an excretory system with a network of tubules throughout the body with openings to the environment and nearby flame cells, whose cilia beat to direct waste

fluids concentrated in the tubules out of the body. The system is responsible for the regulation of dissolved salts and the excretion of nitrogenous wastes. The nervous system consists of a pair of nerve cords running the length of the body with connections between them and a large ganglion or concentration of nerves at the anterior end of the worm, where there may also be a concentration of photosensory and chemosensory cells.

There is neither a circulatory nor respiratory system, with gas and nutrient exchange dependent on diffusion and cell-cell junctions. This necessarily limits the thickness of the body in these organisms, constraining them to be "flat" worms. In addition, most flatworm species are monoecious; typically, fertilization is internal. Asexual reproduction is common in some groups.

Diversity of Flatworms

Platyhelminthes are traditionally divided into four classes: Turbellaria, Monogenea, Trematoda, and Cestoda. The class Turbellaria includes mainly free-living, marine species, although some species live in freshwater or moist terrestrial environments. The ventral epidermis of turbellarians is ciliated which facilitates their locomotion. Some turbellarians are capable of remarkable feats of regeneration: they may regrow the entire body from a small fragment.

Figure $28.3B.1$: **Turbellaria**: Pseudobiceros bedfordi (Bedford's Flatworm), a member of the Turbellaria, is a marine species which uses the epidermis of its belly for locomotion.

The monogeneans are ectoparasites, mostly of fish, with simple life cycles that consist of a free-swimming larva that attaches to a fish to begin transformation to the parasitic adult form. The worms may produce enzymes that digest the host tissues or simply graze on surface mucus and skin particles.

The trematodes, or flukes, are internal parasites of mollusks and many other groups, including humans. Trematodes have complex life cycles that involve a primary host in which sexual reproduction occurs and one or more secondary hosts in which asexual reproduction occurs. The primary host is almost always a mollusk. Trematodes are responsible for serious human diseases including schistosomiasis, a blood fluke.

Figure $28.3B.1$: **Trematodes**: Botulus microporus is a trematode that lives only in the intestinal tract of Lancetfish, *Alepisaurus ferox*.

The cestodes, or tapeworms, are also internal parasites, mainly of vertebrates. Tapeworms live in the intestinal tract of the primary host, remaining fixed by using a sucker on the anterior end, or scolex, of the tapeworm body. The remainder of the tapeworm is composed of a long series of units called proglottids. Each may contain an excretory system with flame cells

and both female and male reproductive structures. Tapeworms do not possess a digestive system; instead, they absorb nutrients from the food matter passing through them in the host's intestine.

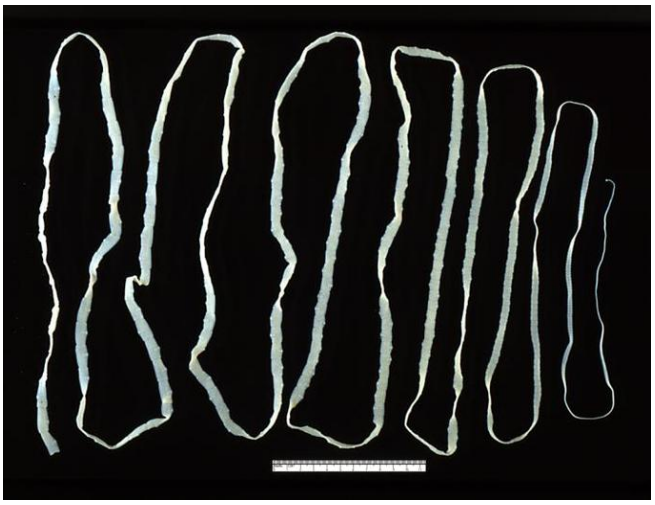

Figure 28.3B.1: **Cestodes**: Taenia saginata, also known as Taeniarhynchus saginata or the beef tapeworm, is a parasite of both cattle and humans. It displays the long series of proglottid subunits characteristic of the species.

28.3C: Phylum Rotifera

Rotifers are microscopic organisms named for a rotating structure (called the corona) at their anterior end that is covered with cilia.

Learning Objectives

- Identify the features of rotifers involved in movement and feeding

Key Points

- The rotifer body form consists of a head (containing the sensory organs in the form of a bi-lobed brain and small eyespots near the corona), the trunk (containing organs), and the foot (which can hold fast).
- The foot of the rotifer secretes a sticky material to help it adhere to surfaces.
- Rotifers are filter feeders that generate a current using the corona to pass food into the mouth, which then passes by digestive and salivary glands into the stomach and intestines.
- Rotifers exhibit sexual dimorphism; the gender of many species is determined by whether the egg is fertilized (and develops into a female) or unfertilized (and develops into a male).

Key Terms

- **pseudocoelomate**: any invertebrate animal with a three-layered body and a pseudocoel
- **mastax**: the pharynx of a rotifer which usually contains four horny pieces that work to crush the food

Phylum Rotifera

The rotifers are a microscopic (about 100 µm to 30 mm) group of mostly-aquatic organisms that get their name from the corona: a rotating, wheel-like structure that is covered with cilia at their anterior end. Although their taxonomy is currently in flux, one treatment places the rotifers in three classes: Bdelloidea, Monogononta, and Seisonidea. The classification of the group is currently under revision, however, as more phylogenetic evidence becomes available. It is possible that the “spiny headed worms” currently in phylum Acanthocephala will be incorporated into this group in the future.

Figure $28.3C.1$: **Rotifers**: A bdelloid rotifer is a member of a class of rotifers found in fresh water and moist soil. The rotifer body consists of a head, a truck, and a foot. They eat by filtering food into the mouth by creating currents with the corona.

The rotifer body form consists of a head (which contains the corona), a trunk (which contains the organs), and the foot. Rotifers are typically free-swimming and truly planktonic organisms, but the toes or extensions of the foot can secrete a sticky material forming a holdfast to help them adhere to surfaces. The head contains sensory organs in the form of a bi-lobed brain and small eyespots near the corona.

The rotifers are filter feeders that will eat dead material, algae, and other microscopic living organisms. Therefore, they are very important components of aquatic food webs. Rotifers obtain food that is directed toward the mouth by the current created from the movement of the corona. The food particles enter the mouth and travel to the mastax (pharynx with jaw-like

structures). Food passes by digestive and salivary glands into the stomach and then into the intestines. Digestive and excretory wastes are collected in a cloacal bladder before being released out the anus.

Rotifers are pseudocoelomates commonly found in fresh water and some salt water environments throughout the world. About 2,200 species of rotifers have been identified. Rotifers are dioecious organisms (having either male or female genitalia) and exhibit sexual dimorphism (males and females have different forms). Many species are parthenogenic and exhibit haplodiploidy, a method of gender determination in which a fertilized egg develops into a female and an unfertilized egg develops into a male. In many dioecious species, males are short-lived and smaller, with no digestive system and a single testis. Females can produce eggs that are capable of dormancy, which protects eggs during harsh environmental conditions.

28.3D: Phylum Nemertea

Nemertea, or ribbon worms, are distinguished by their proboscis, used for capturing prey and enclosed in a cavity called a rhynchocoel.

Learning Objectives

- Identify the key features of the Phylum Nemertea

Key Points

- The Nemertini are mostly bottom-dwelling marine organisms, although some are found in freshwater and terrestrial habitats.
- Most nemerteans are carnivores, some are scavengers, and others have evolved relationships with some mollusks that are benefit the Nemertean but do not harm the mollusk.
- Nemerteans vary greatly in size and are bilaterally symmetrical; they are unsegmented and resemble a flat tube which can change morphological presentation in response to environmental cues.
- Nemertini have a simple nervous system comprised of a ring of four nerve masses called “ganglia” at the anterior end between the mouth and the foregut from which paired longitudinal nerve cords emerge and extend to the posterior end.
- Nemertini are mostly sexually dimorphic, fertilizing eggs externally by releasing both eggs and sperm into the water; a larva may develop inside the resulting young worm and devour its tissues before metamorphosing into the adult.

Key Terms

- **protonephridia**: an invertebrate organ which occurs in pairs and removes metabolic wastes from an animal’s body
- **rhynchocoel**: a cavity which mostly runs above the midline and ends a little short of the rear of the body of a nemertean and extends or retracts the proboscis
- **proboscis**: an elongated tube from the head or connected to the mouth, of an animal

Phylum Nemertea

The Nemertea are colloquially known as ribbon worms. Most species of phylum Nemertea are marine (predominantly benthic or bottom dwellers) with an estimated 900 species known. However, nemertini have been recorded in freshwater and terrestrial habitats as well. Most nemerteans are carnivores, feeding on worms, clams, and crustaceans. Some species are scavengers, while other nemertini species, such as *Malacobdella grossa*, have also evolved commensalistic relationships with some mollusks. Interestingly, nemerteans have almost no predators, two species are sold as fish bait, and some species have devastated commercial fishing of clams and crabs.

Morphology

Ribbon worms vary in size from 1 cm to several meters. They show bilateral symmetry and remarkable contractile properties. Because of their contractility, they can change their morphological presentation in response to environmental cues. Animals in phylum Nemertea also show a flattened morphology: they are flat from front to back, like a flattened tube. In addition, nemertea are soft, unsegmented animals.

Figure 28.3D.1: **Morphology of Nemertea**: A terrestrial Geonemertes, a Nemertean, displaying the flat, ribbon-like body of the organism that is unsegmented.

ue characteristic of this phylum is the presence of a proboscis enclosed in a rhynchocoel. The proboscis serves to e food and may be ornamented with barbs in some species. The rhynchocoel is a fluid-filled cavity that extends from the head to nearly two-thirds of the length of the gut in these animals. The proboscis may be extended or retracted by the retractor muscle attached to the wall of the rhynchocoel.

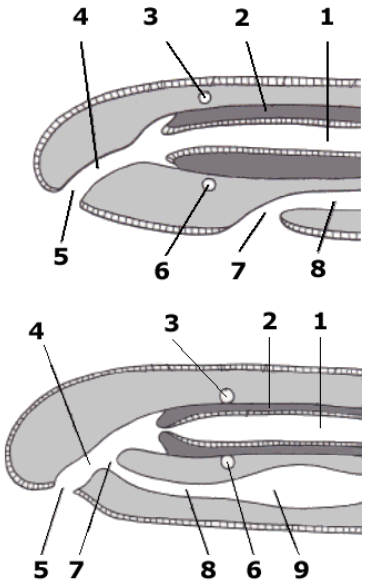

Figure 28.3D.1: **Internal structures of the Nemertini**: This image shows the internal structures of a basic nemertean, including the proboscis and the rhynchocoel: 1: Proboscis 2: Rhynchocoel 3: Dorsal commissure of brain 4: Rhynchodeum 5: Proboscis pore 6: Ventral commissure of brain 7: Mouth 8: Foregut 9: Stomach

Metabolism

The nemertini show a very well-developed digestive system. A mouth opening that is ventral to the rhynchocoel leads into the foregut, followed by the intestine. The intestine is present in the form of diverticular pouches which ends in a rectum that opens via an anus. Gonads are interspersed with the intestinal diverticular pouches, opening outwards via genital pores. A circulatory system consists of a closed loop of a pair of lateral blood vessels. The circulatory system is derived from the coelomic cavity of the embryo. Some animals may also have cross-connecting vessels in addition to lateral ones. Although these are called blood vessels, since they are of coelomic origin, the circulatory fluid is colorless. Some species bear hemoglobin as well as yellow or green pigments. The blood vessels are connected to the rhynchocoel. The flow of fluid in these vessels is facilitated by the contraction of muscles in the body wall. A pair of protonephridia, or primitive kidneys, is present in these animals to facilitate osmoregulation. Gaseous exchange occurs through the skin in the nemertini.

Nervous System

Nemertini have a ganglion or “brain” situated at the anterior end between the mouth and the foregut, surrounding the digestive system as well as the rhynchocoel. A ring of four nerve masses called “ganglia” comprises the brain in these animals. Paired longitudinal nerve cords emerge from the brain ganglia, extending to the posterior end. Ocelli or eyespots are present in pairs, in multiples of two in the anterior portion of the body. It is speculated that the eyespots originate from neural tissue and not from the epidermis.

Reproduction

Animals in phylum Nemertea show sexual dimorphism, although freshwater species may be hermaphroditic. Eggs and sperm are released into the water; fertilization occurs externally. The zygote develops into a special kind of nemertean larvae called a planuliform larva. In some nemertine species, another larva specific to the nemertinis, a pilidium, may develop inside the young worm from a series of imaginal discs. This larval form, characteristically shaped like a deerstalker cap, devours tissues from the young worm for survival before metamorphosing into the adult-like morphology.

E: Phylum Mollusca

Mollusks have a soft body and share several characteristics, including a muscular foot, a visceral mass of internal organs, and a mantle.

Learning Objectives

- Describe the unique anatomical and morphological features of molluscs

Key Points

- A mollusk's muscular foot is used for locomotion and anchorage, varies in shape and function, and can both extend and retract.
- The visceral mass inside the mollusk includes digestive, nervous, excretory, reproductive, and respiratory systems.
- Most mollusks possess a radula, which is similar to a tongue with teeth-like projections, serving to shred or scrape food.
- The mantle is the dorsal epidermis in mollusks; in some mollusks it secretes a chitinous and hard calcareous shell.

Key Terms

- **visceral mass**: the soft, non-muscular metabolic region of the mollusc that contains the body organs
- **mantle**: the body wall of a mollusc, from which the shell is secreted
- **radula**: the rasping tongue of snails and most other mollusks

Phylum Mollusca

Phylum Mollusca is the predominant phylum in marine environments. It is estimated that 23 percent of all known marine species are mollusks; there are around 85,000 described species, making them the second most diverse phylum of animals. The name "mollusca" signifies a soft body; the earliest descriptions of mollusks came from observations of unshelled cuttlefish. Mollusks are predominantly a marine group of animals; however, they are known to inhabit freshwater as well as terrestrial habitats. Mollusks display a wide range of morphologies in each class and subclass. They range from large predatory squids and octopus, some of which show a high degree of intelligence, to grazing forms with elaborately-sculpted and colored shells. In spite of their tremendous diversity, however, they also share a few key characteristics, including a muscular foot, a visceral mass containing internal organs, and a mantle that may or may not secrete a shell of calcium carbonate.

Figure 28.3E.1: **Mollusk shells**: Helix aspersa, a common land snail, has a calcium carbonate shell.

sks have a muscular foot used for locomotion and anchorage that varies in shape and function, depending on the type llusk under study. In shelled mollusks, this foot is usually the same size as the opening of the shell. The foot is a retractable as well as an extendable organ. It is the ventral-most organ, whereas the mantle is the limiting dorsal organ. Mollusks are eucoelomate, but the cavity is restricted to a region around the heart in adult animals. The mantle cavity develops independently of the coelomic cavity.

The visceral mass is present above the foot in the visceral hump. This includes digestive, nervous, excretory, reproductive, and respiratory systems. Mollusk species that are exclusively aquatic have gills for respiration, whereas some terrestrial species have lungs for respiration. Additionally, a tongue-like organ called a radula, which bears chitinous tooth-like ornamentation, is present in many species, serving to shred or scrape food. The mantle (also known as the pallium) is the dorsal epidermis in mollusks; shelled mollusks are specialized to secrete a chitinous and hard calcareous shell.

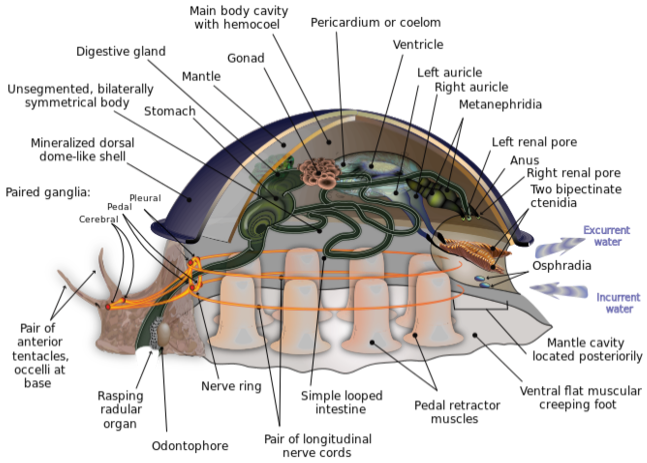

Figure 28.3E.1: **A "generalized mollusk"**: An anatomical diagram of a hypothetical ancestral mollusk, showing features common to many mollusk types.

Most mollusks are dioecious animals where fertilization occurs externally, although this is not the case in terrestrial mollusks, such as snails and slugs, or in cephalopods. In some mollusks, the zygote hatches and undergoes two larval stages, trochophore and veliger, before becoming a young adult; bivalves may exhibit a third larval stage, glochidia.

28.3F: Classification of Phylum Mollusca

The phylum Mollusca includes a wide variety of animals including the gastropods (“stomach foot”), the cephalopods (“head foot”), and the scaphopods (“boat foot”).

Learning Objectives

- Differentiate among the classes in the phylum mollusca

Key Points

- Mollusks can be segregated into seven classes: Aplacophora, Monoplacophora, Polyplacophora, Bivalvia, Gastropoda, Cephalopoda, and Scaphopoda. These classes are distinguished by, among other criteria, the presence and types of shells they possess.
- Class Aplacophora includes worm-like animals with no shell and a rudimentary body structure.
- Members of class Monoplacophora have a single shell that encloses the body.
- Members of class Polyplacophora are better known as “chitons;” these molluscs have a large foot on the ventral side and a shell composed of eight hard plates on the dorsal side.
- Class Bivalvia consists of mollusks with two shells held together by a muscle; these include oysters, clams, and mussels.
- Members of class Gastropoda have an asymmetrical body plan and usually have a shell, which can be planospiral or conispiral. Their key characteristic is the torsion around the perpendicular axis on the center of the foot that is modified for crawling.
- Class Scaphopoda consists of mollusks with a single conical shell through which the head protrudes, and a foot modified into tentacles known as captaculae that are used to catch and manipulate prey.

Key Terms

- **ctenidium**: a respiratory system, in the form of a comb, in some molluscs
- **captacula**: the foot of a Scaphalopod, modified into tentacles for capturing prey
- **nephridium**: a tubular excretory organ in some invertebrates

Classes in Phylum Mollusca

Phylum Mollusca is a very diverse (85,000 species) group of mostly marine species, with a dramatic variety of form. This phylum can be segregated into seven classes: Aplacophora, Monoplacophora, Polyplacophora, Bivalvia, Gastropoda, Cephalopoda, and Scaphopoda.

Class Aplacophora

Class Aplacophora (“bearing no plates”) includes worm-like animals primarily found in benthic marine habitats. These animals lack a calcareous shell, but possess aragonite spicules on their epidermis. They have a rudimentary mantle cavity and lack eyes, tentacles, and nephridia (excretory organs).

Class Monoplacophora

Members of class Monoplacophora (“bearing one plate”) posses a single, cap-like shell that encloses the body. The morphology of the shell and the underlying animal can vary from circular to ovate. A looped digestive system, multiple pairs of excretory organs, many gills, and a pair of gonads are present in these animals. The monoplacophorans were believed extinct and only known via fossil records until the discovery of Neopilina galathaea in 1952. Today, scientists have identified nearly two dozen extant species.

Class Polyplacophora

Animals in the class Polyplacophora (“bearing many plates”) are commonly known as “chitons” and bear an armor-like, eight-plated dorsal shell. These animals have a broad, ventral foot that is adapted for suction to rocks and other substrates, and a mantle that extends beyond the shell in the form of a girdle. Calcareous spines may be present on the girdle to offer protection from predators. Chitons live worldwide, in cold water, warm water, and the tropics. Most chiton species inhabit

intertidal or subtidal zones, and do not extend beyond the photic zone. Some species live quite high in the intertidal zone and are exposed to the air and light for long periods.

Figure $28.3F.1$: **Chiton morphology**: The underside of the gumboot chiton, Cryptochiton stellari, showing the foot in the center, surrounded by the gills and mantle. The mouth is visible to the left in this image.

Class Bivalvia

Bivalvia is a class of marine and freshwater molluscs with laterally compressed bodies enclosed by a shell in two hinged parts. Bivalves include clams, oysters, mussels, scallops, and numerous other families of shells. The majority are filter feeders and have no head or radula. The gills have evolved into ctenidia, specialised organs for feeding and breathing. Most bivalves bury themselves in sediment on the seabed, while others lie on the sea floor or attach themselves to rocks or other hard surfaces.

The shell of a bivalve is composed of calcium carbonate, and consists of two, usually similar, parts called valves. These are joined together along one edge by a flexible ligament that, in conjunction with interlocking “teeth” on each of the valves, forms the hinge.

Figure $28.3F.1$: **Empty shell of a bivalve**: The empty shell of the giant clam, Tridacna gigas. Note the pair of shells that are hinged together, a characteristic of members of the class Bivalvia.

Class Gastropoda

Animals in class Gastropoda (“stomach foot”) include well-known mollusks like snails, slugs, conchs, sea hares, and sea butterflies. Gastropoda includes shell-bearing species as well as species with a reduced shell. These animals are asymmetrical and usually present a coiled shell. Shells may be planospiral (like a garden hose wound up), commonly seen in garden snails, or conispiral (like a spiral staircase), commonly seen in marine conches.

The visceral mass in the shelled species displays torsion around the perpendicular axis on the center of the foot, which is the key characteristic of this group, along with a foot that is modified for crawling. Most gastropods bear a head with tentacles, eyes, and a style. A complex radula is used by the digestive system and aids in the ingestion of food. Eyes may be absent in some gastropods species. The mantle cavity encloses the ctenidia (singluar: ctenidium) as well as a pair of nephridia (singular: nephridium).

Figure $28.3F.1$: **Gastropod foot**: Gastropods, such as this Roman snail, have a large foot that is modified for crawling.

Class Cephalopoda

Class Cephalopoda ("head foot" animals) includes octopi, squids, cuttlefish, and nautilus. Cephalopods are a class of shell-bearing animals as well as mollusks with a reduced shell. They display vivid coloration, typically seen in squids and octopi, which is used for camouflage. All animals in this class are carnivorous predators and have beak-like jaws at the anterior end. All cephalopods show the presence of a very well-developed nervous system along with eyes, as well as a closed circulatory system. The foot is lobed and developed into tentacles and a funnel, which is used as the mode of locomotion. Locomotion in cephalopods is facilitated by ejecting a stream of water for propulsion ("jet" propulsion). Cephalopods, such as squids and octopi, also produce sepia or a dark ink, which is squirted upon a predator to assist in a quick getaway. Suckers are present on the tentacles in octopi and squid. Ctenidia are enclosed in a large mantle cavity serviced by blood vessels, each with its own associated heart. The mantle has siphonophores that facilitate exchange of water.

Figure $28.3F.1$: **Cephalopods**: Cephalopods ("head foot") include this octopus, which ejects a stream of water from a funnel in its body to propel itself through the water.

A pair of nephridia is present within the mantle cavity. Sexual dimorphism is seen in this class of animals. Members of a species mate, then the female lays the eggs in a secluded and protected niche. Females of some species care for the eggs for an extended period of time and may end up dying during that time period. Reproduction in cephalopods is different from other mollusks in that the egg hatches to produce a juvenile adult without undergoing the trochophore and veliger larval stages.

Class Scaphopoda

Members of class Scaphopoda (“boat feet”) are known colloquially as “tusk shells” or “tooth shells,” as evident when examining Dentalium, one of the few remaining scaphopod genera. Scaphopods are usually buried in sand with the anterior opening exposed to water. These animals bear a single conical shell, which has both ends open. The head is rudimentary and protrudes out of the posterior end of the shell. These animals do not possess eyes, but they have a radula, as well as a foot modified into tentacles with a bulbous end, known as captaculae. Captaculae serve to catch and manipulate prey. Ctenidia are absent in these animals.

Figure $28.3F.1$: **The Scaphopods**: The Scaphopods (“boat feet”) include the Antalis vulgaris, the shell of which is depicted here.

28.3G: Phylum Annelida

Annelids include segmented worms, such as leeches and earthworms; they are the most advanced worms as they possess a true coelom.

Learning Objectives

- Describe the morphological and anatomical features of annelids

Key Points

- Annelids are often called “segmented worms” because they possess true segmentation of their bodies, with both internal and external morphological features repeated in each body segment.
- The clitellum is a structure on the anterior portion of the worm that generates mucus to aid in sperm transfer from one worm to another; it also forms a cocoon within which fertilization occurs.
- Most annelids have chitinous hairlike extensions in every segment called chaetae that are anchored in the epidermis, although the number and size of chaetae can vary in the different classes.
- Annelids possess a closed circulatory system, lack a well-developed respiratory system, but have well-developed nervous systems.
- Annelids can either have distinct male and female forms or be hermaphrodites (having both male and female reproductive organs). Earthworms are hermaphrodites and can self-fertilize, but prefer to cross-fertilize if possible.

Key Terms

- **clitellum**: a glandular swelling in the epidermis of some annelid worms; it secretes a viscous fluid in which the eggs are deposited
- **chaeta**: a chitinous bristle of an annelid worm
- **metamerism**: the segmentation of the body into similar discrete units

Phylum Annelida

Phylum Annelida contains the class Polychaeta (the polychaetes) and the class Oligochaeta (the earthworms, leeches, and their relatives). These animals are found in marine, terrestrial, and freshwater habitats, but a presence of water or humidity is a critical factor for their survival, especially in terrestrial habitats. The name of the phylum is derived from the Latin word annellus, which means a small ring. Animals in this phylum show parasitic and commensal symbioses with other species in their habitat. Approximately 16,500 species have been described in phylum Annelida. The phylum includes earthworms, polychaete worms, and leeches. Annelids show protostomic development in embryonic stages and are often called “segmented worms” due to their key characteristic of metamerism, or true segmentation.

Morphology

Annelids display bilateral symmetry and are worm-like in overall morphology. They have a segmented body plan where the internal and external morphological features are repeated in each body segment. Metamerism allows animals to become bigger by adding “compartments,” while making their movement more efficient. This metamerism is thought to arise from identical teloblast cells in the embryonic stage, which develop into identical mesodermal structures. The overall body can be divided into head, body, and pygidium (or tail). The clitellum is a reproductive structure that generates mucus that aids in sperm transfer and gives rise to a cocoon within which fertilization occurs; it appears as a fused band in the anterior third of the animal.

Figure 28.3G.1: **Clitellum**: The clitellum is the reproductive structure of an annelid. It creates mucus that aids in sperm transfer and gives rise to a cocoon within which fertilization occurs. It can be seen in this image as the enlarged band around the animal.

Anatomy

The epidermis is protected by an acellular, external cuticle, but this is much thinner than the cuticle found in the ecdysozoans and does not require periodic shedding for growth. Circular as well as longitudinal muscles are located interior to the epidermis. Chitinous hairlike extensions, anchored in the epidermis and projecting from the cuticle, called setae/chaetae are present in every segment. Annelids show the presence of a true coelom, derived from embryonic mesoderm and protostomy. Hence, they are the most advanced worms. A well-developed and complete digestive system is present in earthworms (oligochaetes) with a mouth, muscular pharynx, esophagus, crop, and gizzard being present. The gizzard leads to the intestine and ends in an anal opening. Each segment is limited by a membranous septum that divides the coelomic cavity into a series of compartments.

Annelids possess a closed circulatory system of dorsal and ventral blood vessels that run parallel to the alimentary canal as well as capillaries that service individual tissues. In addition, these vessels are connected by transverse loops in every segment. These animals lack a well-developed respiratory system; gas exchange occurs across the moist body surface. Excretion is facilitated by a pair of metanephridia (a type of primitive “kidney” that consists of a convoluted tubule and an open, ciliated funnel) that is present in every segment towards the ventral side. Annelids show well-developed nervous systems with a nerve ring of fused ganglia present around the pharynx. The nerve cord is ventral in position, bearing enlarged nodes or ganglia in each segment.

Annelids may be either monoecious, with permanent gonads (as in earthworms and leeches), or dioecious, with temporary or seasonal gonads that develop (as in polychaetes). However, cross-fertilization is preferred in hermaphroditic animals. These animals may also show simultaneous hermaphroditism, participating in simultaneous sperm exchange when they are aligned for copulation.

Earthworms are the most abundant members of the class Oligochaeta, distinguished by the presence of the clitellum as well as few, reduced chaetae (“oligo- = “few”; -chaetae = “hairs”). The number and size of chaetae are greatly diminished in Oligochaeta compared to the polychaetes (poly=many, chaetae = hairs). The many chetae of polychaetes are also arranged within fleshy, flat, paired appendages that protrude from each segment. These parapodia may be specialized for different functions in the polychates. A significant difference between leeches and other annelids is the development of suckers at the anterior and posterior ends and an absence of chaetae. Additionally, the segmentation of the body wall may not correspond to the internal segmentation of the coelomic cavity. This adaptation possibly helps the leeches to elongate when they ingest copious quantities of blood from host vertebrates.

Figure 28.3*G*. 1 : **Leeches**: Unlike earthworms, leeches lack chaetae and have suckers at both ends of the body.

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28.4: Superphylum Ecdysozoa

Topic hierarchy

- 28.4A: Superphylum Ecdysozoa
- 28.4B: Phylum Nematoda
- 28.4C: Phylum Arthropoda
- 28.4D: Subphyla of Arthropoda

28.4A: Superphylum Ecdysozoa

The superphylum Ecdysozoa includes the nematode worms and the arthropods, both of which have a tough external covering called a cuticle.

Learning Objectives

- Discuss the phylogenetic position of Ecdysozoa

Key Points

- The Ecdysozoans are the most diverse group of animals, containing the nematode worms and the arthropods.
- These organisms have an external covering called a cuticle that protects their soft internal organs from water loss and the outside environment.
- After they molt, or shed their cuticle, they grow in size and secrete a new shell; this is called ecdysis.
- The phylogeny of the Ecdysozoans has been the cause of much scientific debate with no definitive consensus in the scientific community.

Key Terms

- **cuticle**: a noncellular protective covering outside the epidermis of many invertebrates and plants
- **coelomate**: any animal possessing a fluid-filled cavity within which the digestive system is suspended.
- **ecdysis**: the shedding of an outer layer of skin in snakes, crustaceans and insects; moulting

Superphylum Ecdysozoa

The superphylum Ecdysozoa contains an incredibly large number of species. This is because it includes two of the most diverse animal groups: Phylum Nematoda (the roundworms) and Phylum Arthropoda (the arthropods). The most distinguishing and prominent feature of Ecdysozoans is their cuticle: a tough, but flexible exoskeleton that protects these animals from water loss, predators, and other aspects of the external environment. All members of this superphylum periodically molt or shed their cuticle as they grow. After molting, they secrete a new cuticle that will last until their next growth phase. The process of molting and replacing the cuticle is called ecdysis, which is the derivation of the superphylum's name.

Figure 28.4A.1: **Molting in arthropods**: This cicada is in the middle of the molting process. The old cuticle splits and the insect climbs out. At this time, the insect's body is very soft. The cicada will then eat the old shell to replace nutrients that would otherwise be lost. This encourages the new shell to harden.

Phylogenetic Hypotheses

There are two main hypotheses about the phylogeny of the Ecdysozoans. The first is called the Articulata hypothesis. This grouping scheme is widely accepted, although some zoologists still hold to the original view that Panarthropoda should be classified with Annelida in a group called the Articulata, and that Ecdysozoa are polyphyletic. Others have suggested that a

possible solution is to regard Ecdysozoa as a sister-group of Annelida, though many scientists consider them unrelated. Inclusion of the roundworms within the Ecdysozoa was initially contested, but since 2003, a broad consensus has formed supporting the Ecdysozoa, placing them in a new set of groupings that include the Ecdysozoa, the Lophotrochozoa, and the Deuterostomia.

The other idea about the phylogeny of the Ecdysozoa is called the coelomate hypothesis. Before Ecdysozoa, one of the prevailing theories for the evolution of the bilateral animals was based on the morphology of their body cavities. There were three types, or grades, of organization: the Acoelomata (no coelom), the Pseudocoelomata (partial coelom), and the Eucoelomata (true coelom). With the introduction of molecular phylogenetics, the coelomate hypothesis was abandoned, although some molecular, phylogenetic support for the Coelomata continued until 2005.

28.4B: Phylum Nematoda

Nematodes are parasitic and free-living worms that are able to shed their external cuticle in order to grow.

Learning Objectives

- Describe the features of animals classified in phylum Nematoda

Key Points

- Nematodes are in the same phylogenetic grouping as the arthropods because of the presence of an external cuticle that protects the animal and keeps it from drying out.
- There are an estimated 28,000 species of nematodes, with approximately 16,000 of them being parasitic.
- Nematodes are tubular in shape and are considered pseudocoelomates because of they do not possess a true coelom.
- Nematodes do not have a well-developed excretory system, but do have a complete digestive system.
- Nematodes possess the ability to shed their exoskeleton in order to grow, a process called ecdysis.

Key Terms

- **exoskeleton**: a hard outer structure that provides both structure and protection to creatures such as insects, Crustacea, and Nematoda

Phylum Nematoda

The Nematoda, similar to most other animal phyla, are triploblastic, possessing an embryonic mesoderm that is sandwiched between the ectoderm and endoderm. They are also bilaterally symmetrical: a longitudinal section will divide them into right and left sides that are symmetrical. Furthermore, the nematodes, or roundworms, possess a pseudocoelom and have both free-living and parasitic forms.

Both the nematodes and arthropods belong to the superphylum Ecdysozoa that is believed to be a clade consisting of all evolutionary descendants from one common ancestor. The name derives from the word ecdysis, which refers to the shedding, or molting, of the exoskeleton. The phyla in this group have a hard cuticle covering their bodies, which must be periodically shed and replaced for them to increase in size.

Phylum Nematoda includes more than 28,000 species with an estimated 16,000 being parasitic in nature. Nematodes are present in all habitats.

Morphology

In contrast with cnidarians, nematodes show a tubular morphology and circular cross-section. These animals are pseudocoelomates; they have a complete digestive system with a distinct mouth and anus. This is in contrast with the cnidarians where only one opening is present (an incomplete digestive system).

The cuticle of Nematodes is rich in collagen and a carbohydrate-protein polymer called chitin. It forms an external “skeleton” outside the epidermis. The cuticle also lines many of the organs internally, including the pharynx and rectum. The epidermis can be either a single layer of cells or a syncytium, which is a multinucleated cell formed from the fusion of uninucleated cells.

The overall morphology of these worms is cylindrical, while the head is radially symmetrical. A mouth opening is present at the anterior end with three or six lips. Teeth occur in some species in the form of cuticle extensions. Some nematodes may present other external modifications such as rings, head shields, or warts. Rings, however, do not reflect true internal body segmentation. The mouth leads to a muscular pharynx and intestine, which leads to a rectum and anal opening at the posterior end. In addition, the muscles of nematodes differ from those of most animals; they have a longitudinal layer only, which accounts for the whip-like motion of their movement.

Figure 28.4*B*.1: **Nematode shape**: Scanning electron micrograph of soybean cyst nematode and its egg. Nematodes are cylindrical in shape, often looking like thin hairs. They possess an exoskeleton that prevents them from drying out. It must be shed (a process called ecdysis) in order for them to grow.

Excretory System

In nematodes, specialized excretory systems are not well developed. Nitrogenous wastes may be lost by diffusion through the entire body or into the pseudocoelom (body cavity), where they are removed by specialized cells. Regulation of water and salt content of the body is achieved by renette glands, present under the pharynx in marine nematodes.

Nervous system

Most nematodes possess four longitudinal nerve cords that run along the length of the body in dorsal, ventral, and lateral positions. The ventral nerve cord is better developed than the dorsal or lateral cords. All nerve cords fuse at the anterior end, around the pharynx, to form head ganglia, or the "brain" of the worm (taking the form of a ring around the pharynx), as well as at the posterior end to form the tail ganglia. In *C. elegans*, the nervous system accounts for nearly one-third of the total number of cells in the animal.

Reproduction

Nematodes employ a variety of reproductive strategies that range from monoecious to dioecious to parthenogenic, depending upon the species under consideration. *C. elegans* is a monoecious species, having development of ova contained in a uterus as well as sperm contained in the spermatheca. The uterus has an external opening known as the vulva. The female genital pore is near the middle of the body, whereas the male's is at the tip. Specialized structures at the tail of the male keep him in place while he deposits sperm with copulatory spicules. Fertilization is internal with embryonic development beginning very soon after fertilization. The embryo is released from the vulva during the gastrulation stage. The embryonic development stage lasts for 14 hours; development then continues through four successive larval stages with ecdysis between each stage (L1, L2, L3, and L4) ultimately leading to the development of a young male or female adult worm. Adverse environmental conditions such as overcrowding and lack of food can result in the formation of an intermediate larval stage known as the dauer larva.

28.4C: Phylum Arthropoda

Arthropods are the largest grouping of animals all of which have jointed legs and an exoskeleton made of chitin.

Learning Objectives

- Describe the morphology of arthropoda

Key Points

- Arthropods include the Hexapoda (insects), the Crustacea (lobsters, crabs, and shrimp), the Chelicerata (the spiders and scorpions), and the Myriapoda (the centipedes and millipedes).
- Arthropods have a segmented body plan that contains fused segments divided into regions called tagma.
- Arthropods have an open circulatory system and can use book gills, book lungs, or tracheal tubes for respiration.

Key Terms

- **tagma**: a specialized grouping of arthropodan segments, such as the head, the thorax, and the abdomen with a common function
- **malpighian tubule**: a tubule that extends from the alimentary canal to the exterior of the organism, excreting water and wastes in the form of solid nitrogenous compounds
- **spiracle**: a pore or opening used (especially by spiders and some fish) for breathing

Phylum Arthropoda

The name “arthropoda” means “jointed legs” (in the Greek, “arthros” means “joint” and “podos” means “leg”); it aptly describes the enormous number of invertebrates included in this phylum. Arthropods dominate the animal kingdom with an estimated 85 percent of known species included in this phylum; many arthropods are as yet undocumented. The principal characteristics of all the animals in this phylum are functional segmentation of the body and presence of jointed appendages. Arthropods also show the presence of an exoskeleton made principally of chitin, which is a waterproof, tough polysaccharide. Phylum Arthropoda is the largest phylum in the animal world; insects form the single largest class within this phylum. Arthropods are eucoelomate, protostomic organisms.

Phylum Arthropoda includes animals that have been successful in colonizing terrestrial, aquatic, and aerial habitats. This phylum is further classified into five subphyla: Trilobitomorpha (trilobites, all extinct), Hexapoda (insects and relatives), Myriapoda (millipedes, centipedes, and relatives), Crustaceans (crabs, lobsters, crayfish, isopods, barnacles, and some zooplankton), and Chelicerata (horseshoe crabs, arachnids, scorpions, and daddy longlegs). Trilobites are an extinct group of arthropods found chiefly in the pre-Cambrian Era that are probably most closely related to the Chelicerata. These are identified based on fossil records.

Figure 28.4*C*.1: **Trilobite fossil**: *Acadoparadoxides*, possibly *A. briareus*, a large trilobite from about 500 million years ago from Morocco, North Africa (Middle Cambrian)

Morphology

A unique feature of animals in the arthropod phylum is the presence of a segmented body and fusion of sets of segments that give rise to functional body regions called tagma. Tagma may be in the form of a head, thorax, and abdomen, or a cephalothorax and abdomen, or a head and trunk. A central cavity, called the hemocoel (or blood cavity), is present; the

open circulatory system is regulated by a tubular, or single-chambered, heart. Respiratory systems vary depending on the group of arthropod. Insects and myriapods use a series of tubes (tracheae) that branch through the body, open to the outside through openings called spiracles, and perform gas exchange directly between the cells and air in the tracheae. Other organisms use variants of gills and lungs. Aquatic crustaceans utilize gills, terrestrial chelicerates employ book lungs, and aquatic chelicerates use book gills. The book lungs of arachnids (scorpions, spiders, ticks, and mites) contain a vertical stack of hemocoel wall tissue that somewhat resembles the pages of a book. Between each of the "pages" of tissue is an air space. This allows both sides of the tissue to be in contact with the air at all times, greatly increasing the efficiency of gas exchange. The gills of crustaceans are filamentous structures that exchange gases with the surrounding water.

Figure 28.4C.1: **Book gills**: The ventral side of a horseshoe crab showing the book gills located near the telson (tail). These gills flap back and forth bringing oxygen to the blood.

Groups of arthropods also differ in the organs used for excretion. Crustaceans possess green glands while insects use Malpighian tubules, which work in conjunction with the hindgut to reabsorb water while ridding the body of nitrogenous waste. The cuticle is the covering of an arthropod. It is made up of two layers: the epicuticle, which is a thin, waxy, water-resistant outer layer containing no chitin; and the chitinous procuticle, which is beneath the epicuticle. Chitin is a tough, flexible polysaccharide. In order to grow, the arthropod must shed the exoskeleton during a process called ecdysis ("to strip off"); this is a cumbersome method of growth. During this time, the animal is vulnerable to predation.

28.4D: Subphyla of Arthropoda

The Phylum Arthropoda includes a wide range of species divided into the subphyla: Hexapoda, Crustacea, Myriapoda, and Chelicerata.

Learning Objectives

- Differentiate among the subphylums hexapoda, myriapoda, crustacea, and chelicerata

Key Points

- The Hexapoda include insects; the Crustacea include lobster, crabs, and shrimp; the Myriapoda include centipedes and millipedes; and the Chelicerata include spiders, scorpions.
- The Hexapoda are the largest grouping of Arthropods, containing the more than one million species of insects, having representatives with six legs and one pair of antennae.
- The Myriapoda are terrestrial, prefering humid environments; they have between 10 and 750 legs.
- The Crustacea are primarily aquatic arthropods, but also include terrestrial forms, which have a cephalothorax covered by a carapace.
- The Chelicerata, which includes the spiders, horseshoe crabs, and scorpions, have mouth parts that are fang-like and used for capturing prey.

Key Terms

- **cephalothorax**: the fused head and thorax of spiders and crustaceans
- **forcipule**: a modified pincer-like foreleg in centipedes, capable of injecting venom

Representatives of Phylum Arthropoda

Subphylum Hexapoda

The name Hexapoda denotes the presence of six legs (three pairs) in these animals, which differentiates them from the number of pairs present in other arthropods. Hexapods are characterized by the presence of a head, thorax, and abdomen, constituting three tagma. The thorax bears the wings as well as six legs in three pairs. Many of the common insects we encounter on a daily basis, including ants, cockroaches, butterflies, and flies, are examples of Hexapoda.

Among the hexapods, the insects are the largest class in terms of species diversity as well as biomass in terrestrial habitats). Typically, the head bears one pair of sensory antennae, mandibles as mouthparts, a pair of compound eyes, and some ocelli (simple eyes), along with numerous sensory hairs. The thorax bears three pairs of legs (one pair per segment) and two pairs of wings, with one pair each on the second and third thoracic segments. The abdomen usually has eleven segments and bears reproductive apertures. Hexapoda includes insects that are winged (like fruit flies) and wingless (like fleas).

Figure 28.4*D*.1: **Insect showing wings and body segments**: *Protaetia fieberi* in flight posture. Hexapods are characterized by having three distinct tagma, or body segments. This beetle is just one of over one million different species of insects that inhabit the Earth.

Subphylum Myriapoda

Subphylum Myriapoda includes arthropods with numerous legs. Although the name is hyperbolic in suggesting that myriad legs are present in these invertebrates, the number of legs may vary from 10 to 750. This subphylum includes 13,000 species; the most commonly-found examples are millipedes and centipedes. All myriapods are terrestrial animals, prefering a humid environment.

Myriapods are typically found in moist soils, decaying biological material, and leaf litter. Centipedes, such as *Scutigera coleoptrata*,are classified as chilopods. These animals bear one pair of legs per segment, mandibles as mouthparts, and are somewhat dorsoventrally flattened. The legs in the first segment are modified to form forcipules (poison claws) that deliver venom to prey such as spiders and cockroaches, as centipedes are predatory. Millipedes bear two pairs of legs per diplosegment, a feature that results from embryonic fusion of adjacent pairs of body segments, are usually rounder in cross-section, and are herbivores or detritivores. Millipedes have visibly more numbers of legs as compared to centipedes, although they do not bear a thousand legs.

Figure $28.4D.1$: **House centipede**: The house centipede (*Scutigera coleoptrata*) is one of the 13,000 species of Myriapoda. They bear one pair of legs per segment and can inject venom. The Myriapods contain the millipedes and centipedes.

Subphylum Crustacea

Crustaceans are the most dominant aquatic arthropods since the total number of marine crustacean species stands at 67,000. However, there are also freshwater and terrestrial crustacean species. Krill, shrimp, lobsters, crabs, and crayfish are all examples of crustaceans. Terrestrial species like the wood lice (*Armadillidium* spp.) (also called pill bugs, rolly pollies, potato bugs, or isopods) are also crustaceans, although the number of non-aquatic species in this subphylum is relatively low.

Figure $28.4D.1$: **Mediterranean green Crab**: This crab (*Carcinus aestuarii*) is one of the 67,000 species of crustaceans inhabiting the world's oceans. Most crustaceans are decapods, having ten legs.

Crustaceans possess two pairs of antennae, mandibles as mouthparts, and biramous ("two branched") appendages: their legs are formed in two parts, as distinct from the uniramous ("one branched") myriapods and hexapods.

Unlike that of the Hexapoda, the head and thorax of most crustaceans is fused to form a cephalothorax, which is covered by a plate called the carapace, thus producing a body structure of two tagma. Crustaceans have a chitinous exoskeleton that is shed by molting whenever the animal increases in size. The exoskeletons of many species are also infused with calcium carbonate, which makes them even stronger than in other arthropods. Crustaceans have an open circulatory system where blood is pumped into the hemocoel by the dorsally-located heart. Hemocyanin and hemoglobin are the respiratory pigments present in these animals.

Subphylum Chelicerata

This subphylum includes animals such as spiders, scorpions, horseshoe crabs, and sea spiders and is predominantly terrestrial, although some marine species also exist. An estimated 77,000 species, found in almost all habitats, are included in subphylum Chelicerata.

The body of chelicerates may be divided into two parts: prosoma and opisthosoma, which are basically the equivalents of cephalothorax (usually smaller) and abdomen (usually larger). A “head” tagmum is not usually discernible. The phylum derives its name from the first pair of appendages, the chelicerae, which are specialized claw-like or fang-like mouthparts. These animals do not possess antennae. The second pair of appendages is known as pedipalps. In some species, such as sea spiders, an additional pair of appendages, called ovigers, is present between the chelicerae and pedipalps.

Chelicerae are used primarily for feeding, but in spiders, these are often modified into fangs that inject venom into their prey before feeding. Members of this subphylum have an open circulatory system with a heart that pumps blood into the hemocoel. Aquatic species have gills, whereas terrestrial species have either trachea or book lungs for gaseous exchange.

Figure $28.4D.1$: **Chelicera of spiders**: This photo shows the chelicera of a spider being held open with a stick. Some chelicerae, such as those found in spiders, are hollow and contain (or are connected to) venom glands which are used to inject venom into prey or a (perceived) threat.

The nervous system in chelicerates consists of a brain and two ventral nerve cords. These animals use external as well as internal fertilization strategies for reproduction, depending upon the species and its habitat. Parental care for the young ranges from absolutely none to relatively-prolonged care.

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28.5: Superphylum Deuterostomia

Topic hierarchy

- 28.5A: Phylum Echinodermata
- 28.5B: Classes of Echinoderms

Echinoderms consist of five distinct classes: sea stars, sea cucumbers, sea urchins and sand dollars, brittle stars, and sea lillies.

- 28.5C: Phylum Chordata

28.5A: Phylum Echinodermata

Echinoderms are invertebrates that have pentaradial symmetry, a spiny skin, a water vascular system, and a simple nervous system.

Learning Objectives

- Describe the characteristics of echinodermata

Key Points

- Echinoderms live exclusively in marine systems; they are widely divergent, with over 7,000 known species in the phylum.
- Echinoderms have pentaradial symmetry and a calcareous endoskeleton that may possess pigment cells that give them a wide range of colors, as well as cells that possess toxins.
- Echinoderms have a water vascular system composed of a central ring of canals that extend along each arm, through which water circulates for gaseous exchange and nutrition.
- Echinoderms have a very simple nervous system, comprised of a nerve ring at the center and five radial nerves extending outward along the arms; there is no structure resembling a brain.
- There are two sexes in echinoderms, which each release their eggs and sperm into the water; here, the sperm will fertilize the eggs.
- Echinoderms can reproduce asexually by regeneration.

Key Terms

- **madreporite**: a lightcolored calcerous opening used to filter water into the water vascular system of echinoderms
- **podocyte**: cells that filter the bodily fluids in echinoderms
- **pentaradial symmetry**: a variant of radial symmetry that arranges roughly equal parts around a central axis at orientations of 72° apart
- **water vascular system**: a hydraulic system used by echinoderms, such as sea stars and sea urchins, for locomotion, food and waste transportation, and respiration
- **ampulla**: the dilated end of a duct

Phylum Echinodermata

Echinodermata are so named owing to their spiny skin (from the Greek “echinos” meaning “spiny” and “dermos” meaning “skin”). This phylum is a collection of about 7,000 described living species. Echinodermata are exclusively marine organisms. Sea stars, sea cucumbers, sea urchins, sand dollars, and brittle stars are all examples of echinoderms. To date, no freshwater or terrestrial echinoderms are known.

Figure 28.5A.1: **Sea stars**: Star stars are among the most familiar of marine invertebrates. They are members of the phylum Echinodermata.

Morphology and Anatomy

Adult echinoderms exhibit pentaradial symmetry and have a calcareous endoskeleton made of ossicles, although the early larval stages of all echinoderms have bilateral symmetry. The endoskeleton is developed by epidermal cells and may possess pigment cells that give vivid colors to these animals, as well as cells laden with toxins. Echinoderms possess a simple digestive system which varies according to the animal's diet. Starfish are mostly carnivorous and have a mouth, oesophagus, two-part pyloric stomach with a pyloric duct leading to the intestine and rectum, with the anus located in the center of the aboral body surface. In many species, the large cardiac stomach can be everted and digest food outside the body. Gonads are present in each arm. In echinoderms such as sea stars, every arm bears two rows of tube feet on the oral side which help in attachment to the substratum. These animals possess a true coelom that is modified into a unique circulatory system called a water vascular system. The more notably distinct trait, which most echinoderms have, is their remarkable powers of regeneration of tissue, organs, limbs, and, in some cases, complete regeneration from a single limb.

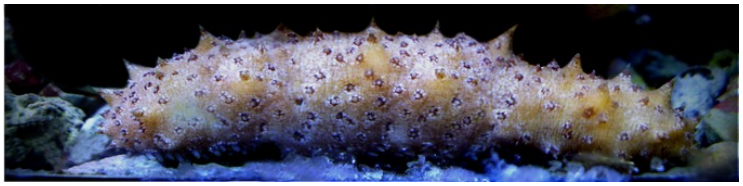

Figure 28.5A.1: **Sea cucumbers**: Sea cucumbers are a member of the phylum Echinodermata which are found on the sea floor worldwide. As with all echinoderms, sea cucumbers have an endoskeleton just below the skin, calcified structures that are usually reduced to isolated microscopic ossicles joined by connective tissue.

Water Vascular System

Echinoderms possess a unique ambulacral or water vascular system, consisting of a central ring canal and radial canals that extend along each arm. Water circulates through these structures and facilitates gaseous exchange as well as nutrition, predation, and locomotion. The water vascular system also projects from holes in the skeleton in the form of tube feet. These tube feet can expand or contract based on the volume of water (hydrostatic pressure) present in the system of that arm.

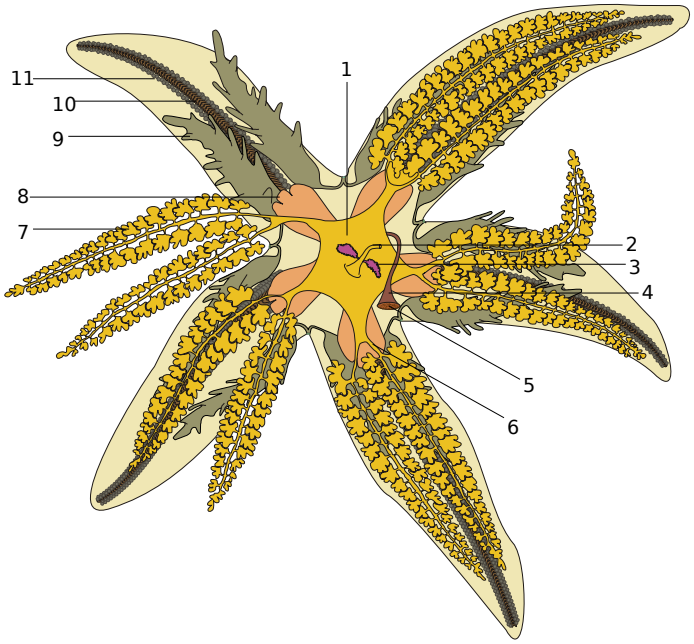

Figure 28.5A.1: **The water vascular system of the sea star**: 1. Pyloric stomach 2. Intestine 3. Rectal gland 4. Stone canal 5. Madreporite 6. Pyloric duct 7. Pyloric cecum 8. Cardiac stomach 9. Gonad 10. Ambulacral plates 11. Ampullae

The madreporite is a light-colored, calcerous opening used to filter water into the water vascular system of echinoderms. Acting as a pressure-equalizing valve, it is visible as a small red or yellow button-like structure (similar to a small wart) on the aboral surface of the central disk of a sea star. Close up, it is visibly structured, resembling a “madrepore” colony. From this, it derives its name. Water enters the madreporite on the aboral side of the echinoderm. From there, it passes into the stone canal, which moves water into the ring canal. The ring canal connects the radial canals (there are five in a pentaradial animal), and the radial canals move water into the ampullae, which have tube feet through which the water moves. By moving water through the unique water vascular system, the echinoderm can move and force open mollusk shells during feeding.

Other Body Systems

The nervous system in these animals is a relatively simple structure with a nerve ring at the center and five radial nerves extending outward along the arms. Structures analogous to a brain or derived from fusion of ganglia are not present in these animals.

Podocytes, cells specialized for ultrafiltration of bodily fluids, are present near the center of echinoderms. These podocytes are connected by an internal system of canals to the madreporite.

Echinoderms are sexually dimorphic and release their eggs and sperm cells into water; fertilization is external. In some species, the larvae divide asexually and multiply before they reach sexual maturity. Echinoderms may also reproduce asexually, as well as regenerate body parts lost in trauma.

28.5B: Classes of Echinoderms

Learning Objectives

- Differentiate among the classes of echinoderms

The phylum echinoderms is divided into five extant classes: Asteroidea (sea stars), Ophiuroidea (brittle stars), Echinoidea (sea urchins and sand dollars), Crinoidea (sea lilies or feather stars), and Holothuroidea (sea cucumbers).

The most well-known echinoderms are members of class Asteroidea, or sea stars. They come in a large variety of shapes, colors, and sizes, with more than 1,800 species known so far. The key characteristic of sea stars that distinguishes them from other echinoderm classes includes thick arms (ambulacra; singular: ambulacrum) that extend from a central disk where organs penetrate into the arms. Sea stars use their tube feet not only for gripping surfaces, but also for grasping prey. Sea stars have two stomachs, one of which can protrude through their mouths and secrete digestive juices into or onto prey, even before ingestion. This process can essentially liquefy the prey, making digestion easier.

Brittle stars belong to the class Ophiuroidea. Unlike sea stars, which have plump arms, brittle stars have long, thin arms that are sharply demarcated from the central disk. Brittle stars move by lashing out their arms or wrapping them around objects and pulling themselves forward. Of all echinoderms, the Ophiuroidea may have the strongest tendency toward 5-segment radial (pentaradial) symmetry. Ophiuroids are generally scavengers or detritivores. Small organic particles are moved into the mouth by the tube feet. Ophiuroids may also prey on small crustaceans or worms. Some brittle stars, such as the six-armed members of the family Ophiactidae, are fissiparous (divide though fission), with the disk splitting in half. Regrowth of both the lost part of the disk and the arms occur, yielding an animal with three large arms and three small arms during the period of growth.

Sea urchins and sand dollars are examples of Echinoidea. These echinoderms do not have arms, but are hemispherical or flattened with five rows of tube feet that help them in slow movement; tube feet are extruded through pores of a continuous internal shell called a test. Like other echinoderms, sea urchins are bilaterans. Their early larvae have bilateral symmetry, but they develop fivefold symmetry as they mature. This is most apparent in the “regular” sea urchins, which have roughly spherical bodies, with five equally-sized parts radiating out from their central axes. Several sea urchins, however, including the sand dollars, are oval in shape, with distinct front and rear ends, giving them a degree of bilateral symmetry. In these urchins, the upper surface of the body is slightly domed, but the underside is flat, while the sides are devoid of tube feet. This “irregular” body form has evolved to allow the animals to burrow through sand or other soft materials.

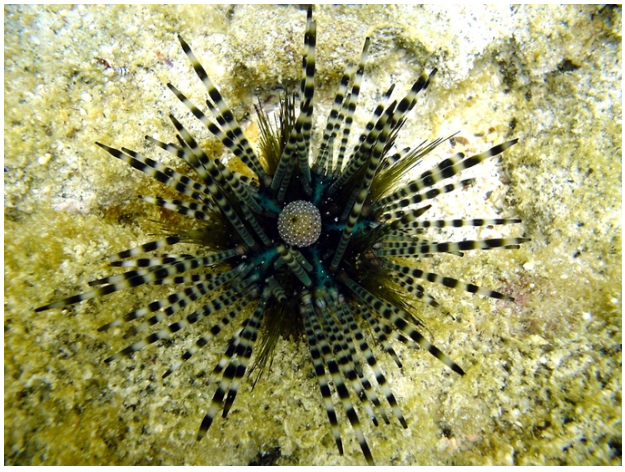

Figure 28.5B.1: **Sea urchins**: Sea urchins do not have arms, but have rows of tube feet that can be extended out of pores of the internal shell.

Sea lilies and feather stars are examples of Crinoidea. Both of these species are suspension feeders. They live both in shallow water and in depths as great as 6,000 meters. Sea lilies refer to the crinoids which, in their adult form, are attached to the sea bottom by a stalk. Feather stars or comatulids refer to the unstalked forms. Crinoids are characterized by a mouth on the top surface that is surrounded by feeding arms. They have a U-shaped gut; their anus is located next to the mouth. Although the basic echinoderm pattern of fivefold symmetry can be recognized, most crinoids have many more than five

arms. Crinoids usually have a stem used to attach themselves to a substrate, but many live attached only as juveniles and become free-swimming as adults.

Figure $28.5B.1$: **Sea lilies**: Sea lilies, like feather stars, have a mouth on their upper surface that is surrounded by arms used for feeding.

Sea cucumbers of class Holothuroidea are extended in the oral-aboral axis and have five rows of tube feet. These are the only echinoderms that demonstrate "functional" bilateral symmetry as adults because the uniquely-extended oral-aboral axis compels the animal to lie horizontally rather than stand vertically. Like all echinoderms, sea cucumbers have an endoskeleton just below the skin: calcified structures that are usually reduced to isolated microscopic ossicles joined by connective tissue. In some species these can sometimes be enlarged to flattened plates, forming armor. In pelagic species, such as Pelagothuria natatrix, the skeleton and a calcareous ring are absent.

Figure $28.5B.1$: **Sea cucumbers**: Sea cucumbers are the only echinoderms that demonstrate "functional" bilateral symmetry as adults, as they lie horizontally as opposed to the vertical axis of other echinoderms.

Key Points

- Sea stars have thick arms called ambulacra that are used for gripping surfaces and grabbing hold of prey.
- Brittle stars have thin arms that wrap around prey or objects to pull themselves forward.

- Sea urchins and sand dollars embody flattened discs that do not have arms, but do have rows of tube feet they use for movement.
- Sea cucumbers demonstrate “functional” bilateral symmetry as adults because they actually lie horizontally rather than stand vertically.
- Sea lilies and feather stars are suspension feeders.

Key Terms

- **ossicle**: a small bone (or bony structure), especially one of the three of the middle ear
- **fissiparous**: of cells that reproduce through fission, splitting into two
- **ambulacrum**: a row of pores for the protrusion of appendages such as tube feet.

28.5C: Phylum Chordata

The phylum Chordata contains all animals that have a dorsal notochord at some stage of development; in most cases, this is the backbone.

Learning Objectives

- Name the features that distinguish the members of the phylum chordata

Key Points

- The phylum chordata is named for the notochord, a longitudinal, flexible rod between the digestive tube and the nerve cord; in vertebrates, this is the spinal column.
- The chordates are also characterized by a dorsal nerve cord, which splits into the brain and spinal cord.
- Chordata contains two clades of invertebrates: Urochordata (tunicates) and Cephalochordata (lancelets), both of which are suspension feeders.
- The phylum chordata includes all animals that share four characteristics, although they might each possess some of them at different stages of their development: a notochord, a dorsal nerve cord, pharyngeal slits, and a postanal tail.
- Chordata contains five classes of animals: fish, amphibians, reptiles, birds, and mammals; these classes are separated by whether or not they can regulate their body temperature, the manner by which they consume oxygen, and their method of reproduction.

Key Terms

- **dorsal nerve cord**: a hollow cord dorsal to the notochord, formed from a part of the ectoderm that rolls, forming a hollow tube.
- **notochord**: a flexible rodlike structure that forms the main support of the body in the lowest chordates; a primitive spine
- **pharyngeal slit**: filter-feeding organs found in non-vertebrate chordates (lancelets and tunicates) and hemichordates living in aquatic environments

Phylum Chordata

Animals in the phylum Chordata share four key features that appear at some stage of their development:

- A notochord, or a longitudinal, flexible rod between the digestive tube and the nerve cord. In most vertebrates, it is replaced developmentally by the vertebral column. This is the structure for which the phylum is named.
- A dorsal nerve cord which develops from a plate of ectoderm that rolls into a tube located dorsal to the notochord. Other animal phyla have solid nerve cords ventrally located. A chordate nerve cord splits into the central nervous system: the brain and spinal cord.
- Pharyngeal slits, which allow water that enters through the mouth to exit without continuing through the entire digestive tract. In many of the invertebrate chordates, these function as suspension feeding devices; in vertebrates, they have been modified for gas exchange, jaw support, hearing, and other functions.
- A muscular, postanal tail which extends posterior to the anus. The digestive tract of most nonchordates extends the length of the body. In chordates, the tail has skeletal elements and musculature, and can provide most of the propulsion in aquatic species.

In some groups, some of these traits are present only during embryonic development. In addition to containing vertebrate classes, the phylum Chordata contains two clades of invertebrates: Urochordata (tunicates) and Cephalochordata (lancelets). However, even though they are invertebrates, they share characteristics with other chordates that places them in this phylum. For example, tunicate larvae have both a notochord and a nerve cord which are lost in adulthood. Most tunicates live on the ocean floor and are suspension feeders. Cephalochordates, or lancelets, have a notochord and a nerve cord (but no brain or specialist sensory organs) and a very simple circulatory system. Lancelets are suspension feeders that feed on phytoplankton and other microorganisms.

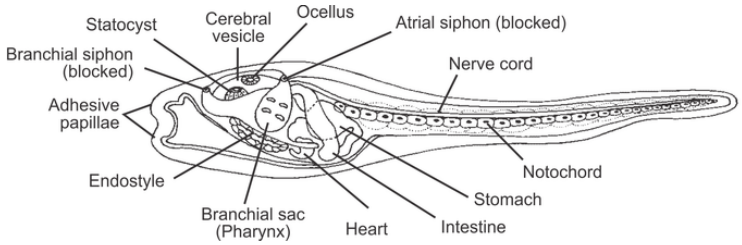

Figure 28.5C.1: **Structures present in a tunicate larva**: While tunicates are invertebrates and may seem very different from the more familiar members of Chordata, the tunicate larva possesses both a notochord and a dorsal nerve cord, although both are lost in adulthood.

The phylum Chordata contains all of the animals that have a rod-like structure used to give them support. In most cases this is the spine or backbone. Within Chordata there are five classes of animals: fish, amphibians, reptiles, birds, and mammals. Three dividing factors separate these classes:

- Regulation of body temperature: animals are either homeothermic (can regulate their internal temperature so that it is kept at an optimum level) or poikilothermic (cannot regulate their internal temperature, the environment affects how hot or cold they are)
- Oxygen Absorption: the way in which oxygen is taken in from the air, which can be through gills, the skin (amphibians), or lungs
- Reproduction: this factor is particularly varied. Animals can be oviparous (lay eggs) or viviparous (birth live young). Fertilization can occur externally or internally. In mammals, the mother produces milk for the young.

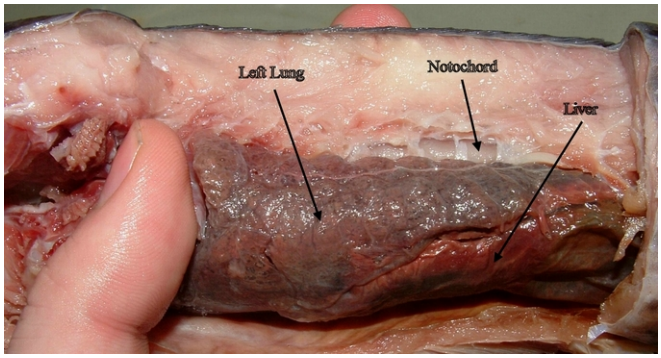

Figure 28.5C.1: **Notochord**: All chordates possess a notochord, or a type of flexible support rod, at some point in their development. In this dissected lungfish, which is a member of the chordates, tissues have been dissected away to display the notochord.

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CHAPTER OVERVIEW

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29.1: CHORDATES

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29.1B: CHORDATES AND THE EVOLUTION OF VERTEBRATES
29.1C: THE EVOLUTION OF CRANIATA AND VERTEBRATA
29.1D: CHARACTERISTICS OF VERTEBRATES
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Gnathostomes, jawed vertebrates, can be divided into two types of fish: Chondrichthyes (cartilaginous fish) or Osteichthyes (bony fish).

29.3: AMPHIBIANS

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29.7: THE EVOLUTION OF PRIMATES

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All primates exhibit adaptations for climbing trees and have evolved into two main groups: Prosimians and Anthropoids.

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29.1: Chordates

Topic hierarchy

- 29.1A: Characteristics of Chordata
- 29.1B: Chordates and the Evolution of Vertebrates
- 29.1C: The Evolution of Craniata and Vertebrata
- 29.1D: Characteristics of Vertebrates

A: Characteristics of Chordata

Animals in the phylum Chordata share four key features: a notochord, a dorsal hollow nerve cord, pharyngeal slits, and a post-anal tail.

Learning Objectives

- Identify the key features of the chordates

Key Points

- These characteristics are only present during embryonic development in some chordates.
- The notochord provides skeletal support, gives the phylum its name, and develops into the vertebral column in vertebrates.
- The dorsal hollow nerve cord develops into the central nervous system: the brain and spine.
- Pharyngeal slits are openings in the pharynx that develop into gill arches in bony fish and into the jaw and inner ear in terrestrial animals.
- The post-anal tail is a skeletal extension of the posterior end of the body, being absent in humans and apes, although present during embryonic development.

Key Terms

- **notochord**: a flexible rodlike structure that forms the main support of the body in the lowest chordates; a primitive spine
- **nerve cord**: a dorsal tubular cord of nervous tissue above the notochord of a chordate
- **pharyngeal slit**: filter-feeding organs found in non-vertebrate chordates (lancelets and tunicates) and hemichordates living in aquatic environments

Characteristics of Chordata

Animals in the phylum Chordata share four key features that appear at some stage during their development (often, only during embryogenesis) (:

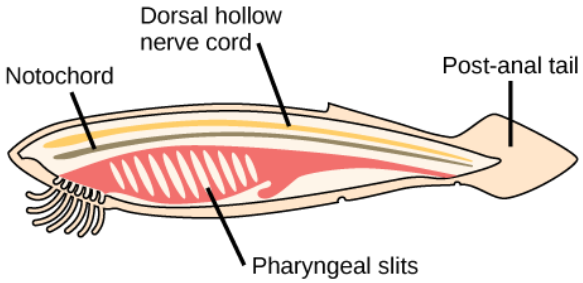

Figure 29.1A.1: **Defining characteristics of chordates**: In chordates, four common features appear at some point during development: a notochord, a dorsal hollow nerve cord, pharyngeal slits, and a post-anal tail.

1. a notochord
2. a dorsal hollow nerve cord
3. pharyngeal slits
4. post-anal tail

Notochord

The chordates are named for the notochord: a flexible, rod-shaped structure that is found in the embryonic stage of all chordates and also in the adult stage of some chordate species. It is located between the digestive tube and the nerve cord, providing skeletal support through the length of the body. In some chordates, the notochord acts as the primary axial support of the body throughout the animal's lifetime.

In vertebrates, the notochord is present during embryonic development, at which time it induces the development of the neural tube which serves as a support for the developing embryonic body. The notochord, however, is replaced by the

ral column (spine) in most adult vertebrates.

al Hollow Nerve Cord

The dorsal hollow nerve cord derives from ectoderm that rolls into a hollow tube during development. In chordates, it is located dorsally (at the top of the animal) to the notochord. In contrast to the chordates, other animal phyla are characterized by solid nerve cords that are located either ventrally or laterally. The nerve cord found in most chordate embryos develops into the brain and spinal cord, which comprise the central nervous system.

Pharyngeal Slits

Pharyngeal slits are openings in the pharynx (the region just posterior to the mouth) that extend to the outside environment. In organisms that live in aquatic environments, pharyngeal slits allow for the exit of water that enters the mouth during feeding. Some invertebrate chordates use the pharyngeal slits to filter food out of the water that enters the mouth. In vertebrate fishes, the pharyngeal slits develop into gill arches, the bony or cartilaginous gill supports.

In most terrestrial animals, including mammals and birds, pharyngeal slits are present only during embryonic development. In these animals, the pharyngeal slits develop into the jaw and inner ear bones.

Post-anal Tail

The post-anal tail is a posterior elongation of the body, extending beyond the anus. The tail contains skeletal elements and muscles, which provide a source of locomotion in aquatic species. In some terrestrial vertebrates, the tail also helps with balance, courting, and signaling when danger is near. In humans and other apes, the post-anal tail is present during embryonic development, but is vestigial as an adult.

29.1B: Chordates and the Evolution of Vertebrates

Chordata contains two subphylums of invertebrates: Urochordata (tunicates) and Cephalochordata (lancelets).

Learning Objectives

- Describe the features and phylogenetic history of lancelets and urochordata

Key Points

- Urochordata (tunicates) and Cephalochordata (lancelets) are invertebrates because they lack a backone.
- Larval tunicates (Urochordata) posses all four structures that classify chordates, but adult tunicates retain only pharyngeal slits.
- Larval tunicates swim for a few days after hatching, then attach to a marine surface and undergo metamorphosis into the sessile adult form.
- Lancelets (Cephalochordata) are marine organisms that possess all features of chordates; they are named Cephalochordata because the notochord extends into the head.
- Lancelets may be the closest-living relatives to vertebrates.

Key Terms

- **Urochordata**: a taxonomic subphylum within the phylum Chordata: the tunicates or sea squirts
- **Cephalochordata**: a taxonomic subphylum within the phylum Chordata: the lancelets
- **sessile**: permanently attached to a substrate; not free to move about; “an attached oyster”

Chordates and the Evolution of Vertebrates

The most familiar group of chordates is the vertebrates. However, in addition to the subphylum Vertebrata, the phylum Chordata also contains two subphylums of invertebrates: Urochordata and Cephalochordata. Members of these groups also possess the four distinctive features of chordates at some point during their development: a notochord, a dorsal hollow nerve cord, pharyngeal slits, and a post-anal tail. Unlike vertebrates, urochordates and cephalochordates never develop a bony backbone.

Urochordata

Members of Urochordata are also known as tunicates. The name tunicate derives from the cellulose-like carbohydrate material, called the tunic, which covers the outer body of tunicates. Although tunicates are classified as chordates, only the larval form possesses all four common structures. Adults only maintain pharyngeal slits and lack a notochord, a dorsal hollow nerve cord, and a post-anal tail.

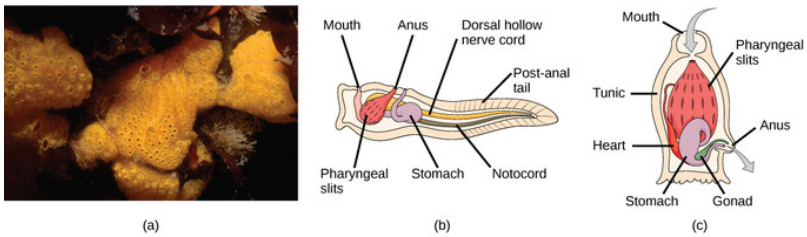

Figure 29.1B.1: **Urochordates**: (a) This photograph shows a colony of the tunicate Botrylloides violaceus. (b) The larval stage of the tunicate possesses all of the features characteristic of chordates: a notochord, a dorsal hollow nerve cord, pharyngeal slits, and a post-anal tail. (c) In the adult stage, the notochord, nerve cord, and tail disappear.

Most tunicates are hermaphrodites. Tunicate larvae hatch from eggs inside the adult tunicate’s body. After hatching, a tunicate larva swims for a few days until it finds a suitable surface on which it can attach, usually in a dark or shaded location. It then attaches via the head to the surface and undergoes metamorphosis into the adult form, at which point the notochord, nerve cord, and tail disappear.

Most tunicates live a sessile existence on the ocean floor and are suspension feeders. The primary foods of tunicates are plankton and detritus. Seawater enters the tunicate’s body through its incurrent siphon. Suspended material is filtered out

of this water by a mucous net (pharyngeal slits) and is passed into the intestine via the action of cilia. The anus empties into the excurrent siphon, which expels wastes and water. Tunicates are found in shallow ocean waters around the world.

Cephalochordata

Members of Cephalochordata possess a notochord, dorsal hollow nerve cord, pharyngeal slits, and a post-anal tail in the adult stage. They do not have a true brain, but the notochord extends into the head, which gives the subphylum its name (“cephalo” is Greek for head). Extinct members of this subphylum include *Pikaia*, which is the oldest known cephalochordate. *Pikaia* fossils were recovered from the Burgess shales of Canada and dated to the middle of the Cambrian age, making them more than 500 million years old.

Extant members of Cephalochordata are the lancelets, named for their blade-like shape. Lancelets are only a few centimeters long and are usually found buried in sand at the bottom of warm temperate and tropical seas. Like tunicates, they are suspension feeders. With notochord and paired muscle blocks, the lancelet and *Pikaia* may belong to the chordate group of animals from which the vertebrates have descended.

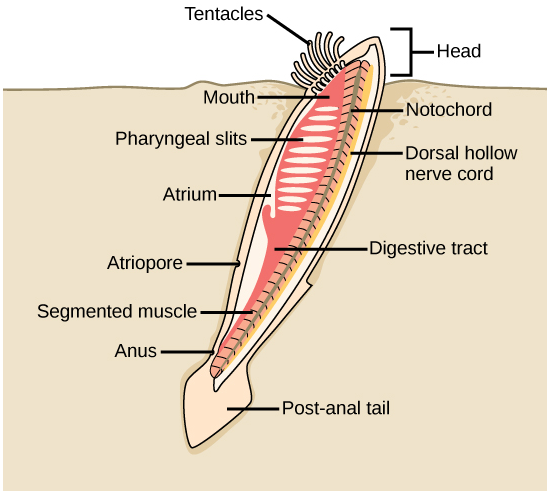

Figure 29.1B.1: **Cephalochrodates**: The lancelet, like all cephalochordates, has a head. Adult lancelets retain the four key features of chordates: a notochord, a dorsal hollow nerve cord, pharyngeal slits, and a post-anal tail. Water from the mouth enters the pharyngeal slits, which filter out food particles. The filtered water then collects in the atrium and exits through the atriopore.

29.1C: The Evolution of Craniata and Vertebrata

Both genomic and fossil evidence suggests that vertebrates evolved from craniates, which evolved from invertebrate chordates.

Learning Objectives

- Explain how genomics informs scientists about chordate evolution

Key Points

- The clade Craniata includes animals that have a cranium: a bony, cartilaginous, or fibrous structure that surrounds the brain, jaw, and facial bones.
- Members of Craniata include the vertebrates and hagfish.
- Genomic evidence suggests that vertebrates diverged from cephalochordates (lancelets), which had previously diverged from urochordates (tunicates).
- Fossil evidence suggests that most vertebrate diversity originated in the Cambrian explosion 540 million years ago.
- Two whole- genome duplications occurred in early vertebrate history.

Key Terms

- **cranium**: the part of the skull enclosing the brain, the braincase
- **genomics**: the study of the complete genome of an organism
- **Cambrian explosion**: the relatively rapid appearance (over a period of many millions of years), around 530 million years ago, of most major animal phyla as demonstrated in the fossil record

Craniata and Vertebrata

The clade Craniata is a subdivision of Chordata. Members of Craniata posses a cranium, which is a bony, cartilaginous, or fibrous structure surrounding the brain, jaw, and facial bones. The clade Craniata includes all vertebrates and the hagfishes (Myxini), which have a cranium but lack a backbone. Hagfish are the only known living animals that have a skull, but not a vertebral column.

Figure 29.1C.1: **Hagfish**: Although it lacks a backbone, the hagfish is a member of the Craniata clade because it possesses a bony skull.

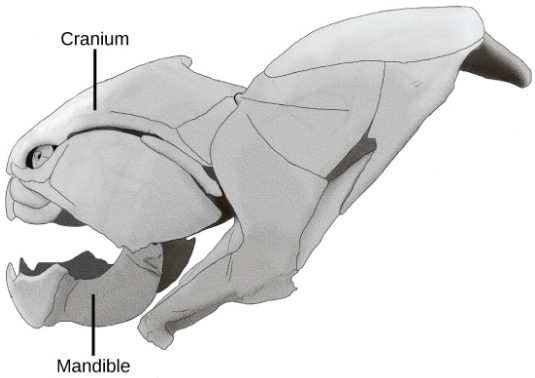

Figure 29.1C.1: **Clade Craniata**: Craniata, including this fish (Dunkleosteus), are characterized by the presence of a cranium, mandible, and other facial bones.

Vertebrates are members of the subphylum Vertebrata, the clade Craniata, and the phylum Chordata. Vertebrates display the four characteristic features of chordates, but they are named for the vertebral column composed of a series of bony vertebrae joined together as a backbone. In adult vertebrates, the vertebral column replaces the embryonic notochord.

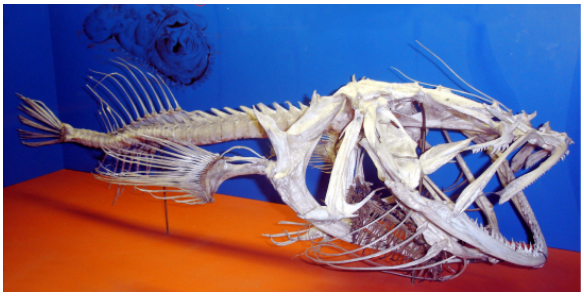

Figure 29.1C.1: **Vertebrates**: Vertebrata are characterized by the presence of a backbone, such as the one that runs through the middle of this fish. All vertebrates are in the Craniata clade and have a cranium.

Vertebrate Evolution

In the phylum Chordata, the closest relatives of the vertebrates are the invertebrate chordates. Based on the molecular analysis of vertebrate and invertebrate genomes (genomics), scientists can determine the evolutionary history of different phylogenetic groups.

According to these genomic analyses, vertebrates appear to be more closely related to the lancelets (cephalochordates) than to the tunicates (urochordates). This suggests that the cephalochordates first diverged from urochordates, and that vertebrates subsequently diverged from the cephalochordates. This hypothesis is further supported by the fossil of a 530 million-year-old organism with a brain and eyes like a vertebrate, but without the skull found in a craniate. A comparison of the genomes of a lancelet, tunicate, lamprey, fish, chicken, and human confirmed that two whole-genome duplications occurred in the early history of the Vertebrata subphylum.

Both fossil and genomic evidence suggests that vertebrates arose during the Cambrian explosion.The Cambrian explosion was the relatively brief span of time during the Cambrian period during which many animal groups appeared and rapidly diversified. Most modern animal phyla originated during the Cambrian explosion.

29.1D: Characteristics of Vertebrates

Vertebrata is a subphlyum of Chordata that is further defined by their bony backbone.

Learning Objectives

- Identify the defining characteristics of vertebrates

Key Points

- As chordates, vertebrates have the same common features: a notochord, a dorsal hollow nerve cord, pharyngeal slits, and a post-anal tail.
- Vertebrates are further differentiated from chordates by their vertebral column, which forms when their notochord develops into the column of bony vertebrae separated by discs.
- Vertebrates are the only chordates that have a brain as part of their central nervous system.

Key Terms

- **vertebral column**: the series of vertebrae that protect the spinal cord; the spinal column
- **chordate**: a member of the phylum Chordata; numerous animals having a notochord at some stage of their development; in vertebrates this develops into the spine
- **notochord**: a flexible rodlike structure that forms the main support of the body in the lowest chordates; a primitive spine

Characteristics of Vertebrates

Vertebrates are members of the subphylum Vertebrata, under the phylum Chordata and under the kingdom Animalia. Animals that possess bilateral symmetry can be divided into two groups, protostomes and deuterostomes, based on their patterns of embryonic development. The deuterostomes, whose name translates as “second mouth,” consist of two phyla: Chordata and Echinodermata. Echinoderms are invertebrate marine animals that have pentaradial symmetry and a spiny body covering; the phylum includes sea stars, sea urchins, and sea cucumbers. The phylum Chordata contains two groups of invertebrate chordates, but the most conspicuous and familiar members of Chordata are the vertebrates.

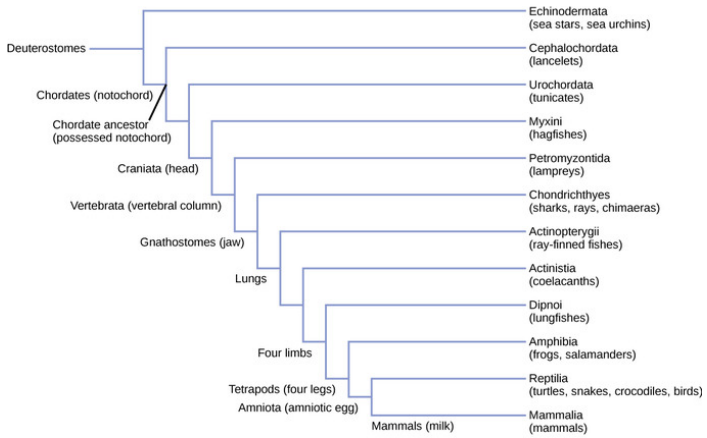

Figure 29.1D.1: **Phylum chordata**: All chordates are deuterostomes, possessing a notochord. Vertebrates are differentiated by having a vertebral column.

As chordates, all vertebrates have a similar anatomy and morphology with the same qualifying characteristics: a notochord, a dorsal hollow nerve cord, pharyngeal slits, and a post-anal tail. However, the subphylum Vertebrata is distinguished from the phylum Chordata by the development of the notochord into a bony backbone. Vertebrates include the amphibians, reptiles, mammals, and birds, as well as the jawless fishes, bony fishes, sharks, and rays.

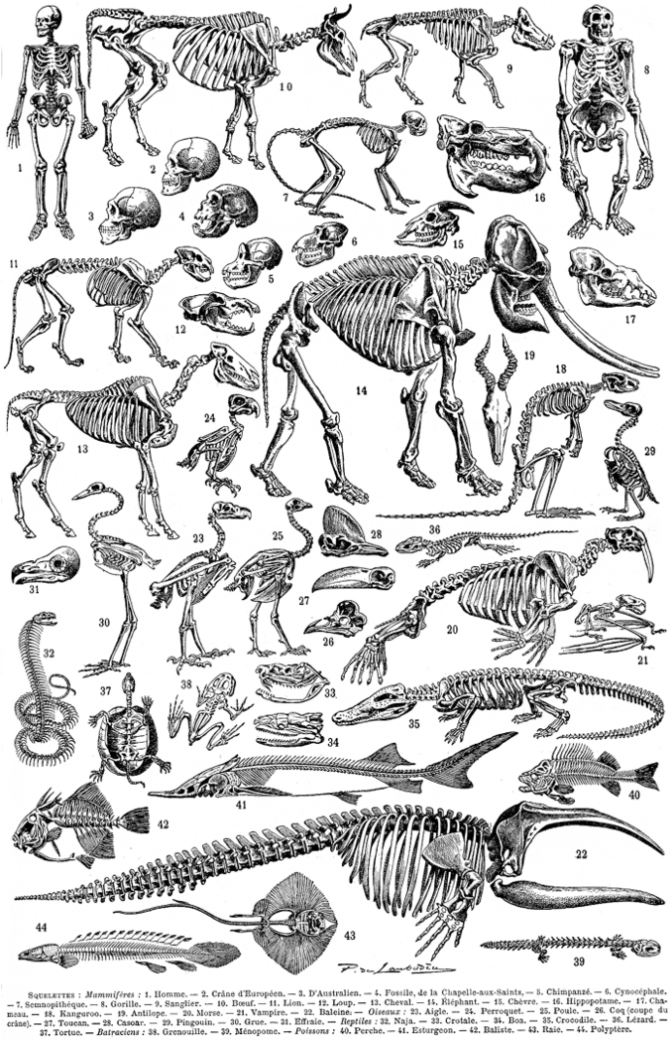

Figure 29.1D.1: **Diversity of vertebrates: animals with backbones**: The subphylum Vertebrata contains all animals that possess backbones, gills, and a central nervous system in at least one phase of development. Vertebrates include amphibians, reptiles, mammals, and birds, as well as the jawless fishes, bony fishes, sharks, and rays.

More than 64,000 species of vertebrates have been described, but the extant vertebrate species represent only a small portion of all the vertebrates that have existed. Vertebrates range in size from the frog species *Paedophryne amauensis* (as small as 7.7 mm (0.3 inch)) to the blue whale (as large as 33 m (110 ft)). Vertebrates comprise about 4 percent of all described animal species; the remainder are invertebrates, which lack backbones.

Anatomy and Morphology

All vertebrates are built along the basic chordate body plan: a stiff rod running through the length of the animal (vertebral column), with a hollow tube of nervous tissue (the spinal cord) above it and the gastrointestinal tract below. In all vertebrates, there is a mouth at anterior end of the animal and an anus before the posterior end of the body. There is a tail posterior to the anus during at least one phase of the animal's development.

The Vertebral Column

Vertebrates are defined by the presence of the vertebral column. In vertebrates, the notochord develops into the vertebral column or spine: a series of bony vertebrae each separated by mobile discs. These vertebrae are always found on the dorsal side of the animal. However, a few vertebrates have secondarily lost their vertebrae and, instead, retain the notochord into adulthood (e.g., the sturgeon fish).

Figure 29.1D.1: **Vertebral column**: A fossilized skeleton of the dinosaur Diplodocus carnegii shows an extreme example of the backbone that characterizes vertebrates.

Central Nervous System

Vertebrates are also the only members of Chordata to possess a brain. In chordates, the central nervous system is based on a hollow nerve tube that runs dorsal to the notochord along the length of the animal. In vertebrates, the anterior end of the nerve tube expands and differentiates into three brain vesicles.

Vertebrate Classification

Vertebrates are the largest group of chordates, with more than 62,000 living species. Vertebrates are grouped based on anatomical and physiological traits. The traditional groups include Agnatha, Chondrichthyes, Osteichthyes, Amphibia, Reptilia, Aves, and Mammalia.

Animals that possess jaws are known as gnathostomes, meaning “jawed mouth.” Gnathostomes include fishes and tetrapods (amphibians, reptiles, birds, and mammals). Tetrapods can be further divided into two groups: amphibians and amniotes. Amniotes are animals whose eggs are adapted for terrestrial living; this group includes mammals, reptiles, and birds. Amniotic embryos, developing in either an externally-shelled egg or an egg carried by the female, are provided with a water-retaining environment and are protected by amniotic membranes.

29.2: Fishes

29.2A: Agnathans: Jawless Fishes

29.2B: Gnathostomes - Jawed Fishes

Gnathostomes, jawed vertebrates, can be divided into two types of fish: Chondrichthyes (cartilaginous fish) or Osteichthyes (bony fish).

29.2A: Agnathans: Jawless Fishes

The superclass Agnatha describes fish that lack jaws and includes the extant species of hagfish and lampreys.

Learning Objectives

- Differentiate between the taxa of jawless fishes

Key Points

- Most agnathans are now extinct, but two branches exist today: hagfishes (not true vertebrates) and lampreys (true vertebrates).
- The earliest jawless fishes were the ostracoderms, which had bony scales as body armor.
- Hagfish are eel-like marine scavengers in the clade Myxini that produce slime and can tie themselves into knots.
- Lampreys are in the clade Petromyzontidae and appear morphologically similar to hagfish, but contain cartilaginous vertebral elements as an adult; thus, they are considered true vertebrates.

Key Terms

- **hagfish**: any of several primitive eellike creatures, of the family Myxinidae, having a sucking mouth with rasping teeth; considered edible in Japan, their skin is used to make a form of leather
- **lamprey**: any long slender primitive eel-like freshwater and saltwater fish of the Petromyzontidae family, having a sucking mouth with rasping teeth, but no jaw
- **agnathan**: a member of the superclass Agnatha of jawless vertebrates

Agnathans: Jawless Fishes

Jawless fishes or agnathans are craniates that represent an ancient vertebrate lineage that arose over one half-billion years ago. “Gnathos” is Greek for “jaw” and the prefix “a” means “without,” so agnathans are “without jaws. ” Most agnathans are now extinct, but two branches still exist today: hagfishes and lampreys. Hagfishes and lampreys are recognized as separate clades, primarily because lampreys are true vertebrates, whereas hagfishes are not. A defining feature of agnathans is the lack of paired lateral appendages or fins.

Some of the earliest jawless fishes were the ostracoderms (Greek for “bone-skin”). Ostracoderms were vertebrate fishes encased in bony armor, unlike present-day jawless fishes, which lack bone in their scales.

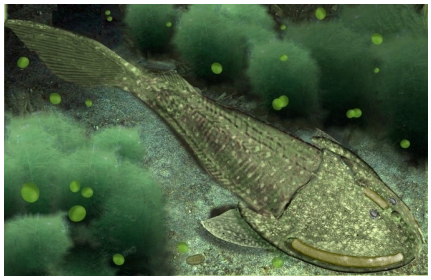

Figure 29.2A.1: **Ostracoderm**: Ostracoderms were some of the earliest jawless fishes and were covered in bony armor. Present-day jawless fishes lack bone in their scales.

Myxini: Hagfishes

The clade Myxini includes at least 20 species of hagfishes. Hagfishes are eel-like scavengers that live on the ocean floor and feed on dead invertebrates, other fishes, and marine mammals. Hagfishes are entirely marine and are found in oceans around the world, except for the polar regions. Hagfish have slime glands beneath the skin that constantly release mucus, allowing them to escape from the grip of predators. Hagfish can also twist their bodies into a knot to gain a mechanical advantage while feeding and are notorious for eating carcasses from the inside out.

Figure 29.2A.1: **Hagfishes**: Pacific hagfish are scavengers that live on the ocean floor. These agnathans are classified as Myxini and do not have a vertebral column.

The skeleton of a hagfish is composed of cartilage, which includes a cartilaginous notochord that runs the length of the body. This notochord provides support to the hagfish's body. Unlike true vertebrates, hagfishes do not replace the notochord with a vertebral column during development. Since they have a cartilaginous skull, they are classified in the clade Craniata.

Petromyzontidae: Lampreys

The clade Petromyzontidae includes approximately 35–40 or more species of lampreys. Lampreys are morphologically similar to hagfishes and also lack paired appendages. However, lampreys develop some vertebral elements as an adult. Their notochord is surrounded by a cartilaginous structure called an arcualia, which may resemble an evolutionarily-early form of the vertebral column.

As adults, lampreys are characterized by a toothed, funnel-like sucking mouth. Many species have a parasitic stage of their life cycle during which they are ectoparasites of fishes. Lampreys live primarily in coastal and fresh waters. They are distributed worldwide, except for the tropics and polar regions. Some species are marine, but all species spawn in fresh water; eggs are fertilized externally. The larvae differ distinctly from the adult form, spending 3 to 15 years as suspension feeders. Once they reach sexual maturity, the adults die within days of reproduction.

Figure 29.2A.1: **Parasitic lampreys**: These parasitic sea lampreys attach to their lake trout host by suction and use their rough tongues to rasp away flesh in order to feed on the trout's blood.

29.2B: Gnathostomes - Jawed Fishes

LEARNING OBJECTIVES

- Differentiate among the types of jawed fishes

Gnathostomes or “jaw-mouths” are vertebrates that possess jaws. One of the most significant developments in early vertebrate evolution was the development of the jaw, which is a hinged structure attached to the cranium that allows an animal to grasp and tear its food. The evolution of jaws allowed early gnathostomes to exploit food resources that were unavailable to the jawless animals. In early evolutionary history, there were gnathostomes (jawed fishes) and agnathans (jawless fishes). Gnathostomes later evolved into all tetrapods (animals with four limbs) including amphibians, birds, and mammals.

Early gnathostomes were jawed fishes that possessed two sets of paired fins, which increased their ability to maneuver accurately. These paired fins were pectoral fins, located on the anterior body, and pelvic fins, on the posterior. The evolution of the jaw combined with paired fins permitted gnathostomes to expand from the sedentary suspension feeding of jawless fishes and become mobile predators. The gnathostomes’ ability to exploit new nutrient sources led to their evolutionary success during the Devonian period. Two early groups of gnathostomes were the acanthodians and placoderms, which arose in the late Silurian period and are now extinct. Most modern gnathostomes belong to the clades Chondrichthyes and Osteichthyes.

Figure $29.2B.1$: **Placoderms**: Dunkleosteous was an enormous placoderm from the Devonian period, 380–360 million years ago. It measured up to 10 meters in length and weighed up to 3.6 tons. As gnathostomes, they were more mobile and could exploit more food resources than the agnathostomes.

Chondrichthyes: Cartilaginous Fishes

The clade Chondrichthyes consists of sharks, rays, and skates, together with sawfishes and a few dozen species of fishes called chimaeras, or “ghost,” sharks. Chondrichthyes are jawed fishes that possess paired fins and a skeleton made of cartilage. This clade arose approximately 370 million years ago in the early or middle Devonian.

Figure $29.2B.1$: **Hammerhead shark**: Hammerhead sharks tend to school during the day and hunt prey at night. As members of Chondrichthyes, their skeletons are composed of cartilage.

Most cartilaginous fishes live in marine habitats, although a few species live in fresh water for part or all of their lives. Most sharks are carnivores that feed on live prey, either swallowing it whole or using their jaws and teeth to tear it into smaller

pieces. Shark teeth probably evolved from the jagged scales that cover their skin called placoid scales. Some species of sharks and rays are suspension feeders that feed on plankton.

Sharks have well-developed sense organs that aid them in locating prey, including a keen sense of smell and electroreception. Organs called ampullae of Lorenzini enable sharks to detect the electromagnetic fields that are produced by all living things, including their prey. Only aquatic or amphibious animals possess electroreception. Sharks, together with most fishes and aquatic and larval amphibians, also have a sense organ called the lateral line, which is used to detect movement and vibration in the surrounding water. It is often considered homologous to "hearing" in terrestrial vertebrates. The lateral line is visible as a darker stripe that runs along the length of a fish's body.

Rays and skates comprise more than 500 species and are closely related to sharks. They can be distinguished from sharks by their flattened bodies, pectoral fins that are enlarged and fused to the head, and gill slits on their ventral surface. Like sharks, rays and skates have a cartilaginous skeleton. Most species are marine and live on the sea floor, with nearly a worldwide distribution.

Osteichthyes: Bony Fishes

Members of the clade Osteichthyes, also called bony fish, are characterized by a bony skeleton. The vast majority of present-day fish belong to this group, which consists of approximately 30,000 species, making it the largest class of vertebrates in existence today.

Nearly all bony fish have an ossified skeleton with specialized bone cells (osteocytes) that produce and maintain a calcium phosphate matrix. A few groups of Osteichthyes, such as sturgeons and paddlefish, have primarily cartilaginous skeletons, but retain some bony elements. The skin of bony fish is often covered by overlapping scales. Skin glands secrete mucus that reduces drag when swimming and aids the fish in osmoregulation. Like sharks, bony fish have a lateral line system that detects vibrations in water. All bony fish use gills for gas exchange. Water is drawn over gills that are located in chambers covered and ventilated by a protective, muscular flap called the operculum. Many bony fish also have a swim bladder, a gas-filled organ that helps to control the buoyancy of the fish.

Bony fish are further divided into two extant clades: Actinopterygii (ray-finned fish) and Sarcopterygii (lobe-finned fish). Actinopterygii, the ray-finned fish include many familiar fish, such as tuna, bass, trout, and salmon, among others. Ray-finned fish are named for their fins that are webs of skin supported by bony spines called rays. In contrast, the fins of Sarcopterygii are fleshy and lobed, supported by bone. Although most members of this clade are extinct, living members include the less-familiar lungfishes and coelacanths. Early Sarcopterygii evolved into modern tetrapods, including reptiles, amphibians, birds, and mammals.

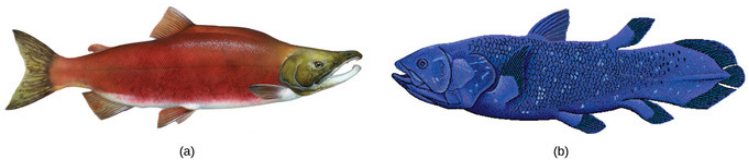

Figure 29.2B.1: **Actinopterygii and Sarcopterygii**: The (a) sockeye salmon (Actinopterygii) and (b) coelacanth (Sarcopterygii) are both bony fishes of the Osteichthyes clade. The coelacanth, sometimes called a lobe-finned fish, was thought to have gone extinct in the Late Cretaceous period, 100 million years ago, until one was discovered in 1938 near the Comoros Islands between Africa and Madagascar.

Key Points

- Early jawed fish (gnathostomes) were able to exploit new nutrient sources because of their jaws and paired fins.
- Chondrichthyes includes all jawed fish with cartilagenous skeletons, such as sharks, rays, skates, and chimaeras.
- Osteichthyes includes all jawed fish with ossified (bony) skeletons; this includes the majority of modern fish.
- Osteichthyes can be further separated into Actinopterygii (the ray-finned fishes) and Sarcopterygii (lobe-finned fishes).
- The majority of modern fish species are actinopterygii, from trout to clownfish.
- Early Sarcopterygii (lobe-finned fishes) evolved into modern tetrapods, including reptiles, amphibians, birds, and mammals.

Key Terms

- **ossified**: composed of bone, which is a calcium phosphate matrix created by special cells called osteoblasts
- **operculum**: a covering flap or lidlike structure in plants and animals, such as a gill cover
- **Chondrichthyes**: a taxonomic class within the subphylum Vertebrata: the cartilaginous fish
- **Osteichthyes**: a taxonomic class within the subphylum vertebrata: the bony fish

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: Amphibians

Topic hierarchy

- 29.3A: Characteristics and Evolution of Amphibians
- 29.3B: Modern Amphibians
- 29.3C: Evolution of Amniotes

29.3A: Characteristics and Evolution of Amphibians

Amphibians evolved from fish 400 million years ago and are characterized by four limbs, moist skin, and sensitive inner ear structures.

LEARNING OBJECTIVES

Describe the evolutionary characteristics that distinguish the amphibian

KEY TAKEAWAYS

Key Points

- Most amphibians are characterized by four well-developed limbs, but some species of salamanders and all caecilians are functionally limbless.
- Amphibians have a moist, permeable skin that is achieved via mucus glands that keep the skin lubricated in order to perform cutaneous respiration.
- All extant adult amphibians are carnivorous; some terrestrial amphibians have a sticky tongue that is used to capture prey.
- Additional characteristics of amphibians include pedicellate teeth, papilla amphibiorum, papilla basilaris, and auricular operculum.
- The tetrapod-like fish, *Tiktaalik roseae*, preceded Acanthostega and lived in a shallow water environment about 375 million years ago.
- All extant adult amphibians are carnivorous, and some terrestrial amphibians have a sticky tongue that is used to capture prey.

Key Terms

- **cutaneous respiration**: exchange of oxygen and carbon dioxide with the environment that takes place through the permeable skin
- **pedicellate teeth**: teeth in which the root and crown are calcified, separated by a zone of noncalcified tissue
- **auricular operculum**: an extra bone in the ear that transmits sounds to the inner ear

Characteristics of Amphibians

As tetrapods, most amphibians are characterized by four well-developed limbs. Some species of salamanders and all caecilians are functionally limbless; their limbs are vestigial. An important characteristic of extant amphibians is a moist, permeable skin that is achieved via mucus glands that keep the skin moist; thus, exchange of oxygen and carbon dioxide with the environment can take place through it (cutaneous respiration). Additional characteristics of amphibians include pedicellate teeth (teeth in which the root and crown are calcified, separated by a zone of noncalcified tissue) and a papilla amphibiorum and papilla basilaris (structures of the inner ear that are sensitive to frequencies below and above 10,00 hertz, respectively). Amphibians also have an auricular operculum, which is an extra bone in the ear that transmits sounds to the inner ear. All extant adult amphibians are carnivorous. Some terrestrial amphibians have a sticky tongue that is used to capture prey.

Evolution of Amphibians

The fossil record provides evidence of the first tetrapods: now-extinct amphibian species dating to nearly 400 million years ago. Evolution of tetrapods from fishes represented a significant change in body plan from one suited to organisms that respired and swam in water, to organisms that breathed air and moved onto land. These changes occurred over a span of 50 million years during the Devonian period. One of the earliest known tetrapods is from the genus Acanthostega. Acanthostega was aquatic; fossils show that it had gills similar to fishes. However, it also had four limbs, with the skeletal structure of limbs found in present-day tetrapods, including amphibians. Therefore, it is thought that Acanthostega lived in shallow waters and was an intermediate form between lobe-finned fishes and early, fully terrestrial tetrapods. What preceded Acanthostega?

In 2006, researchers published news of their discovery of a fossil of a “tetrapod-like fish,” *Tiktaalik roseae*, which seems to be an intermediate form between fishes having fins and tetrapods having limbs. *Tiktaalik* probably lived in a shallow water environment about 375 million years ago.

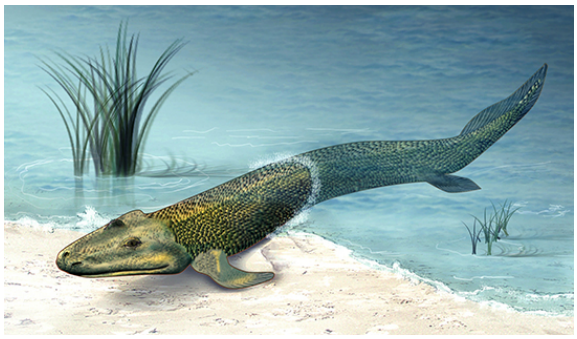

Figure 29.3.A.1: **Tiktaalik roseae**: The recent fossil discovery of Tiktaalik roseae suggests evidence for an animal intermediate to finned fish and legged tetrapods.

The early tetrapods that moved onto land had access to new nutrient sources and relatively few predators. This led to the widespread distribution of tetrapods during the early Carboniferous period: sometimes called the "Age of the Amphibians."

Modern

29.3B: Modern Amphibians

Amphibians can be divided into three groups: Urodela (salamanders), Anura (frogs), and Apoda (caecilians).

Learning Objectives

- Differentiate among the orders of amphibians

Key Points

- Adult salamanders usually have four limbs and a tail, moving by lateral undulation in a fish-like manner while "walking" their arms and legs forward and back.
- The majority of salamanders are lungless; respiration occurs through the skin or through external gills; some terrestrial salamanders have primitive lungs; a few species have both lungs and gills.
- Salamanders utilize internal fertilization after males transfer sperm to the eggs via the spermatophore; there is a prolonged egg phase; metamorphosis occurs before hatching.
- Caecilians are blind, limbless vertebrates that resemble earthworms and are adapted for a soil-burrowing or an aquatic lifestyle.
- Adult frogs use their hind legs to jump; they fertilize externally, laying their shell-less eggs in moist environments.
- Tadpoles (the larval stage of a frog) have gills, a lateral line system, long-finned tails, and lack limbs; when tadpoles become adults, gills, tails, and the lateral line disappear, while an eardrum and lungs develop.

Key Terms

- **lateral undulation**: movement by bending the body from side to side
- **spermatophore**: a capsule or mass created by males, containing sperm and transferred in entirety to the female during fertilization
- **metamorphosis**: a change in the form and often habits of an animal after the embryonic stage during normal development

Modern Amphibians

Amphibia comprises an estimated 6,770 extant species that inhabit tropical and temperate regions around the world. Amphibians can be divided into three clades: Urodela ("tailed-ones"), the salamanders; Anura ("tail-less ones"), the frogs; and Apoda ("legless ones"), the caecilians.

Figure 29.3B.1: **Salamanders**: Most salamanders have legs and a tail, but respiration varies among species.

Urodela: Salamanders

Salamanders are amphibians that belong to the order Urodela. Living salamanders include approximately 620 species, some of which are aquatic, other terrestrial, and some that live on land only as adults. Adult salamanders usually have a generalized tetrapod body plan with four limbs and a tail. They move by bending their bodies from side to side, called lateral undulation, in a fish-like manner while "walking" their arms and legs back and forth. It is thought that their gait is similar to that used by early tetrapods. Respiration differs among the species. The majority of salamanders are lungless, with respiration occurring through the skin or through external gills. Some terrestrial salamanders have primitive lungs; a few species have both gills and lungs.

Unlike frogs, virtually all salamanders rely on internal fertilization of the eggs. The only male amphibians that possess copulatory structures are the caecilians, so fertilization among salamanders typically involves an elaborate and often prolonged courtship. Such a courtship allows the successful transfer of sperm from male to female via a spermatophore. Development in many of the most highly-evolved salamanders, which are fully terrestrial, occurs during a prolonged egg stage, with the eggs guarded by the mother. During this time, the gilled larval stage is found only within the egg capsule, with the gills being resorbed, and metamorphosis being completed, before hatching. Hatchlings resemble tiny adults.

Figure 29.3B.1: **Frogs**: The Australian green tree frog is a nocturnal predator that lives in the canopies of trees near a water source.

Anura: Frogs

Frogs are amphibians that belong to the order Anura. Anurans are among the most diverse groups of vertebrates, with approximately 5,965 species occurring on all of the continents except Antarctica. Anurans have a body plan that is more specialized for movement. Adult frogs use their hind limbs to jump on land. Frogs have a number of modifications that allow them to avoid predators, including skin that acts as camouflage. Many species of frogs and salamanders also release defensive chemicals that are poisonous to predators from glands in the skin.

Figure 29.3B.1: **Frog metamorphosis**: A juvenile frog metamorphoses into a frog. Here, the frog has started to develop limbs, but its tadpole tail is still evident.

Frog eggs are fertilized externally. As with other amphibians, frogs generally lay their eggs in moist environments. This is required as eggs lack a shell and will dehydrate quickly in dry environments. Frogs demonstrate a great diversity of parental behaviors: some species lay many eggs and exhibit little parental care; other species carry eggs and tadpoles on their hind legs or backs. The life cycle of frogs, as with other amphibians, consists of two distinct stages: 1) the larval stage followed by 2) metamorphosis to an adult stage. The larval stage of a frog, the tadpole, is often a filter-feeding herbivore. Tadpoles usually have gills, a lateral line system, long-finned tails, and lack limbs. At the end of the tadpole stage, frogs undergo metamorphosis into the adult form. During this stage, the gills, tail, and lateral line system disappear, and four limbs develop. The jaws become larger and are suited for carnivorous feeding, while the digestive system transforms into the typical short gut of a predator. An eardrum and air-breathing lungs also develop. These changes during metamorphosis allow the larvae to move onto land in the adult stage.

Apoda: Caecilians

An estimated 185 species comprise caecilians, a group of amphibians that belong to the order Apoda. Although they are vertebrates, a complete lack of limbs leads to their resemblance to earthworms in appearance. They are adapted for a soil-burrowing or aquatic lifestyle; they are nearly blind. These animals are found in the tropics of South America, Africa, and Southern Asia. They have vestigial limbs which is evidence that they evolved from a legged ancestor.

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29.3C: Evolution of Amniotes

Modern amniotes, which includes mammals, reptiles, and birds, evolved from an amphibian ancestor approximately 340 million years ago.

Learning Objectives

- Outline the evolution of amniotes

Key Points

- Synapsids include all mammals and therapsids, mammal-like reptiles, from which mammals evolved.
- Sauropsids, which are divided into the anapsids and diapsids, include reptiles and birds.
- The diapsids are divided into lepidosaurs (modern lizards, snakes, and tuataras) and archosaurs (modern crocodiles and alligators, pterosaurs, and dinosaurs).
- Skull structure and number of temporal fenestrae are the key differences between the synapsids, anapsids, and diapsids; anapsids have no temporal fenestrae, synapsids have one, and diapsids have two.
- Turtle classification is still unclear, but based on molecular evidence, they are sometimes classified under diapsids.
- Although birds are considered distinct from reptiles, they evolved from a group of dinosaurs, so considering them separately from reptiles is not phylogenetically accurate.

Key Terms

- **synapsid**: animals that have one opening low in the skull roof behind each eye; includes all living and extinct mammals and therapsids
- **anapsid**: amniote whose skull does not have openings near the temples; includes extinct organisms
- **diapsid**: any of very many reptiles and birds that have a pair of openings in the skull behind each eye
- **temporal fenestrae**: post-orbital openings in the skull of some amniotes that allow muscles to expand and lengthen

Evolution of Amniotes

The first amniotes evolved from their amphibian ancestors approximately 340 million years ago during the Carboniferous period. The early amniotes diverged into two main lines soon after the first amniotes arose. The initial split was into synapsids and sauropsids. Synapsids include all mammals, including extinct mammalian species. Synapsids also include therapsids, which were mammal-like reptiles from which mammals evolved. Sauropsids include reptiles and birds and can be further divided into anapsids and diapsids. The key differences between the synapsids, anapsids, and diapsids are the structures of the skull and the number of temporal fenestrae behind each eye. Temporal fenestrae are post-orbital openings in the skull that allow muscles to expand and lengthen. Anapsids have no temporal fenestrae, synapsids have one, and diapsids have two. Anapsids include extinct organisms and may, based on anatomy, include turtles (Testudines), which have an anapsid-like skull with one opening. However, this is still controversial, and turtles are sometimes classified as diapsids based on molecular evidence. The diapsids include birds and all other living and extinct reptiles.

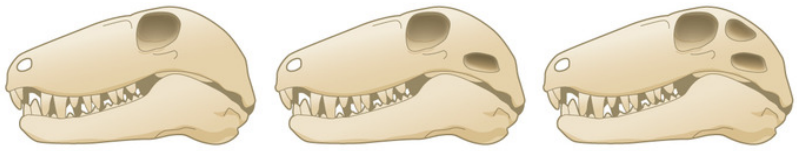

Figure $29.3C.1$: **Tempral fenestrae**: The image illustrates the differences in the skulls and temporal fenestrae of anapsids, synapsids, and diapsids. Anapsids have no openings, synapsids have one opening, and diapsids have two openings.

The diapsids diverged into two groups, the Archosauromorpha ("ancient lizard form") and the Lepidosauromorpha ("scaly lizard form") during the Mesozoic period. The lepidosaurs include modern lizards, snakes, and tuataras. The archosaurs include modern crocodiles and alligators, and the extinct pterosaurs ("winged lizard") and dinosaurs ("terrible lizard"). Clade Dinosauria includes birds, which evolved from a branch of dinosaurs.

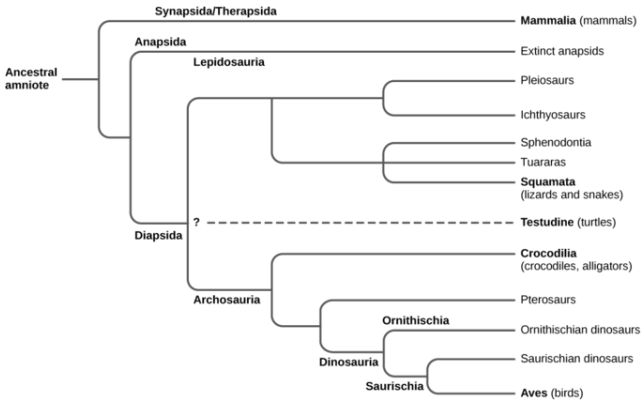

Figure 29.3C.1: **Evolution of amniotes**: This chart shows the evolution of amniotes. The placement of Testudines (turtles) is currently still debated.

In the past, the most common division of amniotes has been into the classes Mammalia, Reptilia, and Aves. Birds are descended, however, from dinosaurs, so this classical scheme results in groups that are not true clades. Birds are considered as a group distinct from reptiles with the understanding that this does not completely reflect phylogenetic history and relationships.

29.4: Reptiles

Topic hierarchy

- 29.4A: Characteristics of Amniotes
- 29.4B: Characteristics of Reptiles
- 29.4C: Evolution of Reptiles
- 29.4D: Modern Reptiles

29.4A: Characteristics of Amniotes

The distinguishing characteristic of amniotes, a shelled egg with an amniotic membrane, allowed them to venture onto land.

Learning Objectives

- Discuss the evolution of amniotes

Key Points

- The amniotes include reptiles, birds, and mammals; shared characteristics between this group include a shelled egg protected by amniotic membranes, waterproof skin, and rib ventilation of the lungs.
- In amniotes, the shell of the egg provides protection for the developing embryo and allows water retention while still being permeable to gas exchange.
- Amniotic eggs contain albumin, which provides the embryo with water and protein, and an egg yolk that supplies the embryo with energy.
- The chorion, amnion, and allantois are key membranes found only in amniotic eggs.
- The chorion facilitates gas exchange between the embryo and the egg's external environment.
- The amnion protects the embryo from mechanical shock and supports hydration, while the allantois stores nitrogenous wastes and facilitates respiration.

Key Terms

- **amnion**: the innermost membrane of the fetal membranes of amniotes; the sac in which the embryo is suspended; protects the embryo from shock and carries out hydration
- **chorion**: allows exchange of oxygen and carbon dioxide between the embryo and the egg's external environment
- **allantois**: membrane in an egg that stores nitrogenous wastes produced by the embryo and facilitates respiration
- **monotreme**: a mammal that lays eggs and has a single urogenital and digestive orifice; only the echidnas and platypuses

Characteristics of Amniotes

The amniotes, reptiles, birds, and mammals, are distinguished from amphibians by their terrestrially-adapted egg, which is protected by amniotic membranes. The evolution of amniotic membranes meant that the embryos of amniotes were now provided with their own aquatic environment, which led to less dependence on water for development, allowing the amniotes to branch out into drier environments. This was a significant development that distinguished them from amphibians, which were restricted to moist environments due their shell-less eggs. Although the shells of various amniotic species vary significantly, they all allow retention of water. The shells of bird eggs are composed of calcium carbonate and are hard, but fragile. The shells of reptile eggs are leathery and require a moist environment. Most mammals do not lay eggs (except for monotremes). Instead, the embryo grows within the mother's body; however, even with this internal gestation, amniotic membranes are still present.

The amniotic egg is the key characteristic of amniotes. In amniotes that lay eggs, the shell of the egg provides protection for the developing embryo while being permeable enough to allow for the exchange of carbon dioxide and oxygen. The albumin, or egg white, provides the embryo with water and protein, whereas the fattier egg yolk is the energy supply for the embryo, as is the case with the eggs of many other animals, such as amphibians. However, the eggs of amniotes contain three additional extra-embryonic membranes: the chorion, amnion, and allantois. Extra-embryonic membranes are those present in amniotic eggs that are not a part of the body of the developing embryo. While the inner amniotic membrane surrounds the embryo itself, the chorion surrounds the embryo and yolk sac. The chorion facilitates exchange of oxygen and carbon dioxide between the embryo and the egg's external environment. The amnion protects the embryo from mechanical shock and supports hydration. The allantois stores nitrogenous wastes produced by the embryo and also facilitates respiration. In mammals, membranes that are homologous to the extra-embryonic membranes in eggs are present in the placenta. Additional derived characteristics of amniotes include waterproof skin, due to the presence of lipids, and costal (rib) ventilation of the lungs.

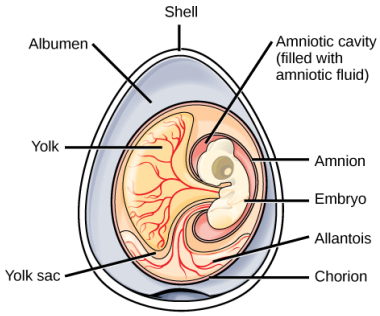

Figure 29.4*A*. 1: **Amniotic eggs**: The key features of an amniotic egg are the chorion, amnion, and allantois.

29.4B: Characteristics of Reptiles

Reptiles are ectothermic tetrapods that lay shelled eggs on land and possess scaly skin and lungs.

Learning Objectives

- Summarize the key adaptations of reptiles

Key Points

- All reptiles, including aquatic ones, lay their eggs on land.
- Reptiles reproduce sexually through internal fertilization; some species are ovoviviparous (lay eggs) and others are viviparous (live birth).
- Because of the development of impermeable, scaly skin, reptiles were able to move onto land since their skin could not be used for respiration in water.
- Reptiles are ectotherms: they depend on their surrounding environment to control their body temperature; this leads to advantages, such as not being dependent on metabolic energy from food for body heat.
- Reptiles are also poikilotherms: animals whose body temperatures vary rather than remain stable.
- Some reptiles go into brumation: a long period during cold weather that consists of no eating and a decreased metabolism.

Key Terms

- **viviparous**: being born alive, as are most mammals, some reptiles, and a few fish (as opposed to being laid as an egg)
- **ovoviviparous**: a mode of reproduction in animals in which embryos develop inside eggs that are retained within the mother's body until they are ready to hatch
- **ectotherm**: a cold-blooded animal that regulates its body temperature by exchanging heat with its surroundings

Characteristics of Reptiles

Reptiles are tetrapods. Limbless reptiles (snakes and other squamates) have vestigial limbs and, as with caecilians, are classified as tetrapods because they are descended from four-limbed ancestors. Reptiles lay on land eggs enclosed in shells. Even aquatic reptiles return to the land to lay eggs. They usually reproduce sexually with internal fertilization. Some species are ovoviviparous, with the eggs remaining in the mother's body until they are ready to hatch. Other species are viviparous, with the offspring born alive.

One of the key adaptations that permitted reptiles to live on land was the development of their scaly skin which contains the protein keratin and waxy lipids, reducing water loss from the skin. Due to this occlusive skin, reptiles cannot use their skin for respiration, as do amphibians; all breathe with lungs.

Reptiles are ectotherms: animals whose main source of body heat comes from the environment. This is in contrast to endotherms, which use heat produced by metabolism to regulate body temperature. In addition to being ectothermic, reptiles are categorized as poikilotherms: animals whose body temperatures vary rather than remain stable. Reptiles have behavioral adaptations to help regulate body temperature, such as basking in sunny places to warm up and finding shady spots or going underground to cool down. The advantage of ectothermy is that metabolic energy from food is not required to heat the body; therefore, reptiles can survive on about 10 percent of the calories required by a similarly-sized endotherm. In cold weather, some reptiles, such as the garter snake, brumate. Brumation is similar to hibernation in that the animal becomes less active and can go for long periods without eating, but differs from hibernation in that brumating reptiles are not asleep or living off fat reserves. Rather, their metabolism is slowed in response to cold temperatures; the animal becomes very sluggish.

Figure 29.4B.1: **Ectotherms**: Reptiles, such as these sunbathing Florida redbelly turtles, are ectotherms: they rely on their environment for body heat.

29.4C: Evolution of Reptiles

Dinosaurs and pterosaurs diverged from early amniotes and dominated the Mesozoic Era.

Learning Objectives

- Outline the evolution of reptiles

Key Points

- Diapsids diverged into archosaurs and lepidosaurs, but these groups did not dominate the ecosystem until the Triassic following the Permian extinction.
- Archosaurs diverged into the dinosaurs and the pterosaurs about 250 million years ago.
- Pterosaurs had the ability to fly because of their wings and hollow bones, a trait convergent to modern birds, but were not ancestral to birds.
- Dinosaurs were quadrupeds or bipeds, carnivorous or herbivorous, and laid eggs.
- It is unknown whether dinosaurs were endothermic or ectothermic, but since birds are endothermic, the dinosaur ancestors of birds were probably endothermic.
- Dinosaurs dominated the Mesozoic Era until the Cretaceous -Tertiary extinction wiped out most of these large-bodied animals.

Key Terms

- **pterosaur**: any of several extinct flying reptiles, of the order Pterosauria, including the pterodactyls
- **Cretaceous-Tertiary extinction**: mass extinction of three-quarters of plant and animal species on earth, including all non-avian dinosaurs, that occurred over a geologically-short period of time 66 million years ago

Evolution of Reptiles

Reptiles originated approximately 300 million years ago during the Carboniferous period. One of the oldest-known amniotes is *Casineria*, which had both amphibian and reptilian characteristics. One of the earliest undisputed reptiles was *Hylonomus*. Soon after the first amniotes appeared, they diverged into three groups (synapsids, anapsids, and diapsids) during the Permian period. The Permian period also saw a second major divergence of diapsid reptiles into archosaurs (predecessors of crocodilians and dinosaurs) and lepidosaurs (predecessors of snakes and lizards). These groups remained inconspicuous until the Triassic period when the archosaurs became the dominant terrestrial group due to the extinction of large-bodied anapsids and synapsids during the Permian-Triassic extinction. About 250 million years ago, archosaurs radiated into the dinosaurs and the pterosaurs.

Although they are sometimes mistakenly called dinosaurs, the pterosaurs were distinct from true dinosaurs. Pterosaurs had a number of adaptations that allowed for flight, including hollow bones (birds also exhibit hollow bones, a case of convergent evolution). Their wings were formed by membranes of skin that attached to the long, fourth finger of each arm and extended along the body to the legs.

Figure 29.4C.1: **Pterosaurs**: Pterosaurs, which existed from the late Triassic to the Cretaceous period (210 to 65.5 million years ago), possessed wings, but are not believed to have been capable of powered flight. Instead, they may have been able to soar after launching from cliffs.

The dinosaurs were a diverse group of terrestrial reptiles with more than 1,000 species identified to date. Paleontologists continue to discover new species of dinosaurs. Some dinosaurs were quadrupeds; others were bipeds. Some were carnivorous, whereas others were herbivorous. Dinosaurs laid eggs; a number of nests containing fossilized eggs have been found. It is not known whether dinosaurs were endotherms or ectotherms. However, given that modern birds are endothermic, the dinosaurs that served as ancestors to birds were probably endothermic as well. Some fossil evidence exists for dinosaurian parental care. Comparative biology supports this hypothesis since the archosaur birds and crocodilians display parental care.

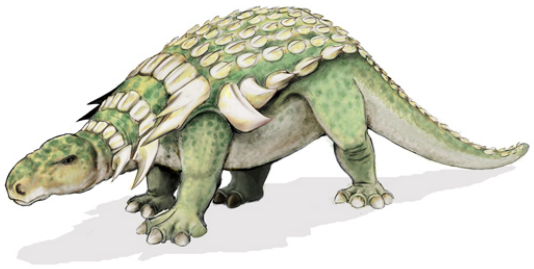

Figure 29.4C.1: **Quadruped dinosaurs**: Edmontonia, an example of an extinct quadruped reptile, was an armored dinosaur that lived in the late Cretaceous period, 145.5 to 65.6 million years ago.

Dinosaurs dominated the Mesozoic Era, which was known as the “Age of Reptiles.” The dominance of dinosaurs lasted until the end of the Cretaceous period, the end of the Mesozoic Era. The Cretaceous-Tertiary extinction resulted in the loss of most of the large-bodied animals of the Mesozoic Era. Birds are the only living descendants of one of the major clades of dinosaurs.

29.4D: Modern Reptiles

Class Reptilia, amniotes that are neither mammals nor birds, has four living clades: Crocodilia, Sphenodontia, Squamata, and Testudine.

Learning Objectives

- Differentiate among the types of modern reptiles

Key Points

- Reptile are amniotes that lay their eggs on land; they have scales or scutes and are ectothermic.
- Crocodilia includes the alligators, crocodiles, and caimans; they are mostly aquatic species, but some are able to move on land because of their semi-erect posture.
- Tuataras are classified as the only group under Sphenodontia; they may be lizard-like, but skull and jaw differences set them apart from true lizards.
- Squamata, the largest group of reptiles, includes the lizards and snakes.
- Snakes, which lack the eyelids and external ears present in lizards, are believed to have evolved from burrowing or aquatic lizards.
- Turtles are grouped under the Testudines; species in this group all have bony or cartilaginous shells.

Key Terms

- **scute**: a horny, chitinous, or bony external plate or scale, as on the shell of a turtle or the skin of crocodiles
- **plastron**: the nearly flat part of the shell structure of a tortoise or other animal, similar in composition to the carapace
- **amniote**: a group of vertebrates having an amnion during the development of the embryo; mammals, birds, and reptiles

Modern Reptiles

Class Reptilia includes many diverse species that are classified into four living clades. These are the 25 species of Crocodilia, 2 species of Sphenodontia, approximately 9,200 Squamata species, and the Testudines, with about 325 species. A reptile is any amniote (a tetrapod whose egg has an additional membrane, originally to allow them to lay eggs on land) that is neither a mammal nor a bird. Unlike mammals, birds, and certain extinct reptiles, living reptiles have scales or scutes (rather than fur or feathers) and are cold-blooded. Modern reptiles inhabit every continent with the exception of Antarctica.

Crocodilia

Crocodilia (“small lizard”) arose with a distinct lineage by the middle Triassic; extant species include alligators, crocodiles, and caimans. Crocodilians are large, solidly built lizard-like reptiles with long flattened snouts and laterally-compressed tails, and eyes, ears, and nostrils at the top of the head. Their skin is thick and covered in non-overlapping scales. They have conical, peg-like teeth and a powerful bite. As with birds, they have a four-chambered heart and a unidirectional system of air flow around the lungs; however, in contrast to birds, they are ectotherms, as are all other reptiles. Crocodilians live throughout the tropics and subtropics of Africa, South America, Southern Florida, Asia, and Australia. They are found in freshwater, saltwater, and brackish habitats, such as rivers and lakes; they spend most of their time in water. Some species are able to move on land due to their semi-erect posture.

Figure 29.4D.1: **Crocodilians**: Crocodilians, such as this Siamese crocodile (Crocodylus siamensis), have large flattened snouts and thick skin.

Sphenodontia

Sphenodontia ("wedge tooth") arose in the Mesozoic era and includes only one living genus, Tuatara, which comprises two species that are found in New Zealand. Tuataras measure up to 80 centimeters and weigh about 1 kilogram. Although quite lizard-like in gross appearance, several unique features of the skull and jaws clearly define them and distinguish the group from the squamates. Their dentition, in which two rows of teeth in the upper jaw overlap one row on the lower jaw, is unique among living species. They are also unusual in having a pronounced photoreceptive eye, dubbed the "third eye", whose current function is a subject of ongoing research, but is thought to be involved in setting circadian and seasonal cycles.

Figure 29.4D.1: **Tuatara**: Tuataras may resemble a lizard but belongs to a distinct lineage, the Sphenodontidae family.

Squamata

Squamata ("scaly") arose in the late Permian; extant species include lizards and snakes. They are most closely-related to tuataras; both groups evolved from a lepidosaurian ancestor. Squamata is the largest extant clade of reptiles. Most lizards differ from snakes by having four limbs, although these have been variously lost or significantly reduced in at least 60 lineages. Snakes lack eyelids and external ears, which are present in lizards. Lizard species range in size from chameleons and geckos, which are a few centimeters in length, to the Komodo dragon, which is about 3 meters in length. Most lizards are carnivorous, but some large species, such as iguanas, are herbivores.

Figure 29.4D.1: **Squamata**: The Squamata, which includes lizards and snakes, are the largest group of reptiles. The garter snake belongs to the genus Thamnophis, the most widely-distributed reptile genus in North America.

Snakes are thought to have descended from either burrowing lizards or aquatic lizards over 100 million years ago. Snakes comprise about 3,000 species and range in size from 10 centimeter-long thread snakes to 10 meter-long pythons and anacondas. All snakes are carnivorous, eating small animals, birds, eggs, fish, and insects. Although variations exist, most snakes have a skull that is very flexible, involving eight rotational joints. They also differ from other squamates by having mandibles (lower jaws) without either bony or ligamentous attachment anteriorly. Having this connection via skin and muscle allows for great expansion of the gape and independent motion of the two sides; both advantages in swallowing big items.

Testudines

Turtles are members of the clade Testudines ("having a shell"). Turtles are characterized by a bony or cartilaginous shell. The shell consists of the ventral surface called the plastron and the dorsal surface called the carapace, which develops from the ribs. The plastron is made of scutes or plates; the scutes can be used to differentiate species of turtles. The two clades of turtles are most easily recognized by how they retract their necks. The dominant group, which includes all North American species, retracts its neck in a vertical S-curve. Turtles in the less speciose clade retract the neck with a horizontal curve.

Figure 29.4D.1: **Testudines**: Testudines include all shelled reptiles, such as the African spurred tortoise (Geochelone sulcata) that lives at the southern edge of the Sahara Desert. It is the third-largest tortoise in the world.

Turtles arose approximately 200 million years ago, predating crocodiles, lizards, and snakes. Similar to other reptiles, turtles are ectotherms. They lay eggs on land, although many species live in or near water. None exhibit parental care. Turtles range in size from the speckled padloper tortoise at 8 centimeters (3.1 inches) to the leatherback sea turtle at 200 centimeters (over 6 feet). The term "turtle" is sometimes used to describe only those species of Testudines that live in the sea, with the terms "tortoise" and "terrapin" used to refer to species that live on land and in fresh water, respectively.

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29.5: Birds

Topic hierarchy

- 29.5A: Characteristics of Birds
- 29.5B: Evolution of Birds

29.5A: Characteristics of Birds

Birds are warm-blooded animals with wings having several adaptations to flight, although not all species can fly.

Learning Objectives

- Summarize the derived characteristics of birds

Key Points

- Birds have down feathers that provide insulation and two types of flight feathers found on the wings: thrust-producing primary feathers at the tip of the wing and lift-providing secondary feathers closer to the body.
- Contour feathers found on the body create a smooth, aerodynamic surface.
- The chest muscles of birds are highly developed as they are responsible for the flapping of the entire wing.
- The two clavicles of birds are fused, forming the furcula or wishbone, which is both flexible and strong enough to support to the shoulder girdle during flapping.
- In order to keep body weight low, birds have pneumatic bones, no urinary bladders, and usually only one ovary.
- Birds have developed an efficient respiratory system using air sacs and unidirectional airflow and a cross-current exchange system with the blood.

Key Terms

- **pneumatic**: having cavities filled with air
- **endothermic**: an animal whose body temperature is regulated by internal factors
- **furcula**: the forked bone formed by the fusion of the clavicles in birds; the wishbone
- **cloaca**: the common duct in fish, reptiles, birds, and some primitive mammals that serves as the anus as well as the genital opening

Characteristics of Birds

Birds are endothermic and, because they fly, they require large amounts of energy, necessitating a high metabolic rate. As with mammals, which are also endothermic, birds have an insulating covering that keeps heat in the body: feathers. Specialized feathers called down feathers are especially insulating, trapping air in spaces between each feather to decrease the rate of heat loss. Certain parts of a bird's body are covered in down feathers; the base of other feathers have a downy portion, while newly-hatched birds are covered in down.

Feathers not only act as insulation, but also allow for flight, enabling the lift and thrust necessary to become airborne. The feathers on a wing are flexible, so the collective feathers move and separate as air moves through them, reducing the drag on the wing. Flight feathers are asymmetrical, which affects airflow over them and provides some of the lifting and thrusting force required for flight. Two types of flight feathers are found on the wings: primary feathers and secondary feathers. Primary feathers are located at the tip of the wing and provide thrust. Secondary feathers are located closer to the body, attach to the forearm portion of the wing, and provide lift. Contour feathers are those found on the body. They help reduce drag produced by wind resistance during flight, creating a smooth, aerodynamic surface allowing air to flow smoothly over the bird's body for efficient flight.

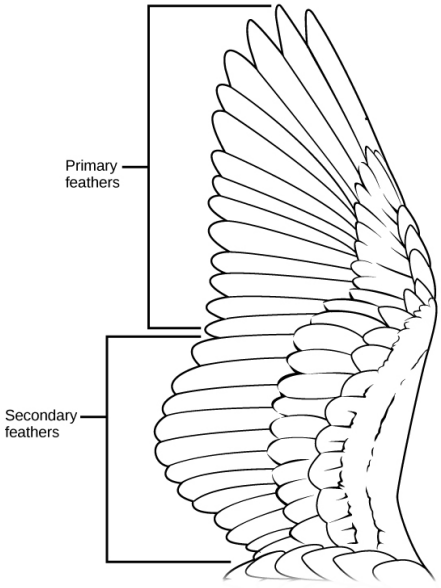

Figure 29.5A.1: **Bird feathers**: Primary feathers are located at the wing tip and provide thrust; secondary feathers are located close to the body and provide lift.

Flapping of the entire wing occurs primarily through the actions of the chest muscles: the pectoralis and the supracoracoideus. These muscles, highly developed in birds and accounting for a higher percentage of body mass than in most mammals, attach to a blade-shaped keel, similar to that of a boat, located on the sternum. The sternum of birds is larger than that of other vertebrates, which accommodates the large muscles required to generate enough upward force to generate lift with the flapping of the wings. Another skeletal modification found in most birds is the fusion of the two clavicles (collarbones), forming the furcula or wishbone. The furcula is flexible enough to bend and provide support to the shoulder girdle during flapping.

An important requirement of flight is a low body weight. As body weight increases, the muscle output required for flying increases. The largest living bird is the ostrich. While it is much smaller than the largest mammals, it is flightless. For birds that do fly, reduction in body weight makes flight easier. Several modifications are found in birds to reduce body weight, including pneumatization of bones. Pneumatic bones are hollow rather than filled with tissue. They contain air spaces that are sometimes connected to air sacs and they have struts of bone to provide structural reinforcement. Pneumatic bones are not found in all birds; they are more extensive in large birds than in small birds. Not all bones of the skeleton are pneumatic, although the skulls of almost all birds are.

Figure 29.5A.1: **Pneumatic bones of birds**: Many birds have hollow, pneumatic bones, which make flight easier.

Other modifications that reduce weight include the lack of a urinary bladder. Birds possess a cloaca: a structure that allows water to be reabsorbed from waste back into the bloodstream. Uric acid is not expelled as a liquid, but is concentrated into urate salts, which are expelled along with fecal matter. In this way, water is not held in the urinary bladder, which would increase body weight. Most bird species possess only one ovary rather than two, further reducing body mass.

The air sacs that extend into bones, making them pneumatic, also join with the lungs and function in respiration. In contrast to mammalian lungs in which air flows in two directions, as it is breathed in and out, airflow through bird lungs travels in one direction. Air sacs allow for this unidirectional airflow, which also creates a cross-current exchange system with the blood. In a cross-current or counter-current system, the air flows in one direction and the blood flows in the opposite direction, creating a very efficient means of gas exchange.

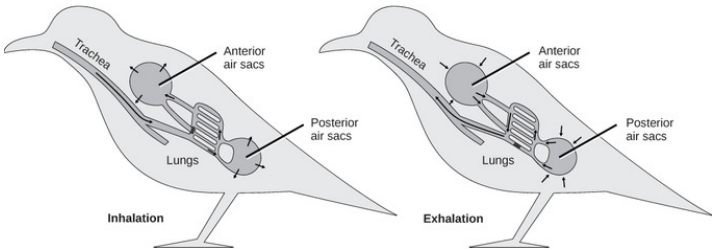

Figure 29.5A.1: **Avian respiration**: Avian respiration is an efficient system of gas exchange with air flowing unidirectionally. During inhalation, air passes from the trachea into posterior air sacs, then through the lungs to anterior air sacs. The air sacs are connected to the hollow interior of bones. During exhalation, air from air sacs passes into the lungs and out the trachea.

29.5B: Evolution of Birds

Modern birds evolved from Saurichia, one of two subgroups of dinosaurs, although it is unclear how flight and/or endothermy arose in birds.

Learning Objectives

- Explain the evolution of birds

Key Points

- Birds have two fenestrations, or openings, in their skulls making them diapsids like crocodiles and dinosaurs.
- Birds did not descend from bird-like dinosaurs (Ornithischia), but rather from a divergent group of lizard-like dinosaurs (Saurischia) called theropods, which were bipedal predators.
- A Jurassic period fossil intermediate to dinosaurs and birds is *Archaeopteryx*, which had teeth like dinosaurs, and feathers modified for flight.
- The arboreal ("tree") hypothesis and the terrestrial ("land") hypothesis are two theories on how flight evolved; these theories propose that wings developed to aid in jumping from branch to branch or to aid in running, respectively.
- It was not until after the extinction of Enantiornithes (a separate evolutionary line of bird-like animals) during the Cretaceous period that the Ornithurae (the evolutionary line of modern birds) became dominant. and prospered.

Key Terms

- **diapsid**: any of very many reptiles and birds that have a pair of openings in the skull behind each eye
- **Archaeopteryx**: a taxonomic genus within the family Archaeopterygidae, known from fossils and widely accepted as the earliest and most primitive known bird
- **fenestration**: an opening in the surface of an organ, etc.

Evolution of Birds

The evolutionary history of birds is still somewhat unclear. Due to the fragility of bird bones, they do not fossilize as well as other vertebrates. Birds are diapsids, meaning they have two fenestrations, or openings, in their skulls. Birds belong to a group of diapsids called the archosaurs, which also includes crocodiles and dinosaurs. It is commonly accepted that birds evolved from dinosaurs.

Dinosaurs were subdivided into two groups, the Saurischia ("lizard like") and the Ornithischia ("bird like"). Despite the names of these groups, it was not the bird-like dinosaurs that gave rise to modern birds. Rather, Saurischia diverged into two groups. One included the long-necked herbivorous dinosaurs, such as Apatosaurus. The second group, bipedal predators called theropods, includes the ancestors of modern birds. This course of evolution is suggested by similarities between theropod fossils and birds, specifically in the structure of the hip and wrist bones, as well as the presence of the wishbone, formed by the fusing of the clavicles.

One important fossil of an animal intermediate to dinosaurs and birds is Archaeopteryx, which is from the Jurassic period and has characteristics of both dinosaurs and birds. Some scientists propose classifying it as a bird, but others prefer to classify it as a dinosaur. The fossilized skeleton of Archaeopteryx looks like that of a dinosaur. It had teeth and birds do not, but it also had feathers modified for flight, a trait associated only with birds among modern animals. Fossils of older, feathered dinosaurs exist, but the feathers do not have the characteristics of flight feathers.

(a) (b)

Figure 29.5B.1: **Bird fossils**: (a) Archaeopteryx lived in the late Jurassic Period around 150 million years ago. It had teeth like a dinosaur, but had (b) flight feathers like modern birds, which can be seen in this fossil.

It is still unclear exactly how flight evolved in birds. Two main theories exist: the arboreal ("tree") hypothesis and the terrestrial ("land") hypothesis. The arboreal hypothesis posits that tree-dwelling precursors to modern birds jumped from branch to branch using their feathers for gliding before becoming fully capable of flapping flight. In contrast to this, the terrestrial hypothesis holds that running was the stimulus for flight, as wings could be used to improve running and then became used for flapping flight. As with the question of how flight evolved, the question of how endothermy evolved in birds still is unanswered. Feathers provide insulation, but this is only beneficial if body heat is being produced internally. Similarly, internal heat production is only viable if insulation is present to retain that heat. It has been suggested that one or the other (feathers or endothermy) evolved in response to some other selective pressure.

During the Cretaceous period, a group known as the Enantiornithes was the dominant bird type. Enantiornithes means "opposite birds," which refers to the fact that certain bones of the feet are joined differently than the way the bones are joined in modern birds. These birds formed an evolutionary line separate from modern birds; they did not survive past the Cretaceous. Along with the Enantiornithes, Ornithurae birds (the evolutionary line that includes modern birds) were also present in the Cretaceous. After the extinction of Enantiornithes, modern birds became the dominant bird, with a large radiation occurring during the Cenozoic Era. Referred to as Neornithes ("new birds"), modern birds are now classified into two groups, the Paleognathae ("old jaw") or ratites (a group of flightless birds including ostriches, emus, rheas, and kiwis) and the Neognathae ("new jaw"), all other birds.

Figure 29.5B.1: **Example of an extinct bird**: *Shanweiniao cooperorum* was a species of Enantiornithes, which evolved separately from modern birds. It did not survive past the Cretaceous period.

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29.6: Mammals

Topic hierarchy

- 29.6A: Characteristics of Mammals
- 29.6B: Evolution of Mammals
- 29.6C: Living Mammals

29.6A: Characteristics of Mammals

Mammalian traits include, among others: specialized glands, modified jaw and inner ear bones, urinary bladder, and hair.

Learning Objectives

- Summarize the distinguishing characteristics of mammals

Key Points

- The various traits which are used to define mammals include: the presence of hair; the integument system which contains specialized secretory glands; the skeletal and muscular systems; the heart and brain structure.
- Mammals contain specialized glands which have various functions: secretion of chemical compounds used for communication; glands that produce milk; glands that produce perspiration used for thermoregulation; and glands that produce sebum, which is used for lubrication.
- Mammals have four-chambered hearts that are defined by the ability to regulate the heart beat with the presence of specialized pacemaker cells.
- A mammal's hair has many purposes, including insulation, sensory perception, protective coloration, and social signaling.
- Mammals possess many unique skeletal structures including a single lower jaw bone that joins the skull at the squamosal bone and three bones in the inner ear.

Key Terms

- **vibrissa**: any of the tactile whiskers on the nose of an animal
- **sebum**: a thick oily substance, secreted by the sebaceous glands of the skin, that consists of fat, keratin and cellular debris
- **diphyodont**: having two successive sets of teeth (deciduous and permanent), one succeeding the other
- **sinoatrial node**: the impulse-generating (pacemaker) tissue located in the right atrium of the heart, and thus the generator of normal sinus rhythm
- **integument**: an outer protective covering such as the feathers or skin of an animal, a rind or shell

Characteristics of Mammals

The presence of hair is one of the most obvious traits of a mammal. Although it is not very extensive on certain species, such as whales, hair has many important functions for mammals. Mammals are endothermic so hair provides insulation to retain heat generated by metabolic work by trapping a layer of air close to the body. Along with insulation, hair can serve as a sensory mechanism via specialized hairs called vibrissae, better known as whiskers. These attach to nerves that transmit information about sensation, which is particularly useful to nocturnal or burrowing mammals. Hair can also provide protective coloration or be part of social signaling, such as when an animal's hair stands "on end. "

Figure 29.6A.1: **Mammalian fur as insulation**: Polar bears use their fur for warmth. While their skin is black, their transparent fur appears white, providing camouflage while hunting and serving as protection by hiding cubs in the snow.

Mammalian integument, or skin, includes secretory glands with various functions. Sebaceous glands produce a lipid mixture called sebum that is secreted onto the hair and skin for water resistance and lubrication. Sebaceous glands are located over most of the body. Eccrine glands produce sweat, or perspiration, which is mainly composed of water. In most mammals, eccrine glands are limited to certain areas of the body; some mammals do not possess them at all. However, in primates, especially humans, sweat figures prominently in thermoregulation, regulating the body through evaporative cooling. Sweat glands are located over most of the body surface in primates. Apocrine glands, or scent glands, secrete substances that are used for chemical communication, such as in skunks. Mammary glands produce milk that is used to feed newborns. While male monotremes and eutherians possess mammary glands, male marsupials do not. Mammary glands are probably modified sebaceous or eccrine glands, but their evolutionary origin is not entirely clear.

The skeletal system of mammals possesses many unique features. The lower jaw of mammals consists of only one bone, the dentary. The jaws of other vertebrates are composed of more than one bone. In mammals, the dentary bone joins the skull at the squamosal bone, while in other vertebrates, the quadrate bone of the jaw joins with the articular bone of the skull. These bones are present in mammals, but they have been modified to function in hearing and form bones in the middle ear. Other vertebrates possess only one middle ear bone, the stapes. Mammals have three: the malleus, incus, and stapes. The malleus originated from the articular bone, whereas the incus originated from the quadrate bone. This arrangement of jaw and ear bones aids in distinguishing fossil mammals from fossils of other synapsids.

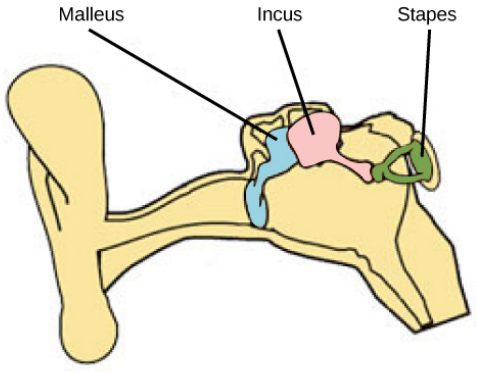

Figure 29.6A.1: **Bones of the mammalian inner ear**: Bones of the mammalian inner ear are modified from bones of the jaw and skull.

The adductor muscle that closes the jaw is composed of two muscles in mammals: the temporalis and the masseter. These allow side-to-side movement of the jaw, making chewing possible, which is unique to mammals. Most mammals have heterodont teeth, meaning that they have different types and shapes of teeth rather than just one type and shape of tooth. Most mammals are diphyodonts, meaning that they have two sets of teeth in their lifetime: deciduous, or “baby” teeth, and permanent teeth. Other vertebrates are polyphyodonts: their teeth are replaced throughout their entire life.

Mammals, like birds, possess a four-chambered heart. Mammals also have a specialized group of cardiac fibers located in the walls of their right atrium called the sinoatrial node, or pacemaker, which determines the rate at which the heart beats. As for blood, mammalian erythrocytes (red blood cells) do not have nuclei, whereas the erythrocytes of other vertebrates are nucleated.

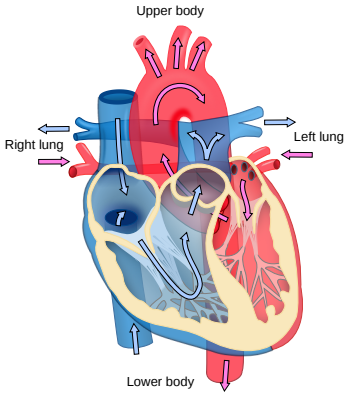

Figure 29.6*A*.1: **Mammalian heart**: Mammals possess a four-chambered heart, with two atria and two ventricles, that circulates blood through the body.

The kidneys of mammals have a portion of the nephron called the loop of Henle or nephritic loop, which allows mammals to produce urine with a high concentration of solutes; higher than that of the blood. Mammals lack a renal portal system: a system of veins that moves blood from the hind or lower limbs and region of the tail to the kidneys. Renal portal systems are present in all other vertebrates except jawless fishes. A urinary bladder is present in all mammals.

Mammalian brains have certain characteristics that differ from other vertebrates. In some, but not all mammals, the cerebral cortex, the outermost part of the cerebrum, is highly folded, allowing for a greater surface area than is possible with a smooth cortex. The optic lobes, located in the midbrain, are divided into two parts in mammals, whereas other vertebrates possess a single, undivided lobe. Eutherian mammals also possess a specialized structure that links the two cerebral hemispheres, called the corpus callosum.

29.6B: Evolution of Mammals

The modern mammals of today are synapsids: descendants of a group called cynodonts which appeared in the Late Permian period.

Learning Objectives

- Outline the evolution of mammals

Key Points

- Synapsids are defined by a single opening in the skull and the fact that they are endothermic.
- Mammals are the only living synapsids, derived from a lineage in the Jurassic period.
- Two groups of mammals include the eutherians, which are closely related to placentals and the metatherians, which are more closely related to the marsupials.
- Mammalian lineages from the Jurassic include Dryolestes, related to placentals and marsupials, and Ambondro, related to monotremes.
- Later synapsids had specialized structures for chewing, including teeth, cheeks that can hold food, and a secondary palate, which gave them the ability to chew and breathe at the same time.

Key Terms

- **eutherian**: the mammals more closely related to animals like humans and rodents than to marsupials
- **metatherian**: belonging or pertaining to the infraclass Metatheria of marsupials

Evolution of Mammals

The evolution of mammals passed through many stages since the first appearance of their synapsid ancestors in the late Carboniferous period. Mammals are synapsids: they have a single opening in the skull. They are the only living synapsids as earlier forms became extinct by the Jurassic period. The early, non-mammalian synapsids can be divided into two groups: the pelycosaurs and the therapsids. Within the therapsids, a group called the cynodonts are thought to be the ancestors of mammals. By the mid-Triassic, there were many synapsid species that looked like mammals. The lineage leading to today's mammals split in the Jurassic. Synapsids from this period include Dryolestes (more closely related to extant placentals and marsupials than to monotremes) as well as Ambondro (more closely related to monotremes). Later, the eutherian and metatherian lineages separated. Metatherians are the animals more closely related to the marsupials, while eutherians are those more closely related to the placentals. Eutherians are distinguished from noneutherians by various features of the feet, ankles, jaws, and teeth. One of the major differences between placental and nonplacental eutherians is that placentals lack epipubic bones, which are present in all other fossil and living mammals (marsupials and monotremes).

Figure $29.6B.1$: **Cynodonts**: Cynodonts, which first appeared in the Late Permian period 260 million years ago, are thought to be the ancestors of modern mammals.

Since Juramaia, the earliest-known eutherian, lived 160 million years ago in the Jurassic, this divergence must have occurred in the same period. After the Cretaceous–Paleogene extinction event wiped out the non-avian dinosaurs (birds are generally regarded as the surviving dinosaurs) and several other mammalian groups, placental and marsupial mammals

diversified into many new forms and ecological niches throughout the Paleogene and Neogene, by the end of which all modern orders had appeared.

The synapsid lineage became distinct from the sauropsid lineage in the late Carboniferous period, between 320 and 315 million years ago. The sauropsids are today's reptiles and birds, along with all the extinct animals more closely related to them than to mammals. This does not include the mammal-like reptiles, a group more closely related to the mammals. Throughout the Permian period, the synapsids included the dominant carnivores and several important herbivores. In the subsequent Triassic period, however, a previously-obscure group of sauropsids, the archosaurs, became the dominant vertebrates. The mammaliaforms appeared during this period; their superior sense of smell, backed up by a large brain, facilitated entry into nocturnal niches with less exposure to archosaur predation. The nocturnal lifestyle may have contributed greatly to the development of mammalian traits such as endothermy and hair. Later in the Mesozoic, after theropod dinosaurs replaced rauisuchians as the dominant carnivores, mammals spread into other ecological niches. For example, some became aquatic, some were gliders, and some even fed on juvenile dinosaurs. Most of the evidence consists of fossils. For many years, fossils of Mesozoic mammals and their immediate ancestors were very rare and fragmentary; however, since the mid-1990s, there have been many important new finds, especially in China. The relatively new techniques of molecular phylogenetics have also shed light on some aspects of mammalian evolution by estimating the timing of important divergence points for modern species. When used carefully, these techniques often, but not always, agree with the fossil record. Although mammary glands are a signature feature of modern mammals, little is known about the evolution of lactation. This is because these soft tissues are not often preserved in the fossil record. Most study of the evolution of mammals centers, rather, around the shapes of the teeth, the hardest parts of the tetrapod body. Other much-studied aspects include the evolution of the middle ear bones, erect limb posture, a bony secondary palate, fur and hair, and warm-bloodedness.

A key characteristic of synapsids is endothermy, rather than the ectothermy seen in most other vertebrates. The increased metabolic rate required to internally-modify body temperature went hand-in-hand with changes to certain skeletal structures. The later synapsids, which had more-evolved characteristics unique to mammals, possess cheeks for holding food and heterodont teeth (specialized for chewing by mechanically breaking down food to speed digestion and releasing the energy needed to produce heat). Chewing also requires the ability to chew and breathe at the same time, which is facilitated by the presence of a secondary palate. It separates the area of the mouth where chewing occurs from the area above where respiration occurs, allowing breathing to proceed uninterrupted during chewing. A secondary palate is not found in pelycosaurs, but is present in cynodonts and mammals. The jawbone also shows changes from early synapsids to later ones. The zygomatic arch, or cheekbone, is present in mammals and advanced therapsids such as cynodonts, but is not present in pelycosaurs. The presence of the zygomatic arch suggests the presence of the masseter muscle, which closes the jaw and functions in chewing.

29.6C: Living Mammals

Living mammals can be classified in three subclasses: eutherian, metatherians and monotremes.

Learning Objectives

- Name and describe the distinguishing features of the three main groups of mammals

Key Points

- Monotremes include the platypus which are defined by their ability to lay eggs instead of giving birth to live young.
- Metatherians are classified as the marsupials which possess a pouch where the premature young reside and nurse while continuing to develop.
- Eutherians are the most common type of mammal and are defined by the presence of a complex placenta which connects the developing fetus to the mother during gestation.

Key Terms

- **marsupial**: a mammal of which the female has a pouch in which it rears its young, which are born immature, through early infancy
- **placental**: a mammal having a placenta; most members of Mammalia

Living Mammals

Living mammals can be classified into three major classes: eutherians, monotremes, and metatherians. The eutherians, or placental mammals, and the metatherians, or marsupials, together comprise the clade of therian mammals. Monotremes form their sister clade.

There are three living species of monotremes: the platypus and two species of echidnas, or spiny anteaters. The leathery-beaked platypus belongs to the family Ornithorhynchidae (“bird beak”), whereas echidnas belong to the family Tachyglossidae (“sticky tongue”). The platypus and one species of echidna are found in Australia; the other species of echidna is found in New Guinea. Monotremes are unique among mammals as they lay eggs rather than giving birth to live young. The shells of their eggs are not like the hard shells of birds, but are leathery, similar to the shells of reptile eggs. All monotremes possess a cloaca which serves as the opening for the intestinal, reproductive, and urinary tracts. Additionally, monotremes have no teeth.

Marsupials are found primarily in Australia,although the opossum is found in North America. Australian marsupials include the kangaroo, koala, bandicoot,Tasmanian devil, and several other species. Most species of marsupials possess a pouch in which the very premature young reside after birth, receiving milk and continuing to develop. Marsupials differ from eutherians in that there is a less complex placental connection. The young are born at an extremely early age and latch onto the nipple within the pouch.

Figure $29.6C.1$: **Tasmanian devil**: The Tasmanian devil is one of several marsupials native to Australia. (credit: Wayne McLean)

Eutherians are the most widespread of the mammals, occurring throughout the world. There are 18 to 20 orders of placental mammals. Some examples are Insectivora, the insect eaters; Edentata, the toothless anteaters; Rodentia, the rodents; Cetacea, the aquatic mammals including whales; Carnivora, carnivorous mammals including dogs, cats, and bears; and Primates, which

includes humans. Eutherian mammals are sometimes called placental mammals because all species possess a complex placenta that connects a fetus to the mother, allowing for gas, fluid, and nutrient exchange. While other mammals possess a less complex placenta or briefly have a placenta, all eutherians possess a complex placenta during gestation.

Figure 29.6C.1: **A red fox**: Red foxes are eutherian (placental) mammals because the mothers nourish their young via a placenta during fetal development. The placenta enables a mother to exchange gases, fluids, and nutrients with the growing embryos.

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29.7: The Evolution of Primates

Topic hierarchy

- 29.7A: Characteristics and Evolution of Primates

All primates exhibit adaptations for climbing trees and have evolved into two main groups: Prosimians and Anthropoids.

- 29.7B: Early Human Evolution
- 29.7C: Early Hominins
- 29.7D: Genus Homo

29.7A: Characteristics and Evolution of Primates

All primates exhibit adaptations for climbing trees and have evolved into two main groups: Prosimians and Anthropoids.

Learning Objectives

- Identify key characteristics of primates

Key Points

- All primates are descended from tree-dwellers, exhibiting adaptations which allow for tree climbing that include: a rotating shoulder joint, separated big toes and thumb for grasping, and stereoscopic vision.
- Other primate characteristics include: having one offspring per pregnancy, claws evolved into flattened nails; and larger brain/body ratio than other mammals, and tendency to hold body upright.
- True primates, ancestral to prosimians, first appear in the fossil record in the Eocene epoch around 55 million years ago; they were similar in form to lemurs.
- Anthropoids ancestral to both Old World and New World monkeys appear in the fossil record in the Oligocene epoch around 35 million years ago.
- Anthropoids ancestral to apes appear in the Miocene epoch around 25 million years ago.
- Apes are divided into two main groups of hominoids: lesser apes or hylobatids (gibbons and siamangs) and great apes (*Pongo*: orangutans, *Gorilla*: gorillas, *Pan*:chimpanzees, and *Homo*: humans).

Key Terms

- **dimorphism**: the occurrence in an animal species of two distinct types of individual
- **adaptive radiation**: the diversification of species into separate forms that each adapt to occupy a specific environmental niche

Characteristics of Primates

All primate species possess adaptations for climbing trees, as they all descended from tree-dwellers. This arboreal heritage of primates has resulted in adaptations that include, but are not limited to: 1) a rotating shoulder joint; 2) a big toe that is widely separated from the other toes and thumbs, that are widely separated from fingers (except humans), which allow for gripping branches; and 3) stereoscopic vision, two overlapping fields of vision from the eyes, which allows for the perception of depth and gauging distance. Other characteristics of primates are brains that are larger than those of most other mammals (larger brain/body ratio than similar-sized non-primates), claws that have been modified into flattened nails, typically only one offspring per pregnancy, and a trend toward holding the body upright.

The Order Primates is divided into two groups: prosimians and anthropoids. Prosimians include the bush babies and pottos of Africa, the lemurs of Madagascar, and the lorises of Southeast Asia. Tarsier, also from Southeast Asia, show some prosimian-like and some anthropoid-like features. Anthropoids include monkeys, apes, and humans. In general, prosimians tend to be nocturnal (in contrast to diurnal anthropoids, excluding the nocturnal *Aotus*, owl monkey) and have a smaller brain/body ratio than anthropoids.

Evolution of Primates

The first primate-like mammals are referred to as proto-primates. They were roughly similar to squirrels and tree shrews in size and appearance. The existing fossil evidence (mostly from North Africa) is very fragmentary. These proto-primates will remain largely mysterious creatures until more fossil evidence becomes available. The oldest known primate-like mammal with a relatively robust fossil record is *Plesiadapis* (although some researchers do not agree that *Plesiadapis* was a proto-primate). Fossils of this primate have been dated to approximately 55 million years ago. Plesiadapiforms had some features of the teeth and skeleton in common with true primates. They were found in North America and Europe in the Cenozoic, going extinct by the end of the Eocene.

The first true primates were found in North America, Europe, Asia, and Africa in the Eocene Epoch. These early primates resembled present-day prosimians such as lemurs. Evolutionary changes continued in these early primates, with larger

brains and eyes, and smaller muzzles being the trend. By the end of the Eocene Epoch, many of the early prosimian species went extinct due either to cooler temperatures or competition from the first monkeys.

Evidence shows that the anthropoid monkeys evolved from prosimians during the Oligocene Epoch. By 35 million years ago, evidence indicates that monkeys were present the Old World (Africa and Asia) and in the New World (South America) by 30 million years ago. New World monkeys are also called Platyrrhini: a reference to their broad noses. Old World monkeys (and apes) are called Catarrhini: a reference to their narrow noses. There is still quite a bit of uncertainty about the origins of the New World monkeys. At the time the platyrrhines arose, the continents of South American and Africa had drifted apart. Therefore, it is thought that monkeys arose in the Old World and reached the New World by drifting on log rafts or mangrove floating 'islands'. Due to this reproductive isolation, New World monkeys and Old World monkeys underwent separate adaptive radiations over millions of years. The New World monkeys are all arboreal, whereas Old World monkeys include arboreal and ground-dwelling species.

Figure 29.7A.1: **Howler monkey**: The howler monkey, a member of the Platyrrhini, is native to Central and South America. It makes a call that sounds like a lion roaring.

Apes evolved from the catarrhines in Africa midway through the Cenozoic during the Miocene epoch, approximately 25 million years ago. Apes are generally larger than monkeys and do not possess a tail. All apes are capable of moving through trees, although many species spend most their time on the ground. Apes are more intelligent than monkeys as they have relatively larger brains proportionate to body size.

The apes are divided into two groups. The lesser apes comprise the family Hylobatidae, including gibbons and siamangs. The great apes include the genera *Pan* (chimpanzees and bonobos), *Gorilla* (gorillas), *Pongo* (orangutans), and *Homo* (humans). The very arboreal gibbons are smaller than the great apes; they have low sexual dimorphism (that is, the genders are not markedly different in size); and they have relatively longer arms used for swinging/brachiating through trees.

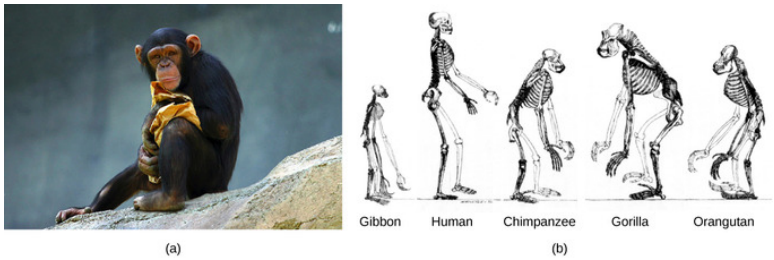

Figure 29.7A.1: **Chimpanzee**: The (a) chimpanzee is one of the great apes. It possesses a relatively large brain and has no tail. (b) All great apes have a similar skeletal structure.

29.7B: Early Human Evolution

Modern humans and chimpanzees evolved from a common hominoid ancestor that diverged approximately 6 million years ago.

Learning Objectives

- List the evolved physical traits used to differentiate hominins from other hominoids

Key Points

- Modern humans are classified as hominins, which also includes extinct bipedal human relatives, such as *Australopithecus africanus*, *Homo habilis* , and *Homo erectus*.
- Few very early (prior to 4 million years ago) hominin fossils have been found so determining the lines of hominin descent is extremely difficult.
- Within the last 20 years, three new genera of hominoids were discovered: *Sahelanthropus tchadensis*, *Orrorin tugenensis*, and *Ardipithecus ramidus* and *kadabba*, but their status in regards to human ancestry is somewhat uncertain.

Key Terms

- **hominin**: the evolutionary group that includes modern humans and now-extinct bipedal relatives
- **hominoid**: any great ape (such as humans) belonging to the superfamily Hominoidea

Human Evolution

The family Hominidae of order Primates includes chimpanzees and humans. Evidence from the fossil record and from a comparison of human and chimpanzee DNA suggests that humans and chimpanzees diverged from a common hominoid ancestor approximately 6 million years ago. Several species evolved from the evolutionary branch that includes humans, although our species is the only surviving member. The term hominin (or hominid) is used to refer to those species that evolved after this split of the primate line, thereby designating species that are more closely related to humans than to chimpanzees. Hominins, who were bipedal in comparison to the other hominoids who were primarily quadrupedal, includes those groups that probably gave rise to our species: *Australopithecus africanus*, *Homo habilis*, and *Homo erectus*, along with non- ancestral groups such as *Australopithecus boisei*. Determining the true lines of descent in hominins is difficult. In years past, when relatively few hominin fossils had been recovered, some scientists believed that considering them in order, from oldest to youngest, would demonstrate the course of evolution from early hominins to modern humans. In the past several years, however, many new fossils have been found. It is possible that there were often more than one species alive at any one time and that many of the fossils found (and species named) represent hominin species that died out and are not ancestral to modern humans. However, it is also possible that too many new species have been named.

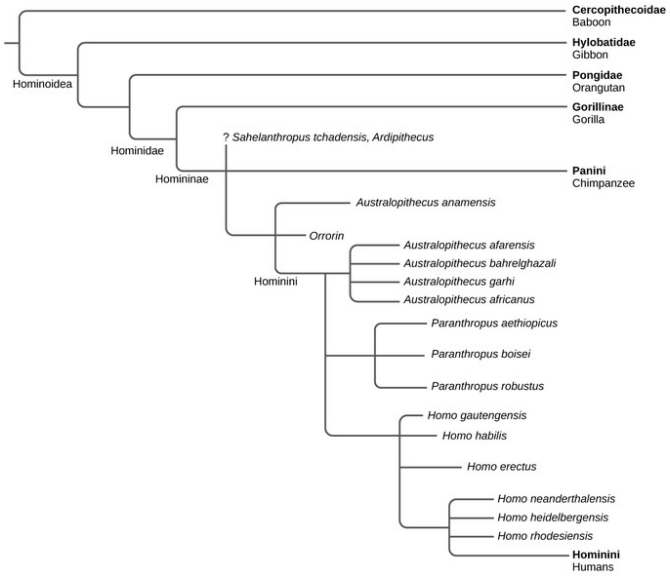

Figure 29.7B.1: **Evolution of modern humans**: This chart shows the evolution of modern humans and includes the point of divergence that occurred between modern humans and the other great apes.

Very Early Hominins

There have been three species of very early hominoids which have made news in the past few years. The oldest of these, *Sahelanthropus tchadensis*, has been dated to nearly seven million years ago. There is a single specimen of this genus, a skull that was a surface find in Chad. The fossil, informally called “Toumai,” is a mosaic of primitive and evolved characteristics. To date, it is unclear how this fossil fits with the picture given by molecular data. The line leading to modern humans and modern chimpanzees apparently bifurcated (divided into branches) about six million years ago. It is not thought at this time that this species was an ancestor of modern humans. It may not have been a hominin.

A second, younger species (around 5.7 million years ago), *Orrorin tugenensis*, is also a relatively-recent discovery, found in 2000. There are several specimens of *Orrorin*. It is not known whether *Orrorin* was a human ancestor, but this possibility has not been ruled out. Some features of *Orrorin* are more similar to those of modern humans than are the australopiths, although *Orrorin* is much older.

A third genus, *Ardipithecus ramidus (*4.4 million years ago), was discovered in the 1990s. The scientists who discovered the first fossil found that some other scientists did not believe the organism to be a biped (thus, it would not be considered a hominid). In the intervening years, several more specimens of *Ardipithecus*, including a new species, *Ardipithecus kadabba* (5.6 million years ago), demonstrated that they were bipedal. Again, the status of this genus as a human ancestor is uncertain, but, given that it was bipedal, it was a hominin.

29.7C: Early Hominins

The hominin *Australopithecus* evolved 4 million years ago and is believed to be in the ancestral line of the genus *Homo*.

Learning Objectives

- Describe the physical characteristics of the Australopiths and compare them to those of modern humans

Key Points

- The early hominin *Australopithecus* displayed various characteristics which show more similarity to the great apes than to modern humans: great sexual dimorphism, small brain size in comparison to body mass, larger canines and molars, and a prognathic jaw.
- *Australopithecus africanus* lived between 2 and 3 million years ago and had a larger brain than *A. afarensis*, but was still less than one-third the size of the modern human brain.
- The gracile australopiths had a relatively slender build and teeth that were suited for soft food and may have had a partially carnivorous diet, while the robust australopiths probably ate tough vegetation.

Key Terms

- **dentition**: the type, number and arrangement of the normal teeth of an organism or of the actual teeth of an individual
- **sexual dimorphism**: a physical difference between male and female individuals of the same species
- **bipedalism**: the habit of standing and walking on two feet

Early Hominins: Genus *Australopithecus*

Australopithecus ("southern ape") is a genus of hominin that evolved in eastern Africa approximately 4 million years ago and became extinct about 2 million years ago. This genus is of particular interest to us as it is thought that our genus, genus *Homo*, evolved from *Australopithecus* about 2 million years ago. *Australopithecus* had a number of characteristics that were more similar to the great apes than to modern humans. For example, sexual dimorphism was more exaggerated than in modern humans. Males were up to 50 percent larger than females, a ratio that is similar to that seen in modern gorillas and orangutans. In contrast, modern human males are approximately 15 to 20 percent larger than females. The brain size of *Australopithecus* relative to its body mass was also smaller than modern humans and more similar (although larger) to that seen in the great apes. A key feature that *Australopithecus* had in common with modern humans was bipedalism, although it is likely that *Australopithecus* also spent time in trees. Hominin footprints, similar to those of modern humans, found in Laetoli, Tanzania, are dated to 3.6 million years ago. They show that hominins at the time of *Australopithecus* were walking upright.

There were a number of *Australopithecus* species, often referred to as australopiths. *Australopithecus anamensis* lived about 4.2 million years ago. More is known about another early species, *Australopithecus afarensis*, which lived between 3.9 and 2.9 million years ago. This species demonstrates a trend in human evolution: the reduction of the dentition and jaw in size. *A. afarensis* had smaller canines and molars compared to apes, but these were larger than those of modern humans. Its brain size was 380–450 cubic centimeters, approximately the size of a modern chimpanzee brain. It also had prognathic jaws, which is a relatively longer jaw than that of modern humans. In the mid-1970s, the fossil of an adult female *A. afarensis* was found in the Afar region of Ethiopia, dated to 3.24 million years ago. The fossil, which is informally called "Lucy," is significant because it was the most complete australopith fossil found, with 40 percent of the skeleton recovered.

Figure 29.7C.1: **Adult Female *Australopithecus afarensis***: This adult female *Australopithecus afarensis* skeleton, nicknamed Lucy, was discovered in the mid 1970s.

Figure 29.7C.1: **Skull comparison: *Australopithecus afarensis* vs modern humans**: The skull of (a) *Australopithecus afarensis*, an early hominid that lived between three and four million years ago, resembled that of (b) modern humans, but was smaller with a sloped forehead and prominent jaw.

Australopithecus africanus lived between 2 and 3 million years ago. It had a slender build and was bipedal, but had robust arm bones and, as with other early hominids, may have spent significant time in trees. Its brain was larger than that of *A. afarensis* at 500 cubic centimeters, which is slightly less than one-third the size of modern human brains. Two other species, *Australopithecus bahrelghazali*and *Australopithecus garhi*, have been added to the roster of australopiths in recent years.

A Dead End

While most australopiths had a relatively slender, gracile build and teeth suited for soft food, there were also australopiths of a more robust build, dating to approximately 2.5 million years ago. These hominids were larger and had large grinding teeth. Their molars show heavy wear, suggesting that they had a coarse and fibrous vegetarian diet as opposed to the partially carnivorous diet of the more gracile australopiths. They include *Australopithecus robustus* of South Africa, and *Australopithecus aethiopicus* and *Australopithecus boisei* of East Africa. These hominids became extinct more than 1 million years ago and are not thought to be ancestral to modern humans, but rather members of an evolutionary branch on the hominin tree that left no descendants.

29.7D: Genus Homo

The human genus *Homo*, which includes modern humans as well as extinct human relatives, appeared around 2.3 million years ago.

Learning Objectives

- Compare and contrast the evolution and characteristics associated with the various Homo species: Homo habilis, erectus, and sapiens

Key Points

- *Homo erectus*, appearing 1.8 million years ago, was the first hominin species to migrate out of East Africa, use fire, and hunt.
- Compared to *Homo habilis*, *Homo erectus* was more similar to modern humans due to its height and weight, brain size, limited sexual dimorphism, and downward-facing nostrils.
- Archaic *Homo sapiens* had a similar brain size to modern humans (*Homo sapiens sapiens*), but, unlike modern humans, they had a thick skull, prominent brow ridge, and a receding chin.
- The multiregional hypothesis of modern human origins states that there is an unbroken line of evolution involving regional adaptations and gene flow from *H. erectus* to *H. sapiens sapiens*.
- The recent out of Africa hypothesis of modern human origins states that *H. sapiens sapiens* arose in Africa between 100,000 – 200,000 years, left Africa around 60,000 years ago, and replaced all archaic humans, with very little inter-breeding.
- All men today inherited a Y chromosome from a male that lived in Africa about 140,000 years ago.

Key Terms

- **Homo habilis**: (“handy man”) an extinct taxonomic species within the genus Homo that had long arms and may have used stone tools
- **Homo erectus**: (“upright man) extinct species of hominin that appeared 1.8 million years ago; the first hominin to use fire, hunt, and have a home base
- **Homo sapiens**: evolved from H. erectus starting about 500,000 years ago; humans

Early Hominins: Genus *Homo*

The human genus, *Homo*, first appeared around 2.3 million years ago. For many years, fossils of a species called *Homo habilis* were the oldest examples in the genus *Homo*, but in 2010, a new species called *Homo gautengensis* was proposed that may be older, although it is not well accepted. In comparison to *Australopithecus africanus*, *H. habilis* had a number of features more similar to modern humans. *H. habilis* had a jaw that was less prognathic (forward projection of the jaw) than the australopiths and a larger brain, at 600–750 cubic centimeters. However, *H. habilis* retained some features of older hominin species, such as long arms. The name *H. habilis* means “handy man,” which is a reference to the stone tools that have been found with its remains.

H. erectus appeared approximately 1.8 million years ago. It is believed to have originated in East Africa and was the first hominin species to migrate out of Africa. Fossils of *H. erectus* have been found in India, China, Java, and Europe, and were known in the past as “Java Man” or “Peking Man.” *H. erectus* had a number of features that were more similar to modern humans than those of *H. habilis*. *H. erectus* was larger in size than earlier hominins, reaching heights up to 1.85 meters and weighing up to 65 kilograms, sizes similar to those of modern humans. Its degree of sexual dimorphism was less than earlier species, with males being 20 to 30 percent larger than females, which is close to the size difference seen in our species. *H. erectus* had a larger brain than earlier species at 775–1,100 cubic centimeters, which compares to the 1,130–1,260 cubic centimeters seen in modern human brains. *H. erectus* also had a nose with downward-facing nostrils similar to modern humans, rather than the forward facing nostrils found in other primates. Longer, downward-facing nostrils allow for the warming of cold air before it enters the lungs and may have been an adaptation to colder climates. Artifacts found with fossils of *H. erectus* suggest that it was the first hominin to use fire, hunt, and have a home base. *H. erectus* is generally thought to have lived until about 50,000 years ago.

Homo erectus

Homo erectus had a prominent brow and a nose that pointed downward rather than forward.

Humans: *Homo sapiens*

A number of species, sometimes called archaic *Homo sapiens*, apparently evolved from *H. erectus* starting about 500,000 years ago. These archaic *H. sapiens* had a brain size similar to that of modern humans, averaging 1,200–1,400 cubic centimeters. They differed from modern humans by having a thick skull, a prominent brow ridge, and a receding chin. Some of these populations survived until 30,000–10,000 years ago, overlapping with anatomically-modern humans.

Figure 29.7*D*.1: ***Homo sapiens neanderthalensis* Tools**: The *Homo sapiens neanderthalensis* used tools and may have worn clothing.

There is considerable debate about the origins of anatomically-modern humans or *Homo sapiens sapiens*. As discussed earlier, *H. erectus* migrated out of Africa and into Asia and Europe in the first major wave of migration about 1.5 million years ago. The multiregional hypothesis holds that humans first arose near the beginning of the Pleistocene two million years ago and subsequent human evolution has been within a single, continuous human species. This species encompasses archaic human forms such as *Homo erectus* and Neanderthals as well as modern forms, which evolved worldwide to the diverse populations of modern *Homo sapiens sapiens*. The hypothesis contends that humans evolve through a combination of adaptation within various regions of the world and gene flow between those regions. Proponents of multiregional origin point to fossil and genomic data and continuity of archaeological cultures as support for their hypothesis.

The primary alternative hypothesis is the recent African origin of modern humans, which holds that modern humans arose in Africa around 100,000–200,000 years ago, moving out of Africa around 50,000–60,000 years ago to replace archaic

human forms with limited interbreeding: at least once with Neanderthals and once with Denisovans.

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CHAPTER OVERVIEW

3: BIOLOGICAL MACROMOLECULES

3.1: CARBOHYDRATES

3.1A: CARBOHYDRATE MOLECULES
3.1B: IMPORTANCE OF CARBOHYDRATES
3.2: LIPID MOLECULES

3.2A: LIPID MOLECULES
3.2B: WAXES
Waxes are nonpolar lipids that plants and animals use for protection and have many functions in society.

3.2C: PHOSPHOLIPIDS
3.2D: STEROIDS
3.3: PROTEINS

3.3A: TYPES AND FUNCTIONS OF PROTEINS
3.3B: AMINO ACIDS
An amino acid contains an amino group, a carboxyl group, and an R group, and it combines with other amino acids to form polypeptide chains.

3.3C: PROTEIN STRUCTURE
3.3D: DENATURATION AND PROTEIN FOLDING
Denaturation is a process in which proteins lose their shape and, therefore, their function because of changes in pH or temperature.

3.4: NUCLEIC ACIDS

3.4A: DNA AND RNA
DNA and RNA are nucleic acids that carry out cellular processes, especially the regulation and expression of genes.

3.4B: THE DNA DOUBLE HELIX
3.4C: DNA PACKAGING
3.4D: TYPES OF RNA

Carbohydrates

Topic hierarchy

- 3.1A: Carbohydrate Molecules
- 3.1B: Importance of Carbohydrates

3.1A: Carbohydrate Molecules

Carbohydrates are essential macromolecules that are classified into three subtypes: monosaccharides, disaccharides, and polysaccharides.

Learning Objectives

- Describe the structure of mono-, di-, and poly-saccharides

Key Points

- Monosaccharides are simple sugars made up of three to seven carbons, and they can exist as a linear chain or as ring-shaped molecules.
- Glucose, galactose, and fructose are monosaccharide isomers, which means they all have the same chemical formula but differ structurally and chemically.
- Disaccharides form when two monosaccharides undergo a dehydration reaction (a condensation reaction); they are held together by a covalent bond.
- Sucrose (table sugar) is the most common disaccharide, which is composed of the monomers glucose and fructose.
- A polysaccharide is a long chain of monosaccharides linked by glycosidic bonds; the chain may be branched or unbranched and can contain many types of monosaccharides.

Key Terms

- **isomer**: Any of two or more compounds with the same molecular formula but with different structure.
- **dehydration reaction**: A chemical reaction in which two molecules are covalently linked in a reaction that generates H2O as a second product.
- **biopolymer**: Any macromolecule of a living organism that is formed from the polymerization of smaller entities; a polymer that occurs in a living organism or results from life.

Carbohydrates can be represented by the stoichiometric formula $(CH_2O)_n$, where n is the number of carbons in the molecule. Therefore, the ratio of carbon to hydrogen to oxygen is 1:2:1 in carbohydrate molecules. The origin of the term “carbohydrate” is based on its components: carbon (“carbo”) and water (“hydrate”). Carbohydrates are classified into three subtypes: monosaccharides, disaccharides, and polysaccharides.

Monosaccharides

Monosaccharides (mono- = “one”; sacchar- = “sweet”) are simple sugars. In monosaccharides, the number of carbons usually ranges from three to seven. If the sugar has an aldehyde group (the functional group with the structure R-CHO), it is known as an aldose, and if it has a ketone group (the functional group with the structure RC(=O)R’), it is known as a ketose. Depending on the number of carbons in the sugar, they also may be known as trioses (three carbons), pentoses (five carbons), and or hexoses (six carbons). Monosaccharides can exist as a linear chain or as ring-shaped molecules; in aqueous solutions they are usually found in ring forms.

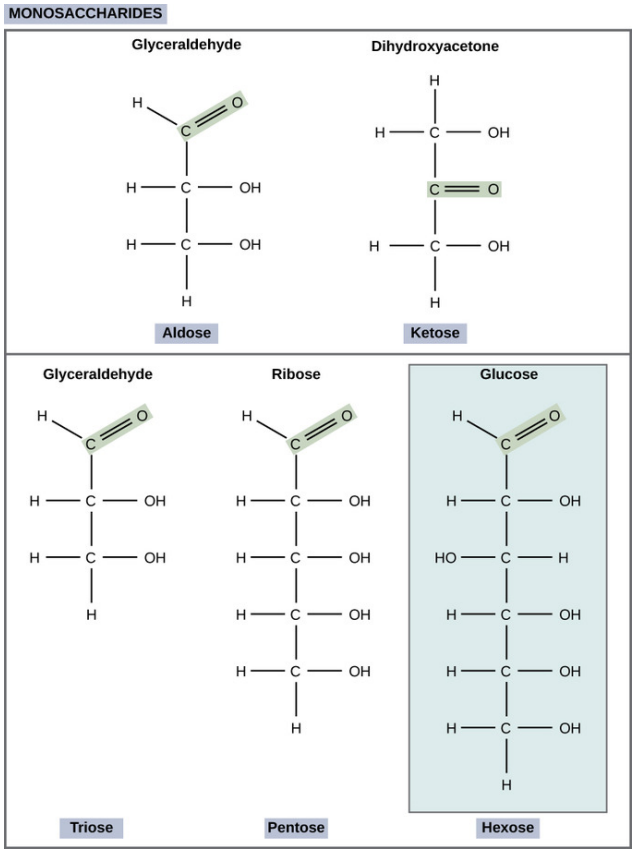

Figure 3.1*A*. 1: **Monosaccharides**: Monosaccharides are classified based on the position of their carbonyl group and the number of carbons in the backbone. Aldoses have a carbonyl group (indicated in green) at the end of the carbon chain, and ketoses have a carbonyl group in the middle of the carbon chain. Trioses, pentoses, and hexoses have three, five, and six carbon backbones, respectively.

Common Monosaccharides

Glucose ($C_6H_{12}O_6$) is a common monosaccharide and an important source of energy. During cellular respiration, energy is released from glucose and that energy is used to help make adenosine triphosphate (ATP). Plants synthesize glucose using carbon dioxide and water, and glucose, in turn, is used for energy requirements for the plant.

Galactose (a milk sugar) and fructose (found in fruit) are other common monosaccharides. Although glucose, galactose, and fructose all have the same chemical formula ($C_6H_{12}O_6$), they differ structurally and stereochemically. This makes them different molecules despite sharing the same atoms in the same proportions, and they are all isomers of one another, or isomeric monosaccharides. Glucose and galactose are aldoses, and fructose is a ketose.

Disaccharides

Disaccharides (di- = “two”) form when two monosaccharides undergo a dehydration reaction (also known as a condensation reaction or dehydration synthesis). During this process, the hydroxyl group of one monosaccharide combines with the hydrogen of another monosaccharide, releasing a molecule of water and forming a covalent bond. A covalent bond formed between a carbohydrate molecule and another molecule (in this case, between two monosaccharides) is known as a glycosidic bond. Glycosidic bonds (also called glycosidic linkages) can be of the alpha or the beta type.

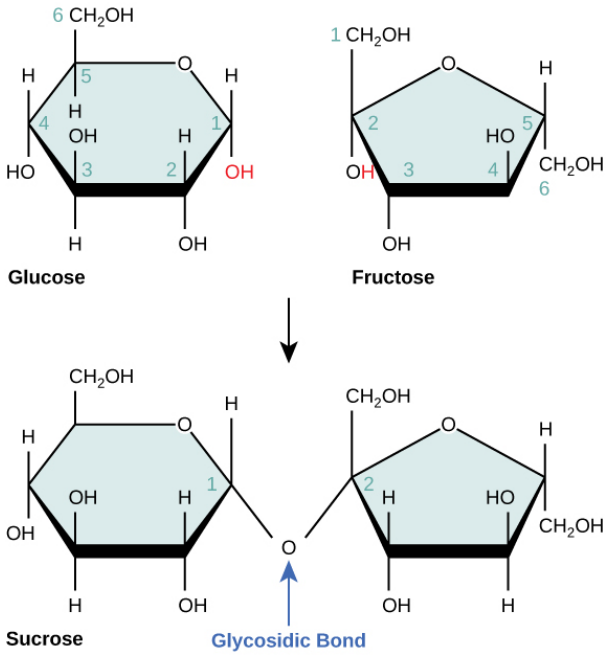

Figure 3.1*A*. 1: **Disaccharides**: Sucrose is formed when a monomer of glucose and a monomer of fructose are joined in a dehydration reaction to form a glycosidic bond. In the process, a water molecule is lost. By convention, the carbon atoms in a monosaccharide are numbered from the terminal carbon closest to the carbonyl group. In sucrose, a glycosidic linkage is formed between carbon 1 in glucose and carbon 2 in fructose.

Common Disaccharides

Common disaccharides include lactose, maltose, and sucrose. Lactose is a disaccharide consisting of the monomers glucose and galactose. It is found naturally in milk. Maltose, or malt sugar, is a disaccharide formed by a dehydration reaction between two glucose molecules. The most common disaccharide is sucrose, or table sugar, which is composed of the monomers glucose and fructose.

Polysaccharides

A long chain of monosaccharides linked by glycosidic bonds is known as a polysaccharide (poly- = “many”). The chain may be branched or unbranched, and it may contain different types of monosaccharides. Starch, glycogen, cellulose, and chitin are primary examples of polysaccharides.

Plants are able to synthesize glucose, and the excess glucose is stored as starch in different plant parts, including roots and seeds. Starch is the stored form of sugars in plants and is made up of glucose monomers that are joined by α1-4 or 1-6 glycosidic bonds. The starch in the seeds provides food for the embryo as it germinates while the starch that is consumed by humans is broken down by enzymes into smaller molecules, such as maltose and glucose. The cells can then absorb the glucose.

Common Polysaccharides

Glycogen is the storage form of glucose in humans and other vertebrates. It is made up of monomers of glucose. Glycogen is the animal equivalent of starch and is a highly branched molecule usually stored in liver and muscle cells. Whenever blood glucose levels decrease, glycogen is broken down to release glucose in a process known as glycogenolysis.

Cellulose is the most abundant natural biopolymer. The cell wall of plants is mostly made of cellulose and provides structural support to the cell. Cellulose is made up of glucose monomers that are linked by β 1-4 glycosidic bonds. Every other glucose monomer in cellulose is flipped over, and the monomers are packed tightly as extended long chains. This gives cellulose its rigidity and high tensile strength—which is so important to plant cells.

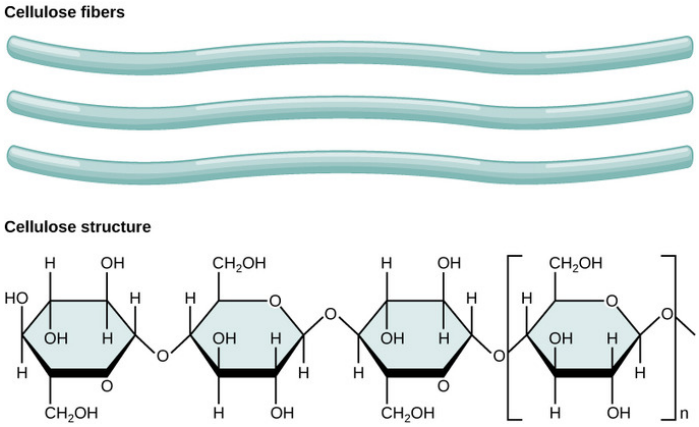

Figure 3.1A.1: **Polysaccharides**: In cellulose, glucose monomers are linked in unbranched chains by β 1-4 glycosidic linkages. Because of the way the glucose subunits are joined, every glucose monomer is flipped relative to the next one resulting in a linear, fibrous structure.

Carbohydrate Function

Carbohydrates serve various functions in different animals. Arthropods have an outer skeleton, the exoskeleton, which protects their internal body parts. This exoskeleton is made of chitin, which is a polysaccharide-containing nitrogen. It is made of repeating units of N-acetyl-β-d-glucosamine, a modified sugar. Chitin is also a major component of fungal cell walls.

3.1B: Importance of Carbohydrates

Carbohydrates are a major class of biological macromolecules that are an essential part of our diet and provide energy to the body.

Learning Objectives

- Describe the benefits provided to organisms by carbohydrates

Key Points

- Carbohydrates provide energy to the body, particularly through glucose, a simple sugar that is found in many basic foods.
- Carbohydrates contain soluble and insoluble elements; the insoluble part is known as fiber, which promotes regular bowel movement, regulates the rate of consumption of blood glucose, and also helps to remove excess cholesterol from the body.
- As an immediate source of energy, glucose is broken down during the process of cellular respiration, which produces ATP, the energy currency of the cell.
- Since carbohydrates are an important part of the human nutrition, eliminating them from the diet is not the best way to lose weight.

Key Terms

- **carbohydrate**: A sugar, starch, or cellulose that is a food source of energy for an animal or plant; a saccharide.
- **glucose**: a simple monosaccharide (sugar) with a molecular formula of C6H12O6; it is a principal source of energy for cellular metabolism
- **ATP**: A nucleotide that occurs in muscle tissue, and is used as a source of energy in cellular reactions, and in the synthesis of nucleic acids. ATP is the abbreviation for adenosine triphosphate.

Benefits of Carbohydrates

Biological macromolecules are large molecules that are necessary for life and are built from smaller organic molecules. One major class of biological macromolecules are carbohydrates, which are further divided into three subtypes: monosaccharides, disaccharides, and polysaccharides. Carbohydrates are, in fact, an essential part of our diet; grains, fruits, and vegetables are all natural sources of carbohydrates. Importantly, carbohydrates provide energy to the body, particularly through glucose, a simple sugar that is a component of starch and an ingredient in many basic foods.

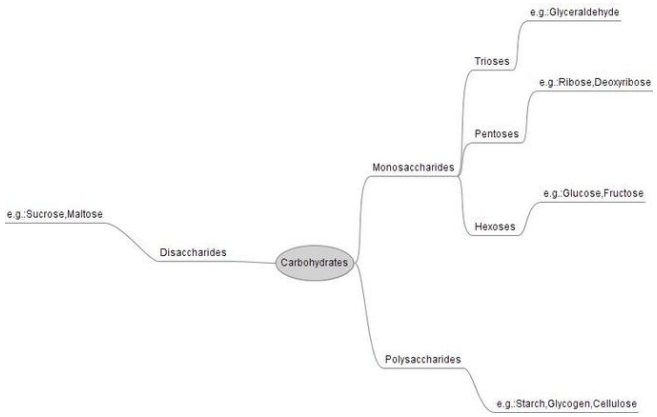

Figure 3.1*B*.1: **Carbohydrates**: Carbohydrates are biological macromolecules that are further divided into three subtypes: monosaccharides, disaccharides, and polysaccharides. Like all macromolecules, carbohydrates are necessary for life and are built from smaller organic molecules.

Carbohydrates in Nutrition

Carbohydrates have been a controversial topic within the diet world. People trying to lose weight often avoid carbs, and some diets completely forbid carbohydrate consumption, claiming that a low-carb diet helps people to lose weight faster.

However, carbohydrates have been an important part of the human diet for thousands of years; artifacts from ancient civilizations show the presence of wheat, rice, and corn in our ancestors' storage areas.

Carbohydrates should be supplemented with proteins, vitamins, and fats to be parts of a well-balanced diet. Calorie-wise, a gram of carbohydrate provides 4.3 Kcal. In comparison, fats provide 9 Kcal/g, a less desirable ratio. Carbohydrates contain soluble and insoluble elements; the insoluble part is known as fiber, which is mostly cellulose. Fiber has many uses; it promotes regular bowel movement by adding bulk, and it regulates the rate of consumption of blood glucose. Fiber also helps to remove excess cholesterol from the body. Fiber binds and attaches to the cholesterol in the small intestine and prevents the cholesterol particles from entering the bloodstream. Then cholesterol exits the body via the feces. Fiber-rich diets also have a protective role in reducing the occurrence of colon cancer. In addition, a meal containing whole grains and vegetables gives a feeling of fullness. As an immediate source of energy, glucose is broken down during the process of cellular respiration, which produces adenosine triphosphate (ATP), the energy currency of the cell. Without the consumption of carbohydrates, the availability of "instant energy" would be reduced. Eliminating carbohydrates from the diet is not the best way to lose weight. A low-calorie diet that is rich in whole grains, fruits, vegetables, and lean meat, together with plenty of exercise and plenty of water, is the more sensible way to lose weight.

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3.2: Lipid Molecules

Topic hierarchy

- 3.2A: Lipid Molecules
- 3.2B: Waxes

Waxes are nonpolar lipids that plants and animals use for protection and have many functions in society.

- 3.2C: Phospholipids
- 3.2D: Steroids

3.2A: Lipid Molecules

Fats and oils, which may be saturated or unsaturated, can be unhealthy but also serve important functions for plants and animals.

Learning Objectives

- Differentiate between saturated and unsaturated fatty acids

Key Points

- Fats provide energy, insulation, and storage of fatty acids for many organisms.
- Fats may be saturated (having single bonds) or unsaturated (having double bonds).
- Unsaturated fats may be cis (hydrogens in same plane) or trans (hydrogens in two different planes).
- Olive oil, a monounsaturated fat, has a single double bond whereas canola oil, a polyunsaturated fat, has more than one double bond.
- Omega-3 fatty acid and omega-6 fatty acid are essential for human biological processes, but they must be ingested in the diet because they cannot be synthesized.

Key Terms

- **hydrogenation**: The chemical reaction of hydrogen with another substance, especially with an unsaturated organic compound, and usually under the influence of temperature, pressure and catalysts.
- **ester**: Compound most often formed by the condensation of an alcohol and an acid, by removing water. It contains the functional group carbon-oxygen double bond joined via carbon to another oxygen atom.
- **carboxyl**: A univalent functional group consisting of a carbonyl and a hydroxyl functional group (-CO.OH); characteristic of carboxylic acids.

Fats have important functions, and many vitamins are fat soluble. Fats serve as a long-term storage form of fatty acids and act as a source of energy. They also provide insulation for the body.

Glycerol and Fatty Acids

A fat molecule consists of two main components: glycerol and fatty acids. Glycerol is an alcohol with three carbons, five hydrogens, and three hydroxyl (OH) groups. Fatty acids have a long chain of hydrocarbons with a carboxyl group attached and may have 4-36 carbons; however, most of them have 12-18. In a fat molecule, the fatty acids are attached to each of the three carbons of the glycerol molecule with an ester bond through the oxygen atom. During the ester bond formation, three molecules are released. Since fats consist of three fatty acids and a glycerol, they are also called triacylglycerols or triglycerides.

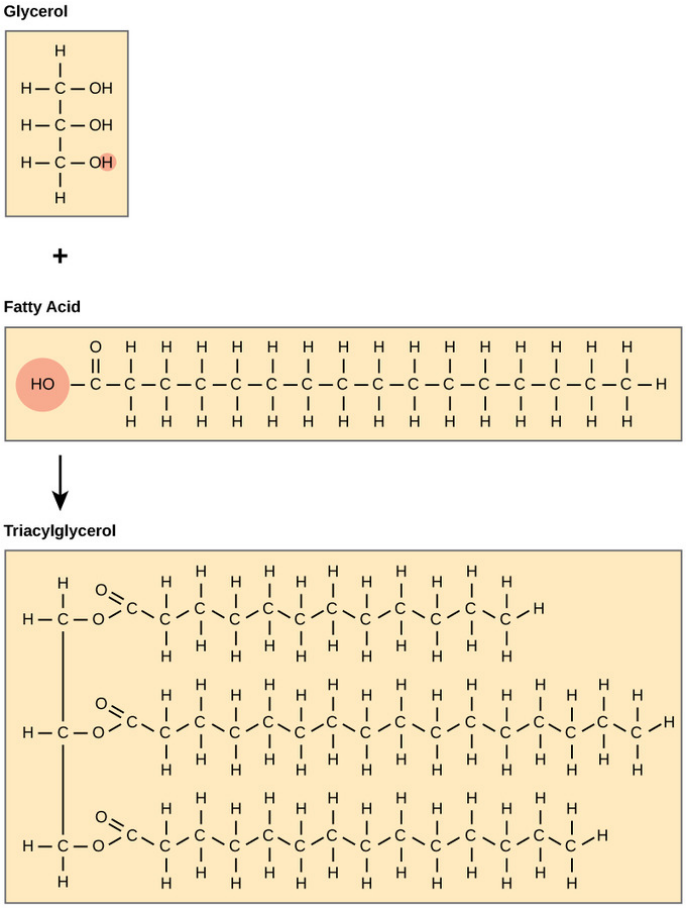


Figure 3.2A.1: **Triacylglycerols**: Triacylglycerol is formed by the joining of three fatty acids to a glycerol backbone in a dehydration reaction. Three molecules of water are released in the process.

Saturated vs. Unsaturated Fatty Acids

Fatty acids may be saturated or unsaturated. In a fatty acid chain, if there are only single bonds between neighboring carbons in the hydrocarbon chain, the fatty acid is said to be saturated. Saturated fatty acids are saturated with hydrogen since single bonds increase the number of hydrogens on each carbon. Stearic acid and palmitic acid, which are commonly found in meat, are examples of saturated fats.

When the hydrocarbon chain contains a double bond, the fatty acid is said to be unsaturated. Oleic acid is an example of an unsaturated fatty acid. Most unsaturated fats are liquid at room temperature and are called oils. If there is only one double bond in the molecule, then it is known as a monounsaturated fat; e.g. olive oil. If there is more than one double bond, then it is known as a polyunsaturated fat; e.g. canola oil. Unsaturated fats help to lower blood cholesterol levels whereas saturated fats contribute to plaque formation in the arteries.

Unsaturated fats or oils are usually of plant origin and contain cis unsaturated fatty acids. Cis and trans indicate the configuration of the molecule around the double bond. If hydrogens are present in the same plane, it is referred to as a cis fat; if the hydrogen atoms are on two different planes, it is referred to as a trans fat. The cis double bond causes a bend or a "kink" that prevents the fatty acids from packing tightly, keeping them liquid at room temperature.

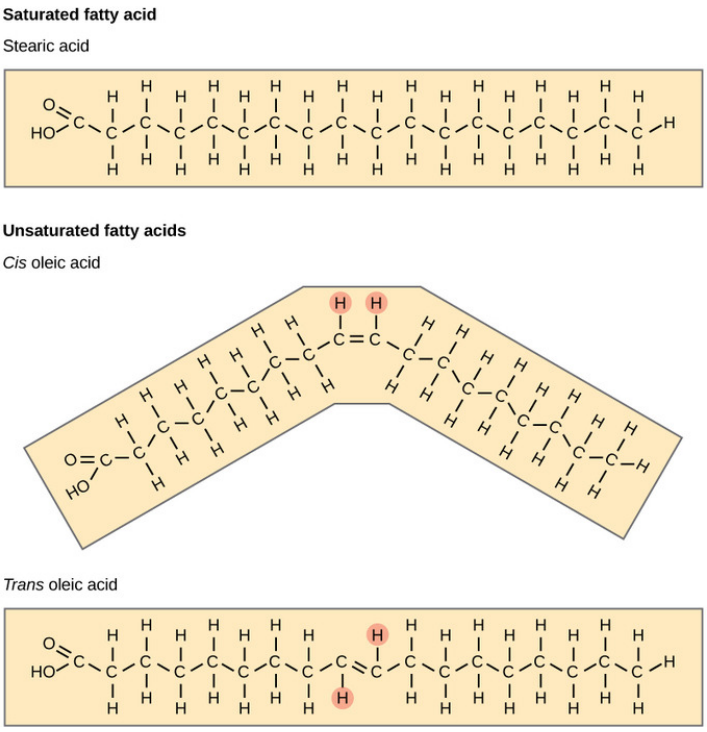


Figure 3.2*A*.1: **Fatty Acids**: Saturated fatty acids have hydrocarbon chains connected by single bonds only. Unsaturated fatty acids have one or more double bonds. Each double bond may be in a cis or trans configuration. In the cis configuration, both hydrogens are on the same side of the hydrocarbon chain. In the trans configuration, the hydrogens are on opposite sides. A cis double bond causes a kink in the chain.

Trans Fats

In the food industry, oils are artificially hydrogenated to make them semi-solid and of a consistency desirable for many processed food products. During this hydrogenation process, gas is bubbled through oils to solidify them, and the double bonds of the cis-conformation in the hydrocarbon chain may be converted to double bonds in the trans-conformation.

Margarine, some types of peanut butter, and shortening are examples of artificially-hydrogenated trans fats. Recent studies have shown that an increase in trans fats in the human diet may lead to an increase in levels of low-density lipoproteins (LDL), or “bad” cholesterol, which in turn may lead to plaque deposition in the arteries, resulting in heart disease. Many fast food restaurants have recently banned the use of trans fats, and food labels are required to display the trans fat content.

Essential Fatty Acids

Essential fatty acids are fatty acids required for biological processes, but not synthesized by the human body. Consequently, they have to be supplemented through ingestion via the diet and are nutritionally very important. Omega-3 fatty acid, or alpha-linoleic acid (ALA), falls into this category and is one of only two fatty acids known to be essential for humans (the other being omega-6 fatty acid, or linoleic acid). These polyunsaturated fatty acids are called omega-3 because the third carbon from the end of the hydrocarbon chain is connected to its neighboring carbon by a double bond. Salmon, trout, and tuna are good sources of omega-3 fatty acids.

Research indicates that omega-3 fatty acids reduce the risk of sudden death from heart attacks, reduce triglycerides in the blood, lower blood pressure, and prevent thrombosis by inhibiting blood clotting. They also reduce inflammation and may help reduce the risk of some cancers in animals.

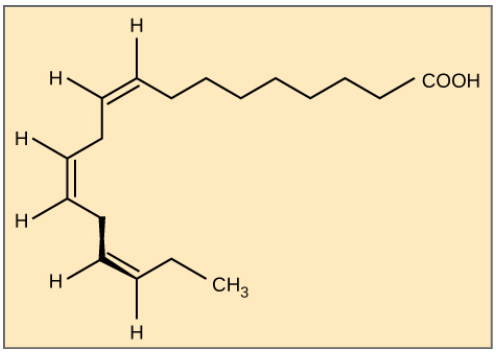


Figure 3.2A.1: **Omega Fatty Acids**: Alpha-linolenic acid is an example of an omega-3 fatty acid. It has three cis double bonds and, as a result, a curved shape. For clarity, the carbons are not shown. Each singly bonded carbon has two hydrogens associated with it, also not shown.

3.2B: Waxes

Waxes are nonpolar lipids that plants and animals use for protection and have many functions in society.

Learning Objectives

- Describe the roles played by waxes

Key Points

- Natural waxes are typically esters of fatty acids and long chain alcohols.
- Animal wax esters are derived from a variety of carboxylic acids and fatty alcohols.
- Plant waxes are derived from mixtures of long-chain hydrocarbons containing functional groups.
- Because of their hydrophobic nature, waxes prevent water from sticking on plants and animals.
- Synthetic waxes are derived from petroleum or polyethylene and consist of long-chain hydrocarbons that lack functional groups.
- Synthetic and waxes are used in adhesives, cosmetics, food, and many other commercial products.

Key Terms

- **paraffin wax**: A waxy white solid hydrocarbon mixture used to make candles, wax paper, lubricants, and sealing materials.
- **polyethylene**: A polymer consisting of many ethylene monomers bonded together; used for kitchenware, containers etc.

Waxes

Waxes are a type of long chain nonpolar lipid. Natural waxes are typically esters of fatty acids and long chain alcohols. Waxes are synthesized by many animals and plants. Animal wax esters are typically derived from a variety of carboxylic acids and fatty alcohols. The composition of a wax depends not only on the species, but also on the geographic location of the organism. The best known animal wax is beeswax, but other insects secrete waxes as well. A major component of beeswax is the ester myricyl palmitate, which bees use for constructing honeycombs. Spermaceti is also a wax that occurs in large amounts in the oil of a sperm whale's head. One of its main constituents is cetyl palmitate, an ester of a fatty acid and fatty alcohol. Plant waxes are derived from mixtures of long-chain hydrocarbons containing functional groups such as alkanes, fatty acids, alcohols, diols, ketones, and aldehydes. Plants also use waxes as a protective coating to control evaporation and hydration and to prevent them from drying out. Waxes are valuable to both plants and animals because of their hydrophobic nature. This makes them water resistant, which prevents water from sticking on surfaces.

Figure $3.2B.1$: **Plant Waxes**: Waxy coverings on some leaves are used as protective coatings.

Unlike most natural waxes, which are esters, synthetic waxes consist of long-chain hydrocarbons lacking functional groups. Paraffin wax is a type of synthetic wax derived from petroleum and refined by vacuum distillation. Synthetic waxes may also be obtained from polyethylene. Millions of of these waxes are produced annually, and they are used in adhesives, cosmetics, sealants and lubricants, insecticides, and UV protection. They are also used in foods like chewing gum.

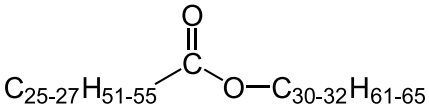


Figure 3.2B.1: **Generic structure formula of bee waxes**: Ester myricyl palmitate is a major component of beeswax.

3.2C: Phospholipids

Phospholipids are amphipathic molecules that make up the bilayer of the plasma membrane and keep the membrane fluid.

Learning Objectives

- Describe phospholipids and their role in cells

Key Points

- Phospholipids consist of a glycerol molecule, two fatty acids, and a phosphate group that is modified by an alcohol.
- The phosphate group is the negatively-charged polar head, which is hydrophilic.
- The fatty acid chains are the uncharged, nonpolar tails, which are hydrophobic.
- Since the tails are hydrophobic, they face the inside, away from the water and meet in the inner region of the membrane.
- Since the heads are hydrophilic, they face outward and are attracted to the intracellular and extracellular fluid.
- If phospholipids are placed in water, they form into micelles, which are lipid molecules that arrange themselves in a spherical form in aqueous solutions.

Key Terms

- **micelle**: Lipid molecules that arrange themselves in a spherical form in aqueous solutions.
- **amphipathic**: Describing a molecule, such as a detergent, which has both hydrophobic and hydrophilic groups.

Defining Characteristics of Phospholipids

Phospholipids are major components of the plasma membrane, the outermost layer of animal cells. Like fats, they are composed of fatty acid chains attached to a glycerol backbone. Unlike triglycerides, which have three fatty acids, phospholipids have two fatty acids that help form a diacylglycerol. The third carbon of the glycerol backbone is also occupied by a modified phosphate group. However, just a phosphate group attached to a diacylglycerol does not qualify as a phospholipid. This would be considered a phosphatidate (diacylglycerol 3-phosphate), the precursor to phospholipids. To qualify as a phospholipid, the phosphate group should be modified by an alcohol. Phosphatidylcholine and phosphatidylserine are examples of two important phospholipids that are found in plasma membranes.

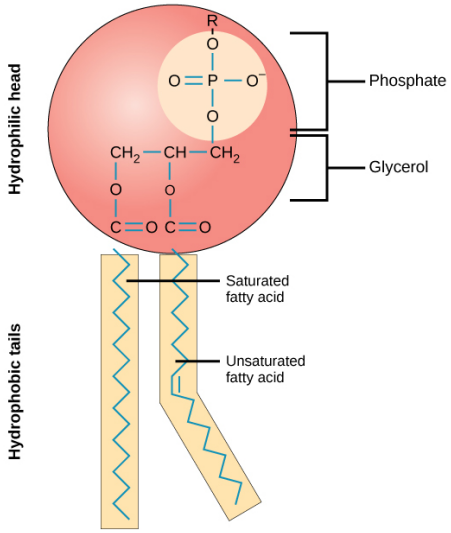


Figure 3.2C.1: **Phospholipid Molecule**: A phospholipid is a molecule with two fatty acids and a modified phosphate group attached to a glycerol backbone. The phosphate may be modified by the addition of charged or polar chemical groups. Two chemical groups that may modify the phosphate, choline and serine, are shown here. Both choline and serine attach to the phosphate group at the position labeled R via the hydroxyl group indicated in green.

Structure of a Phospholipid Molecule

A phospholipid is an amphipathic molecule which means it has both a hydrophobic and a hydrophilic component. A single phospholipid molecule has a phosphate group on one end, called the “head,” and two side-by-side chains of fatty acids that make up the lipid “tails. ” The phosphate group is negatively charged, making the head polar and hydrophilic, or “water loving.” The phosphate heads are thus attracted to the water molecules in their environment.

The lipid tails, on the other hand, are uncharged, nonpolar, and hydrophobic, or “water fearing.” A hydrophobic molecule repels and is repelled by water. Some lipid tails consist of saturated fatty acids and some contain unsaturated fatty acids. This combination adds to the fluidity of the tails that are constantly in motion.

Phospholipids and Biological Membranes

The cell membrane consists of two adjacent layers of phospholipids, which form a bilayer. The fatty acid tails of phospholipids face inside, away from water, whereas the phosphate heads face the outward aqueous side. Since the heads face outward, one layer is exposed to the interior of the cell and one layer is exposed to the exterior. As the phosphate groups are polar and hydrophilic, they are attracted to water in the intracellular fluid.

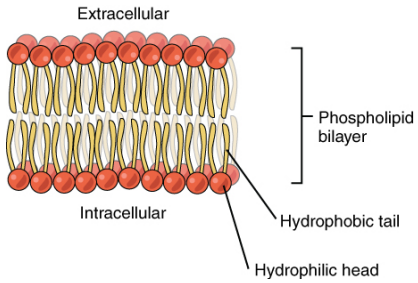


Figure $3.2C.1$: **Phospholipid Bilayer**: The phospholipid bilayer consists of two adjacent sheets of phospholipids, arranged tail to tail. The hydrophobic tails associate with one another, forming the interior of the membrane. The polar heads contact the fluid inside and outside of the cell.

Because of the phospholipds’ chemical and physical characteristics, the lipid bilayer acts as a semipermeable membrane; only lipophilic solutes can easily pass the phospholipd bilayer. As a result, there are two distinct aqueous compartments on each side of the membrane. This separation is essential for many biological functions, including cell communication and metabolism.

Membrane Fluidity

A cell’s plasma membrane contain proteins and other lipids (such as cholesterol) within the phospholipid bilayer. Biological membranes remain fluid because of the unsaturated hydrophobic tails, which prevent phospholipid molecules from packing together and forming a solid.

If a drop of phospholipids is placed in water, the phospholipids spontaneously form a structure known as a micelle, with their hydrophilic heads oriented toward the water. Micelles are lipid molecules that arrange themselves in a spherical form in aqueous solution. The formation of a micelle is a response to the amphipathic nature of fatty acids, meaning that they contain both hydrophilic and hydrophobic regions.

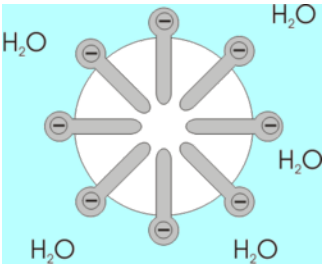


Figure $3.2C.1$: **Micelles**: An example of micelles in water.

3.2D: Steroids

Steroids, like cholesterol, play roles in reproduction, absorption, metabolism regulation, and brain activity.

Learning Objectives

- Describe some functions of steroids

Key Points

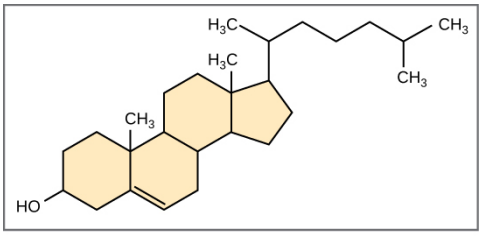
- Steroids are lipids because they are hydrophobic and insoluble in water, but they do not resemble lipids since they have a structure composed of four fused rings.
- Cholesterol is the most common steroid and is the precursor to vitamin D, testosterone, estrogen, progesterone, aldosterone, cortisol, and bile salts.
- Cholesterol is a component of the phospholipid bilayer and plays a role in the structure and function of membranes.
- Steroids are found in the brain and alter electrical activity in the brain.
- Because they can tone down receptors that communicate messages from neurotransmitters, steroids are often used in anesthetic medicines.

Key Terms

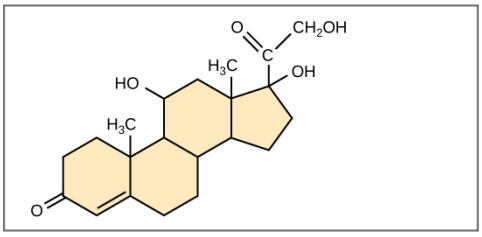
- **neurotransmitter**: any substance, such as acetylcholine or dopamine, responsible for sending nerve signals across a synapse between two neurons
- **osmoregulation**: the homeostatic regulation of osmotic pressure in the body in order to maintain a constant water content
- **hormone**: any substance produced by one tissue and conveyed by the bloodstream to another to affect physiological activity

Structure of Steroid Molecules

Unlike phospholipids and fats, steroids have a fused ring structure. Although they do not resemble the other lipids, they are grouped with them because they are also hydrophobic and insoluble in water. All steroids have four linked carbon rings, and many of them, like cholesterol, have a short tail. Many steroids also have the –OH functional group, and these steroids are classified as alcohols called sterols.



Cholesterol



Cortisol

Figure 3.2D.1: **Steroid Structures**: Steroids, such as cholesterol and cortisol, are composed of four fused hydrocarbon rings.

Cholesterol

Cholesterol is the most common steroid and is mainly synthesized in the liver; it is the precursor to vitamin D. Cholesterol is also a precursor to many important steroid hormones like estrogen, testosterone, and progesterone, which are secreted by the gonads and endocrine glands. Therefore, steroids play very important roles in the body's reproductive system. Cholesterol also plays a role in synthesizing the steroid hormones aldosterone, which is used for osmoregulation, and cortisol, which plays a role in metabolism.

Cholesterol is also the precursor to bile salts, which help in the emulsification of fats and their absorption by cells. It is a component of the plasma membrane of animal cells and the phospholipid bilayer. Being the outermost structure in animal cells, the plasma membrane is responsible for the transport of materials and cellular recognition; and it is involved in cell-to-cell communication. Thus, steroids also play an important role in the structure and function of membranes.

It has also been discovered that steroids can be active in the brain where they affect the nervous system, These neurosteroids alter electrical activity in the brain. They can either activate or tone down receptors that communicate messages from neurotransmitters. Since these neurosteroids can tone down receptors and decrease brain activity, steroids are often used in anesthetic medicines.

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3.3: Proteins

Topic hierarchy

3.3A: Types and Functions of Proteins

3.3B: Amino Acids

An amino acid contains an amino group, a carboxyl group, and an R group, and it combines with other amino acids to form polypeptide chains.

3.3C: Protein Structure

3.3D: Denaturation and Protein Folding

Denaturation is a process in which proteins lose their shape and, therefore, their function because of changes in pH or temperature.

: Types and Functions of Proteins

Proteins perform many essential physiological functions, including catalyzing biochemical reactions.

Learning Objectives

- Differentiate among the types and functions of proteins

Key Points

- Proteins are essential for the main physiological processes of life and perform functions in every system of the human body.
- A protein's shape determines its function.
- Proteins are composed of amino acid subunits that form polypeptide chains.
- Enzymes catalyze biochemical reactions by speeding up chemical reactions, and can either break down their substrate or build larger molecules from their substrate.
- The shape of an enzyme's active site matches the shape of the substrate.
- Hormones are a type of protein used for cell signaling and communication.

Key Terms

- **amino acid**: Any of 20 naturally occurring α-amino acids (having the amino, and carboxylic acid groups on the same carbon atom), and a variety of side chains, that combine, via peptide bonds, to form proteins.
- **polypeptide**: Any polymer of (same or different) amino acids joined via peptide bonds.
- **catalyze**: To accelerate a process.

Types and Functions of Proteins

Proteins perform essential functions throughout the systems of the human body. These long chains of amino acids are critically important for:

- catalyzing chemical reactions
- synthesizing and repairing DNA
- transporting materials across the cell
- receiving and sending chemical signals
- responding to stimuli
- providing structural support

Proteins (a polymer) are macromolecules composed of amino acid subunits (the monomers). These amino acids are covalently attached to one another to form long linear chains called polypeptides, which then fold into a specific three-dimensional shape. Sometimes these folded polypeptide chains are functional by themselves. Other times they combine with additional polypeptide chains to form the final protein structure. Sometimes non-polypeptide groups are also required in the final protein. For instance, the blood protein hemogobin is made up of four polypeptide chains, each of which also contains a heme molecule, which is ring structure with an iron atom in its center.

Proteins have different shapes and molecular weights, depending on the amino acid sequence. For example, hemoglobin is a globular protein, which means it folds into a compact globe-like structure, but collagen, found in our skin, is a fibrous protein, which means it folds into a long extended fiber-like chain. You probably look similar to your family members because you share similar proteins, but you look different from strangers because the proteins in your eyes, hair, and the rest of your body are different.

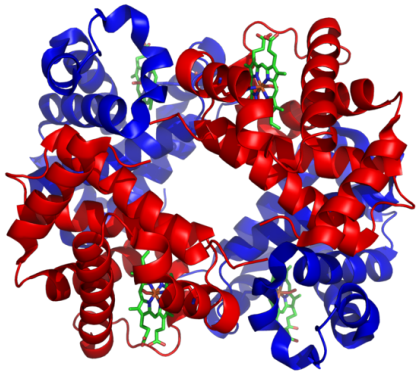

Figure 3.3A.1: **Human Hemoglobin**: Structure of human hemoglobin. The proteins' α and β subunits are in red and blue, and the iron-containing heme groups in green. From the protein data base.

Because form determines function, any slight change to a protein's shape may cause the protein to become dysfunctional. Small changes in the amino acid sequence of a protein can cause devastating genetic diseases such as Huntington's disease or sickle cell anemia.

Enzymes

Enzymes are proteins that catalyze biochemical reactions, which otherwise would not take place. These enzymes are essential for chemical processes like digestion and cellular metabolism. Without enzymes, most physiological processes would proceed so slowly (or not at all) that life could not exist.

Because form determines function, each enzyme is specific to its substrates. The substrates are the reactants that undergo the chemical reaction catalyzed by the enzyme. The location where substrates bind to or interact with the enzyme is known as the active site, because that is the site where the chemistry occurs. When the substrate binds to its active site at the enzyme, the enzyme may help in its breakdown, rearrangement, or synthesis. By placing the substrate into a specific shape and microenvironment in the active site, the enzyme encourages the chemical reaction to occur. There are two basic classes of enzymes:

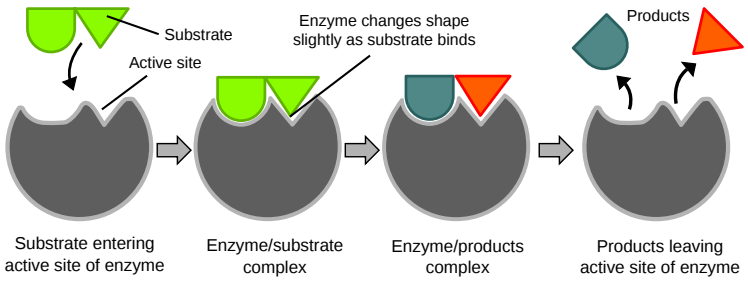

Figure 3.3A.1: **Enzyme reaction**: A catabolic enzyme reaction showing the substrate matching the exact shape of the active site.

- Catabolic enzymes: enzymes that break down their substrate
- Anabolic enzymes: enzymes that build more complex molecules from their substrates

Enzymes are essential for digestion: the process of breaking larger food molecules down into subunits small enough to diffuse through a cell membrane and to be used by the cell. These enzymes include amylase, which catalyzes the digestion carbohydrates in the mouth and small intestine; pepsin, which catalyzes the digestion of proteins in the stomach; lipase, which catalyzes reactions need to emulsify fats in the small intestine; and trypsin, which catalyzes the further digestion of proteins in the small intestine.

Enzymes are also essential for biosynthesis: the process of making new, complex molecules from the smaller subunits that are provided to or generated by the cell. These biosynthetic enzymes include DNA Polymerase, which catalyzes the synthesis of new strands of the genetic material before cell division; fatty acid synthetase, which the synthesis of new fatty

for fat or membrane lipid formation; and components of the ribosome, which catalyzes the formation of new ptides from amino acid monomers.

Hormones

Some proteins function as chemical-signaling molecules called hormones. These proteins are secreted by endocrine cells that act to control or regulate specific physiological processes, which include growth, development, metabolism, and reproduction. For example, insulin is a protein hormone that helps to regulate blood glucose levels. Other proteins act as receptors to detect the concentrations of chemicals and send signals to respond. Some types of hormones, such as estrogen and testosterone, are lipid steroids, not proteins.

Other Protein Functions

Proteins perform essential functions throughout the systems of the human body. In the respiratory system, hemoglobin (composed of four protein subunits) transports oxygen for use in cellular metabolism. Additional proteins in the blood plasma and lymph carry nutrients and metabolic waste products throughout the body. The proteins actin and tubulin form cellular structures, while keratin forms the structural support for the dead cells that become fingernails and hair. Antibodies, also called immunoglobins, help recognize and destroy foreign pathogens in the immune system. Actin and myosin allow muscles to contract, while albumin nourishes the early development of an embryo or a seedling.

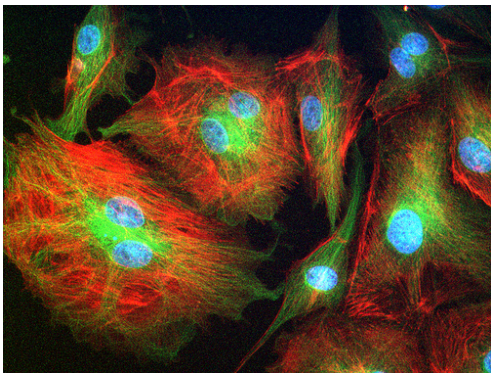

Figure 3.3.A.1: **Tubulin**: The structural protein tubulin stained red in mouse cells.

3.3B: Amino Acids

Learning Objectives

- Describe the structure of an amino acid and the features that confer its specific properties

Structure of an Amino Acid

Amino acids are the monomers that make up proteins. Each amino acid has the same fundamental structure, which consists of a central carbon atom, also known as the alpha (α) carbon, bonded to an amino group (NH_2), a carboxyl group (COOH), and to a hydrogen atom. In the aqueous environment of the cell, the both the amino group and the carboxyl group are ionized under physiological conditions, and so have the structures $-NH_3^+$ and $-COO^-$, respectively. Every amino acid also has another atom or group of atoms bonded to the central atom known as the R group. This R group, or side chain, gives each amino acid proteins specific characteristics, including size, polarity, and pH.

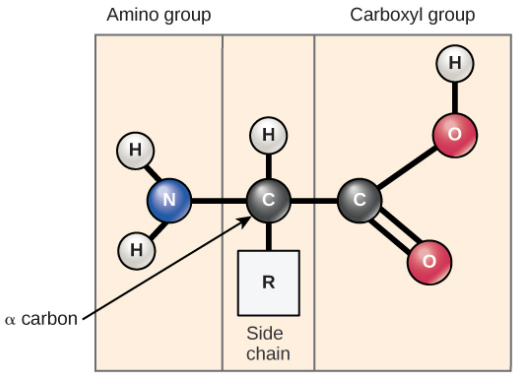

Figure 3.3B.1: **Amino acid structure**: Amino acids have a central asymmetric carbon to which an amino group, a carboxyl group, a hydrogen atom, and a side chain (R group) are attached. This amino acid is unionized, but if it were placed in water at pH 7, its amino group would pick up another hydrogen and a positive charge, and the hydroxyl in its carboxyl group would lose and a hydrogen and gain a negative charge.

Types of Amino Acids

The name “amino acid” is derived from the amino group and carboxyl-acid-group in their basic structure. There are 21 amino acids present in proteins, each with a specific R group or side chain. Ten of these are considered essential amino acids in humans because the human body cannot produce them and they must be obtained from the diet. All organisms have different essential amino acids based on their physiology.

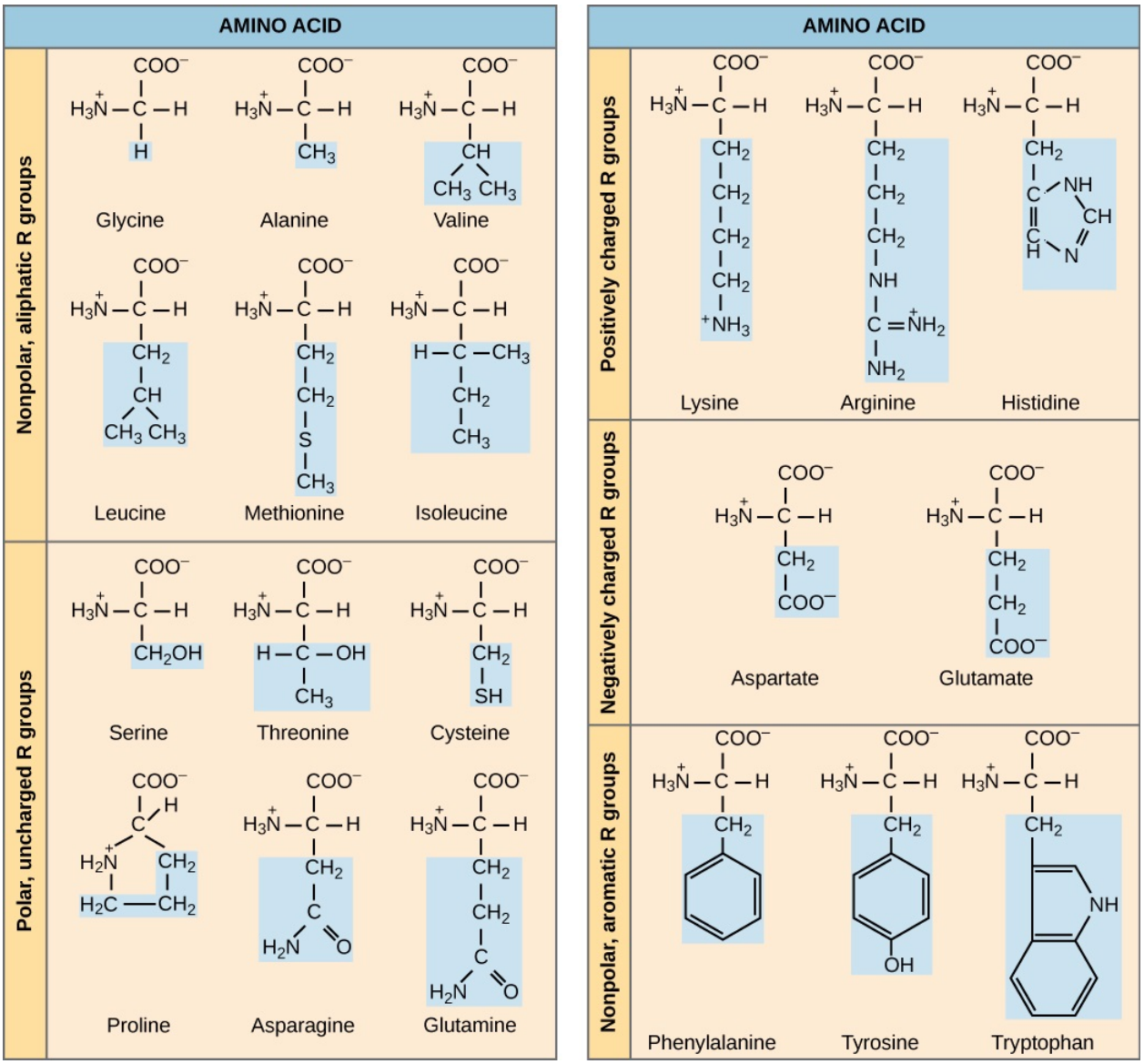

Figure 3.3*B*.1: **Types of amino acids**: There are 21 common amino acids commonly found in proteins, each with a different R group (variant group) that determines its chemical nature. The 21st amino acid, not shown here, is selenocysteine, with an R group of -CH_2-SeH.

Characteristics of Amino Acids

Which categories of amino acid would you expect to find on the surface of a soluble protein, and which would you expect to find in the interior? What distribution of amino acids would you expect to find in a protein embedded in a lipid bilayer?

The chemical composition of the side chain determines the characteristics of the amino acid. Amino acids such as valine, methionine, and alanine are nonpolar (hydrophobic), while amino acids such as serine, threonine, and cysteine are polar (hydrophilic). The side chains of lysine and arginine are positively charged so these amino acids are also known as basic (high pH) amino acids. Proline is an exception to the standard structure of an amino acid because its R group is linked to the amino group, forming a ring-like structure.

Amino acids are represented by a single upper case letter or a three-letter abbreviation. For example, valine is known by the letter V or the three-letter symbol val.

Peptide Bonds

The sequence and the number of amino acids ultimately determine the protein's shape, size, and function. Each amino acid is attached to another amino acid by a covalent bond, known as a peptide bond. When two amino acids are covalently attached by a peptide bond, the carboxyl group of one amino acid and the amino group of the incoming amino acid combine and release a molecule of water. Any reaction that combines two monomers in a reaction that generates H_2O as one of the products is known as a dehydration reaction, so peptide bond formation is an example of a dehydration reaction.

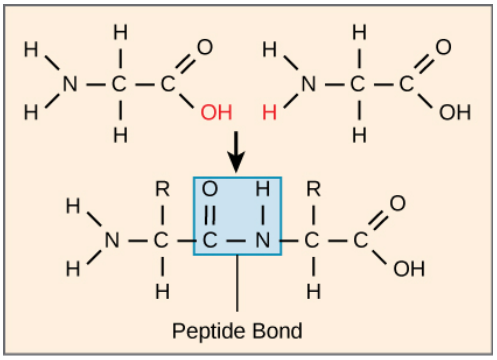

Figure 3.3B.1: Peptide bond formation: Peptide bond formation is a dehydration synthesis reaction. The carboxyl group of one amino acid is linked to the amino group of the incoming amino acid. In the process, a molecule of water is released.

Polypeptide Chains

The resulting chain of amino acids is called a polypeptide chain. Each polypeptide has a free amino group at one end. This end is called the N terminal, or the amino terminal, and the other end has a free carboxyl group, also known as the C or carboxyl terminal. When reading or reporting the amino acid sequence of a protein or polypeptide, the convention is to use the N-to-C direction. That is, the first amino acid in the sequence is assumed to the be one at the N terminal and the last amino acid is assumed to be the one at the C terminal.

Although the terms polypeptide and protein are sometimes used interchangeably, a polypeptide is technically any polymer of amino acids, whereas the term protein is used for a polypeptide or polypeptides that have folded properly, combined with any additional components needed for proper functioning, and is now functional.

Key Points

- Each amino acid contains a central C atom, an amino group (NH2), a carboxyl group (COOH), and a specific R group.
- The R group determines the characteristics (size, polarity, and pH) for each type of amino acid.
- Peptide bonds form between the carboxyl group of one amino acid and the amino group of another through dehydration synthesis.
- A chain of amino acids is a polypeptide.

Key Terms

- **amino acid**: Any of 20 naturally occurring α-amino acids (having the amino, and carboxylic acid groups on the same carbon atom), and a variety of side chains, that combine, via peptide bonds, to form proteins.
- **R group**: The R group is a side chain specific to each amino acid that confers particular chemical properties to that amino acid.
- **polypeptide**: Any polymer of (same or different) amino acids joined via peptide bonds.

3.3C: Protein Structure

Each successive level of protein folding ultimately contributes to its shape and therefore its function.

Learning Objectives

- Summarize the four levels of protein structure

Key Points

- Protein structure depends on its amino acid sequence and local, low-energy chemical bonds between atoms in both the polypeptide backbone and in amino acid side chains.
- Protein structure plays a key role in its function; if a protein loses its shape at any structural level, it may no longer be functional.
- Primary structure is the amino acid sequence.
- Secondary structure is local interactions between stretches of a polypeptide chain and includes α-helix and β-pleated sheet structures.
- Tertiary structure is the overall the three-dimension folding driven largely by interactions between R groups.
- Quarternary structures is the orientation and arrangement of subunits in a multi-subunit protein.

Key Terms

- **antiparallel**: The nature of the opposite orientations of the two strands of DNA or two beta strands that comprise a protein's secondary structure
- **disulfide bond**: A bond, consisting of a covalent bond between two sulfur atoms, formed by the reaction of two thiol groups, especially between the thiol groups of two proteins
- **β-pleated sheet**: secondary structure of proteins where N-H groups in the backbone of one fully-extended strand establish hydrogen bonds with C=O groups in the backbone of an adjacent fully-extended strand
- **α-helix**: secondary structure of proteins where every backbone N-H creates a hydrogen bond with the C=O group of the amino acid four residues earlier in the same helix.

The shape of a protein is critical to its function because it determines whether the protein can interact with other molecules. Protein structures are very complex, and researchers have only very recently been able to easily and quickly determine the structure of complete proteins down to the atomic level. (The techniques used date back to the 1950s, but until recently they were very slow and laborious to use, so complete protein structures were very slow to be solved.) Early structural biochemists conceptually divided protein structures into four "levels" to make it easier to get a handle on the complexity of the overall structures. To determine how the protein gets its final shape or conformation, we need to understand these four levels of protein structure: primary, secondary, tertiary, and quaternary.

Primary Structure

A protein's primary structure is the unique sequence of amino acids in each polypeptide chain that makes up the protein. Really, this is just a list of which amino acids appear in which order in a polypeptide chain, not really a structure. But, because the final protein structure ultimately depends on this sequence, this was called the primary structure of the polypeptide chain. For example, the pancreatic hormone insulin has two polypeptide chains, A and B.

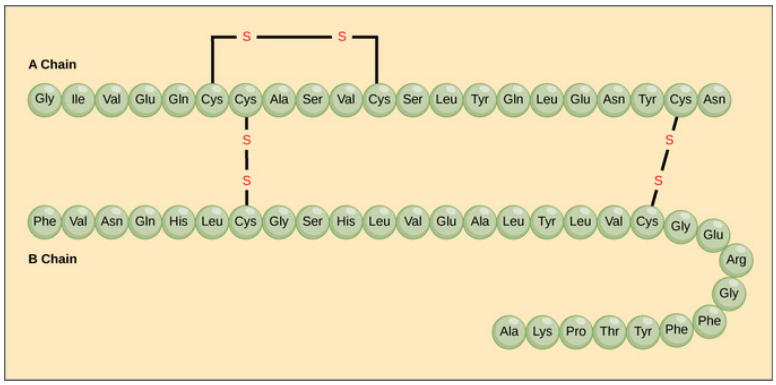

Figure 3.3C.1: **Primary structure**: The A chain of insulin is 21 amino acids long and the B chain is 30 amino acids long, and each sequence is unique to the insulin protein.

The gene, or sequence of DNA, ultimately determines the unique sequence of amino acids in each peptide chain. A change in nucleotide sequence of the gene's coding region may lead to a different amino acid being added to the growing polypeptide chain, causing a change in protein structure and therefore function.

The oxygen-transport protein hemoglobin consists of four polypeptide chains, two identical α chains and two identical β chains. In sickle cell anemia, a single amino substitution in the hemoglobin β chain causes a change the structure of the entire protein. When the amino acid glutamic acid is replaced by valine in the β chain, the polypeptide folds into an slightly-different shape that creates a dysfunctional hemoglobin protein. So, just one amino acid substitution can cause dramatic changes. These dysfunctional hemoglobin proteins, under low-oxygen conditions, start associating with one another, forming long fibers made from millions of aggregated hemoglobins that distort the red blood cells into crescent or "sickle" shapes, which clog arteries. People affected by the disease often experience breathlessness, dizziness, headaches, and abdominal pain.

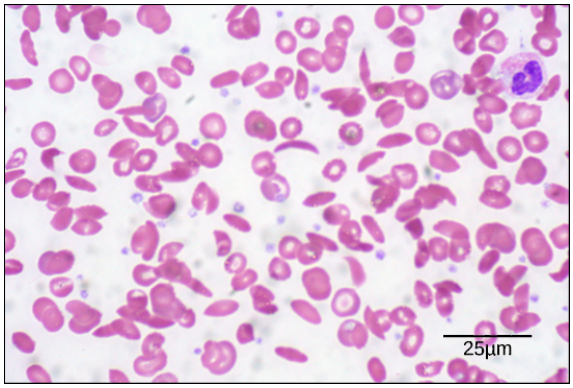

Figure 3.3C.1: **Sickle cell disease**: Sickle cells are crescent shaped, while normal cells are disc-shaped.

Secondary Structure

A protein's secondary structure is whatever regular structures arise from interactions between neighboring or near-by amino acids as the polypeptide starts to fold into its functional three-dimensional form. Secondary structures arise as H bonds form between local groups of amino acids in a region of the polypeptide chain. Rarely does a single secondary structure extend throughout the polypeptide chain. It is usually just in a section of the chain. The most common forms of secondary structure are the α-helix and β-pleated sheet structures and they play an important structural role in most globular and fibrous proteins.

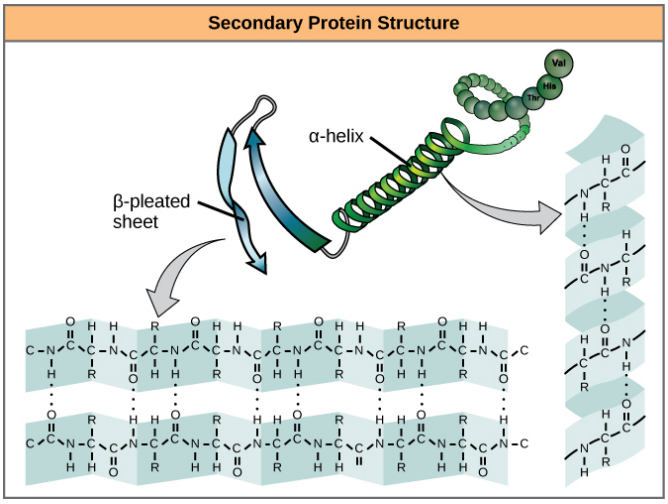

Figure 3.3C.1: **Secondary structure**: The α-helix and β-pleated sheet form because of hydrogen bonding between carbonyl and amino groups in the peptide backbone. Certain amino acids have a propensity to form an α-helix, while others have a propensity to form a β-pleated sheet.

In the α-helix chain, the hydrogen bond forms between the oxygen atom in the polypeptide backbone carbonyl group in one amino acid and the hydrogen atom in the polypeptide backbone amino group of another amino acid that is four amino acids farther along the chain. This holds the stretch of amino acids in a right-handed coil. Every helical turn in an alpha helix has 3.6 amino acid residues. The R groups (the side chains) of the polypeptide protrude out from the α-helix chain and are not involved in the H bonds that maintain the α-helix structure.

In β-pleated sheets, stretches of amino acids are held in an almost fully-extended conformation that "pleats" or zig-zags due to the non-linear nature of single C-C and C-N covalent bonds. β-pleated sheets never occur alone. They have to held in place by other β-pleated sheets. The stretches of amino acids in β-pleated sheets are held in their pleated sheet structure because hydrogen bonds form between the oxygen atom in a polypeptide backbone carbonyl group of one β-pleated sheet and the hydrogen atom in a polypeptide backbone amino group of another β-pleated sheet. The β-pleated sheets which hold each other together align parallel or antiparallel to each other. The R groups of the amino acids in a β-pleated sheet point out perpendicular to the hydrogen bonds holding the β-pleated sheets together, and are not involved in maintaining the β-pleated sheet structure.

Tertiary Structure

The tertiary structure of a polypeptide chain is its overall three-dimensional shape, once all the secondary structure elements have folded together among each other. Interactions between polar, nonpolar, acidic, and basic R group within the polypeptide chain create the complex three-dimensional tertiary structure of a protein. When protein folding takes place in the aqueous environment of the body, the hydrophobic R groups of nonpolar amino acids mostly lie in the interior of the protein, while the hydrophilic R groups lie mostly on the outside. Cysteine side chains form disulfide linkages in the presence of oxygen, the only covalent bond forming during protein folding. All of these interactions, weak and strong, determine the final three-dimensional shape of the protein. When a protein loses its three-dimensional shape, it will no longer be functional.

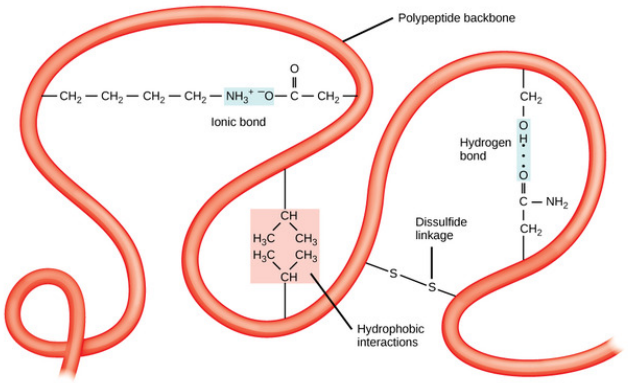

Figure $3.3C.1$: **Tertiary structure**: The tertiary structure of proteins is determined by hydrophobic interactions, ionic bonding, hydrogen bonding, and disulfide linkages.

Quaternary Structure

The quaternary structure of a protein is how its subunits are oriented and arranged with respect to one another. As a result, quaternary structure only applies to multi-subunit proteins; that is, proteins made from more than one polypeptide chain. Proteins made from a single polypeptide will not have a quaternary structure.

In proteins with more than one subunit, weak interactions between the subunits help to stabilize the overall structure. Enzymes often play key roles in bonding subunits to form the final, functioning protein.

For example, insulin is a ball-shaped, globular protein that contains both hydrogen bonds and disulfide bonds that hold its two polypeptide chains together. Silk is a fibrous protein that results from hydrogen bonding between different β-pleated chains.

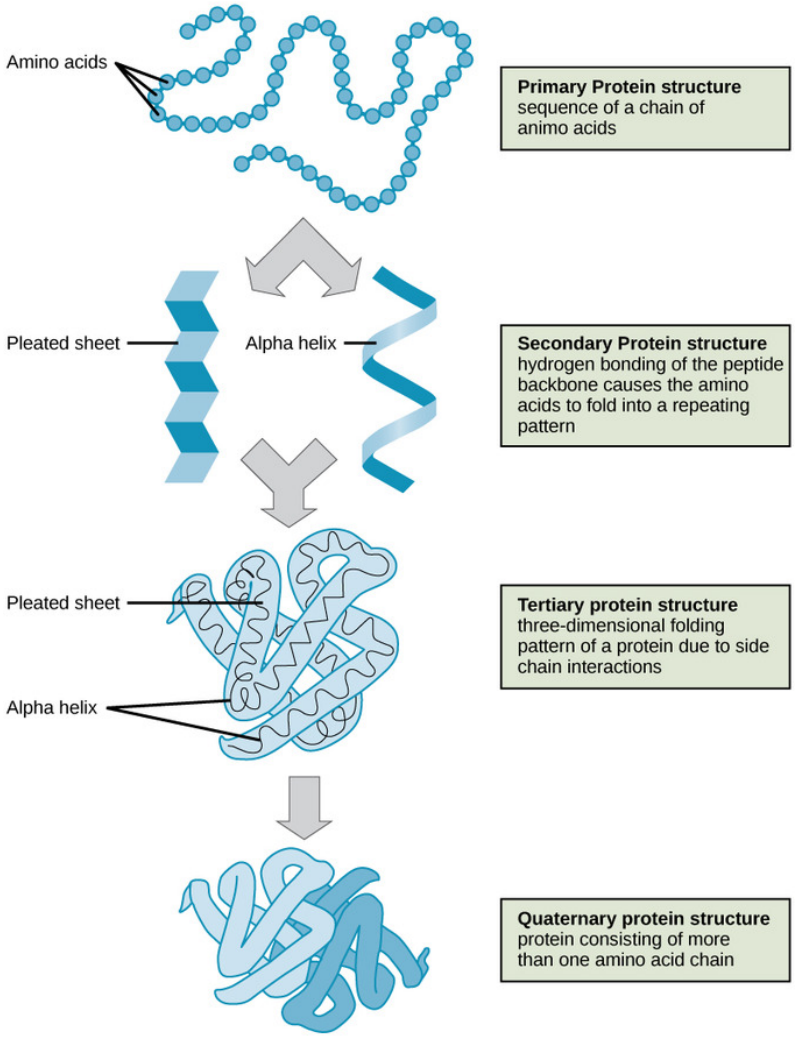

Figure $3.3C.1$: **Four levels of protein structure**: The four levels of protein structure can be observed in these illustrations.

3.3D: Denaturation and Protein Folding

LEARNING OBJECTIVES

- Discuss the process of protein denaturation

Each protein has its own unique sequence of amino acids and the interactions between these amino acids create a specify shape. This shape determines the protein's function, from digesting protein in the stomach to carrying oxygen in the blood.

Changing the Shape of a Protein

If the protein is subject to changes in temperature, pH, or exposure to chemicals, the internal interactions between the protein's amino acids can be altered, which in turn may alter the shape of the protein. Although the amino acid sequence (also known as the protein's primary structure) does not change, the protein's shape may change so much that it becomes dysfunctional, in which case the protein is considered denatured. Pepsin, the enzyme that breaks down protein in the stomach, only operates at a very low pH. At higher pHs pepsin's conformation, the way its polypeptide chain is folded up in three dimensions, begins to change. The stomach maintains a very low pH to ensure that pepsin continues to digest protein and does not denature.

Because almost all biochemical reactions require enzymes, and because almost all enzymes only work optimally within relatively narrow temperature and pH ranges, many homeostatic mechanisms regulate appropriate temperatures and pH so that the enzymes can maintain the shape of their active site.

Reversing Denaturation

It is often possible to reverse denaturation because the primary structure of the polypeptide, the covalent bonds holding the amino acids in their correct sequence, is intact. Once the denaturing agent is removed, the original interactions between amino acids return the protein to its original conformation and it can resume its function. However, denaturation can be irreversible in extreme situations, like frying an egg. The heat from a pan denatures the albumin protein in the liquid egg white and it becomes insoluble. The protein in meat also denatures and becomes firm when cooked.

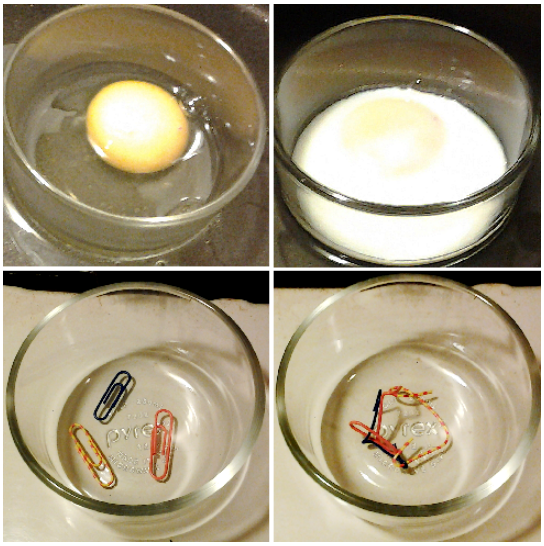

Figure $3.3D.1$: **Denaturing a protein is occasionally irreversible**: (Top) The protein albumin in raw and cooked egg white. (Bottom) A paperclip analogy visualizes the process: when cross-linked, paperclips ('amino acids') no longer move freely; their structure is rearranged and 'denatured'.

Chaperone proteins (or chaperonins) are helper proteins that provide favorable conditions for protein folding to take place. The chaperonins clump around the forming protein and prevent other polypeptide chains from aggregating. Once the target protein folds, the chaperonins disassociate.

Key Points

- Proteins change their shape when exposed to different pH or temperatures.
- The body strictly regulates pH and temperature to prevent proteins such as enzymes from denaturing.
- Some proteins can refold after denaturation while others cannot.
- Chaperone proteins help some proteins fold into the correct shape.

Key Terms

- **chaperonin**: proteins that provide favorable conditions for the correct folding of other proteins, thus preventing aggregation
- **denaturation**: the change of folding structure of a protein (and thus of physical properties) caused by heating, changes in pH, or exposure to certain chemicals

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3.4: Nucleic Acids

Topic hierarchy

- 3.4A: DNA and RNA

DNA and RNA are nucleic acids that carry out cellular processes, especially the regulation and expression of genes.

- 3.4B: The DNA Double Helix
- 3.4C: DNA Packaging
- 3.4D: Types of RNA

3.4A: DNA and RNA

Learning Objectives

- Describe the structure of nucleic acids and the types of molecules that contain them

Types of Nucleic Acids

The two main types of nucleic acids are deoxyribonucleic acid (DNA) and ribonucleic acid (RNA). DNA is the genetic material found in all living organisms, ranging from single-celled bacteria to multicellular mammals. It is found in the nucleus of eukaryotes and in the chloroplasts and mitochondria. In prokaryotes, the DNA is not enclosed in a membranous envelope, but rather free-floating within the cytoplasm.

The entire genetic content of a cell is known as its genome and the study of genomes is genomics. In eukaryotic cells, but not in prokaryotes, DNA forms a complex with histone proteins to form chromatin, the substance of eukaryotic chromosomes. A chromosome may contain tens of thousands of genes. Many genes contain the information to make protein products; other genes code for RNA products. DNA controls all of the cellular activities by turning the genes “on” or “off. ”

The other type of nucleic acid, RNA, is mostly involved in protein synthesis. In eukaryotes, the DNA molecules never leave the nucleus but instead use an intermediary to communicate with the rest of the cell. This intermediary is the messenger RNA (mRNA). Other types of RNA—like rRNA, tRNA, and microRNA—are involved in protein synthesis and its regulation.

Nucleotides

DNA and RNA are made up of monomers known as nucleotides. The nucleotides combine with each other to form a polynucleotide: DNA or RNA. Each nucleotide is made up of three components:

1. a nitrogenous base
2. a pentose (five-carbon) sugar
3. a phosphate group

Each nitrogenous base in a nucleotide is attached to a sugar molecule, which is attached to one or more phosphate groups.

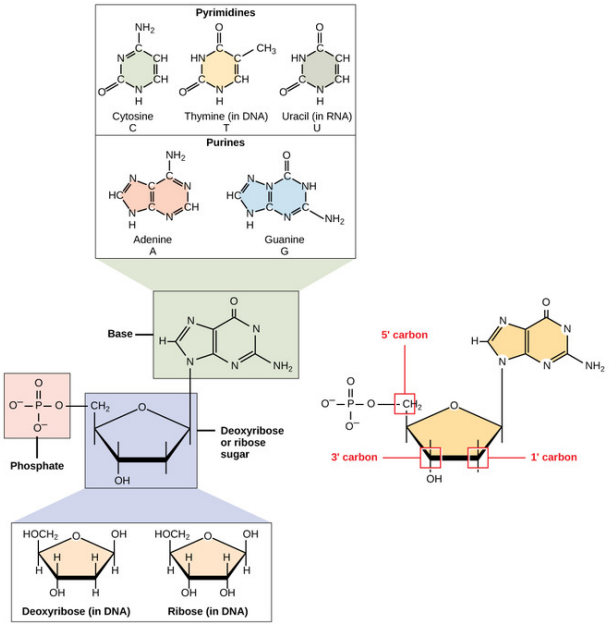

Figure 3.4*A*.1: **DNA and RNA**: A nucleotide is made up of three components: a nitrogenous base, a pentose sugar, and one or more phosphate groups. Carbon residues in the pentose are numbered 1′ through 5′ (the prime distinguishes these residues from those in the base, which are numbered without using a prime notation). The base is attached to the 1′ position of the ribose, and the phosphate is attached to the 5′ position. When a polynucleotide is formed, the 5′ phosphate of the incoming nucleotide attaches to the 3′ hydroxyl group at the end of the growing chain. Two types of pentose are found in nucleotides, deoxyribose (found in DNA) and ribose (found in RNA). Deoxyribose is similar in structure to ribose, but it has an H instead of an OH at the 2′ position. Bases can be divided into two categories: purines and pyrimidines. Purines have a double ring structure, and pyrimidines have a single ring.

Nitrogenous Base

The nitrogenous bases are organic molecules and are so named because they contain carbon and nitrogen. They are bases because they contain an amino group that has the potential of binding an extra hydrogen, and thus, decreasing the hydrogen ion concentration in its environment, making it more basic. Each nucleotide in DNA contains one of four possible nitrogenous bases: adenine (A), guanine (G) cytosine (C), and thymine (T).

Adenine and guanine are classified as purines. The primary structure of a purine consists of two carbon-nitrogen rings. Cytosine, thymine, and uracil are classified as pyrimidines which have a single carbon-nitrogen ring as their primary structure. Each of these basic carbon-nitrogen rings has different functional groups attached to it. In molecular biology shorthand, the nitrogenous bases are simply known by their symbols A, T, G, C, and U. DNA contains A, T, G, and C whereas RNA contains A, U, G, and C.

Five-Carbon Sugar

The pentose sugar in DNA is deoxyribose and in RNA it is ribose. The difference between the sugars is the presence of the hydroxyl group on the second carbon of the ribose and hydrogen on the second carbon of the deoxyribose. The carbon atoms of the sugar molecule are numbered as 1′, 2′, 3′, 4′, and 5′ (1′ is read as “one prime”).

Phosphate Group

The phosphate residue is attached to the hydroxyl group of the 5′ carbon of one sugar and the hydroxyl group of the 3′ carbon of the sugar of the next nucleotide, which forms a 5′3′ phosphodiester linkage. The phosphodiester linkage is not formed by simple dehydration reaction like the other linkages connecting monomers in macromolecules: its formation involves the removal of two phosphate groups. A polynucleotide may have thousands of such phosphodiester linkages.

Key Points

- The two main types of nucleic acids are DNA and RNA.

- Both DNA and RNA are made from nucleotides, each containing a five-carbon sugar backbone, a phosphate group, and a nitrogen base.
- DNA provides the code for the cell ‘s activities, while RNA converts that code into proteins to carry out cellular functions.
- The sequence of nitrogen bases (A, T, C, G) in DNA is what forms an organism’s traits.
- The nitrogen bases A and T (or U in RNA) always go together and C and G always go together, forming the 5′-3′ phosphodiester linkage found in the nucleic acid molecules.

Key Terms

- **nucleotide**: the monomer comprising DNA or RNA molecules; consists of a nitrogenous heterocyclic base that can be a purine or pyrimidine, a five-carbon pentose sugar, and a phosphate group
- **genome**: the cell’s complete genetic information packaged as a double-stranded DNA molecule
- **monomer**: A relatively small molecule which can be covalently bonded to other monomers to form a polymer.

3.4B: The DNA Double Helix

The DNA double helix looks like a twisted staircase, with the sugar and phosphate backbone surrounding complementary nitrogen bases.

Learning Objectives

- Describe the structure of DNA

Key Points

- The structure of DNA is called a double helix, which looks like a twisted staircase.
- The sugar and phosphate make up the backbone, while the nitrogen bases are found in the center and hold the two strands together.
- The nitrogen bases can only pair in a certain way: A pairing with T and C pairing with G. This is called *base pairing*.
- Due to the *base pairing*, the DNA strands are *complementary* to each other, run in opposite directions, and are called antiparallel strands.

Key Terms

- **mutation**: any error in base pairing during the replication of DNA
- **sugar-phosphate backbone**: The outer support of the ladder, forming strong covalent bonds between monomers of DNA.
- **base pairing**: The specific way in which bases of DNA line up and bond to one another; A always with T and G always with C.

A Double-Helix Structure

DNA has a double-helix structure, with sugar and phosphate on the outside of the helix, forming the sugar-phosphate backbone of the DNA. The nitrogenous bases are stacked in the interior in pairs, like the steps of a staircase; the pairs are bound to each other by hydrogen bonds. The two strands of the helix run in opposite directions. This *antiparallel* orientation is important to DNA replication and in many nucleic acid interactions.

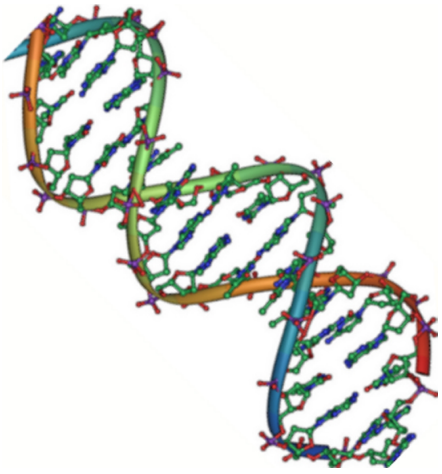

Figure $3.4B.1$: **DNA is a Double Helix**: Native DNA is an antiparallel double helix. The phosphate backbone (indicated by the curvy lines) is on the outside, and the bases are on the inside. Each base from one strand interacts via hydrogen bonding with a base from the opposing strand.

Base Pairs

Only certain types of base pairing are allowed. This means Adenine pairs with Thymine, and Guanine pairs with Cytosine. This is known as the base complementary rule because the DNA strands are complementary to each other. If the sequence of one strand is AATTGGCC, the complementary strand would have the sequence TTAACCGG.

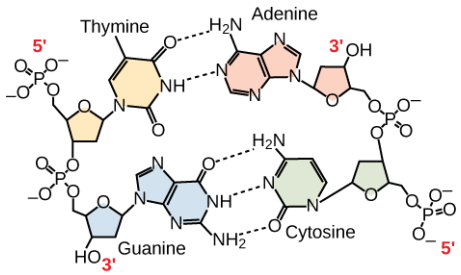

Figure 3.4B.1: **Antiparallel Strands**: In a double stranded DNA molecule, the two strands run antiparallel to one another so one is upside down compared to the other. The phosphate backbone is located on the outside, and the bases are in the middle. Adenine forms hydrogen bonds (or base pairs) with thymine, and guanine base pairs with cytosine.

DNA Replication

During DNA replication, each strand is copied, resulting in a daughter DNA double helix containing one parental DNA strand and a newly synthesized strand. At this time it is possible a mutation may occur. A mutation is a change in the sequence of the nitrogen bases. For example, in the sequence AATTGGCC, a mutation may cause the second T to change to a G. Most of the time when this happens the DNA is able to fix itself and return the original base to the sequence. However, sometimes the repair is unsuccessful, resulting in different proteins being created.

3.4C: DNA Packaging

DNA packaging is an important process in living cells. Without it, a cell is not able to accommodate the large amount of DNA that is stored inside.

Learning Objectives

- Describe how DNA is packaged differently in prokaryotes and eukaryotes

Key Points

- In eukaryotic cells, DNA and RNA synthesis occur in a different location than protein synthesis; in prokaryotic cells, both these processes occur together.
- DNA is “supercoiled” in prokaryotic cells, meaning that the DNA is either under-wound or over-wound from its normal relaxed state.
- In eukaryotic cells, DNA is wrapped around proteins known as histones to form structures called nucleosomes.

Key Terms

- **nucleosomes**: The fundamental subunit of chromatin, composed of a little less than two turns of DNA wrapped around a set of eight proteins called histones.
- **histones**: The chief protein components of chromatin, which act as spools around which DNA winds.

A eukaryote contains a well-defined nucleus, whereas in prokaryotes the chromosome lies in the cytoplasm in an area called the nucleoid. In eukaryotic cells, DNA and RNA synthesis occur in a separate compartment from protein synthesis. In prokaryotic cells, both processes occur together. What advantages might there be to separating the processes? What advantages might there be to having them occur together?

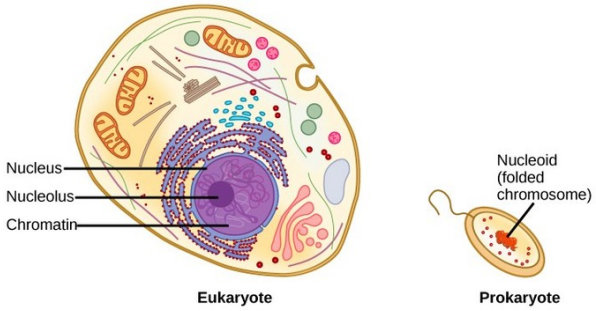

Figure 3.4C.1: **Eukaryotic and prokaryotic cells**: A eukaryote contains a well-defined nucleus, whereas in prokaryotes, the chromosome lies in the cytoplasm in an area called the nucleoid.

The size of the genome in one of the most well-studied prokaryotes, *E.coli*, is 4.6 million base pairs (approximately 1.1 mm, if cut and stretched out). So how does this fit inside a small bacterial cell? The DNA is twisted by what is known as supercoiling. Supercoiling means that DNA is either under-wound (less than one turn of the helix per 10 base pairs) or over-wound (more than 1 turn per 10 base pairs) from its normal relaxed state. Some proteins are known to be involved in the supercoiling; other proteins and enzymes such as DNA gyrase help in maintaining the supercoiled structure.

Eukaryotes, whose chromosomes each consist of a linear DNA molecule, employ a different type of packing strategy to fit their DNA inside the nucleus. At the most basic level, DNA is wrapped around proteins known as histones to form structures called nucleosomes. The histones are evolutionarily conserved proteins that are rich in basic amino acids and form an octamer. The DNA (which is negatively charged because of the phosphate groups) is wrapped tightly around the histone core. This nucleosome is linked to the next one with the help of a linker DNA. This is also known as the “beads on a string” structure. This is further compacted into a 30 nm fiber, which is the diameter of the structure. At the metaphase stage the chromosomes are at their most compact, approximately 700 nm in width, and are found in association with scaffold proteins.

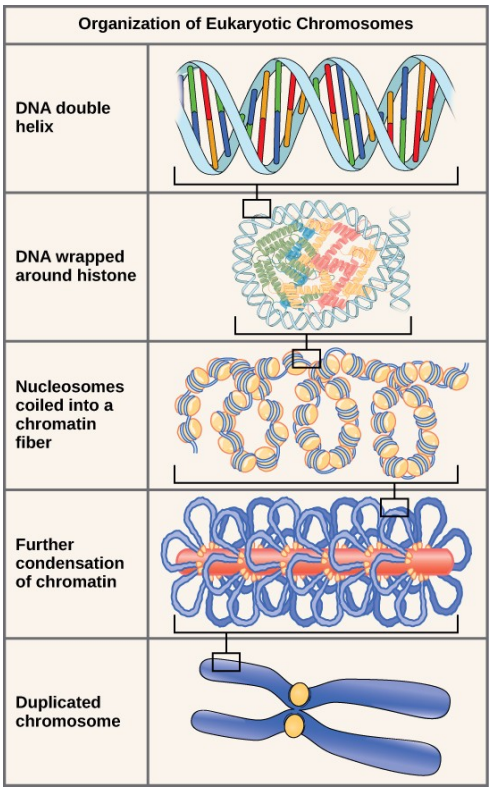

Figure 3.4C.1: **Eukaryotic chromosomes**: These figures illustrate the compaction of the eukaryotic chromosome.

In interphase, eukaryotic chromosomes have two distinct regions that can be distinguished by staining. The tightly packaged region is known as heterochromatin, and the less dense region is known as euchromatin.

Heterochromatin usually contains genes that are not expressed, and is found in the regions of the centromere and telomeres. The euchromatin usually contains genes that are transcribed, with DNA packaged around nucleosomes but not further compacted.

3.4D: Types of RNA

RNA is the nucleic acid that makes proteins from the code provided by DNA through the processes of transcription and translation.

Learning Objectives

- Describe the structure and function of RNA

Key Points

- The nitrogen bases in RNA include adenine (A), guanine (G), cytosine (C), and uracil (U).
- Messenger RNA (mRNA) carries the code from the DNA to the ribosomes, while transfer RNA (tRNA) converts that code into a usable form.
- Ribosomes are the sites where tRNA and rRNA assemble proteins.
- RNA differs from DNA in that it is single stranded, has uracil instead of thymine, carries the code for making proteins instead of directing all of the cell ‘s functions, and has ribose as its five-carbon sugar instead of deoxyribose.

Key Terms

- **codon**: a sequence of three adjacent nucleotides, which encode for a specific amino acid during protein synthesis or translation
- **transcription**: the synthesis of RNA under the direction of DNA

RNA Structure and Function

The two main types of nucleic acids are deoxyribonucleic acid (DNA) and ribonucleic acid (RNA). DNA is the genetic material found in all living organisms and is found in the nucleus of eukaryotes and in the chloroplasts and mitochondria. In prokaryotes, the DNA is not enclosed in a membranous envelope.

The other type of nucleic acid, RNA, is mostly involved in protein synthesis. Just like in DNA, RNA is made of monomers called nucleotides. Each nucleotide is made up of three components: a nitrogenous base, a pentose (five-carbon) sugar called ribose, and a phosphate group. Each nitrogenous base in a nucleotide is attached to a sugar molecule, which is attached to one or more phosphate groups.

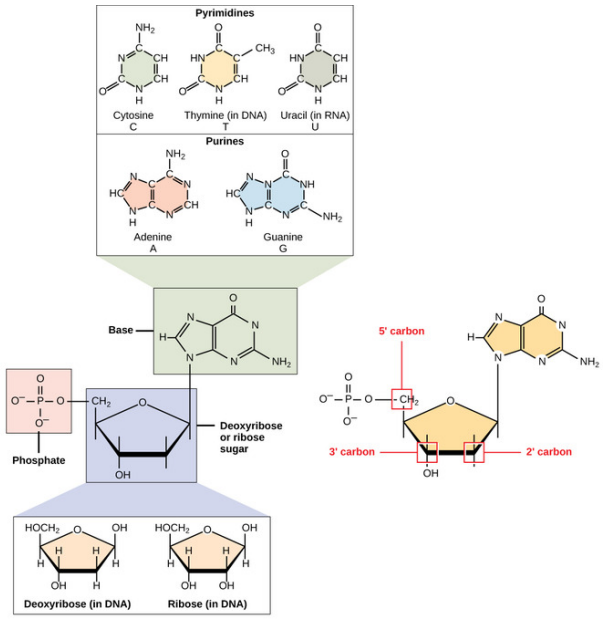

Figure 3.4*D*.1: **RNA Structure**: A nucleotide is made up of three components: a nitrogenous base, a pentose sugar, and one or more phosphate groups. Carbon residues in the pentose are numbered 1′ through 5′ (the prime distinguishes these residues from those in the base, which are numbered without using a prime notation). The base is attached to the 1′ position of the ribose, and the phosphate is attached to the 5′ position. When a polynucleotide is formed, the 5′ phosphate of the incoming nucleotide attaches to the 3′ hydroxyl group at the end of the growing chain. Two types of pentose are found in nucleotides, deoxyribose (found in DNA) and ribose (found in RNA). Deoxyribose is similar in structure to ribose, but it has an H instead of an OH at the 2′ position. Bases can be divided into two categories: purines and pyrimidines. Purines have a double ring structure, and pyrimidines have a single ring.

In RNA, the nitrogenous bases vary slightly from those of DNA. Adenine (A), guanine (G), and cytosine (C) are present, but instead of thymine (T), a pyrimidine called uracil (U) pairs with adenine. RNA is a single stranded molecule, compared to the double helix of DNA.

The DNA molecules never leave the nucleus but instead use an intermediary to communicate with the rest of the cell. This intermediary is the messenger RNA (mRNA). When proteins need to be made, the mRNA enters the nucleus and attaches itself to one of the DNA strands. Being complementary, the sequence of nitrogen bases of the RNA is opposite that of the DNA. This is called transcription. For example, if the DNA strand reads TCCAAGTC, then the mRNA strand would read AGGUUCAG. The mRNA then carries the code out of the nucleus to organelles called ribosomes for the assembly of proteins.

Once the mRNA has reached the ribosomes, they do not read the instructions directly. Instead, another type of RNA called transfer RNA (tRNA) needs to translate the information from the mRNA into a usable form. The tRNA attaches to the mRNA, but with the opposite base pairings. It then reads the sequence in sets of three bases called codons. Each possible three letter arrangement of A,C,U,G (e.g., AAA, AAU, GGC, etc) is a specific instruction, and the correspondence of these instructions and the amino acids is known as the “genetic code.” Though exceptions to or variations on the code exist, the standard genetic code holds true in most organisms.

The ribosome acts like a giant clamp, holding all of the players in position, and facilitating both the pairing of bases between the messenger and transfer RNAs, and the chemical bonding between the amino acids. The ribosome has special subunits known as ribosomal RNAs (rRNA) because they function in the ribosome. These subunits do not carry instructions for making a specific proteins (i.e., they are not messenger RNAs) but instead are an integral part of the ribosome machinery that is used to make proteins from mRNAs. The making of proteins by reading instructions in mRNA is generally known as ” translation.”

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CHAPTER OVERVIEW

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30.1: The Plant Body

Topic hierarchy

- 30.1A: Plant Tissues and Organ Systems

30.1A: Plant Tissues and Organ Systems

Plants are made up of meristematic and permanent tissues and are supported by shoot and root organ systems.

Learning Objectives

- Differentiate among the types of plant tissues and organs

Key Points

- There are two types of plant tissues: meristematic tissue found in plant regions of continuous cell division and growth, and permanent (or non-meristematic) tissue consisting of cells that are no longer actively dividing.
- Meristems produce cells that differentiate into three secondary tissue types: dermal tissue which covers and protects the plant, vascular tissue which transports water, minerals, and sugars and ground tissue which serves as a site for photosynthesis, supports vascular tissue, and stores nutrients.
- Vascular tissue is made of xylem tissue which transports water and nutrients from the roots to different parts of the plant and phloem tissue which transports organic compounds from the site of photosynthesis to other parts of the plant.
- The xylem and phloem always lie next to each other forming a structure called a vascular bundle in stems and a vascular stele or vascular cylinder in roots.
- Parts of the shoot system include the vegetative parts, such as the leaves and the stems, and the reproductive parts, such as the flowers and fruits.

Key Terms

- **meristem**: the plant tissue composed of totipotent cells that allows plant growth
- **parenchyma**: the ground tissue making up most of the non-woody parts of a plant
- **xylem**: a vascular tissue in land plants primarily responsible for the distribution of water and minerals taken up by the roots; also the primary component of wood
- **phloem**: a vascular tissue in land plants primarily responsible for the distribution of sugars and nutrients manufactured in the shoot
- **tracheid**: elongated cells in the xylem of vascular plants that serve in the transport of water and mineral salts

Plant Tissues

Plants are multicellular eukaryotes with tissue systems made of various cell types that carry out specific functions. Plant tissue systems fall into one of two general types: meristematic tissue and permanent (or non-meristematic) tissue. Cells of the meristematic tissue are found in meristems, which are plant regions of continuous cell division and growth. Meristematic tissue cells are either undifferentiated or incompletely differentiated; they continue to divide and contribute to the growth of the plant. In contrast, permanent tissue consists of plant cells that are no longer actively dividing.

Meristematic tissues consist of three types, based on their location in the plant. Apical meristems contain meristematic tissue located at the tips of stems and roots, which enable a plant to extend in length. Lateral meristems facilitate growth in thickness or girth in a maturing plant. Intercalary meristems occur only in monocots at the bases of leaf blades and at nodes (the areas where leaves attach to a stem). This tissue enables the monocot leaf blade to increase in length from the leaf base; for example, it allows lawn grass leaves to elongate even after repeated mowing.

Meristems produce cells that quickly differentiate, or specialize, and become permanent tissue. Such cells take on specific roles and lose their ability to divide further. They differentiate into three main types: dermal, vascular, and ground tissue. Dermal tissue covers and protects the plant. Vascular tissue transports water, minerals, and sugars to different parts of the plant. Ground tissue serves as a site for photosynthesis, provides a supporting matrix for the vascular tissue, and helps to store water and sugars.

Plant tissues are either simple (composed of similar cell types) or complex (composed of different cell types). Dermal tissue, for example, is a simple tissue that covers the outer surface of the plant and controls gas exchange. Vascular tissue is an example of a complex tissue. It is made of two specialized conducting tissues: xylem and phloem. Xylem tissue transports water and nutrients from the roots to different parts of the plant. It includes three different cell types: vessel elements and tracheids (both of which conduct water) and xylem parenchyma. Phloem tissue, which transports organic

compounds from the site of photosynthesis to other parts of the plant, consists of four different cell types: sieve cells (which conduct photosynthates), companion cells, phloem parenchyma, and phloem fibers. Unlike xylem-conducting cells, phloem-conducting cells are alive at maturity. The xylem and phloem always lie adjacent to each other. In stems, the xylem and the phloem form a structure called a vascular bundle; in roots, this is termed the vascular stele or vascular cylinder.

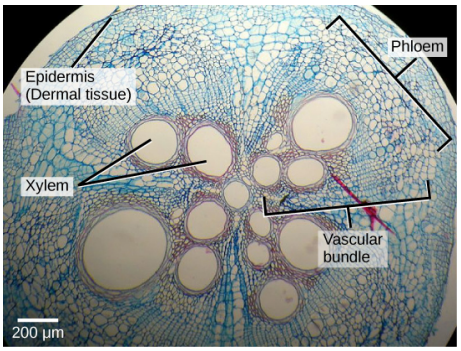

Figure 30.1*A*. 1: **Cross section of a squash stem showing a vascular bundle**: This light micrograph shows a cross section of a squash (Curcurbita maxima) stem. Each teardrop-shaped vascular bundle consists of large xylem vessels toward the inside and smaller phloem cells toward the outside. Xylem cells, which transport water and nutrients from the roots to the rest of the plant, are dead at functional maturity. Phloem cells, which transport sugars and other organic compounds from photosynthetic tissue to the rest of the plant, are living. The vascular bundles are encased in ground tissue and surrounded by dermal tissue.

Plant Organ Systems

In plants, just as in animals, similar cells working together form a tissue. When different types of tissues work together to perform a unique function, they form an organ; organs working together form organ systems. Vascular plants have two distinct organ systems: a shoot system and a root system. The shoot system consists of two portions: the vegetative (non-reproductive) parts of the plant, such as the leaves and the stems; and the reproductive parts of the plant, which include flowers and fruits. The shoot system generally grows above ground, where it absorbs the light needed for photosynthesis. The root system, which supports the plants and absorbs water and minerals, is usually underground.

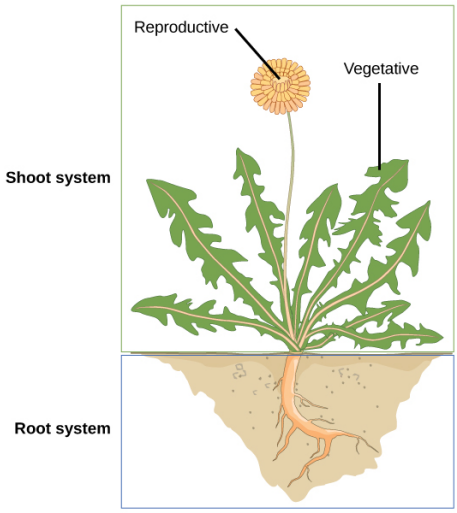

Figure 30.1*A*. 1: **Example plant organ systems**: The shoot system of a plant consists of leaves, stems, flowers, and fruits. The root system anchors the plant while absorbing water and minerals from the soil.

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30.2: Stems

Topic hierarchy

- 30.2A: Functions of Stems
- 30.2B: Stem Anatomy
- 30.2C: Primary and Secondary Growth in Stems
- 30.2D: Stem Modifications

30.2A: Functions of Stems

A stem connects the roots to the leaves, provides support, stores food, and holds the leaves, flowers, and buds.

Learning Objectives

- Summarize the main function and basic structure of stems

Key Points

- Most stems are found above ground, but some of them grow underground.
- Stems can be either unbranched or highly branched; they may be herbaceous or woody.
- Stems connect the roots to the leaves, helping to transport water, minerals, and sugars to different parts of the plant.
- Plant stems always have nodes (points of attachments for leaves, roots, and flowers) and internodes (regions between nodes).
- The petiole is the stalk that extends from the stem to the base of the leaf.
- An axillary bud gives rise to a branch or a flower; it is usually found in the axil: the junction of the stem and petiole.

Key Terms

- **node**: points of attachment for leaves, aerial roots, and flowers
- **internode**: a section of stem between two stem nodes
- **petiole**: stalk that extends from the stem to the base of the leaf
- **axillary bud**: embryonic shoot that lies at the junction of the stem and petiole that gives rise to a branch or flower

Stems

Stems are a part of the shoot system of a plant. They may range in length from a few millimeters to hundreds of meters. They also vary in diameter, depending on the plant type. Stems are usually above ground, although the stems of some plants, such as the potato, also grow underground. Stems may be herbaceous (soft) or woody in nature. Their main function is to provide support to the plant, holding leaves, flowers, and buds; in some cases, stems also store food for the plant. A stem may be unbranched, like that of a palm tree, or it may be highly branched, like that of a magnolia tree. The stem of the plant connects the roots to the leaves, helping to transport absorbed water and minerals to different parts of the plant. The stem also helps to transport the products of photosynthesis (i.e., sugars) from the leaves to the rest of the plant.

Plant stems, whether above or below ground, are characterized by the presence of nodes and internodes. Nodes are points of attachment for leaves, aerial roots, and flowers. The stem region between two nodes is called an internode. The stalk that extends from the stem to the base of the leaf is the petiole. An axillary bud is usually found in the axil (the area between the base of a leaf and the stem) where it can give rise to a branch or a flower. The apex (tip) of the shoot contains the apical meristem within the apical bud.

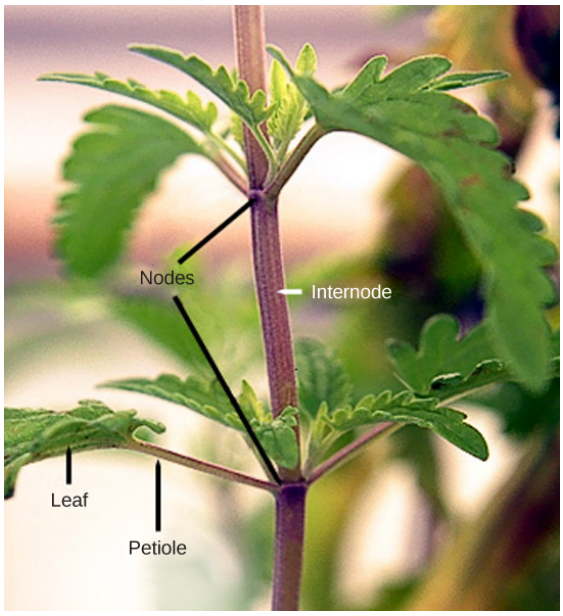

Figure 30.2*A*.1: **Parts of a stem**: Leaves are attached to the plant stem at areas called nodes. An internode is the stem region between two nodes. The petiole is the stalk connecting the leaf to the stem. The leaves just above the nodes arise from axillary buds.

30.2B: Stem Anatomy

The stem's anatomy consists of three tissue systems that work together to support, protect, and aid in nourishing the plant.

Learning Objectives

- Summarize the roles of dermal tissue, vascular tissue, and ground tissue

Key Points

- The stem has three simple cell types: the parenchyma, collenchyma, and sclerenchyma cells that are responsible for metabolic functions, repairing and healing wounds, and storing starch.
- The stem is composed of three tissue systems that include the epidermis, vascular, and ground tissues, all of which are made from the simple cell types..
- The xylem and phloem carry water and nutrients up and down the length of the stem and are arranged in distinct strands called vascular bundles.
- The epidermis is a single layer of cells that makes up the dermal tissue covering the stem and protecting the underlying tissue. Woody plants have an extra layer of protection on top of the epidermis made of cork cells known as bark.
- The vascular tissue of the stem consists of the complex tissues xylem and phloem which carry water and nutrients up and down the length of the stem and are arranged in distinct strands called vascular bundles.
- Ground tissue helps support the stem and is called pith when it is located towards the middle of the stem and called the cortex when it is between the vascular tissue and the epidermis.

Key Terms

- **collenchyma**: a supporting ground tissue just under the surface of various leaf structures formed before vascular differentiation
- **sclerenchyma**: a mechanical, supportive ground tissue in plants consisting of aggregates of cells having thick, often mineralized walls
- **sclereid**: a reduced form of sclerenchyma cells with highly-thickened, lignified walls
- **lignin**: a complex, non-carbohydrate, aromatic polymer present in all wood
- **stoma**: a pore found in the leaf and stem epidermis used for gaseous exchange
- **trichome**: a hair- or scale-like extension of the epidermis of a plant
- **xylem**: a vascular tissue in land plants primarily responsible for the distribution of water and minerals taken up by the roots; also the primary component of wood
- **phloem**: a vascular tissue in land plants primarily responsible for the distribution of sugars and nutrients manufactured in the shoot
- **tracheid**: elongated cells in the xylem of vascular plants that serve in the transport of water and mineral salts
- **pith**: the soft spongy substance in the center of the stems of many plants and trees
- **cortex**: the tissue of a stem or root that lies inward from the epidermis, but exterior to the vascular tissue
- **parenchyma**: the ground tissue making up most of the non-woody parts of a plant

Stem Anatomy

The stem and other plant organs are primarily made from three simple cell types: parenchyma, collenchyma, and sclerenchyma cells. Parenchyma cells are the most common plant cells. They are found in the stem, the root, the inside of the leaf, and the pulp of the fruit. Parenchyma cells are responsible for metabolic functions, such as photosynthesis. They also help repair and heal wounds. In addition, some parenchyma cells store starch.

Figure 30.2B.1: **Parenchyma cells in plants**: The stem of common St John's Wort (Hypericum perforatum) is shown in cross section in this light micrograph. The central pith (greenish-blue, in the center) and peripheral cortex (narrow zone 3–5 cells thick, just inside the epidermis) are composed of parenchyma cells. Vascular tissue composed of xylem (red) and phloem tissue (green, between the xylem and cortex) surrounds the pith.

Collenchyma cells are elongated cells with unevenly-thickened walls. They provide structural support, mainly to the stem and leaves. These cells are alive at maturity and are usually found below the epidermis. The "strings" of a celery stalk are an example of collenchyma cells.

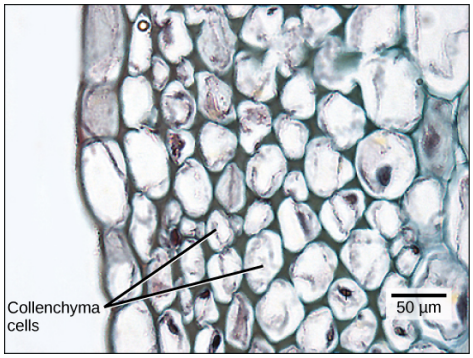

Figure 30.2B.1: **Collenchyma cells in plants**: Collenchyma cell walls are uneven in thickness, as seen in this light micrograph. They provide support to plant structures.

Sclerenchyma cells also provide support to the plant, but unlike collenchyma cells, many of them are dead at maturity. There are two types of sclerenchyma cells: fibers and sclereids. Both types have secondary cell walls that are thickened with deposits of lignin, an organic compound that is a key component of wood. Fibers are long, slender cells; sclereids are smaller-sized. Sclereids give pears their gritty texture. Humans use sclerenchyma fibers to make linen and rope.

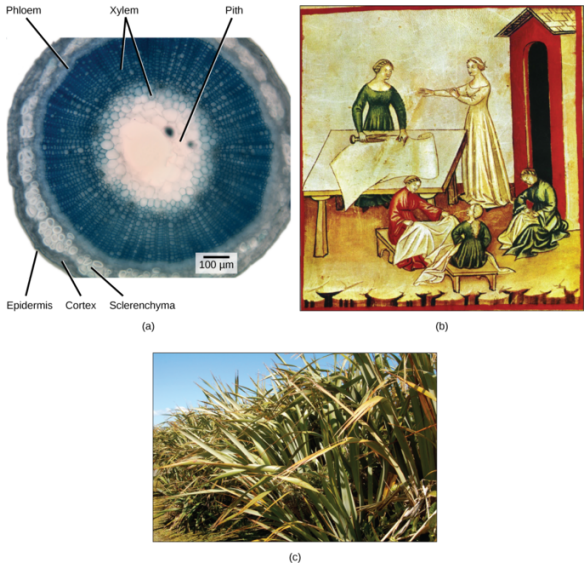

Figure $30.2B.1$: **Sclerenchyma cells in plants**: The central pith and outer cortex of the (a) flax stem are made up of parenchyma cells. Inside the cortex is a layer of sclerenchyma cells, which make up the fibers in flax rope and clothing. Humans have grown and harvested flax for thousands of years. In (b) this drawing, fourteenth-century women prepare linen. The (c) flax plant is grown and harvested for its fibers, which are used to weave linen, and for its seeds, which are the source of linseed oil.

As with the rest of the plant, the stem has three tissue systems: dermal, vascular, and ground tissue. Each is distinguished by characteristic cell types that perform specific tasks necessary for the plant's growth and survival.

Dermal Tissue

The dermal tissue of the stem consists primarily of epidermis: a single layer of cells covering and protecting the underlying tissue. Woody plants have a tough, waterproof outer layer of cork cells commonly known as bark, which further protects the plant from damage. Epidermal cells are the most-numerous and least-differentiated of the cells in the epidermis. The epidermis of a leaf also contains openings, known as stomata, through which the exchange of gases takes place. Two cells, known as guard cells, surround each leaf stoma, controlling its opening and closing and, thus, regulating the uptake of carbon dioxide and the release of oxygen and water vapor. Trichomes are hair-like structures on the epidermal surface. They help to reduce transpiration (the loss of water by aboveground plant parts), increase solar reflectance, and store compounds that defend the leaves against predation by herbivores.

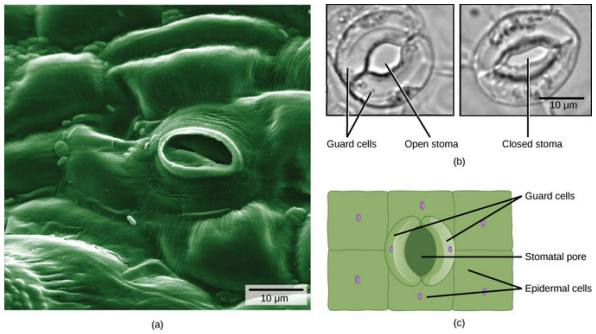

Figure $30.2B.1$: **Stomata**: Openings called stomata (singular: stoma) allow a plant to take up carbon dioxide and release oxygen and water vapor. The (a) colorized scanning-electron micrograph shows a closed stoma of a dicot. Each stoma is flanked by two guard cells that regulate its (b) opening and closing. The (c) guard cells sit within the layer of epidermal cells.

Vascular Tissue

The xylem and phloem that make up the vascular tissue of the stem are arranged in distinct strands called vascular bundles, which run up and down the length of the stem. Both are considered complex plant tissue because they are composed of more than one simple cell type that work in concert with each other. When the stem is viewed in cross section, the vascular

bundles of dicot stems are arranged in a ring. In plants with stems that live for more than one year, the individual bundles grow together and produce the characteristic growth rings. In monocot stems, the vascular bundles are randomly scattered throughout the ground tissue.

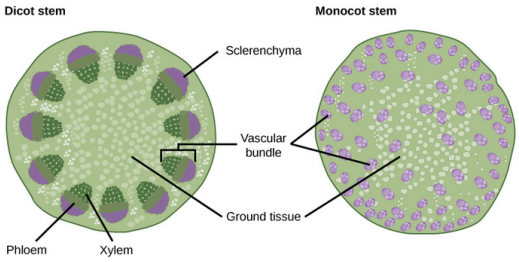

Figure $30.2B.1$: **Vascular bundles**: In (a) dicot stems, vascular bundles are arranged around the periphery of the ground tissue. The xylem tissue is located toward the interior of the vascular bundle; phloem is located toward the exterior. Sclerenchyma fibers cap the vascular bundles. In (b) monocot stems, vascular bundles composed of xylem and phloem tissues are scattered throughout the ground tissue.

Xylem tissue has three types of cells: xylem parenchyma, tracheids, and vessel elements. The latter two types conduct water and are dead at maturity. Tracheids are xylem cells with thick secondary cell walls that are lignified. Water moves from one tracheid to another through regions on the side walls known as pits where secondary walls are absent. Vessel elements are xylem cells with thinner walls; they are shorter than tracheids. Each vessel element is connected to the next by means of a perforation plate at the end walls of the element. Water moves through the perforation plates to travel up the plant.

Phloem tissue is composed of sieve-tube cells, companion cells, phloem parenchyma, and phloem fibers. A series of sieve-tube cells (also called sieve-tube elements) are arranged end-to-end to create a long sieve tube, which transports organic substances such as sugars and amino acids. The sugars flow from one sieve-tube cell to the next through perforated sieve plates, which are found at the end junctions between two cells. Although still alive at maturity, the nucleus and other cell components of the sieve-tube cells have disintegrated. Companion cells are found alongside the sieve-tube cells, providing them with metabolic support. The companion cells contain more ribosomes and mitochondria than do the sieve-tube cells, which lack some cellular organelles.

Ground Tissue

Ground tissue is mostly made up of parenchyma cells, but may also contain collenchyma and sclerenchyma cells that help support the stem. The ground tissue towards the interior of the vascular tissue in a stem or root is known as pith, while the layer of tissue between the vascular tissue and the epidermis is known as the cortex.

30.2C: Primary and Secondary Growth in Stems

Plants undergo primary growth to increase length and secondary growth to increase thickness.

Learning Objectives

- Distinguish between primary and secondary growth in stems

Key Points

- Indeterminate growth continues throughout a plant's life, while determinate growth stops when a plant element (such as a leaf) reaches a particular size.
- Primary growth of stems is a result of rapidly-dividing cells in the apical meristems at the shoot tips.
- Apical dominance reduces the growth along the sides of branches and stems, giving the tree a conical shape.
- The growth of the lateral meristems, which includes the vascular cambium and the cork cambium (in woody plants), increases the thickness of the stem during secondary growth.
- Cork cells (bark) protect the plant against physical damage and water loss; they contain a waxy substance known as suberin that prevents water from penetrating the tissue.
- The secondary xylem develops dense wood during the fall and thin wood during the spring, which produces a characteristic ring for each year of growth.

Key Terms

- **lenticel**: small, oval, rounded spots upon the stem or branch of a plant that allow the exchange of gases with the surrounding atmosphere
- **periderm**: the outer layer of plant tissue; the outer bark
- **suberin**: a waxy material found in bark that can repel water

Growth in Stems

Growth in plants occurs as the stems and roots lengthen. Some plants, especially those that are woody, also increase in thickness during their life span. The increase in length of the shoot and the root is referred to as primary growth. It is the result of cell division in the shoot apical meristem. Secondary growth is characterized by an increase in thickness or girth of the plant. It is caused by cell division in the lateral meristem. Herbaceous plants mostly undergo primary growth, with little secondary growth or increase in thickness. Secondary growth, or "wood", is noticeable in woody plants; it occurs in some dicots, but occurs very rarely in monocots.

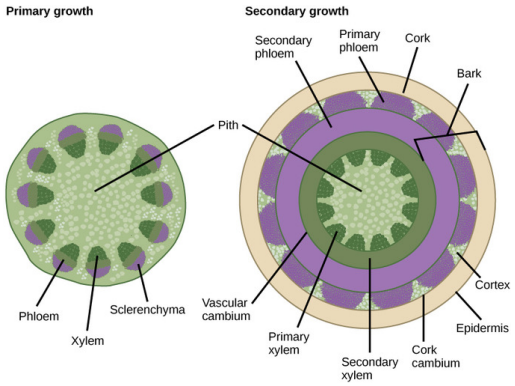

Figure 30.2C.1: **Primary and secondary growth**: In woody plants, primary growth is followed by secondary growth, which allows the plant stem to increase in thickness or girth. Secondary vascular tissue is added as the plant grows, as well as a cork layer. The bark of a tree extends from the vascular cambium to the epidermis.

Some plant parts, such as stems and roots, continue to grow throughout a plant's life: a phenomenon called indeterminate growth. Other plant parts, such as leaves and flowers, exhibit determinate growth, which ceases when a plant part reaches a particular size.

Primary Growth

Most primary growth occurs at the apices, or tips, of stems and roots. Primary growth is a result of rapidly-dividing cells in the apical meristems at the shoot tip and root tip. Subsequent cell elongation also contributes to primary growth. The growth of shoots and roots during primary growth enables plants to continuously seek water (roots) or sunlight (shoots).

The influence of the apical bud on overall plant growth is known as apical dominance, which diminishes the growth of axillary buds that form along the sides of branches and stems. Most coniferous trees exhibit strong apical dominance, thus producing the typical conical Christmas tree shape. If the apical bud is removed, then the axillary buds will start forming lateral branches. Gardeners make use of this fact when they prune plants by cutting off the tops of branches, thus encouraging the axillary buds to grow out, giving the plant a bushy shape.

Secondary Growth

The increase in stem thickness that results from secondary growth is due to the activity of the lateral meristems, which are lacking in herbaceous plants. Lateral meristems include the vascular cambium and, in woody plants, the cork cambium. The vascular cambium is located just outside the primary xylem and to the interior of the primary phloem. The cells of the vascular cambium divide and form secondary xylem (tracheids and vessel elements) to the inside and secondary phloem (sieve elements and companion cells) to the outside. The thickening of the stem that occurs in secondary growth is due to the formation of secondary phloem and secondary xylem by the vascular cambium, plus the action of cork cambium, which forms the tough outermost layer of the stem. The cells of the secondary xylem contain lignin, which provides hardiness and strength.

In woody plants, cork cambium is the outermost lateral meristem. It produces cork cells (bark) containing a waxy substance known as suberin that can repel water. The bark protects the plant against physical damage and helps reduce water loss. The cork cambium also produces a layer of cells known as phelloderm, which grows inward from the cambium. The cork cambium, cork cells, and phelloderm are collectively termed the periderm. The periderm substitutes for the epidermis in mature plants. In some plants, the periderm has many openings, known as lenticels, which allow the interior cells to exchange gases with the outside atmosphere. This supplies oxygen to the living- and metabolically-active cells of the cortex, xylem, and phloem.

Figure $30.2C.1$: **Example of lenticels**: Lenticels on the bark of this cherry tree enable the woody stem to exchange gases with the surrounding atmosphere.

Annual Rings

The activity of the vascular cambium gives rise to annual growth rings. During the spring growing season, cells of the secondary xylem have a large internal diameter; their primary cell walls are not extensively thickened. This is known as early wood, or spring wood. During the fall season, the secondary xylem develops thickened cell walls, forming late wood, or autumn wood, which is denser than early wood. This alternation of early and late wood is due largely to a seasonal decrease in the number of vessel elements and a seasonal increase in the number of tracheids. It results in the formation of an annual ring, which can be seen as a circular ring in the cross section of the stem. An examination of the number of annual rings and their

nature (such as their size and cell wall thickness) can reveal the age of the tree and the prevailing climatic conditions during each season.

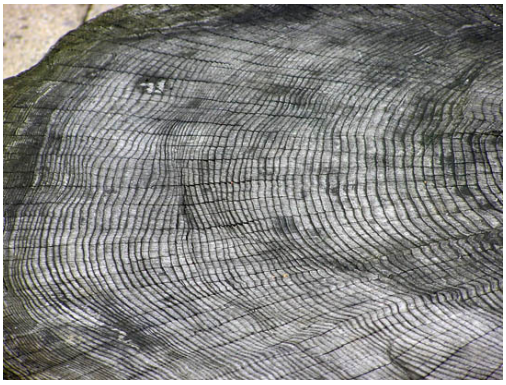

Figure $30.2C.1$: **Annual growth rings**: The rate of wood growth increases in summer and decreases in winter, producing a characteristic ring for each year of growth. Seasonal changes in weather patterns can also affect the growth rate. Note how the rings vary in thickness.

30.2D: Stem Modifications

Stem modifications, either aboveground, underground, or aerial, enable plants to survive in particular habitats and environments.

Learning Objectives

- Explain the reasons for stem modifications

Key Points

- Modified stems that grow horizontally underground are either rhizomes, from which vertical shoots grow, or fleshier, food-storing corms.
- New plants can arise from the nodes of stolons and runners (an aboveground stolon): stems that run parallel to the ground, or just below the surface.
- Potatoes are examples of tubers: the swollen ends of stolons that may store starch.
- The stem modification that has enlarged fleshy leaves emerging from the stem or surrounding the base of the stem is called a bulb; it is also used to store food.
- Aerial modifications of stems include tendrils, thorns, bulbils, and cladodes..

Key Terms

- **stolon**: a shoot that grows along the ground and produces roots at its nodes; a runner
- **tuber**: a fleshy, thickened, underground stem of a plant, usually containing stored starch, as for example a potato or arrowroot
- **cladode**: green branches of limited growth which have taken up the functions of photosynthesis
- **rhizome**: a horizontal underground stem of some plants that sends out roots and shoots from its nodes
- **corm**: a short, vertical, swollen underground stem of a plant that serves as a storage organ to enable the plant to survive winter or other adverse conditions such as drought
- **bulb**: the bulb-shaped root portion of a plant such as a tulip, from which the rest of the plant may be regrown
- **tendril**: a thin, spirally-coiling stem that attaches a plant to its support
- **thorn**: a sharp, protective spine of a plant
- **bulbil**: a bulb-shaped bud in the place of a flower or in a leaf axil

Stem Modifications

Some plant species have modified stems that are especially suited to a particular habitat and environment. A rhizome is a modified stem that grows horizontally underground; it has nodes and internodes. Vertical shoots may arise from the buds on the rhizome of some plants, such as ginger and ferns. Corms are similar to rhizomes, except they are more rounded and fleshy (such as in gladiolus). Corms contain stored food that enables some plants to survive the winter. Stolons are stems that run almost parallel to the ground, or just below the surface, and can give rise to new plants at the nodes. Runners are a type of stolon that runs above the ground and produces new clone plants at nodes at varying intervals: strawberries are an example. Tubers are modified stems that may store starch, as seen in the potato. Tubers arise as swollen ends of stolons, and contain many adventitious or unusual buds (familiar to us as the “eyes” on potatoes). A bulb, which functions as an underground storage unit, is a modification of a stem that has the appearance of enlarged fleshy leaves emerging from the stem or surrounding the base of the stem, as seen in the iris.

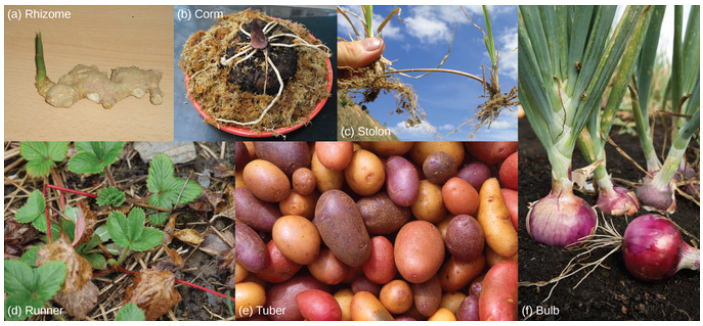

Figure 30.2D.1: **Stem modifications**: Stem modifications enable plants to thrive in a variety of environments. Shown are (a) ginger (Zingiber officinale) rhizomes, (b) a carrion flower (Amorphophallus titanum) corm (c) Rhodes grass (Chloris gayana) stolons, (d) strawberry (Fragaria ananassa) runners, (e) potato (Solanum tuberosum) tubers, and (f) red onion (Allium) bulbs.

Modifications to the aerial stems, vegetative buds, and floral buds of plants perform functions such as climbing, protection, and synthesis of food vegetative propagation. Aerial modifications of stems include the following:

Figure 30.2D.1: **Aerial modifications of stems**: Found in southeastern United States, (a) buckwheat vine (Brunnichia ovata) is a weedy plant that climbs with the aid of tendrils. This one is shown climbing up a wooden stake. (b) Thorns are modified branches.

- Tendrils are slender, twining strands that enable a plant (like the buckwheat vine) to seek support by climbing on other surfaces. These may develop from either the axillary bud or the terminal bud of the stem.
- Thorns are modified branches appearing as hard, woody, sharp outgrowths that protect the plant; common examples include roses, osage orange, and devil's walking stick.
- Bulbils are axillary buds that have become fleshy and rounded due to storage of food. They become detached from the plant, fall on ground and develop into a new plant.
- Cladodes are green branches of limited growth (usually one internode long) which have taken up the functions of photosynthesis.

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30.3: Roots

Topic hierarchy

- 30.3A: Types of Root Systems and Zones of Growth
- 30.3B: Root Modifications

30.3A: Types of Root Systems and Zones of Growth

The root tip has three main zones: a zone of cell division, a zone of elongation, and a zone of maturation.

Learning Objectives

- Describe the three zones of the root tip and summarize the role of each zone in root growth

Key Points

- Root tips ultimately develop into two main types of root systems: tap roots and fibrous roots.
- The growing root tip is protected by a root cap.
- Within the root tip, cells differentiate, actively divide, and increase in length, depending on in which zone the cells are located.
- Dividing cells make up the zone of cell division in a germinating plant.
- The newly-forming root increases in size in the zone of elongation.
- Differentiating cells make up the zone of cell maturation.

Key Terms

- **radicle**: the rudimentary shoot of a plant that supports the cotyledons in the seed and from which the root is developed downward; the root of the embryo
- **meristem**: the plant tissue composed of totipotent cells that allows plant growth
- **germination**: the beginning of vegetation or growth from a seed or spore

Types of Root Systems

There are two main types of root systems. Dicots have a tap root system, while monocots have a fibrous root system, which is also known as an adventitious root system. A tap root system has a main root that grows down vertically, from which many smaller lateral roots arise. Dandelions are a common example; their tap roots usually break off when these weeds are pulled from the ground; they can regrow another shoot from the remaining root. A tap root system penetrates deep into the soil. In contrast, a fibrous root system is located closer to the soil surface where it forms a dense network of roots that also helps prevent soil erosion (lawn grasses are a good example, as are wheat, rice, and corn). Some plants have a combination of tap roots and fibrous roots. Plants that grow in dry areas often have deep root systems, whereas plants that grow in areas with abundant water are likely to have shallower root systems.

Figure 30.3A.1: **Main types of root systems**: (a) Tap root systems have a main root that grows down, while (b) fibrous root systems consist of many small roots.

Zones of the Root Tip

Root growth begins with seed germination. When the plant embryo emerges from the seed, the radicle of the embryo forms the root system. The tip of the root is protected by the root cap, a structure exclusive to roots and unlike any other plant structure. The root cap is continuously replaced because it is easily damaged as the root pushes through soil. The root tip can be divided into three zones: a zone of cell division, a zone of elongation, and a zone of maturation. The zone of cell division is closest to the root tip and is made up of the actively-dividing cells of the root meristem, which contains the undifferentiated cells of the germinating plant. The zone of elongation is where the newly-formed cells increase in length, thereby lengthening the root. Beginning at the first root hair is the zone of cell maturation where the root cells differentiate into specialized cell types. All three zones are in approximately the first centimeter of the root tip.

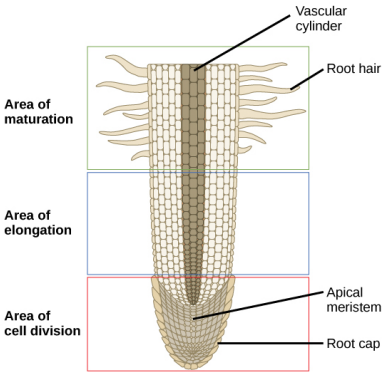

Figure 30.3A.1: **Zones of the root tip**: A longitudinal view of the root reveals the zones of cell division, elongation, and maturation. Cell division occurs in the apical meristem.

30.3B: Root Modifications

Plants have a wide variety of roots for functions as diverse as structural support, food storage, and parasitism.

Learning Objectives

- Explain the reasons for root modifications

Key Points

- Storage roots, which include a large number of edible vegetables such as potatoes and carrots, are some of the most commonly-known types of modified roots.
- Aerial roots encompass a variety of shapes, yet function similarly as structural support for the plant.
- Parasitic plants have special haustorial roots that allow the plant to absorb nutrients from a host plant.

Key Terms

- **succulent**: having fleshy leaves or other tissues that store water
- **epiphyte**: a plant that grows on another, using it as a physical support but neither obtaining nutrients from it nor causing it any damage if also offering no benefit

Root Modifications

Plants have different root structures for specific purposes. There are many different types of specialized roots, but two of the more familiar types of roots include aerial roots and storage roots. Aerial roots grow above the ground, typically providing structural support. Storage roots (for example, taproots and tuberous roots) are modified for food storage.

Aerial roots are found in many different kinds of plants, offering varying functions depending on the location of the plant. Epiphytic roots are a type of aerial root that enable a plant to grow on another plant in a non-parasitic manner. The banyan tree begins as an epiphyte, germinating in the branches of a host tree. Aerial prop roots develop from the branches and eventually reach the ground, providing additional support. Over time, many roots will come together to form what appears to be a trunk. The epiphytic roots of orchids develop a spongy tissue to absorb moisture and nutrients from any organic material on their roots. In screwpine, a palm-like tree that grows in sandy tropical soils, aerial roots develop to provide additional support that help the tree remain upright in shifting sand and water conditions.

Figure $30.3B.1$: **Aerial roots**: The (a) banyan tree, also known as the strangler fig, begins life as an epiphyte in a host tree. Aerial roots extend to the ground, supporting the growing plant, which eventually strangles the host tree. The (b) screwpine develops aerial roots that help support the plant in sandy soils.

Storage roots, such as carrots, beets, and sweet potatoes, are examples of roots that are specially modified for storage of starch and water. They usually grow underground as protection from plant-eating animals. Some plants, however, such as leaf succulents and cacti, store energy in their leaves and stems, respectively, instead of in their roots.

Figure 30.3B.1: **Storage roots**: Many vegetables, such as carrots and beets, are modified roots that store food and water.

Other examples of modified roots are aerating roots and haustorial roots. Aerating roots, which rise above the ground, especially above water, are commonly seen in mangrove forests that grow along salt water coastlines. Haustorial roots are often seen in parasitic plants such as mistletoe. Their roots allow the plants to absorb water and nutrients from other plants.

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30.4: Leaves

Topic hierarchy

- 30.4A: Leaf Structure and Arrangment
- 30.4B: Types of Leaf Forms
- 30.4C: Leaf Structure, Function, and Adaptation

30.4A: Leaf Structure and Arrangment

Most leaves have similar essential structures, but differ in venation patterns and leaf arrangement (or phyllotaxy).

Learning Objectives

- Sketch the basic structure of a typical leaf

Key Points

- Each leaf typically has a leaf blade (lamina), stipules, a midrib, and a margin.
- Some leaves have a petiole, which attaches the leaf to the stem; leaves that do not have petioles are directly attached to the plant stem and are called sessile leaves.
- The arrangement of veins in a leaf is called the venation pattern; monocots have parallel venation, while dicots have reticulate venation.
- The arrangement of leaves on a stem is known as phyllotaxy; leaves can be classified as either alternate, spiral, opposite, or whorled.
- Plants with alternate and spiral leaf arrangements have only one leaf per node.
- In an opposite leaf arrangement, two leaves connect at a node. In a whorled arrangement, three or more leaves connect at a node.

Key Terms

- **petiole**: stalk that extends from the stem to the base of the leaf
- **lamina**: the flat part of a leaf; the blade, which is the widest part of the leaf
- **stipule**: small green appendage usually found at the base of the petiole

Structure of a Typical Leaf

Each leaf typically has a leaf blade called the lamina, which is also the widest part of the leaf. Some leaves are attached to the plant stem by a petiole. Leaves that do not have a petiole and are directly attached to the plant stem are called sessile leaves. Leaves also have stipules, small green appendages usually found at the base of the petiole. Most leaves have a midrib, which travels the length of the leaf and branches to each side to produce veins of vascular tissue. The edge of the leaf is called the margin.

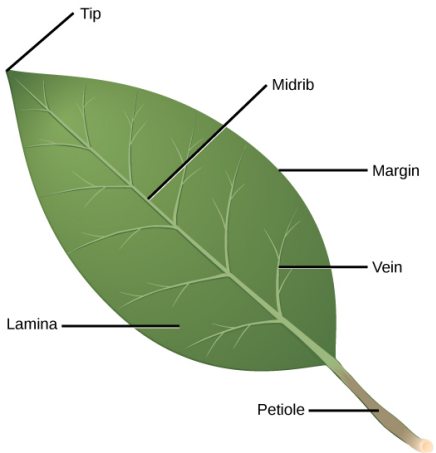


Figure 30.4A.1: **Parts of a leaf**: A leaf may seem simple in appearance, but it is a highly-efficient structure. Petioles, stipules, veins, and a midrib are all essential structures of a leaf.

Within each leaf, the vascular tissue forms veins. The arrangement of veins in a leaf is called the venation pattern. Monocots and dicots differ in their patterns of venation. Monocots have parallel venation in which the veins run in straight lines across the length of the leaf without converging. In dicots, however, the veins of the leaf have a net-like appearance, forming a pattern known as reticulate venation. *Ginkgo biloba* is an example of a plant with dichotomous venation.

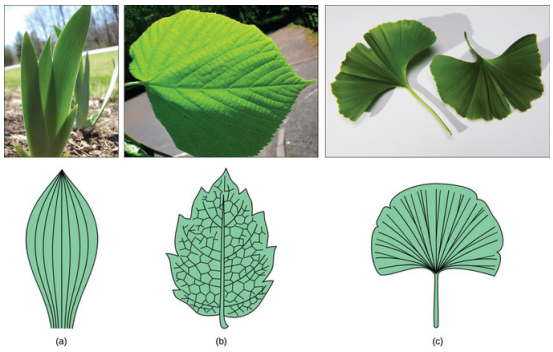


Figure 30.4*A*.1: **Venation patterns**: (a) Tulip (*Tulipa*), a monocot, has leaves with parallel venation. (b) The netlike venation in this linden (*Tilia cordata*) leaf distinguishes it as a dicot. (c) The Ginkgo biloba tree has dichotomous venation.

Leaf Arrangement

The arrangement of leaves on a stem is known as phyllotaxy. The number and placement of a plant's leaves will vary depending on the species, with each species exhibiting a characteristic leaf arrangement. Leaves are classified as either alternate, spiral, opposite, or whorled. Plants that have only one leaf per node have leaves that are said to be either alternate or spiral. Alternate leaves alternate on each side of the stem in a flat plane, and spiral leaves are arranged in a spiral along the stem. In an opposite leaf arrangement, two leaves arise at the same point, with the leaves connecting opposite each other along the branch. If there are three or more leaves connected at a node, the leaf arrangement is classified as whorled.

30.4B: Types of Leaf Forms

Leaves may be categorized as simple or compound, depending on how their blade (or lamina) is divided.

Learning Objectives

- Differentiate among the types of leaf forms

Key Points

- In a simple leaf, the blade is completely undivided; leaves may also be formed of lobes where the gaps between lobes do not reach to the main vein.
- In a compound leaf, the leaf blade is divided, forming leaflets that are attached to the middle vein, but have their own stalks.
- The leaflets of palmately-compound leaves radiate outwards from the end of the petiole.
- Pinnately-compound leaves have their leaflets arranged along the middle vein.
- Bipinnately-compound (double-compound) leaves have their leaflets arranged along a secondary vein, which is one of several veins branching off the middle vein.

Key Terms

- **simple leaf**: a leaf with an undivided blade
- **compound leaf**: a leaf where the blade is divided, forming leaflets
- **palmately compound leaf**: leaf that has its leaflets radiating outwards from the end of the petiole
- **pinnately compound leaf**: a leaf where the leaflets are arranged along the middle vein

Leaf Form

There are two basic forms of leaves that can be described considering the way the blade (or lamina) is divided. Leaves may be simple or compound.

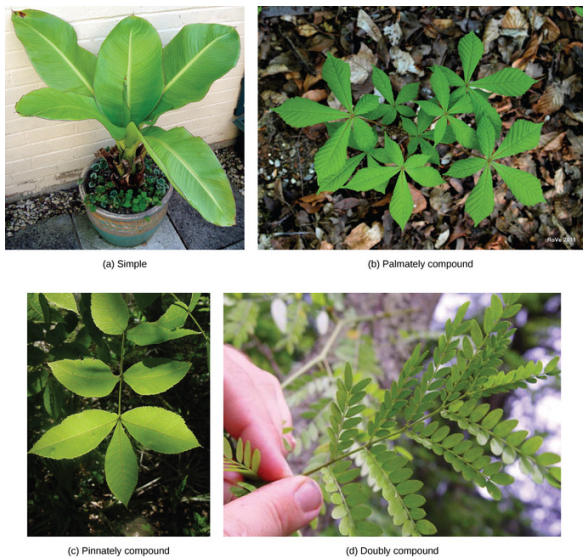


Figure 30.4B.1: **Simple and compound leaves**: Leaves may be simple or compound. In simple leaves, the lamina is continuous. (a) The banana plant (*Musa sp.*) has simple leaves. In compound leaves, the lamina is separated into leaflets. Compound leaves may be palmate or pinnate. (b) In palmately compound leaves, such as those of the horse chestnut (*Aesculus hippocastanum*), the leaflets branch from the petiole. (c) In pinnately compound leaves, the leaflets branch from the midrib, as on a scrub hickory (*Carya floridana*). (d) The honey locust has double compound leaves, in which leaflets branch from the veins.

In a simple leaf, such as the banana leaf, the blade is completely undivided. The leaf shape may also be formed of lobes where the gaps between lobes do not reach to the main vein. An example of this type is the maple leaf.

In a compound leaf, the leaf blade is completely divided, forming leaflets, as in the locust tree. Compound leaves are a characteristic of some families of higher plants. Each leaflet is attached to the rachis (middle vein), but may have its own stalk. A palmately compound leaf has its leaflets radiating outwards from the end of the petiole, like fingers off the palm of a hand. Examples of plants with palmately compound leaves include poison ivy, the buckeye tree, or the familiar house plant *Schefflera sp.* (commonly called “umbrella plant”). Pinnately compound leaves take their name from their feather-like appearance; the leaflets are arranged along the middle vein, as in rose leaves or the leaves of hickory, pecan, ash, or walnut trees. In a pinnately compound leaf, the middle vein is called the midrib. Bipinnately compound (or double compound) leaves are twice divided; the leaflets are arranged along a secondary vein, which is one of several veins branching off the middle vein. Each leaflet is called a “pinnule”. The pinnules on one secondary vein are called “pinna”. The silk tree (*Albizia*) is an example of a plant with bipinnate leaves.

30.4C: Leaf Structure, Function, and Adaptation

Leaves have many structures that prevent water loss, transport compounds, aid in gas exchange, and protect the plant as a whole.

Learning Objectives

- Describe the internal structure and function of a leaf

Key Points

- The epidermis consists of the upper and lower epidermis; it aids in the regulation of gas exchange via stomata.
- The epidermis is one layer thick, but may have more layers to prevent transpiration.
- The cuticle is located outside the epidermis and protects against water loss; trichomes discourage predation.
- The mesophyll is found between the upper and lower epidermis; it aids in gas exchange and photosynthesis via chloroplasts.
- The xylem transports water and minerals to the leaves; the phloem transports the photosynthetic products to the other parts of the plant.
- Plants in cold climates have needle-like leaves that are reduced in size; plants in hot climates have succulent leaves that help to conserve water.

Key Terms

- **trichome**: a hair- or scale-like extension of the epidermis of a plant
- **cuticle**: a noncellular protective covering outside the epidermis of many invertebrates and plants
- **mesophyll**: the inner tissue (parenchyma) of a leaf, containing many chloroplasts.

Leaf Structure and Function

The outermost layer of the leaf is the epidermis. It consists of the upper and lower epidermis, which are present on either side of the leaf. Botanists call the upper side the adaxial surface (or adaxis) and the lower side the abaxial surface (or abaxis). The epidermis aids in the regulation of gas exchange. It contains stomata, which are openings through which the exchange of gases takes place. Two guard cells surround each stoma, regulating its opening and closing. Guard cells are the only epidermal cells to contain chloroplasts.

The epidermis is usually one cell layer thick. However, in plants that grow in very hot or very cold conditions, the epidermis may be several layers thick to protect against excessive water loss from transpiration. A waxy layer known as the cuticle covers the leaves of all plant species. The cuticle reduces the rate of water loss from the leaf surface. Other leaves may have small hairs (trichomes) on the leaf surface. Trichomes help to avert herbivory by restricting insect movements or by storing toxic or bad-tasting compounds. They can also reduce the rate of transpiration by blocking air flow across the leaf surface.

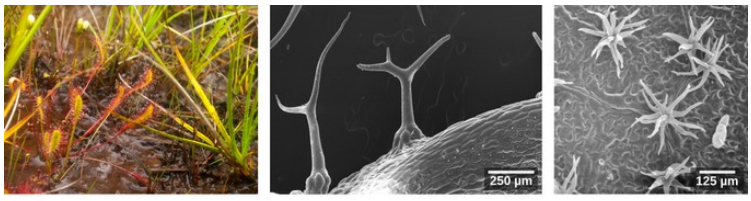
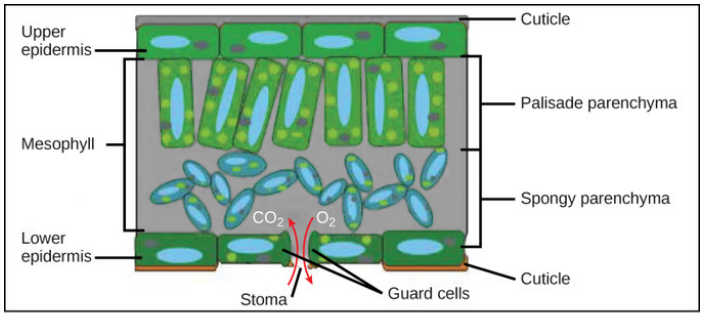


Figure 30.4C.1: **Trichomes**: Trichomes give leaves a fuzzy appearance as in this (a) sundew (*Drosera sp.*). Leaf trichomes include (b) branched trichomes on the leaf of *Arabidopsis lyrata* and (c) multibranched trichomes on a mature *Quercus marilandica* leaf.

Below the epidermis of dicot leaves are layers of cells known as the mesophyll, or “middle leaf.” The mesophyll of most leaves typically contains two arrangements of parenchyma cells: the palisade parenchyma and spongy parenchyma. The palisade parenchyma (also called the palisade mesophyll) aids in photosynthesis and has column-shaped, tightly-packed cells. It may be present in one, two, or three layers. Below the palisade parenchyma are loosely-arranged cells of an irregular shape. These are the cells of the spongy parenchyma (or spongy mesophyll). The air space found between the

spongy parenchyma cells allows gaseous exchange between the leaf and the outside atmosphere through the stomata. In aquatic plants, the intercellular spaces in the spongy parenchyma help the leaf float. Both layers of the mesophyll contain many chloroplasts.



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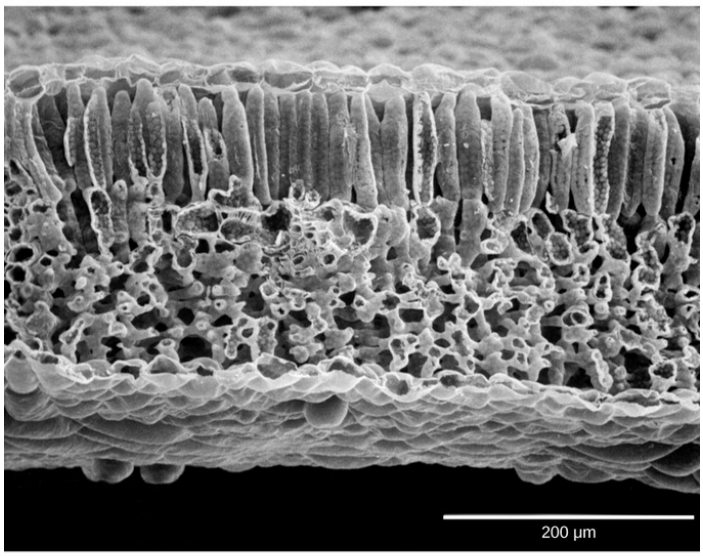

(b)

Figure 30.4C.1: **Mesophyll**: (a) (top) The central mesophyll is sandwiched between an upper and lower epidermis. The mesophyll has two layers: an upper palisade layer and a lower spongy layer. Stomata on the leaf underside allow gas exchange. A waxy cuticle covers all aerial surfaces of land plants to minimize water loss. (b) (bottom) These leaf layers are clearly visible in the scanning electron micrograph. The numerous small bumps in the palisade parenchyma cells are chloroplasts. The bumps protruding from the lower surface of the leaf are glandular trichomes.

Similar to the stem, the leaf contains vascular bundles composed of xylem and phloem. The xylem consists of tracheids and vessels, which transport water and minerals to the leaves. The phloem transports the photosynthetic products from the leaf to the other parts of the plant. A single vascular bundle, no matter how large or small, always contains both xylem and phloem tissues.

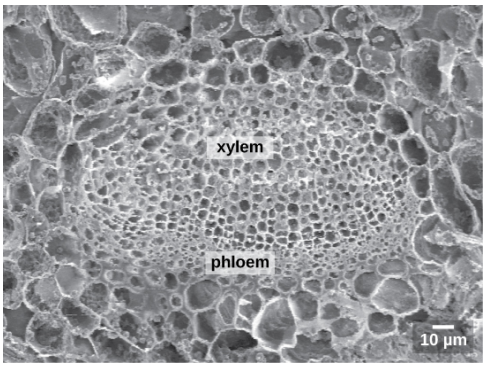

Figure 30.4C.1: **Xylem and phloem**: This scanning electron micrograph shows xylem and phloem in the leaf vascular bundle.

Leaf Adaptations

Coniferous plant species that thrive in cold environments, such as spruce, fir, and pine, have leaves that are reduced in size and needle-like in appearance. These needle-like leaves have sunken stomata and a smaller surface area, two attributes that aid in reducing water loss. In hot climates, plants such as cacti have succulent leaves that help to conserve water. Many aquatic plants have leaves with wide lamina that can float on the surface of the water; a thick waxy cuticle on the leaf surface that repels water.

Topic hierarchy

30.5A: Meristems

Plant meristematic tissues are cells that divide in order to give rise to various organs of the plant and keep the plant growing.

30.5B: Genetic Control of Flowers

30.5A: Meristems

LEARNING OBJECTIVES

- Discuss the attributes of meristem tissue and its role in plant development and growth

The adult body of vascular plants is the result of meristematic activity. Plant meristems are centers of mitotic cell division, and are composed of a group of undifferentiated self-renewing stem cells from which most plant structures arise. Meristematic cells are also responsible for keeping the plant growing. The Shoot Apical Meristem (SAM) gives rise to organs like the leaves and flowers, while the Root Apical Meristem (RAM) provides the meristematic cells for the future root growth. The cells of the shoot and root apical meristems divide rapidly and are considered to be indeterminate, which means that they do not possess any defined end fate. In that sense, the meristematic cells are frequently compared to the stem cells in animals, which have an analogous behavior and function.

Meristem tissue and plant development

Meristematic tissues are cells or group of cells that have the ability to divide. These tissues in a plant consist of small, densely packed cells that can keep dividing to form new cells. Meristematic tissue is characterized by small cells, thin cell walls, large cell nuclei, absent or small vacuoles, and no intercellular spaces.

Meristematic tissues are found in many locations, including near the tips of roots and stems (apical meristems), in the buds and nodes of stems, in the cambium between the xylem and phloem in dicotyledonous trees and shrubs, under the epidermis of dicotyledonous trees and shrubs (cork cambium), and in the pericycle of roots, producing branch roots. The two types of meristems are primary meristems and secondary meristems.

Meristem Zones

The apical meristem, also known as the “growing tip,” is an undifferentiated meristematic tissue found in the buds and growing tips of roots in plants. Its main function is to trigger the growth of new cells in young seedlings at the tips of roots and shoots and forming buds. Apical meristems are organized into four zones: (1) the central zone, (2) the peripheral zone, (3) the medullary meristem and (3) the medullary tissue.

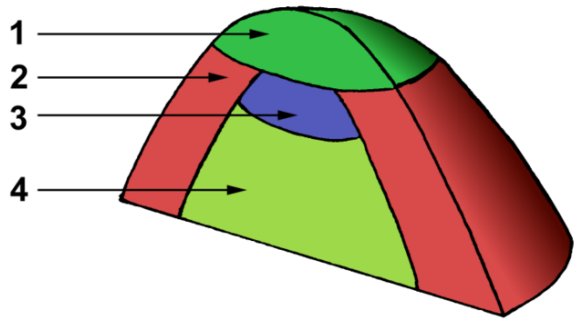

Figure 30.5A.1: **Meristematic zones**: Each zone of the apical meristem has a particular function. Pictured here are the (1) central zone, (2) peripheral zone, (3) medullary meristem and (3) medullary tissue.

Figure 30.5A.1: **Apical meristem**: The apical meristem, pictured in the center of the leaves of this image, is also termed the "growing tip". Its main function is to begin growth of new cells in young seedlings at the tips of roots and shoots (forming buds, among other things).

The central zone is located at the meristem summit, where a small group of slowly dividing cells can be found. Cells of this zone have a stem cell function and are essential for meristem maintenance. The proliferation and growth rates at the meristem summit usually differ considerably from those at the periphery. Surrounding the central zone is the peripheral zone. The rate of cell division in the peripheral zone is higher than that of the central zone. Peripheral zone cells give rise to cells which contribute to the organs of the plant, including leaves, inflorescence meristems, and floral meristems.

An active apical meristem lays down a growing root or shoot behind itself, pushing itself forward. They are very small compared to the cylinder-shaped lateral meristems, and are composed of several layers, which varies according to plant type. The outermost layer is called the tunica, while the innermost layers are cumulatively called the corpus.

Key Points

- Mitotic cell division happens in plant meristems, which are composed of a group of self-renewing stem cells from which most plant structures arise.
- The cells of the shoot and root apical meristems divide rapidly and are "indeterminate", which means that they are not designed for any specific end goal.
- The Shoot Apical Meristem (SAM) gives rise to organs like the leaves and flowers, while the Root Apical Meristem (RAM) provides cells for future root growth.
- Meristematic tissue has a number of defining features, including small cells, thin cell walls, large cell nuclei, absent or small vacuoles, and no intercellular spaces.
- The apical meristem (the growing tip) functions to trigger the growth of new cells in young seedlings at the tips of roots and shoots and forming buds.
- The apical meristem is organized into four meristematic zones: (1) central zone, (2) peripheral zone, (3) medullary meristem and (3) medullary tissue.

Key Terms

- **meristem**: the plant tissue composed of totipotent cells that allows plant growth
- **undifferentiated**: describes tissues where the individual cells have not yet developed mature or distinguishing features, or describes embryonic organisms where the organs cannot be identified
- **apical**: situated at the growing tip of the plant or its roots, in comparison with intercalary growth situated between zones of permanent tissue

30.5B: Genetic Control of Flowers

A variety of genes control flower development, which involves sexual maturation and growth of reproductive organs as shown by the ABC model.

Learning Objectives

- Diagram the ABC model of flower development and identify the genes that control that development

Key Points

- Flower development describes the process by which angiosperms (flowering plants) produce a pattern of gene expression in meristems that leads to the appearance of a flower; the biological function of a flower is to aid in reproduction.
- In order for flowering to occur, three developments must take place: (1) the plant must reach sexual maturity, (2) the apical meristem must transform from a vegetative meristem to a floral meristem, and (3) the plant must grow individual flower organs.
- These developments are initiated using the transmission of a complex signal known as florigen, which involves a variety of genes, including CONSTANS, FLOWERING LOCUS C and FLOWERING LOCUS T.
- The last development (the growth of the flower's individual organs) has been modeled using the ABC model of flower development.
- Class A genes affect sepals and petals, class B genes affect petals and stamens, class C genes affect stamens and carpels.

Key Terms

- **sepal**: a part of an angiosperm, and one of the component parts of the calyx; collectively the sepals are called the calyx (plural calyces), the outermost whorl of parts that form a flower
- **stamen**: in flowering plants, the structure in a flower that produces pollen, typically consisting of an anther and a filament
- **verticil**: a whorl; a group of similar parts such as leaves radiating from a shared axis
- **biennial**: a plant that requires two years to complete its life cycle
- **whorl**: a circle of three or more leaves, flowers, or other organs, about the same part or joint of a stem
- **apical meristem**: the tissue in most plants containing undifferentiated cells (meristematic cells), found in zones of the plant where growth can take place at the tip of a root or shoot.
- **angiosperm**: a plant whose ovules are enclosed in an ovary
- **perennial**: a plant that is active throughout the year or survives for more than two growing seasons
- **primordium**: an aggregation of cells that is the first stage in the development of an organ

Genetic Control of Flowers

Flower development is the process by which angiosperms produce a pattern of gene expression in meristems that leads to the appearance of a flower. A flower (also referred to as a bloom or blossom) is the reproductive structure found in flowering plants. There are three physiological developments that must occur in order for reproduction to take place:

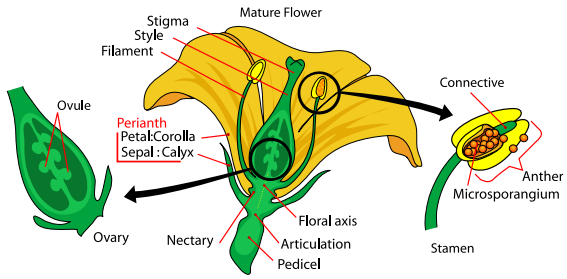

Figure 30.5B.1: **Anatomy of a flower**: Mature flowers aid in reproduction for the plant. In order to achieve reproduction, the plant must become sexually mature, the apical meristem must become a floral meristem, and the flower must develop its individual reproductive organs.

1. the plant must pass from sexual immaturity into a sexually mature state
2. the apical meristem must transform from a vegetative meristem into a floral meristem or inflorescence
3. the flowers individual organs must grow (modeled using the ABC model)

Flower Development

A flower develops on a modified shoot or axis from a determinate apical meristem (determinate meaning the axis grows to a set size). The transition to flowering is one of the major phase changes that a plant makes during its life cycle. The transition must take place at a time that is favorable for fertilization and the formation of seeds, hence ensuring maximal reproductive success. In order to flower at an appropriate time, a plant can interpret important endogenous and environmental cues such as changes in levels of plant hormones and seasonable temperature and photoperiod changes. Many perennial and most biennial plants require vernalization to flower.

Genetic Control of Flower Development

When plants recognize an opportunity to flower, signals are transmitted through florigen, which involves a variety of genes, including CONSTANS, FLOWERING LOCUS C and FLOWERING LOCUS T. Florigen is produced in the leaves in reproductively favorable conditions and acts in buds and growing tips to induce a number of different physiological and morphological changes.

From a genetic perspective, two phenotypic changes that control vegetative and floral growth are programmed in the plant. The first genetic change involves the switch from the vegetative to the floral state. If this genetic change is not functioning properly, then flowering will not occur. The second genetic event follows the commitment of the plant to form flowers. The sequential development of plant organs suggests that a genetic mechanism exists in which a series of genes are sequentially turned on and off. This switching is necessary for each whorl to obtain its final unique identity.

ABC Model of Flower Development

In the simple ABC model of floral development, three gene activities (termed A, B, and C-functions) interact to determine the developmental identities of the organ primordia (singular: primordium) within the floral meristem. The ABC model of flower development was first developed to describe the collection of genetic mechanisms that establish floral organ identity in the Rosids and the Asterids; both species have four verticils (sepals, petals, stamens and carpels), which are defined by the differential expression of a number of homeotic genes present in each verticil.

In the first floral whorl only A-genes are expressed, leading to the formation of sepals. In the second whorl both A- and B-genes are expressed, leading to the formation of petals. In the third whorl, B and C genes interact to form stamens and in the center of the flower C-genes alone give rise to carpels. For example, when there is a loss of B-gene function, mutant flowers are produced with sepals in the first whorl as usual, but also in the second whorl instead of the normal petal formation. In the third whorl the lack of B function but presence of C-function mimics the fourth whorl, leading to the formation of carpels also in the third whorl.

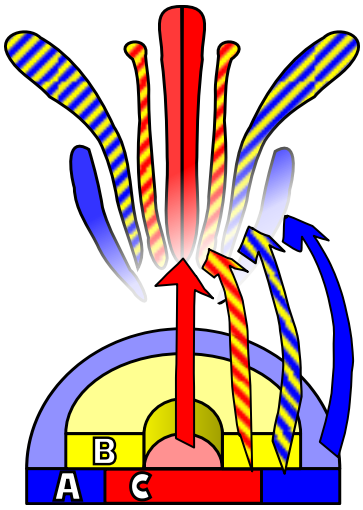

Figure 30.5B.1: **ABC model of flower development**: Class A genes (blue) affect sepals and petals, class B genes (yellow) affect petals and stamens, class C genes (red) affect stamens and carpels.

Most genes central in this model belong to the MADS-box genes and are transcription factors that regulate the expression of the genes specific for each floral organ.

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